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Sinusoidal Oscillators and Waveform Generators using Modern Electronic Circuit Building Blocks

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Preface

Sinusoidal oscillators and waveform generators have numerous applications in electronics, instrumentation, measurement, communications, control systems, and signal processing, due to which they have continued to remain a dominant and popular topic of research in the Circuits and Systems literature. Consequently, well over 1500 research papers have so far been published on the analysis, synthesis, and design of oscillators and wave form generators in various international journals. By contrast, only a handful of books have so far been written on oscillators, which suffer from one or more of the following limitations: (1) a number of books are more than a decade old; (2) several of the books deal with very specific types of sinusoidal oscillators only and, hence, have a very limited coverage; (3) a number of books deal with only general issues related to oscillators; (4) even those books which have been written after 2004 do not deal with oscillators and waveform generators using new electronic circuit building blocks which find very prominent space in modern analog circuits journals; (5) as far as could be known, there is no book written so far on non-sinusoidal relaxation oscillators or waveform generators. By contrast, the present monograph is intended to cover a wide variety of sinusoidal oscillators and waveform generators, using a variety of modern electronic circuit building blocks, which do not appear to have been dealt in any of the available books so far. This monograph is intended to provide the following:

- Single-source reference on sinusoidal oscillators and waveform generators using classical as well as modern electronic circuit building blocks (such as operational transconductance amplifiers, current conveyors and their numerous variants, current feedback operational amplifiers, differential difference amplifiers, four-terminal floating nullors, unity gain voltage/current followers, operational transresistance amplifiers, current differencing buffered amplifiers, current differencing transconductance amplifiers, current follower transconductance amplifiers, voltage differencing inverting buffered amplifier, voltage differencing differential input buffered amplifiers, and numerous others).

- State-of-the-art review of a large variety of sinusoidal oscillators and non-sinusoidal waveform generators/relaxation oscillators.
- A catalogue of over 600 chosen topologies of oscillators and waveform generators, from amongst those evolved during the last four decades, with their design details and their salient performance features/limitations highlighted.
- A comprehensive collection of well over 1500 references on oscillators and waveform generators for readers interested in further studies.
- A number of interesting research problems in almost every chapter of the monograph for the research-oriented readers.
- A useful reference for educators, students, researchers, practicing engineers, and hobbyists who have an interest in the design of sinusoidal oscillators and non-sinusoidal waveform generators/relaxation oscillators.

Lastly, we must acknowledge that in a monograph based upon over 1500 published research papers, there might have been some inadvertent omissions of some references; however, the same is not intentional. Aggrieved authors, whose works might have been omitted, are most welcome to bring to our attention (using the e-mail ID: senani@ieee.org) any missing reference(s) which we would surely like to include in the next edition of this monograph. Any other suggestions are also most welcome!

New Delhi, India
July 07, 2015

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Acknowledgements

After having written a monograph on *Current Feedback Operational Amplifiers and Their Applications* and another on *Current Conveyors*: *Variants*, *Applications and Hardware Implementations*, both published by Springer, it was rather obvious for the first author to think about writing a monograph on *Sinusoidal Oscillators*—a topic on which he and his several collaborators have worked extensively and quite intimately. Having convinced ourselves about this, we then set out to write this monograph and proposed the same to Charles Glaser, the Executive Editor, Springer, who gave us the signal to go ahead.

Let us admit that writing this monograph did not turn out to be an easy task! All the four authors went through a lot of difficulties and turmoil in their personal lives, one by one, during the entire period in which the preparation of the manuscript took place. Nevertheless, with the kind support and understanding of Charles Glaser, we somehow persisted and completed the assignment, though somewhat later than anticipated.

First of all, the authors gratefully acknowledge the contribution of their respective family members for their continued encouragement, moral support, and understanding shown during the preparation of this monograph. The authors thankfully acknowledge the facilities provided by the Analog Signal Processing Research Lab. (where this entire project was carried out), Division of Electronics and Communication Engineering and Netaji Subhas University of Technology, New Delhi, where the first author (R.S.) works. The authors would like to thank Prof. Maneesha Gupta, Dr. Tarun Rawat, and Prof. A. K. Singh for their moral support. Thanks are also due to Jessica Lauffer, Jeffrey Taub, and Nicole Lowary for their understanding and to Shashi Rawat for her support in the preparation of the manuscript.

The authors have been involved in teaching a number of ideas contained in the present monograph to their students in the UG courses on *Linear Integrated Circuits* and *Bipolar and CMOS Analog Integrated Circuits* and PG courses on *Signal Acquisition and Conditioning* and *Advanced Network Synthesis*, during which a persistent query from our students has been as to *in which book the material taught to them could be found?* We thank our numerous students for this and do hope that this monograph provides an answer to their query.

New Delhi, India

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Contents

Abbreviations

A/D	Analog to digital
ABB	Active building block
AD	Analog devices
ADC	Analog to digital converter
AGC	Automatic gain control
AGPE	All grounded passive elements
AM	Analog multiplier
AP	All pass
APF	All-pass filter
BDI	Bilinear discrete integrator
BDI-DOCC	Balanced dual input–dual output current conveyor
BE	Band elimination
BiCMOS	Bipolar complementary metal oxide semiconductor
BIT	Built-in testing
BJT	Bipolar junction transistor
BO-CCII	Balanced output-current conveyor, second generation
BO-COA	Balanced-output current operational amplifier
BOICCII	Balanced output inverting current conveyor, second generation
BOOA	Balanced output op-amps
BO-OTA	Balanced-output-operational transconductance amplifier
BO-VOA	Balanced-output voltage mode operational amplifier
BP	Band pass
BPF	Band-pass filter
BR	Band reject
BS	Band stop
BSF	Band stop filter
BW	Bandwidth
C/f	Capacitance-to-frequency
C/T	Capacitance-to-time period

CAB	Configurable analog block
CB	Complementary bipolar
CC	Current conveyor
CCC	Composite current conveyor
CC-CBTA	Current controlled current backward transconductance amplifier
CC-CC	Current-controlled current conveyor
CC-CFOA	Current controlled current feedback operational amplifier
CCCCTA	Current controlled current conveyor transconductance amplifier
CC-CDBA	Current-controlled current differencing buffered amplifier
CC-CD-CCC	Current-controlled current differencing current copy conveyor
CC-CDTA	Current-controlled current differencing transconductance amplifier
CC-CFA	Current-controlled current feedback amplifier
CC-CFOA	Current-controlled current feedback operational amplifier
CCCII	Controlled current conveyor, second generation
CCCS	Current controlled current source
CCDDCC	Current controlled differential difference current conveyor
CCDDCC	Current controllable differential difference current conveyor
CCDDCCTA	Current controlled differential difference current conveyor transconductance amplifier
CCI	Current conveyor, first generation
CCII	Current conveyor, second generation
CCIII	Current conveyor, third generation
CCO	Current controlled oscillator
CCTA	Current controlled transconductance amplifier
CCVS	Current controlled voltage source
CCW	Counterclockwise
CDA	Complimentary differential amplifier
CDBA	Current differencing buffered amplifier
CDIBA	Current differencing inverting buffered amplifier
CDTA	Current differencing transconductance amplifier
CDU	Current differencing unit
CE	Characteristic equation
CF	Current follower
CFA	Current feedback amplifier
CFBCCII	Controlled fully balanced current conveyor, second generation
CFC	Current feedback conveyor
CFOA	Current feedback operational amplifier
CFTA	Current follower transconductance amplifier
CG-CCCTA	Current gain controlled current controlled transconductance amplifier
CM	Current mode; also, current mirror
CMOS	Complementary metal oxide semiconductor
CMRR	Common mode rejection ratio

CO	Condition of oscillation
COA	Current mode operational amplifier or current-mode output
CPFSK	Continuous phase frequency shift keying
CR	Current repeater
CTTA	Current through transconductance amplifier
CVC	Current voltage conveyor
CW	Clockwise
D/A	Digital to analog
DAC	Digital to analog converter
DBTA	Differential-input buffered transconductance amplifier
DC	Direct current
DCC	Differential current conveyor
DCC-CFA	Double current-controlled current feedback amplifier
DCCCTA	Differential current controlled conveyor transconductance amplifier
DCFDCCII	Digitally controlled fully differential current conveyor, second generation
DCVC	Differential current voltage conveyor
DDA	Differential difference amplifiers
DDCC	Differential difference current conveyor
DDCCC	Differential difference complimentary current conveyor
DDCCFA	Differential difference complimentary current feedback amplifier
DDCCTA	Differential difference current conveyor transconductance amplifiers
DDOFA	Differential difference operational floating amplifier
DDOMA	Differential difference operational mirrored amplifier
DIBO-COA	Differential input balanced output-current operational amplifier
DIBO-OTA	Differential input balanced output operational transconductance amplifier
DIBO-VOA	Differential input balanced output-voltage mode operational amplifier
DI-COA	Differential input current mode operational amplifier
DIDO	Differential input differential output
DI-OTA	Differential input-operational transconductance amplifier
DISO	Differential-input-single-output
DI-VOA	Differential input voltage (mode) operational amplifier
DOCC	Dual output current conveyor
DO-DVCC	Dual-output-differential voltage current conveyor
DOICCII	Dual output inverting current conveyor, second generation
DPDT	Double-pole double-throw
DRAM	Dynamic random access memory
DVCC	Differential voltage current conveyor
DVCC+	Differential voltage current conveyor (positive-type)

DVCCC	Differential voltage complimentary current conveyor
DVCCCTA	Differential voltage current-controlled conveyor transconductance amplifier
DVCCII	Differential voltage current conveyor, second generation
DVCCS	Differential voltage controlled current source
DVCCTA	Differential voltage current conveyor transconductance amplifier
DVCFA	Differential voltage current feedback amplifier
DVCFOA	Differential voltage current feedback operational amplifier
DVCVS	Differential voltage controlled voltage source
DVTA	Differential voltage transconductance amplifier
DXCCII	Dual-X current conveyor, second generation
ECC	Extended current conveyor
ECC	Electronically controlled current conveyor
ECCII	Electronically tunable current conveyor, second generation
ECO	Explicit current output
ELIN	Externally linear but internally nonlinear
FAC	Floating admittance converter
FBCCII	Fully balanced current conveyor, second generation
FBDDA	Fully balanced differential difference amplifier
FC	Floating capacitance
FCC	Floating current conveyor
FCCNR	Floating current controlled negative resistance
FCCPR	Floating current controlled positive resistance
FDCC	Fully differential current conveyor
FDCCII	Fully differential current conveyor, second generation
FDCFOA	Fully differential current feedback operational amplifier
FDNC	Frequency-dependent negative conductance
FDNR	Frequency-dependent negative resistance
FDPR	Frequency-dependent positive resistance
FET	Field effect transistor
FGPIC/ FGPII	Floating generalized positive immittance converter/inverter
FI	Floating immittance or floating inductance or floating impedance
FM	Frequency modulation
FO	Frequency of oscillation
FPAA	Field programmable analog array
FPBW	Full power bandwidth
FPGA	Field programmable gate array
FSK	Frequency shift keying
FTFN	Four-terminal-floating-nullor
FVCR	Floating voltage controlled resistor
GBP	Gain bandwidth product
GC	Grounded capacitor
GCC	Generalized current conveyor

GCFTA	Generalized current follower transconductance amplifier
GFTC	Generalized frequency/time period converter
GI	Grounded impedance
GIC	Generalized impedance converter
GNIC	Generalized negative impedance converter
GNII	Generalized negative impedance inverter
GPIC	Generalized positive impedance converter
GPII	Generalized positive impedance inverter
GVC	Generalized voltage conveyor
HP	High pass
HPF	High-pass filter
IC	Integrated circuit
ICC	Inverting current conveyor
ICCII	Inverting current conveyor, second generation
ICCIII	Inverting current conveyor, third generation
INIC	Current inversion negative impedance converter
KHN	Kerwin–Huelsman–Newcomb
L/f	Inductance-to-frequency
L/T	Inductance-to-time period
LC	Inductance-capacitance
LDI	Lossless discrete integrator
LHS	Left hand side
LNA	Low noise amplifier
LP	Low pass
LPF	Low-pass filter
MCC-CDTA	Modified current controlled current differencing transconductance amplifier
MCCCII	Multi-output controlled current conveyor, second generation
MCCIII	Modified current conveyor, third generation
MCFOA	Modified current feedback operational amplifier
MDAC	Multiplying digital-to-analog converter
MDCC	Modified differential current conveyor
MDO-DDCC	Modified dual output-differential difference current conveyor
MICCII	Modified inverting current conveyor, second generation
MIDCC	Multiple input differential current conveyor
MIMO	Multiple-input–multiple-output
MISO	Multiple-input–single-output
MMCC	Multiplication-mode current conveyor
MOCC	Multiple output current conveyor
MO-CCCA	Multiple output current-controlled current amplifier
MO-CCCDTA	Multi-output-current-controlled current differencing transconductance amplifier
MO-CC-CTTA	Multiple output current controlled current through transconductance amplifier

MOCCII	Multiple output current conveyor, second generation
MOCF	Multiple output current follower
MOSFET	Metal oxide semiconductor field effect transistor
MOTA	Multi-output operational transconductance amplifier
MRC	MOS resistive circuit
MSO	Multi-phase sinusoidal oscillator
MTC	Mixed translinear cell
NAM	Nodal admittance matrix
NE	Node equation
NF	Notch filter
NIC	Negative impedance converter
NMOS	N-type metal oxide semiconductor
OC	Operational conveyor
OCC	Operational current conveyor
OFA	Operational floating amplifier
OFC	Operational floating conveyor
OFCC	Operational floating current conveyor
OLTF	Open loop transfer function
OMA	Operational mirrored amplifier
OTA	Operational transconductance amplifier
OTA-C	Operational-transconductance-amplifier-capacitor
OTRA	Operational transresistance amplifier
PCA	Programmable current amplifier
PIC	Positive impedance converter
PII	Positive impedance inverter
PLL	Phase locked loop
PM	Phase modulation
PMOS	P-type metal oxide semiconductor
QO	Quadrature oscillator
R/f	Resistance-to-frequency
R/T	Resistance-to-time period
RC	Resistance-capacitance
RHS	Right hand side
SCCO	Single-capacitor-controlled oscillator
SCIC	Summing current immittance converter
SCO	Switched capacitor oscillator
SEC	Single element controlled
SECO	Single-element-controlled oscillator
SFG	Signal flow graph
SIFO	Single input five output
SIMO	Single input multiple output
SIO	Switched current oscillator
SISO	Single input single output
SR	Slew rate

SRC	Single resistance controlled
SRCO	Single-resistance-controlled oscillator
SVIC	Summing voltage immittance convertor
TA	Transconductance amplifier
TAC	Transconductance and capacitance
TAM	Trans-admittance-mode
TCCII	Transconductance current conveyor, second generation
THD	Total harmonic distortion
TI	Texas Instruments
TIM	Trans-impedance-mode
TL	Trans-linear
TO-ICCII	Triple output-inverting current conveyor, second generation
TX-TZ CCII	Two-X two-Z current conveyor, second generation
UCC	Universal current conveyor
UGA	Unity gain amplifier
UGB	Unity gain buffer
UGC	Unity gain cell
UGDA	Unity gain differential amplifier
UGS	Unity gain summer
UVC	Universal voltage conveyor
VC	Voltage conveyor
VCC	Voltage-controlled capacitance
VCCS	Voltage-controlled-current-source
VCFI	Voltage controlled floating impedance
VCG	Voltage and current gain
VCG-CCII	Voltage and current gain current conveyor, second generation
VCL	Voltage controlled inductance
VCO	Voltage-controlled oscillator
VCR	Voltage-controlled-resistance
VCVS	Voltage-controlled voltage source
VCZ	Voltage-controlled impedance
VDCC	Voltage differencing current conveyor
VD-DIBA	Voltage differencing differential input buffered amplifier
VDIBA	Voltage differencing inverting buffered amplifier
VDTA	Voltage differencing transconductance amplifier
VF	Voltage follower
VFO	Variable frequency oscillator
VLF	Very low frequency
VLFO	Very low frequency oscillator
VLSI	Very large scale integrated circuits
VM	Voltage mirror; also voltage-mode
VMQO	Voltage mode quadrature oscillator
VNIC	Voltage inversion negative impedance converter
VOA	Voltage (mode) operational amplifier voltage-mode output

WBO	Wien Bridge oscillator
WCDMA	Wide-band code division multiple access
ZC-CCCITA	Z-copy current controlled current inverting transconductance amplifier
ZC-CCCITA	Z-copy current-controlled current inverting transconductance amplifier
ZC-CDU	Z-copy-current differencing unit
ZC-CG-CDBA	Z-copy current gain current differencing buffered amplifier
ZC-CG-CDTA	Z-copy current gain current differencing transconductance amplifier
ZC-CG-VDCC	Z-copy current gain voltage differencing current conveyor

Part I
Introductory Chapter

Chapter 1
Basic Sinusoidal Oscillators and Waveform Generators Using IC Building Blocks

Abstract This chapter discusses the basic principle of generating sinusoidal oscillators, reviews the classical sinusoidal oscillators and nonsinusoidal waveform generators, enumerates a number of other sinusoidal oscillator topologies, and outlines some basic methods of oscillator analysis and synthesis.

1.1 Introduction

Sinusoidal oscillators and nonsinusoidal waveform generators play an essential role in various instrumentation, measurement, communication, control, and other electronic systems, and therefore, discussion of a number of classical sinusoidal oscillators is an important topic dealt with in almost all standard text and reference books on electronics and electronic circuits (for instance, see [1–6]).

However, during the past four decades, a large number of circuits and techniques have been advanced by various researchers for realizing sinusoidal oscillators, using a variety of active devices and active circuit building blocks such as bipolar transistors, FETs, IC op-amps, operational transconductance amplifiers (OTA), current conveyors (CC), current feedback op-amps (CFOA), and numerous others. The analog circuit's literature is flooded with a huge number of papers on oscillators and their various aspects. Concurrently, a lot of effort has gone on the analysis, synthesis, and design of oscillators together with or the study of a variety of related aspects such as amplitude stabilization and control, start-up phenomenon, as well as a number of important performance-related issues like amplitude stability, frequency stability, jitter, phase noise, etc. (for instance, see [7–109]). Whereas several authors have concentrated on evolving systematic methods for realizing sinusoidal oscillators, a number of researchers have also worked on evolving design strategies or methods for improving practical performance of RC-active oscillators.

This chapter is concerned with the discussion about well-known as well as some not-so-well-known RC sinusoidal oscillators and nonsinusoidal waveform generators using op-amps in case of the former and using op-amps and IC 555 timer in case of the latter. The remaining chapters of this monograph will gradually unfold a

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variety of circuit configurations and types of oscillators and waveform generators employing modern electronic circuit building blocks.

1.2 Classical Sinusoidal Oscillators

Sinusoidal oscillators, being closed active RC circuits without functional input (other than the DC power supplies used for biasing the active device(s)), can be analyzed in a number of ways. From the classical theory of feedback oscillators, the sinusoidal oscillator, which is usually made from an amplifier and a frequency-selective RC circuit or LC circuit arranged in a positive feedback, can be analyzed by using Barkhausen criterion. A typical block diagram model of an oscillator shown in Fig. 1.1 contains an amplifier of gain A and a frequency-selective feedback network having gain β.

According to Barkhausen criterion, in order that such a system can generate sinusoidal oscillations, the necessary conditions are $|A\beta| = 1$ and $\angle A\beta = 0°$ or an integral multiple of 360°.

It is well known that a number of classical oscillators such as a Wien bridge oscillator, RC phase-shift oscillator, twin-T oscillator, and bridge-T oscillators can be readily seen to be belonging to the general block diagram of Fig. 1.1.

1.2.1 *Wien Bridge Oscillator*

In case of the Wien bridge oscillator (Fig. 1.2a), the op-amp is configured as non-inverting amplifier, whereas the feedback network consisting of two resistors and two capacitors is a second-order band-pass filter such that at the center frequency of the band-pass filter given by

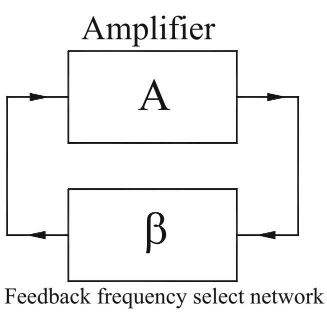


Fig. 1.1 The basic topology of the classical sinusoidal oscillators

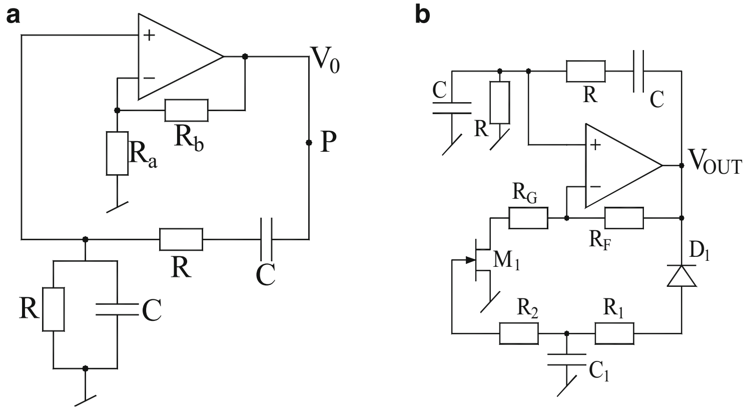


Fig. 1.2 Wien bridge oscillator (WBO). (**a**) The basic WBO, (**b**) WBO with amplitude control [76]

$$\omega_0 = \frac{1}{RC} \tag{1.1}$$

the phase shift contributed by the band-pass filter becomes exactly zero. Since non-inverting amplifier does not have a phase shift between its input and output, this makes the total phase shift around the loop exactly equal to zero. On the other hand, at the frequency ω_o, the passive RC band-pass filter has a voltage gain (attenuation) $= \frac{1}{3}$ so that the non-inverting amplifier must have a voltage gain equal to 3 so that at the frequency ω_o, the total loop gain becomes unity. This can be shown mathematically as follows:

With the feedback loop opened at node "P," the open-loop transfer function of the circuit of Fig. 1.2a is found to be

$$\frac{V_o}{V_{in}} = \frac{\left(1 + \frac{R_b}{R_a}\right) sCR}{s^2C^2R^2 + 3sCR + 1} \tag{1.2}$$

From the above function, equating $|A\beta| = 1$ and $\angle A\beta = 0$ leads to

$$\frac{\omega CR\left(1 + \frac{R_b}{R_a}\right)}{\sqrt{\left(1 - \omega^2C^2R^2\right)^2 + (3\omega CR)^2}} = 1 \tag{1.3}$$

and

$$\phi = \tan^{-1}(\infty) - \tan^{-1}\left(\frac{3\omega CR}{1 - \omega^2C^2R^2}\right) = 0 \tag{1.4}$$

From the above, therefore, the condition of the oscillation is given by

$$\left(1 + \frac{R_b}{R_a}\right) = 3 \tag{1.5}$$

whereas the frequency of oscillation is given by Eq. (1.1).

It is worth pointing out that the linear analysis of the oscillators, as above, does not throw any light on the amplitude of the oscillations; however, since the transfer characteristics of the amplifier is having a saturation type nonlinearity, it limits the amplitude level ultimately. Thus, to get an effective stabilization of the amplitude, an additional automatic gain control (AGC) loop is required to be added to the basic circuit. One such arrangement has been shown here in Fig. 1.2b. In this scheme, the generated sinusoidal signal is rectified and filtered, and the resulting DC voltage is applied at the gate of an FET used as a voltage-controlled resistor (VCR) to control the gain of the amplifier such that the oscillation magnitude can be kept at a constant level by decreasing or increasing the gain of the non-inverting amplifier depending upon whether the output sinusoidal signal amplitude is increasing or decreasing.

1.2.2 RC Phase-Shift Oscillators

We next consider the classical RC phase-shift oscillator as shown in Fig. 1.3a and its RC:CR transformed version (excluding the resistors which appear as ratio in the transfer function of the original circuit) shown in Fig. 1.3b [1–6].

In the circuit of Fig. 1.3a, the op-amp inverting amplifier provides a phase shift of 180°, whereas the third-order RC circuit provides the remaining phase shift of 180° at a specific frequency

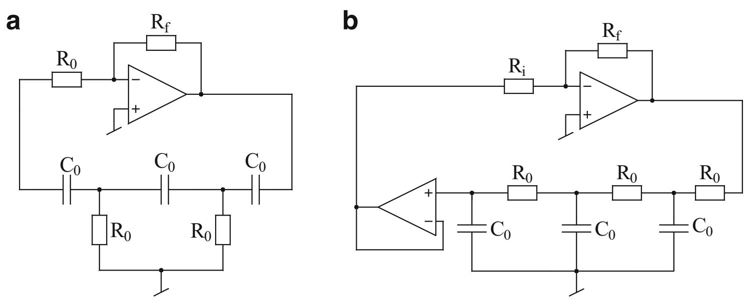


Fig. 1.3 RC phase-shift oscillators [1–6]. (**a**) The basic circuit, (**b**) RC:CR transformed version

$$\omega_o = \frac{1}{\sqrt{6}C_0R_0} \tag{1.6}$$

thereby making the total phase shift around the loop as 360°. Note that the feedback RC circuit is, in fact, a third-order high-pass filter. It is interesting to note that the third grounded resistor of the CR ladder has been connected to the virtual ground input of the op-amp instead of the actual ground node. However, while, on one the hand, this resistor assumes the role of the third resistor of the CR ladder, on the other hand, this resistor, in conjunction with the op-amp's feedback resistor R_f, constitutes the inverting amplifier having voltage gain equal to $\left(-\frac{R_f}{R_0}\right)$. However, at the frequency ω_o, the RC feedback circuit has the voltage gain $= \frac{1}{29}$ so that to make the loop gain equal to unity, the inverting amplifier must provide a gain of 29. This is verified by the analysis as follows:

The open-loop transfer function of the oscillator circuit of Fig. 1.3a is found to be

$$T(s) = \frac{-\left(\frac{R_f}{R_0}\right)\left(s^3C_0^3R_0^3\right)}{s^3C_0^3R_0^3 + 6s^2C_0^2R_0^2 + 5sC_0R_0 + 1} \tag{1.7}$$

from where the characteristic equation (CE) of the closed-loop circuit can be written as {1-T(s)} = 0, thereby leading to the equation

$$\left(1 + \frac{R_f}{R_0}\right)s^3C_0^3R_0^3 + 6s^2C_0^2R_0^2 + 5sC_0R_0 + 1 = 0 \tag{1.8}$$

Equation (1.8) is of the general form

$$a_3s^3 + a_2s^2 + a_1s + a_0 = 0 \tag{1.9}$$

from where the condition of oscillation (CO) is given by

$$\frac{a_1}{a_3} = \frac{a_0}{a_2}, \text{ thus, leading to} \left(\frac{R_f}{R_0}\right) = 29 \tag{1.10}$$

while the frequency of oscillation (FO) is given by

$$\omega_0 = \sqrt{\frac{a_1}{a_3}} = \sqrt{\frac{a_0}{a_2}} = \frac{1}{\sqrt{6}R_0C_0} \tag{1.11}$$

In the RC:CR transformed version of Fig. 1.3b, the frequency-selective feedback circuit is a third-order RC low-pass filter, and a voltage follower is required to ensure that no current flows into the inverting amplifier. It is worth mentioning that most of the text/reference books and IC manufacturer's data sheets show this circuit

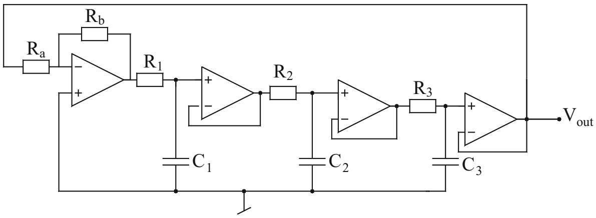


Fig. 1.4 Buffered RC feedback oscillators

without this buffer, assuming that the current taken by the inverting amplifier is negligibly small which is not very well justified. An analysis of this circuit yields the open-loop transfer function:

$$T(s) = -\left(\frac{R_{\mathrm{f}}}{R_i}\right)\frac{1}{s^3C_0^3R_0^3 + 5\,s^2C_0^2R_0^2 + 5s\,C_0R_0 + 1} \tag{1.12}$$

The CO remains the same as in Eq. (1.10), while FO of this oscillator is given by the following equation:

$$\omega_0 = \frac{\sqrt{6}}{C_0R_0} \tag{1.13}$$

Yet another variant of the classical RC phase-shift oscillator (described in Texas Instruments application note [76] *without the fourth buffer*) is the so-called buffered RC phase-shift oscillator shown in Fig. 1.4.

The analysis of the circuit leads to the following open-loop transfer function:

$$T(s) = -\left(\frac{R_{\mathrm{b}}}{R_{\mathrm{a}}}\right)\frac{1}{(1 + sC_0R_0)^3} \text{ for } \mathrm{R_1 = R_2 = R_3 = R_0} \text{ and } \mathrm{C_1 = C_2 = C_3 = C_0} \tag{1.14}$$

from where, through a straightforward analysis, the CO and FO are found to be

$$\frac{R_{\mathrm{b}}}{R_{\mathrm{a}}} = 8 \text{ and } \omega_0 = \frac{\sqrt{3}}{C_0R_0} \tag{1.15}$$

This oscillator can be best implemented by a quad op-amp IC. Lastly, yet another RC phase-shift oscillator which employs four identical low-pass RC sections rather than three is the Bubba oscillator of which a slightly modified form obtained with one more buffer is shown in Fig. 1.5. It may be pointed out that the version shown here differs from that shown in [76] in terms of one additional buffer employed here, though for convenient practical realizability, the circuit with a single quad op-amp IC, this additional buffer may be dispensed with.

In this circuit, each RC circuit contributes a phase shift of 45°. The CO and FO for this may be easily verified to be

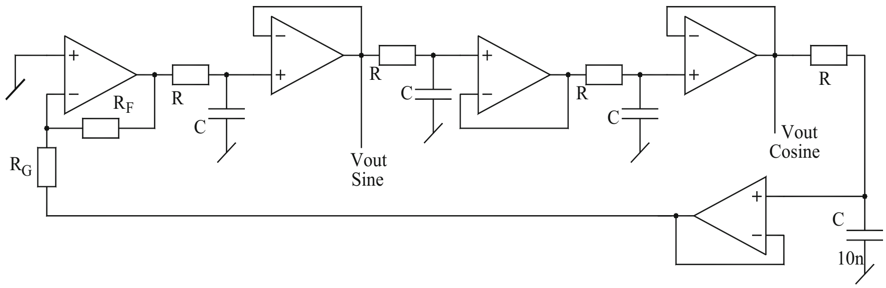


Fig. 1.5 Bubba oscillator with an additional buffer incorporated after the fourth RC section

$$\frac{R_F}{R_G} = 4 \text{ and } \omega_0 = \frac{1}{RC} \tag{1.16}$$

When compared among them, it is obvious that the Bubba oscillator is the one which requires the least value of the gain. Furthermore, even this gain requirement can be distributed over all the op-amps to reduce distortion. Thus, this oscillator is the one which gives the least distortion coupled with good frequency stability (which is to be elaborated further subsequently), though this improved performance comes at the cost of more number of op-amps than the classical single-op-amp-based RC phase-shift oscillators.

1.2.3 Colpitts and Hartley Oscillators

While active RC oscillators are suitable for low to medium frequencies, for generating high-frequency signals, usually LC oscillators are preferred. Two popular high-frequency oscillators are Colpitts and Hartley oscillators both of which employ LC resonant circuits as the frequency-selective networks along with an amplifier which could be a BJT amplifier, an FET multiplier, or the one made from IC op-amps.

The Colpitts oscillator is an LC oscillator which contains a tuned tank circuit consisting of one inductor and two capacitors in the form of a three terminal circuit; the two capacitors therein are actually making a capacitive voltage divider. An operational transconductance amplifier (OTA)-based implementation of the Colpitts oscillator is shown in Fig. 1.6a, wherein the three terminal LC networks are connected in such a manner that between two nodes of the three terminal LC circuits, an inverting transconductance amplifier of gain $-G_m$ is connected, whereas the common node of the two capacitors is connected to ground. The Colpitts oscillator results in the production of sinusoidal signal of good purity.

By routine circuit analysis, the oscillation frequency of the Colpitts oscillators is found to be

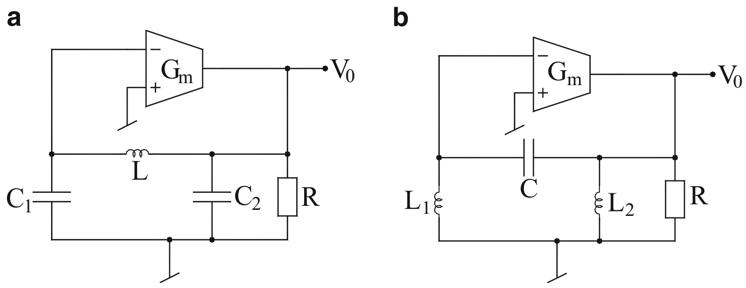


Fig. 1.6 Two popular high-frequency LC oscillators. (**a**) Colpitts oscillator, (**b**) Hartley oscillator

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{(C_1 + C_2)}{LC_1C_2}} \tag{1.17}$$

while the condition required to maintain oscillations is given by

$$G_m R \geq \frac{C_1}{C_2} \tag{1.18}$$

The Hartley oscillator uses an alternative form of the LC tank which can be obtained from that of Colpitts oscillator by interchange inductor by capacitor and the two capacitors by two inductors as shown in Fig. 1.6b.

For the Hartley oscillator, the oscillation frequency is given by

$$f_0 = \frac{1}{2\pi\sqrt{L_T C}} \tag{1.19}$$

where the total inductance L_T is $(L_1 + L_2)$ if the two inductors are wound on separate cores and would be equal to $(L_1 + L_2 + 2\text{M})$ when the two inductors are wound on a common core, in which case M represents the mutual inductance. The condition of oscillation is found to be

$$G_m R \geq \frac{L_1}{L_2} \tag{1.20}$$

1.2.4 A Family of Canonic Single-Op-Amp Oscillators

Bhattacharyya, Sundaramurthy, and Swamy [39] carried out a systematic study for evolving all possible canonic oscillators using a single op-amp by a systematic consideration of all possible two resistors-two capacitors canonic three terminal RC oscillators. A set of 12 circuits were found – however, only six of them were shown to be realizable with a conventional op-amp configured as a non-inverting amplifier. These circuit configurations are shown in Fig. 1.7.

In all the above cases, the frequency of oscillation is given by

$$\omega_0 = \frac{1}{\sqrt{C_1 C_2 R_1 R_2}} \tag{1.21}$$

The condition of oscillation for the circuits of Fig. 1.7a, b, and c is given by

$$(2R_4 - R_3) \leq 0 \text{ for } R_1 = R_2 = R \text{ and } C_1 = C_2 = C \tag{1.22}$$

On the other hand, for the circuits of Fig. 1.7d, e, and f, it is given by

$$(R_4 - 2R_3) \leq 0 \text{ for } R_1 = R_2 = R \text{ and } C_1 = C_2 = C \tag{1.23}$$

Subsequently, Boutin [44] carried out a systematic derivation of all possible canonic single-op-amp four resistors-two capacitors oscillators and came to the conclusion that only twelve such circuits are possible. Out of these 12 oscillators, 6 circuits are already derived in [38] which have been shown in Fig. 1.7; the remaining six oscillator configurations are shown here in Fig. 1.8.

An important point to be noticed is that whereas the Wien bridge and some other oscillators require gain of 3 to maintain sustained oscillations, several others require only half of this gain. Consequently, their maximum frequency range of operation will be nearly twice that of the Wien bridge oscillator for a given gain-bandwidth product of the op-amp employed.

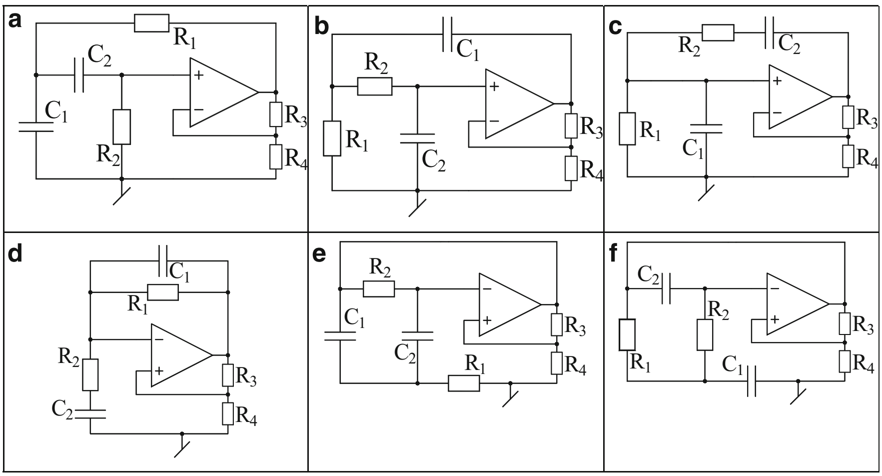


Fig. 1.7 (**a–f**) The family of canonic single-op-amp oscillators derived by Bhattacharyya, Sundaramurthy, and Swamy [39]

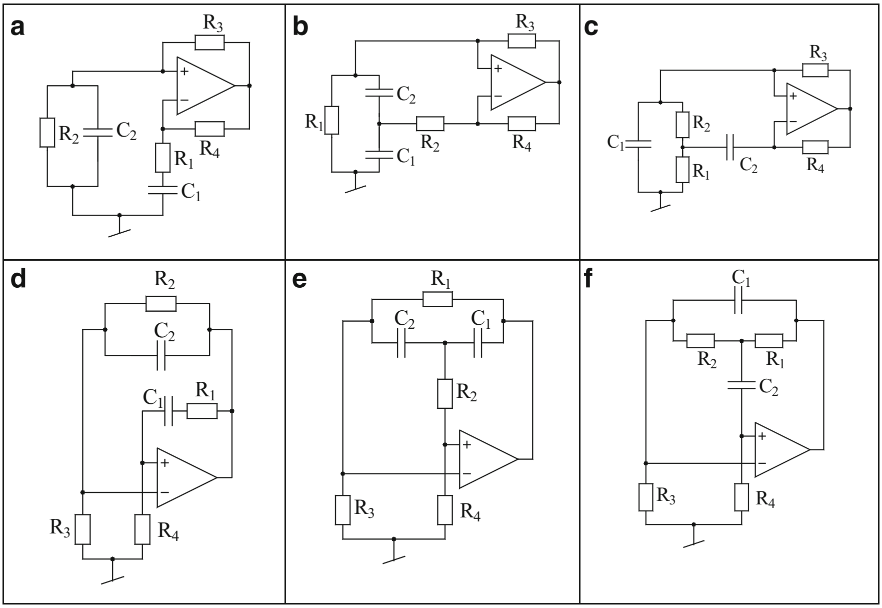


Fig. 1.8 (**a–f**) Some of the canonic single-op-amp sinusoidal oscillators from among those proposed by Boutin [44]

1.2.5 Twin-T Oscillators

The classical twin-T oscillators employ, along with an amplifier, two T-networks connected in parallel which are high-pass and low-pass networks exhibiting a frequency response which shows a null at a certain frequency ω_0 subject to the fulfillment of appropriate relationship between the RC component values (Fig. 1.9a).

At exactly the null frequency, the circuit has positive feedback to the non-inverting terminal of the op-amp, and the circuit thus oscillates at this frequency ω_0.

By a straightforward analysis of the circuit, the open-loop transfer function is found to be

$$T(s) = \frac{-1}{s^2C^2R^2} \text{ for } R_1 = R_2 = R, C_1 = C_2 = C, R_3 = \frac{R}{2} \text{ and } C_3 = 2C \quad (1.24)$$

Thus, the circuit oscillates at a frequency

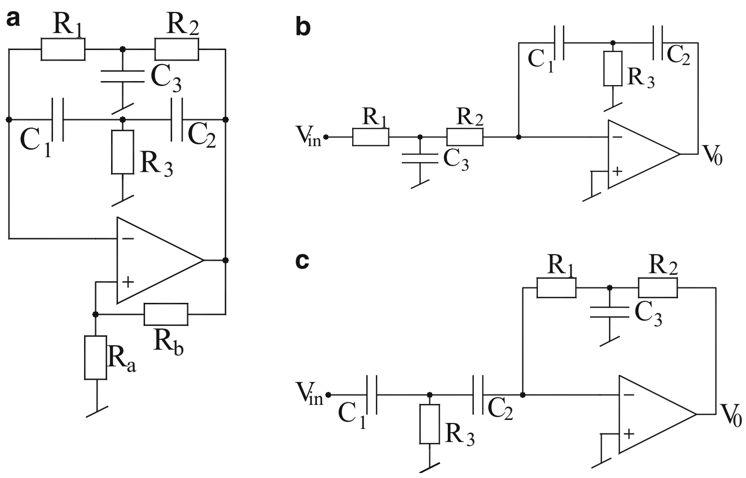


Fig. 1.9 Twin-T oscillators. (**a**) The basic twin-T oscillator configuration, (**b**) the double integrator, (**c**) the double differentiator

$$\omega_0 = \frac{1}{RC} \tag{1.25}$$

It is interesting to mention that a simplified version of the twin-T oscillator can give rise to two other op-amp circuits, namely, the inverting double integrator of Fig. 1.9b and the inverting double differentiator of Fig. 1.9c.

In the first case, the open-loop transfer function is found to be

$$T(s) = -\frac{[s(C_1 + C_2)R_3 + 1]}{s^2C_1C_2R_3\left[sC_3R_1R_2 + {}^2R_1 + R_2\right]} \tag{1.26}$$

With $C_1 = C_2 = C$, $C_3 = 2C$, $R_1 = R_2 = R$, $R_3 = R/2$, the transfer function reduces to $T(s) = -1/s^2C^2R^2$ which represents an inverting double integrator.

In the latter case, the transfer function is given by

$$T(s) = -\frac{s^2C_1C_2R_3[sC_3R_1R_2 + R_1 + R_2]}{[s(C_1 + C_2)R_3 + 1]} \tag{1.27}$$

With $C_1 = C_2 = C$, $C_3 = 2C$, $R_1 = R_2 = R$, $R_3 = R/2$, the transfer function reduces to $T(s) = -s^2C^2R^2$ which represents an inverting double differentiator.

Thus, the twin-T oscillator with an infinite gain op-amp can also be obtained from either of these two circuits by returning their inputs to the output terminal of the op-amps. Consequently, in both cases, exactly the same characteristic equation (CE) $(s^2C^2R^2 + 1) = 0$ results from which the oscillation frequency is given by $\omega_0 = 1/RC$.

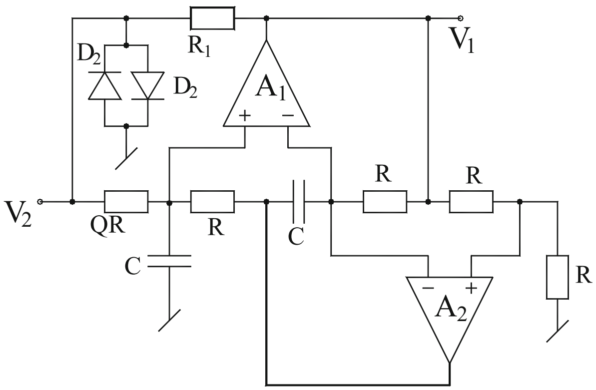


Fig. 1.10 The band-pass filter-tuned oscillator [6]

1.2.6 A Band-Pass Filter-Tuned Oscillator

Another popular op-amp oscillator is the so-called band-pass filter-tuned oscillator, shown in Fig. 1.10, well documented in standard texts on electronics (for instance, see [2]). The circuit employs a simulated inductor using a generalized impedance converter (GIC) made from two op-amps along with four identical resistors equal to R and a capacitor, thereby simulating an inductor of value $L = CR^2$, which in conjunction with another capacitor of value C and resistor of value QR makes the band-pass filter. The hard limiter is made from resistors R_1 and the two back-to-back connected diodes D_1 and D_2. Output of the limiter is feed back to the input of the band-pass filter. Assuming that the output of the limiter is symmetrical square wave, when this is proposed by a high Q band-pass filter, the result is a sine wave of frequency equal to the $\omega_0 = 1/cR$ fundamental frequency of the square wave which would be a sine wave of frequency. The quality factor of the filter determines the spectral priority of the generated sine waves.

1.3 Quadrature and Multiphase Sinusoidal Oscillators

Quadrature oscillators find numerous applications in communication and measurements. Some common applications of quadrature oscillators in communication are in quadrature mixers and single sideband modulators. On the other hand, in measurement applications, quadrature oscillators are useful in vector generators and selective voltmeters. In the following, we highlight a number of circuits and techniques of realizing quadrature oscillators of varying complexity using IC op-amps.

1.3.1 Quadrature Oscillators

The simplest way of creating a quadrature oscillator is to cascade two inverting integrators along with an inverting amplifier and close the loop or alternatively create a closed-loop circuit containing one inverting and one non-inverting integrator. A classical state-variable oscillator, which was also employed in analog computer simulations in the era of *analog computers*, is based upon the first idea and is shown in Fig. 1.11a.

For this circuit, the open-loop function is readily found as

$$T(s) = \left(-\frac{R_4}{R_3}\right)\left(-\frac{1}{sC_1R_1}\right)\left(-\frac{1}{sC_2R_2}\right) \tag{1.28}$$

thereby leading to the closed-loop CE as

$$s^2 + \frac{R_4}{C_1C_2R_1R_2R_3} = 0 \tag{1.29}$$

from which the frequency of oscillation is found to be

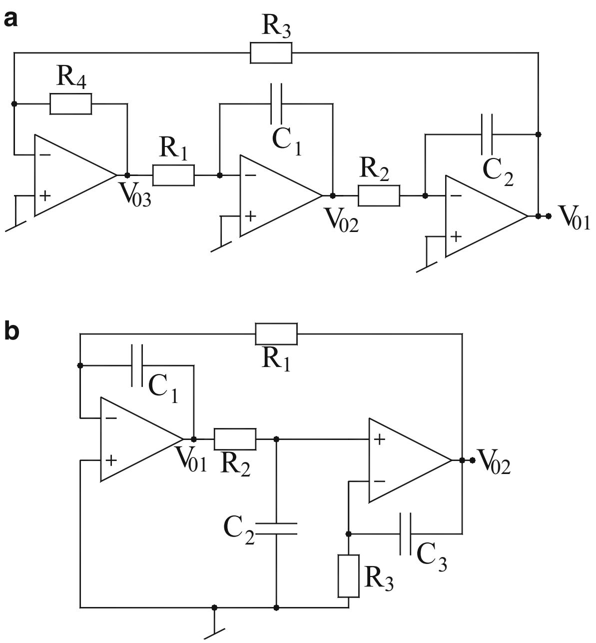


Fig. 1.11 Quadrature oscillator topologies (**a**) using three inverting stages and (**b**) using one inverting and one non-inverting integrator

$$\omega_0 = \sqrt{\frac{R_4}{C_1 C_2 R_1 R_2 R_3}} \tag{1.30}$$

Clearly, it is seen that the circuit does not have any way to move the roots of the CE slightly to the right half of s-plane as would be needed to ensure that oscillations do not fail to build up. However, curiously, this is an exceptional oscillator which does not appear to fail to start in spite of this *apparent* difficulty!

The second circuit (Fig. 1.11b) has the same underlying principle except that the non-inverting integrator is made from a cascade of a passive RC section and a non-inverting op-amp-RC stage. By inspection, the open-loop transfer function for this circuit can be written as

$$T(s) = \left(-\frac{1}{sC_1R_1}\right)\left(\frac{1}{sC_2R_2 + 1}\right)\left(\frac{1 + sC_3R_3}{sC_3R_3}\right) \tag{1.31}$$

If we take $C_2R_2 = C_3R_3$, the FO is then given by

$$\omega_0 = \sqrt{\frac{1}{C_1 C_2 R_1 R_2}} \tag{1.32}$$

RC oscillators with precise 90° phase shift between two low impedance output signals can also be made from a cascade of a single-op-amp all-pass filter along with an inverting integrator as in Fig. 1.12a or with an all-pass filter along with differentiator as in Fig. 1.12b.

When the two time constants are chosen to be equal, then the two voltage signals are not only in quadrature but also have equal amplitudes. Furthermore, in such a case, the amplitude stabilization has no interaction with oscillation frequency. For the circuit of Fig. 1.12a, the CE is given by

$$s^2 C_1 C_2 R_1 R_2 + sC_2(R_2 - R_1) + 1 = 0 \tag{1.33}$$

from which the condition of oscillation (CO) and frequency of oscillation (FO) can be found as

$$\text{CO}: (R_2 - R_1) \leq 0 \text{ and FO}: \omega_0 = \sqrt{\frac{1}{C_1 C_2 R_1 R_2}} \tag{1.34}$$

For the circuit of Fig. 1.12b, the CE is obtained as

$$s^2 C_1 C_2 R_1 R_2 + s(C_1R_1 - C_2R_2) + 1 = 0 \tag{1.35}$$

Thus, CO and FO are given by

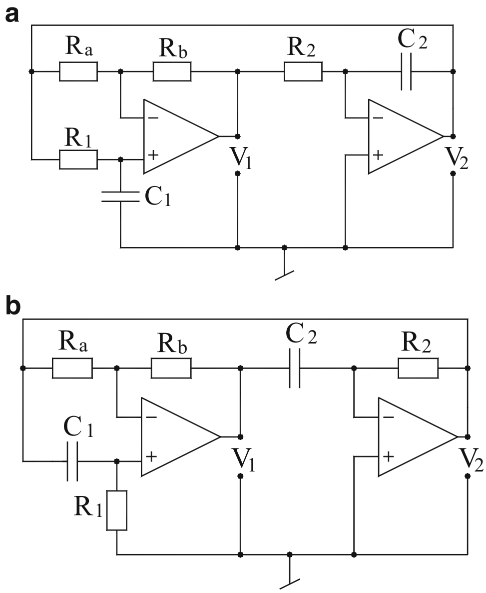


Fig. 1.12 Two alternative quadrature oscillator topologies (**a**) using an all-pass filter and an inverting integrator and (**b**) using an all-pass filter and an inverting differentiator

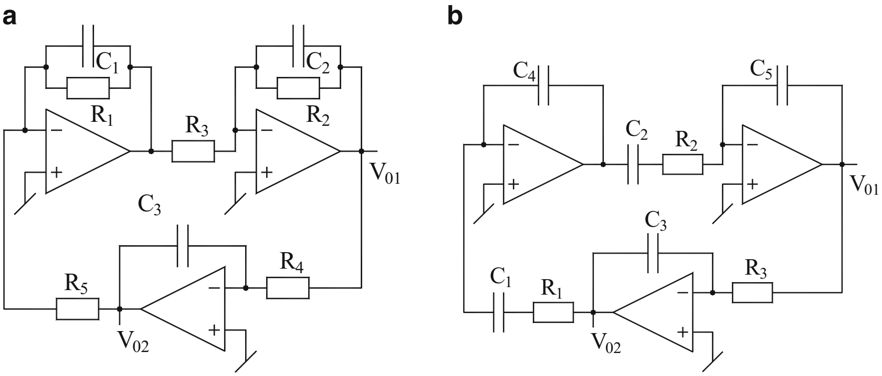


Fig. 1.13 (**a**, **b**) Quadrature oscillators based on third-order CE proposed by Horng [95]

$$\text{CO}: (C_1R_1 - C_2R_2) \leq 0 \text{ and FO}: \omega_0 = \sqrt{\frac{1}{C_1C_2R_1R_2}} \tag{1.36}$$

Horng [95] proposed two new quadrature oscillator circuits based on third-order characteristic equation (CE). These circuits are shown in Fig. 1.13. The CE of the first circuit (Fig. 1.13a) is given by

$$s^3C_1C_2C_3R_1R_2R_3R_4R_5 + s^2C_3R_3R_4R_5(C_1R_1 + C_2R_2) + sC_3R_3R_4R_5 + R_1R_2 = 0 \tag{1.37}$$

from which CO and FO are found to be

$$\mathrm{CO}: R_3R_4R_5 = \frac{C_1C_2R_1^2R_2^2}{C_3(C_1R_1 + C_2R_2)} \text{ and FO}: \omega_0 = \frac{1}{\sqrt{C_1C_2R_1R_2}} \tag{1.38}$$

It is found that outputs V_{01} and V_{02} are in quadrature since the voltage transfer function between them is found to be

$$\frac{V_{02}(s)}{V_{01}(s)} = -\frac{1}{sC_3R_4} \tag{1.39}$$

On the other hand, the CE for the circuit of Fig. 1.13b is found to be

$$s^3C_1C_2C_3C_4C_5R_1R_2R_3 + s^2C_3C_4C_5R_3(C_1R_1 + C_2R_2) + sC_3C_4C_5R_3 + C_1C_2 = 0 \tag{1.40}$$

In this case, the CO and FO are found to be

$$\mathrm{CO}: R_3 = \frac{{C_1}^2{C_2}^2R_1R_2}{C_3CC_5(C_1R_1 + C_2R_2)} \text{ and FO}: \omega_0 = \frac{1}{\sqrt{C_1C_2R_1R_2}} \tag{1.41}$$

In this circuit also V_{02} and V_{01} are in quadrature as the transfer function between them is given by

$$\frac{V_{02}(s)}{V_{01}(s)} = -\frac{1}{sC_3R_3} \tag{1.42}$$

It is easy to see from the above equations that CO and FO are orthogonally controllable in both the circuits. The workability of these circuits has been confirmed [95], by constructing them using LF351 op-amps, and it has been possible to obtain variable frequency quadrature signals from both the circuits over a range of one decade (up to a frequency of 140 KHz). It may be noted that since LF351 is an FET-input op-amp, not having a DC path for the inverting input terminal of one of the op-amps in the second circuit does not pose any difficulties which would have arisen had the general-purpose op-amp μA741 been used to implement these circuits.

1.3.2 Multiphase Oscillators

Like quadrature oscillators, multiphase oscillators (**n**-phase oscillators) providing three, four, or six sinusoidal signals having equal phase shift between them given by 360°/**n** also have potential applications in communication and some power electronic systems. For example, 4-phase oscillators are useful in realizing subharmonic mixers to reduce noise and distortion, while in general, multiphase oscillators find applications in power electronic circuits and subharmonically pumped frequency conversion circuits. In the following, we discuss some prominent methods of generating multiphase sinusoidal oscillations using IC op-amps.

A scheme of generating multiphase sinusoidal oscillations using op-amps was advanced by Gift [71] in which "**n**" identical first-order low-pass filter stages are cascaded along with an inverting amplifier in a closed loop, with the output of each first-order stage also available explicitly in its original form as well as inverting form through "**n**" number of inverting amplifier stages. This general scheme is shown in Fig. 1.14a, whereas a special case of the general ***n***-phase structure with $\mathbf{n} = 3$ takes the form as shown in Fig. 1.14b.

If each first-order low-pass filter section is characterized by ($K/1 + sT$), where K is the maximum gain and T is the time constant of the circuit, the loop gain for the general nth-order configuration can then be written as

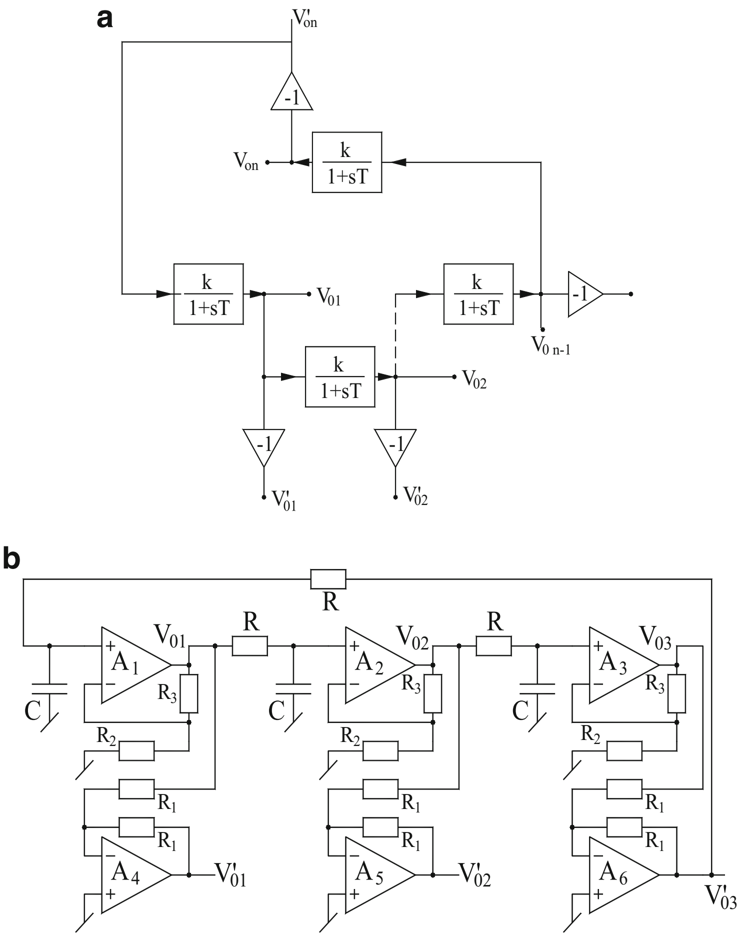


Fig. 1.14 Multiphase sinusoidal oscillator based on first-order non-inverting low-pass filters proposed by Gift [71]. (**a**) General scheme, (**b**) a special case for $n = 3$

$$\frac{V_{0n}^{/}}{V_{\text{in}}} = -\left(\frac{K}{1+sT}\right)^{n}; \text{where } T = RC \text{ and } K = 1 + \frac{R_3}{R_2} \tag{1.43}$$

Applying Barkhausen criterion of oscillation, therefore, one can write

$$(1 + j\omega T)^{n} + K^{n} = 0 \tag{1.44}$$

From the above, one can find the oscillation frequency and the gain required from each section for specific values of n. For instance, for $n = 3$, equating the real and imaginary parts to zero, one can obtain the CO and FO as

$$\text{CO} : K = 2 \text{ and FO} : f_0 = \frac{\sqrt{3}}{2\pi T} \tag{1.45}$$

Note that by means of three inverting amplifiers added to the structure, the circuit is capable of providing 6-phase oscillations. The workability of this circuit has been verified [71] by testing the circuit using op-amps with higher gain bandwidth products such as Harris HA2544 having GBP of 50 MHz, and it has been found that the errors in frequency generated by the circuit of Fig. 1.14b (designed for 125 KHz) were within 2 % of the theoretical values.

Gift [72] demonstrated that other similar schemes can be employed to generate multiphase oscillations for even/odd order "**n**" using a cascade of first-order inverting low-pass filters connected in a loop (see Fig. 1.15a). An exemplary implementation for the third-order 3-phase oscillator from [72] is shown in Fig. 1.15b.

It is easy to see that oscillations will occur at a frequency ω_0 according to

$$\left(\frac{-K}{1+sT}\right)^{n}_{s=j\omega_0} = 1 \tag{1.46}$$

Equation (1.46) can be rewritten as

$$(1 + j\omega_0 T)^{n} + (-1)^{n+1} K^{n} = 0 \tag{1.47}$$

It is found that the above equation has a solution only if n is odd ($n \geq 3$). For $n = 3$, solving the above equation by simultaneously equating real and imaginary parts equal to zero gives, the CO and FO which are found to be the same as for the previous circuit.

Yet another technique of designing even and odd phase sinusoidal oscillators using all-pass filters was presented by Gift [75]. The scheme for n-phase sinusoidal signal generation shown in Fig. 1.16 involves a cascade of n first-order phase lag-type all-pass filters connected in a closed loop, whereas, by using additionally n-inverting amplifiers, each one from the output of each all-pass filter leads to a modified scheme capable of generating $2n$ even-phase sinusoidal oscillators.

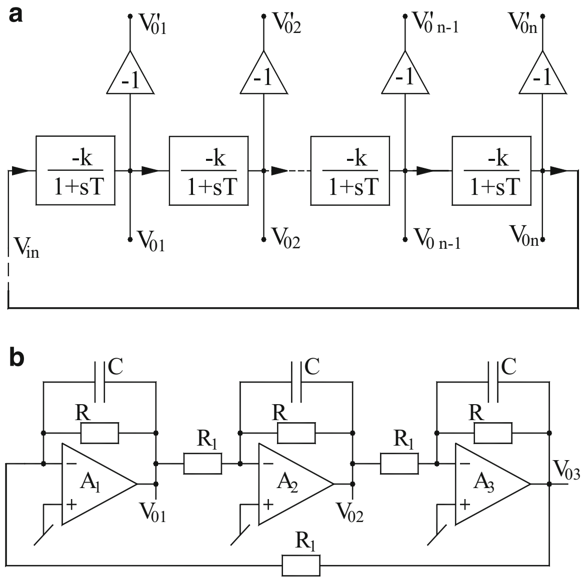


Fig. 1.15 Another method for multiphase oscillator realization [72]. (**a**) The generalized scheme for n-phase oscillator realization using inverting low-pass stages, (**b**) a 3-phase oscillator

For a general n-phase case, the Barkhausen criterion gives

$$T(j\omega) = \left(\frac{1 - j\omega\tau}{1 + j\omega\tau}\right)^n = 1; \quad \tau = \mathrm{RC} \tag{1.48}$$

Clearly, $|T(j\omega_0)| = 1$ is satisfied automatically due to all the first-order sections being all-pass filters with gain of unity. On the other hand, $\angle T(j\omega_0) = -2\pi$ leads to the equation

$$n\left(-2\tan^{-1}(\omega_0\tau)\right) = -2\pi \tag{1.49}$$

from which the FO is found to be

$$f_0 = \frac{1}{2\pi RC}\tan\left(\frac{\pi}{n}\right) \tag{1.50}$$

whereas the phase shift produced by each stage is given by

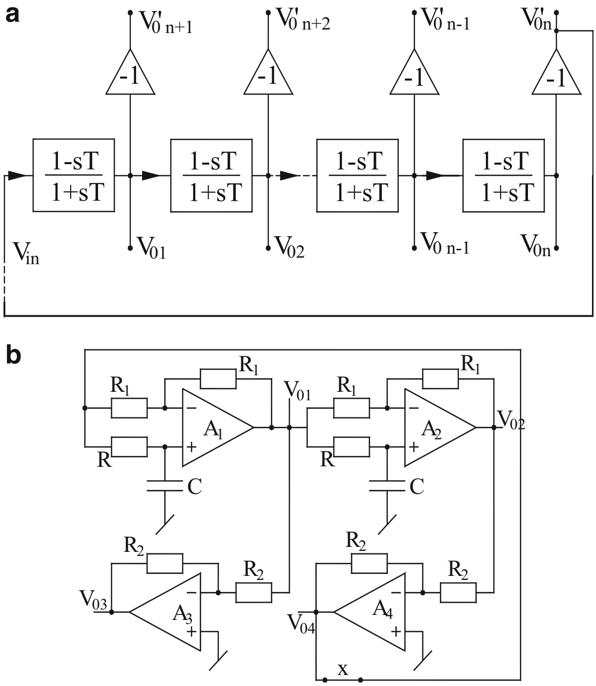


Fig. 1.16 Multiphase oscillator realization using first-order all-pass sections [75]. (**a**) The general scheme, (**b**) an exemplary 4-phase oscillator

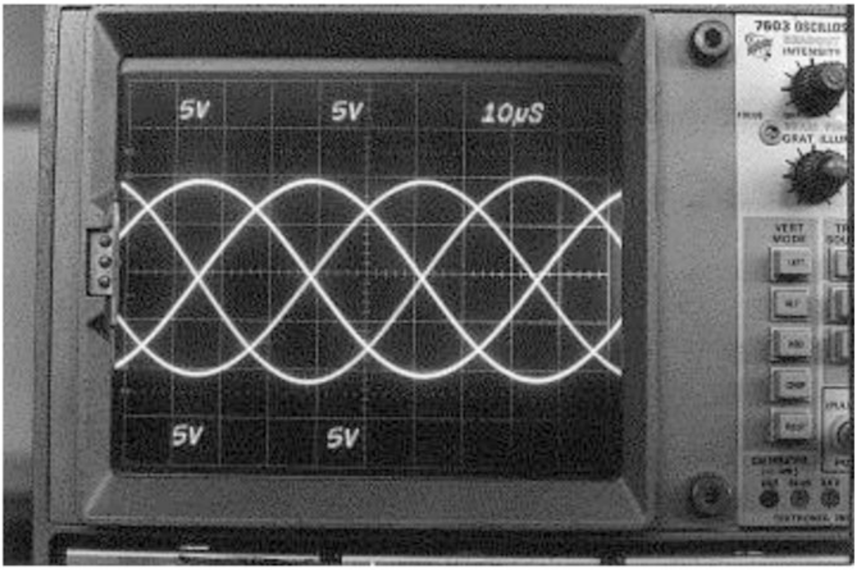
$$\phi = -\frac{2\pi}{n} \tag{1.51}$$

The workability of the proposed method and circuits has been verified in [75] by realizing the various circuits with op-amps Harris HA2544 C for frequencies up to 125 KHz with an error in the realized frequencies being no more than 1 %. Some sample results have been shown in Fig. 1.17.

1.4 Some Other Sinusoidal Oscillator Topologies

In the earlier section, we have discussed some well-known classical oscillators. In this section, we highlight some other oscillators both employing a single op-amp and those requiring two to three op-amps which are closely related to classical concepts but are not so well known and hence, to the best knowledge of the authors, are not yet documented in contemporary books.

a



b

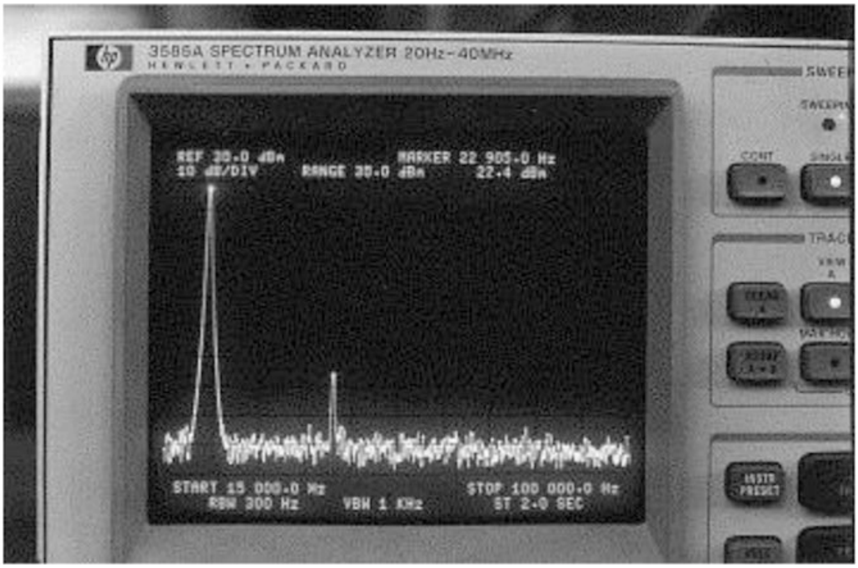


Fig. 1.17 Experimental results of the 4-phase oscillator [75]. (**a**) A typical output waveform of the circuit exhibiting 4-phase oscillations having 90° phase shift at 22.89KHz, (**b**) frequency spectrum of one of the outputs at $f_0 = 22.89$ KHz

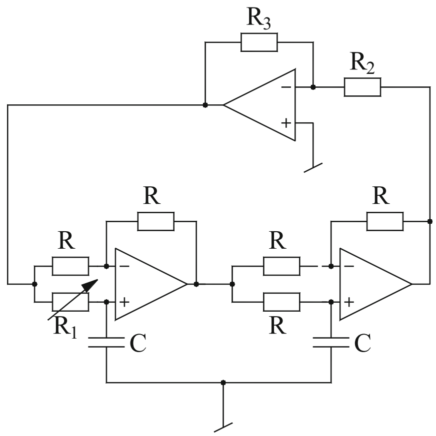


Fig. 1.18 Phase-shift oscillator based on all-pass filters [33]

1.4.1 An Oscillator Based Upon All-Pass Filters

An interesting variable frequency oscillator was proposed by Comer [33] using two first-order all-pass filters along with an inverting amplifier stage – all connected in a closed loop as shown in Fig. 1.18. While the popular Wien bridge and phase-shift oscillators compulsorily require ganged variable resistors for tuning if variable frequency is desired, there could be circuits in which the amplitude will inevitably change excessively if frequency change be possible through a single element. The circuit of Fig. 1.17 overcomes this problem. A straightforward analysis of this circuit shows that the loop gain or the open-circuit transfer function of this circuit is given by

$$T(s) = -\left(\frac{R_3}{R_2}\right)\left(\frac{1 - sCR_1}{1 + sCR_1}\right)\left(\frac{1 - sCR}{1 + sCR}\right) \tag{1.52}$$

From the above, it turns out that the magnitude of the loop gain is given by

$$|T(j\omega)| = \frac{R_3}{R_2} \tag{1.53}$$

It is, therefore, seen that the magnitude of the loop gain is independent of R_1; on the other hand, since the CO would be the one in which the magnitude of loop gain becomes unity, it would be therefore $R_2 = R_3$, whereas the FO with $C_1 = C_2$ would be given by

$$\omega_0 = \frac{1}{C}\sqrt{\left(\frac{1}{RR_1}\right)} \tag{1.54}$$

Thus, it is seen that the frequency of oscillation of this circuit can be varied simply by varying R_1 maintaining essentially the constant amplitude.

1.4.2 Two-Section Multiple Op-Amp Oscillators

An oscillator with two similar RC sections with both resistors grounded (which could be a preferable situation when electronic control of FO is employed by replacing these with matched FETs controlled by a common control voltage) was proposed by Ganguly [37] and is shown here in Fig. 1.19. By a straightforward analysis, the FO is found to be

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{C_1C_2R_1R_2}} \tag{1.55}$$

whereas with $R_1 = R_2 = R$, $C_1 = n\,C_2$; $R_A = R_B = R_C = R_D$, the CO is given by

$$K = \frac{n+1}{n-1} \text{ where } K = 1 + \frac{R_y}{R_x} \tag{1.56}$$

Vosper [56] suggested that the oscillator proposed by Ganguly [37] as described above can be modified as shown here in Fig. 1.20 leading to the characteristic equation of closed-loop system given by

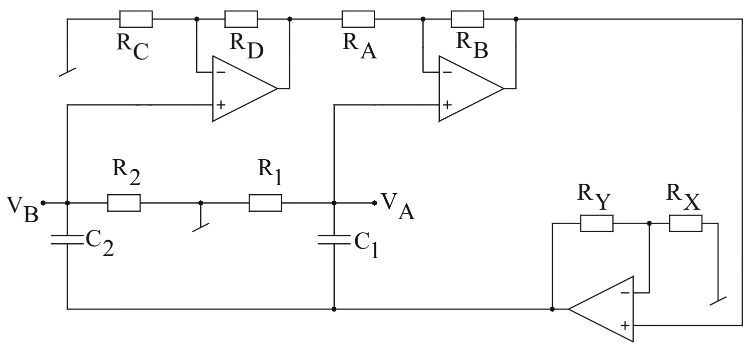
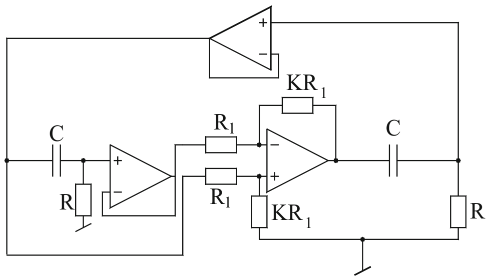


Fig. 1.19 An oscillator proposed by Ganguly [37] having two identical RC sections

Fig. 1.20 An alternate two high-pass section-based oscillator [56]



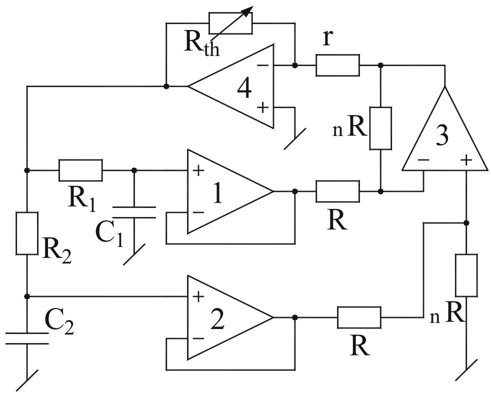


Fig. 1.21 Variable frequency oscillators with two identical RC sections and grounded capacitors proposed by Ganguly and Ganguly [57]

$$s^2 + s\frac{[2 - K]}{\tau} + \frac{1}{\tau^2} = 0; \quad \tau = \mathrm{RC} \tag{1.57}$$

From where the FO can be obtained as

$$f_0 = \frac{1}{2\pi\tau} \tag{1.58}$$

And the gain of the differential amplifier to maintain oscillations being required to be $K = 2$. It was demonstrated that the frequency stability factor of this modified version is found to be $S_F = 1$ which is larger than the original design of Ganguly [37].

Around the same time, Ganguly and Ganguly [57] also proposed another two sections of RC oscillator which employed as many as four op-amps along with two identical RC sections having both capacitors grounded as preferred for IC implementation (shown here in Fig. 1.21). With $\tau_1 = C_1R_1$ and $\tau_2 = C_2R_2$, also $\tau_2 = m/\tau_1$, and gain $K = (m + 1)/(m - 1)$, the FO is given by

$$\omega_0 = \frac{1}{\sqrt{m}RC} \tag{1.59}$$

For this circuit, the frequency stability factor is found to be

$$S_F = 2\frac{\sqrt{(m)}}{m + 1} \tag{1.60}$$

It is interesting to note that for $m = 2$ this oscillator is more stable than the classical Wien bridge oscillator.

1.5 Some Common Methods of Analyzing Sinusoidal Oscillators

Although many methods of analyzing linear sinusoidal oscillators have been advanced in the literature, of which the method is based upon, the use of Barkhausen criterion on the open-loop gain function is quite well known; in the following, we describe three other simple and popular methods of analyzing oscillator circuits all of which have been extensively employed by various researchers in presenting their new proposals of oscillator configurations during the last three decades.

For the sake of convenience, we illustrate some easy and popular methods of analysis through the example of Wien bridge oscillator which is reproduced here again in Fig. 1.22a.

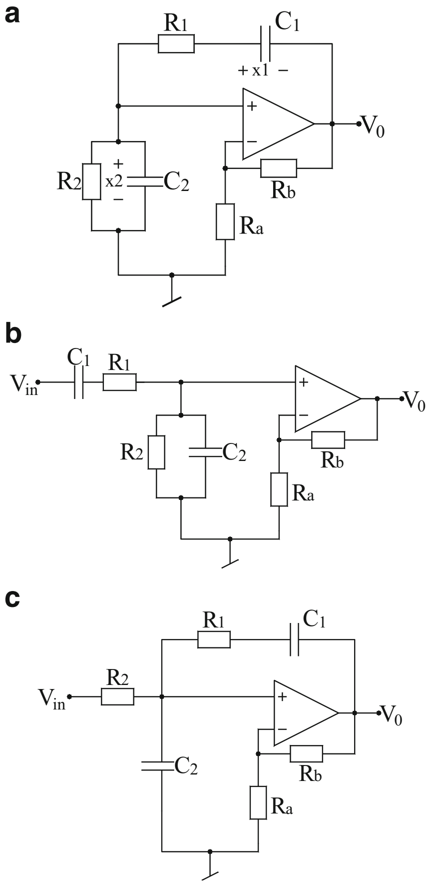


Fig. 1.22 (**a**–**c**) The Wien bride oscillator and two different ways of analyzing it

1.5.1 Analysis Based Upon the Closed-Loop Characteristic Equation

It may be recalled that as per the traditional analysis, the circuit can be analyzed to find the CO and FO by determining its open-loop transfer function (OLTF) T (s) from which the Barkhausen criterion is applied by equating $|T(jw)| = 1$ and $\angle T(jw) = 0°$ or integral multiple of 360°.

Alternatively, from the above determined OLTF, the characteristic equation (CE) of the closed circuit (i.e., the original circuit) can be found by equating $T(s) = 1$. The CO is then the condition under which the roots of the CE can be placed on the imaginary axis of the s-plane (for sustained sinusoidal oscillations) and slightly in the right half of the s-plane (for ensuring the building up of the oscillations). In the present case (Fig. 1.22a), by breaking the feedback loop by removing series of RC branch from the op-amp output, the OLTF is given by

$$T(s) = \frac{\left(1 + \frac{R_b}{R_a}\right) sC_1R_2}{s^2C_1C_2R_1R_2 + s(C_1R_1 + C_2R_2 + C_1R_2) + 1} \tag{1.61}$$

which, as per the procedure outlined above, leads to the CE:

$$s^2C_1C_2R_1R_2 + s\left\{(C_1R_1 + C_2R_2 + C_1R_2) - C_1R_2\left(1 + \frac{R_b}{R_a}\right)\right\} + 1 = 0 \tag{1.62}$$

From the above CE, it is clearly seen that the CO for the circuit is given by

$$\left(1 + \frac{R_b}{R_a}\right) \geq \left(1 + \frac{R_1}{R_2} + \frac{C_2}{C_1}\right) \tag{1.63}$$

whereas the FO is given by

$$\omega_0 = \frac{1}{\sqrt{C_1C_2R_1R_2}} \tag{1.64}$$

which are seen to be the same (with inequality sign removed from Eq. (1.63)) as obtained from the application of Barkhausen criterion.

1.5.2 Analysis by Finding CE by Ungrounding Any Element(s)/Terminal(s)

Another way to analyze an oscillator circuit is to open the circuit by ungrounding any or all grounded terminals or grounded elements or any combination thereof.

Thus, in the present case, such an ungrounding can be carried out by ungrounding the resistor R_2 or R_a or C_2 or all of R_2, R_a, and C_2. Let the OLTF in such a case be given by $H(s) = V_0/V_{in} = N(s)/D(s)$. Since the original circuit is obtained back by connecting to ground the elements/terminals ungrounded to find this OLTF, it follows that the CE of the circuit (and hence, the short circuit natural frequency of the circuit) can be found by making $V_{in} = 0$, i.e., $D(s) = 0$. In the present case, by ungrounding the resistor R_2 (Fig. 1.22c), the transfer function H(s) is found to be

$$H(s) = \frac{N(s)}{D(s)} = \frac{(sC_1R_1 + 1)}{s^2C_1C_2R_1R_2 + s\left\{C_1R_1 + C_2R_2 - \frac{R_b}{R_a}C_1R_2\right\} + 1} \tag{1.65}$$

Thus, the CE of the closed circuit is given by $D(s) = 0$ which is seen to be exactly the same as obtained by the earlier method; see Eq. (1.62).

1.5.3 State Variable Analysis of Sinusoidal Oscillators

In this method, it is not necessary to open the feedback loop and unground any elements/terminals to find out the closed circuit CE. The same can be determined without any of these operations simply by formulating state equations of the given oscillator circuit, *as it is*. It is well known that a second-order sinusoidal oscillator is an autonomous system for which the state equations can be as written as in the following general form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \tag{1.66}$$

From the above state equations, the characteristic equation of the circuit is given by $\det[sI - A] = 0$, which upon simplification leads to the following CE:

$$s^2 - (a_{11} + a_{22})s + (a_{11}a_{22} - a_{12}a_{21}) = 0 \tag{1.67}$$

From the above CE, the CO and FO are given by

$$\text{CO} : (a_{11} + a_{22}) \geq 0, \ \text{FO} : \omega_0 = \sqrt{a_{11}a_{22} - a_{12}a_{21}} \tag{1.68}$$

To demonstrate the application of the above theory, consider the circuit of the Wien bridge oscillator yet again. Defining the two state variables as the capacitor voltages (Fig. 1.22a), the following state equations can be written for this circuit:

$$\begin{bmatrix} \dfrac{\mathrm{d}x_1}{\mathrm{d}t} \\ \dfrac{\mathrm{d}x_2}{\mathrm{d}t} \end{bmatrix} = \begin{bmatrix} -\dfrac{1}{R_1C_1} & -\dfrac{R_b}{R_aR_1C_1} \\ +\dfrac{1}{C_2R_1} & +\dfrac{1}{C_2}\left(\dfrac{R_b}{R_aR_1} - \dfrac{1}{R_2}\right) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \tag{1.69}$$

From the above state equations, the CO and FO are, therefore, found to be

$$\text{CO} : \frac{R_b}{R_a} = \frac{C_2}{C_1} + \frac{R_1}{R_2} \text{ and FO} : \omega_0 = \frac{1}{\sqrt{C_1C_2R_1R_2}} \tag{1.70}$$

which are exactly the same as obtained from Eqs. (1.65) and (1.62) earlier.

From the above example, it is clear that this method can be applied to any given sinusoidal oscillator circuit and does not involve in any way altering the topology of the circuit by opening any feedback loop or by ungrounding any grounded element/terminal and, therefore, is a fairly general method of analysis.

1.6 Oscillator Synthesis Using ±RLC Models

Negative resistances are known to be useful circuit elements in oscillator design. A shunt RLC circuit of the type shown in Fig. 1.23a is a popular model used both for the synthesis and analysis of sinusoidal oscillator circuits.

The genesis of sinusoidal oscillators in this circuit could be intuitively understood as follows. With $-R_{02}$ and R_{01} omitted, if an impulse current $i(t) = \delta(t)$ is injected into the parallel LC tank circuit, and if L_o and C_o are ideal (lossless) a cosine, voltage is created by the circuit since $v(t) = (1/C_0)\cos\left(1/\sqrt{L_0C_0}\ t\right)$ and the circuit would oscillate at a frequency $\omega_0 = 1/\sqrt{L_0C_0}$. However, in practice, both inductor and capacitor are lossy, and if the resistor R_{01} representing these losses is taken into account, the impulse response would be an exponentially decaying sinusoid with the oscillation frequency still given by $\omega_0 = 1/\sqrt{L_0C_0}$. If L_o be the inductance simulated by active network (say by an op-amp-based gyrator or GIC circuit) as shown in Fig. 1.23b, the *sudden switching ON* of the DC power supply would tantamount to the application of an impulse, in response to which the circuit would generate an exponentially decaying sinusoidal voltage signal v(t). This could be corrected, and the design of the oscillator in Fig. 1.23b would be

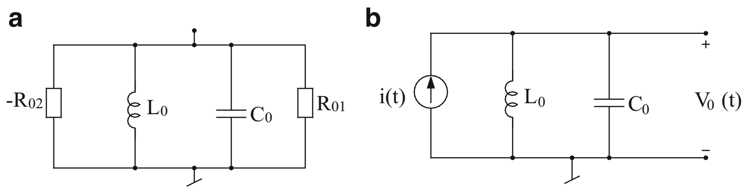


Fig. 1.23 (**a**, **b**) RLC models for sinusoidal oscillator synthesis

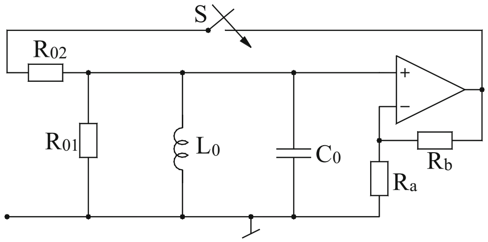


Fig. 1.24 Conversion of a band-pass filter (BPF) into an oscillator

completed if a negative R_{o2} is added to the circuit to compensate for the losses of the inductor and the capacitor represented by R_{o1}. With effect of R_{o1} exactly compensated (cancelled by $-R_{o2}$), it is obvious that the circuit should generate a constant amplitude sine wave voltage signal $v(t)$ having frequency $\omega_0 = 1/\sqrt{L_0 C_0}$.

In practice, however, it would be useful to have the net equivalent of the R_{01} and $-R_{02}$ to be slightly negative to enforce the starting of oscillations. Also there should be some additional mechanism in the circuit by which the amplitude of the oscillators generated can be kept to a desired level in the absence of which amplitude of oscillations will be limited only by the saturation characteristics (i.e., ultimate values of V_{sat} reaching utmost $\pm V_{CC}$) of the op-amp.

It is interesting to point out that the above-described RLC model could be alternatively arrived at by an alternative process also. It is known that a band-pass filter can be easily converted into a sinusoidal oscillator by putting an amplifier in cascade with this and then closing the feedback loop by shorting the input and output of the resulting active band-pass filter. Recall that the classical Wien bridge oscillator is nothing but a second-order RC band-pass filter followed by a non-inverting amplifier and then with its input and output shorted.

Let us construct such a circuit by having a second-order RLC band-pass filter and an op-amp non-inverting circuit as shown in Fig. 1.24.

A standard analysis reveals that the open-loop transfer function of Fig. 1.24 is given by

$$T(s) = \frac{K\left(\frac{s}{C_0 R_{02}}\right)}{s^2 + s\left(\frac{1}{C_0}\right)\left(\frac{1}{R_{01}} + \frac{1}{R_{02}}\right) + \frac{1}{C_0 L_0}}; K = \left(1 + \frac{R_b}{R_a}\right) \tag{1.71}$$

from which the characteristic equation is found to be

$$s^2 + s\left(\frac{1}{C_0}\right)\left\{\frac{1}{R_{01}} + \frac{1}{R_{02}}(1 - K)\right\} + \frac{1}{C_0 L_0} = 0 \tag{1.72}$$

leading to the CO and FO given by

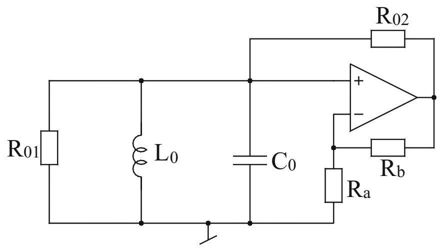


Fig. 1.25 An alternative representation of the BPF of the Fig. 1.23 with loop closed

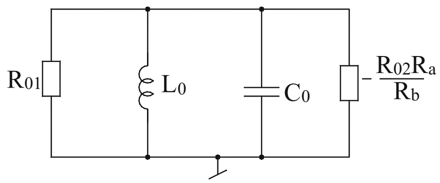


Fig. 1.26 RLC shunt resonator circuit obtained from Fig. 1.24

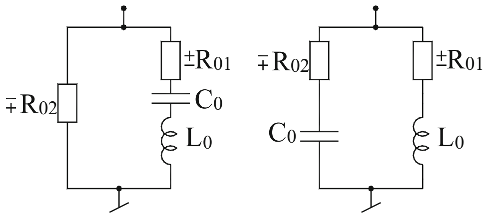


Fig. 1.27 An exemplary oscillator implementation based on RLC resonator

$$\text{CO}: \left\{\frac{1}{R_{01}} + \frac{1}{R_{02}}(1-K)\right\} = 0 \text{ and FO}: \omega_0 = \sqrt{\frac{1}{L_0C_0}} \tag{1.73}$$

If now the circuit be rearranged as shown in Fig. 1.25, it may be observed that since the last part of the circuit using an op-amp and three resistors, in fact, simulates a negative resistance of value $-(R_{02}R_a/R_b)$, it follows that the circuit turns out to be exactly the same as the RLC shunt resonator of Fig. 1.26.

We now show that the model shown in Fig. 1.26 can lead to many practical op-amp circuits in a number of ways. For example, one way of implementing this model could be to realize the inductor L_0 by a generalized impedance converter (GIC) circuit and realizing the negative resistance by a single-op-amp circuit configured as negative-impedance converter (NIC). The resulting circuit is shown in Fig. 1.27.

Through a routine circuit analysis, the CO and FO of this oscillator are given by

$$\text{CO}: (-R_{01}R_b + R_{02}R_a) = 0 \text{ and FO}: \omega_0 = \sqrt{\frac{r_4}{r_1r_3r_5C_2C_0}} \tag{1.74}$$

It is interesting to observe that this method of synthesizing an oscillator can give rise to a variety of circuits depending upon how the various components of the

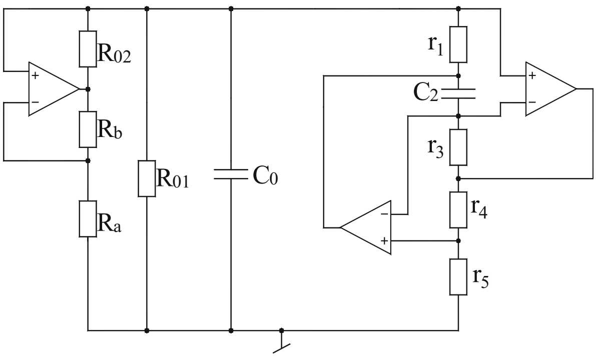


Fig. 1.28 An economical implementation of RLC resonator-based oscillator

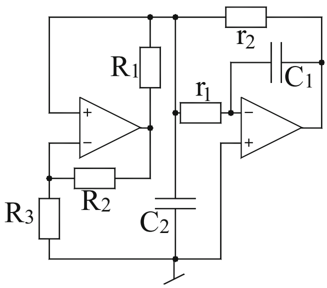


Fig. 1.29 Two alternative ±RLC models suitable for oscillator synthesis

model are combined and simulated by appropriate op-amp RC circuits. For example, an alternative oscillator with only two op-amps results when the parallel RL part of the circuit is simulated by Ford-Girling [103] circuit and then the negative resistance is simulated by another op-amp connected as NIC. The resulting circuit takes the form of as shown in Fig. 1.28.

The CO and FO for this circuit are given by

$$\text{CO}:\left\{\left(\frac{1}{r_1}+\frac{1}{r_2}\right)-\frac{R_2}{R_1R_3}\right\}=0 \text{ and FO}:\omega_0=\frac{1}{\sqrt{C_1C_2r_1r_2}} \tag{1.75}$$

Two other models which can be used to synthesize an oscillator could be shown in Fig. 1.29.

It is obvious that by using different op-amp circuits to simulate the various impedance branches of these models, one can derive a number of different op-amp oscillators.

1.7 Nonsinusoidal Waveform Generators Using IC Op-Amps, IC Timers, and Op-Amp Timer Combinations

In the remaining part of this chapter, we present some well-known relaxation oscillators and waveform generators using IC op-amps, IC 555 timer, and the combinations thereof.

1.7.1 The Op-Amp-Based Schmitt Trigger and the Astable Multivibrator

An astable multivibrator employing an op-amp comparator is used to generate symmetrical square wave signal which is required in many applications. For example, if a square wave generator is available, one can obtain triangular waveform by integrating square wave, and from a triangular waveform, one can create a sinusoidal waveform by an appropriate wave shaping circuit such as a triangular to sine wave converter. Thus, a square wave generator may constitute an important building block of a function generator.

An astable multivibrator using an op-amp comparator with positive feedback is shown in Fig. 1.30a, where op-amp along with resistors R_B and R_A is configured as a Schmitt trigger. By analysis, it can be shown that this circuit can generate square wave whose time period can be set by proper selection of component values R, C, R_B, and R_A.

For the analysis of this circuit, let us assume that the output $V_0 = + V_{sat}$. It is, therefore, seen that the input signal at the non-inverting input of the op-amp comparator will be $+\beta V_{sat}$, where $\beta = (R_A/(R_A + R_B))$. Therefore, the capacitor voltage $V_c(t)$ will try to charge exponentially towards $+V_{sat}$ by time constant RC. $V_c(t)$ will continue to grow until it becomes slightly more than $+\beta V_{sat}$ at which time the comparator will switch to $-V_{sat}$. The reference voltage at the non-inverting input terminal of the comparator will now change to $-\beta V_{sat}$. Since the capacitor voltage has already reached a value equal to $+\beta V_{sat}$, it will now try to discharge exponentially towards $-V_{sat}$. This will continue until V_c attains a value equal to $+\beta V_{sat}$ at which time the comparator will again switch back to $+V_{sat}$. This process continues and the resulting waveforms of the capacitor voltage V_c and comparator's output V_0 will be as shown in Fig. 1.30b.

It may be noted that since the charging and discharging are taking place with the same RC network, the resulting square wave will be having $T_H = T_L$. Also, the capacitor voltage will be alternating between $-\beta V_{sat}$ and $+\beta V_{sat}$, whereas the output square wave will be alternating between $+V_{sat}$ and $-V_{sat}$.

The frequency of the output waveforms can be calculated as follows. The general equation for $V_c(t)$ can be written as

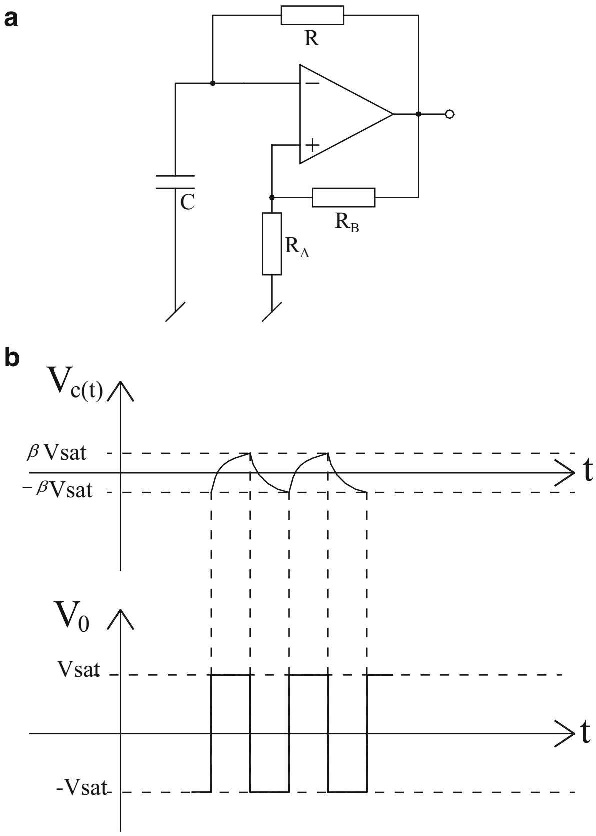


Fig. 1.30 Astable multivibrator using an op-amp comparator. (**a**) Circuit diagram, (**b**) various associated waveforms

$$V_c(t) = A + Be^{-\frac{t}{RC}} \tag{1.76}$$

where the constants A and B can be determined by using the following conditions: at $t = 0$, $V_c\ (0) = +\beta V_{sat}$; at $t = \infty$, $V_c\ (\infty) = +V_{sat}$; and for $t = T_H$, $V_c\ (T_H) = -\beta V_{sat}$. Using these conditions, it is found that

$T_H = T_L = RC\ln\left(\frac{1+\beta}{1-\beta}\right)$, and hence, the frequency of oscillation of the generated waves can be obtained as

$$f = \frac{1}{2RC\ln\left(\frac{1+\beta}{1-\beta}\right)} \tag{1.77}$$

1.7.2 Square/Triangular Waveform Generator

The square/triangular waveform generator employs a Schmitt trigger cascaded with an inverting integrator as shown in Fig. 1.31. The output of Schmitt trigger (V_{01}) is a

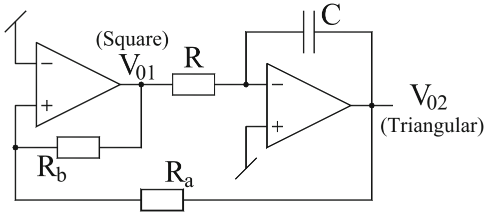


Fig. 1.31 The square/triangular waveform generator

square wave; hence, the integrator will generate the triangular wave (V_{02}). The detailed analysis of this circuit is well documented in [1–6].

1.7.3 The Monostable Multivibrator

The classical single-op-amp monostable multivibrator is shown in Fig. 1.32. As shown in Fig. 1.32a, C_1, R_1, and D_2 constitute a pulse shaping circuit. A monostable circuit, as the name implies, has a permanent stable state. It can be easily deduced that this permanent stable state is $V_0 = V_{\text{sat}} = \text{HIGH}$ (with D_1 clamped at diode's cut-in voltage V_{D1}); if β is enough to make $+\beta V_{\text{sat}} > V_{\text{D1}}$, the output of the circuit would be $V_0 = +V_{\text{sat}} = \text{HIGH}$ state. Even if the circuit accidentally starts from $V_0 = -V_{\text{sat}}$, the signal at the non-inverting input is $-\beta V_{\text{sat}}$ and capacitor C_0 charges until voltage across it becomes more negative than $-\beta V_{\text{sat}}$ at which time $V_0(t)$ switches to $+V_{\text{sat}}$ and would remain so for all times to come until forced to change this state, by applying an external trigger voltage (V_{tr}).

Upon the application of an external signal (V_{e}) resulting in a short (narrow) negative-going trigger pulse (V_{tr}) at $t = 0$, the $V_0(t)$ and $V_{\text{co}}(t)$ undergo changes as depicted in Fig. 1.32b. The time period during which waveform remains LOW ($-V_{\text{sat}}$) is the time taken by C_0 to change from V_{D1} volts to $+\beta V_{\text{sat}}$ and can be calculated from the following general equation:

$$V_{\text{c}}(t) = A + Be^{-\frac{t}{\tau}}; \quad \tau = \text{RC}_0 \tag{1.78}$$

Considering that with $t = 0$, $V_{\text{co}}(0) = V_{\text{D1}}$, and for $t = \infty$, $V_{\text{co}}(\infty) = -V_{\text{sat}}$, one can easily find that

$$V_{\text{C0}}(t) = -V_{\text{sat}} + (V_{\text{D1}} + V_{\text{sat}})e^{-\frac{t}{\tau}} \tag{1.79}$$

For $t = T$, $V_{co}(\text{T}) = -\beta V_{sat}$. Hence, we find the pulse width (T)

$$T = C_0 R \ln \frac{V_{sat} + V_{D1}}{V_{sat}(1 - \beta)} \tag{1.80}$$

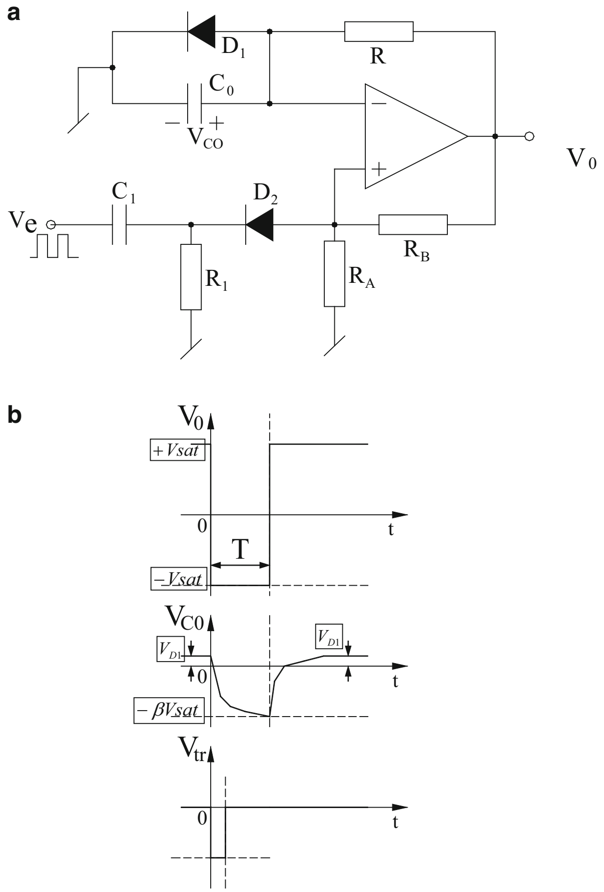


Fig. 1.32 Monostable multivibrator using an op-amp comparator. (**a**) Circuit diagram, (**b**) its associated waveforms

1.7.4 Synthesis of Waveform Generators in Phase Plane

Pimentel [32] proposed a method of systematically synthesizing electronic circuits to generate different waveforms like triangular/square wave and sawtooth/pulse waveforms through the phase plane representations. The proposition for generating square/triangular waveforms is based upon the system of differential equations given by

$\dot{x}_1 = \omega\, sign\,(x_2)$ and $\dot{x}_2 = -\omega\, sign\,(x_1)$, where x_1 and x_2 are the phase variables $(x_1 = \omega x$ and $x_2 = \omega \dot{x})$ and are shown in Fig. 1.33. The circuit realization employs two comparators (used to generate signum functions) and two inverting integrators as shown in Fig. 1.33.

It is seen that the circuit generates two square waves being 90° apart in phase and two triangular waves also having 90° phase shift: the former from the outputs of the two comparators and the latter from the outputs of the two integrators. The time

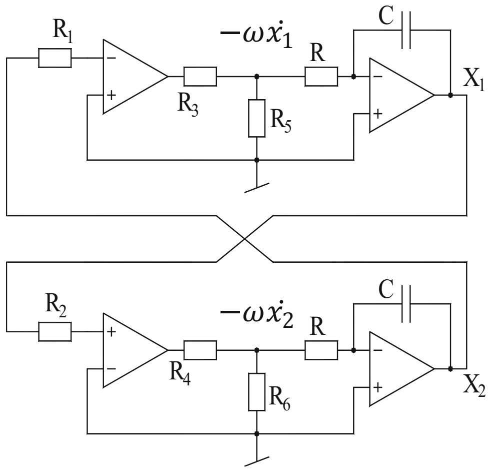


Fig. 1.33 Square/triangular waveform generator proposed by Pimentel [32] ($R_1 = R_2 = 8.2$ kΩ, $R_3 = R_4 = 9$ kΩ, $R_5 = R_6 = 1$ kΩ)

needed for the signal x_1 to reach to zero value from its peak value V is $T/4$, where T is the period of the triangular wave. Hence,

$$V = \frac{1}{RC}\left(\frac{T}{4}\right) \tag{1.81}$$

As a consequence, the frequency of the signals generated by this circuit is given by

$$f = \frac{1}{4RCV} \tag{1.82}$$

Similarly, the differential equations to realize sawtooth generator are given by

$$\dot{x}_1 = \omega \text{ sign}\,(x_2) \text{ and } \dot{x}_2 = -\omega \text{ pos}\,(x_1) \tag{1.83}$$

where pos(x) function is defined as

$$\text{pos}(x) = \left\{\begin{matrix} \infty & \text{for} & x > 0 \\ 0 & \text{for} & x = 0 \\ -1 & \text{for} & x < 0 \end{matrix}\right\} \tag{1.84}$$

The pos(x) function required in the second equation is approximated by an op-amp in an open-loop configuration, where two diodes are used at the output node in order to clamp the output voltage at approximately 1.2 V. The resulting final circuit is shown in Fig. 1.34. In this case, the time needed by the signal x_2 to reach the zero value from its maximum positive value V is $T/2$, where T is the period of oscillation. As a consequence, one can write

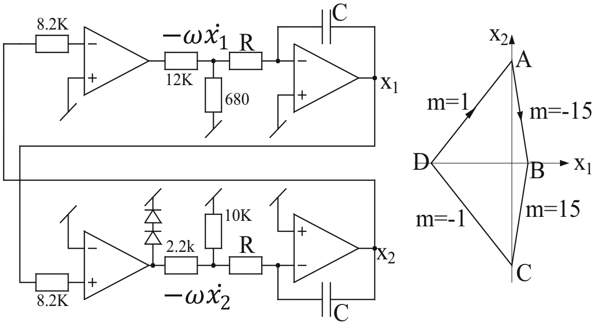


Fig. 1.34 Pimentel's circuit configuration for producing sawtooth waveforms [32]

$$V = \frac{1}{RC}\left(\frac{T}{2}\right) \tag{1.85}$$

Hence, $T = 2RCV$, and therefore, the frequency of the generated waveforms is given by

$$f = \frac{1}{2RCV} \tag{1.86}$$

Requirement of matched components and the frequency limitations of the op-amps employed are the two main limitations of these circuits. Nevertheless, it would appear that with the use of modern comparators and op-amps exhibiting large gain-bandwidth products as well as high slew rates, the implemented circuits can be useful in generating higher-frequency waveforms than those demonstrated in [32].

1.7.5 *Quadrature Oscillators for Generating Square and Triangular Waveforms*

A quite similar concept and implementation was formally proposed *independently* by Kaplan, Har-Zahav, and Blau [34]. They argued that since a sinusoidal quadrature oscillator is comprised of a closed loop of an inverting and a non-inverting integrator and can be described by a conservative system of equations, namely,

$$\dot{x} = \omega\, y \text{ and } \dot{y} = -\omega\, x \tag{1.87}$$

where ω is the positive frequency parameter, it follows that a similar method can be employed to generate quadrature square/triangular waveforms by employing the following differential equations:

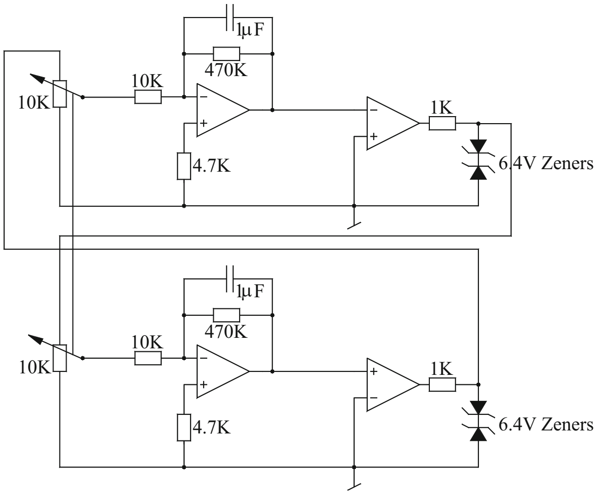


Fig. 1.35 Quadrature oscillators for realizing square/triangular waveforms proposed by Kaplan, Har-zahav, and Blau [34]

$$\dot{x} = \alpha \operatorname{sign}(y) \text{ and } \dot{y} = -\alpha \operatorname{sign}(x) \tag{1.88}$$

An implementation of the above set of equations is shown in Fig. 1.35 which was actually implemented by using LM324 quad op-amps: two op-amps to realize the two comparators and the remaining two op-amps to realize the integrators required for implementing the intended system of differential equations. The circuit generates different frequencies by changing the setting of the potentiometers.

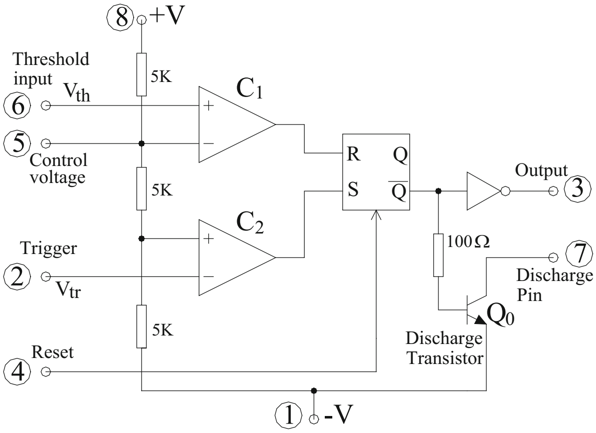
This method can be used to generate more complex waveforms by using additional circuits. For example, a trapezoidal waveform can be generated by adding the two triangular waveforms, while a staircase waveform can be generated by adding the two square waves.

1.8 Multivibrators and Waveform Generators Using IC 555 Timer

The IC 555 timer is a very popular and cheap general-purpose active building block which is useful in generating a variety of waveforms and in various timing applications. By using external components, in conjunction with op-amps and other devices, it can be used to realize many interesting functions such as voltage-controlled oscillators (VCO) and variable-duty-cycle VCOs. The block diagram of IC 555 timer is shown in Fig. 1.36 which can be used to explain its various modes of operations.

The various operating modes of the IC 555 timer can be explained as follows. Let $+V = V_{cc}$, $-V =$ ground (0 Volts); consider now the following cases:

Fig. 1.36 A simplified block diagram of IC 555 timer



Case 1: When $V_{tr} < V_{cc}/3$ and $V_{th} < 2V_{cc}/3$, then output of comparator C_2 goes High, and output of comparator C_1 remains Low which sets the flip-flop resulting in V_{03} going HIGH. However, $\overline{Q}$ is low (≈0.2 Volts for + 5 Volts power supply), and hence, the discharge transistor Q_0 is OFF and pin 7 acts like an OPEN circuit.

Case 2: When $V_{th} > 2V_{cc}/3$ and $V_{tr} > V_{cc}/3$, the output of C_1 changes to High, and the output of C_2 goes low which resets the flip-flop leading to V_{03} going LOW. However, since Q is HIGH (≈4.5 Volts for + 5 Volts power supply), hence discharge transistor Q_0 is saturated and pin 7 acts as a SHORT circuit.

Case 3: When $V_{tr} > V_{cc}/3$ and simultaneously $V_{th} < 2\mathrm{V}_{cc}/3$, then the output of the comparators C_1 and C_2 are both low; hence, V_{03} remains the same as the previous state ($Q_{n+1} = Q_n$). This is called the “memory state” of the timer. The functions and roles of the various terminals of the 555 timer are as follows:

8, 1: $+V$, and $-V$ *but* could also be $+V$ and ground.
3: Output: HIGH ≈ $+V$, Low ≈ $-V$ or 0.
7: Termed as “discharge pin,” it discharges or charges an external capacitor.
4: A LOW on this terminal resets the timer, regardless of the other inputs; to prevent it from happening, it is usually wired to $+V$.
5: This is called *control voltage* terminal; a voltage/resistor connected here can change threshold and t*rigger levels* and, hence, the operation of the timer. This can be done in the following ways. If we connect a 5 kΩ resistor between 5 and 8, then $(V_{th})_{ref} = (10/12.5)$ volts and $(V_{tr})_{ref} = (5/12.5)$ volts, and thus, the reference levels of the comparators are changed. If we connect terminal 1 to $-V$ and terminal 5 to $+V$, then $(V_{th})_{ref} = +V$ and $(V_{tr})_{ref} = 0$ V! When not in use, 5 pin is connected to a capacitor from 5 to 1, thereby making a low-pass filter to filter-out the ripples and spikes of power supply which may otherwise result in false triggering of the timer.
2: Trigger input (V_{tr}).
6: Threshold input (V_{th}).

We now explain some basic application circuits incorporating IC timer.

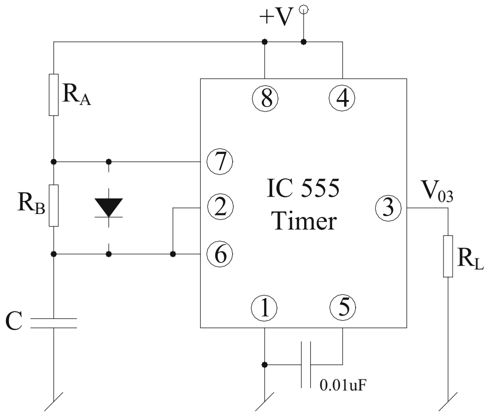


Fig. 1.37 Basic astable multivibrator using 555 timer

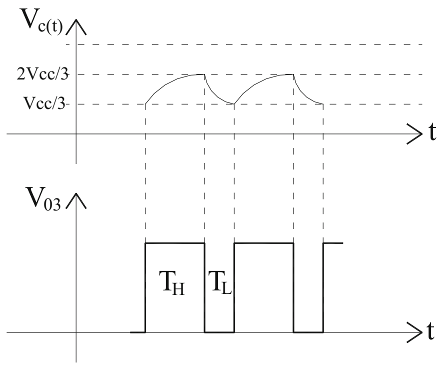


Fig. 1.38 Various waveforms of the astable multivibrator of Fig. 1.36

1.8.1 Astable Multivibrators

An astable multivibrator is a relaxation oscillator, which does not have any permanent stable state. It alternates between two quasi-stable states and is, therefore, suitable for generating a square or rectangular waveform. Such circuits can be made from two BJTs, inverters, op-amps, or IC 555 timer. We show here how it can be implemented using IC 555 timer (Fig. 1.37).

The operation of the circuit can be explained as follows. Assume V_{03} to be HIGH, so that pin 7 acts as open circuit and C charges with a time constant $\{(R_A + R_B)\ C\}$ towards $+V$. When $V_c(t)$ becomes $> \frac{2}{3}V_{cc}$, V_{03} goes Low, pin 7 is short, and at which instant, capacitor discharges through pin 7 with the time constant (R_BC) till $V_c(t)$ becomes $< \frac{1}{3}V_{cc}$ at which instant V_{o3} again goes high (see Fig. 1.38). This sequence goes on repeating such that $V_c(t)$ is confined between $\frac{1}{3}V_{cc}$ and $\frac{2}{3}V_{cc}$ and V_{03} alternates between High and Low as shown in the diagram. Since charging and the discharging times are not the same, clearly, $T_H \neq T_L$. Also, note that $(T_H)_{1st\ cycle} \neq (T_H)$ sequent cycles, and hence, the circuit has "first cycle timing error."

An expression for the time period of the output waveform can be determined as follows. The equation for the capacitor charging can be written in the following general form:

$$V_{\mathrm{c}}(t) = A + Be^{\frac{-t}{(R_{\mathrm{A}}+R_{\mathrm{B}})C}} \tag{1.89}$$

here A and B are constants which can be determined from the following conditions:

$t = 0$; $V_{\mathrm{c}}(0) = V_{\mathrm{CC}}/3$ and at $t = \infty$, $V_{\mathrm{c}}(\infty) = V_{\mathrm{CC}}$; thus the values of A and B can be obtained as $A = V_{\mathrm{CC}}$ and $B = 2V_{\mathrm{CC}}/3$. Therefore, Eq. (1.89) now becomes

$$V_{\mathrm{c}}(t) = V_{\mathrm{CC}} - \frac{2}{3}V_{\mathrm{CC}}\,e^{\frac{-t}{(R_{\mathrm{A}}+R_{\mathrm{B}})C}} \tag{1.90}$$

Now at $t = T_{\mathrm{H}}$, $V_{\mathrm{C}}(t) = \frac{2}{3}V_{\mathrm{cc}}$; hence, the value of T_{H} can be calculated as

$$T_{\mathrm{H}} = (R_{\mathrm{A}} + R_{\mathrm{B}})C\ln 2 \tag{1.91}$$

Similarly, one can find that considering the equation for discharge of the capacitor with time constant CR_{B}, one can find T_{L} as

$$T_{\mathrm{L}} = R_{\mathrm{B}}C\ln 2 \tag{1.92}$$

The frequency of the output waveform is, therefore, given by

$$f = \frac{1}{(R_{\mathrm{A}} + 2R_{\mathrm{B}})C\ln 2} \tag{1.93}$$

and the duty cycle of the output waveform is given by

$$\delta = \frac{(R_{\mathrm{A}} + R_{\mathrm{B}})}{(R_{\mathrm{A}} + 2R_{\mathrm{B}})} \tag{1.94}$$

Note that even if one selects $R_{\mathrm{A}} = R_{\mathrm{B}}$, the duty cycle cannot be made 50 %. However, the duty cycle can be made 50 % by shunting the resistor R_{B} by a diode as shown in Fig. 1.39.

This makes $T_{\mathrm{H}} = R_{\mathrm{A}}C \ln 2$ and $T_{\mathrm{L}} = R_{\mathrm{B}}C \ln 2$ (assuming ideal diode), so that the duty cycle is given by

$$\delta = \frac{R_{\mathrm{A}}}{R_{\mathrm{A}} + R_{\mathrm{B}}} \tag{1.95}$$

which would become 50 % for $R_{\mathrm{A}} = R_{\mathrm{B}}$.

In yet another scheme, the diode is avoided altogether and still one can obtain 50 % duty cycle. Such a circuit is shown in Fig. 1.40.

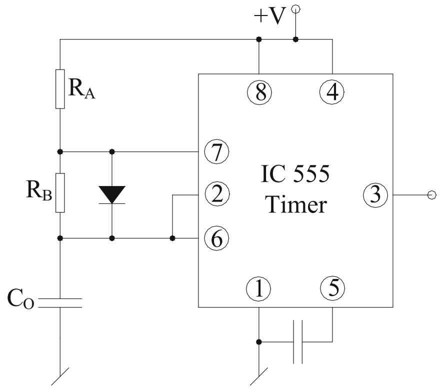


Fig. 1.39 Astable multivibrator with 50 % duty cycle

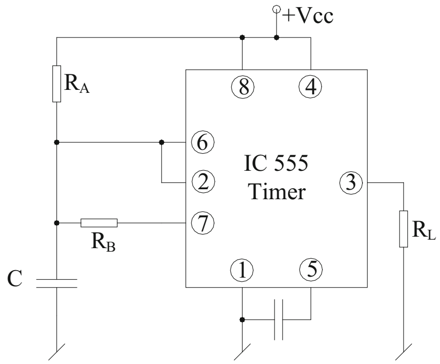


Fig. 1.40 Alternative astable multivibrator with 50 % duty cycle

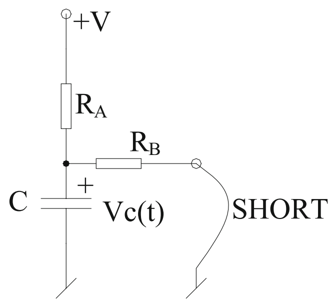


Fig. 1.41 Equivalent circuit for the case when pin 7 acts as "short"

As with the earlier circuit, when V_{03} is high, pin 7 is open and the charging of the capacitor takes place through $R_A C$; hence, T_H remains $T_H = R_A C \ln 2$. For the duration for which V_{03} is Low and pin 7 short, the time T_L can be calculated as follows. The discharge equivalent circuit (Fig. 1.41) can be made as follows:

Before the situation depicted in the equivalent circuit arises, the following apply: at $t = 0$, $V_c(0) = \frac{2}{3}V_{CC}$; then, the general equation for the capacitor voltage can be written as

$$V_c(t) = A + B\,e\frac{-t}{\tau},\ \text{where}\ \tau = \frac{CR_A R_B}{(R_A + R_B)} \tag{1.96}$$

Substitution of the initial condition in Eq. (1.96) gives

$A + B = \frac{2}{3}V_{CC}$; now, if there were no control, the capacitor voltage could have gone at $t = \infty$ up to the ultimate value

$V_C(t) = \left(\frac{R_B}{R_A + R_B}\right)V_{CC}$ due to which from Eq. (1.96), we obtain

$A = \left(\frac{R_B}{R_A + R_B}\right)V_{CC}$; hence, one finds $B = \frac{(2R_A - R_B)}{3(R_A + R_B)}V_{CC}$, and the required equation is, therefore,

$$V_c(t) = \left(\frac{R_B}{R_A + R_B}\right)V_{CC} + \frac{(2R_A - R_B)}{3(R_A + R_B)}V_{CC}\,e^{\frac{-t}{\tau}} \tag{1.97}$$

Now, for $t = T_L$, $V_c\ (t) = V_{cc}/3$; hence, substituting these values in Eq. (1.97), we find

$V_{CC}\left\{\frac{1}{3} - \frac{R_B}{R_A + R_B}\right\} = \frac{(2R_A - R_B)V_{CC}}{3(R_A + R_B)}e^{\frac{-T_L}{\tau}}$ which is simplified to

$(R_A - 2R_B) = (2R_A - R_B)\,e^{\frac{-T_L}{\tau}}$ which can further be simplified to give the time period during which the waveform is LOW and is finally found to be

$$T_L = \frac{CR_A R_B}{(R_A + R_B)}\ln\left\{\frac{2R_A - R_B}{R_A - 2R_B}\right\}; R_A > 2R_B \tag{1.98}$$

Furthermore, $T_H = C\,R_A\,\ln 2$ (as usual); hence, 50 % duty cycle will be attained if

$$\frac{CR_A R_B}{(R_A + R_B)}\ln\left\{\frac{2R_A - R_B}{R_A - 2R_B}\right\} = CR_A \ln 2 \tag{1.99}$$

With some algebraic manipulations, the above equation can be rewritten as

$$\frac{2K - 1}{K - 2} = 2^{1+K}\text{where}\ \frac{R_A}{R_B} = K \tag{1.100}$$

Clearly, Eq. (1.100) does not have a closed solution, but it can be solved by *hit and trial*, by picking up some value of K and then seeing that RHS equals LHS. This happens at $K = 2.3$; hence, if we take $R_B = R$, then R_A should be 2.3 R.

A circuit for an alternative astable multivibrator having symmetrical output swings and not having the first cycle timing error is shown in Fig. 1.42. Here, the basic trick to avoid the first cycle timing error is to ensure that whether in the first

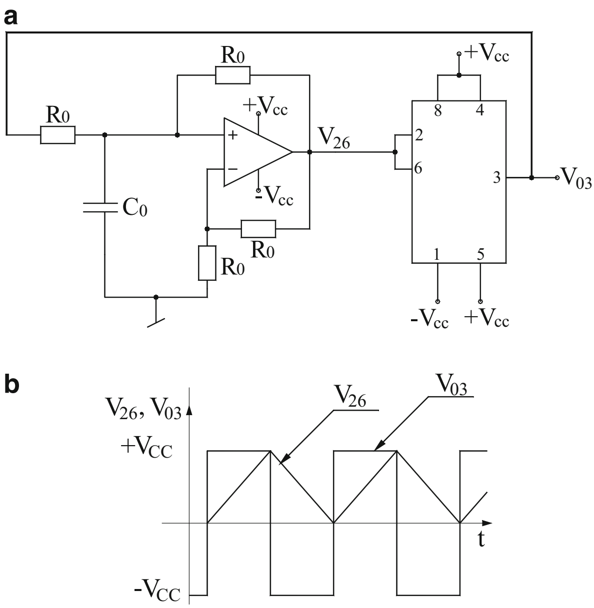


Fig. 1.42 Astable multivibrator free from *first-cycle timing error*

cycle or in the subsequent cycle the triangular waveform generated by the circuit has the same start and end points. This is achieved by connecting terminal 1 of the timer to $-\mathrm{V}_{\mathrm{CC}}$ and terminal 5 to $+V_{\mathrm{CC}}$. Looking into the internal block diagram, it can be easily figured out that this would change the reference levels of the internal comparators (which are normally $V_{\mathrm{CC}}/3$ and $2V_{\mathrm{CC}}/3$) to 0 and $+V_{\mathrm{CC}}$, respectively. As a consequence, when the power supply is switched ON, the capacitor voltage is initially zero which makes $V_{26}=0$; thus, if initially V_{03} is high, the non-inverting integrator made from the op-amp results in a positive going ramp at V_{26} rising from zero to a maximum value of $+V_{\mathrm{CC}}$ with the slope of ramp being $2V_{\mathrm{sat}}/R_0C_0$ (where $V_{\mathrm{sat}} \cong +V_{\mathrm{CC}}$) after which the timer's output V_{03} goes low $(-V_{\mathrm{sat}} \cong -V_{\mathrm{CC}})$, the $-V_{\mathrm{sat}}$ is fed back to non-inverting integrator which makes V_{26} a negative going ramp having numerical value of the slope still the same. This V_{26} can go down only up to zero volts. This sequence of operation is repetitive resulting in a triangular wave alternating between zero and $+V_{\mathrm{CC}}$ and square wave alternating between $-V_{\mathrm{CC}}$ and $+V_{\mathrm{CC}}$. The time period in which V_{03} remains high is given by $T_{\mathrm{H}} = \frac{V_{\mathrm{CC}}-0}{\frac{2V_{\mathrm{sat}}}{R_0C_0}}$. Obviously, T_{L} is the same as T_{H}; therefore, the frequency of generated waveforms is given by $f = \frac{1}{(T_{\mathrm{H}}+T_{\mathrm{L}})} = \frac{1}{R_0C_0}$. Rathore [104] suggested that such a scheme can be used to measure external capacitance or the time period of any unknown non-inverting integrator under test.

Rathore [46] demonstrated that a versatile astable multivibrator can be made from two IC 555 timers both operating in monostable mode such that one triggers the other. It is also shown that such a circuit can be used in many interesting applications such as analog division and some unusual function generation.

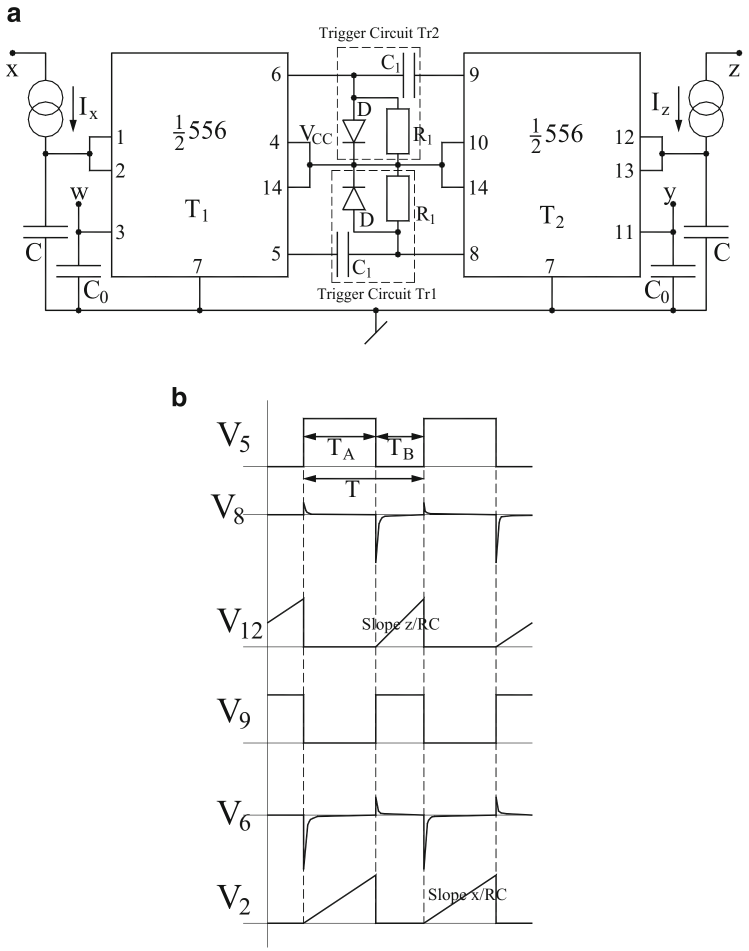


Fig. 1.43 A versatile astable multivibrator proposed by Rathore [46]. (**a**) Circuit configuration, (**b**) various generated waveforms

Considering now the circuit of Fig. 1.43a, it may be noted that each of the timer is connected in monostable mode and triggers the other. Here I_x and I_z provide currents proportional to signal voltages, i.e., $I_x = \mathbf{X}/R$ and $I_z = \mathbf{Z}/R$ where R is constant.

The operation of this circuit can be explained as follows: initially let us assume that the discharge transistor inside the left timer is OFF; correspondingly, V_5 will be high and consequently the left capacitor C starts charging linearly with the rate **X**/RC. When the voltage V_2 reaches the potential **W**, the discharge transistor of the left timer becomes ON (short circuited to ground). This allows the discharging of the capacitor rapidly through the internal saturated transistor. In turn, V_5 simultaneously goes low; now, the trailing edge of V_5 through the trigger circuit T_{r2} triggers the second timer (one on the right side); consequently, the discharge transistor of the right timer now goes off and V_9 goes high. Due to this, the right capacitor C starts charges linearly at a rate **Z**/RC. Now when the voltage V_{12} reaches the potential y, the discharge transistor of right timer becomes on, thereby allowing this capacitor to discharge rapidly through the internal saturated transistor. Simultaneously, V_9 has gone low; therefore, the trailing edge of the voltage waveform V_9 triggers the left timer through the trigger circuit T_{r1}. Thus, it follows that the circuit in fact works as an astable multivibrator with complementary outputs available from pin 5 of the first timer and pin 9 of the second timer. These waveforms are shown in Fig. 1.43b.

Since the ramp voltage appearing at pin 2 changes from zero to a voltage level **W** with the rate **X**/RC, it follows that the time taken in doing so is given by

$T_A = RC$ (**W**/**X**). Similarly, it is easy to see that T_B would be given by $T_B = RC$ (**Y**/**Z**). Consequently, the time period T and the frequency of the two output waveforms generated by this circuit are given by

$$T = 2\mathrm{RC}\ (\mathbf{W}/\mathbf{X} + \mathbf{Y}/\mathbf{Z}) = 1/f \tag{1.101}$$

It may be noted that if no external signals are connected at nodes where **W** and **Y** are applied, the potential at these pins would be $2V_{cc}/3$; hence, this should be substituted in place of **W** and **Y** in the relevant equations above.

The following special cases/applications of the proposed circuit are worth considering now:

1. Two different voltage ratios **W**/**X** and **Y**/**Z** can be converted simultaneously into two different time periods by the circuit.
2. If one takes **W** = **Y** = **U** and **X** = **Z** = **V**, it follows that the time period of the output square wave is then given by

$$T = 2RC(\mathbf{U}/\mathbf{V}) = 1/f \tag{1.102}$$

   and the circuit, thus, acts as an analog divider whose result can be displayed digitally by counting the number of clock pulses in the period T or the number of output pulses over a fixed duration. With respect to **U**, the circuit acts as a linear voltage-to-time converter, while with respect to voltage **V**, it becomes a linear voltage-to- frequency converter. For **U** = **V**, the circuit functions as a conventional astable multivibrator with complementary outputs.
3. By appropriate choice of various control voltages, the circuit can be easily operated as a variable duty cycle while maintaining constant frequency.

1.8.2 Monostable Multivibrators

A monostable multivibrator has one (mono) permanent stable state. The circuit remains in this permanent state if left undisturbed. On the application of an external

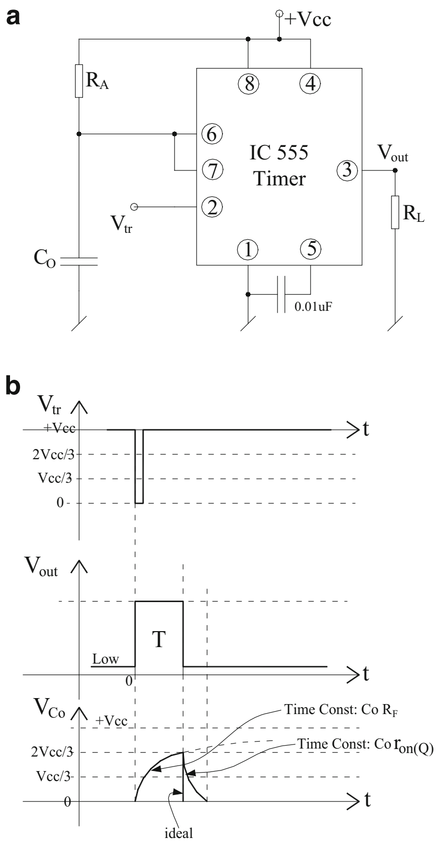


Fig. 1.44 Monostable multivibrators using IC 555 timer. (**a**) The basic circuit, (**b**) various waveforms of the monostable multivibrator

trigger input, the circuit comes out of this permanent state and assumes the other possible state for the duration of time "*T*" dictated by an external RC circuit, after which it goes back to its permanent state. A monostable multivibrator, thus, generates a "single pulse" of duration "T" in response to a short (narrow) trigger pulse and is also called a "one shot."

"One shots" can be implemented in a variety of ways using BJTs, inverters, and op-amps and are also available in IC form, e.g., IC74123. In the following, we show how a "one shot" can be realized from an IC 555 timer.

Consider the circuit shown in Fig. 1.44.

The permanent stable state of the circuit is $Vo_3 = \text{LOW}$, so that pin 7 is SHORT and it maintains the capacitor shorted to zero volts. However, even if the circuit accidentally starts from V_{03} as HIGH, eventually after some time, it will switch to LOW and would continue to remain so until triggered.

The circuit operation can now be explained as follows (see Fig. 1.44a, b). At $t = 0$, a narrow negative going pulse having amplitude lower than $V_{cc}/3$ is applied at V_{tr}.

Now since, $V_{tr} < V_{cc}$,/3 pin 7 opens up and capacitor C_0 charges exponentially towards $+V_{cc}$ and continues till $V_{co} > 2V_{cc}/3$ at which time, timer output goes

LOW. Now, pin 7 is shorted to ground and V_{co} becomes zero. Thus, the circuit comes back to its permanent stable state and will remain so unless another trigger pulse is applied to start the operation all over again.

The time period during which waveform remains HIGH can be determined as follows. The general equation for the capacitor voltage can be written as given in Eq. (1.78) in which constants A and B can be found by applying the conditions : at $t=0$, $V_{co}(0)=0$ and at $t=\infty$, $V_{co}(\infty)=+V_{cc}$ which gives the values of A and B as $A=V_{cc}$ and $B=-V_{cc}$, so that we have

$$V_c(t) = V_{CC}\left(1 - e^{-\frac{t}{R_A C_0}}\right) \tag{1.103}$$

Now, for $t=T$, $V_{co}(T)=+2V_{cc}/3$; hence, it is found that $T = R_A C_0 \ln 3$.

1.8.3 Sawtooth Waveform Generators

A modified form of the basic monostable multivibrator is shown in Fig. 1.45a, where the capacitor C_0 charges *linearly* through the current source comprising of the transistor Q-R_0-R_A-R_B combination. The various waveforms exploring the operation of this circuit are shown in Fig. 1.45b.

The current I_0 through R_0 (Fig. 1.45a) can be calculated as

$$I_0 = I_c \cong I_e = \frac{V_{CC} - V_e}{R_0} \tag{1.104}$$

Neglecting the base current of the transistor Q, we can write

$$V_e = V_{BE} + \left(\frac{R_B}{R_A + R_B}\right) V_{CC} \tag{1.105}$$

Therefore, I_0 can be obtained as $I_0 = \left\{\frac{V_{CC} R_A}{(R_A + R_B) R_0} - \frac{V_{BE}}{R_0}\right\}$. Since, $q = C_0 v_{C0}$

we can write $\frac{dq}{dt} = C_0 \frac{dv_{C0}}{dt}$ which yields $\frac{I_0}{C_0} = \frac{dv_{C0}}{dt}$. Now, if $T=$ time taken by the capacitor to charge to a voltage $(2/3)V_{cc}$, it is found to be

$$T = \frac{\left({}^2\!/_3 V_{CC} - 0\right)}{\left({}^{d}v_{C0}/_{dt}\right)} = \frac{2V_{CC}C_0}{3I_0} = \frac{2V_{CC}C_0R_0}{3\left(\frac{V_{CC}R_A}{R_A+R_B} - V_{BE}\right)} \tag{1.106}$$

An alternative implementation of a sawtooth generator using external current-controlled time period is shown in Fig. 1.46. For this circuit, the time period is found to be $T = \frac{2V_{CC}C_0}{3I_0}$.

Fig. 1.45 A modified form of monostable circuit for sawtooth waveform generation. (**a**) Circuit configuration, (**b**) various waveforms

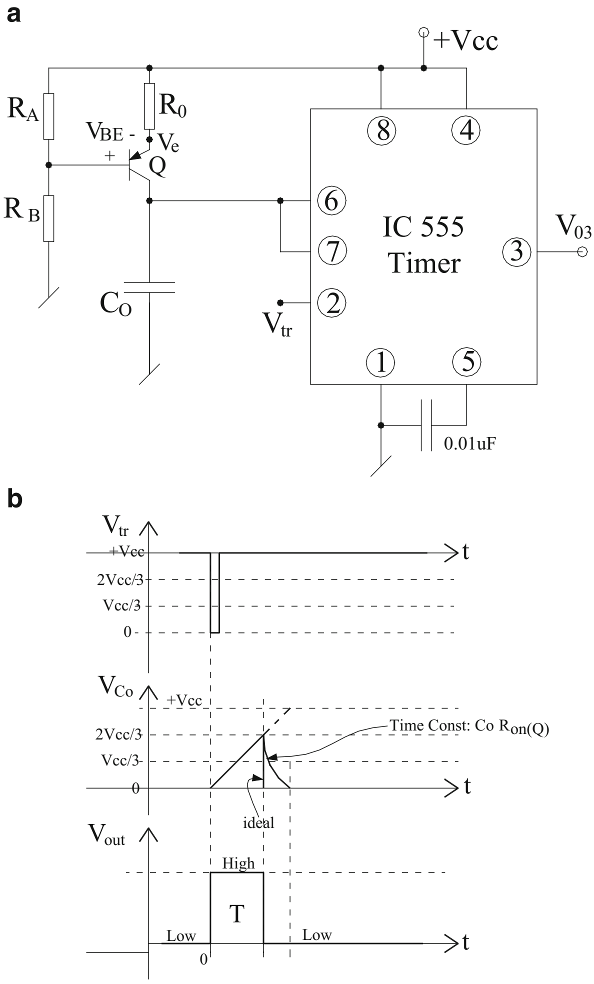
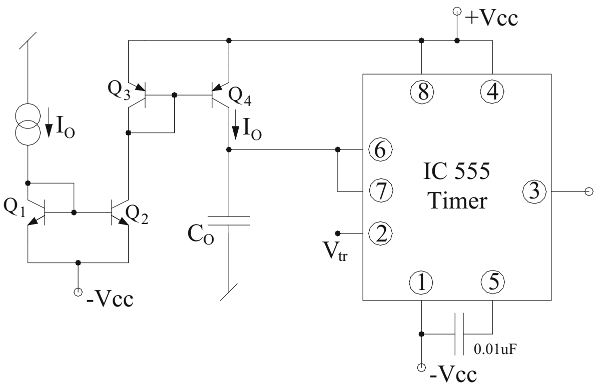


Fig. 1.46 An alternative sawtooth wave generator with current-controlled time period



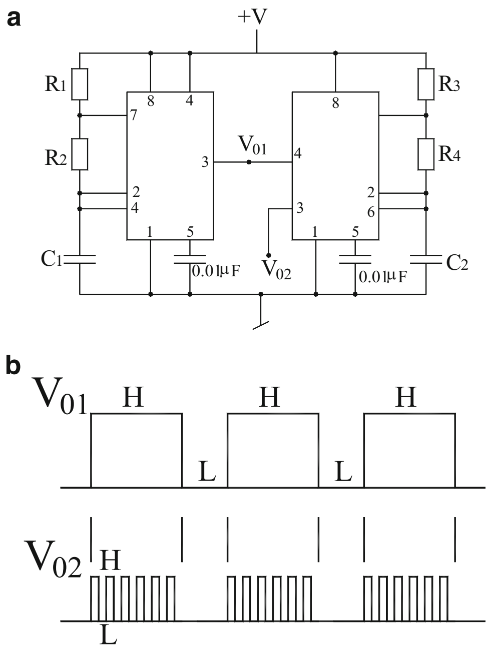


Fig. 1.47 Tone-burst generator using IC 555 timers. (**a**) Circuit diagram, (**b**) generated waveforms

1.8.4 Tone-Burst Generator

A tone-burst generator is a circuit which creates a tone after a predefined time period repetitively. This can be generated in several ways such as by applying a slowly varying square wave as one input along with fast varying square wave as another signal to a multiplier. An alternative simpler method not requiring two square wave generators and a multiplier is to use the schematic of Fig. 1.47a which is essentially a cascade of two astable multivibrators made from IC 555 timer.

It may be noted that whereas the reset terminal 4 of the first timer is wired to $+V$ (to disable its operation), on the other hand the reset terminal 4 of the second timer is effectively used to ensure that the second astable multivibrator operates only when the output V_{01} of the first timer goes LOW. Thus, if the multivibrator is designed to produce a slowly varying square wave, whereas the second timer has component chosen in such a way that this circuit generates a much higher frequency waveform, then the output V_{02} will appear as shown in Fig. 1.47b, thereby making the complete circuit a tone-burst generator.

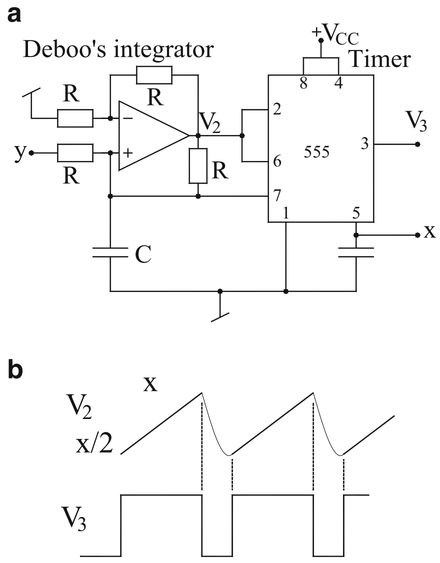


Fig. 1.48 Analog divider with digital output proposed by Rathore [42]; the circuit can also function as a VCO with external input signal *x* removed. (**a**) Circuit diagram, (**b**) generated waveforms

1.8.5 Voltage-Controlled Oscillators

We now present three circuits which employ an IC 555 timer along with an integrator to produce interesting functions.

These circuits are shown in Figs.1.48, 1.49, and 1.50 out of which those of Figs.1.48 and 1.49 were proposed by Rathore [42], while the circuit of Fig. 1.50 was proposed years earlier (although not so well known and recognized in literature) by Bhaskar Rao [105]. The first circuit of Fig. 1.48a can be operated in two different modes of operations. When two external signals x and y are applied as shown in the diagram, it can be used to generate a square wave whose time period is proportional to the ratio of the two signals (x/y) and the circuit functions as an analog divider with a digital output whose time period can be measured by counting the number of clock pulses during the time period equal to the pulse width of the generated output waveform. When the external signal x is removed, the circuit will generate voltage-controlled oscillations.

The operation of this circuit can be explained as follows: let us assume that the output of the timer is HIGH so that discharge transistor is in cutoff, thereby pin 7 acting as an open circuit. Accordingly, the Deboo's non-inverting integrator made from the op-amp integrates the signal x, and the output of the integrator is a positive going ramp with the slope ($2y$/CR). It may be noted by looking into the internal circuit diagram of IC 555 timer that when a signal x is applied on the control voltage terminal 5, the two internal comparators inside the timer have fixed references equal to "x/2" and "x" only. Correspondingly, the voltage V_{26} at the combined terminals 2 and 6 would remain confined to the two levels "x/2" and "x" only. Thus, the voltage V_{26} can rise only up to a maximum value equal to x when the timer output

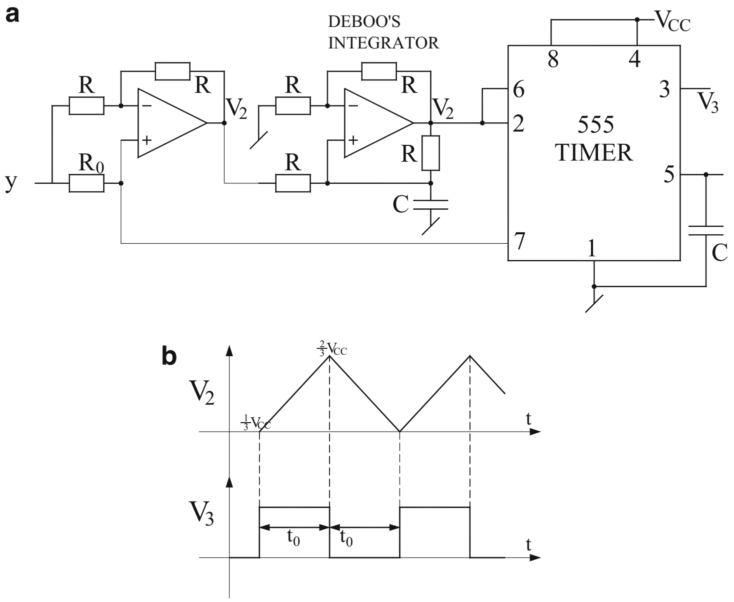


Fig. 1.49 Voltage-controlled oscillator proposed by Rathore [42]. (**a**) Circuit diagram, (**b**) associated waveforms

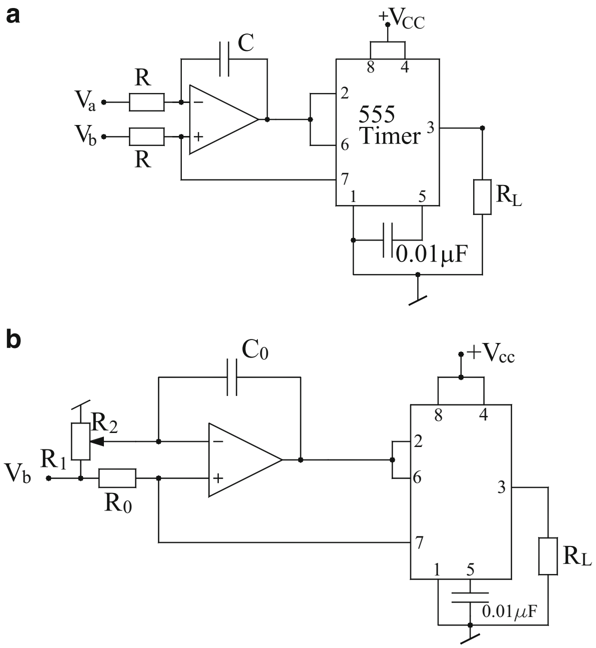


Fig. 1.50 A variable duty cycle VCO proposed by Bhaskara Rao [105]. (**a**) Duty cycle adjustment through the ratio of two voltages, (**b**) duty cycle variable through the ratio of two resistors

goes LOW and the status of discharge transistor and pin 7 changes. Discharge transistor is saturated and pin 7 acts as SHORT, thereby discharging the capacitor through the saturated transistor. Accordingly, the output V_{26} does not come down abruptly to zero but decreases exponentially to zero with a very small time constant ($C\ r_{\text{on}}$), where r_{on} is the on-resistance of the discharge transistor.

When V_{26} is just coming down from the level ($x/2$), the timer output again goes HIGH and this cycle of event repeats itself. Since V_{26} rises by a potential ($x/2$) with the rate ($2y$/RC), the time T_{H} is given by

$$T_{\text{H}} = \frac{RC}{4}\left(\frac{x}{y}\right) \tag{1.107}$$

Hence, the pulse width is proportional to the ratio of the two external voltages.

If the external signal x is not applied, then voltage V_{26} would be confined to the two threshold levels $V_{\text{cc}}/3$ and $2V_{\text{cc}}/3$ only. Accordingly, the total change in voltage V_{26} is $V_{\text{cc}}/3$ which takes place with the rate ($2y$/RC) from where the time taken is $T_{\text{H}} = \frac{RCV_{\text{cc}}}{6y}$ and the frequency of oscillation is therefore given by $f = \frac{1}{T_{\text{H}}+T_{\text{L}}}$. If $T_{\text{H}} >> T_{\text{L}}$, the frequency of oscillation will be approximated as $f \cong \frac{1}{T_{\text{H}}} = \frac{6y}{RCV_{\text{cc}}}$.

Alternatively, if T_{L} can be made equal to T_{H}, the oscillation frequency will become $f = \frac{3y}{RCV_{\text{cc}}}$.

The circuit, thus, with the external signal x applied on pin 5 removed, functions as a linear voltage-controlled oscillator (VCO). An alternative VCO with more precisely defined expression for oscillation frequency, which can also generate a symmetrical square as well as triangular waveforms with 50 % duty cycle, was also proposed by Rathore [42] and is shown in Fig. 1.49.

In this configuration, the circuit made from first amplifier acts as a unity gain amplifier with gain $+1$ or -1 depending upon whether pin 7 is acting as an OPEN circuit or SHORT circuit (which in turn depends upon whether the timer output is HIGH or LOW). Thus, in response to the external control voltage signal V_{c}, the first op-amp circuit yields an output equal to $+V_{\text{c}}$ or $-V_{\text{c}}$. The second op-amp along with the RC components is configured as a Deboo's non-inverting integrator with the transfer function given by (+2/sCR). Thus, the output of the integrator V_{26} can be either a positive going ramp with the slope ($2V_{\text{c}}$/CR) or a negative ramp with a slope $-(2V_{\text{c}}/\text{CR})$, while the timer output will be alternating between the voltages HIGH and LOW. Since the numerical value of the slopes of the positive and negative ramps is equal in magnitude, a symmetrical square wave results at the output of the timer, while a symmetrical triangular wave alternating between $V_{\text{cc}}/3$ and $2V_{\text{cc}}/3$ appears at the output of the integrator as shown in Fig. 1.49b.

By observing the waveforms, it is readily deduced that

$$T_{\text{H}} = T_{\text{L}} = \frac{\frac{2V_{\text{cc}}}{3} - \frac{V_{\text{cc}}}{3}}{\frac{2V_{\text{c}}}{CR}} = \frac{CRV_{\text{cc}}}{6V_{\text{c}}} \tag{1.108}$$

from where the frequency of the generated waveforms is given by

$$f = \frac{1}{T_{\mathrm{H}} + T_{\mathrm{L}}} = \frac{3V_{\mathrm{c}}}{CRV_{\mathrm{cc}}} \tag{1.109}$$

Thus, f is a linear function of the control voltage V_{C}, and the circuit, thus, acts as a linear voltage-controlled oscillator (VCO).

An interesting circuit, which not only is a linear VCO but also has a variable duty cycle which is obtained by adjusting a potentiometer, was proposed as early as in 1972 by Bhaskara Rao [105] and is shown here in Fig. 1.50.

By a straightforward analysis, assuming the output of the timer to be HIGH thereby pin 7 acting as an OPEN circuit, the output of the op-amp-RC circuit can be written in frequency domain as

$$V_{26}(s) = \frac{V_{\mathrm{b}}}{s} + \frac{1}{s^2 CR}(V_{\mathrm{b}} - V_{\mathrm{a}}) \tag{1.110}$$

which can be written in time domain as

$$V_{26}(t) = V_{\mathrm{b}} + \left(\frac{V_{\mathrm{b}} - V_{\mathrm{a}}}{CR}\right)t \tag{1.111}$$

Thus, with $V_{\mathrm{b}} > V_{\mathrm{a}}$, the output of the op-amp and, hence, V_{26} will be a positive ramp with the slope $(V_{\mathrm{b}} - \mathrm{V_a})/\mathrm{CR}$. On the other hand, when the timer output goes LOW and the pin 7 acts as SHORT, the op-amp-RC circuit behaves as an inverting integrator for the control signal V_{a} (with V_{b} becoming redundant) and the output of the op-amp is now a negative going ramp with slope $-V_{\mathrm{a}}/\mathrm{CR}$. As a consequence like the earlier described circuit, the output of the op-amp circuit is confined between $2V_{\mathrm{cc}}/3$ and $V_{\mathrm{cc}}/3$, while the output of the timer is alternating between HIGH and LOW. The time period during which the output waveform is HIGH is given by

$$T_{\mathrm{H}} = \frac{\frac{2V_{\mathrm{cc}}}{3} - \frac{V_{\mathrm{cc}}}{3}}{\frac{V_{\mathrm{b}} - V_{\mathrm{a}}}{CR}} = \frac{V_{\mathrm{cc}}CR}{3(V_{\mathrm{b}} - V_{\mathrm{a}})} \tag{1.112}$$

Similarly, the time period under which the waveform remains LOW can be shown to be

$$T_{\mathrm{L}} = \frac{V_{\mathrm{cc}}CR}{3V_{\mathrm{a}}} \tag{1.113}$$

In view of the above, the duty cycle of the output waveform is found to be

$$\delta = \frac{T_{\mathrm{H}}}{T_{\mathrm{H}} + T_{\mathrm{L}}} = \left(\frac{V_{\mathrm{a}}}{V_{\mathrm{b}}}\right) \tag{1.114}$$

The expression for the frequency of oscillation is found to be

$$f = \frac{3V_{\mathrm{a}}(V_{\mathrm{b}} - V_{\mathrm{a}})}{V_{\mathrm{cc}}C_{\mathrm{R}}V_{\mathrm{b}}} \tag{1.115}$$

An interesting modification of the this circuit is obtained by removing the external voltage V_a and deriving this voltage by potentiometer connecting from V_b to ground as shown in Fig. 1.50b. The duty cycle δ and frequency of oscillation f of this modified circuit can be determined easily by applying Thevenin theorem. Thus, Thevenin's equivalent consisting of an equivalent Thevenin's voltage V_a and an equivalent Thevenin's resistance R can be easily found to be

$$V_{\mathrm{a}} = \left(\frac{R_2}{R_1 + R_2}\right)V_b \text{ and } R = \frac{R_1R_2}{R_1 + R_2} \tag{1.116}$$

Substituting this value, the modified expressions for the duty cycle δ and the frequency of oscillation f are therefore given by

$$\delta = \frac{R_2}{R_1 + R_2} \text{ and } f = \frac{3v_{\mathrm{b}}}{V_{\mathrm{cc}}c_0(R_1 + R_2)} \tag{1.117}$$

From the above, therefore, it is seen that the circuit is still a linear VCO, due to f being a linear function of V_b. The duty cycle of the resulting waveform can be easily varied by adjusting the potentiometer (i.e., by changing the ratio of resistances R_1 and R_2 without affecting the frequency of oscillation) since f involves some of the R_1 and R_2 which is the total resistance of the potentiometer and hence constant.

1.9 Specialized Square Wave Generators for Measurement Applications

Many circuits have been presented in literature for generating square wave signals whose frequency or time period is a linear function of a single resistance, capacitance, or inductance. Such circuits have been evolved with the idea that such frequency/time-period converters can be employed for the measurement of physical quantities sensed by impedance-type transducers by using the well-known techniques of digital methods of time-period/frequency measurement.

A linear inductance to time period converter proposed by Senani [38] is shown in Fig. 1.51.

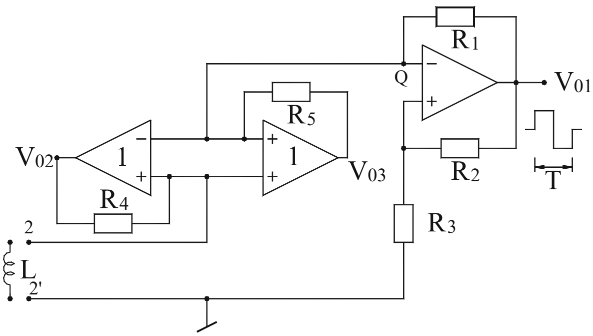


Fig. 1.51 A linear inductance-to-time-period converter proposed by Senani [38]

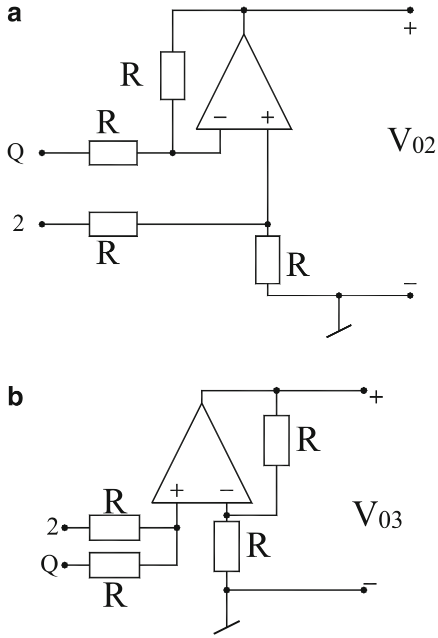


Fig. 1.52 (**a**) Realization of UGDA, (**b**) realization of UGS [38]

The circuit employs an op-amp Schmitt trigger, a unity gain differential amplifier (UGDA), and another unity gain summer (UGS) such that the UGDA-UGS-based active gyrator circuit converts the inductance into a grounded capacitance of value $C_Q = L/R_4R_5$. Realizations of UGDA and UGS are shown in Fig. 1.52. Thus, the circuit generates a square wave whose time period is ultimately given by

$$T = \mathrm{K}\mathbf{L}, \text{ where } K = \frac{2R_1 \ln\left(1 + \frac{2R_3}{R_2}\right)}{R_4R_5} \tag{1.118}$$

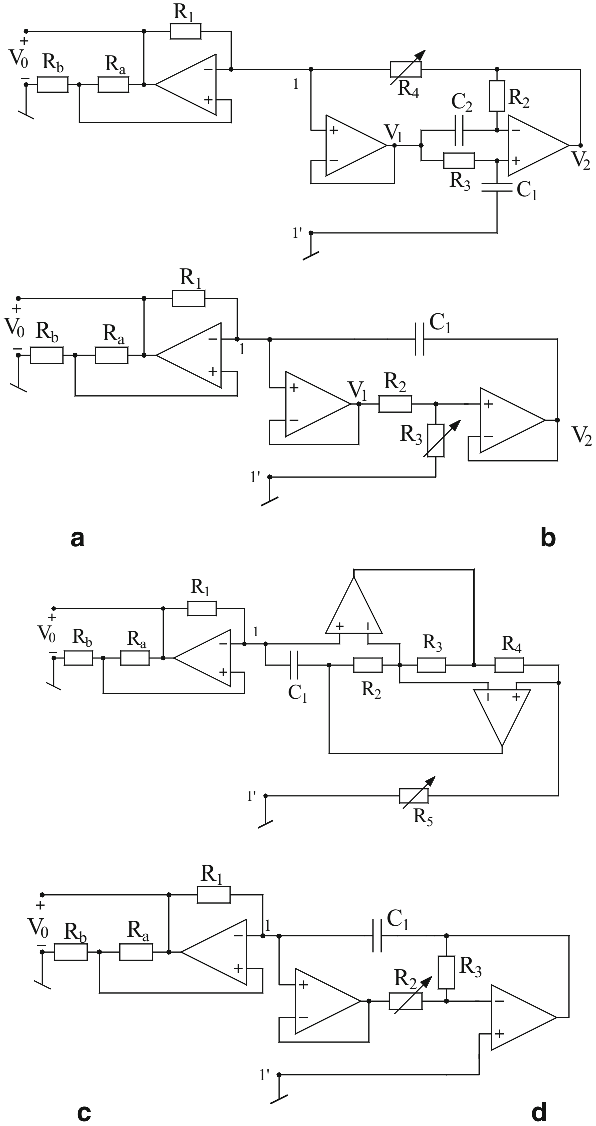


Fig. 1.53 (**a–d**) Some L/T converters using different capacitance simulators [40]

A similar technique was employed in [40] to systematically evolve a number of linear resistance to frequency converters by employing different networks of converting a resistance linearly into a grounded capacitance. Figure 1.53 shows four such circuits from [40]. In the first circuit (Fig. 1.53a), the simulated capacitance is given by $C = C_1R/R_4$ (for $R_2 = R_3 = R$, $C_2 = C_1$), thereby leading to the frequency of the square wave generated by the circuit as a linear function of resistor R_4. In the R/f converter of Fig. 1.53 (b), the amplifier A_3 along with the remaining

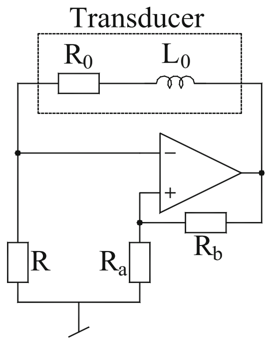


Fig. 1.54 The circuit proposed by Goras and Marcuta [106] for linear inductance-to-time-period conversion

passive components simulates a capacitance whose value is given by $C = C_1/(1 + R_3/R_2)$ which results in the frequency of the output square wave being linearly dependent on R_3. In the third circuit (Fig. 1.53c), Antoniou's generalized impedance converter (GIC) has been employed as a simulated capacitor such that the oscillation frequency becomes a linear function of R_5. In Fig. 1.53d, a capacitance multiplier circuit has been employed which multiplies the capacitor C_1 by a factor $(1 + R_3/R)$ so that, subject to the condition $R_3/R_2 >> 1$, the resistance R_2 is linearly converted into the frequency of the output square wave.

When commercially available IC operational amplifier μ741 is implemented, all the four circuits exhibited good-quality square waves and excellent linearity between the converted resistance and the obtained frequency over a frequency range 0.1–1.2 KHz.

Subsequent to the publication of the linear inductance to time period converters as given in Fig. 1.53 which requires three op-amps, Goras and Marcuta [106] highlighted a simpler alternative circuit shown here in Fig. 1.54 which requires only one op-amp and could also accommodate the effect of losses of an inductive transducer or any other inductance to be measured.

There are many situations which require grounding of one terminal of the transducer or the impedance to be measured. For example, in many cases where measurement of physical variables sensed by impedance-type transducer (inductive, capacitive, or resistive), it is often desirable to have one end of the transducer connected to ground [53]. Also when inductance-to-time-period (L/T), capacitance-to-time-period (C/T), or resistance-to-time-period (R/T) converters are intended to measure simulated inductance, capacitance, or resistance, quite often one of the terminals between which such simulated impedance is available is the ground terminal.

In reference [53], it was demonstrated that any appropriate modification of the circuits of [53] is possible to derive simplified circuit configurations which make it possible to ground the transducer or the impedance to be measured. It was also demonstrated that this modified approach is also applicable to devise a linear R/T converter having the same features. Furthermore, incorporating Antoniou's GIC leads to a configuration which can act as a generalized frequency/time-period converter (GFTC) as shown in Fig. 1.56.

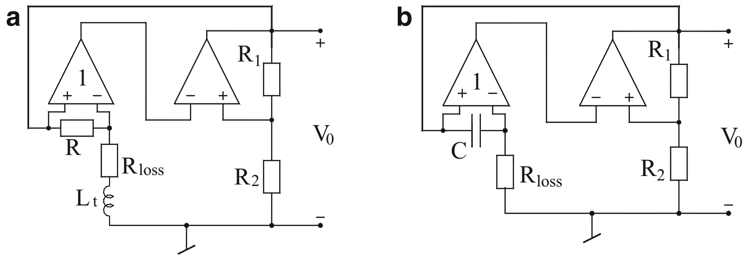


Fig. 1.55 The modified converters proposed by Senani [53]. (**a**) linear L/T converter. (**b**) linear R/T converter

For the modified L/T converter obtained by appropriate addition of the UGDA, the time period of the square wave generated by the circuit of Fig. 1.55a is given by

$$T = \frac{L_t}{(R + R_{\text{loss}})}\left\{\ln\frac{1+\beta\frac{V_{01}}{V_{02}}}{1-\beta} + \ln\frac{1+\beta\frac{V_{02}}{V_{01}}}{1-\beta}\right\} \tag{1.119}$$

where $\beta = R_2/(R_1 + R_2)$ and V_{01} and V_{02} are the positive and negative saturation levels of the op-amp comparator. Thus, it is seen that, in contrast to the Goras-Marcuta circuit [106], the modified circuit presented above can have one end of the transducer connected to ground as well as can account for the losses of the transducer.

Similarly, in the configuration shown in Fig. 1.55b, a linear R/T converter is achieved, with the time period of the generated square wave being given by

$$T = KCR_{\text{t}} \tag{1.120}$$

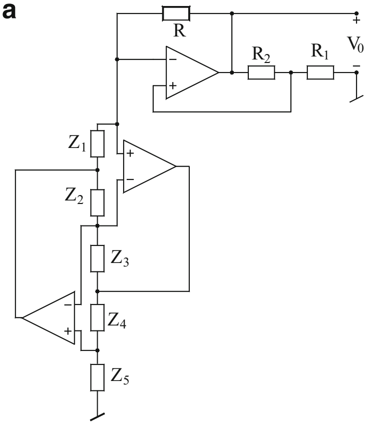
where R_t is the resistance of transducer or any other resistance to be measured and K is given by

$$K = \left\{\ln\frac{1+\beta\frac{V_{01}}{V_{02}}}{1-\beta} + \ln\frac{1+\beta\frac{V_{02}}{V_{01}}}{1-\beta}\right\} \tag{1.121}$$

It is worthwhile to mention that classical astable multivibrator in its original form is already a linear C/T converter which can easily accommodate the losses of the capacitive transducer since the expression for the time period of the square wave generated by this circuit, with losses of the capacitance/capacitive transducer accounted, is given by

$$T = KC_{\text{t}}\left\{\frac{RR_{\text{loss}}}{R + R_{\text{loss}}}\right\} \tag{1.122}$$

a



b

No	Types of conversion	Selection of elements					Time period/ frequency	Converted variable
		Z_1	Z_2	Z_3	Z_4	Z_5		
1	R/f	C_1	r_2	r_3	r_4	r_5	$f = (\frac{Kr_2}{RC_1 r_2 r_4}) r_5$	r_5
2	L/f	C_1	r_2	C_3	r_4	L_5	$f = (\frac{K}{RC_1 C_3 r_2 r_4}) L_5$	L_5
3	C/T	r_1	r_2	r_3	r_4	C_5	$T = (\frac{Rr_2 r_4}{Kr_1 r_3}) C_5$	C_5
4	R/T	r_1	r_2	C_3	r_4	r_5	$T = (\frac{RC_3 r_4}{Kr_1 r_5}) r_2$	r_2
5	L/T	r_1	L_2	r_3	r_4	r_5	$T = (\frac{Rr_4}{Kr_1 r_3 r_5}) L_2$	L_2
6	C/f	r_1	C_2	C_3	r_4	C_5	$f = (\frac{Kr_1}{RC_3 C_5 r_4}) C_2$	C_2

Fig. 1.56 Generalized frequency/time-period converter [53]. (**a**) GFTC, (**b**) various special cases

The last part of Fig. 1.56 contains Antoniou's GIC which makes the resulting circuit a generalized frequency/time-period converter which is capable of providing all the six possible linear conversion, namely, L/T, R/T, C/T, inductance to frequency (L/f), R/f, and capacitance to frequency (C/f) conversion in accordance with the appropriate selection of various impedances in the circuit as outlined in Fig. 1.56b.

It may, however, be mentioned that the grounding of the impedance to be measured is possible only in case of R/f, L/f, and C/T conversions only. To the best of authors' knowledge, the problems of evolving an alternative GFTC circuit, providing the feature of grounding of transducer in all the six cases, have not been resolved in the open literature till the writing of this monograph.

1.10 IC Function Generators

Although a number of IC function generators can be found in IC manufacturer's data sheets and catalogue, here we briefly describe two of the relatively more popular ones, namely, the LM566 and ICL8038 function generators.

1.10.1 LM566 VCO

LM566 is a versatile function generator IC which can be used to generate square and triangular waveforms and can also be used as a voltage-controlled oscillator (VCO). VCOs find applications in several areas such as frequency modulation (FM) and pulse modulation (PM) and are also a basic building block of phase-locked loops (PLL). On the other hand, a VCO can also be used as variable frequency signal generator by itself.

A typical connection of the IC 566 VCO as a basic square/triangular wave generator is shown here in Fig. 1.57, where the control voltage is derived by setting a potentiometer. With appropriate values of the timing resistor and capacitor connected as shown, the circuit simultaneously generates a square wave and triangular wave with the frequency of the generated waveform given by

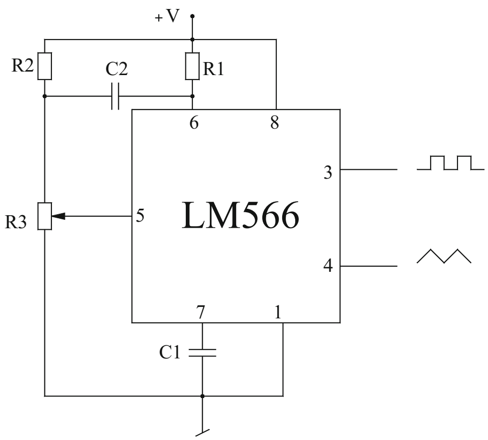


Fig. 1.57 A typical connection of the IC 566 VCO as a basic square/triangular wave generator

$$f_0 = \frac{2.4(V^+ - V_5)}{RCV^+} \tag{1.123}$$

Thus, the frequency of the output waveform can be either adjusted using an external control voltage or can be set using an external resistor and capacitor. Some typical applications where LM566 IC can be usefully employed are signal generators, FM modulators, FSK modulators, and tone generators. The LM566 IC can be operated from a single supply or a dual supply. While using a single supply, the supply voltage range is from 10 to 24 V.

1.10.2 ICL8038 IC Function Generator

ICL8038 function generator is another versatile IC which can produce a square wave, a triangular wave, and a sine wave simultaneously whose frequency and duty cycle (in case of square wave) can be adjusted by applying appropriate signals or connecting a potentiometer between specified pins of the IC. A simplified block diagram of this is shown in Fig. 1.58a, whereas a typical connection using external RC components is shown in Fig. 1.58b.

As can be seen from the block diagram, the internal circuit contains a constant current source I which would charge the external capacitor C to be connected between pin 10 and pin 11. Each comparator has a fixed reference voltage connected to one of its input terminals. The two outputs of the comparators are applied to a flip-flop with complementary outputs such that one of them controls the switch S which when closed forces a current equal to I in the capacitor C but in the reverse direction. Another output of the flip-flop is passed through a buffer and creates the square wave output terminal. Since the capacitor is charged by a current either +I or −I in either case, the voltage across the capacitor rises linearly, thereby resulting in either a positive ramp or a negative ramp, the slop of the ramp being equal to (I/C). Thus, the voltage at node 10 is a symmetrical triangular wave which is limited to $+V_{\mathrm{ref1}}$ and $-V_{\mathrm{ref2}}$. This output is taking to a buffer and is made available as triangular waveform at pin 3. Lastly, through the internal triangular to sine wave shaper (converter), this triangular waveform is converted into sinusoidal signal at pin 2. The DC power supply for IC 8038 could be a dual power supply connected between terminals 6 and 11 or it could be a single power supply connected at terminal 6 in which case terminal 11 is grounded. Typical circuit connections using a pull-up resistor of 10K, a potentiometer of 1K for duty cycle, and frequency adjustment with pins 7 and 8 shorted and external timing capacitor connected from pin 10–11 for the generation of a square, triangular, and sinusoidal signal are shown in Fig. 1.58b.

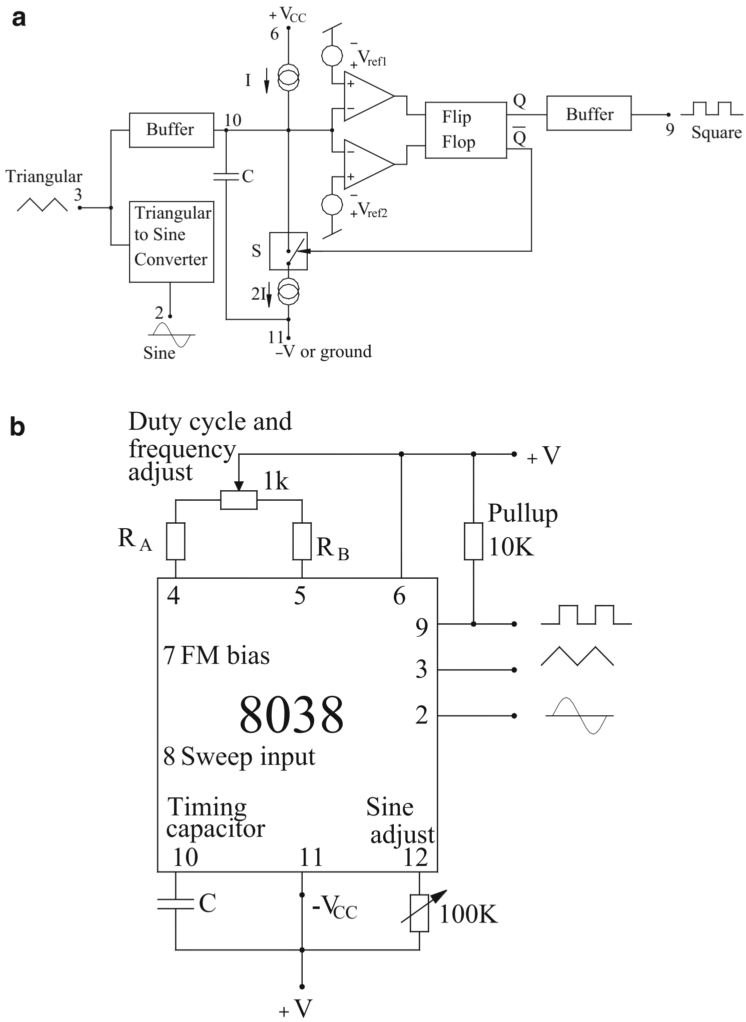


Fig. 1.58 ICL8038 function generator. (**a**) Simplified block diagram, (**b**) a typical connection showing external component connections

1.11 Concluding Remarks

In this introductory chapter, a number of basic sinusoidal oscillator circuits and relaxation oscillators/nonsinusoidal waveform generators have been described. Since the focus of the present monograph is on realization of the abovementioned classes of circuits using modern building blocks, the transistor level implementations as well as FET-based realizations of classical Wien bridge, RC phase-shift, Colpitts, and Hartley oscillators have not been included which are well documented in almost all standard text books/reference books dealing with Electronics and Electronic circuits. Thus, the major emphasis has been on IC op-amp-based sinusoidal oscillators due to the commercial availability of a number of general-purpose op-amps (like UA741/LF356) as well as ease of realization offered by them vis-a-vis their discrete counterparts realized with either BJTs or FETs. In the area of nonsinusoidal oscillators, most basic circuits using IC op-amps as comparators as well as those realizable from the versatile IC 555 timer have been elaborated.

Although the major intention of this chapter has been to cover most of the classical circuits in both categories, a number of sinusoidal oscillators realized with multiple number of op-amps as well as a number of waveform generators realizable by combinations of IC timers and IC op-amps were also included. The authors of this monograph have been teaching most of the material covered in this chapter in a number of basic courses on Analog Integrated Circuits for the past three decades or more, and therefore, in the opinion of the authors, all the circuits described in this chapter appear to have acquired the status of being *classical* circuits though we do acknowledge that some of these may not necessarily turnout to be so well known to many readers and instructors of various courses dealing with electronic circuits elsewhere.

Subsequent chapters of this monograph will unfold oscillators and nonsinusoidal waveform generators/relaxation oscillators realized with either a specific building block or a specific class of building block. However, in doing so, the major emphasis has been to highlight the salient features of the chosen circuits. In doing so, however, it has not been possible to detail out the complete derivation or complete analysis of all the circuits chosen, because of a large number of circuits (over 600) intended to be covered in this monograph. Nevertheless, wherever considered appropriate and necessary, complete derivations are given in some cases; in others, either the key steps of the derivation are given or only the key results are given, assuming that wherever the reader requires more details, he/she can always go to the concerned original source (reference).

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Part II
Various kinds of Sinusoidal Oscillators

Chapter 2
Single-Element-Controlled and Other Varieties of Op-Amp Sinusoidal Oscillators

Abstract This chapter presents a comprehensive treatment on the evolution of single-element-controlled oscillators (SECOs) in general and single-resistance-controlled oscillators (SRCOs) in particular. The discussion includes a wide variety of op-amp-based oscillators such as single-op-amp-based SRCOs, grounded-capacitor (GC) SRCOs, and unity gain amplifier-based SRCOs. Also included are oscillators with linear tuning laws, varieties of active-R oscillators, and op-amp-based voltage-controlled oscillators (VCOs).

2.1 Introduction

In view of the versatility of the IC op-amps, coupled with the easy availability of hundreds of their varieties as *off-the-shelf* integrated circuits (ICs), they have been extensively used and relied upon to realize both sinusoidal oscillators and relaxation oscillators [1–136]. Several researchers have demonstrated that the operational frequency range of op-amp oscillators made from general-purpose op-amps like μA741 can be extended beyond the audio frequency range by either involving active-compensated building blocks and the so-called composite amplifiers (for instance, see 12, 44, 54, 68, 78, 82–85, 88, 91, 96) or by resorting to the use of the *active-R* design methodology [4, 13, 18, 22, 24, 31, 32, 35, 37, 38, 45, 47, 60, 65, 79, 81, 86, 89, 92, 97, 99–103, 106, 108, 114, 115, 117, 130] wherein the otherwise parasitic op-amp compensation pole of the internally compensated type of op-amps (such as μA741 of LF356) is treated as a useful parameter in the design, thereby resulting in circuits which can operate satisfactorily over several hundreds of KHz and, in some specific topologies, even up to frequencies nearing the gain-bandwidth product of the op-amps used!

In this chapter, we present a variety of op-amp-based sinusoidal oscillators with particular emphasis on single-element-controlled ones. However, several other types of oscillators such as those possessing *linear* tuning laws, as well as those which permit easy conversion into voltage-controlled oscillators (VCO), have also been dealt with.

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2.2 Some Earlier Variable-Frequency Single-Op-Amp Oscillators

Since none of the classical oscillators using op-amps, namely, the Wien bridge oscillator, the RC phase-shift oscillators, and the twin-T oscillators, are capable of providing the control of frequency of oscillation through a single-variable element (either a single-variable capacitor or a single-variable resistor, preferably the latter), the attention of researchers was initially focused on the most popular Wien bridge oscillator to find out whether any modification of this classical circuit could lead to single-element control. Two pioneering works in this direction were those of Shivprasada [3] and Dutta Roy [11].

Shivprasada [3] demonstrated that modifying the parallel RC branch of the frequency-selective RC network of the Wien bridge oscillator, by inserting a variable resistor in series with the capacitor as shown in Fig. 2.1a, leads to the frequency control through this additional resistor.

By a routine analysis, the frequency of oscillation is found to be

$$\omega_0 = \frac{1}{\sqrt{C_1 C_2 R_1 R_2}} \left(\frac{1}{\left[1 - m(1+m)\dfrac{C_1 R_1}{C_2 R_2} \right]} \right)^{1/2} \tag{2.1}$$

whereas the condition for sustained oscillations is found to be

$$K = 1 + \frac{R_2}{R_1} + (1+m)\frac{C_1}{C_2} \tag{2.2}$$

If we let

$$\omega_n = \frac{1}{R_1 C_1} = \frac{n}{R_2 C_2} \tag{2.3}$$

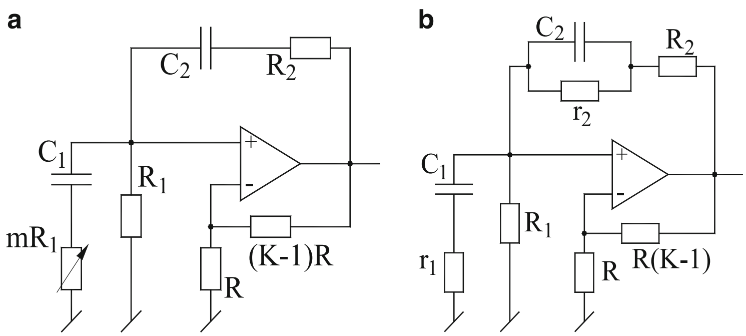


Fig. 2.1 The modified versions of the Wien bridge oscillator for single-element control. (**a**) Shivprasada's modification [3]. (**b**) Dutta Roy's modification [11]

then Eq. (2.1) can be rewritten as

$$\omega_0 = \frac{\omega_n}{\sqrt{(n - m - m^2)}} \tag{2.4}$$

Based upon the above, it was claimed in [3] that the oscillation frequency can be varied from some minimum value to a very high value (theoretically, even infinite) by proper selection of the parameter **m**, with the highest attainable frequency being limited only by (i) the loading between the amplifier and the phase-shift network and (ii) the finite GBP of the op-amp employed. The modified Wien bridge oscillator was shown to be useful in the design of frequency-shift-keyed oscillator and swept frequency oscillator. However, from an inspection of the Eqs. (2.2) and (2.4), it is clear that even though frequency can be varied by varying the parameter **m**, the condition of oscillation is also dependent upon **m** and therefore the circuit *does not possess noninteracting controls*!

Subsequently, Dutta Roy [11] demonstrated that a more generalized feedback network in place of the conventional feedback network provides further flexibility of obtaining variable-frequency oscillations (see Fig. 2.1b).

With $R_1C_1 = R_2C_2 = RC$, $n = R_2/r_2$, and $m = r_1/R_1$, the expressions for the oscillation frequency and the condition of oscillation for this circuit were found to be [11]

$$\omega_0 = \left(\frac{1}{RC}\right)\sqrt{\frac{[1 - n(1 + n)]}{[1 - m(1 + m)]}} \tag{2.5}$$

$$K = 1 + \frac{[1 + (n + m + 1)]}{[mn + 1]} \tag{2.6}$$

Out of the four possible cases investigated for different ranges of values of the parameters **m** and **n**, it was shown therein [11] that in two of the cases, good-quality variable-frequency oscillations were obtainable over a decade at frequencies lower than 100 Hz and above 10 KHz, respectively, while in the remaining two cases, the frequency range obtainable was two octaves and one octave, respectively. However, from Eqs. (2.5) and (2.6), it may be noted that *noninteracting controls of oscillation frequency and condition of oscillation are not available even in this generalized structure*.

2.3 Two-Op-Amp-Based Single-Resistance-Controlled Oscillators (SRCOs)

Since the initial developments on the modification of the classical Wien bridge oscillator did not yield any single-resistance-controlled oscillator *without any constraints*, clearly, the problem of devising a good single-op-amp oscillator configuration with independent control for both condition of oscillation (CO) and frequency of oscillation (FO) remained an unresolved problem till about 1979. Meanwhile, a number of ideas were proposed by several researchers which established that an oscillator possessing both of these desirable properties could, however, be made from two op-amps *without any constraints*. In this section, we outline two such *op-amp-based* single-element-controlled oscillators.

2.3.1 Oscillator Realization Using the Concept of FDNR

The first method to create an SRCO, *without any constraints* using no more than two op-amps, came from Genin [7] who demonstrated that if the Bruton's transformation [8] is applied on a parallel RLC resonator, consisting of an inductor, a capacitor, a positive resistor, and a negative resistor (thereby representing a sinusoidal oscillator), this transforms the inductor into a resistor, the resistor into a capacitor, the negative resistor into a negative capacitor and finally, and the capacitor into a *frequency-dependent-negative resistance* (FDNR) – an element having input impedance of the form $Z(s) = 1/Ds^2$. The transformed model, thus, realizes an SRCO. Genin demonstrated [7] that the parallel combination of the negative capacitor and FDNR can be realized by a two-op-amp circuit such that connecting a parallel RC across it results in an SRCO. Genin's oscillator is shown here in Fig. 2.2. The CO and FO for this oscillator can be easily derived as

$$\mathrm{CO}: \mu \leq \frac{\lambda C_2}{C_2 + (1+\lambda)C_0} \quad \text{and} \quad \mathrm{FO}: \omega_0 = \sqrt{\frac{(1+\lambda)}{C_1 C_2 R_0 R}} \tag{2.7}$$

Thus, FO can be varied independently by the grounded resistance R_0.

Senani [119] demonstrated that Genin's [7] approach can be modified and applied to other forms of RLC resonators also. When this is done, the modified transformed resonators, using Bruton's transformation [8], do result in alternative two-op-amp-based oscillators providing independent control of CO and FO. Consider now one of the resonators from [119], as shown in Fig. 2.3a. If the Bruton's transformation is applied on this model, the transformed circuit turns out to be as shown in Fig. 2.3b. An op-amp circuit can now be devised from this model by simulating the series CD branch by a single-op-amp circuit shown in dotted box,

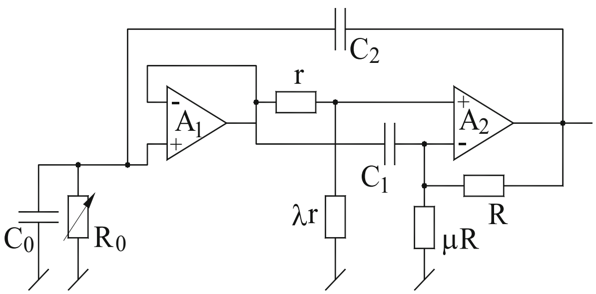


Fig. 2.2 Single-resistance-controlled oscillator using Genin's approach [7] based on the concept of FDNR

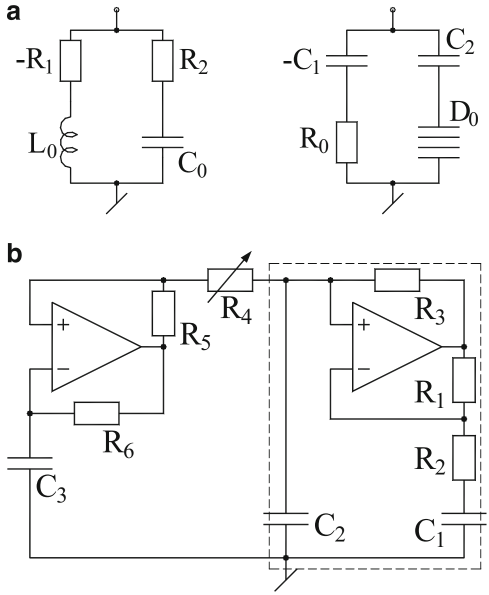


Fig. 2.3 Derivation of two-op-amp SRCO using the concept of FDNR. (**a**) Alternative RCD model derived from RLC model for SRCO synthesis. (**b**) An exemplary synthesized circuit [119]

whereas the series RC part can be realized by positive resistor R_4 in series with a negative capacitance $-C_1$ which is realized by an op-amp configured as an NIC and a grounded-capacitor C_3. Since the series CD branch is realizable subject to the conditions $C_1 = C_2 = C_0$ and $R_1 = R_3 = R$, the CO and FO of the resulting circuit of Fig. 2.3b are given by

$$C_0 = R_6 C_3 / R_5 \tag{2.8}$$

$$f_0 = \frac{1}{2\pi C} \sqrt{\frac{1}{R_2 R_4}} \tag{2.9}$$

It is seen that f_0 can be varied through either R_2 or R_4 and that CO is also independently adjustable by R_5 or R_6.

This circuit is noncanonic due to the employment of three capacitors but offers the advantage of having all capacitors grounded which is a preferable feature from the viewpoint of IC implementation [120, 121].

2.3.2 *Single-Resistance-Controlled/Voltage-Controlled Oscillators (VCOs)*

The second successful implementation of a two-op-amp SRCO came from Hribsek and Newcomb [10] who proposed two circuits for realizing single-resistance-controlled oscillators. These circuits are shown in Fig. 2.4. The CO for both the circuits is fulfilled by having the gain of the non-inverting amplifier $K_1 = 4$. The FO for these circuits is then given by (Fig. 2.5a, b)

$$\omega_0 = \frac{1}{RC}\sqrt{2 - 4K_2} \quad \text{for the Fig. 2.5a} \tag{2.10}$$

and

$$\omega_0 = \frac{1}{RC}\sqrt{\frac{2}{1 - 4K_2}} \quad \text{for the Fig. 2.5b} \tag{2.11}$$

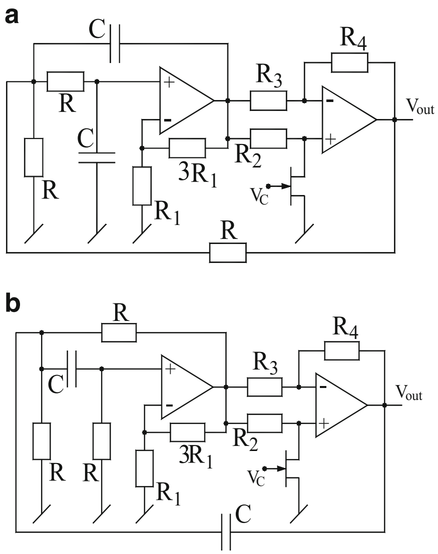


Fig. 2.4 VCOs proposed by Hribsek and Newcomb [10]. (**a**) VCO based upon a low-pass filter. (**b**) VCO based upon a band-pass filter

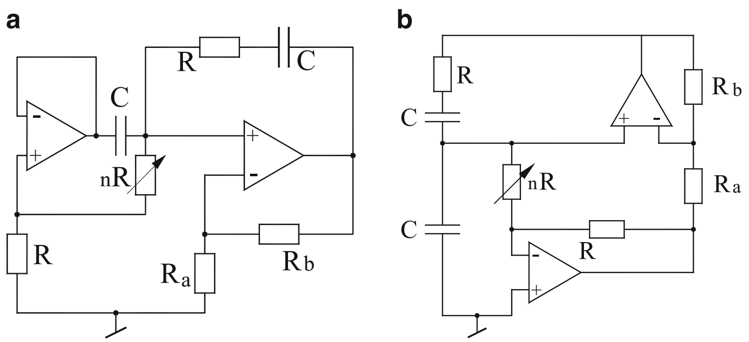


Fig. 2.5 Two different modified versions of the Wein bridge oscillator proposed by Williams [41] for obtaining single-resistance control of the oscillation frequency

where

$$K_2 = -\left(\frac{1}{R_3}\right)\frac{R_2R_4 - R_3R_{\mathrm{DS}}}{R_2 + R_{\mathrm{DS}}} \quad \text{and} \quad R_{\mathrm{DS}} = \frac{1}{\beta[V_{\mathrm{C}} + V_{\mathrm{P}}]} \tag{2.12}$$

V_{P} being the pinch-off voltage for depletion devices and negative of the threshold voltage for enhancement-mode devices and β being a constant dependent on the properties of the channel.

2.3.3 Modified Single-Element-Controlled Wien Bridge Oscillators

It was shown by Williams [41] that the classical Wien bridge oscillator can be modified in a number of ways, by appropriately incorporating one more op-amp along with only one additional resistor in the circuit, to enable obtaining the control of FO through a single-variable resistance. Two of the Williams propositions are shown here in Fig. 2.5, both of which have the expression of the frequency of oscillation as

$$\omega_0 = \frac{1}{\sqrt{n}CR} \tag{2.13}$$

whereas to fulfill the condition of oscillation, the resistors R_{b} and R_{a} are required to be equal.

It is, thus, seen that the FO can be continuously varied through the variable resistance **n**R without affecting CO which is $R_{\mathrm{b}} = R_{\mathrm{a}}$ and, therefore, independent of **n**.

2.3.4 Two-Op-Amp SRCO Employing Simulated Inductors

In retrospection, it is not difficult to arrive at other alternative means of systematically generating two-op-amp-based circuits which can realize SRCOs. An exemplary circuit of this type can be readily obtained from the ± RLC model by simulating the parallel RL part by Senani's single-op-amp single-resistance-tunable grounded inductor circuit from [48] and then connecting a capacitor C_{o} and a negative resistance $-R_{\mathrm{o}}$ (simulated by a single-op-amp negative-impedance converter) across the input port of such a circuit. A circuit conforming to this formulation is shown in Fig. 2.6a.[1] The equivalent resistance R_{eq} and equivalent inductance L_{eq} of the inductor simulator is given by

[1] This circuit, however, has not been published in the open literature earlier.

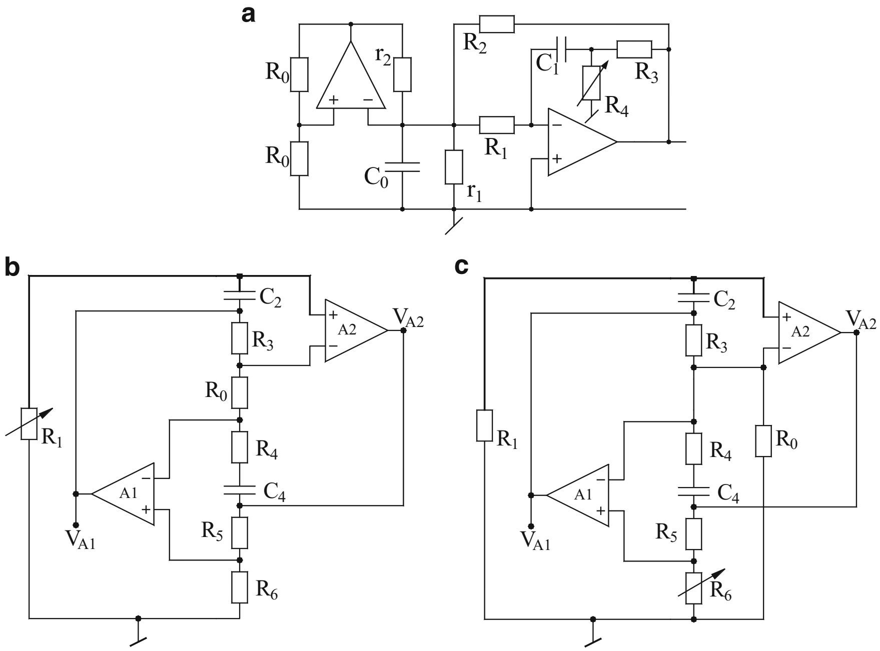


Fig. 2.6 Some exemplary SRCOs using only two op-amps employing simulated inductors and modified GICs. (**a**) Oscillator using a single-op-amp simulated inductor proposed by Senani [48]. (**b**) An SRCO using Wangenheim's modified GIC [111]. (**c**) An SRCO using Senani's modified GIC [135]

$$\frac{1}{R_{\mathrm{eq}}}=\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{R_3}{R_1R_2}\right);\ \ L_{\mathrm{eq}}=\left(\frac{C_1R_1R_2}{1+\dfrac{R_3}{R_4}}\right) \tag{2.14}$$

The second op-amp configured as a negative-impedance converter (NIC) creates a negative resistance $-R_{\mathrm{o}}$. Thus, an oscillator results by connecting a capacitor $\mathrm{C_o}$across the circuit. The CO and FO for this oscillator are given by

$$\mathrm{CO}:-\frac{1}{R_0}=-\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{R_3}{R_1R_2}\right) \tag{2.15}$$

$$\mathrm{FO}:\omega_0=\sqrt{\frac{\left(1+\dfrac{R_3}{R_4}\right)}{C_0C_1R_1R_2}} \tag{2.16}$$

It is, therefore, seen that CO and FO are independently controllable, the former by R_{o} and the latter by R_4.

Two other interesting two-op-amp single-element-controlled oscillators can be formulated from two modifications of the classical GIC, one proposed by Wangenheim [111] and the other proposed by Senani [135]. Out of the various possibilities of designing SRCOs using these modified GICs, two specific circuits are shown here in Fig. 2.6b, c. The CO and FO of these GIC-based oscillators are given by

For the oscillator of Fig. 2.6b ($R_5 = R_6$)

$$\mathrm{CO}: (R_4 - R_0) \leq 0; \quad \mathrm{FO}: \omega_0 = \sqrt{\frac{1}{C_2 C_4 R_1 R_3}} \tag{2.17}$$

For the SRCO of Fig. 2.6c

$$\mathrm{CO}: (C_2 R_1 R_3 - C_4 R_0 R_4) \leq 0; \quad \mathrm{FO}: \omega_0 = \sqrt{\frac{R_0 R_6}{C_2 C_4 R_1 R_3 (R_0 R_5 - R_4 R_6)}} \tag{2.18}$$

In this case also, FO is independently controllable by R_5 and/or R_6.

2.4 Single-Op-Amp-Based Single-Capacitor-Controlled Oscillator

The first clear-cut circuit which could provide single-element-controlled oscillation frequency, without any constraints, was proposed by Soliman and Awad [21], although it required a variable capacitor to do so. This circuit is shown in Fig. 2.7.

By straightforward analysis, the conditions required to produce sinusoidal oscillations are found to be

$$\mathrm{CO}: \frac{C_1}{C_2} = 2 + \frac{R_3}{R_1}; \quad \text{provided } R_1 = R_2; R_3 = R_4 \tag{2.19}$$

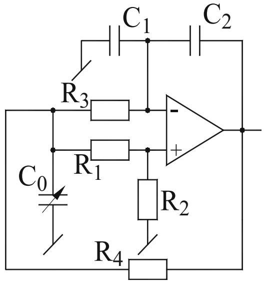


Fig. 2.7 Single-capacitor-controlled oscillator proposed by Soliman and Awad [21]

The FO is then given by

$$f_0 = \frac{1}{2\pi R_3 \sqrt{2C_0 C_2}} \tag{2.20}$$

It is, thus, seen that the oscillation frequency can be varied through a single-variable capacitor C_0.

2.5 Single-Op-Amp-Based SRCOs

After the reporting of the single-op-amp single-capacitance-controlled oscillator described in the previous section in 1978, it was apparent that a more preferable circuit would be one which could provide noninteracting controls of condition of oscillation (CO) as well as frequency of oscillation (FO) through separate variable resistors. Such a circuit was proposed by Senani [26] in 1979 which has been widely cited by other researchers between1979 and 2012.[2] This circuit gave rise to many subsequent investigations which would be described in the next subsections.

2.5.1 Single-Op-Amp-Based Single-Resistance-Controlled Oscillator

Senani's single-resistance-controlled oscillator (SRCO) quoted above has been presented here in Fig. 2.8. In retrospection, this circuit can be shown to be obtainable from Genin's circuit [7] by removing the op-amp configured as unity gain voltage follower and adding one resistor in parallel with the capacitor C_1. Moreover, it may be noted that while the Genin's circuit implements a parallel RCD resonator comprised of a resistor, a positive capacitor, a negative capacitor, and a frequency-dependent negative impedance characterized by input impedance $Z(s) = 1/Ds^2$, Senani's circuit (with R_1 omitted) is also equivalent to the same parallel resonators albeit with different values of the equivalent impedances. A straightforward analysis of Senani's circuit, with R_1 deleted and capacitors taken equal valued, i.e., $C_1 = C_2 = C$, reveals that the values of the equivalent elements are given by

[2] It was brought to the attention of the first author, only at the time of finalizing Chap. 5 of the monograph [122] (during 13–17 September 2012), that a similar single-op-amp SRCO employing only five resistors and two capacitors had earlier been proposed by Soliman and Awad in 1978 in [23]. Curiously, this reference remained unnoticed by all researchers working on SRCOs, e.g., those of [29, 42, 51, 53, 131, 137–140], all of whom cited [26] but apparently were completely unaware about the existence of [23].

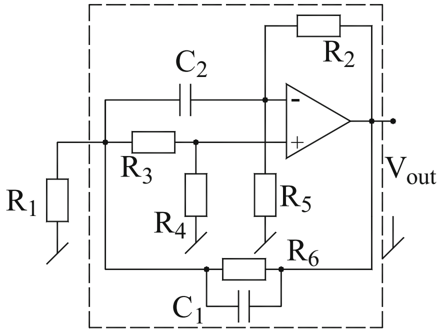


Fig. 2.8 Senani's [26] single-op-amp SRCO without constraints

$$
\begin{aligned}
\frac{1}{R_{\mathrm{eq}}} &= \frac{[R_5(R_3+R_6)-R_2R_4]}{(R_3+R_4)R_5R_6} && \text{(a)}\\
C_{\mathrm{eq1}} &= \frac{CR_3(R_2+2R_6)}{(R_3+R_4)R_6} && \text{(b)}\\
-C_{\mathrm{eq2}} &= -\frac{CR_2R_4}{(R_3+R_4)R_5} && \text{(c)}\\
D_{\mathrm{eq}} &= \frac{C^2R_2R_3}{(R_3+R_4)} && \text{(d)}
\end{aligned}
\tag{2.21}
$$

A single-resistance-controlled oscillator is realizable by setting $1/R_{\mathrm{eq}}=0$ and connecting the resistor R_1. The condition of oscillation is determined by setting $C_{\mathrm{eq1}}=C_{\mathrm{eq2}}$, whereas the frequency of oscillation is given by $f_0=\frac{1}{2\pi\sqrt{RD}}$. Using Eq. (2.21), FO and CO are found to be

$$\frac{R_4}{R_3}=\frac{2R_5}{R_2}+\frac{R_5}{R_6};\quad \text{with } C_1=C_2=C \tag{2.22}$$

$$f_0=\frac{1}{2\pi C}\sqrt{\left[\left(1+\frac{R_3}{R_6}+\frac{R_3+R_4}{R_1}-\frac{R_2R_4}{R_5R_6}\right)/R_2R_3\right]} \tag{2.23}$$

It is worth mentioning that this circuit uses one less op-amp as compared to Genin's circuit and one less capacitor as compared to Genin's circuit as well as Ahmed's single-op-amp low-component-count sine-wave generators [17] and provides independent control of oscillation frequency through a single grounded resistor in contrast to restricted control in the circuits of [17]. The following features of this circuit are noteworthy:

1. Use as a VCO: by replacing R_1 by an FET (used a voltage-controlled resistance), the circuit can be used as a VCO. In this mode, the circuit is more economical than Hribsek-Newcomb circuit [10] as well as Sundaramurthy-Bhattacharyya-Swamy

[16] circuit because of the requirement of only one op-amp, in contrast to the preceding circuits requiring two and three op-amps, respectively.

2. Use as a very low-frequency oscillator: since the expression for FO contains a difference term, the circuit can be used as a very low-frequency oscillator (VLFO) by choosing the component value as $R_2 = R_3 = R_5 = R_6 = R$, $R_4 = 3R$, $R_1 = R/n$, thereby leading to the modified expression for FO as

$$f_0 = \frac{1}{2\pi RC}(4n-1)^{1/2} \tag{2.24}$$

From the above, it is seen that by appropriate choice of **n**, the circuit can be used to generate very low frequencies without having to use large-valued RC components.

2.5.2 Identification and Design of Single-Amplifier SRCOs

Pyara, Dutta Roy, and Jamuar [51] identified a set of single-op-amp SRCOs by formulating a generalized two-port network containing nine admittances and a single op-amp from where they derived the CO and single-element controllability in terms of the Y-parameters of the passive network and the finite frequency-dependent gain of the op-amp. Through a rigorous search, they generated a family of 14 SRCOs, out of which as many as eight were SRCOs using infinite-gain op-amp, whereas the remaining six circuits were single-capacitor networks which function as oscillator incorporating the dominant pole of the op-amp as a parameter incorporated in the design. Out of these the eight SRCOs are displayed in Fig. 2.9. Considering equal-valued capacitors for the structures of Circuit 1 to Circuit 4, the CO and FO for these SRCOs are obtained as

$$\text{Circuit 1 CO}: (2R_2R_8 - R_6R_7) \le 0; \text{FO}: f_0 = \frac{1}{2\pi C}\sqrt{\frac{R_2+R_4+R_6}{R_2R_4R_7}} \tag{2.25}$$

$$\text{Circuit 2 CO}: (2R_4R_5 - R_6R_7) \le 0; \text{FO}: f_0 = \frac{1}{2\pi C}\sqrt{\frac{R_2+R_4+R_6}{R_2R_4R_7}} \tag{2.26}$$

$$\text{Circuit 3 CO}: (2R_3R_8 - R_6R_7) \le 0; \text{FO}: f_0 = \frac{1}{2\pi C}\sqrt{\frac{R_1+R_3+R_7}{R_1R_3R_6}} \tag{2.27}$$

$$\text{Circuit 4 CO}: (2R_1R_5 - R_6R_7) \le 0; \text{FO}: f_0 = \frac{1}{2\pi C}\sqrt{\frac{R_1+R_3+R_7}{R_1R_3R_6}} \tag{2.28}$$

while for the SRCO Circuit 5 to Circuit 8, the CO and FO are given by

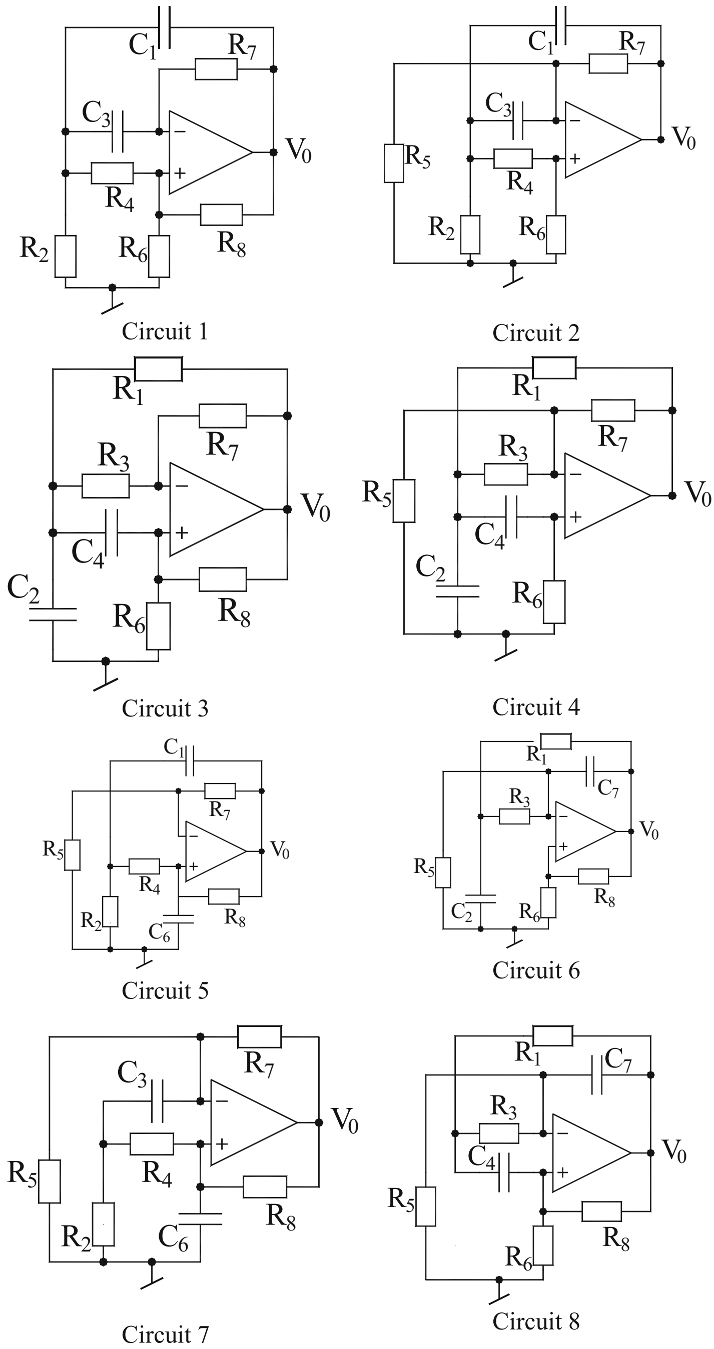


Fig. 2.9 The eight SRCOs using single op-amp [51]

Circuit 5 ($R_2 = R_8$)

$$\begin{aligned}
&\text{CO}: (C_6R_5 - C_1R_7) \le 0;\\
&\text{FO}: \omega_0 = \sqrt{\left(\frac{R_7}{C_1C_6R_2}\right)\left\{\frac{1}{R_4}\left(\frac{1}{R_7} - \frac{1}{R_5}\right) - \frac{1}{R_2R_5}\right\}}
\end{aligned} \tag{2.29}$$

Circuit 6 ($R_1 = R_5$)

$$\begin{aligned}
&\text{CO}: (C_7R_8 - C_2R_6) \le 0;\\
&\text{FO}: \omega_0 = \sqrt{\left(\frac{R_6}{C_2C_7R_1}\right)\left\{\frac{1}{R_3}\left(\frac{1}{R_6} - \frac{1}{R_8}\right) - \frac{1}{R_1R_8}\right\}}
\end{aligned} \tag{2.30}$$

Circuit 7 ($R_4 = R_5$)

$$\begin{aligned}
&\text{CO}: (C_6R_8 - C_3R_7) = 0;\\
&\text{FO}: \omega_0 = \sqrt{\left(\frac{R_6}{C_3C_6R_4}\right)\left\{\frac{1}{R_2}\left(\frac{1}{R_7} - \frac{1}{R_8}\right) - \frac{1}{R_4R_8}\right\}}
\end{aligned} \tag{2.31}$$

Circuit 8 ($R_3 = R_8$)

$$\begin{aligned}
&\text{CO}: (C_7R_5 - C_4R_6) \le 0;\\
&\text{FO}: \omega_0 = \sqrt{\left(\frac{R_6}{C_4C_7R_3}\right)\left\{\frac{1}{R_1}\left(\frac{1}{R_6} - \frac{1}{R_5}\right) - \frac{1}{R_3R_5}\right\}}
\end{aligned} \tag{2.32}$$

The following observations were made: (1) Out of the eight circuits, the two capacitor SRCOs of Circuits 2 and 3 have the highest frequency stability factor. (2) Furthermore, in principle, all SRCOs are convertible into VCOs by replacing the frequency-controlling resistor by an FET; however, much better results were obtained in case of Circuits 2 and 7 wherein one end of the frequency-controlling resistance is grounded. (3) The incorporation of a feedback-type amplitude-controlling circuit was also found to be much easier in case of Circuits 2 and 4 where the condition of oscillation is set by a grounded resistor.

The amplitude-controlling feedback circuit was made up of a precision rectifier and a filter which generates a DC signal proportional to peak amplitude of the oscillator which is then compared with a fixed reference voltage, and the error signal thus obtained is used to control the resistance of an FET (replacing the condition setting resistance). It is thus interesting to observe that it is Circuit 2 of the eight generated circuits which was found to be advantageous in all the three cases described above (Circuit 2 is, in fact, the same as Senani's single-op-amp SRCO presented in [26] and described here in Sect. 2.5.1).

2.5.3 Derivation of Single-Op-Amp SRCOs Using Boutin's Transformations

Boutin [123] demonstrated a procedure of systematically synthesizing eight SRCO circuits, starting from one of the eight SRCOs derived by Pyara, Dutta Roy, and Jamuar [51]. Boutin's synthesis procedure consists of the following steps: (1) Starting from any given SRCO circuit, a second oscillator is synthesized by interchanging the polarity of both the input and output terminals of the passive network. This amounts to interchanging the polarity of both the input and output terminals of the op-amp. (2) To obtain the two remaining oscillator circuits belonging to the same family, interchange the input and output ports of the passive network. This is equivalent to interchanging the input and output ports of the op-amp on each of the two previous oscillators.

Thus, the application of step (1) on the Circuits **a** and **b** of Fig. 2.10 results in the Circuits **c** and **d**, respectively. On the other hand, the application of step (2) to the Circuits **a** to **d** of Fig. 2.10 results in the remaining Circuits **e** to **h**.

The CO and FO of these SRCOs are given by:

For Circuits **a**, **c**, **e**, and **g** ($C_1 = C_3 = C$)

$$\mathrm{CO}:(2R_2R_8 - R_6R_7) = 0; \mathrm{FO}: f_0 = \frac{1}{2\pi C}\sqrt{\frac{R_2 + R_4 + R_6}{R_2R_4R_7}} \tag{2.33}$$

For Circuits **b**, **d**, **f,** and **h** ($R_1 = R_5$)

$$\begin{aligned} &\mathrm{CO}:(C_7R_8 - C_2R_6) = 0; \\ &\mathrm{FO}:\omega_0 = \sqrt{\left(\frac{R_6}{C_2C_7R_1}\right)\left\{\frac{1}{R_3}\left(\frac{1}{R_6} - \frac{1}{R_8}\right) - \frac{1}{R_1R_8}\right\}} \end{aligned} \tag{2.34}$$

An obvious advantage of this methodology is that this procedure preserves the nature and the value of all the admittances because of which the CO and the FO for Circuits **c**, **e**, and **g** are the same as those of Circuits **a**, and similarly, the CO and FO for Circuits **d**, **f**, and **h** are the same of that Circuit **b**.

2.5.4 Bandopadhyaya's SRCO and Williams' Simplified Version

Bandopadhyaya [5] presented an interesting modification of the classical Wien bridge oscillator (WBO) (Fig. 2.11) in which he embedded an extra op-amp stage consisting of an op-amp, an additional resistor R_1, and an additional capacitor C_1 while maintaining equal RC components and a non-inverting amplifier of gain of

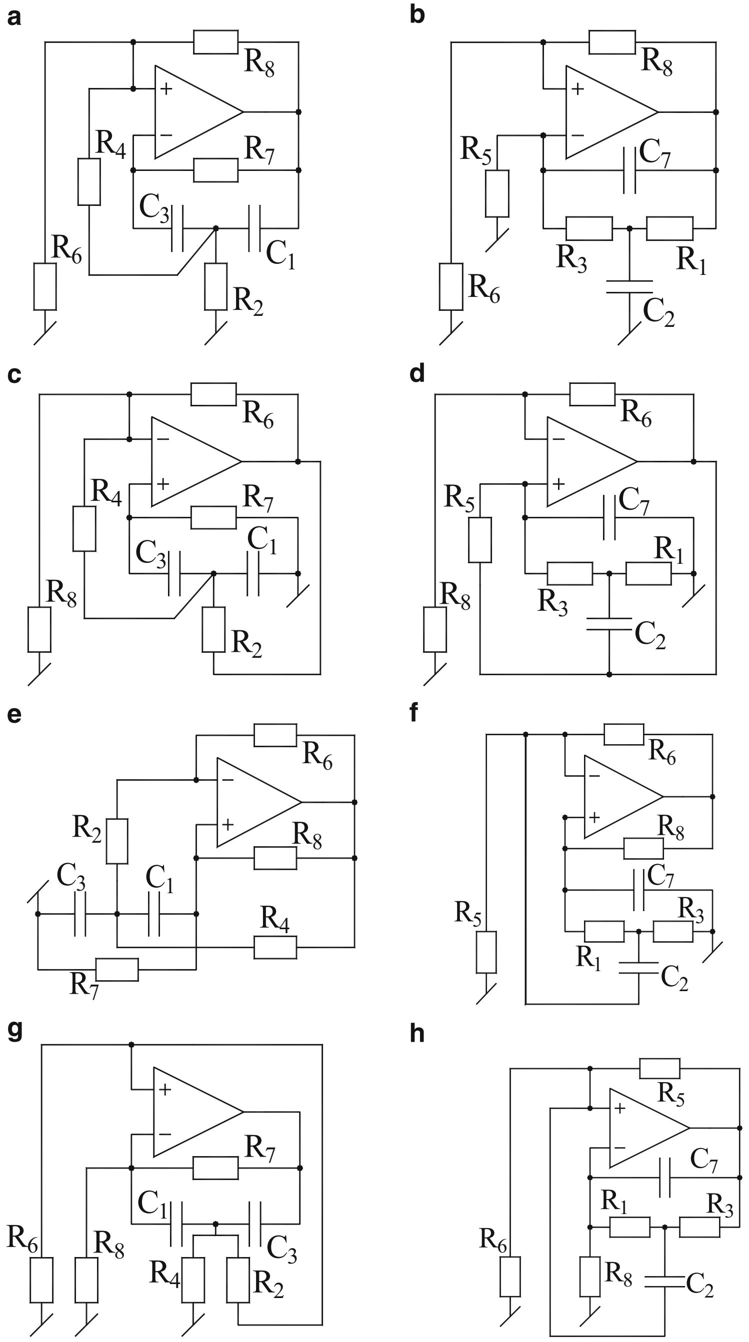


Fig. 2.10 (**a–h**) The eight SRCOs synthesized as per Boutin's method [123]

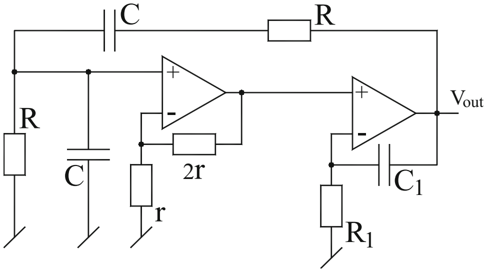


Fig. 2.11 Bandopadhyaya's [5] modification of the classic WBO for obtaining single-resistance control

three in the remaining part of the WBO. This intuitive modification resulted in the oscillation frequency as

$$f_0 = \frac{1}{2\pi RC}\sqrt{\left(1 - \frac{3RC}{R_1C_1}\right)} \tag{2.35}$$

Thus, f_0 is controllable by a single-variable resistance R_1.

Subsequently, Williams [6] demonstrated that using nullator-norator models, the circuit of Fig. 2.11 can, in fact, be reduced to a modified single-op-amp version which, however, retains the single-resistance control intact (of course, with the values of two resistors slightly modified). Williams' simplification is illustrated here in Fig. 2.12a–c.

From the nullor model, it may be noted that the nullor along with the two resistors r and 2r creates a voltage at the junction of the series RC and parallel RC which is one third of the voltage existing on the junction of R_1 and C_1. Exactly the same voltage can be created by removing these elements and modifying the *RC* branches as shown in Fig. 2.12c. Finally, the resistor R_b needs to be added to provide the DC path necessary to allow input bias current to flow into the inverting input terminal of the op-amp. Its value is a compromise to minimize disturbance to the AC operation while providing adequate DC stability [6].

2.5.5 SRCOs: A Network Synthetic Approach

Dutta Roy and Pyara [29] formulated a general topology to systematically generate all possible SRCOs of the type described in the previous section which is shown in Fig. 2.13.

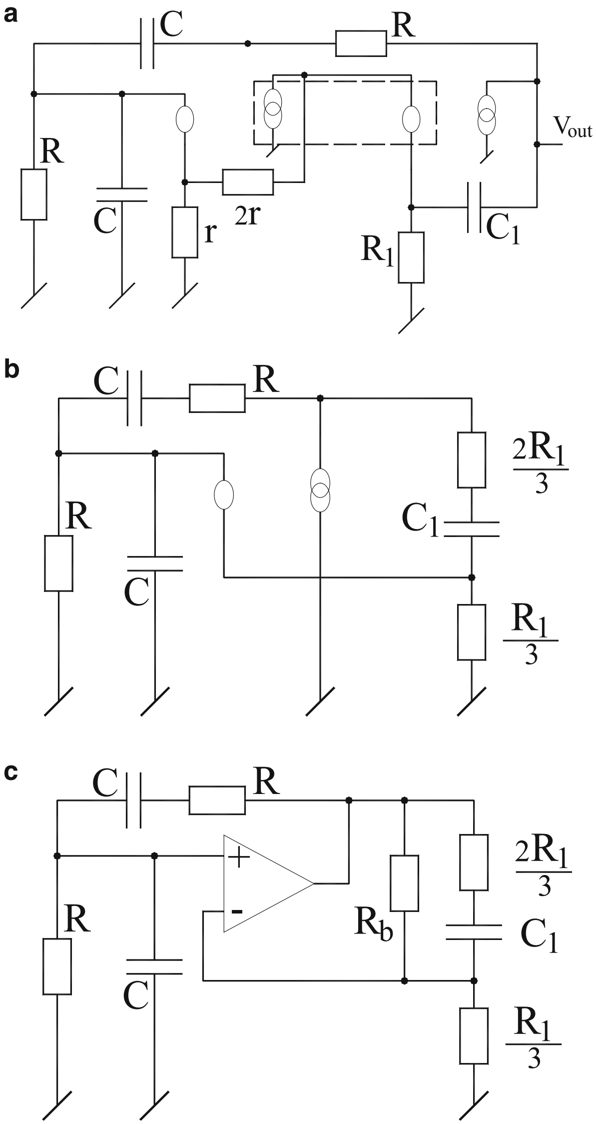


Fig. 2.12 Williams' [6] single-op-amp SRCO derived from Bandopadhyaya's two-op-amp SRCO. (**a**) The nullor model of Bandopadhyaya's SRCO. (**b**) Simplification of the nullor model. (**c**) The op-amp implementation

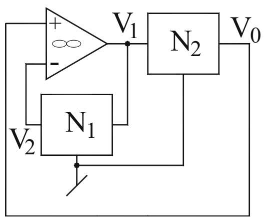


Fig. 2.13 Dutta Roy and Pyara's [29] generalized single-op-amp scheme to synthesize SRCOs

In this scheme, a synthetic approach is applied to Williams' simplified version of Bandopadhyaya's circuit. It is found that the open-loop transfer function (OLTF) of the circuit is given by

$$T_1(s) = \frac{3RC(sC_1R_1 + 1)}{R_1C_1\left(s^2C^2R^2 + 3sCR + 1\right)} \tag{2.36}$$

which is responsible for the genesis of the oscillator having the expression for the oscillation frequency as

$$f_0 = \frac{1}{2\pi RC}\sqrt{1 - \frac{3RC}{C_1R_1}} \tag{2.37}$$

From Eqs. (2.36) and (2.37), it can be easily deduced that the CO is independent of R_1C_1 and that frequency of oscillation is controllable by a single-variable resistance R_1. By routine analysis if the transfer functions of the two-port RC networks N_1 and N_2 are assumed to be $T_1(s)$ and $T_2(s)$, respectively, it is found that the OLTF of the network of the general schematic of Fig. 2.13 is given by

$$T^{'}(s) = T_2(s)/T_1(s) \tag{2.38}$$

The problem of synthesizing a single-op-amp SRCO as per the general schematic now reduces to finding appropriate two-port RC networks having $T_1(s)$ and $T_2(s)$ resulting in $T'(s)$ becoming the same as $T(s)$. Dutta Roy-Pyara [29] showed three possible function pairs satisfying the requirement which lead to the three new SRCOs shown in Fig. 2.14.

Out of the three circuits of Fig. 2.14, those of Fig. 2.14a, b are characterized by the FO:

$$f_0 = \frac{1}{2\pi RC}\sqrt{1 - \frac{R}{R_1}} \tag{2.39}$$

whereas the last one (Fig. 2.14c) has FO given by

$$f_0 = \frac{1}{2\pi RC}\sqrt{1 - \frac{4R}{R_1}} \tag{2.40}$$

With µA741-type op-amps and RC components of 5 and 10 % tolerances, respectively, a frequency range in the ratio 1:3 was found to be obtainable with these derived SRCOs with almost negligible distortion [29].

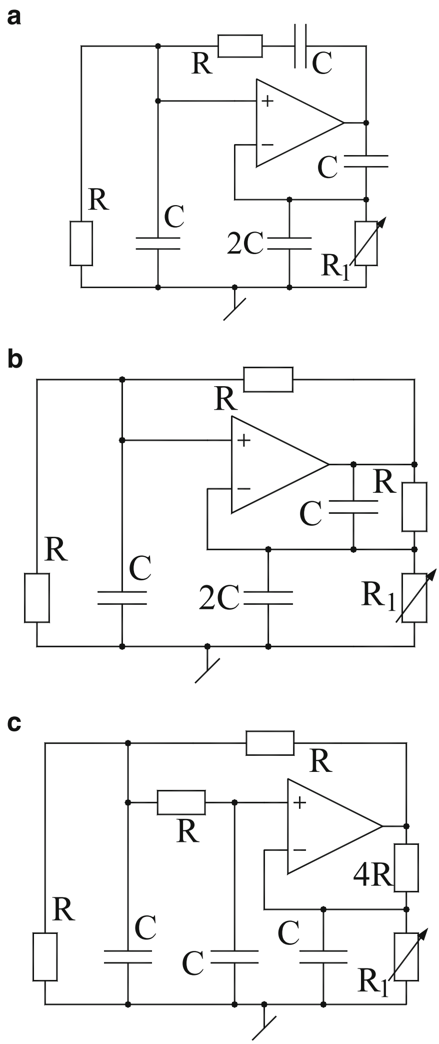


Fig. 2.14 (**a**–**c**) Three new single-op-amp SRCOs derived by Dutta Roy and Pyara [29] through a *network synthetic* approach

2.5.6 The Complete Family of Single-Op-Amp SRCOs

Whereas several authors considered the problem of realizing an SRCO using no more than a single op-amp as has already been described in the previous section, it would appear from the works of Dutta Roy, Jamuar, and Pyara [51] and Boutin [123] as if only a total of eight circuits are all which fulfill the intended requirements. However, it was Bhattacharyya and Darkani [53] who demonstrated conclusively following a very systematic and unified approach that not eight but sixteen single-op-amp SRCOs exist satisfying the intended objectives and that this set is complete, i.e., no more canonic single-op-amp SRCOs, other than these, are possible. They found that these 16 circuits can be categorized into four categories according to their dependence on the frequency-controlling resistor. The circuits derived by Bhattacharyya and Darkani [53] are shown in Fig. Single-resistance-controlled oscillators (SRCOs):single-Op-amp-based:2.15, and their CO, FO, and other constraints are shown in Table 2.1.

All the 16 SRCOs were tested in laboratory [53] and were found to behave almost as predicted by theory. It may be noted that three of these circuits, namely, the Circuits a, f, and i, have the frequency-controlling resistor grounded and, hence, these circuits are more suitable for use as VCOs by replacing this resistor by an FET used as a VCR.

2.6 SRCOs Using Grounded Capacitors

A feature of SRCOs very often sought after has the possibility of employing both grounded capacitors (GCs), which are desirable from the viewpoint of IC implementation and easy absorption of parasitic capacitances into these external capacitors [16, 36, 58]. A number of researchers have proposed SRCO structures employing both grounded capacitors. In this section, we highlight some prominent configurations, proposed by various researchers, possessing this feature.

2.6.1 Three-Op-Amp SRCO Employing Grounded Capacitors

An SRCO circuit employing both grounded capacitors was first proposed by Sundaramurthy, Bhattacharyya, and Swamy [16], which required three op-amps to accomplish the intended objective (see Fig. 2.16).

The circuit can be seen to be composed of the following subcircuits: an inverting summing amplifier with gains δ_0, δ_1, and δ_2; two first-order RC low-pass sections; and an inverting amplifier of gain β_1 and a non-inverting amplifier of gain β_2. From a straightforward analysis of this circuit, the CO is found to be

$$C_1R_1 + C_2R_2(1 - \delta_1\beta_1) = 0 \tag{2.41}$$

whereas the FO is found to be

$$\omega_0 = \sqrt{\frac{(1 - \beta_1\delta_1 - \beta_1\beta_2\delta_2)}{R_1R_2C_1C_2}} \tag{2.42}$$

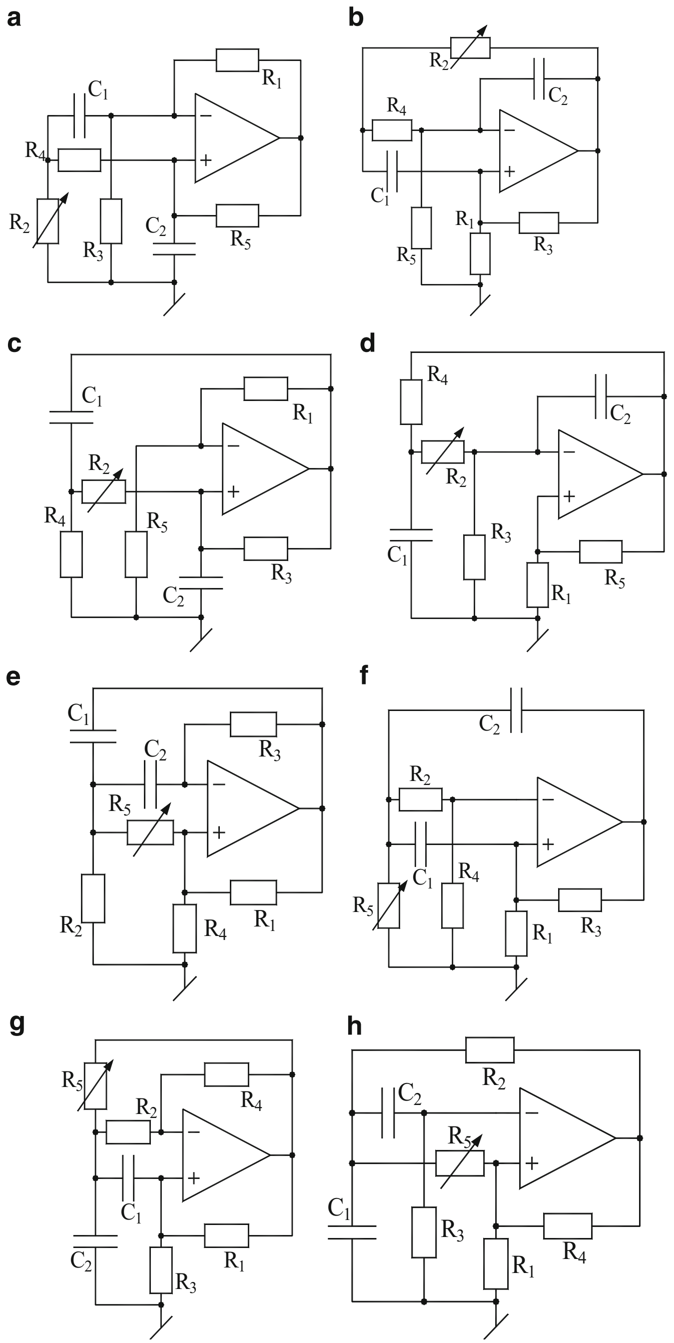


Fig. 2.15 (**a–p**) The complete family of 16 single-op-amp canonic SRCOs derived by Bhattacharyya and Darkani [53]

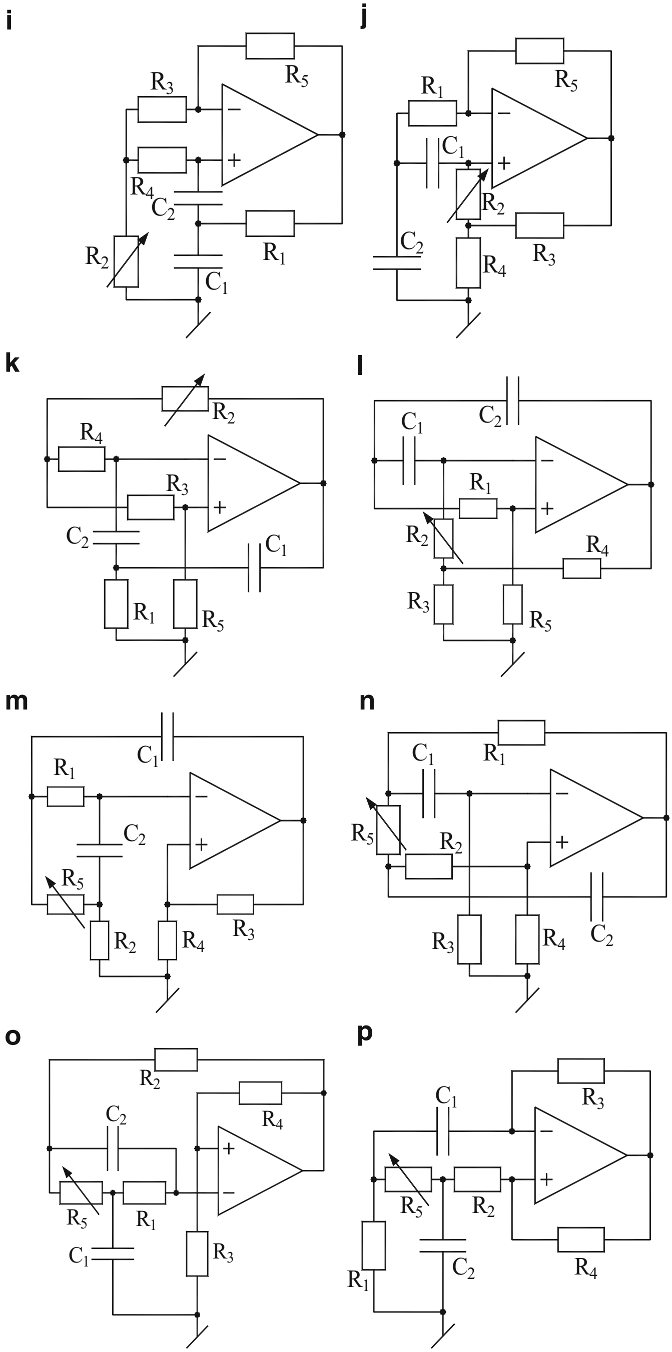


Fig. 2.15 (continued)

Table 2.1 CO, FO, and other design constraints of the 16 single-op-amp canonic SRCOs derived by Bhattacharyya and Darkani [53]

Fig. 2.15a, b, c, d	CO: $(C_2R_5 - C_1R_1) = 0$ for $R_3 = R_4$ FO: $\omega_0 = \left[\dfrac{1 - \dfrac{R_1(R_2+R_4)}{R_3R_5}}{C_1C_2R_2R_4}\right]^{1/2}$ with $R_3 = R_4 = R$, $C_2 = C$, $\dfrac{R_5}{R_1} = \dfrac{C_1}{C_2} = K$, then FO: $\omega_0 = \dfrac{1}{KRC}\left[\dfrac{R}{R_2}(K-1) - 1\right]^{1/2}$ CO: $R > \dfrac{R_2}{K-1}$, $K > 1$
Fig. 2.15e, f, g, h	CO: $\left\{\dfrac{C_1+C_2}{R_3R_4} - \dfrac{C_2}{R_1R_2}\right\} = 0$ and FO: $\omega_0 = \left(\dfrac{R_2+R_4+R_5}{C_1C_2R_2R_3R_5}\right)^{\frac{1}{2}}$ with $C_1 = C_2 = C$, $R_2 = R_3 = R_4 = R$, then CO: $2R_1 = R$ and FO: $\omega_0 = \dfrac{1}{RC}\left(\dfrac{2R+R_5}{R_5}\right)^{\frac{1}{2}}$
Fig. 2.15i, j, k, l	CO: $\left\{\dfrac{C_1+C_2}{R_5R_4} - \dfrac{C_2}{R_1R_3}\right\} = 0$ FO: $\omega_0 = \left(\frac{R_3}{C_1C_2R_1(R_1R_2+R_2R_3+R_3R_4)}\right)^{\frac{1}{2}}$ with $C_1 = C_2 = C$, $R_1 = R_3 = R_4 = R$, then CO: $2R = R_5$ and FO: $\omega_0 = \dfrac{1}{RC}\left(\dfrac{R}{2R_2+R}\right)^{\frac{1}{2}}$
Fig. 2.15m, n, o, p	CO: $R_1 = R_2 = R$, $\dfrac{R_4}{R_3} = \dfrac{C_1}{C_2} = K > 1$, $\dfrac{R_1R_2}{KR_1 - R_2} < R_5$ FO: $\omega_0 = \left(\frac{R_4}{C_1C_2(R_1R_4R_5 - R_2R_3R_5 - R_1R_2R_3)}\right)^{\frac{1}{2}}$ with $R_1 = R_2 = R$, $C_1 = \mathrm{KC}$, $C_2 = C$ then CO: $R_5 > \dfrac{R}{K-1}$ and FO: $\omega_0 = \dfrac{1}{RC}\left(\dfrac{R}{R_5(K-1) - R}\right)^{\frac{1}{2}}$

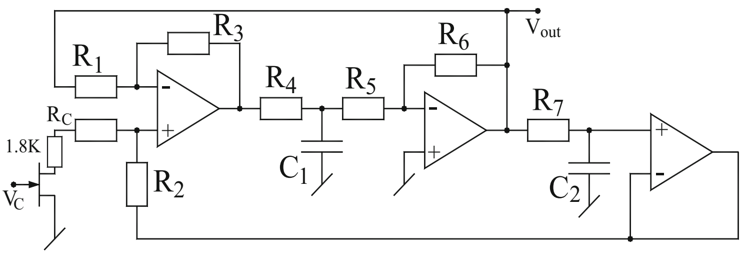


Fig. 2.16 A single-resistance-controlled/voltage-controlled oscillator with grounded capacitors proposed by Sundaramurthy, Bhattacharyya, and Swamy [16]

To keep $\omega_0 > 0$, there is an additional constraint to be fulfilled as

$$(1 - \beta_1\delta_1 - \beta_1\beta_2\delta_0) > 0 \tag{2.43}$$

A circuit was constructed [16] employing ML741-type op-amps, FET μMPF102, and RC components with 1 and 2 % tolerances which demonstrated that voltage-controlled oscillations over a decade range were easily obtainable with change in amplitude less than 5 % over this frequency range.

2.6.2 Two-Op-Amp-GC SRCO

Whereas the Sundaramurthy-Bhattacharyya-Swamy [16] circuit required three op-amps, Senani [36, 40] came up with a new circuit possessing single-resistance-control and employing two grounded capacitors but requiring only two op-amps. Senani's circuit is shown here in Fig. 2.17.

A straightforward analysis of this circuit yields

$$\mathrm{CO}: \frac{R_3}{R_4} = \frac{R_1}{R_2}\left(1 + \frac{C_2}{C_1}\right) \tag{2.44}$$

$$\mathrm{FO}:\ f_o = \frac{1}{2\pi\sqrt{R_1R_2C_1C_2}}\left(\frac{R_3R_5}{R_4R_6}\right)^{1/2} \tag{2.45}$$

The following features of this circuit are worth mentioning:

1. Due to the frequency-controlling resistor R_6being grounded, the circuit is readily convertible into a VCO by replacing this resistance by an FET using a voltage-controlled resistance (VCR).
2. Since the expression for FO contains a resistive scale factor $\sqrt{(R_3R_5/R_4R_6)}$, the circuit can be easily used for generating very low-frequency (VLF) oscillations without having to use large RC component values.
3. Since both the capacitors used are connected to ground, the circuit is suitable from the point of view of integrated circuit implementation [16, 36, 58].

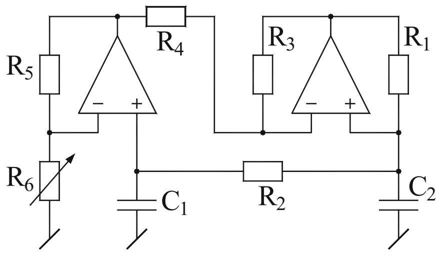


Fig. 2.17 Senani's SRCO employing both grounded capacitors [36]

2.6.3 Single-Op-Amp SRCOs Employing All Grounded Capacitors

Proceeding logically, it was, thus, obvious that the next development should have been to attempt to derive a circuit for realizing SRCOs with both grounded capacitors but using no more than one op-amp.

Indeed, the first single-op-amp SRCO employing grounded *capacitors*, *but employing three of them rather than the minimum number of two*, was proposed by Kaliyugavaradan [42], which is shown here in Fig. 2.18. An analysis of this circuit reveals the CO and FO to be

$$C_3R_4 = C_1R_1 + C_2R_2 + C_2R_1 + R_1R_2\left(\frac{C_2}{R_5} + \frac{C_1}{R_6}\right) \tag{2.46}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{\frac{1}{R_6}(R_1 + R_2) + \frac{R_1}{R_5}\left(1 + \frac{R_2}{R_6}\right) - \left(\frac{R_4}{R_3}\right)}{C_1C_2R_1R_2}} \tag{2.47}$$

It is, thus, seen that f_0 is independently adjustable with single grounded resistor R_3. A close examination of Eqs. (2.34) and (2.35) reveals[3] that either R_5 or R_6 can be omitted from the circuit without altering the essential properties but simplifying the design, although as rightly pointed out in [42], retaining both the resistors has the advantage of having all the three grounded capacitors with a parallel resistance across each one of them, thereby retaining the essential features of the circuit intact even if lossy capacitors are employed to realize the circuit! The two cases pointed out are:

1. With R_5 omitted, the modified CO and FO are given by

$$C_3R_4 = C_1R_1 + C_2R_2 + C_2R_1 + R_1R_2\left(\frac{C_1}{R_6}\right) \tag{2.48}$$

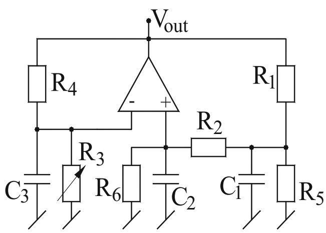


Fig. 2.18 Single-op-amp SRCO employing grounded capacitors proposed by Kaliyugavaradan [42]

[3] Apparently, not explicitly recognized in [42]

$$f = \frac{1}{2\pi}\sqrt{\frac{\frac{1}{R_6}(R_1+R_2) - \left(\frac{R_4}{R_3}\right)}{C_1C_2R_1R_2}} \tag{2.49}$$

2. On the other hand, with R_6 omitted, the modified CO and FO are given by:

$$C_3R_4 = C_1R_1 + C_2R_2 + C_2R_1 + R_1R_2\left(\frac{C_2}{R_5}\right) \tag{2.50}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{\left(\frac{R_1}{R_5}\right) - \left(\frac{R_4}{R_3}\right)}{C_1C_2R_1R_2}} \tag{2.51}$$

Thus, in all the cases, the FO can be independently varied by varying R_3 which does not affect the CO, since R_3 does not appear in CO. In a practical oscillator constructed using $C_1 = C_2 = C_3 = 0.01$ μF, $R_1 = R_2 = R_5 = R_6 = 2.2$ KΩ, and $R_4 = 11$ KΩ, it was found that with R_3 varied from above 2.75 KΩ to 1 MΩ, FO could be varied from 14.4 KHz down to a value less than 10 Hz [42].

2.6.4 *Single-Op-Amp-Two-GC SRCO*

While Kaliyugavaradan's [42] circuit realized an SRCO with a single op-amp and all grounded capacitors, it was clearly noncanonic due to employment of three capacitors for a second-order dynamics. Singh [39] presented a single-op-amp SRCO with only two grounded capacitors as an alternative to Senani's two-op-amp-GC SRCO [36]; the realization conditions were, however, not so simple as in the case of Senani's circuit [36] or for that matter Kaliyugavaradan's circuit [42]. Singh's [39] circuit is shown in Fig. 2.19.

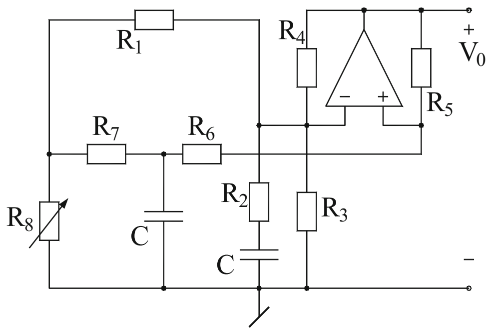


Fig. 2.19 Singh's single-op-amp SRCO requiring only two GCS [39]

As per [39], assuming the resistor values to be

$$R_1 = R_2 = R_3 = R_4 = R; R_5 = 9R, R_6 = 3R \quad \text{and} \quad R_7 = 5R/4 \tag{2.52}$$

the FO is, then, given by

$$f = \frac{1}{2\pi CR}\sqrt{\frac{2}{15}\left(\frac{R}{R_8} - \frac{21}{2}\right)} \tag{2.53}$$

Singh [39] did not substantiate the workability of his circuit either by hardware results or by simulations; neither has anybody else ever verified the practical validity of this circuit, but it can be seen that in spite of reducing the number of grounded capacitors by one, Singh's circuit is much more complex than Kaliyuga-varadan's circuit [42] which, in spite of requiring three capacitors, has relatively simpler design equations, although it does not provide independent control of CO.

2.6.5 A Family of Single-Op-Amp-Two-GC SRCOs

A comprehensive study for the generation of variable-frequency oscillators (VFOs) employing only two grounded capacitors (GCs) and no more than a single op-amp was, in fact, carried out by Darkani and Bhattacharyya [58]. They systematically derived a set of 12 canonic VFOs which provide tunability of FO through a single-variable resistance. These circuits are shown in Fig. 2.20, whereas the design equations for these circuits are shown in Table 2.2.

Of these 12 circuits, Circuit **i** of Table 2.3 has the advantage of providing frequency control through a single grounded resistor, that too without any constraints. This grounded resistance control is attractive for voltage-controlled oscillator realization by replacing the frequency-controlling resistor R_v by an FET used as a VCR. Furthermore, all the circuits can be designed with equal-valued grounded capacitors which are attractive to the viewpoint of IC implementation. The workability of all the circuits was confirmed by the authors of [58] through experimental results.

2.7 Scaled-Frequency Oscillators

Scaled-frequency oscillators [15, 87], in particular, the very low-frequency (VLF) oscillators, are useful in many biomedical, geophysical, and control instrumentation systems. An easy method of devising sinusoidal oscillators suitable for generating VLF oscillations is to have a difference term in the expression for FO such that by

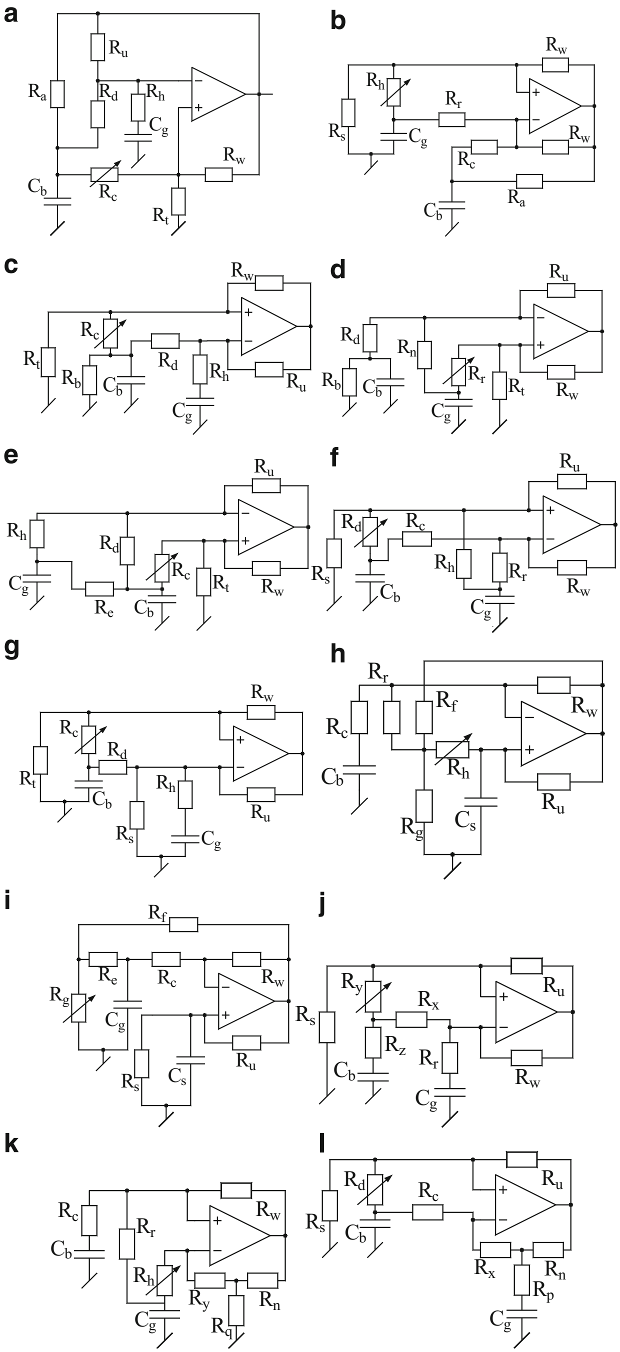


Fig. 2.20 (**a–l**) Single-op-amp SRCOs employing both grounded capacitors derived by Darkani and Bhattacharyya [58]

Table 2.2 The design equations for the VFOs of Fig. 2.20

No.	Selection of RC elements	Oscillation frequency ω_o	Constraint
a	$C_b = C_g = C$ $R_t = R_w = R_a = R_d = R$; $R_u = R/\sqrt{2}$ $R_h = R/(1+\sqrt{2})$; $R_c = R_v$ $K_1 = (1+\sqrt{2})$; $K_2 = [3+(5/\sqrt{2})]$	$\frac{1}{RC}\sqrt{\left(\frac{K_1R+K_2R_v}{R-\sqrt{2}R_v}\right)}$	$R_v < R/\sqrt{2}$
b	$C_b = C_g = C$ $R_c = R_w = R_a = R_R = R$; $R_u = R/12$ $R_s = R/9$; $R_h = R_v$	$\frac{3}{RC}\sqrt{\left(\frac{3(R_v+R)}{R-15R_v}\right)}$	$R_v < R/15$
c	$C_b = C_g = C$ $R_b = R/4$, $R_h = R/8$, $R_t = R/12$, $R_w = R_u/3$, $R_d = R$, $R_c = R_v$	$\frac{8}{RC}\sqrt{\left(\frac{2(R+3R_v)}{R-15R_v}\right)}$	$R_v < R/15$
d	$C_b = C_g = C$ $R_b = R_d = R$, $R_w = R/2$, $R_t = R/4$, $R_h = 7R$, $R_u = 3R$	$\frac{1}{RC}\sqrt{\left(\frac{2(7R+R_v)}{7R-20R_v}\right)}$	$R_v < 7R/20$
e	$C_b = C_g = C$ $R_u = Re = R$, $R_w = R/2$, $R_t = R/3$, $R_h = R/5$, $R_d = 21R$, $R_c = R_v$	$\frac{3}{RC}\sqrt{\left(\frac{(126R+111R_v)}{21R-149R_v}\right)}$	$R_v < 21R/149$
f	$C_b = C_g = C$, $R_w = R_h = R$, $R_u = R/3$, $R_r = R_s = R/2$, $Rc = 2R$, $R_d = R_v$	$\frac{3}{RC}\sqrt{\left(\frac{(4R+2R_v)}{2R-9R_v}\right)}$	$R_v < 2R/9$
g	$C_b = C_g = C$, $R_s = R_u = R$, $R_t = R/4$, $R_w = R_h = R/2$, $R_d = 3R/2$, $R_c = R_v$	$\frac{1}{RC}\sqrt{\left(\frac{(12R+8R_v)}{3R-10R_v}\right)}$	$R_v < 3R/10$
h	$C_s = C_g = C$ $R_w = R_u = R_f = R_r = R_g = R_c = R$; $R_h = R_v$	$\frac{1}{RC}\sqrt{\left(\frac{(R-R_v)}{R+4R_v}\right)}$	$R_v < R$
i	$C_s = C_b = C$ $R_w = R_e = R_f = R_c = R$, $R_s = R/3$, $R_u = R/5$, $R_g = R_v$	$\frac{1}{RC}\sqrt{\left(\frac{(R+12R_v)}{R+2R_v}\right)}$	No constraint
j	$C_b = C_g = C$ $R_s = R_r = R_u = R$, $R_z = R/2$, $R_w = 5R/2$, $R_x = 7R/2$, $R_y = R_v$	$\frac{2}{RC}\sqrt{\left(\frac{(7R+2R_v)}{R-68R_v}\right)}$	$R_v < R/68$
k	$C_b = C_g = C$ $R_y = R_n = R_e = R_q = R$; $R_r = 7R$ $R_w = 4R$, $R_h = R_v$	$\frac{1}{RC}\sqrt{\left(\frac{(7R+R_v)}{21R-25R_v}\right)}$	$R_v < 21R/25$
l	$C_b = C_g = C$ $R_u = R_x = R_p = R_s = R$; $R_c = 4R$ $R_n = 2R$; $R_d = R_v$	$\frac{1}{RC}\sqrt{\left(\frac{(4R+R_v)}{4R-9R_v}\right)}$	$R_v < 4R/9$

keeping this term small, VLF oscillations can be generated. Thus, this method requires the expression for the FO to be of type:

$$f_0 = \frac{1}{2\pi RC}\sqrt{(1-n)} \tag{2.54}$$

where **n** is the frequency-controlling resistor ratio. The sensitivity of f_0 with respect to the parameter **n** is found to be

Table 2.3 CO and FO for the various circuits of Fig. 2.29

FO	CO
$\frac{1}{(R_2R_3 + R_3R_5 + R_5R_2)C_1C_6}$	$R_2R_3R_5C_4 = (R_2R_3 + R_3R_5 + R_5R_2)\{R_2C_1 + C_6(R_3 + R_5)\}$
$\frac{(C_2 + C_3 + C_5)}{R_1R_6C_2C_3C_5}$	$\frac{C_2C_3C_5}{R_4} = (C_2 + {}^2C_3 + C_5)\left[\frac{C_3C_5}{R_4} + \frac{C_2}{R_6}(C_3 + C_5)\right]$
$\frac{1}{R_1R_4C_2C_6}$	$\frac{C_5}{R_1R_4} = \left[\frac{C_2}{R_3R_4} + \frac{C_6}{R_1R_3}\right]$
$\frac{1}{R_2R_6C_1C_4}$	$\frac{C_1C_4}{R_5} = C_3\left(\frac{C_1}{R_6} + \frac{C_4}{R_2}\right)$
$\frac{1}{R_4R_6C_3C_5}$	$\frac{C_3C_5}{R_1} = C_2C_3\left(\frac{1}{R_6} + \frac{1}{R_4}\right)$
$\frac{1}{R_3R_5C_4C_6}$	$\frac{C_1}{R_3R_5} = \frac{1}{R_2R_3}(C_4 + C_6)$
$\frac{1}{R_1R_2C_4C_3}$	$\frac{C_5}{R_1R_2} = \frac{1}{R_6}\left(\frac{C_3}{R_1} + \frac{C_4}{R_1} + \frac{C_3}{R_2}\right)$
$\frac{1}{R_4R_3C_1C_2}$	$\frac{C_1C_2}{R_5} = C_6\left(\frac{C_1}{R_3} + \frac{C_1}{R_4} + \frac{C_2}{R_3}\right)$
$\frac{1}{R_1R_3C_4C_2}$	$\frac{C_5}{R_3R_1} = \frac{1}{R_6}\left(\frac{C_2}{R_1} + \frac{C_4}{R_1} + \frac{C_2}{R_3}\right)$
$\frac{1}{R_4R_2C_1C_3}$	$\frac{C_1C_3}{R_5} = C_6\left(\frac{C_1}{R_2} + \frac{C_1}{R_4} + \frac{C_2}{R_3}\right)$

$$S_n^{f_0} = \frac{n}{f_0}\frac{\partial f_0}{\partial n} = -\frac{n}{2(1-n)} \tag{2.55}$$

which will be quite large when as per Eq. (2.54) $(1-n)$ is required to be as small as possible to enable VLF generation.

It was proposed by Senani [49] that another possible approach to generate VLF oscillators could be to synthesize an oscillator in which the tuning law takes the form

$$f = \frac{\sqrt{k}}{2\pi RC} \tag{2.56}$$

where "k" is the frequency-controlling resistor ratio, so that the oscillation frequency can be scaled down by making the frequency-controlling resistor ratio k as small as possible while keeping the other resistance values used in the circuit within the applicable lowest and highest values. It may be noted that in this case, the relevant sensitivity coefficient would turn out to be

$$S_k^{f_0} = 1/2 \tag{2.57}$$

and is, thus, seen to be very low.

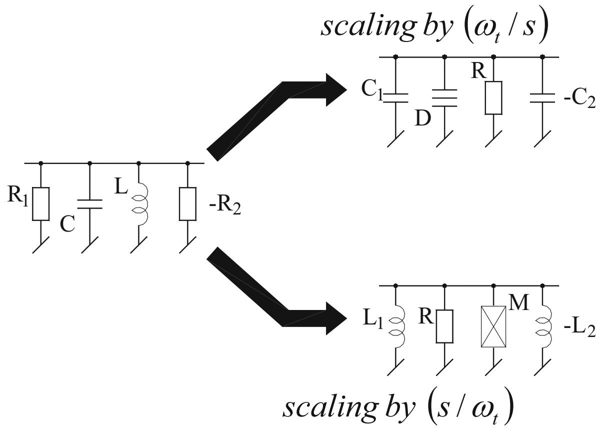


Fig. 2.21 Derivation of ±CRD and ±LRM networks from an ±RLC network [49]

Senani [49] demonstrated that oscillators inherently possessing an additional resistive scale factor of the kind employed in equation (2.56) can be systematically synthesized starting from a passive ±RLC resonator and applying on it a frequency-dependent scaling by multiplying all impedances by the complex frequency scaling factor ω_t/s as shown in Fig. 2.21.

Note that multiplying all impedances by the scaling factor ω_t/s results in a ±CRD model, whereas applying the s/ω_t leads to a model containing a positive resistor R, a positive inductor L_1, a negative inductor $-L_2$, and an element having $Z(s) = Ms^2$ which can be called a frequency-dependent negative conductance (FDNC) or *superinductor*. The characteristic equation of this circuit can be easily found to be

$$s^2 + sR\left(\frac{1}{L_1} - \frac{1}{L_2}\right) + \frac{R}{M} = 0 \tag{2.58}$$

where the condition of oscillation is given by

$$(L_2 - L_1) \leq 0 \tag{2.59}$$

and the frequency of oscillation is given by

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{R}{M}} \tag{2.60}$$

A dimensional check of the above equation indicates that since an R already appears in the numerator, M should be proportional to C^2r^3. Assuming, as an example, $M = C_1C_2r_1r_2\,r_3$, it therefore follows that a typical ±LRM oscillator would have the expression for the FO of the type:

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{C_1 C_2 r_1 r_2}\left(\frac{R}{r_3}\right)} \tag{2.61}$$

Hence, it follows that as compared to oscillators based on ±RLC model or ±CRD models, those based on the ±LRM models would be inherently capable of providing an extra frequency scaling term which makes them naturally suitable for generating VLF oscillations while ensuring low-sensitivity properties, as explained earlier.

An exemplary ±LRM oscillator was introduced by Senani [49] itself which is shown here in Fig. 2.22.

Senani and Bhaskar [87] subsequently demonstrated that such oscillators can, in fact, be eventually realized even from single-op-amp networks. Four such oscillators were presented in [87] which are shown here in Fig. 2.23.

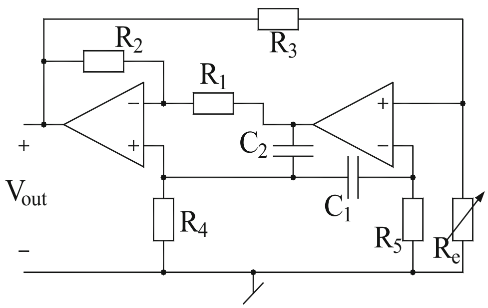


Fig. 2.22 An exemplary ±LRM oscillator proposed by Senani [49]

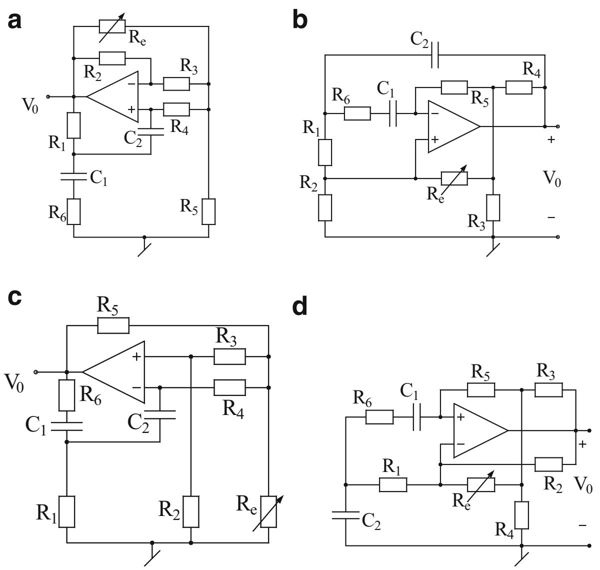


Fig. 2.23 ±LRM model-based single-op-amp single-resistance-controlled VLF oscillators proposed by Senani and Bhaskar [87]

All the four oscillators of Fig. 2.23 are characterized by the same characteristic equation from which the CO and FO are found to be

$$R_6\left(\frac{C_1}{C_2}\right) + R_1\left(1 + \frac{C_1}{C_2}\right) = \frac{R_2 R_4}{R_3} \tag{2.62}$$

$$f_0 = \frac{1}{2\pi\sqrt{C_1 C_2 R_1 R_4}}\left\{\frac{R_e/R_5}{1 + (R_2/R_3)}\right\}^{1/2} \tag{2.63}$$

provided that

$$R_2 R_4 R_6 = R_1[R_4 R_5 + R_3(R_4 + R_5 + R_6)] \tag{2.64}$$

From the above equation, it may be seen that the FO can be controlled independently by the single-variable resistance R_e.

All the four circuits were tested using μA741-type op-amps and performed as predicted by the theory. In one version of the circuit of Fig. 2.23 c, realized with $C_1 = C_2 = 0.2$ μF, $R_1 = R_3 = R_4 = R_6 = (1/3)$ MΩ, and $R_2 = 1.01$ MΩ, with R_e varied from 195 Ω to 96.55 KΩ and the op-amp biased with ±15 V DC, it was found to realize sinusoidal oscillations of frequency as low as 38.9 mHz. The experimental results, thus, confirmed the workability of the circuits. Out of the four circuits, the variant of Fig. 2.23 c is obviously a better version due to having the frequency-controlling resistor grounded which is especially attractive for obtaining voltage-controlled oscillations by replacing this resistor by an FET used as VCR.

2.8 Sinusoidal Oscillators Exhibiting Linear Tuning Laws

All of the oscillators described in this chapter so far have the FO either proportional to $\sqrt{R_e}$ or $1/\sqrt{R_e}$ or proportional to $\sqrt{C_e}$ or $1/\sqrt{C_e}$ where R_e and C_e are the single resistor and single capacitor, respectively, through which the FO is supposed to be controlled independently without affecting the CO which may be controllable through another variable element in the circuit. Senani [56] introduced two new types of sinusoidal oscillators which showed that it is possible to devise configurations which bring two most desirable *linear* tuning laws of the forms $f \propto R_e$ and $f_0 \propto C_e$ into the domain of sinusoidal oscillators. The circuits presented in [56] are shown in Fig. 2.24.

By straightforward analysis, it can be observed that looking onto the terminal "P" and ground, the equivalent of the impedance simulated by the circuit of Fig. 2.24a turns out to be a parallel combination of a positive resistance, a positive inductance, a negative inductance, and a super-inductance (FDNC) characterized by $Z(s) = Ms^2$, and the oscillator thus belongs to the class of ±LRM oscillators discussed in the previous section.

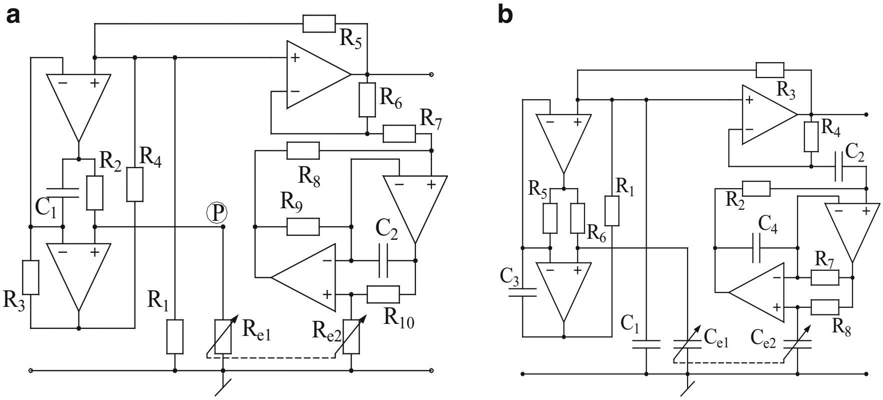


Fig. 2.24 Sinusoidal oscillators providing *linear* tuning laws [56]. (**a**) Oscillator providing f_0 proportional to $\sqrt{R_{e1}R_{e2}}$. (**b**) Oscillator providing f_0 proportional to $\sqrt{C_{e1}C_{e2}}$

The CO and FO for the circuit of Fig. 2.24a are given by

$$\frac{R_6}{R_5} = \frac{R_7}{R_1} + \frac{C_1R_2R_3R_8R_{e2}}{C_2R_4R_9R_{10}R_{e1}} \tag{2.65}$$

$$f_{01} = \frac{1}{2\pi}\sqrt{\frac{R_4R_8R_{e1}R_{e2}}{C_1C_2R_1R_2R_3R_7R_9R_{10}}} \tag{2.66}$$

If R_{e1} and R_{e2} are realized by the ganged variable resistors, i.e., R_{e1} s$=R_{e2}=R_e$, the FO is modified to be

$$f_{01} = \frac{R_e}{2\pi\sqrt{C_1C_2R_1R_2}}\left(\frac{R_4R_8}{R_3R_7R_9R_{10}}\right)^{1/2} \tag{2.67}$$

On the other hand, for the circuit of Fig. 2.24b which is obtainable from that of Fig. 2.24a by the application of RC-CR transformation, the CO and FO are given by (assuming $C_{e1}=C_{e2}=C_e$)

$$\frac{R_4}{R_3} = \frac{C_1}{C_2} + \frac{R_2R_6R_7}{R_1R_5R_8} \tag{2.68}$$

$$f_{01} = \frac{C_e}{2\pi\sqrt{C_1C_2R_1R_2}}\left(\frac{R_6R_8}{C_3C_4R_5R_7}\right)^{1/2} \tag{2.69}$$

The oscillators with linear tuning laws can be attractive as test oscillators as well as transducer oscillators which in conjunction with two identical resistive or capacitive transducers can provide linear conversion of resistance or capacitance of the transducer into frequency.

The first oscillator, because of containing two identical grounded resistors for frequency control, appears very appropriate for conversion into a VCO where two identical FETs derived by a common gate control voltage can be advantageously employed for voltage control of the oscillation frequency.

The circuits, when implemented with μA741-type op-amps biased with ± 9 V DC supply, exhibited good performance with frequency linearly tunable with R_e over a decade with nearly constant magnitude and % THD in the range 3–7 % which was reduced considerably with the application of an amplitude stabilization circuitry. The circuit, in VLF mode, could easily generate a frequency as low as 0.1 Hz while employing RC components of normal values.

Subsequently, in [75, 76] elaborate methods were presented for the systematic synthesis of such circuits; the interested readers are referred to [75, 76] for further reading. However, in [76] it was demonstrated that if the condition of having the two frequency-controlling resistors grounded be relaxed then the oscillators with linear tuning laws can be realized even with three op-amps.

One such circuit from [76] is shown here in Fig. 2.25. The CO and FO of this reduced-component oscillator are given by

$$\frac{R_4}{R_3} = 2\left(\frac{R_5}{R_2}\right) \quad \text{for} \quad C_1 = C_2 = C \tag{2.70}$$

$$f_0 = \frac{R_e}{2\pi C}\sqrt{\frac{(R_3 + R_4)}{R_1 R_2 R_3 R_6 R_7}} \quad \text{with} \quad R_{e1} = R_{e2} = R_e \tag{2.71}$$

Lastly, it may be pointed out that although these four-op-amp and three-op-amp configurations were advanced in 1985–1987, till date any op-amp realizations of these new types of oscillators employing a smaller number of op-amps (less than three) have not been reported in literature, and this constitutes an interesting possibility. Realizations of such oscillators, if attempted with other building blocks such as CCs or CFOAs, may be expected to reduce the total passive component count as well as active element count. These aspects appear worth investigating further.

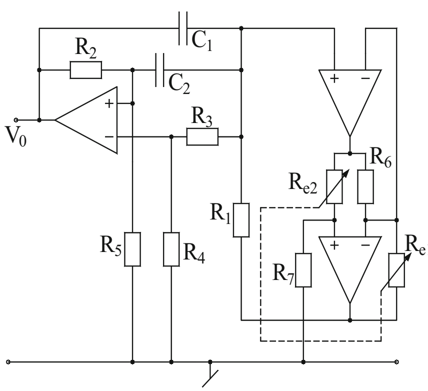


Fig. 2.25 An exemplary oscillator providing linear tuning laws realized with a reduced number of op-amps proposed by Senani [76]

2.9 SRCOs Using Unity Gain Amplifiers

In 1985, Senani proposed [57] a new type of oscillator based upon the use of the op-amps as unity gain voltage followers. Compared with other types of op-amp-based oscillators, these unity gain amplifier (UGA)-based oscillators were capable of generating sinusoidal oscillations over a larger frequency range since the bandwidth of the op-amp configured as a unity gain amplifier is the maximum possible (equal to the gain-bandwidth product of the op-amps employed). The genesis of such oscillators, as demonstrated in [71], stems from the introduction and application of four novel *network transformations* on the classical LC tank circuit as shown in Fig. 2.26.

In [71], four new network transformations were introduced which make it possible to incorporate even nonideal immittance simulation networks directly as elements in the design of higher-order filters. Since lossy inductors and FDNRs can be realized using smaller number of active and passive elements than their ideal counterparts, it was shown that this approach results in higher-order filter designs using both unity gain amplifiers and negative second-generation current conveyors (CCII-) which require a far smaller number of active and passive elements than required in the design of the same passive filters using lossless simulated impedance networks. These four network transformations involved multiplying all the impedances of a given prototype RLC filter by a frequency-dependent scaling function which can take four possible forms, as follows:

Transformation T-1: multiply all impedances by the scaling function $F(s) = (a_0 + a_1 s)$.

Transformation T-2: multiply all impedances by the scaling function $F(s) = 1/(b_0 + b_1 s)$.

Transformation T-3: multiply all impedances by the scaling function $F(s) = a_1 s/(b_0 + b_1 s)$.

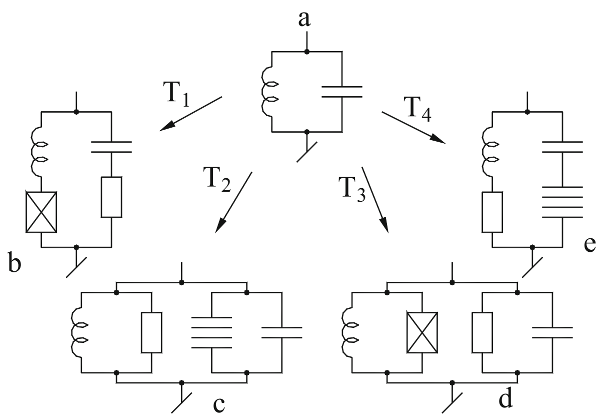


Fig. 2.26 Generation of new kinds of immittance models for synthesizing sinusoidal oscillators as proposed by Senani [71]

Transformation T-4: multiply all impedances by the scaling function $F(s) = (a_0 + a_1 s)/b_1 s$.

In the present case, the application of these four transformations on the parallel LC tank circuit (which models a sinusoidal oscillator having constant amplitude) results in four different models from which new oscillator circuits can be synthesized [71]. The step-by-step realization of new UGA-based sinusoidal oscillators resulting from the application of Senani's transformation T_4 is shown in Fig. 2.27.

In the ±LRM oscillator model of Fig. 2.26e, if the series RL impedance is simulated by the network of Fig. 2.27a and the CD branch is simulated by the RC-CR transformed version of this circuit shown in Fig.2.27b, the resulting oscillator takes the form as shown in Fig. 2.27c. It may, however, be noted that in this

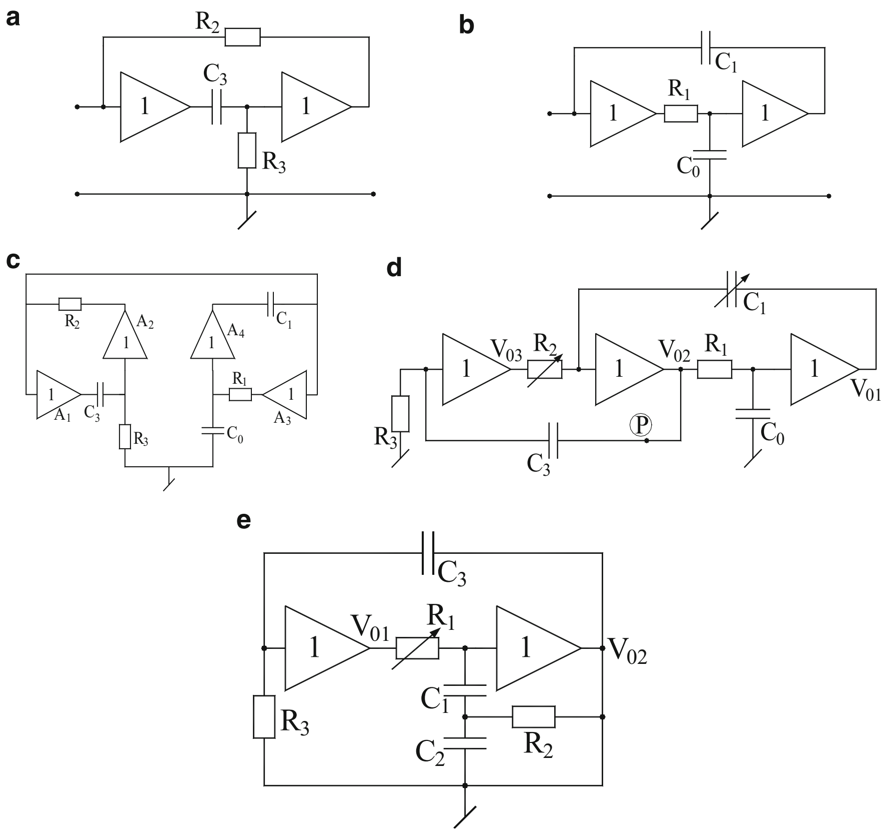


Fig. 2.27 Derivation of SRCOs using only unity gain voltage followers as active elements. (**a**) A circuit for simulation of a single-resistance tunable lossy (series RL) impedance [125]. (**b**) A circuit for simulation of a single-resistance-controlled CD impedance obtained by RC-CR transformation of the inductance simulation of [125]. (**c**) A VF-based oscillator derived through the proposed procedure [71]. (**d**) A simplified version of the SRCO using only three VFs [71]. (**e**) An SRCO using only two VFs [57, 71]

circuit, out of the VFs A_1 and A_3, one is redundant since the function of both the VFs can be performed by only a single VF. Thus, eliminating one redundant VF, the oscillator may be simplified as shown in Fig. 2.27d which is realizable with only three VFs. This circuit is characterized by the following CO and FO:

$$C_3R_3 = C_0R_1 \tag{2.72}$$

$$f_0 = \frac{1}{2\pi\sqrt{C_0C_1R_1R_2}} \tag{2.73}$$

The CO and FO both are adjustable independent of each other, the former by R_3 and/or C_3 and the latter by R_2 and/or C_1.

A further reduced-component version of the circuit employing no more than two VFs can be obtained by simulating the series RL impedance by the circuit of Fig. 2.26a but by simulating the series CD branch by an RC-CR transformed version of the single-op-amp lossy inductance simulation of Prescott [124] which needs only a single VF. The simplified version of the resulting circuit is realizable with only two VFs and is shown in Fig. 2.27e. The CO and FO for this circuit are given by

$$C_3R_3 = R_2(C_1 + C_2) \tag{2.74}$$

$$f_0 = \frac{1}{2\pi\sqrt{C_1C_2R_1R_2}} \tag{2.75}$$

Thus, it may be noted that in this reduced version also, the CO is adjustable by R_3 and/or C_3, while the FO can be independently varied without disturbing the CO, by a single-variable resistance R_1.

Before proceeding further, it is worthwhile to consider the various features of these circuits in some detail.

Note that due to the use of more than one UGA, the circuit of Fig. 2.27d, e offers some degree of freedom in selecting the appropriate output terminal for these oscillators. An inspection of the circuit of Fig. 2.27d reveals that V_{03} is the high-pass filtered version of V_{02}, whereas V_{01} is the low-pass filtered version of V_{02}. The open-loop-transfer function (OLTF) at V_{02} is found to be

$$T_1(s) = \left(\frac{sC_0R_1 + 1}{sC_3R_3 + 1}\right)\left\{\frac{sC_3R_3}{s^2C_0C_1R_1R_2 + sC_0R_1 + 1}\right\} \tag{2.76}$$

from which it is found that subject to condition $(C_0R_1) = (C_3R_3)$, the above function represents a band-pass filter so that when the loop is closed, the signal generated undergoes an inherent band-pass filtering before reaching at V_{02}. In conclusion, although three outputs are available, waveforms at V_{01} and V_{02} will be having comparatively lesser harmonic distortion due to inherent filtering of higher-order harmonics. In case of the circuit of Fig. 2.27e, V_{01} is the high-pass filtered form of V_{02}, whereas OLTF with R_1 disconnected from output of UGA A_3 is found to be

$$T_2(s) = \frac{V_{02}(s)}{V_{\text{in}}(s)} = \frac{sC_3R_3[1 + sR_2(C_1 + C_2)]}{(1 + sC_3R_3)[s^2C_1C_2R_1R_2 + sR_2(C_1 + C_2) + 1]} \tag{2.77}$$

Thus, in this case also, subject to $C_3R_3 = R_2(C_1 + C_2)$, the function is a BPF, whereas V_{01} is a high-pass filtered version of V_{02}. Hence, out of the two outputs available, that at V_{02} could be better (i.e., lesser harmonic content).

Frequency stability of an oscillator is an important figure of merit. For determining the frequency stability factors for the oscillators derived here, the various open-loop transfer functions are determined wherefrom the phase function $\phi(u)$, $u = \omega/\omega_0$ is determined. Subsequently from $\phi(u)$, S_F is determined by using the classical definition of S_F, viz.,

$$S_F = \left.\frac{\partial\phi(u)}{\partial u}\right|_{u=1} \tag{2.78}$$

where $u = \omega/\omega_0$ is the normalized frequency. In the following, the calculated S_F for various circuits are presented. For the circuit of Fig. 2.27d with $R_1 = R_3 = R$, $R_2 = R/n$, and $C_1 = Co = C_3 = C$, it is found that

$$\phi(u) = \frac{\pi}{2} - \tan^{-1}\left(\frac{u/\sqrt{n}}{1 - u^2}\right) \tag{2.79}$$

from where, using Eq. (2.79), one obtains

$$S_F = 2\sqrt{n} \tag{2.80}$$

For the circuit of Fig. 2.27e, the S_F is obtained with $R_2 = R_3 = R$, $C_1 = C_2 = C$, $C_3 = 2C$, and $R_1 = nR$ and is given by

$$S_F = \sqrt{n} \tag{2.81}$$

Finally, if both the series RL and the series CD branch are simulated by single-VF-based circuits, the former by the Prescott inductance simulation circuit and the latter by the RC-CR transformed version of Prescott circuit [124], then after removing one redundant VF, the final circuit is realizable by only a single VF and turns out to be as shown in Fig. 2.28.

The CO and FO for this circuit are given by

$$R_1\left(\frac{1}{R_2} + \frac{1}{R_3}\right) = \frac{C_3}{C_0 + C_1} \tag{2.82}$$

$$f = \frac{1}{2\pi}\left[\left(\frac{1}{C_0} + \frac{1}{C_1}\right)\frac{1}{C_3R_2R_3}\right]^{1/2} \tag{2.83}$$

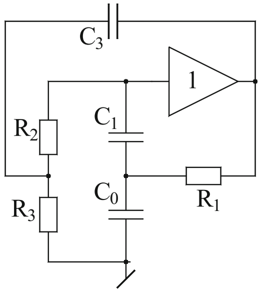


Fig. 2.28 An oscillator using a single VF [71]

In this case however, although the CO can be independently adjusted by the resistor R_1, unfortunately, the independent control of FO is not available. Later, we will show that although single-VF-based SRCO does not appear to be feasible, the circuit providing potentiometric control of the oscillations is, nevertheless, feasible [126].

Subsequent to the disclosure of new SRCOs using op-amps as unity gain amplifiers by Senani [57], Abuelma'atti [66] came up with a catalog of ten two-UGA (realized with op-amps)-based sinusoidal oscillator circuits which are shown here in Fig. 2.29, all of which employ two UGAs, three resistors, and three capacitors. *However*, *none of these circuits provide independent single-element control of the oscillation frequency*!

The CO and FO of these circuits are shown in Table 2.3.

All the ten circuits were built [66] using μA741-type op-amps and were found to be workable till a maximum frequency of around 295 KHz.

Later on, Boutin [64] presented an interesting synthesis procedure for deriving a sinusoidal oscillator employing a single UGA, leading to the circuit of Fig. 2.30.

It must be mentioned that the first author of this monograph had alternatively derived this circuit in [126] as follows. If instead of resorting to the realization schemes of Figs. 2.26 and 2.27, the whole series RLM branch of model of Fig. 2.26b is replaced by a network such as the one shown in Fig. 2.31a, this results in an interesting circuit which requires only a single UGA (Fig. 2.31b).

In this way however, the property of single-element tunability is lost. For the oscillator based on these circuits (see Fig. 2.31b), the CO is given by

$$C_0 = \left(\frac{1}{R_1 + R_2 + R_3}\right)\left\{\frac{1}{C_1R_1} + \frac{1}{C_2R_2} + \frac{1}{C_1R_2} + \frac{1}{C_2R_3}\right\} \tag{2.84}$$

and the FO is given by

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{C_1C_2}\left(\frac{1}{R_1R_2} + \frac{1}{R_2R_3} + \frac{1}{R_1R_3}\right)} \tag{2.85}$$

As earlier, an alternative RC-CR transformed version is possible which is shown in Fig. 2.31c, and for this version of the circuit, the CO is found to be

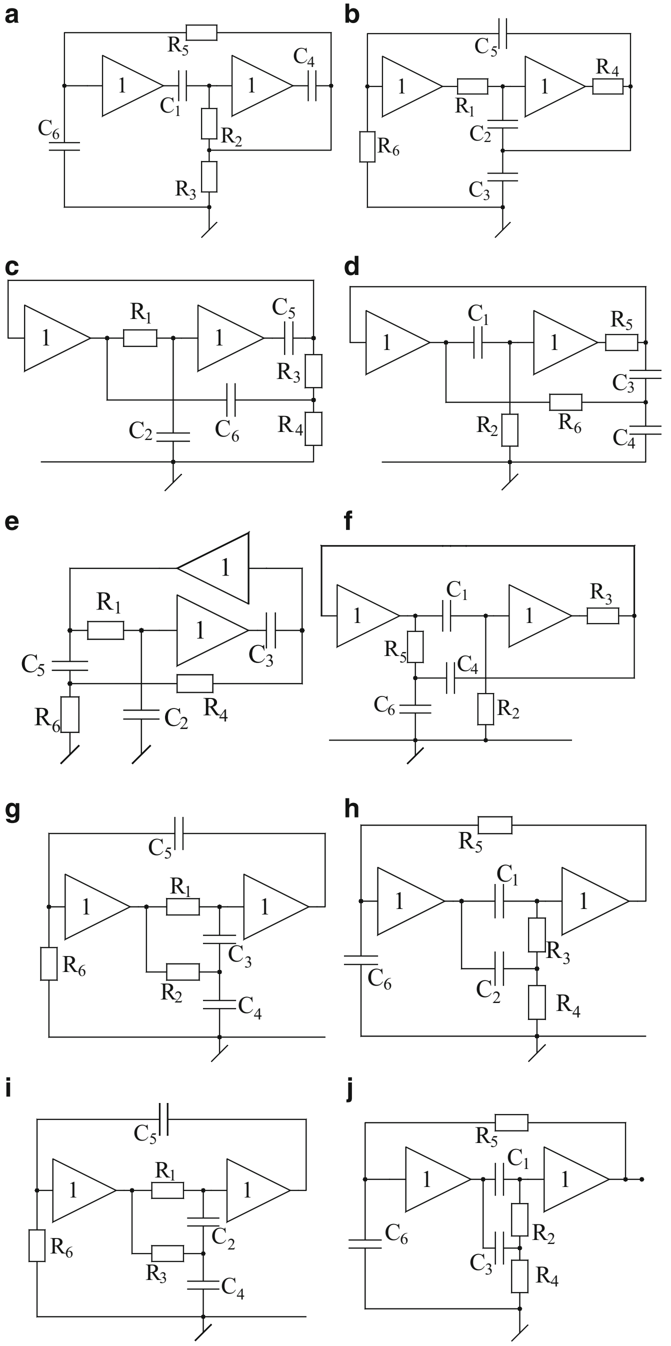


Fig. 2.29 Ten other UGA-based SRCOs presented by Abuelma'atti [66]

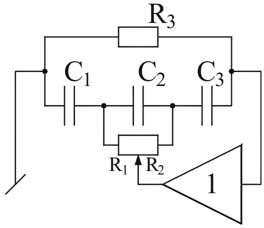


Fig. 2.30 Boutin's single UGA oscillator [64]

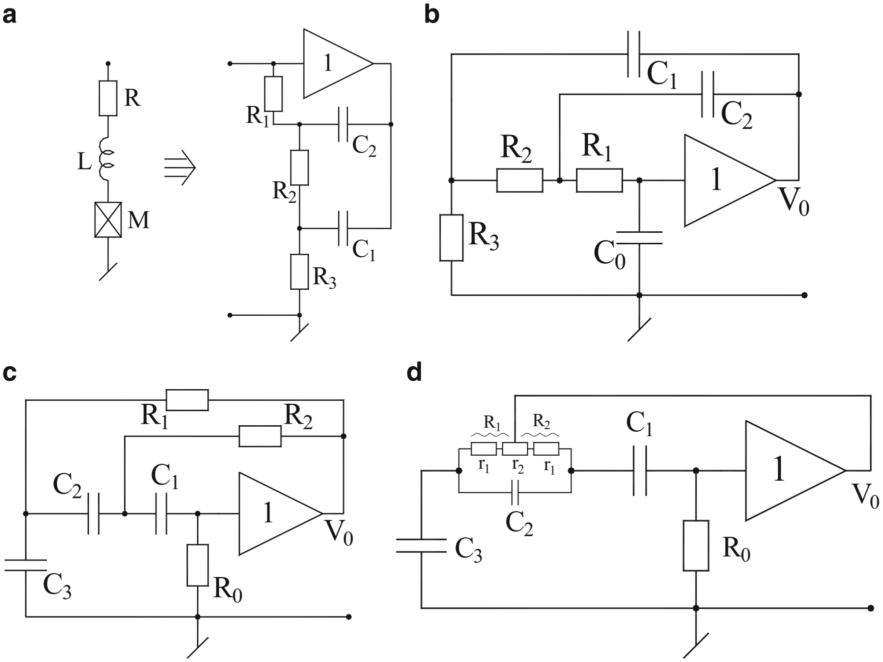


Fig. 2.31 Alternative derivation of the single-VF oscillator with single potentiometric control of oscillation frequency. (**a**) A circuit for simulating series-R L M branch with a single UGA. (**b**) An oscillator implementing the model of Fig.2.26b with a single UGA. (**c**) RC-CR transformed oscillator corresponding to that of Fig. 2.31b. (**d**) Obtaining single *potentiometric* control in the circuit of Fig. 2.31c

$$R_0 = \left\{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right\}(C_1R_1 + C_2R_2 + C_2R_1 + C_3R_2) \tag{2.86}$$

whereas the FO is given by

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{R_1R_2(C_1C_2 + C_2C_3 + C_1C_3)}} \tag{2.87}$$

Note that in both cases, the CO is independently adjustable, through *C*o and *R*o, respectively, but the oscillation frequency is not single-element controllable in either of them. It will now be shown that a slight modification in the latter circuit (Fig. 2.31d) really brings back the feature of independent tunability of FO. Letting $C_1 = C_2 = C_3 = C$, Eqs. (2.74) and (2.75) reduce to

$$R_0 = 6(R_1 + R_2) \tag{2.88}$$

and

$$f = \frac{1}{2\pi C\sqrt{3R_1R_2}} \tag{2.89}$$

Now as R_1 and R_2 share a common node, they can both be realized through a single potentiometer so that by adjusting the potentiometer, it is possible to change the ratio of R_1/R_2 while keeping $(R_1 + R_2)$ and hence Eq. (2.74) invariant. For example, letting $r_1 = 1$ KΩ, $r_2 = 10$ KΩ gives $(R_1 + R_2) = 12$ KΩ, but by adjustment of the potentiometer, the ratio **n** ($R_1/R_2 = n, R_2 = R, R_1 = nR$) can be changed from 1/11 to 11 as a consequence of which

$$f = \frac{1}{2\pi CR\sqrt{3n}} \tag{2.90}$$

can be varied through a single potentiometer over a decade.

Lastly, it must be pointed out that all only-UGA/VF-based oscillators presented in this section are, *curiously*, third-order oscillators. This leads to an interesting question: Is a second-order SRCO using only two capacitors along with any arbitrary number of resistors and unity gain VFs at all possible? The answer to this appears to be negative, though a rigorous and formal mathematical proof of this contention has never been published[4] in the open literature and appears to be an interesting problem for further investigations.

[4] Although some considerations toward this end were made in R. Senani, "On the realizability of canonic second-order sinusoidal oscillators using only voltage followers," manuscript # ELL-56789, September 1985, *unpublished*

2.10 Oscillators with Extended Operational Frequency Range Using Active Compensation and Composite Amplifiers

During 1977–1990, extensive research activity was witnessed on the issue of designing op-amp-based circuits such as VCVS, VCCS, summers, integrators, and filters with improved high-frequency performance using an idea which was called *active compensation*. The basic idea was to reduce primarily the phase error and sometimes both phase errors and normalized magnitude error in the frequency response of a given op-amp-RC circuit, caused by the finite frequency-dependent gain of the op-amp modeled by a one-pole roll-off characteristic. This was done either by modifying the feedback path in the given circuit by adding one or more op-amps along with a few resistors or by designing altogether a new op-amp building block capable of providing reduced phase error or reduced phase error as well as reduced magnitude error. It was demonstrated by a number of researchers that when such active-compensated building blocks are substituted in the given op-amp circuit, the performance of the overall circuit improved in terms of the frequency range of operation. For example, a phase-corrected active-compensated Tow-Thomas biquad could be operated satisfactorily over a much wider frequency range as compared to the original uncompensated Tow-Thomas biquad. As a thumb rule, the application of active compensation in RC-op-amp circuits extended the frequency range of operation of such circuits by one order of magnitude. In view of the extensive work done on active compensation and their applications in designing improved VCVS, VCCS, integrators, and filter structures, it was, therefore, not surprising that several researchers also investigated whether active compensation techniques could lead to the extension of the frequency range of op-amp-RC sinusoidal oscillators. It was demonstrated by several researchers [12, 44, 54, 68, 78, 82–85, 88, 91, 96] that, indeed, this was feasible.

In this section, we present the details of two representative works dealing with the extension of the operational frequency range of oscillators using active compensation techniques. For more detailed study of such compensated oscillators, the reader is referred to [12, 44, 54, 68, 78, 82–85, 88, 91, 96].

The design of active-compensated networks is based upon the one-pole model of the op-amps. The open-loop voltage gain (A_v) of the op-amp is characterized by $A_v(s) = \frac{A_0\omega_p}{s+\omega_p} \approx \frac{A_0\omega_p}{s} \cong \frac{\omega_t}{s}$; for $\omega >> \omega_p$ where $A_0\omega_p = \omega_t$ is the gain-bandwidth product of the op-amp, with A_0 being the DC gain and ω_p being the dominant pole frequency. For μA741-type op-amp the typical values of the various parameters are $f_p = \frac{\omega_p}{2\pi} \cong 5$ Hz, $A_0 = 2 \times 10^5$, and $f_t = \frac{\omega_t}{2\pi} \cong 1$ MHz. On the other hand, the two-pole model of the op-amp gain is characterized by $A_v(s) = \frac{A_0\omega_{p1}\omega_{p2}}{(s+\omega_{p1})(s+\omega_{p2})}$, where $\omega_{p2}$is the second pole of the op-amp which, for $\mu A741$-type op-amp, occurs at a frequency around 1.8 MHz.

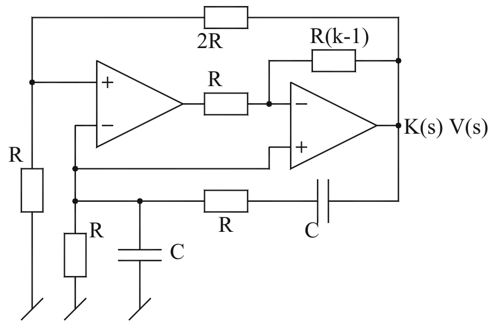


Fig. 2.32 The *active-compensated* WBO proposed by Budak and Nay [44]

If the classical Wien bridge oscillator (WBO) is designed to generate high frequencies, the roots of the characteristic equation no longer remain on the imaginary axis due to the finite frequency-dependent gain of the op-amp employed. By replacing the conventional non-inverting amplifier (of nominal gain of 3) by an appropriate active-compensated non-inverting amplifier (designed with two or more op-amps), it is expected that the magnitude and the phase of the compensated amplifier may make it possible to keep the poles on the imaginary axis up to a much higher frequency than is normally possible with the conventional single-op-amp-based WBO.

In Fig. 2.32 we show the *active-compensated* WBO obtained by employing such a technique proposed by Budak and Nay [44]. A straightforward analysis of this circuit reveals the following:

The gain of the second-order passive RC feedback circuit is given by $\beta(s) = \dfrac{s\omega_0}{s^2 + 3\omega_0 s + \omega_0^2}$, where ω_0 =1/RC, while the active-compensated k-gain amplifier, assuming matched op-amps with their gains represented as ω_t/s, has its nonideal gain function given by

$K(s) = \dfrac{s\tau + \frac{k-1}{k}}{s^2\tau^2 + \frac{\tau}{k}s + \frac{k-1}{3k}}$ where $\tau = 1/\omega_t$ is the time constant of the op-amps.

At $s = j\omega$, the phase function and the magnitude squared of the associated loop gain of the oscillator are given by, respectively

$$\phi = \tan^{-1}\frac{\omega\tau}{\frac{k-1}{k}} - \tan^{-1}\frac{\omega\tau/k}{\frac{k-1}{3k} - (\omega\tau)^2} + \frac{\pi}{2} - \tan^{-1}\frac{3\omega\omega_0}{\omega_0^2 - \omega^2} \tag{2.91}$$

$$|K(j\omega)\beta(j\omega)|^2 = \frac{(\omega\tau)^2 + (k - 1/k)^2}{\left[\frac{k-1}{3k} - (\omega\tau)^2\right]^2 + \left(\frac{\omega\tau}{k}\right)^2}\left(\frac{(\omega\omega_0)^2}{(\omega_0 - \omega)^2 + 9(\omega\omega_o)^2}\right) \tag{2.92}$$

Keeping in mind that the FO has to be near to ω_0 and further assuming that

$$\omega\tau \ll \frac{|k-1|}{k}, \sqrt{\frac{|k-1|}{3k}}, \frac{|k-1|}{3} \tag{2.93}$$

it is found that the approximate value of the frequency at which the total phase shift around the loop becomes zero is given by

$$\omega \cong \frac{\omega_0}{\sqrt{1+3\left(\frac{3-k}{k-1}\right)\omega_0\tau}} = \omega_{0\ \text{non-ideal}} \tag{2.94}$$

On the other hand, the condition required to force the loop gain equal to unity turns out to be independent of both τ and ω_0 and is found to be simply $k = \sqrt{3}$.

Substituting this value in Eq. (2.94), the nonideal frequency of oscillation is found to be

$$\omega_{0\text{-non-ideal}} = \frac{\omega_0}{\sqrt{1+3\sqrt{3}\omega_0\tau}} \tag{2.95}$$

From the above, it can be estimated that for $\omega_0\tau = 0.01$, the reduction in the oscillation frequency from its nominal value is only about 3 %. The superiority of the oscillator design with $k = \sqrt{3}$ has been substantiated by experimental results [44].

Several authors have proposed techniques of improving the performance of conventional op-amp RC oscillators by using the so-called composite amplifiers [12, 44, 54, 68, 78, 82–85, 88, 91, 96]. A composite amplifier is composed of two to three op-amps in such a way that externally it resembles the characteristics of an ideal op-amp like infinite input impedance, infinite-gain and zero-output impedance, etc., but, in addition, offers an extended frequency range of operation than a single op-amp. Although several authors have described methods of improving the high-frequency performance of WBO using composite amplifiers, here we present a configuration devised by Awad [78] (Fig. 2.33).

To understand the mechanism of improving the operational frequency range of the classical WBO using composite amplifiers, we may recollect that while the second-order passive feedback network has the transfer function $\beta(s)$ as already worked out above, the single-op-amp realized non-inverting amplifier has its transfer function as

$$K(s) = \frac{K_0}{a + sK_0/\omega_t} \tag{2.96}$$

so that the nonideal closed-loop characteristic equation of the WBO, taking due cognizance of the amplifier dynamics, is given by

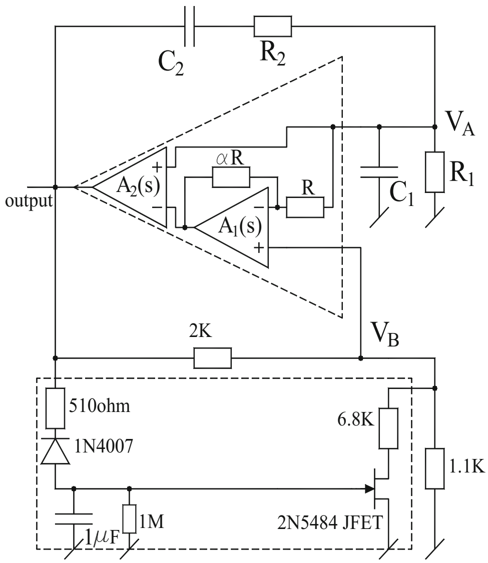


Fig. 2.33 An improved Wien bridge oscillator using a composite amplifier as proposed by Awad [78]

$$K_0\left(\frac{s}{\omega_t}\right)^3+\left(1+3K_0\left(\frac{\omega_0}{\omega_t}\right)\right)\left(\frac{s}{\omega_t}\right)^2$$
$$+\left[(3-K_0)\left(\frac{\omega_0}{\omega_t}\right)+K_0\left(\frac{\omega_0}{\omega_t}\right)^2\right]\left(\frac{s}{\omega_t}\right)+\left(\frac{\omega_0}{\omega_t}\right)^2=0 \tag{2.97}$$

The value of required K_0 for sustained oscillations, assuming $(\omega_0/\omega_t) << 1$, is found to be

$$K_0 \cong 3+27\left(\frac{\omega_0}{\omega_t}\right)^2 \tag{2.98}$$

whereas the nonideal oscillation frequency is found to be

$$\omega_{0(\text{non-ideal})} \cong \omega_0\left(1-\frac{3}{2}\left(\frac{\omega_0}{\omega_t}\right)\right) \tag{2.99}$$

It is, thus, seen that due to the finite GBP of the op-amp employed, the real gain required from the non-inverting amplifier is slightly more than the theoretical value of 3, whereas the nonideal oscillation frequency would be less than the ideal value of ω_0.

This situation can be improved by replacing the conventional one op-amp non-inverting amplifier by a two-op-amp-based composite amplifier as shown in Fig. 2.33 in the dotted box.

A straightforward analysis in this case indicates that, firstly, the input-output relation of the composite amplifier itself is given by

$$V_0(s) = \frac{A_2(s)(1 + A_1(s))(1+\alpha)}{A_1(s) + 1 + \alpha} V_\mathrm{A}(s) - \frac{A_1(s)A_2(s)(1+\alpha)}{A_1(s) + 1 + \alpha} V_\mathrm{B}(s) \qquad (2.100)$$

Assuming matched op-amps, the nonideal gain function of the composite non-inverting amplifier is then found to be

$$K(s) = \frac{K_0\left(1 + \frac{s}{\omega_\mathrm{t}}\right)}{1 + \frac{K_0}{(1+\alpha)}\left(\frac{s}{\omega_\mathrm{t}}\right) + K_0\left(\frac{s}{\omega_\mathrm{t}}\right)^2} \qquad (2.101)$$

The nonideal FO of the WBO with composite amplifier as shown in Fig. 2.33 is found to be

$$\omega_{0(\text{non-ideal})} \cong \omega_0\left[1 - \frac{3}{2}\left(\frac{K - 1 - \alpha}{(1+\alpha)\left(\frac{\omega_0}{\omega_\mathrm{t}}\right)}\right)\right] \qquad (2.102)$$

Thus, by choosing an appropriate value of $\boldsymbol{\alpha}$, the error in the oscillation frequency can be minimized and reduced as compared to that present in case of the conventional WBO.

Alternatively, for an acceptable maximum value of error, the corresponding realizable frequency would be more in the WBO with non-inverting amplifier realized with a *composite amplifier*, as compared to the WBO realized with a conventional single-op-amp non-inverting amplifier.

Experimental results described in [78] have proved the validity of the theory described above. Using μA747-type op-amps biased with ±15 V DC power supplies, it has been reported in [78] that the magnitude of the error in the frequency varied from 3.29 % at 1 KHz to 4.2 % at 92 KHz for WBO realized with composite amplifier as compared to 3.29 % at 1 KHz to 31.42 % at 92 KHz for the original WBO.

Thus, the use of composite amplifier significantly improved the operational frequency range of the WBO. However, on the other hand, the output amplitude varied and, hence, an automatic gain control (AGC) was essential to be incorporated as shown in Fig. 2.33.

2.11 Active-R, Partially Active-R, and Active-C Oscillators Using Op-Amp Compensation Poles

In the 1970s, Rao and Srinivasan [128] and Soderstrand [129] *independently* pioneered the idea of designing active filters without external capacitors by employing the pole of the internally compensated type of op-amps such as μA741 or μA747 as a useful parameter in the design, thereby leading to circuits which employed only resistors and op-amps and were, therefore, named *active-R* filters. This approach was widely investigated by a number of researchers [4, 13, 18, 22, 24, 31, 35, 37, 38, 45, 47, 60, 65, 79, 86, 89, 92, 97, 99–102, 106, 108, 114, 115, 117, 119, 123–125, 127, 130–133]. It may, however, be understood that the term *active-R* is a misnomer since the compensating capacitor inside the op-amp is still part of the circuit dynamics. The leading analog designer Gilbert cautioned [20] that the basic premise of the so-called active-R designs that the dominant pole of the internally compensated type of op-amps is (1) simple, (2) accurate, and (3) temperature independent *are not correct* and, hence, the claims of suitability of the active-R filters for integrated circuit implementation are not sound. Secondly, the implication that the on-chip capacitors are somehow not available for use in a more direct fashion was also misplaced. It was argued that good stability and matching of MOS capacitors forming the poles in conjunction with accurate thin- or thick-film resistors, which can be trimmed to realize filter characteristics of high quality and stability, could be a more viable alternative to the active-R filters. The proponents of the active-R approach [32] counterargued that it is not the pole but the gain-bandwidth product (GBP) which is the parameter employed in this approach and that the tolerances in GBP are of the same order as the tolerances in the passive elements in monolithic integrated circuits to which domain the active-R circuits were intended. Furthermore, it was suggested that variations in GBP can be measured and schemes exist [32] for compensation of these variations.

In spite of this debate, what is interesting is to observe that the active-R approach continued to attract attention of circuit designers and was even extended to other functions such as simulation of inductors and FDNRs, resonators, and sinusoidal oscillators. An outcome of this burst of activity was the evolution of many interesting topologies of oscillators which either eliminated the use of external capacitors completely (*active-R* oscillators) or at least reduced one capacitor for designing a second-order oscillator (resulting in the so-called partially active-R oscillators). Since the GBP of the op-amps can be measured by indirect methods, viz., either by setting up a resonator based upon an active-R-simulated capacitor or active-R-simulated inductor and then measuring the resonant frequency, this information can be readily used to design an active-R sinusoidal oscillator.

Thus, active-R and partially active-R oscillators are of interest from the viewpoint of extending the operating frequency range of op-amp-based oscillators beyond the range possible with op-amp-RC oscillators or active-compensated op-amp RC oscillators. Furthermore, there could be many applications where the requirements of wave shape and frequency stability may not be very stringent such as in transmitter tone generators [45]. In such high-frequency applications, active-R oscillators appear to be attractive choices.

Due to the abovementioned reasons, the authors of this monograph believe that active-R sinusoidal oscillators have sufficient applications and do deserve to be discussed. With this motivation, in the following, we describe some prominent developments which have taken place in the design of active-R, partially active-R, and active-C oscillators.

In the subsequent subsections, a number of prominent active-R and partially active-R oscillators utilizing op-amp pole have been presented. The realization of all such oscillators is based upon mostly a single-pole model of the op-amp, while some researchers have also considered a more elaborate two-pole model.

2.11.1 Three-Op-Amp Active-R Oscillators

An interesting third-order continuously tunable sinusoidal oscillator without external capacitor was presented by Bhattacharyya and Natarajan [18] as shown in Fig. 2.34. By a straightforward analysis of this oscillator circuit, assuming the integrator model of the op-amp as explained earlier, the CO and FO are found to be

$$\mathrm{CO}: \frac{\omega_{t1}}{\omega_{t3}} = \frac{c}{a} \quad \text{and} \quad \mathrm{FO}: \omega_0 = \sqrt{bc\omega_{t3}\omega_{t2}} = \sqrt{ab\omega_{t1}\omega_{t2}} \tag{2.103}$$

where ω_{t1}, ω_{t2} and ω_{t3} are the gain-bandwidth products of the op-amps A_1, A_2, and A_3, respectively. For a more elaborate analysis of this circuit, the reader is referred to [27, 28].

It is, thus, readily seen that the FO is independently controllable by changing resistor ratio b. It is also easy to visualize that resistor ratios can remain stable over a wide range of temperature variation, and moreover temperature-compensated op-amps are commercially available such as LM324, with the use of regulated power supplies to bias the op-amp (thereby ensuring no variation in the gain-bandwidth product of the op-amps is supposed to change when power supply

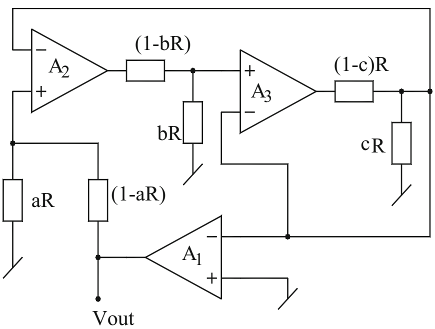


Fig. 2.34 Active-R oscillator proposed by Bhattacharyya and Natarajan [18]

voltage changes). In view of this, it appears that temperature-stable oscillations can be obtained from the circuit easily. Furthermore, tuning of the frequency is easily achievable through a single grounded resistor which can be made voltage dependent by replacing it with an FET, thereby making realization of a voltage-controlled oscillation (VCO) possible.

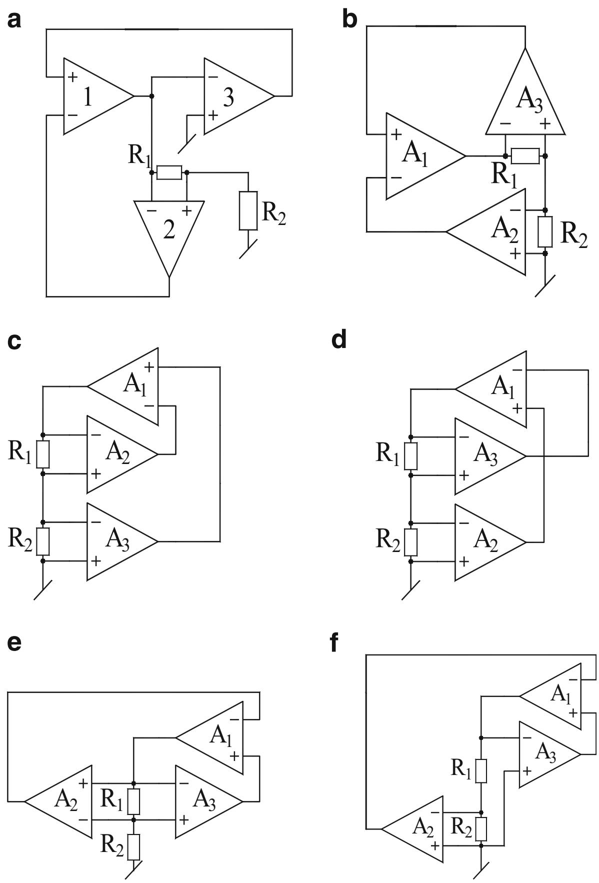
Using LM 741 op-amp and setting the values of *a* and *c*, variable-frequency oscillations were generated by this circuit by changing the parameter b from 0.01 to 0.1. The circuit was generated sinusoidal oscillations over a wide range of frequency.

Sanyal, Sarker, and Nandi [89] presented a family of six three-op-amp-based active-R sine-wave oscillators realizable with only three op-amps and a bare minimum of only two resistors. These circuits are shown in Fig. 2.35.

Using the integrator model of the op-amps and assuming matched op-amps, the characteristic equation of the circuit of Fig. 2.35a is given by

$$s^2 + K\omega_t^2 = 0 \tag{2.104}$$

Fig. 2.35 (**a–f**) The family of three-op-amp active-R oscillators proposed Sanyal, Sarker, and Nandi [89]



which leads to the expression for oscillation frequency as $\omega_0 = \omega_t\sqrt{K}$, where $K = R_2/(R_1 + R_2)$ and $\omega_t = A_0\omega_p$ are the gain-bandwidth product of the op-amps.

Using a two-pole model of the op-amp gain A_v and assuming matched op-amps, the characteristic equation of the first circuit is found to be

$$\begin{aligned} &s^4 + 2s^3(\omega_{p2} + \omega_{p1}) + s^2\left(\omega_{p1}^2 + 4\omega_{p1}\omega_{p2} + \omega_{p2}^2\right) \\ &\quad + 2s\omega_{p1}\omega_{p2}(\omega_{p1} + \omega_{p2}) + \beta(\omega_{p1}\omega_{p2})^2 = 0 \end{aligned} \tag{2.105}$$

where

$$\beta = \left(1 + KA_0^2\right) \tag{2.106}$$

The above equation can be approximated by an equivalent second-order equation by neglecting the fourth-order term and substituting $s^3 \approx -\omega_0^2 s$. Such an approximation leads to the following approximation of the CE:

$$s^2 - 2\sigma s + \omega_n^2 = 0 \tag{2.107}$$

where $\sigma = \frac{(\omega_0^2 - \omega_{p1}\omega_{p2})(\omega_{p1} + \omega_{p2})}{\omega_{p1}^2 + \omega_{p2}^2 + 4\omega_{p1}\omega_{p2}}$; $\omega_n = \frac{\omega_{p1}\omega_{p2}\beta^{1/2}}{\sqrt{\omega_{p1}^2 + \omega_{p2}^2 + 4\omega_{p1}\omega_{p2}}}$ and

$$\omega_p = \left(\omega_n^2 - \sigma^2\right)^{1/2} \tag{2.108}$$

Thus, with $\sigma \geq 0$, ω_p becomes the FO and the CO is given by

$$A_0 \geq \frac{1}{\sqrt{Ku}} \quad \text{where} \quad u = \frac{\omega_{p1}}{\omega_{p2}} < 1 \tag{2.109}$$

It may be seen that the above condition is not difficult to satisfy. For instance, if one selects μA741 type of op-amps with $A_0 = 2 \times 10^5$, $f_{p1} = 5$ Hz, and $f_{p2} = 1.8$ Hz, sustained oscillations will be ensured for all $K < 1.33 \times 10^5$. Since K represents the attenuation factor decided by the ratio of two resistors employed, its maximum value is, obviously, unity.

For brevity, the CO and FO and other design constraints for all the circuits are summarized in Table 2.4.

These circuits were tested [89] with μA741-type op-amps biased with ±15 V DC power supply. Good-quality sine waves were obtainable from all the six circuits over a frequency range 3–700 KHz with op-amps having a gain-bandwidth product of 0.99 MHz, $f_{p1} = 5$ Hz, and $f_{p2} = 1.8$ MHz. Proportionally higher-frequency generation up to about 2 MHz was obtained in the proposed design when LF356 op-amps were employed which have a gain-bandwidth product of 5 MHz. These circuits were shown to be useful for continuous-phase frequency-shift keying (CPFSK) applications.

Table 2.4 Design equations for the various three-op-amp active-*R* oscillators of Fig. 2.35.

Circuit	Oscillation frequency	Condition of oscillation
a	$\omega_t\sqrt{K};\ 0 \le K \le 1$	$A_0 > 1/\sqrt{Ku}$
b	$\omega_t\sqrt{(1-2K)};\ 0 \le K \le 1/2$	$A_0 > 1/\sqrt{(1-2K)u}$
c, d	$\omega_t\sqrt{(2K-1)};\ 1/2 \ge K \ge 1$	$A_0 > 1/\sqrt{(2K-1)u}$
d	$\omega_t\sqrt{2(1-K)};\ 0 \le K \le 1$	$A_0 > 1/\sqrt{2(1-K)u}$
e	$\omega_t\sqrt{(1-K)};\ 0 \le K \le 1$	$A_0 > 1/\sqrt{(1-K)u}$

These oscillator circuits were tested [89] using uA741-type op-amps, and high-frequency oscillations over a fairly wide range were obtainable as predicted by theory. The frequency of oscillation could be varied by the grounded resistor R_2.

2.11.2 Two-Op-Amp Active-R Sinusoidal Oscillators

In the previous section, we described a set of oscillators which were all third-order oscillators. However, for realizing an oscillator, only two op-amps acting as integrators, along with a number of resistors, would suffice. Several researchers have described such two-op-amp active-R oscillators. In the following, we present some selected two-op-amp oscillators employing op-amp poles with integrator approximation.

While presenting an active-R realization of bilinear RL impedance, Nandi [24] presented its application in realizing an external capacitor-less oscillator which is shown here in Fig. 2.36. By a straightforward analysis, using integrator models for the two op-amps, the CO and FO are found to be

$$a = \frac{(1-K)}{\left[\left(\frac{\omega_{t1}}{\omega_{t2}}\right) + K\right]} \quad \text{and} \quad \omega_0 = \sqrt{\frac{aK\omega_{t1}\omega_{t2}}{1+a}} \tag{2.110}$$

It may be seen that although an oscillator is realizable, in general, CO and FO are not independently tunable and that is a limitation of this circuit. However, if one chooses K to be sufficiently small, independent frequency control is possible. It has been reported [24] that for a circuit built from μA741 op-amps with $\omega_{t1} = \omega_{t2} = 2\pi \times 10^6$, it was possible to realize sinusoidal oscillations variable from 24–100 KHz by adjusting K in the range 0.001–0.02 with the CO reducing to a $\approx$1 almost being independent of K.

Sanyal, Sarker, and Nandi [79] presented yet another two-op-amp four-resistor circuit which is shown in Fig. 2.37. Although the authors have carried out a very elaborate analysis considering two-pole models for the gains of both the operational amplifiers which are supposed to be identical, a single-pole analysis, however, shows the CO and FO being given by

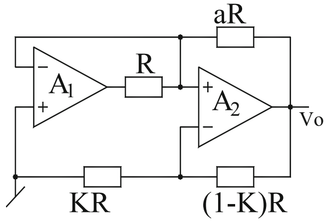


Fig. 2.36 An active-R oscillator proposed by Nandi [24]

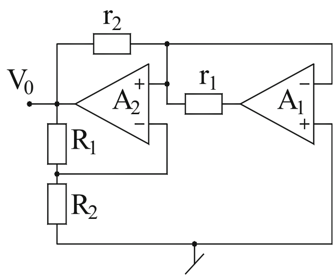


Fig. 2.37 Active-R oscillator proposed in [79]

$$a = \frac{(1-k)}{(1+k)} \cong 1; k << 1 \quad \text{and} \quad \omega_0 = \omega_t \sqrt{\frac{k}{2}} \tag{2.111}$$

Thus, with only four resistors and two op-amps, independent control of FO, without affecting CO, looks difficult to achieve.

Because of the above reasons, there have been many studies and configurations which employ two op-amps and seven to nine resistors such as [47] to make an oscillator providing independent control for FO. A circuit employing only six resistors to achieve these objectives was reported by Venkataramani and Venkateshwaran [127] and is shown here in Fig. 2.38.

An analysis of this circuit, assuming $\alpha = R_2/(R_1 + R_2)$, $\beta = R_4/(R_3 + R_4)$, and $\delta = R_6/(R_5 + R_6)$, reveals that the FO and CO are given by

$$(\delta - \alpha) = 0 \quad \text{and} \quad f_0 = f_t \sqrt{(\beta - \alpha\delta)} \tag{2.112}$$

The circuit can be adjusted to produce sinusoidal oscillations by adjusting α or δ, while frequency can be adjusted by β without affecting the condition of oscillation. Since the GBP of the op-amps is dependent upon the DC bias voltage, the frequency of the sine waves can be controlled by varying the DC bias supply voltage. Alternatively, the oscillation frequency can be varied through an external control voltage by replacing the grounded resistor by an FET used as a VCR (see [93, 94, 104] and the references cited therein).

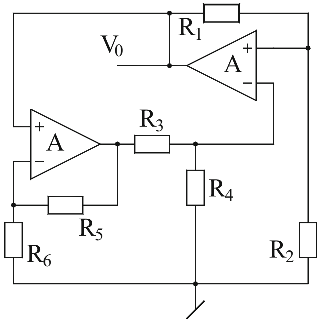


Fig. 2.38 An active-R sinusoidal oscillator proposed by Venkataramani and Venkateshwaran [127]

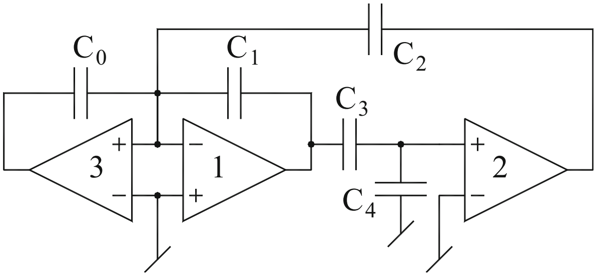


Fig. 2.39 MOS-compatible active-C oscillator employing op-amp poles, proposed by Khan-Ahmed [52]

2.11.3 Active-C Sinusoidal Oscillators

With the need for fully integrated sinusoidal oscillators implementable in CMOS technology along with CMOS digital circuits, MOS-switched-capacitor oscillators were developed. Such oscillators will be dealt in a subsequent chapter of this monograph. The so-called active-C sinusoidal oscillators were evolved [46, 52, 55] to overcome some limitations of the switched-capacitor oscillators such as limited high-frequency performance, aliasing errors, and switching noise while retaining their advantage, namely, the suitability for CMOS implementation. With this motivation, a number of authors have proposed MOS-compatible active-C oscillators which employ only CMOS op-amps and MOS capacitors and are designed by incorporating the op-amp compensation poles as a useful parameter rather than being treated as a parasitic effect. One such active-C oscillator proposed by Khan-Ahmed [52] is shown here in Fig. 2.39.

Straightforward analysis of this circuit, employing integrator models of the three op-amps, shows that its CO and FO are given by, respectively

$$\frac{\omega_{t1}}{\omega_{t2}} = \frac{C_0}{C_1} \quad \text{and} \quad \omega_0 = \sqrt{a_2 K \omega_{t1} \omega_2} \tag{2.113}$$

where

$$K = C_3/(C_3 + C_4) \quad \text{and} \quad a_2 = C_2/(C_0 + C_1 + C_2) \tag{2.114}$$

From the above, it is seen that the independent tuning of the FO is possible through the capacitor ratios a_2 or K, respectively, by varying C_3 or C_4 without disturbing CO.

The workability of this circuit was verified [52] by employing RCA3140 MOSFET-input op-amps and capacitors of 1 % tolerance. The circuit was designed to function as a variable-frequency oscillator from 10 KHz to 1.14 MHz by choosing the various capacitors as $C_0 = 110$ pF, $C_2 = 44$ pF, and $C = 33$ pF to 100 nF variable. The oscillator was found to work satisfactorily over a wide range of frequencies.

2.11.4 Partially *Active-R Oscillators*

There have been many studies of designing *partially active-R* oscillators using two op-amps, a capacitor and a few resistors (for instance, see 4, 13, 31, 97, 106, 108, 115, 117, 134). Needless to say, with op-amp gains modeled as integrators, all such circuits are essentially third-order partial active-R circuits. In this section, we present a simple sinusoidal third-order oscillator employing only two op-amps, a resistor and a grounded capacitor which was proposed by Senani [97]. The quoted circuit from [97] is shown here in Fig. 2.40.

Using an integrator model of the two op-amps, assuming different gain-bandwidth product, the characteristic equation of the circuit is found to be

$$Ts^3 + s^2 + \omega_{t2}s + \omega_{t1}\omega_{t2} = 0; \quad \text{where} \quad T = R_0C_0 \tag{2.115}$$

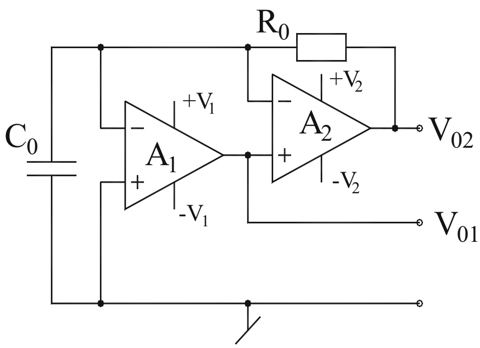


Fig. 2.40 A simple sinusoidal oscillator using op-amp compensation poles introduced by Senani [97]

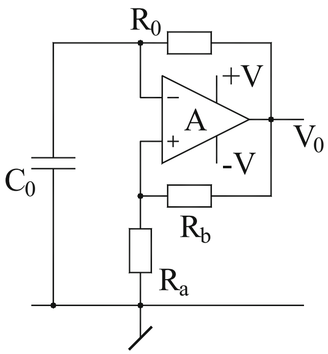


Fig. 2.41 The classical astable multivibrator as a partially active-R sinusoidal oscillator (originally proposed by Anandamohan [31])

from which the CO and FO are given by

$$\left(\omega_{t1} - \frac{1}{T}\right) \leq 0 \quad \text{and} \quad f_0 = \sqrt{f_{t1} f_{t2}} \tag{2.116}$$

It may be seen that the CO can be adjusted by the resistor R_0, while the frequency can be varied through ω_{t2} which, in turn, can be varied by changing the DC bias supplies $\pm V_2$ since the gain-bandwidth products of the op-amps can be varied by varying the power supply voltages.

Experimental results of this circuit using μA741 op-amps with measured values of GBPs as $f_{t1} = 0.988$ MHz and $f_{t2} = 0.935$ MHz, $\pm V_1 = \pm 15$ V, $C_0 = 100$ pF, and R_0 realized by a variable resistor demonstrated that with $\pm V_2$ varied from ±2.5–16 V, the oscillation frequency could be varied from 502 to 941 KHz.

Since in conjunction with the use of the op-amp pole approximated as an integrator, only a single capacitor and a few resistors should be sufficient to make a sinusoidal oscillator, there have been several studies on evolving single-op-amp-based partially active-R oscillators requiring only one external capacitor and a few resistors (for instance, see 4, 13, 31, 97, 106, 108, 115, 117 and the references cited therein).

From among the variety of single-op-amp single-capacitor partially active-R sinusoidal oscillators known in literature and quoted above, the most elegant and versatile appears to be the one presented by Rao and Srinivasan [4][5] which is shown here in Fig. 2.41. It may be noted that the circuit is same as the classical astable multivibrator which by proper design of the circuit can be turned around to work as a sinusoidal oscillator. It was first demonstrated in 1979 by Anandamohan [31] that this circuit can also be tailored to generate sine waves. With the op-amp open-loop

[5] The above-described mode of operation of this circuit was first reported by Rao and Srinivasan [4] in 1973 and then by Anandamohan in 1979 [31]. Ironically, more than two decades later since the publication of [4], this circuit was published again in [106], obviously somehow missing to take cognizance of not only [4] but also that of [31]!

gain A_v modeled as an integrator, it can be easily deduced that the closed-loop characteristic equation of this circuit is given by

$$s^2 + \left(\frac{1}{C_0R_0} - \beta\omega_t\right)s + \frac{(1-\beta)\omega_t}{C_0R_0} = 0 \tag{2.117}$$

where $\beta = R_a/(R_a + R_b)$ and is seen to be less than unity. It, therefore, follows that if β is chosen such that $(1 - \beta\omega_t C_0 R_0) = 0$, the circuit would function as a sine-wave generator with oscillation frequency given by

$$\omega_0 = \sqrt{\frac{(1-\beta)\omega_t}{C_0R_0}} \tag{2.118}$$

On the other hand, when the pole is placed deep into the right half of the s-plane by choosing $\beta\omega_t C_0 R_0 \gg 1$, the op-amp will move into saturation, and the circuit dynamics would be that of a first-order system with saturation-type nonlinearity for which the square-wave mode of the operation of the circuit is well known and well documented in literature; see Refs. [1–6] of Chap. 1 of this monograph.

We close this section by presenting a very *unusual* sine-wave generator which was reported by Lucas [13] in 1977. Lucas's circuit, shown in Fig. 2.42 with the component values as shown, oscillates and generates sine waves of frequency 3.8 KHz. What appears interesting is that neither a one-pole model of the op-amp gain nor a more elaborate two-pole model, along with the incorporation of the nonideal differential input resistance and/or fine nonzero output resistance, predicts or throws any light on the sinusoidal oscillator nature of this circuit!

Should one then incorporate the nonideal time delay along with a two-pole model and also consider the nonlinear model of the op-amp including saturation nonlinearity to determine and predict its function as a sinusoidal oscillator? To the best knowledge of the authors, any convincing analysis of this circuit has not appeared in technical literature, and in this context, this circuit presents an interesting problem for research.

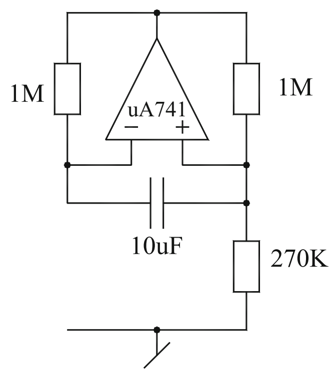


Fig. 2.42 The *unusual* sine-wave generator reported by Lucas [13]

2.12 Op-Amp-Based VCOs with *Linear* Tuning Laws

Voltage-controlled oscillators (VCOs) exhibiting linear tuning laws are useful in several applications. A convenient method to realize such VCOs is to embed two analog multipliers appropriately in a second-order op-amp RC filter of appropriate form and then close the loop. A simple general scheme to generate linear sinusoidal VCOs was presented in [98] leading to three VCO structures which are shown in Fig. 2.43.

By straightforward analysis the characteristic equation of the circuit of Fig.2.43a is found to be

$$f_0 = \left(\frac{v_c}{v_{ref}}\right)\frac{1}{2\pi}\sqrt{\frac{1}{R_1R_2C_1C_2}} \tag{2.119}$$

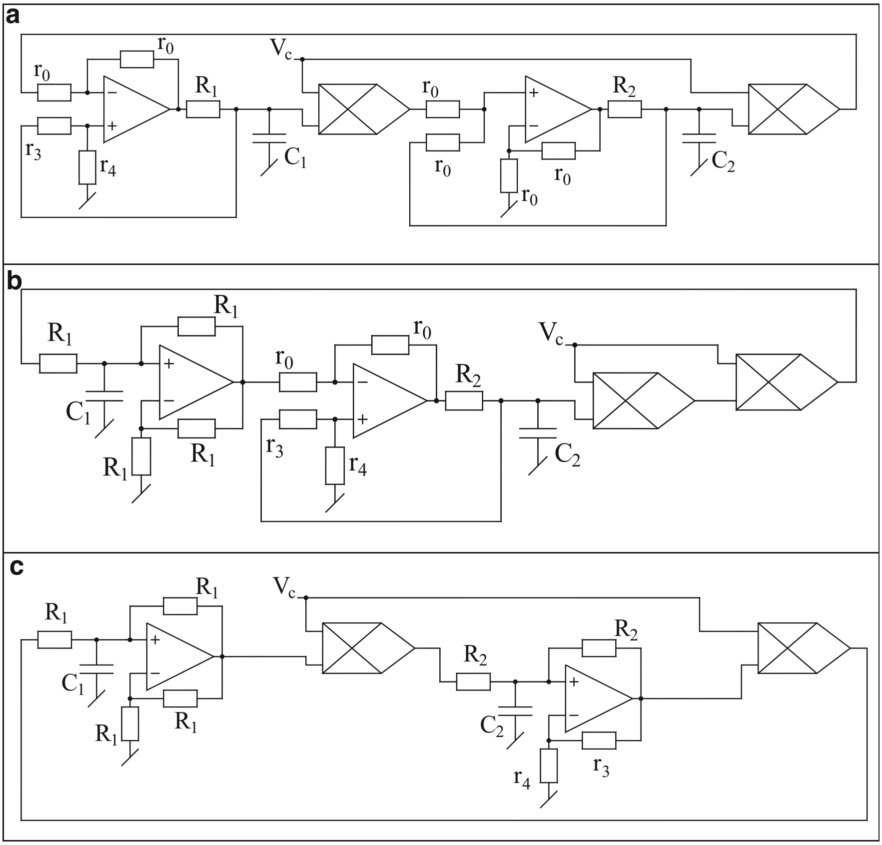


Fig. 2.43 Active-R linear sinusoidal VCOs proposed by Senani, Bhaskar, and Tripathi [98]

whereas the condition of oscillation is found to be

$$1 - \frac{2}{\left(1 + \frac{r_3}{r_4}\right)} \leq 0 \tag{2.120}$$

Thus, it is seen that the FO is a linear function of the control voltage V_c as intended, while the CO can be adjusted by the resistor ratio r_3/r_4 without affecting FO.

The circuit of Fig. 2.43b is also characterized by the same Eqs. (2.107) and (2.108). On the other hand, the last circuit of Fig. 2.43c has FO and CO given by

$$f_0 = \left(\frac{v_c}{v_{\mathrm{ref}}}\right)\frac{1}{2\pi}\sqrt{\frac{1}{R_1R_2C_1C_2}} \tag{2.121}$$

$$(r_4 - r_3) \leq 0 \tag{2.122}$$

Subsequently, Senani and Bhaskar [107] demonstrated that using op-amp compensation poles, sinusoidal VCOs with linear tuning laws can also be evolved using active-R technique. Their circuits are shown in Fig. 2.44. For the generalized scheme of Fig. 2.44a, the CO and FO are found to be

$$[k_3k_4\omega_{t2} - k_1k_2\omega_{t1}] = 0 \tag{2.123}$$

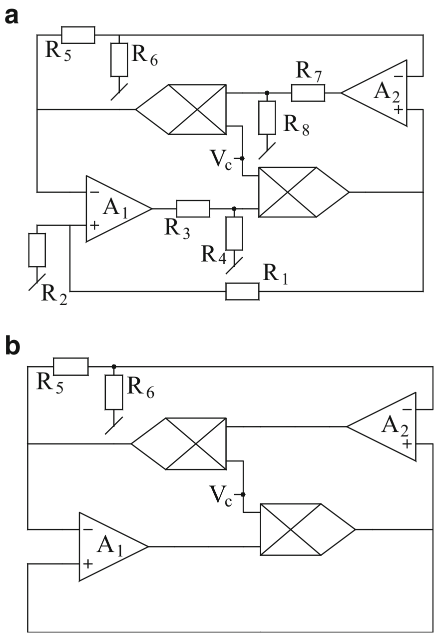


Fig. 2.44 Active-R linear sinusoidal VCOs proposed by Senani and Bhaskar [107]. (**a**) A generalized scheme. (**b**) A special low-component version of the general structure

$$\omega_0 = \left(\frac{V_c}{V_{ref}}\right)[k_2 k_4 \omega_{t1}\omega_{t2}(1 - k_1 k_3)]^{1/2} \tag{2.124}$$

$$\text{where } k_i = \frac{R_{2i}}{R_{2i} + R_{2i-1}}; \quad \text{for all } i,\ i = 1 - 4 \tag{2.125}$$

In [107], it has been shown that as many as 11 linear VCOs can be derived from this general scheme; however, to conserve space here, we demonstrate one specific special case which is shown in Fig. 2.44b. For this circuit the CO and FO are found to be

$$[k_3\omega_{t2} - \omega_{t1}] = 0 \tag{2.126}$$

$$\omega_0 = \left(\frac{V_c}{V_{ref}}\right)[\omega_{t1}\omega_{t2}(1 - k_3)]^{1/2} \tag{2.127}$$

Thus, after setting the oscillations through k_3, the oscillation frequency can be linearly controlled through the external control voltage V_c.

While all the above-described schemes of Fig. 2.43 and Fig. 2.44 do provide linear VCOs with independent controls, they need as many as two op-amps. A question, thus, arises whether a circuit possessing such properties can be devised even with a single op-amp or not. An affirmative answer to this was provided by Bhaskar and Tripathi [116] who presented two such single-op-amp-two-AM linear VCOs which are shown here in Fig. 2.45.

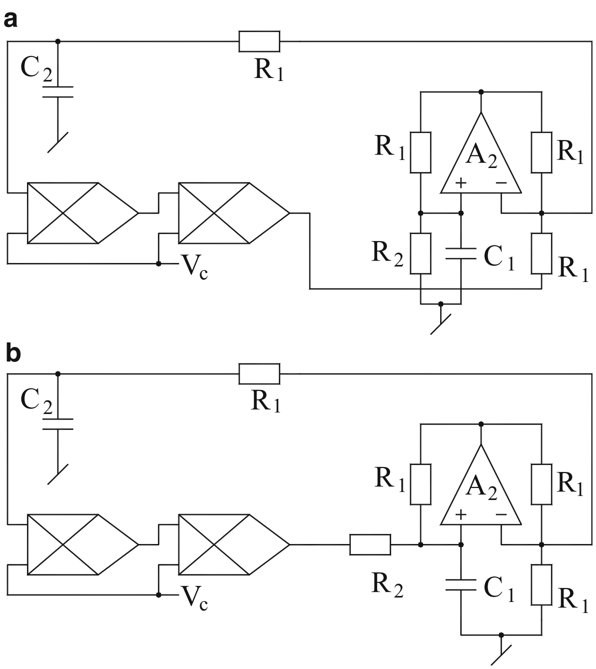


Fig. 2.45 Op-amp RC linear VCOs requiring only one op-amp as proposed by Bhaskar and Tripathi [116]

The first circuit can be seen to be a variant of Wien bridge oscillator in which a multiplicative feedback dependent on an external control signal has been appropriately devised by tapping a voltage from the common node of the series resistor and capacitor and feeding back the same, through an additional resistor R, to the inverting input of the op-amp. In the second scheme, an additional resistor has been added in parallel with the series branch, but the parallel RC branch has been modified by ungrounding the resistor R and putting a multiplicative voltage signal of appropriate value at its ungrounded end. Both the modifications result in the VCOs possessing the intended properties, as can be confirmed by straightforward analysis. For both the circuits, the CO and FO are found to be

$$(C_1 - C_2) = 0 \tag{2.128}$$

$$f_0 = \left(\frac{V_\mathrm{c}}{V_\mathrm{ref}}\right)\frac{1}{2\pi R\sqrt{C_1 C_2}}; \quad \text{for } R_1 = R_2 = R \tag{2.129}$$

It is, thus, seen that the circuits can be adjusted to oscillate through a variable capacitance, while the oscillation frequency can be *linearly* varied through the external control voltage (V_c).

Both the circuits have been tested for their workability and practicability using μA741-type IC op-amps and MPY534-type AMs with RC components 1 of 5 % tolerance, with biasing voltage for both op-amps and AMs taken as ±15 V DC. - Good-quality sine waves with linear tunability of oscillation frequency over a range of about a decade confirmed the workability of these VCOs. Lastly, it may be pointed out that single-op-amp VCO providing linear control of FO but providing CO control through a single-variable resistance (preferably grounded) has still not been discovered.

2.13 Concluding Remarks

In this chapter the main focus was on presenting sinusoidal oscillators in which oscillation frequency can be controlled independently either through a single-variable element (a variable capacitor or a variable resistor) or through a voltage (which could be either a single external control voltage or the supply voltage of the active devices employed). In doing so, however, an attempt has been made to maintain a historical perspective of sequential developments which have taken place in this exciting area of analog circuit research. Consequently, we have dealt with single-element-controlled oscillators (SECOs) realizable with three op-amps, subsequently culminating into SECOs realizable with only a single op-amp. On the other hand, we have also taken into consideration VCO structures realizable with op-amps and multipliers as well as those realizable from SRCOs by replacing the frequency-controlling resistor(s) by an FET used as a VCR. Finally, the so-called active-R and partially active-R oscillators have also been dealt with whose realization is based upon the consideration of op-amp compensation pole(s) as useful parameter (s) in the design. Lastly, two interesting unresolved problems pertaining to a specific partially active-R oscillator reported by Lucas [13] and the other related to realizability of a single-op-amp-AM VCO with CO control through a single-variable resistance have been pointed out.

It is hoped that this exposition not only gives an overview of the developments on op-amp-based SECOs but also highlights the key contributions made in this area which may be helpful to the circuit designers for choosing an appropriate topology for a specific application at hand and at the same time also provides some ideas for research-oriented readers.

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Chapter 3
Electronically Controllable OTA-C and G_m-C Sinusoidal Oscillators

Abstract OTA-C oscillators and G_m-C oscillators are attractive due to a number of advantageous features provided by them, namely, electronically controllable frequency of oscillation as well as condition of oscillation and suitability for integrated circuit implementation in both bipolar and CMOS technologies since OTAs do not require any resistors and are implementable exclusively using transistors only. Furthermore, in circuits containing OTAs or transconductors (G_m), all functions can be invariably performed without requiring any passive resistors (of course, wherever needed, resistor(s) can always be simulated by OTA(s)/transconductor (s)). In this chapter, we present significant developments taken place in the area of realization of electronically controllable sinusoidal oscillators using OTA-C and G_m-C circuits.

3.1 Introduction

The operational transconductance amplifier (OTA) came into existence in 1969 when it was anticipated that due to its versatility and the capability of providing electronic tunability of its transconductance gain, it may make the voltage mode op-amp *obsolete* [1]. It was demonstrated by Bialko and Newcomb [2] that in conjunction with the capacitors, the differential voltage-controlled current source (DVCCS) is capable of realizing all components needed for linear circuit construction in integrated circuit form.

Subsequent to the pioneering work of Bialko and Newcomb [2], the integrated differential voltage-controlled current source/differential voltage-controlled voltage source (DVCCS/DVCVS) attracted a lot of attention in the realization of synthetic impedances [3, 4] and sinusoidal oscillators [5, 6]. Nandi [3] demonstrated that a lossless grounded inductance can be realized by using one DVCCS, one unity gain DVCVS, a grounded capacitor, and a resistor. Similarly, a grounded FDNR can be realized by adding one more integrator stage (consisting of one additional DVCCS and one additional grounded capacitor) to the inductor simulation circuit. It was also demonstrated that adding one external grounded capacitor across the lossless inductor and an external grounded resistor across the ideal

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FDNR resulted in the realization of sinusoidal oscillators. All these circuits, however, required resistor(s) also. Nandi [4] presented another oscillator circuit using two DVCCS, two grounded capacitors, one unity gain amplifier, and two resistors, but the circuit suffers from the drawback of not having any control over the condition of oscillation. Yet another circuit, proposed by Nandi [5], employed one DVCCS, two unity gain amplifiers, two capacitors (one of which was floating), and a resistor and suffered from the same drawback. However, Saha and Nandi [6] presented a sine wave oscillator with single-resistance control (with the control of condition of oscillation also) employing two DVCCS, two grounded capacitors, and two resistors.

The true potential of the integrated DVCCS or the OTA was appreciated only due to subsequent research; it became clear that it is possible to realize various analog functions exclusively using OTAs and capacitor(s) only without requiring any external resistor(s). Such OTA-C realizations were considered to be suitable for IC implementation because of complete elimination of resistors from the circuit. It may be mentioned that an OTA can be implemented in bipolar or CMOS technology using exclusively transistors only without requiring even single resistor. Consequently, OTA-C or G_m-C circuits started attracting worldwide attention of the researchers for electronically tunable, fully integrated analog signal processing/ signal generation applications. An excellent tutorial on above filter design using OTAs was presented by Geiger and Sanchez and Sinencio [7]. Subsequently, many authors presented a number of topologies and methods for designing active filters [1–7], analog multipliers, etc., using OTA-C or Gm-C circuits. Together with these, several dozens of papers have also been published on the realization of sinusoidal oscillators using OTAs (for instance, see [8–82] and the references cited therein).

This chapter considers the realization of electronically controllable sinusoidal oscillators using OTAs/transconductors and presents some representative circuits for the realization of a variety of oscillators.

3.2 OTA-C Sinusoidal Oscillators

The interest toward developing circuit topologies of OTA-based sinusoidal oscillators was motivated by the following anticipated advantages of OTA-based oscillators over conventional op-amp-based sinusoidal oscillators:

1. *Linear* electronical tunability of the frequency of oscillation; since the transconductance g_m of a bipolar OTA is a linear function of the external DC bias current I_B (since $g_m = I_B/2V_T$), the resulting oscillator will have frequency of oscillation of the type $f_0 = \frac{g_m}{2\pi C}$, thereby providing $f_0 \propto I_B$.
2. Relatively higher operational frequency range.
3. Suitability for IC implementation due to complete absence of passive resistors and the circuits realizable with only transistors and very low-valued capacitors

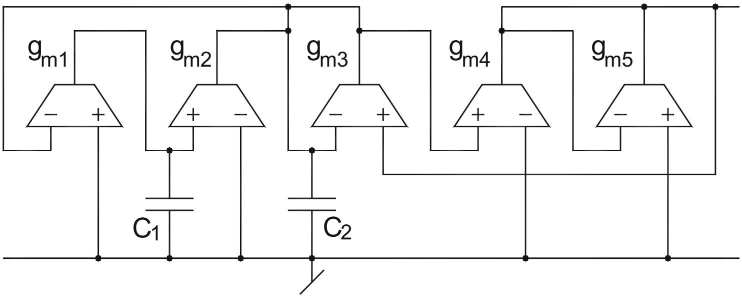
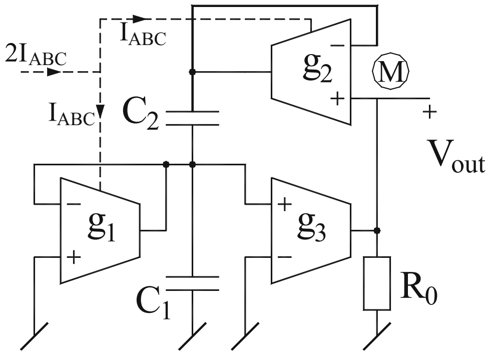


Fig. 3.1 The OTA-C sinusoidal oscillator proposed by Abuelma'atti and Almaskati [8]

Fig. 3.2 OTA-based linearly tunable Wien bridge oscillator proposed by Senani and Kumar [9]



(of the order of a few pF) all of which hold the promise of accomplishing much smaller chip area when implemented in integrated circuit (IC) form.

Probably the first entirely OTA-C oscillator was the one proposed by Abuelma'atti and Almaskati [8] employing as many as five OTAs and two grounded capacitors and providing control of both condition of oscillation (CO) and frequency of oscillation (FO) independent of each other. A drawback of this circuit, however, was that one of the OTAs therein (having transconductance g_{m3}) was clearly redundant as it neither appeared in CO nor in FO. However, this redundant-appearing OTA could not be taken out from the circuit by any means to reduce the number of OTAs to four. This circuit is shown in Fig. 3.1.

Senani and Kumar [9] presented an OTA analogue of the classical Wien bridge oscillator which provides similar features employing only three OTAs, two capacitors (one floating and one grounded), and a grounded resistor (which can also be simulated by one OTA). The CO and FO of this oscillator (Fig. 3.2) can be found as

$$\mathrm{CO} : g_{m3}R_0 = 3 \tag{3.1}$$

$$\mathrm{FO} : f_0 = \frac{1}{2\pi}\sqrt{\frac{g_{m1}g_{m2}}{C_1C_2}} \tag{3.2}$$

If the DC bias currents are made equal, i.e., $I_{B1} = I_{B2} = I_B$, then f_0 is linearly controllable through I_B without effecting CO which can be independently adjusted by R_0. This circuit generates good-quality sine waves over 100 kHz.

It is simple logic to postulate that to have completely independent CO and FO, a circuit with only four OTAs should suffice. Also for obtaining independent control of CO and FO, only three transconductances are really needed such that two of them set the CO while the third one (together with any one of the two transconductance appearing in CO) constitutes the FO and controls it. Thus, the stage was clearly set to search for new topologies of OTA-C oscillators using no more than three/four OTAs along with two grounded capacitors. In the following sections, we give an account of the key developments taken place in the evolution of such three-/four-OTA-based sinusoidal oscillators.

3.2.1 Four-OTA-C Grounded-Capacitor Oscillators

In [10], a simple method of generating a four-OTA-based sinusoidal oscillator is described. A direct way to create such an oscillator is to simulate the parallel ±RLC resonator model of Fig. 3.3a from which it is deduced that the circuit represents an oscillator with CO given by $R_0 \leq R_0'$ and FO: $f_0 = \frac{1}{2\pi\sqrt{L_0 C_0}}$. An OTA implementation of this model is shown in Fig. 3.3b, where OTA_1 and OTA_2 along with the

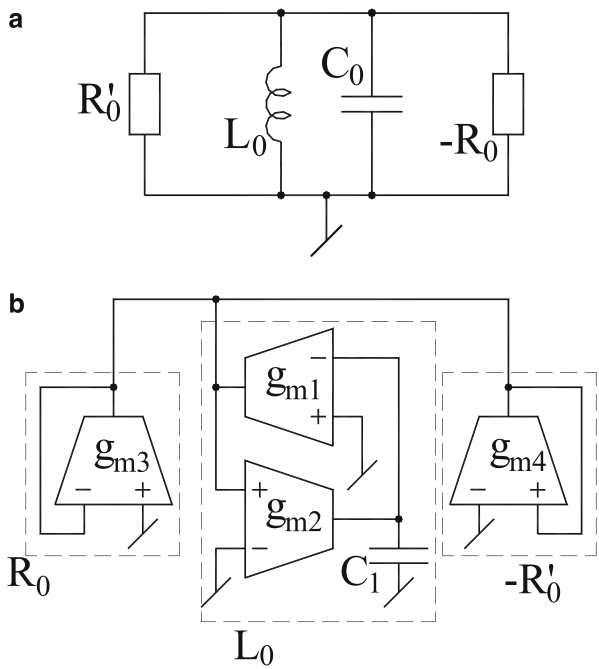


Fig. 3.3 (**a**) The parallel ±RLC resonator model. (**b**) An OTA-C oscillator based upon the model of Fig. 3.3(**a**)

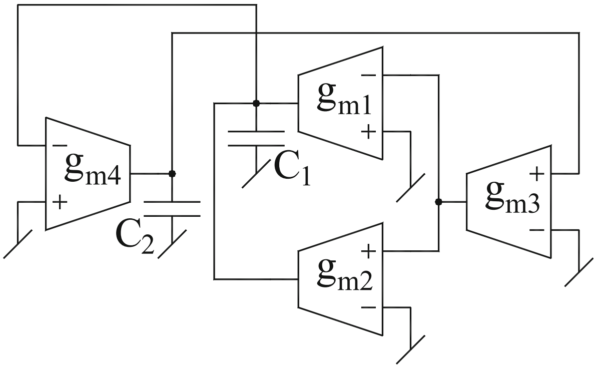


Fig. 3.4 A four-OTA-C sinusoidal oscillator with noninteracting controls [11]

capacitor C_1 represent the lossless grounded inductor having a value $L_0 = C_1/g_{m1}g_{m2}$, OTA$_3$ is configured as a positive resistor with $R_0' = \frac{1}{g_{m3}}$, and the OTA$_4$ configured as a negative resistance with $R_0 = \frac{-1}{g_{m4}}$. As a consequence, the resulting oscillator, as shown in Fig. 3.3b, is characterized by the following:

$$\mathrm{CO}: g_{m3} \le g_{m4} \tag{3.3}$$

$$\mathrm{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{g_{m1}g_{m2}}{C_1C_2}} \tag{3.4}$$

Another four-OTA-C sinusoidal oscillator providing noninteracting controls was presented in [11] which is shown in Fig. 3.4. The CO and FO of this circuit can be expressed by

$$\mathrm{CO}: g_{m1} \le g_{m2} \tag{3.5}$$

$$FO: f_0 = \frac{1}{2\pi}\sqrt{\frac{g_{m3}g_{m4}}{C_1C_2}} \tag{3.6}$$

It is thus seen that both CO and FO are independently adjustable: the former by g_{m1} or g_{m2} and the latter by g_{m3} and/or g_{m4}. Furthermore, if g_{m3} and g_{m4} are made equal by providing the same value of DC bias current in the respective OTAs, i.e., $I_{B3} = I_{B4} = I_B$, then f_0 is given by

$$f_0 = \frac{I_B}{4\pi V_T}\sqrt{\frac{1}{C_1C_2}} \tag{3.7}$$

Thus, the circuit provides a linear control of FO through the external bias current I_B and can be considered as a linear current-controlled oscillator (CCO).

It may be mentioned that this circuit, although not derived in this manner, was first proposed by Linares-Barranco, Rodriguez-Vazquez, Huertas, Sanchez-Sinencio, and Hoyle in [11]. Subsequently, Linares-Barranco, Rodriguez-Vazquez, Sanchez-Sinencio, and Huertas demonstrated [12] that by proper design, this configuration can be used to design oscillators implementable in CMOS technology for a frequency as high as 10 MHz. Furthermore, an alternative four-OTA and two-capacitor oscillator was proposed by Abuelma'atti and Almaskati [13], but their circuit suffered from the drawback of employing one floating capacitor.

Since only one four-OTA and two-grounded-capacitor oscillator (Fig. 3.4) was known with fully noninteracting linear control of CO and FO, Bhaskar and Senani [14] proposed three more four-OTA and two-capacitor sinusoidal oscillators which have the same features as in [11]. Out of these three circuits, one circuit is four-OTA and two-grounded-capacitor oscillator, while the other two structures employ four OTAs, one grounded, and one floating capacitor. These circuits are shown in Fig. 3.5. To derive these oscillator circuits, an alternative approach is highlighted to obtain CMOS implementations (using linear transconductor) which requires a considerably reduced number of components as compared to the traditional OTA-based approach applied in [11, 12]. The workability of CMOS versions of these oscillators was verified by PSPICE simulations with $L = 16$ μm, $W = 63$ μm for NMOS transistors and for PMOS transistors $L = 10$ μm, and $W = 100$ μm. The CMOS model parameters were selected to be the same as in Park and Schaumann [15].

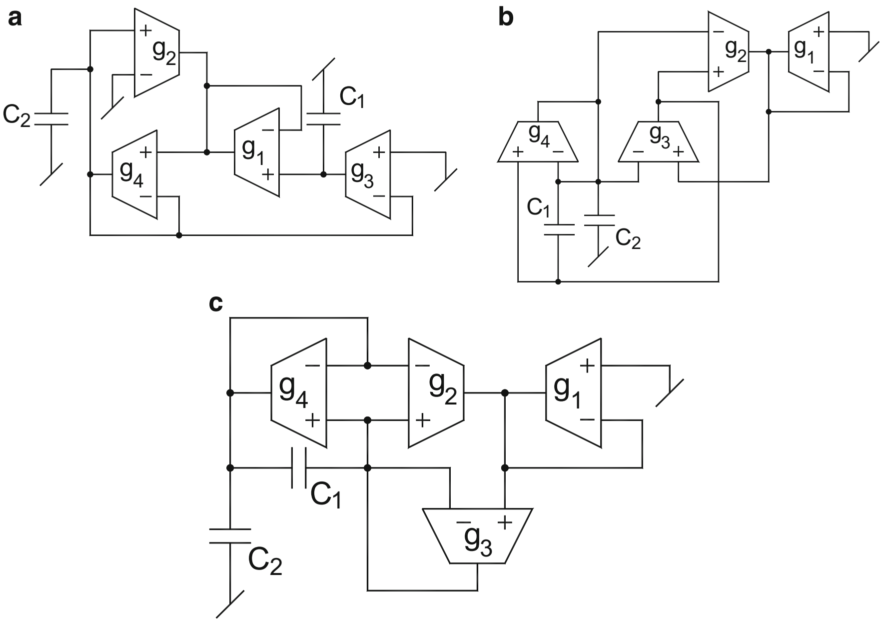


Fig. 3.5 (**a–c**) Four-OTA and two-capacitor sinusoidal oscillators providing fully noninteracting controls of CO and FO proposed by Bhaskar and Senani [14]

3.2.2 Three-OTA-C Oscillators

Since a minimal realization of an OTA-C oscillator capable of providing independent control of both CO and FO should be possible with only three OTAs, the first such oscillator circuits were proposed almost at the same time by Senani [16] and by Abuelma'atti [17]. These are shown here in Figs. 3.6 and 3.8, respectively. The CO and FO for the circuit of Fig. 3.6 (with $C_1 = C_2 = \mathrm{C}$) are given by

$$\mathrm{CO}:(g_{\mathrm{m3}} - g_{\mathrm{m2}}) = 0 \quad \text{and} \quad \mathrm{FO}: f_0 = \frac{1}{2\pi C}\sqrt{g_{\mathrm{m1}} g_{\mathrm{m3}}} \tag{3.8}$$

If the transconductance of the three OTAs is controlled simultaneously through a single external bias current $\mathrm{I_B}$ as shown in the arrangement of Fig. 3.7, then FO is reduced to $f_0 = \frac{g_{\mathrm{m}}}{2\pi C}$, and since $g_{\mathrm{m}i} = \frac{I_{\mathrm{B}i}}{2V_{\mathrm{T}}}$, $i = 1$–3. The oscillation frequency can be expressed as $f_0 = \frac{I_{\mathrm{B}}}{4\pi C V_{\mathrm{T}}}$. Therefore, f_0 is linearly programmable through an external current I_{B} without disturbing the CO.

The CO and FO for the oscillator structure of Fig. 3.8 can be obtained by

$$\mathrm{CO}:(g_{\mathrm{m2}} - g_{\mathrm{m1}}) \le 0 \quad \text{and} \quad \mathrm{FO}: f_0 = \frac{1}{2\pi C}\sqrt{g_{\mathrm{m1}} g_{\mathrm{m3}}} \tag{3.9}$$

Fig. 3.6 Three-OTA-two-grounded-capacitor oscillator proposed by Senani [16]

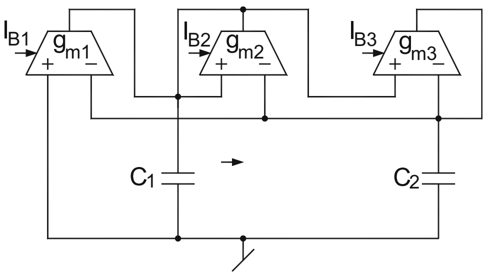
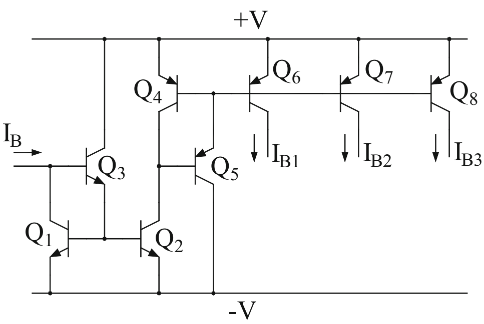


Fig. 3.7 Integratable circuit for the simultaneous control of three transconductance through single external current I_{B}



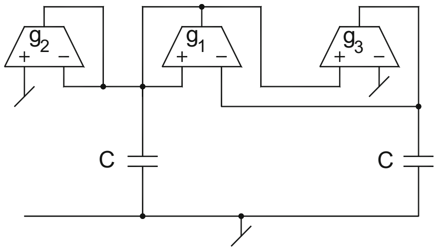


Fig. 3.8 Three-OTA-two-grounded capacitor oscillator proposed by Abuelma'atti [17]

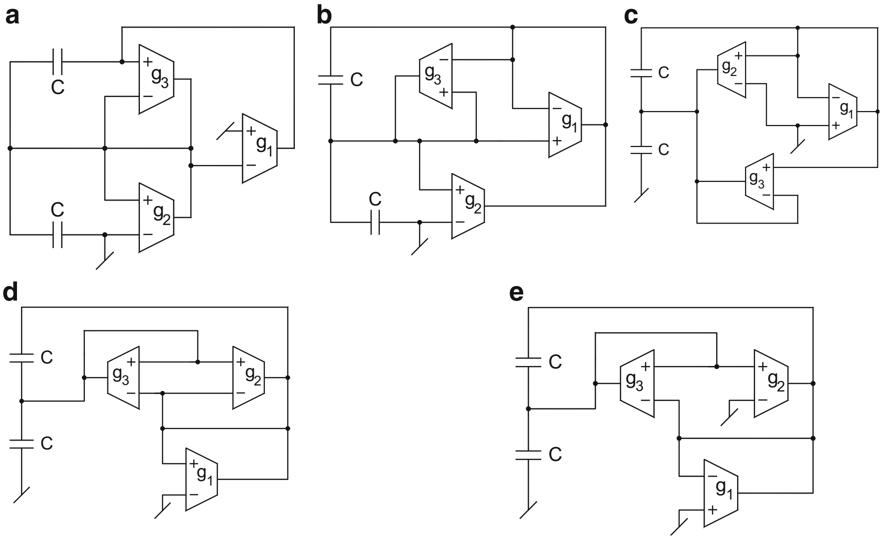


Fig. 3.9 (**a–e**) All possible oscillator circuits for one grounded and one floating capacitor with a common node between them

Subsequently, Senani and his group, through a systematic procedure, evolved the complete set of all possible canonic three-OTA-three-capacitor oscillators in [18–20]. Some circuits with only two OTAs were also proposed, but they require one resistor also which could be implemented with one (grounded) or two (when floating) resistors, thereby being equivalent to three-/four-OTA-based oscillators. Figure 3.9 presents all possible three-OTA-two-capacitor oscillators with noninteracting controls of CO and FO employing one grounded and one floating capacitor with a common node between them. The CO for the oscillator circuits shown in Fig. 3.9a and b is CO : $(g_{m1} - g_{m2}) \le 0$, while for the circuits of Fig. 3.9c–e, it can be expressed as $(2g_{m1} - g_{m2}) \le 0$. The FO for the oscillators shown in Fig. 3.9a, c–e is FO : $f_0 = \frac{1}{2\pi C}\sqrt{g_{m1}g_{m3}}$ and for Fig. 3.9b $f_0 = \frac{1}{2\pi C}\sqrt{g_{m2}g_{m3}}$.

Figure 3.10 shows all possible oscillator structures providing independent control of both CO and FO corresponding to one grounded and one floating capacitor

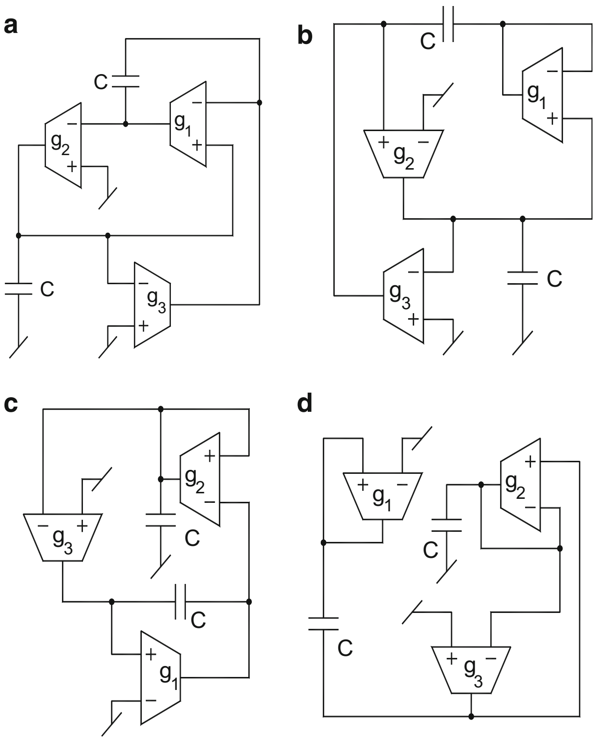


Fig. 3.10 (**a–d**) OTA-C oscillators employing one floating and one grounded capacitor having no common node between them

having no common node between them. The CO and FO for these configurations are given by

$$\mathrm{CO}:(g_{\mathrm{m1}}-g_{\mathrm{m3}})\le 0 \quad \text{and} \quad \mathrm{FO}: f_0=\frac{1}{2\pi C}\sqrt{g_{\mathrm{m2}}g_{\mathrm{m3}}} \tag{3.10}$$

OTA-C oscillator circuits with both capacitors floating having no common node between them are shown in Fig. 3.11 with the following CO and FO:

$$\mathrm{CO}:(g_{\mathrm{m2}}-2g_{\mathrm{m1}})\le 0 \quad \text{and} \quad FO: f_0=\frac{1}{2\pi C}\sqrt{g_{\mathrm{m2}}g_{\mathrm{m3}}} \tag{3.11}$$

The workability of all the oscillator structures has been verified experimentally employing CA3080 IC OTAs and programmable current sources for the bias current controls. It was found that the performance of oscillators of Figs. 3.6, 3.8, 3.9, and 3.11 has been better than those in Fig. 3.10. The oscillator circuits of Figs. 3.6 and 3.8 are particularly useful due to the employment of grounded capacitors which are advantageous for CMOS IC implementation and can absorb capacitive shunt parasitics [15].

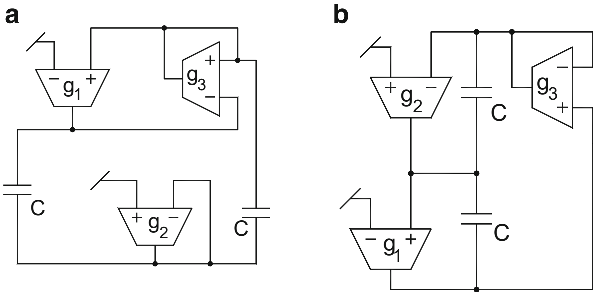


Fig. 3.11 (**a**, **b**) OTA-C oscillator circuits with both capacitor floating having no common node between them

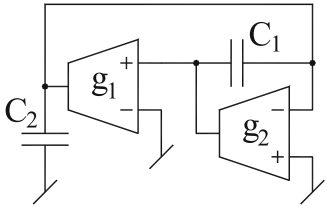


Fig. 3.12 Two-OTA-C oscillator proposed by Abuelma'atti [21]

3.2.3 *Two-OTA-C Oscillators*

Abuelma'atti [21] claimed to have reported two OTA-C oscillators each employing two OTAs and two capacitors only. One of these circuits is shown in Fig. 3.12. It has gone unnoticed in [21] that the second circuit therein is easily obtainable from the first one by renaming the two OTAs. Hence, in spite of apparently two circuits, there is, in fact, only one distinct circuit. The characterizing equations for the circuit shown in Fig. 3.12 are given by

$$\mathrm{CO}:(g_{\mathrm{m2}}-g_{\mathrm{m1}})\leq 0 \quad \text{and} \quad \mathrm{FO}:f_0=\frac{1}{2\pi}\sqrt{\frac{g_{\mathrm{m1}}g_{\mathrm{m2}}}{C_1C_2}} \tag{3.12}$$

3.2.4 *OTA-C Quadrature Oscillators*

Quadrature oscillators play an important role in many applications in instrumentation, signal processing, and communication systems. Ahmed, Khan, and Minhaj [22] introduced two electronically controllable quadrature oscillator circuits consisting of the cascade of an all-pass section and a non-inverting integrator. The first structure employs two OTAs, one buffer, and three capacitors which are shown in Fig. 3.13. The CO and FO of this quadrature oscillator are given by

$$\mathrm{CO}:\frac{g_1}{g_2}=\frac{C_1}{C_3} \quad \text{and} \quad \mathrm{FO}:f_0=\frac{1}{2\pi}\sqrt{\frac{g_1g_2}{C_1C_2}} \tag{3.13}$$

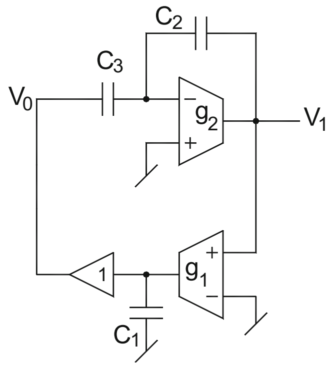


Fig. 3.13 An OTA-C quadrature oscillators proposed by Ahmed, Khan, and Minhaj [22]

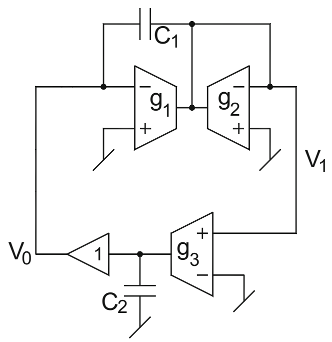


Fig. 3.14 Another OTA-C quadrature oscillators proposed by Ahmed, Khan, and Minhaj [22]

Thus, only CO is independently tunable through the capacitor C_3. The second circuit employing three OTAs, two capacitors, and a buffer is shown in Fig. 3.14. This oscillator structure is characterized by the following CO and FO:

$$\mathrm{CO}: \frac{g_2}{g_3} = \frac{C_1}{C_2} \quad \text{and} \quad \mathrm{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{g_1 g_2}{C_1 C_2}} \tag{3.14}$$

From Eq. (3.14), it is clear that the CO and FO are both independently tunable, the former by g_3 and the latter through g_1. Both the oscillators have been tested using CA3080 OTAs to establish the workability. The circuits were designed to provide the tunability over three decades (300 Hz to 300 kHz).

Since transconductance of the bipolar OTA is temperature dependent, quadrature oscillators proposed by Ahmed, Khan, and Minhaj [22] will be temperature sensitive. Although in [23] a bias circuit with a current linearly proportional to temperature was presented for the temperature compensation of OTA-based circuits, transconductance of the OTA can only be programmed accurately up to two decades only. To overcome above disadvantages, Kumwachara and Surakampontorn [24] proposed an integrable temperature-insensitive g_m–RC

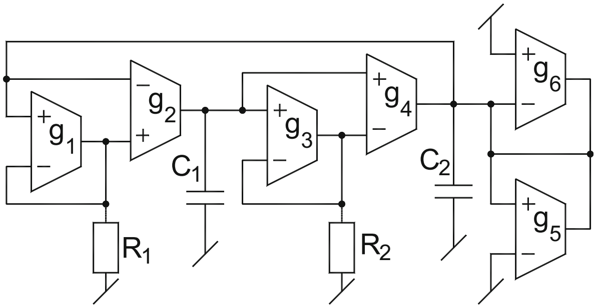


Fig. 3.15 Temperature-compensated g_m –RC quadrature oscillator proposed by Kumwachara and Surakampontorn [24]

quadrature oscillator employing bipolar OTAs with all grounded passive components. This oscillator is shown in Fig. 3.15. The CO and FO are given by

$$\text{CO}:(g_6-g_5)\leq 0 \quad \text{or} \quad (I_{B6}-I_{B5})\leq 0 \tag{3.15}$$

and

$$\text{FO}:f_0=\frac{1}{2\pi}\sqrt{\frac{g_2g_4}{C_1C_2R_1R_2g_1g_3}} \tag{3.16}$$

Now, if we choose $R_1=R_2=R$, $C_1=C_2=C$, $g_1=g_3=I_{B1}/2V_T$, and $g_2=g_4=I_{B2}/2V_T$, the expression for the frequency of oscillation will become

$$\omega_0=\frac{1}{RC}\left(\frac{g_2}{g_1}\right)=\frac{1}{RC}\left(\frac{I_{B2}}{I_{B1}}\right) \tag{3.17}$$

Thus, from Eqs. (3.15) and (3.17), it is obvious that CO can be independently established without disturbing FO which is controllable linearly through the DC bias current $I_{B2}=I_{B4}$. The performance of this circuit is temperature insensitive.

The circuit of Fig. 3.15 was bredboarded using bipolar OTAs from LM13600 with 1 % tolerance passive elements. The passive components used were $R_1=R_2=10\ \text{k}\Omega$, $C_1=C_2=0.5$ nF, $I_{B1}=I_{B3}=500$ μA, $I_{B2}=I_{B4}=785$ μA, $I_{B6}=105$ μA, $I_{B5}=100$ μA, and DC power supplies of ±10 V. Good-quality quadrature outputs were observed for 47 kHz while on simultaneous variation of bias currents $I_{B2}=I_{B4}$ from 1 to 1000 μA, FO could be tuned for three decades (64 Hz to 64 kHz).

Galan, Carvajal, Torralba, Munoz, Ramirez, and Angulo [25] presented a low-power low-voltage OTA-C quadrature voltage-controlled oscillator (VCO) with large tuning range (from 1 to 25 MHz) using a class-AB linear OTA based on the flipped voltage follower. This OTA-C VCO was fabricated in a standard 0.8 μm CMOS process using 2 V DC supply voltage. The amplitude of the generated waves was stabilized by the inherent transconductor nonlinear characteristic. The power consumption over the entire tuning range varied from 1.05 to 1.58 mW. This circuit is shown in Fig. 3.16.

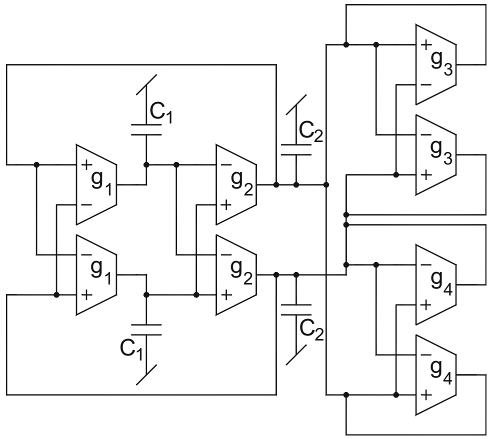


Fig. 3.16 OTA-C quadrature VCO proposed by Galan, Carvajal, Torralba, Munoz, Ramirez, and Angulo [25]

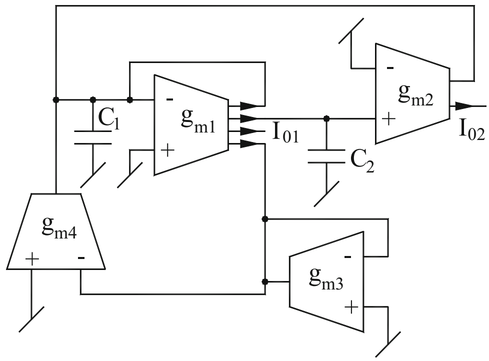


Fig. 3.17 Current-mode quadrature oscillator proposed by Summart, Thongsopa, and Jaikla [26]

In [26], an OTA-based current-mode quadrature sinusoidal oscillator with independent control of CO and FO was introduced by Summart, Thongsopa, and Jaikla. Their structure employs two simple OTAs, one dual output OTA, one four output OTA, and two grounded capacitors. The CO and FO of this quadrature oscillator shown in Fig. 3.17 are

$$\mathrm{CO}: g_{\mathrm{m3}} \leq g_{\mathrm{m4}} \quad \text{and} \quad FO: f_0 = \frac{1}{2\pi}\sqrt{\frac{g_{\mathrm{m1}} g_{\mathrm{m2}}}{C_1 C_2}} \tag{3.18}$$

The quadrature current outputs can be obtained as

$$\frac{I_{\mathrm{o2}}(s)}{I_{\mathrm{o1}}(s)} = \frac{g_{\mathrm{m2}}}{sC_2} \tag{3.19}$$

3.3 OTA-RC Oscillators

Whereas the major attention of researchers has been on the evolution of resistorless OTA-C oscillators as they combine the two important properties, namely, the electronic tunability of oscillation frequency and compatibility for IC implementation in both bipolar and CMOS technologies, nevertheless, many researchers have also investigated OTA-RC oscillator configurations which although appear to be deficient in respect of the suitability for integration due to the introduction of several resistors, however, due to the incorporation of multiple output type OTAs, in several instances, this class of oscillators has also been found to possess some interesting features. It is due to this reason that OTA-RC oscillators have also been investigated in literature and therefore, in this section, we highlight some prominent structures belonging to this class.

3.3.1 Two-OTA-RC Oscillators

The harmonic oscillator structure as shown in Fig. 3.18 employs one single-input single-output OTA, one single-input three-output OTA, two grounded capacitors, and one grounded resistor which was presented by Sotner, Jerabek, Petrzela, Dostal, and Vrba [27]. The CO and FO of this OTA-RC oscillator are given by

$$\text{CO}:(C_2 - C_1 g_{m2} R) \leq 0 \quad \text{and} \quad \text{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{g_{m2}(g_{m1}R - 1)}{C_1 C_2 R}} \tag{3.20}$$

This *on-chip integrable* oscillator can be implemented because resistance R can be simulated by OTA.

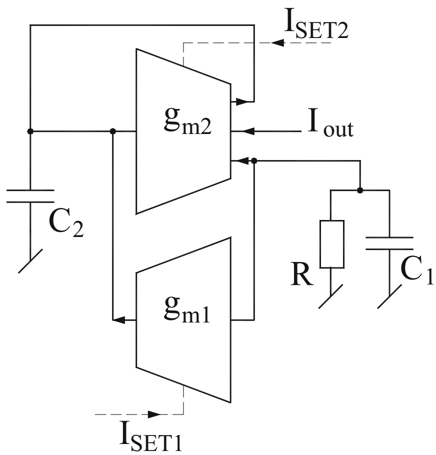


Fig. 3.18 OTA-RC oscillator proposed by Sotner, Jerabek, Petrzela, Dostal, and Vrba [27]

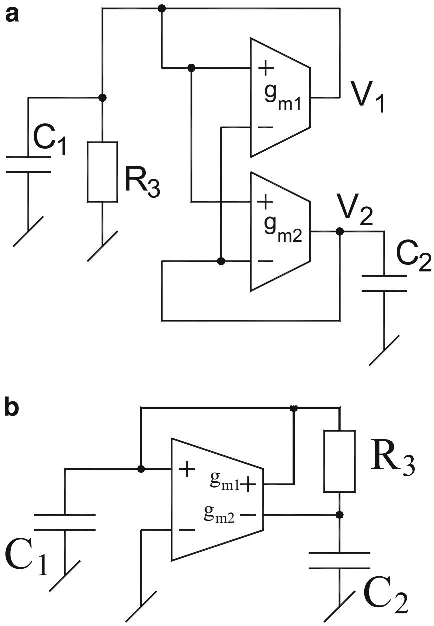


Fig. 3.19 (**a**) OTA-RC oscillator employing two single-output OTAs [28]. (**b**) OTA-RC oscillator employing single dual current output OTA [28]

3.3.2 Single-OTA RC Oscillators

In 2002, Chang and Liao [28] introduced two OTA-RC oscillator configurations which are shown in Fig. 3.19. The first OTA-RC oscillator as presented in Fig. 3.19a employs a single dual-output OTA, two grounded capacitors, and a grounded resistor. The CO and FO of this oscillator can easily be derived as

$$\text{CO}:\left[C_1 g_{\text{m2}} + C_2\left(\frac{1}{R_3} - g_{\text{m1}}\right)\right] \leq 0 \quad \text{and} \quad \text{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{g_{\text{m2}}}{C_1 C_2 R_3}} \tag{3.21}$$

The second OTA-RC oscillator uses one single dual-output OTA[1], two grounded capacitors, and one floating resistor as shown in Fig. 3.19b. This circuit is characterized by the following CO and FO:

$$\text{CO}:\left\{\left(\frac{C_1 + C_2}{R_3}\right) - C_2 g_{\text{m1}}\right\} \leq 0 \quad \text{and} \quad \text{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{g_{\text{m2}} - g_{\text{m1}}}{C_1 C_2 R_3}} \tag{3.22}$$

[1] This dual current output OTA was suggested to be implemented in [28] by using two single-output OTAs in parallel connection and that is how the same has been shown here in Fig. 3.19a

From Eqs. (3.21) and (3.22), it is seen that the oscillator of Fig. 3.19a qualifies for orthogonal control of CO and FO, while the circuit of Fig. 3.19b can provide independent and electronic control of FO.

HSPICE simulation results were given using LT 1228 OTA model (Linear-Tech Co.) biased with ±5 V DC power supplies. For the circuit of Fig. 3.19a, the components used were $C_1 = C_2 = C$and varied from 0.1 to 50 nF, $g_{m1} = 5$ mS, $g_{m2} = 4$ mS, and $R_3 = 1$ kΩ while for the oscillator of Fig. 3.19b, $C_1 = C_2 = 0.1$ nF, $R_3 = 1$ kΩ, $g_{m1} = 2$ mS, and g_{m2} were varied from 3–10 mS. It was possible to generate sinusoidal waveforms of frequency 1.5 MHz for the oscillator of Fig. 3.19a and a frequency of 3.18 MHz for the oscillator of Fig. 3.19b.

3.4 *Active-Only* OTA-Based Oscillators

In chapter 2 of this book, we have discussed a variety of oscillator circuits using op-amps such as active-RC, active-R, and active-C. The main problem with active-only capacitor-less and resistorless oscillator structures is that the CO is not controllable and FO can be adjusted through the variation of DC power supply. Therefore, active-only oscillator circuits which use only active elements and can provide independent control of both CO and FO are advantageous in respect of (1) their monolithic implementation in either bipolar or CMOS technologies, (2) programmability, or (3) wide range of operational frequency. Motivated with the development of such circuits, many researchers have contributed a number of oscillator structures of this kind. The first active-only sinusoidal oscillator was introduced by Abuelma'atti and Alzaher [29]. This oscillator circuit employs four OTAs and two op-amps and is shown in Fig. 3.20. Assuming that the open-loop voltage gain of an internally compensated op-amps is given by $A_v(s) = \frac{A_0\omega_p}{s+\omega_p} \cong \frac{B_i}{s}$;

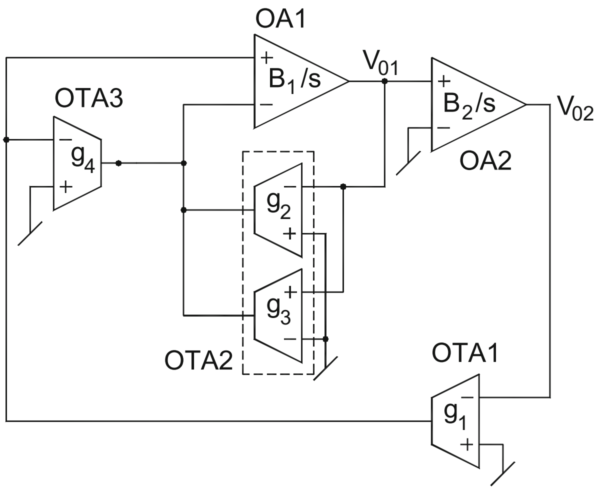


Fig. 3.20 Active-only oscillator proposed by Abuelma'atti [29]

$i = 1 - 2$ where $B = A_0\omega_{\mathrm{p}} =$ gain-bandwidth product of the op-amp, through a routine circuit analysis, the CO and FO for this configuration are obtained as

$$\mathrm{CO}: (g_2 - g_3) \leq 0 \quad \text{and} \quad FO: f_0 = \frac{1}{2\pi}\sqrt{\frac{g_1 B_1 B_2}{g_4}} \tag{3.23}$$

Thus, from the above equation, it is obvious that both CO and FO are electronically independently controllable and that the circuit is temperature insensitive. The other notable feature of this oscillator is that quadrature outputs are available from the nodes having output voltages V_{01} and V_{02} from two low output impedance nodes.

Abuelma'atti and Al-Qahtani [30] presented two more active-only sinusoidal oscillators which are shown in Fig. 3.21. Taking into account the input resistance R_{i}, the input capacitance C_{i}, the output resistance R_0, and output capacitance C_0, respectively, of the OTAs and similarly considering the input resistance Rx of the x-port, parasitic impedance Z_{y} (consisting $R_{\mathrm{y}} \| 1/sC_{\mathrm{y}}$) at y-terminal, and the parasitic impedance Z_{p} (consisting $R_{\mathrm{p}} \| 1/sC_{\mathrm{p}}$) at z-port for the CFOA, a straightforward analysis of the circuit of Fig. 3.21a reveals the following characteristic equation (CE):

$$Y_1 Y_3 (1 + R_{\mathrm{x}} Y_2) - g_4 Y_2 = 0 \tag{3.24}$$

$$\begin{aligned} &\text{where } Y_1 = \left(sC_1 + \frac{1}{R_1} + g_1\right), \quad Y_2 = \left(sC_2 + \frac{1}{R_2} - g_2\right) \text{ and} \\ &Y_3 = \left(sC_3 + \frac{1}{R_3} + g_3\right) \text{ and} \\ &C_1 = C_{01} + C_{04} + C_{i1} + C_y, \quad C_2 = C_{02} + C_{i2}, \quad C_3 = C_{03} + C_p + C_{i3} \\ &R_1 = R_{01}\|R_{04}\|R_y\|R_{i1}, \quad R_2 = R_{02}\|R_{i2} \text{ and } R_3 = R_{03}\|R_{i3}\|R_p \end{aligned} \tag{3.25}$$

now assuming $Y_2 R_x \ll 1$, $\frac{1}{R_1} << g_1$, $\frac{1}{R_2} << g_2$ and $\frac{1}{R_3} << g_3$; then, the CE can be expressed by the following equation:

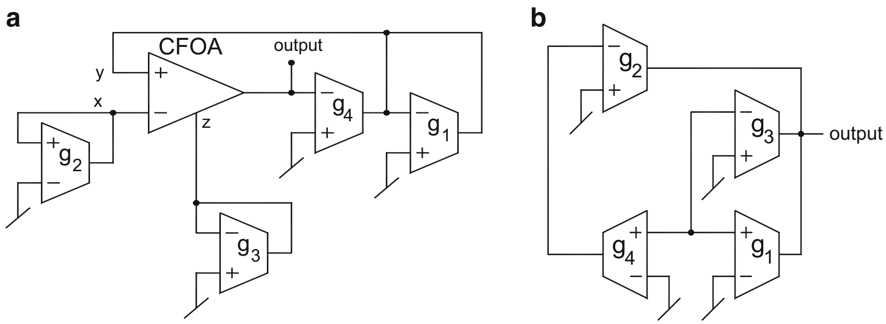


Fig. 3.21 (**a**, **b**) Modified active-only oscillators proposed by Abuelma'atti and Al-Qahtani [30]

$$s^2C_1C_2 + s(C_1g_3 + C_3g_1 - C_2g_4) + g_1g_3 + g_2g_4 = 0 \tag{3.26}$$

Thus, from Eq. (3.26), CO and FO can be given by

$$\mathrm{CO}: (C_1g_3 + C_3g_1 - C_2g_4) \leq 0 \tag{3.27}$$

$$\mathrm{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{g_1g_3 + g_2g_4}{C_1C_3}} \tag{3.28}$$

Therefore, from Eqs. (3.27) and (3.28), it is clear that CO can be controlled independently by C_2 and FO can be varied electronically by g_{m2}.

Similarly, the CE for the oscillator structure of Fig. 3.21b can be determined by the following equation:

$$\left(sC_2 + \frac{1}{R_2}\right)\left(sC_1 + \frac{1}{R_1} + g_3 - g_1\right) + g_2g_4 = 0 \tag{3.29}$$

$$\begin{gathered}\text{where } R_1 = R_{01}||R_{03}||R_{i1}||R_{i3}||R_{02}||R_{i4},\ \ R_2 = R_{04}||R_{i2} \\ C_2 = C_{04} + C_{i2} \text{ and } C_1 = C_{01} + C_{02} + C_{03} + C_{i1} + C_{i3} + C_{i4}\end{gathered} \tag{3.30}$$

again for $\frac{1}{R_1} << \ (g_3 - g_1)$, the CO and FO can be obtained as

$$CO: (g_3 - g_1) \leq 0 \quad \text{and} \quad FO: f_0 = \frac{1}{2\pi}\sqrt{\frac{g_2g_4}{C_1C_2}} \tag{3.31}$$

Thus, the circuit is capable of providing fully uncoupled CO and FO.

Both the oscillator configurations have been simulated using OTA macro model with $R_i = 65$ kΩ, $R_0 = 63$ MΩ, and $C_0 = 7.5$ pF and $R_x = 50$ Ω, $R_y = 10$ MΩ, $R_p = 3$ MΩ, and $C_p = 4.5$ pF for CFOA macro model. For the values of $g_1 = g_2 = g_3 = 0.1$ mS, $g_4 = 1.45$ mS, the circuit of Fig. 3.21a generates the sinusoidal waves of 2.742 MHz, while the second oscillator of Fig. 3.21b generates 77.6 MHz waves for $g_1 = g_2 = 6$ mS and $g_3 = g_4 = 7$ mS.

Figure 3.22 shows another OTA-based sinusoidal oscillator in which three OTAs, one grounded capacitor, and an op-amp are being employed by Abuelma'atti [31]. Assuming that the open-loop voltage gain ($A_1(s)$) of an internally compensated op-amp is approximated by $A_1(s) \cong \frac{B_1}{s}$, where B_1 is the gain-bandwidth product of the op-amp and ideal OTAs, then by routine circuit analysis, the CO and FO for this oscillator are obtained as

$$\mathrm{CO}: (g_2 - g_3) \leq 0 \quad \text{and} \quad \mathrm{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{g_1B_1}{C_1}} \tag{3.32}$$

Minaei and Cicekoglu [32] presented an active-only quadrature oscillator employing OTAs and op-amps. This circuit is shown in Fig. 3.23. Assuming

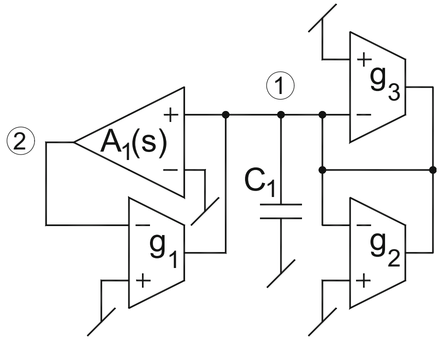


Fig. 3.22 OTA-based sinusoidal oscillator proposed by Abuelma'atti [31]

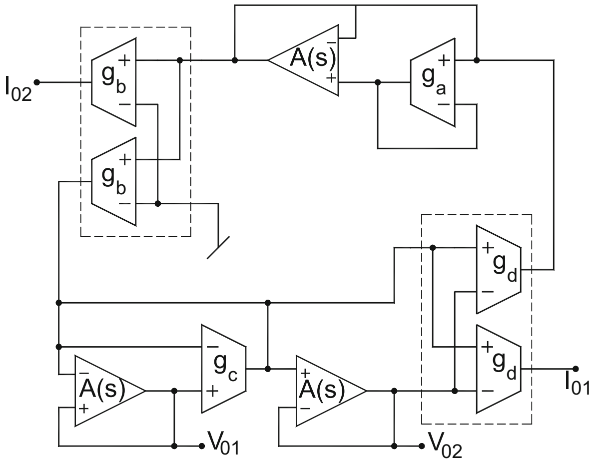


Fig. 3.23 Dual-mode quadrature oscillator proposed by Minaei and Cicekoglu [32]

ideal OTAs and the open-loop voltage gain of the op-amps $A(s) = B/s$, the CE of this quadrature oscillator is given by

$$s^2 g_a g_c + s(g_a g_c - g_b g_d)B + g_b g_d B^2 = 0 \tag{3.33}$$

Thus, from Eq. (3.33), CO and FO are given as

$$\text{CO} : (g_a g_c - g_b g_d) \leq 0 \quad \text{and} \quad \text{FO} : f_0 = \frac{B}{2\pi}\sqrt{\frac{g_b g_d}{g_a g_c}} \tag{3.34}$$

The two voltages V_{01} and V_{02} can be expressed as $\frac{V_{02}}{V_{01}} = -\frac{s-B}{s+B}$ and currents I_{01} and I_{02} by

$$\frac{I_{02}}{I_{01}} = \left(\frac{g_b}{g_a}\right)\left(\frac{B}{s}\right) \tag{3.35}$$

Thus, the circuit functions as dual-mode quadrature oscillator.

3.5 Electronically Controlled Current-Mode Oscillators Using MO-OTAs

Tsukutani, Sumi, and Fukui [33] proposed two configurations for OTA-C sinusoidal oscillators, each of which employed three multiple output OTAs (MO-OTA) and provided three explicit current outputs. These circuits are shown in Fig. 3.24. The CO and FO for these circuits are as follows:

$$\text{Circuit of Fig 3.24a} \quad \text{CO} : (g_1C_3 - g_3C_2) \leq 0 \quad \text{and} \quad FO : f_0 = \frac{1}{2\pi}\sqrt{\frac{g_1g_2}{C_1C_2}} \tag{3.36}$$

$$\text{Circuit of Fig 3.24b} \quad \left(\frac{g_2}{C_2} + \frac{g_1}{C_1} = \frac{g_3}{C_3}\right) \quad \text{and} \quad \text{FO} : f_0 = \frac{1}{2\pi}\sqrt{\frac{g_1g_2}{C_1C_2}} \tag{3.37}$$

It may be noted that one of these circuits, namely, that of Fig. 3.24b, does not have independent controllability of CO and FO, on the other hand, although both the circuits employ two grounded capacitors as desired for integrated circuit implementation. However, since both the circuits employ three capacitors, they are noncanonic.

Subsequently, Bhaskar, Abdalla, and Senani [34] proposed five such three OTA circuits (see Fig. 3.25) which, in contrast to the circuits of [33], use no more than two grounded capacitors. In addition, these circuits not only provide independent controls of both CO and FO but also provide quadrature outputs. It may be pointed that circuits providing quadrature outputs have numerous applications in communications for quadrature mixers and single-side band generators and in instrumentation for vector generators for selective voltmeters [35]. A straightforward analysis of these circuits reveals their CO and FO turnout to be as given in Table 3.1.

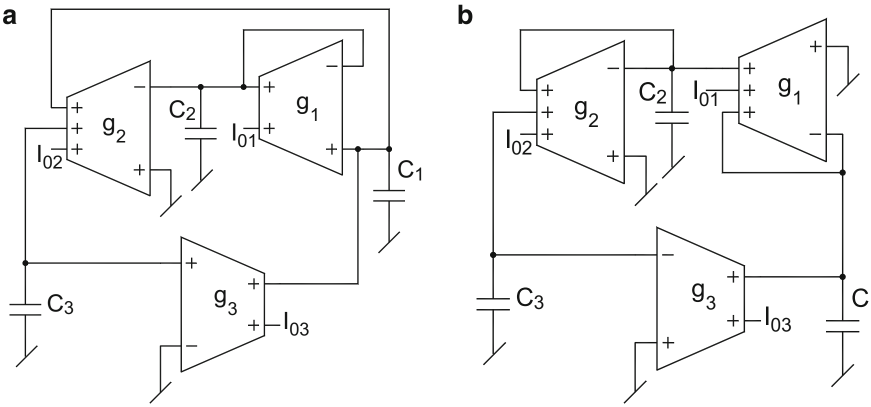


Fig. 3.24 (**a**, **b**) Current-mode oscillators employing MO-OTAs [33]

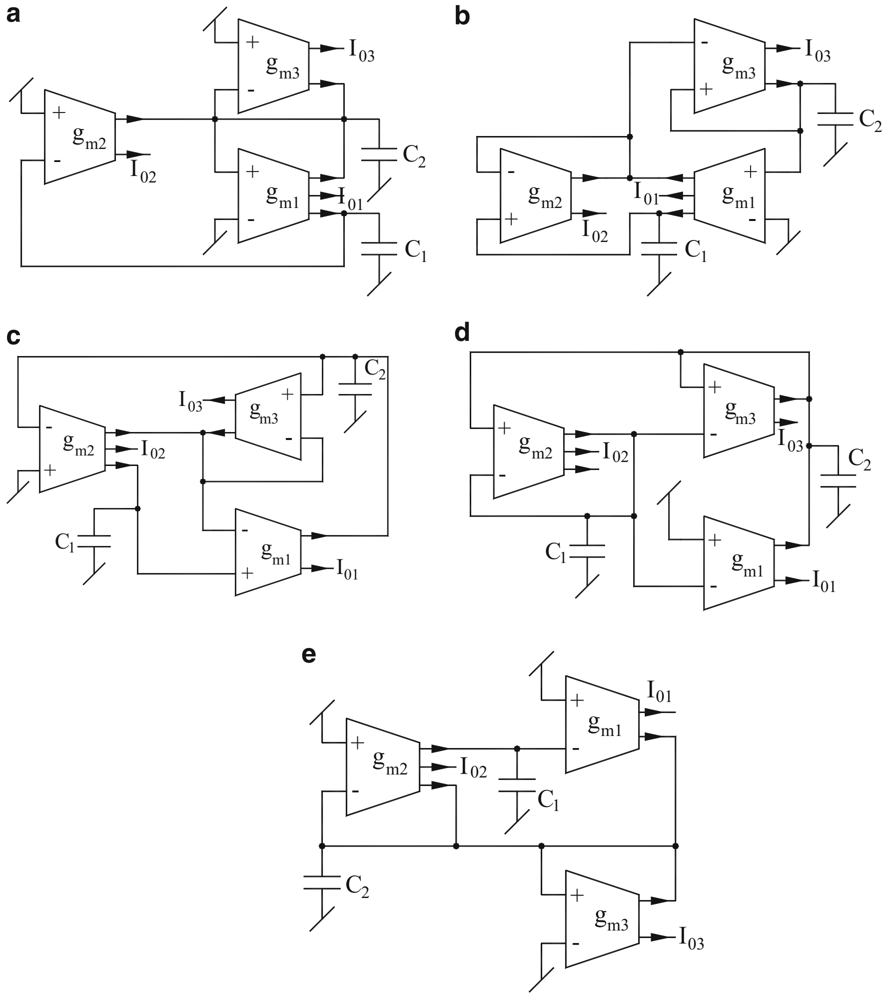


Fig. 3.25 (**a**–**e**) Current-mode oscillators using MO-OTAs proposed by Bhaskar, Abdalla, and Senani [34]

Table 3.1 CO, FO, and the relation between various outputs for the oscillators of Fig. 3.25

Circuit No.	CO	FO	Availability of quadrature outputs
Figure 3.25a	$(g_{m3} - g_{m1}) \leq 0$	$\frac{1}{2\pi}\sqrt{\frac{g_{m1}g_{m2}}{C_1C_2}}$	$\frac{I_{02}}{I_{01}} = -\frac{g_{m2}}{sC_2}, \frac{I_{02}}{I_{03}} = \frac{g_{m2}g_{m1}}{sC_2g_{m3}}$
Figure 3.25b	$(g_{m2} - g_{m1}) \leq 0$	$\frac{1}{2\pi}\sqrt{\frac{g_{m3}g_{m2}}{C_1C_2}}$	$\frac{I_{03}}{I_{01}} = \frac{g_{m3}}{sC_1}, \frac{I_{03}}{I_{02}} = -\frac{g_{m3}}{sC_1}$, for $g_{m1} = g_{m2}$
Figure 3.25c	$(g_{m1} - g_{m2}) \leq 0$	$\frac{1}{2\pi}\sqrt{\frac{g_{m3}g_{m2}}{C_1C_2}}$	$\frac{I_{03}}{I_{01}} = -\frac{g_{m3}}{sC_1}, \frac{I_{03}}{I_{02}} = \frac{g_{m3}}{sC_1}$, for $g_{m1} = g_{m2}$
Figure 3.25d	$(C_2g_{m3} - C_1g_{m1}) \leq 0$	$\frac{1}{2\pi}\sqrt{\frac{g_{m1}g_{m2}}{C_1C_2}}$	$\frac{I_{01}}{I_{02}} = -\frac{g_{m1}}{sC_2}$
Figure 3.25e	$(g_{m2} - g_{m3}) \leq 0$	$\frac{1}{2\pi}\sqrt{\frac{g_{m1}g_{m2}}{C_1C_2}}$	$\frac{I_{01}}{I_{02}} = -\frac{g_{m1}}{sC_1}, \frac{I_{01}}{I_{03}} = \frac{g_{m1}g_{m2}}{sC_1g_{m3}}$,

From Table 3.1, it may be seen that, except the circuit of Fig. 3.25d in which CO is a function of two transconductance as well as both the capacitors, in the remaining circuits, CO is governed by only two of the three transconductances whereas the FO is independently controllable by the remaining transconductance. Furthermore, whereas in circuits of Fig. 3.25b, c, quadrature signals are available subject to the equality of two transconductances, in the case of circuits of Fig. 3.25a, d, e, no such constraints are needed.

The validity of these configurations has been tested through SPICE simulations based upon CMOS OTAs using 0.5 μm process parameters.

3.6 CMOS Implementation of OTA-C Oscillators

Although all the OTA-C oscillators described in this chapter can be implemented in CMOS technology, most of the OTA-C oscillator circuits proposed by numerous researchers have used bipolar OTAs for their implementations. However, to implement entire complex system on a single chip, the use of CMOS or BiCMOS technology is desirable for their high-frequency operation and compatibility with CMOS digital circuits. From the available literature, it is found that only a few authors have dealt with explicit use of CMOS OTAs for implementing their propositions of OTA-C oscillators (see [12, 14, 32, 36]). In this section, therefore, some comments are now in order about CMOS OTA-C oscillators and CMOS transconductor-C oscillators.

While the usage of a single-output OTA was more prevalent in earlier literature, quite a number of authors subsequently found and demonstrated that a multiple output OTA (MO-OTA) provides more flexibility with more degrees of freedom (such as availability of multiple number of explicit outputs) in designing sinusoidal oscillators. A CMOS implementation of such a MO-OTA can be readily obtained by extending the internal architecture particularly by replicating the output of the CMOS current mirrors through current repeater structures; one such implementation is shown in Fig. 3.26.

It may be mentioned that an analogous architecture of an MO-OTA suitable for implementation in bipolar technology is not difficult to conceive. As an example, a bipolar OTA with three current outputs with very high output impedances can be easily obtained by employing current repeaters based upon Wilson current mirror as shown in Fig. 3.27.

Some authors have also attempted to bring simplicity in their designs in which case they have often resorted to the use of much simpler CMOS OTA architecture requiring no more than 6-MOSFETs (shown in Fig. 3.28) for which the transconductance (g_m) is given by $g_m = \sqrt{I_{ss}(\mu_0 C_{ox})\frac{W}{L}}$, where symbols have their usual meaning.

We have also described a number of active-only oscillators wherein an internally compensated type op-amp with a roll-off characteristic has been assumed.

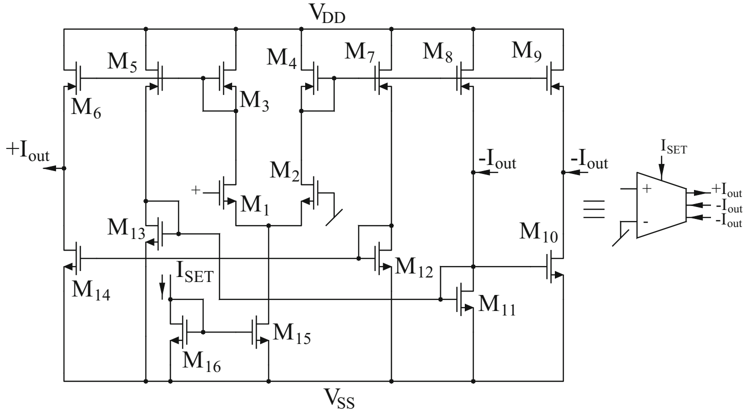


Fig. 3.26 CMOS implementation of MO-OTA

Fig. 3.27 A bipolar MO-OTA

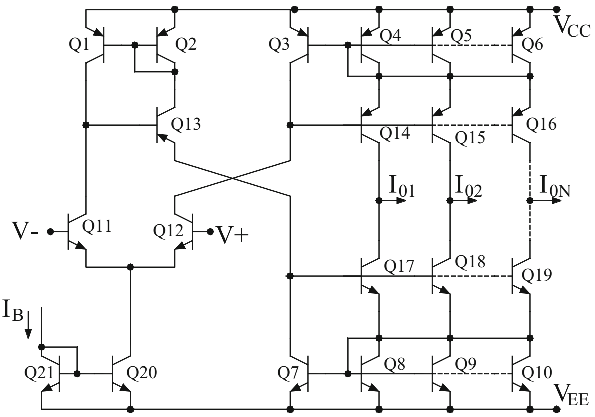


Fig. 3.28 A simple CMOS OTA

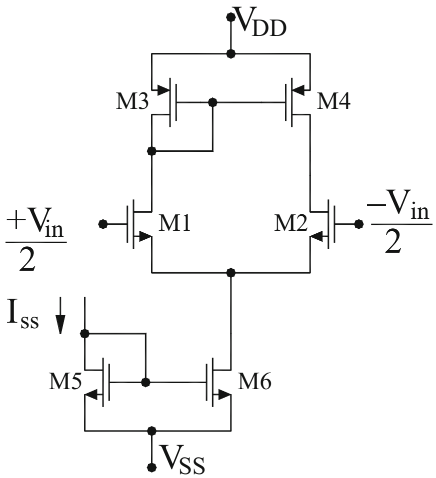
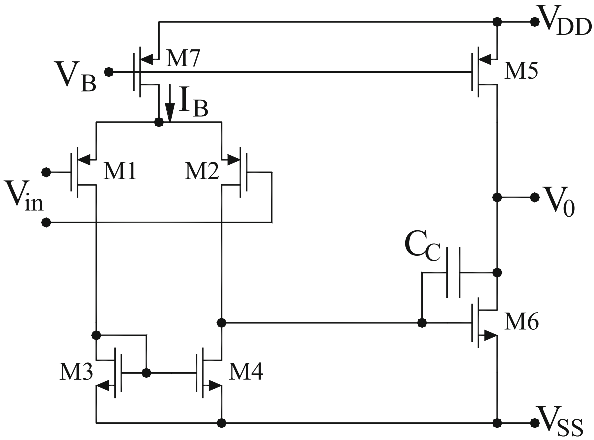


Fig. 3.29 A simple CMOS op-amp



The simplest circuit to design such a CMOS op-amp is shown in Fig. 3.29 which has been employed by several researchers (for instance, see [32]). On the other hand, there have been a number of efforts to devise high-performance bipolar and CMOS OTAs which attain certain desirable features even if at the cost of more number of MOSFETs.

In case of bipolar OTAs, two designs are particularly noteworthy [23, 80]. Surakampontorn, Riewruja, Kumwachara, and Fongsamut [23] proposed a simple and integrable temperature compensation scheme which can be applied on translinear CC-based circuits and also on OTAs. The proposed method uses a new bias circuit which has a current that is directly proportional to the absolute temperature and can also be varied electronically. The modified bipolar OTA then has a transconductance which depends upon the ratio of two currents and a resistor R used in the circuit and is therefore completely insensitive to any changes in the temperature.

Chung, Son, and Kim [80] have presented another temperature-stable linear bipolar OTA architecture, the details of which may be seen in [80].

On the other hand, a number of authors have developed improved CMOS OTA architectures using a variety of techniques and circuits; see [25, 36, 72].

In [12], the authors have used the linearized CMOS OTA configuration of Nedungadi and Geiger [36] which is shown in Fig. 3.30.

This OTA circuit consists of source-coupled NMOS pair $M_1 - M_2$ biased by a current proportional to the square of differential input voltage $V_{id} = (V_1 - V_2)$. This current is created by the cross-coupled structure $M_3 - M_6$ which is coupled with a level shifter M_7. Through proper selection of aspect ratios (W/L) of $M_1 - M_2$ and cross-coupled MOSFETs, the nonlinearities of the input stage get cancelled over a wide differential input voltage. MOSFETs M_8 to M_{31} are used to bias the input differential stage and to add device currents to determine the overall output current I_{out}. Assuming that the input devices are operated in saturation region and the current mirrors are of unity gain, the expression for I_{out} is given by

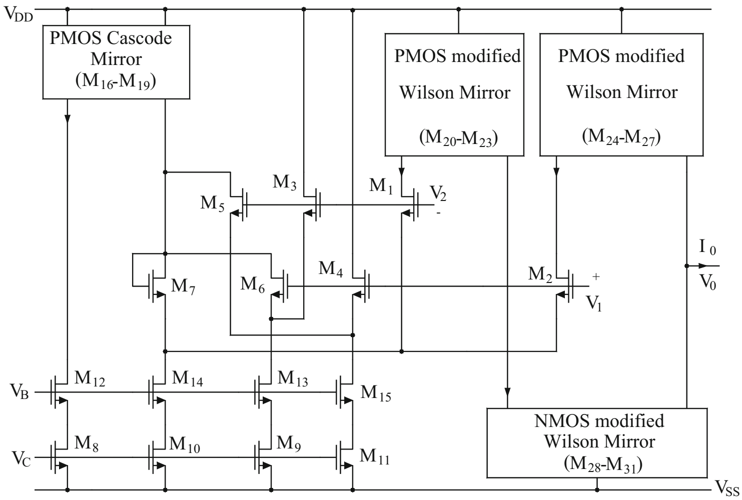


Fig. 3.30 Linearized CMOS OTA proposed by Nedungadi and Geiger [36]

$$I_{\text{out}} = g_{\text{m}} V_{\text{id}} = K(V_{\text{C}} - V_{\text{TN}})V_{\text{id}} \tag{3.38}$$

where K is a constant which depends on process parameters and the geometries of devices $M_1 - M_6$ and $M_8 - M_{11}$ and V_{TN} is the threshold voltage of NMOSFETS. The value of g_m from Eq. (3.38) can be obtained as

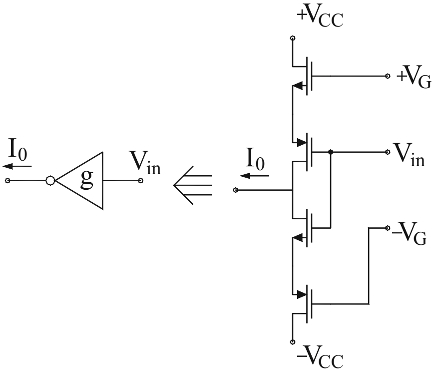
$$g_{\text{m}} = K(V_{\text{C}} - V_{\text{TN}}) \tag{3.39}$$

Thus, from Eq. (3.39), it is clear that g_m of this OTA is controllable through the DC control voltage V_C. The details of the design of differential input stage and the dynamic range of V_{id} over which i-v characteristics remain essentially straight lines are available in [37]. The OTA circuit of Fig. 3.30 was fabricated employing a standard 3 μm double-poly p-well CMOS process. The aspect ratios selected were $W/L = 2$ for $M_1 - M_2$ and $M_5 - M_7$, while for $M_3 - M_4$, the $W/L = 4$. The substrates and sources of all these MOSFETS were tied together. The rest of the PMOS transistors were in common p-well connected to negative DC power supply V_{SS}. The substrate of all PMOSFETS was connected to positive DC power supply V_{DD}. This fabricated OTA circuit occupies an area of $220 \times 700\ \mu m^2$.

Bhaskar and Senani [14] used a CMOS transconductor proposed by Park and Schaumann [15] to realize a number of linearly tunable CMOS-compatible OTA-C oscillators with noninteracting controls. This CMOS transconductor from [15] is shown in Fig. 3.31. The characterizing equation and the value of transconductance g[2] is given by

[2] For the detailed analysis of this transconductor, the reader is referred to [15].

Fig. 3.31 CMOS transconductor proposed by Park and Schaumann [15]



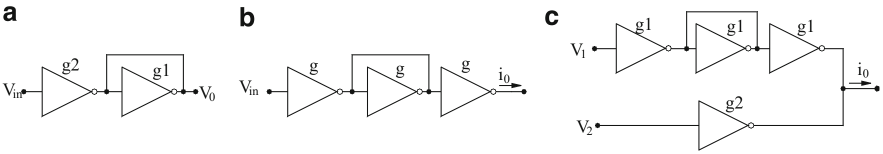


Fig. 3.32 Realization of basic circuits using transconductors

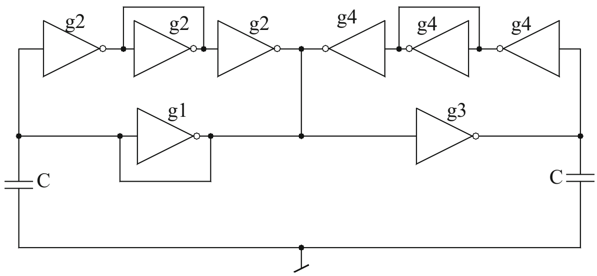
$i_0 = -gV_{\text{in}}$, where $g = 4k_{\text{eff}}\left\{V_{\text{G}} - \frac{1}{2}\left(V_{\text{TN1}} + V_{\text{TN3}} + \left|V_{\text{TP2}}\right| + \left|V_{\text{TP4}}\right|\right)\right.$ and $k_{\text{eff}} = k_{\text{N}}k_{\text{P}}/\left(\sqrt{k_{\text{N}}} + \sqrt{k_{\text{P}}}\right)^2$ and $k_{\text{N,P}} = \frac{1}{2}\left(\mu_{\text{eff}}C_{\text{OX}}\frac{W}{L}\right)\Big|_{\text{N,P}}$ and μ_{eff} are effective mobilities of electrons and holes and C_{OX} is the gate oxide capacitance per unit area and the symbols have their usual meaning.

Bhaskar and Senani [14] presented three four-OTA and two-capacitor-based oscillators providing noninteracting controls of CO and FO. It was then proposed that instead of using CMOS OTAs, if one uses instead a single-ended CMOS transconductor proposed by Park and Schaumann [15], this would result in a CMOS compatible oscillator with reduced number of components. The basic subcircuits needed for successful implementation of this approach are the inverting amplifier (1), the non-inverting transconductor (2), and the differential transconductor (3) whose realization in terms of the basic inverting transconductor is shown in Fig. 3.32.

The authors [14] presented three new OTA-C oscillators; one of which had the attractive features of employing both GCs as preferred for integrated circuit implementation. Using the single-ended transconductor-based building blocks of Fig. 3.31, a transconductor-based realization of the quoted 4-OTA-C oscillator was obtained which is shown in Fig. 3.33.

It may be noted that this realization would be employing only 32 MOSFETs which is a smaller number as compared to the total number of transistors required in 4-OTA-C oscillator.

Fig. 3.33 Realization of an oscillator using transconductors by Bhaskar and Senani [14]



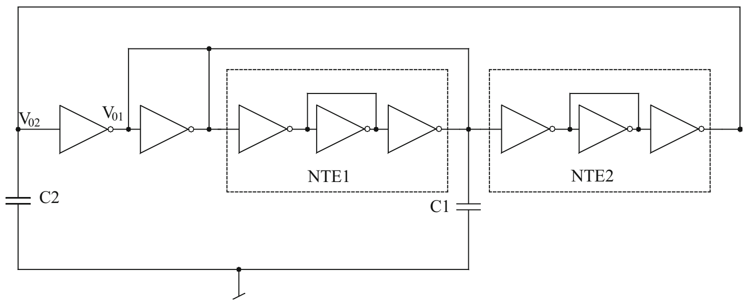


Fig. 3.34 Quadrature oscillator using transconductors by Khan, Ahmed, and Minhaj [68]

Subsequently,[3] about eight years later, Khan, Ahmed, and Minhaj [68] also employed exactly the same CMOS transconductors and proposed another eight transconductors-two GC oscillator which is shown in Fig. 3.34.

Other than the circuits presented here, a variety of other CMOS OTA architectures, CMOS op-amps (with one-pole roll-off characteristic), as well as CMOS transconductors are available in literature which can be employed for implementation of the OTA-C oscillators described here, in CMOS technology.

3.7 Concluding Remarks

This chapter has elaborated an important class of electronically tunable oscillator based upon OTA-C technique and some of its variants. This class of oscillators is important because of the electronic tunability as well as their suitability for complete integration both in bipolar and CMOS technology. We have presented only some representative circuits in various categories such as OTA-C oscillators,

[3] It is ironical that Khan, Ahmed, and Minhaj [68] in their publication do not mention [14] at all where this approach was first proposed in literature.

OTA-RC oscillators, and external capacitor-less OTA-based oscillators employing op-amp and OTA combinations with op-amp pole used as design parameter.

Although a large number of OTA-based circuits have been proposed in literature, the large repertoire of circuits [8–82] still appears to have left several questions unanswered. For instance, no study has yet conclusively been established as to which one of the four OTA-C oscillators or which one of the three OTA-C oscillators is the best from the viewpoint of practical performance criteria such as maximum operating frequency range, percentage THD, power consumption, minimum possible power supply voltage acceptable, frequency stability properties, etc. Therefore, an investigation into this aspect appears to be a worthwhile problem for investigation. Furthermore, the methods of systematic synthesis of OTA-C oscillators using pathological models have only recently started being investigated [77] and therefore constitute another worthwhile area for further investigations.

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Chapter 4
Sinusoidal Oscillators Using Current Conveyors

Abstract This chapter discusses the evolution of sinusoidal oscillators using the new active building blocks known as Current Conveyors by Sedra and Smith in 1968–1970. Although fixed frequency oscillators as well as variable frequency oscillators both have been discussed, a major emphasis has been on the so-called single-element-controlled oscillators. A large variety of CC-based sinusoidal oscillators have been evolved so far which include both single CC-based as well as two or more CC-based oscillators, including quadrature and multiphase oscillators. Oscillators employing all grounded passive elements as well as those providing explicit current output have also been discussed.

4.1 Introduction

This chapter is concerned with the realization of oscillators using CCs a wide variety of which have been advanced during the last 35 years or so [1–62].

In Chaps. 1 and 2 of this monograph, we deal with op-amp based sinusoidal oscillators. In this context, it is understood that with the advent of the integrated circuit (IC) op-amps, it was a logical development to devise op-amp-based configurations for realizing a number of classical sinusoidal oscillator topologies such as Wien bridge oscillator, RC-phase shift oscillator and twin-T oscillators etc. However, a common difficulty with the mentioned topologies was that they could not provide variable frequency oscillations by varying a single variable element. Hribsek and Newcomb were the first to propose a single-resistance-controlled oscillator intended to realize a voltage-controlled oscillator using two IC op-amps [63]. Since then, researchers soon directed attention towards devising new topologies for single-element-controlled oscillators (SECO)/ and single-resistance-controlled oscillators (SRCO) employing a single op-amp along with a canonical number of passive components, namely only two capacitors and at most five resistors. A number of such single-op-amp SRCOs have since been discovered, a detailed account of which has already been given in Chap. 2.

Since the introduction of the current conveyors as alternative and attractive building blocks for analog circuit design, a number of research groups around the

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world also started looking for the possibility of realizing sinusoidal oscillators using CCs. It was found that a canonic SECO/SRCO using a single CC can be realized with only five passive components, namely either three resistors or two capacitors or with three capacitors and two resistors. Furthermore, researchers also focused on realization of quadrature and multiphase sinusoidal oscillators, however, the research on these is still continuing. Because of the flexibility offered by variable frequency oscillators, this chapter focuses only on SECOs/SRCOs realizable with the basic types of current conveyors, namely CCI± and CCII± only.

In the subsequent sections of this chapter, we focus on the single-CC-based SECOs/SRCOs and a number of other varieties of sinusoidal oscillator configurations evolved over the past 35 years [1–62], [64–90] using the basic three-terminal types of CCs.

4.2 Single-CC SRCOs

A number of early current conveyor based oscillators were based upon classical Wien bridge oscillator and its variants using op-amp for instance see Refs. [1, 24, 35, 37]. Martinez–Celma–Gutierrez [24] described a number of Wein type of oscillators using CCII+, out of which two interesting circuits each employing only a single CCII+, three resistors and two capacitors are shown in Fig. 4.1.

The CO and FO of these oscillator circuits are given by the following expressions:

For the Fig. 4.1a

$$\mathrm{CO}:\left(1+\frac{C_2}{C_1}\right)=R_1\left(\frac{1}{R_2}+\frac{1}{R_\mathrm{A}}\right)\quad\text{and}\quad \mathrm{FO}:\omega_0=\frac{1}{\sqrt{R_1C_1C_2R_2}}\tag{4.1}$$

For the Fig. 4.1b

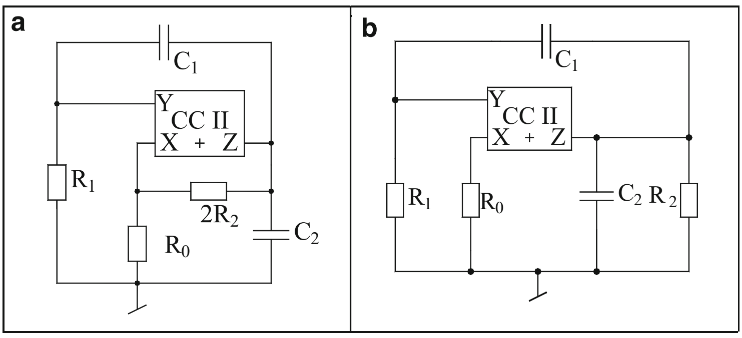


Fig. 4.1 (**a**, **b**) Wein-type oscillators using CCII+ proposed by Martinez–Celma–Gutierrez [24]

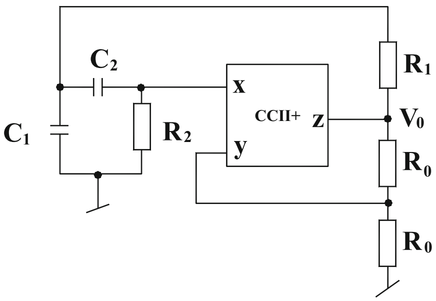


Fig. 4.2 Soliman's single-capacitor-controlled oscillator [3]

$$\left(1+\frac{C_2}{C_1}\right)=R_1\left(-\frac{1}{R_2}+\frac{1}{R_A}\right) \quad \text{and} \quad \text{FO}: \quad \omega_0=\frac{1}{\sqrt{R_1 C_1 C_2 R_2}} \tag{4.2}$$

Subsequent to the development of CC-based Wien bridge oscillators, a number of other single-CC based single frequency [11, 12, 15, 20] as well as variable frequency [3–10, 13, 16, 19, 20][1] oscillators were reported in literature.

In the following, we discuss mainly variable frequency oscillators capable of providing frequency control through a single-variable-element (a capacitance or resistance).

Soliman [3] presented a single CCII+ based oscillator whose frequency of oscillation (FO) is controllable through a single grounded capacitor (GC) without disturbing the condition of oscillation (CO). This circuit is shown in Fig. 4.2. The CO and FO of this configuration are given by:

$$\text{CO}: \frac{1}{R_2}=\frac{1}{R_0}+\frac{2}{R_1} \tag{4.3}$$

$$\text{FO}: \quad \omega_0=\frac{1}{R_1}\sqrt{\frac{2}{C_1 C_2}} \tag{4.4}$$

Thus, CO can be controlled by R_2 and FO then also can be tuned by C_1 or C_2 independently. Although not explicitly pointed out therein, this circuit is obtained from an active gyrator circuit also proposed by the same author [65] by terminating the input port of the gyrator into a GC which becomes the frequency controlling capacitor.

In the same year, Soliman also proposed [4] another single CCII+ based circuit which could realize an ideal FDNR, a lossy FDNR, a parallel tuned circuit or an oscillator from the same structure subject to the fulfillment of appropriate conditions in terms of RC elements used in the circuit. This circuit is shown in Fig. 4.3.

[1] Apart from a single CC, this circuit, however, also incorporates an OTA.

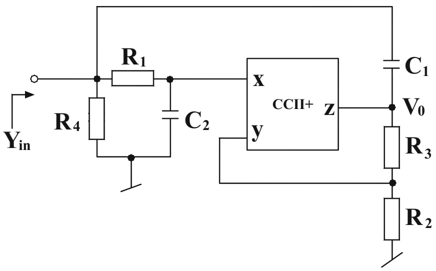


Fig. 4.3 A canonic active-RC circuit capable of realizing an ideal FDNR, a lossy FDNR, parallel tuned circuit and an oscillator proposed by Soliman [4]

When designed a sinusoidal oscillator, the CO and FO for this circuit are given by:

$$\text{CO}: \quad C_2R_2\left(1+\frac{R_1}{R_4}\right) - C_1R_1\left(1+\frac{2R_3}{R_1}+\frac{R_2+R_3}{R_4}\right) = 0 \tag{4.5}$$

$$\text{FO}: \quad f_0 = \frac{1}{2\pi C_1}\sqrt{\left\{\frac{\frac{1}{R_4}-\frac{1}{R_3}}{R_1\left(2+\frac{R_1}{R_3}\right)}\right\}} \tag{4.6}$$

Subject to the fulfillment of the following conditions:

$$C_2 = C_1\left(1+\frac{R_3}{R_2}\right), \quad R_2 = (R_1+R_3) \quad \text{and} \quad R_3 > R_4 \tag{4.7}$$

If the component values are chosen such that $R_1 = R_3$, $R_2 = 2R_1$, $R_1 > R_4$, and $C_2 = (3/2)C_1$, then oscillation frequency is given by

$$f_0 = \frac{1}{2\pi C_1R_1}\sqrt{\left\{\frac{1}{3}\left(\frac{R_1}{R_4}-1\right)\right\}} \tag{4.8}$$

Thus, f_o can be controlled by varying a single grounded resistance R_4. FO can also be controlled by an external voltage if the frequency controlling resistor is replaced by an FET working in the triode region. However, the circuit does not provide independent control of CO through a single variable resistance.

A single-CC SRCO which does not have the above mentioned limitation was proposed by Senani [5]. This circuit is shown here as Fig. 4.4. Subject to the fulfillment of the condition

$$\frac{C_1}{C_2} = 1 + R_3\left(\frac{1}{R_1}+\frac{1}{R_2}\right) \tag{4.9}$$

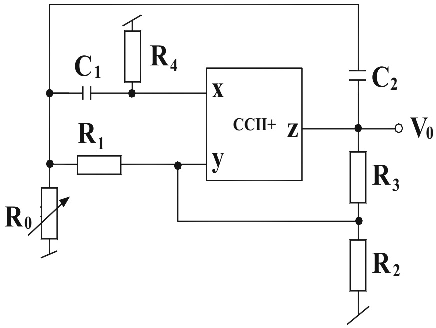


Fig. 4.4 Senani's single-resistance-controlled oscillator. [5]

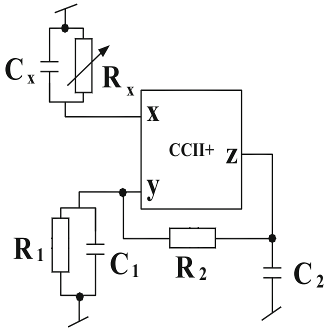


Fig. 4.5 The SRCO proposed by Jana and Nandi [6]

the CO and FO for this circuit are:

$$CO : R_4 = R_2 \tag{4.10}$$

$$\text{FO} : \omega_0 = \left(\frac{R_2}{2C_1C_2R_1R_3R_0} \right)^{1/2} \tag{4.11}$$

Thus, in this circuit, CO can be satisfied by adjusting the grounded resistor R_4 (without effecting FO) while the FO is also independently controllable by another single variable grounded resistance R_0 (which does not appear either in Eqs. (4.10) or (4.11)).

It is well known that in integrated circuit realization, it is desirable to have all the capacitors grounded as this not only results in the simplification of the fabrication process but also helps in absorbing the parasitic capacitances of the active device into the external capacitors. While the previously described single CC SECOs/SRCOs do not have both the capacitors grounded, a single-CC SRCO, with all grounded capacitors (although three in numbers), was presented by Jana and Nandi [6] and is shown in Fig. 4.5.

The CO and FO for this circuit from Ref. [6] are given by:

$$C_{\mathrm{x}} = C_2\left(1 + \frac{R_2}{R_1}\right) + C_1 \tag{4.12}$$

$$\omega_0 = \sqrt{\frac{R_x - R_1}{R_1 R_2 R_x C_1 C_2}} \tag{4.13}$$

Thus, the capacitance C_{x} can be used to set the circuit to produce sinusoidal oscillations whereas the oscillation frequency can be varied by varying R_{x} subject to maintaining $R_{\mathrm{x}} > R_1$. Chong and Smith [7] carried out a detailed evaluation of the effects of non-zero offset current at terminal-Z of the CCs on the operation of CC-based sinusoidal oscillators. It was revealed that to reduce the effect of non-zero output offset current, the oscillator circuit proposed by Jana and Nandi [6] needs to be modified by removing the resistor R_1 connected across C_1 and instead of this, putting another resistance R_0 in parallel with the capacitor C_2. With this modification, CO and FO get modified to:

$$\mathrm{CO}: C_{\mathrm{x}} = C_1\left(1 + \frac{R_2}{R_0}\right) + C_2 \tag{4.14}$$

$$\mathrm{FO}: \omega_0 = \sqrt{\frac{R_{\mathrm{x}} - R_0}{R_0 R_2 R_x C_1 C_2}} \tag{4.15}$$

It is worth pointing out that although one still requires R_{x} to be larger than $R_0$for low frequency oscillations, it is possible to choose R_2 to be very large without affecting the DC offset voltage at terminal-Z. This is an advantage of the suggested modification in the oscillator of Jana and Nandi [6].

After these intuitive propositions, a number of systematic studies were conducted by several researchers such as by Abuelma'atti and Humood [8–10] and Celma, Martinez, and Carlosena [16] to derive complete set of single-CC-based canonic SECOs/SRCOs. It was eventually found that only five passive components are really necessary along with a single CC, to make either a single-capacitor-controlled oscillator (SCCO) or SRCO; two resistors and three capacitors for the former and two capacitors and three resistors in case of the latter. An exemplary SCCO/SRCO (both derivable from each other through RC:CR transformation) is shown in Fig. 4.6. The SRCO of Fig. 4.5 was independently devised by Abuelma'atti and Humood [10] as well as by Celma, Martinez, and Carlosena [16].

The CO and FO for this circuit are given by

$$\frac{R}{R_2} = 1 + \frac{C_2}{C_1} \tag{4.16}$$

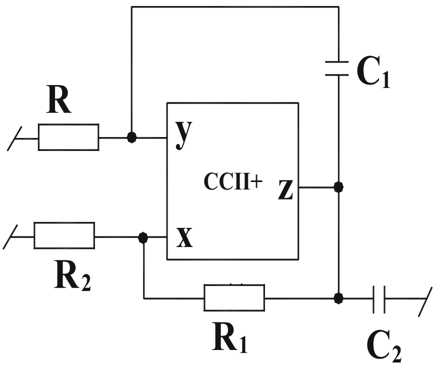


Fig. 4.6 Canonic SRCO using a CCII+ proposed by Abuelma'amatti and Humood [10] and also by Celma, Martinez, and Carlosena [16]

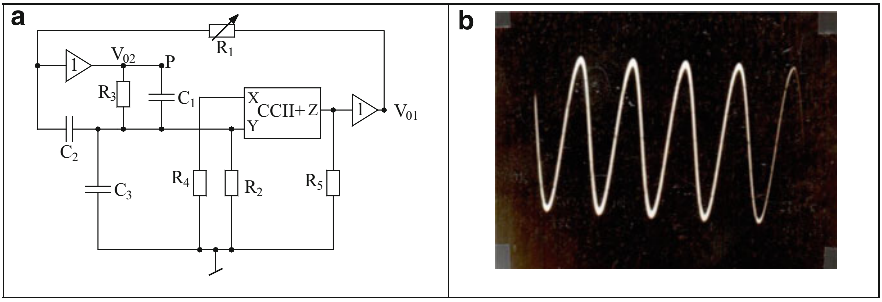


Fig. 4.7 (**a**) SRCO realization using single IC PA630 proposed by Senani and Singh [14] (**b**) A typical wave form generated by the circuit; $f_0 = 312$ kHz

and

$$\omega_0 = \sqrt{\frac{2}{C_1 C_2 R R_1}} \tag{4.17}$$

from where it is seen that CO is controllable by R_2 whereas FO is independently adjustable by R_1.

A CCII based SRCO suitable for implementation with the commercially available IC CC PA 630 was presented by Senani and Singh [14] which is shown here in Fig. 4.7. The expressions for CO and FO for this circuit are derived as:

$$\text{CO}: \left(1 - \frac{R_4}{R_5}\right) \leq 0 \quad \text{provided} \quad R_3(C_2 + C_3) = C_1 R_2 \tag{4.18}$$

$$\omega_0 = \frac{1}{\sqrt{R_1 C_2 C_3 R_3}} \tag{4.19}$$

The workability of this circuit was verified using PA630 CCs with component values selected as $C_1 = C_2 = C_3 = 1$ nF (and 470 pF), $R_2 = 2$ kΩ, $R_3 = 1$ KΩ, $R_1 =$ variable from 1 to 100 kΩ for generating $f_0 = 312$ kHz. By varying R_1 in the quoted range f_0 was found to vary as predicted by theory. A typical wave form generated by this circuit is shown in Fig. 4.7b.

This circuit has the following advantageous features: (1) since PA 630 contains one CCII+ and two on chip voltage buffers, the circuit is implementable by a single IC (2) non-interacting single resistance control of FO and (3) excellent frequency stability properties (the S_F is found to be $\sqrt{n}$ where n is the frequency controlling resistor ratio).

4.3 SRCOs Employing Grounded Capacitors

Sinusoidal oscillator configurations employing all grounded capacitors are attractive not only from the view point of IC implementation but also due to the fact that they can easily accommodate the parasitic capacitances of the active devices employed. It is worth pointing out that using the traditional op-amps, a number of circuits are known in literature which employ three/two op-amps along with two grounded capacitors to realize SRCOs. However, similar circuits with a single op-amp are although possible but the required design constraints are very complex and so are the resulting tuning laws.

During the research on CC-based oscillators it was found that the current conveyors outperform the op-amps in this respect in that not only such circuits can be realized using CCs while avoiding complex design equations but also the resulting circuits employ invariably the least possible number of passive components. We highlight some exemplary circuits exhibiting these features from amongst many such circuits advanced by various researchers, for instance, [2, 14, 18, 31, 42, 44, 48, 52, 55, 63].

In this section, first we present two minimum-component oscillators which employ only two CCIIs, only two resistors and both grounded capacitors. Such circuits were proposed by Horng, Chang, and Lee [31] and are shown in Fig. 4.8. Both the circuits are characterized by the same characteristic equation and, therefore, have the same CO and FO which are given respectively by:

$$s^2 C_1 C_2 R_1 R_2 + s R_1 (C_2 - C_1) + 1 = 0 \tag{4.20}$$

$$\mathrm{CO} : C_1 = C_2 \tag{4.21}$$

and

$$\mathrm{FO} : \omega_0 = \sqrt{\frac{1}{C_1 C_2 R_1 R_2}} \tag{4.22}$$

In these circuits, the only point of inconvenience of adjusting the CO through a variable capacitor, which is not a very convenient option.

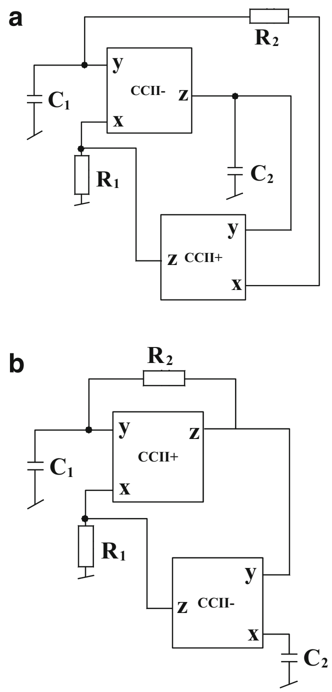


Fig. 4.8 (**a**, **b**) Minimum-component oscillators proposed by Horng, Chang, and Lee [31]

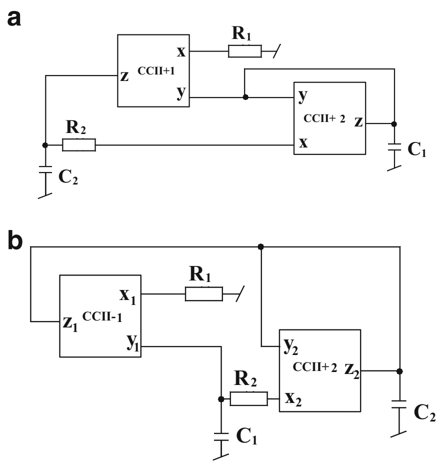


Fig. 4.9 Grounded capacitors oscillators (**a**) Proposed by Horng [42] (**b**) Proposed by Fongsamut, Anuntahirunrat, Kumwachara, and Surakampontorn [48]

Two other circuits, using exactly the same number of active and passive components and also governed by exactly the same CO and FO are shown in Fig. 4.9a, b and were proposed by Horng [42] and Fongsamut, Anuntahirunrat, Kumwachara, and Surakampontorn [48] respectively.

CCIIs as α_1 and α_2 and the voltage gains as β_1 and β_2, the nonideal expressions for the FO of the two circuits are found to be:

$$\mathrm{FO}: \omega_0 = \sqrt{\frac{\beta_1 \alpha_1 \alpha_2}{C_1' C_2' R_1' R_2'}} \quad \text{for the Fig. 4.9a} \tag{4.23}$$

and

$$\omega_0 = \sqrt{\frac{\beta_1 \alpha_1 \beta_2}{C_1' C_2' R_1' R_2'}} \quad \text{for the Fig. 4.9b} \tag{4.24}$$

where

$$\begin{aligned} &C_1^{/} = C_1 + C_{\mathrm{y1}}, \quad C_2^{/} = C_2 + C_{\mathrm{y2}} + C_{\mathrm{z1}} + C_{\mathrm{z2}}, \\ &R_1^{/} = R_1 + R_{\mathrm{x1}} \quad \text{and} \quad R_2^{/} = R_2 + R_{x2} \quad \text{for the Fig. 4.9a} \end{aligned} \tag{4.25}$$

$$\begin{aligned} &C_1^{/} = C_1 + C_{\mathrm{y1}} + C_{\mathrm{y2}} + C_{\mathrm{z2}}, \quad C_2^{/} = C_2 + C_{\mathrm{z1}}, \\ &R_1^{/} = R_1 + R_{\mathrm{x1}} \quad \text{and} \quad R_2^{/} = R_2 + R_{\mathrm{x2}} \quad \text{for the Fig. 4.9b} \end{aligned} \tag{4.26}$$

Since the value of voltage gain β is much closer to unity than the current gain α, therefore by comparing the above equations, it is turns out that the FO of the circuit of Fig. 4.7b would be in a closer agreement of the theoretical value than that of the circuit of Fig. 4.7a. The validity of this contention has been confirmed in Ref. [48] by using the translinear current conveyor as well as by using CCII+ implemented from AD844 and CCII− implemented from two AD844s.

The circuits of Fig. 4.9 suffer from the difficulty of adjusting the CO through a variable capacitance. However, it appears that if an SRCO could be synthesized with at least three resistors such that one of the two resistors can control the CO and the remaining third resistor controls the FO then in such a circuit this difficulty would not feature. One such circuit using a CCII− and a voltage follower was proposed by Bhaskar and Senani [18][2] and is shown in Fig. 4.10.

The CO and FO for this circuit are given by

$$\mathrm{CO}: \frac{R_1}{R_2} = \frac{C_2}{C_1} \tag{4.27}$$

and

[2] During the writing of this monograph it has come to the notice of the authors that a slightly different variant of this circuit was proposed earlier by Nandi and Nandi [29].

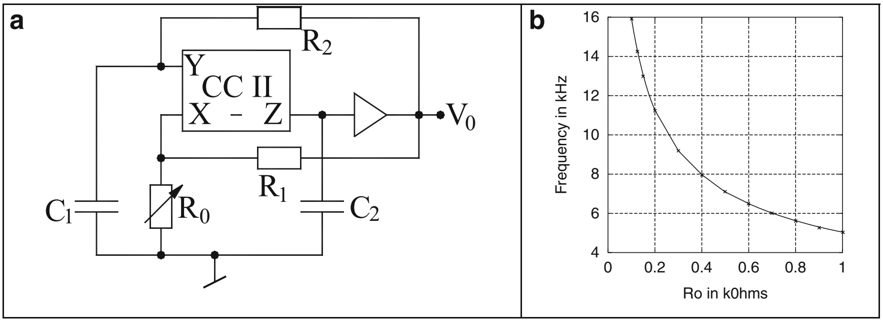


Fig. 4.10 (**a**) SRCO proposed by Bhaskar and Senani [18] (**b**) Variation of the f_0 with respect to variations in R_0

$$\text{FO}: f_o = \frac{1}{2\pi}\sqrt{\frac{1}{R_0R_2C_1C_2}} \tag{4.28}$$

An alternative version of this circuit is also possible with the same functionalities but with slightly altered expression for CO when the resistor R_2, instead of being connected to V_0, is returned to the Z-terminal of the CCI−. The CO of the resulting SRCO is then given by

$$\text{CO}: \frac{R_1}{R_2} = \left(\frac{C_2}{C_1} + 1\right) \tag{4.29}$$

The expression for the FO however, does not change with this modification and remains the same as earlier.

A remarkable property of this circuit is that with $R_1 = R_2 = R$; $C_1 = C_2 = C$ and $R_3 = R/n$, the frequency stability factor S^{F}, for $n \gg 1$ can be approximated as $S^{\mathrm{F}} \cong 2\sqrt{n}$ which can be made quite large (in fact this value is the largest stability factor attainable by any SRCO as demonstrated in Chap. 2 of this monograph.

Because of the availability of the control of FO through a grounded resistor R_3 the FO can be easily made voltage-controllable by replacing this frequency controlling resistor R_3 by an FET used as a voltage controlled resistance.

The workability of this circuit has been verified by making CCI− from LF356 type op-amp in supply current sensing mode implemented through current mirrors made from CA3096E mixed transistor arrays and the voltage buffer made from LM310N with component values chosen as $R_1 = 10$ kΩ, $C_1 = C_2 = 10$ nF with CO adjusted by a variable resistance R_1 and FO tuning through a variable resistance R_0. The variation of f_0 with respect to R_0 is given in Fig. 4.10b. The various features of the circuit have been verified and the capability to realize voltage-controlled oscillator was also verified by using the simplified single buffer based linearized VCR of Ref. [63].

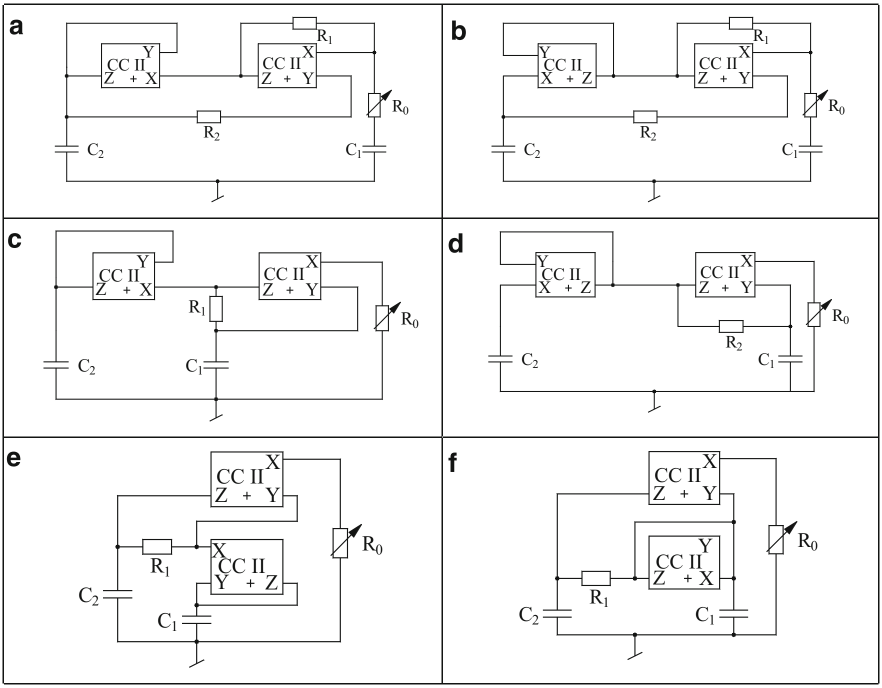


Fig. 4.11 (**a–f**) Grounded resistor controlled oscillators proposed by Nandi [57]

An alternative possibility of implementing this circuit is by using IC CC, PA630A. This however, requires additional current mirrors to enable the realization of a CCI−. In this case, however, the VCO version can be implemented without requiring any additional voltage followers because of the availability of two-on chip buffers in the PA630A CC chip.

In Ref. [57] Nandi proposed a class of two CCII+ based SRCOs (see Fig. 4.11) employing both GCs which provide independent single resistance control of FO through a single grounded variable resistor. The expressions for CO and FO for these circuits are as follows:

For the circuits of Fig. 4.11a, b

$$\text{CO}: \frac{R_2}{R_1} = \frac{1}{2}\left(1 + \frac{C_2}{C_1}\right) \quad \text{and} \quad \text{FO}: \omega_0 = \frac{1}{\sqrt{R_0 C_1 C_2 R_2}} \tag{4.30}$$

For the circuits of Fig. 4.11c–f

$$\text{CO}: \frac{C_2}{C_1} = 1 \quad \text{and} \quad \text{FO}: \omega_0 = \frac{1}{\sqrt{R_0 C_1 C_2 R_2}} \tag{4.31}$$

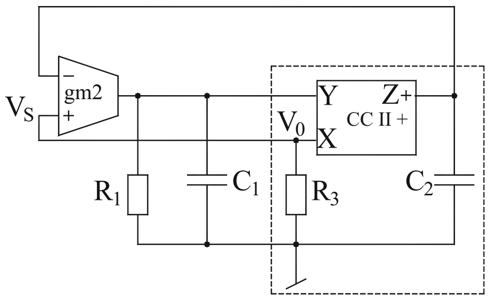


Fig. 4.12 An exemplary two integrator loop based oscillators proposed by Martinez–Sanz [47]

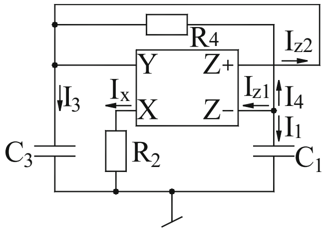


Fig. 4.13 Oscillator realizable with single dual-output variable gain CCII [40]

Martinez–Sanz [47] considered the generation of two integrator loop oscillators using OTAs, CCs and combination thereof. An exemplary circuit is shown in Fig. 4.12, the CO and FO for which are given by:

$$\mathrm{CO} : (1 - R_1 g_{m2}) \leq 0 \tag{4.32}$$

$$\mathrm{FO} : \sqrt{\frac{g_{m2}}{C_1 C_2 R_3}} \tag{4.33}$$

The circuit of Fig. 4.13b has the advantage of providing electronic controllability of FO.

From the preceding discussion it would appear that whereas there exist SRCOs with both GCs using two or three CCs, to the best knowledge of the authors such a circuit using only a single CCII has still not reported in the literature so far, although such circuits using three GCs are known, for instance see Jana–Nandi [6]. However, this circuit employs three grounded capacitors, hence, is non-canonic. On the other hand, a circuit using a single CC and two GCs was proposed by Papazoglou and Karybakas [40] and is shown in Fig. 4.13, its characteristic equations:

$$\mathrm{CO} : R_2 = \left(\frac{C_1}{C_1 + C_3}\right) R_4 \quad \text{and}$$

$$\mathrm{FO} : \omega_0 = \left(\frac{1}{C_1 R_4}\right) \sqrt{\frac{(h-1)(C_1 + C_3)}{C_3}} \tag{4.34}$$

where h is the current-gain factor of the CCII through which the FO would be electronically controllable. An electronically controllable CCII configuration was also suggested by the same authors [40]. Through SPICE simulations the authors demonstrated that frequencies of the order of 40 MHz are easily realizable with their circuit. However, this circuit is not a SRCO of the kind we have been elaborating so far. Hence, the problem of devising a SRCO using only one CC and both GCs remains unresolved!

4.4 SRCOs Employing All Grounded Passive Elements

Sinusoidal Oscillator circuits employing all grounded passive components are of interest due to three basic reasons. Firstly, having all resistors and capacitors grounded is attractive from the point of view of integrated circuit implementation of an oscillator. Secondly, all resistors being grounded is useful from the point of view of electronic control of the oscillation frequency since the frequency controlling grounded resistor(s) can be easily replaced by FET/MOSFET-based linear voltage-controlled-resistances (VCR) [62] thereby making the oscillation frequency voltage-controllable. Thirdly, if the resistor controlling the condition of oscillation is also grounded, this is useful in incorporating of an automatic feedback system for stabilizing/varying the amplitude since this condition setting grounded resistor can also be replaced by a grounded VCR as part of the feedback control circuit.

Motivated by the above mentioned advantages, several CC-based oscillators have been reported in literature which employ all grounded passive elements (AGPE), for instance see Refs. [17, 22, 25, 26, 28–30, 32, 38, 47, 51, 54, 58]. In the following, we discuss some representative circuits out of these.

Two oscillator configurations each employing two CCI− along with three GCs and two grounded resistors with the provision of independent control of CO and FO, have introduced by Abuelma'atti–Al-Ghumaiz [28] are shown in Fig. 4.14. The expressions for CO and FO for these oscillators are given by:

$$\mathrm{CO}: R_1 = R_4 \quad \text{and} \quad \mathrm{FO}: \omega_0 = \frac{1}{\sqrt{R_3 C_1 C_2 R_4}} \quad \text{for the Fig. 4.14a} \tag{4.35}$$

$$\mathrm{CO}: R_2 = R_3 \quad \text{and} \quad \mathrm{FO}: \omega_0 = \frac{1}{\sqrt{R_1 C_3 C_4 R_2}} \quad \text{for the Fig. 4.14b} \tag{4.36}$$

The workability of these circuits was confirmed in Ref. [28] by using the CCI− realization proposed by Senani [88].

Abuelma'atti–Al-Ali–Ahsan [25] presented a programmable CII-based oscillator employing three GCs and two grounded resistors. Their proposition is shown in Fig. 4.15a and is characterized by the following CO and FO:

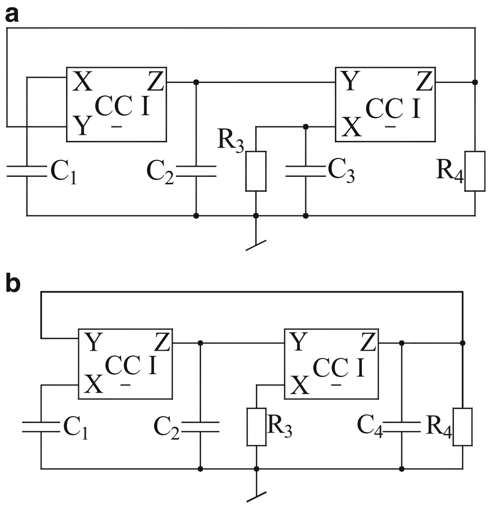


Fig. 4.14 (**a**, **b**) CCI-based oscillators using all grounded elements proposed by Abuelma'atti and Al-Ghumaiz [28]

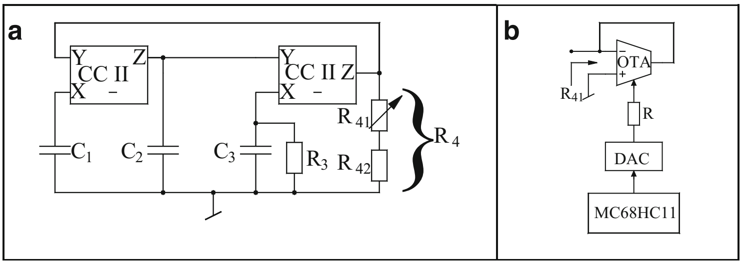


Fig. 4.15 (**a**) CCI-based sinusoidal oscillator (**b**) OTA-based realization of a programmable grounded resistor [25]

$$C_2 = C_3 \quad \text{and} \quad \text{FO} : \omega_0 = \frac{1}{\sqrt{R_3 C_1 C_2 R_4}} \tag{4.37}$$

The digital control of frequency was obtained by using an OTA as a programmable grounded resistor using MC68HC11 μP and a D/A converter which is shown in Fig. 4.15b.

Abuelma'atti [19] combined an OTA along with a CCII to produce new kind of sinusoidal oscillators which possessed the advantage of employing all grounded capacitors and grounded resistors see Fig. 4.16. Two of the circuits are SRCOs while the third one has CO adjustable by a variable capacitor but has the advantage of providing an electronic control of FO through g_m of the OTA which is itself controllable through an external DC bias current. The CO and FO of these circuits of Fig. 4.16 are as follows:

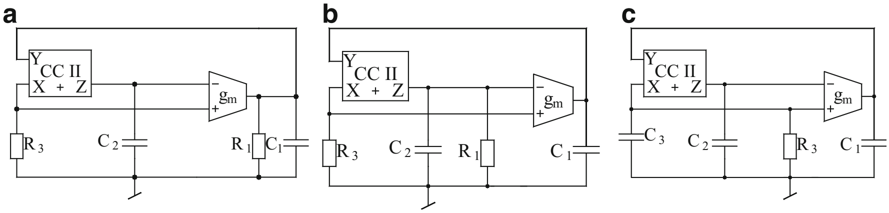


Fig. 4.16 (**a**–**c**) Current-controlled CCII-OTA based RC oscillators proposed by Abuelma'atti [19]

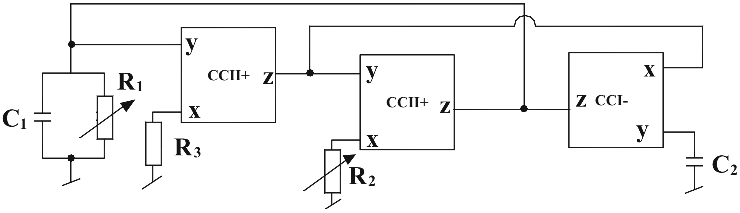


Fig. 4.17 SRCO employing AGPE proposed by Chang [17]

For the Fig. 4.16a

$$\mathrm{CO}: R_1 g_{\mathrm{m}1} = 1;\ \mathrm{FO}: \omega_0 = \sqrt{\frac{g_{\mathrm{m}1}}{C_1 C_2 R_3}} \tag{4.38}$$

For the Fig. 4.16b

$$\mathrm{CO}: R_2 g_{\mathrm{m}1} = \frac{C_1}{C_2};\ \ \mathrm{FO}: \omega_0 = \sqrt{\frac{g_{\mathrm{m}1}(R_2 - R_3)}{C_1 C_2 R_3 R_2}} \tag{4.39}$$

For the Fig. 4.16c

$$\mathrm{CO}: C_2 = C_3;\ \mathrm{FO}: \omega_0 = \sqrt{\frac{g_{\mathrm{m}1}}{C_1 C_2 R_3}} \tag{4.40}$$

A circuit which although requires three CCs and does not have the advantage of providing a buffered outlet but has the advantage of employing all grounded passive elements (AGPE) was proposed by Chang [17] and is shown in Fig. 4.17.

The circuit is characterized by the following CO and FO:

$$\mathrm{CO}: R_1 = R_3 \tag{4.41}$$

$$\mathrm{FO}: \omega_0 = \sqrt{\frac{1}{C_1 C_2 R_2 R_3}} \tag{4.42}$$

While Chang's circuit required three CCs to achieve the intended properties, Liu [22] came up with a novel configuration which could achieve all the features of Chang's circuit employing only two CCs. This circuit is shown in Fig. 4.18 and is characterized by:

$$\mathrm{CO}: R_1 = R_3 \tag{4.43}$$

$$\mathrm{FO}: \omega_0 = \sqrt{\frac{1}{C_1 C_2 R_1 R_2}} \tag{4.44}$$

This circuit lets the control of CO through R_3 and that of FO by R_2. Also, the frequency stability factor is found to be as $S^{\mathrm{F}} \cong 2\sqrt{n}$ which can be made quite large by making the resistor ratio **n** large as in the circuit of Ref. [18], and the circuit enjoys excellent frequency stability.

Another similar circuit with five grounded passive elements and using CCI− instead of CCIIs, was advanced by Abuelma'atti and Al-Ghumaiz [28] and is shown in Fig. 4.19. This circuit has the following CO and FO:

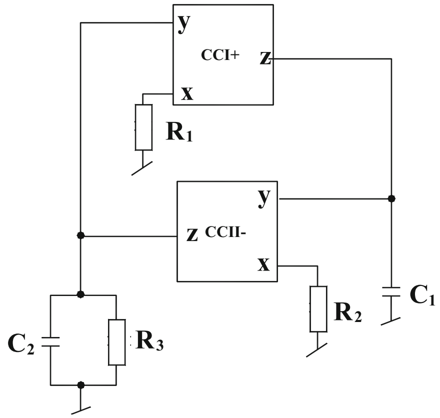


Fig 4.18 Liu's SRCO with AGPE [22]

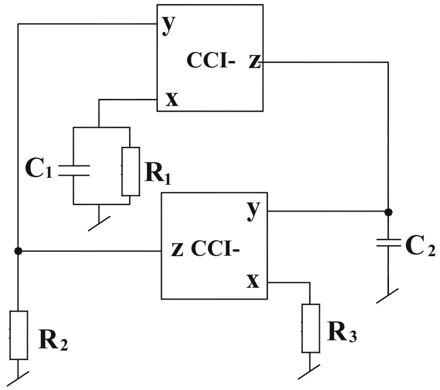


Fig. 4.19 SRCOs proposed by Abuelma'atti and Al-Ghumaiz [28]

$$\text{CO}: R_1 = R_2;\ \text{FO}: \omega_0 = \sqrt{\frac{1}{C_1 C_2 R_2 R_3}} \tag{4.45}$$

It may be mentioned that the authors of Ref. [28] had formulated a general configuration containing two CCI− and four grounded admittances of which eight special cases were considered using three/two resistors and two/three capacitors. A careful inspection, however, reveals that contrary to the claims made in Ref. [28] there are only four *distinctly different* cases therein. In the Table 1 of Ref. [28] it may be seen that circuits numbered 1 and 3 are same and so are circuit numbered 2 and 4. Likewise, special cases 5 and 6 are the same as well as 7 and 8 are the same. This can be easily checked by swapping the "first CCI− along with Y_1 and Y_4" by "second CCI− along with admittances Y_3 and Y_2". Equation (7.30) of Ref. [28] can be obtained by selecting Y_1 as parallel combination of R_1 and C_1, Y_2 as capacitor, Y_3 as resistor and Y_4 as resistor respectively.

An interesting circuit which also employs two CCII+ and only five grounded external passive components and yet is able to take cognizance of the X-port parasitic input resistance R_x and Z-port parasitic impedance consisting of R_p in parallel with C_p for both the current conveyors was proposed by Abuelma'atti and Khan [30] and is shown here in Fig. 4.20.

A straightforward analysis of this circuit shows that the CO and the FO are given by:

$$\text{CO}: R_2 R_4 \cong R_1 R_x,\quad \text{FO}: \omega_0^2 = \frac{R_1(R_3 + R_x) - R_2 R_4}{C_p\left(C_z + C_y\right) R_1 R_3 R_x (R_2 + R_4)} \tag{4.46}$$

Thus, the interesting features of this circuit are: requirement of only a single grounded capacitor and incorporation of all the parasitic impedances of the CCs, a priori, in the design. The latter feature means that this circuit can be designed to generate frequencies much higher than those possible with other two-CC-based

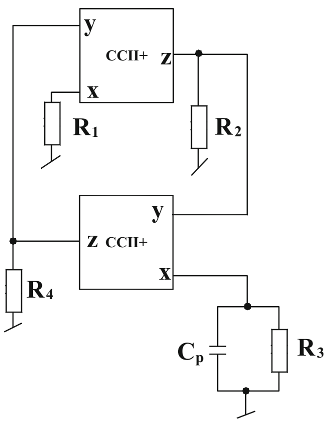


Fig. 4.20 Sinusoidal oscillator proposed by Abuelma'atti and Khan [30]

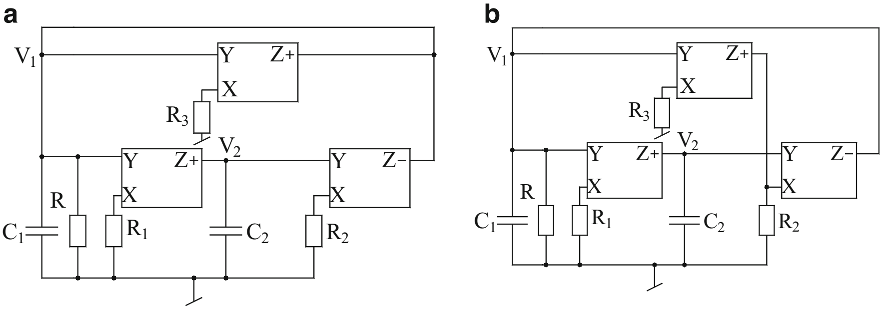


Fig. 4.21 Grounded-capacitor and grounded resistor oscillators proposed by Soliman [38]

circuits described earlier. This was demonstrated by the authors of Ref. [30] by realizing CCII+s using AD844 ICs from which oscillation frequencies as high as 11.11 MHz could be generated easily.

In other publication, Soliman [38] presented a systematic synthesis of oscillators using CCIIs are AGPE based upon the well-known two-integrator loop oscillator topology. Two exemplary circuits derived from Ref. [38] are shown here in Fig. 4.21. The expressions for CO and FO for these oscillators are found to be:

$$\mathrm{CO}: R_3 = R \quad \text{and} \quad \mathrm{FO}: \omega_0 = \frac{1}{\sqrt{R_1 C_1 C_2 R_2}} \tag{4.47}$$

The workability of both the circuits have been confirmed through SPICE simulations with the component values chosen as $C_1 = C_2 = 1$ nF, $R_1 = R_2 = 1$ kΩ. R_3 being variable from 0.95–1 kΩ to get the oscillation frequency of 153.8 kHz.

In Fig. 4.21 we showed three CCIIs based oscillators employing all grounded RC components such that the components which appear in the CO do not appear in the FO and vice versa. Consider now the oscillator of Fig. 4.22. This oscillator configuration is one of four circuits possessing such features as derived by Martinez–Sabadell–Aldea–Celma [36]. This circuit is characterized by the following CO and FO:

$$\mathrm{CO}: R_1 = R_\mathrm{A} \quad \text{and} \quad \mathrm{FO}: \omega_0 = \frac{1}{\sqrt{R_3 C_1 C_2 R_2}} \tag{4.48}$$

4.5 Quadrature and Multiphase Sinusoidal Oscillators

Sinusoidal oscillators known as quadrature oscillators are capable of generating two sinusoidal signals having a phase difference of 90 degrees (in quadrature) and are useful in several applications in communication area such as in quadrature mixers and single sideband modulators. In measurement applications, quadrature oscillators are useful in vector generators and selective voltmeters. In addition, four-phase

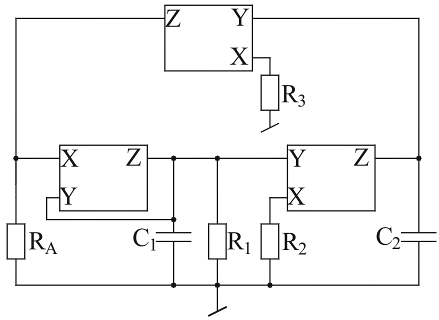


Fig. 4.22 Oscillator proposed by Martinez–Sabadell–Aldea–Celma [36]

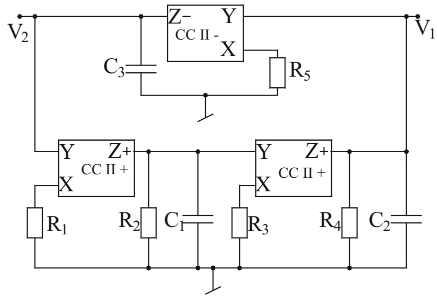


Fig. 4.23 Quadrature oscillator proposed by Horng–Hou–Chang–Chung–Tang–Wen [46]

quadrature oscillators find applications in sub-harmonic mixers to reduce noise and intermodulation distortion. Lastly, multiphase oscillators may find applications in power electronic systems and sub-harmonically pumped frequency conversion circuits.

A number of quadrature and multiphase oscillator topologies using CCs have been reported in literature, for example, [21, 23, 27, 33, 34, 45, 46, 49, 50, 53]. In the following, we discuss a number of quadrature and multiphase oscillators.

While a majority of researchers focused on only second order SRCOs with AGPE, three third-order quadrature oscillators employing AGPEs were reported by Horng–Hou–Chang–Chung–Tang–Wen [46]. One exemplary circuit from this set is shown here in Fig. 4.23 which has the advantage that X-terminals of all the three CCs have only resistors connected to ground thereby making it possible to incorporate the X-port parasitic resistances into the external resistors. In addition, the outputs V_2 and V_1 are in quadrature. Through SPICE simulation based upon AD 844 the authors have demonstrated that the circuits can be used to generate oscillations in the frequency range of several hundred kHz.

Consider now the quadrature oscillator proposed by Horng [45], shown in Fig. 4.24.

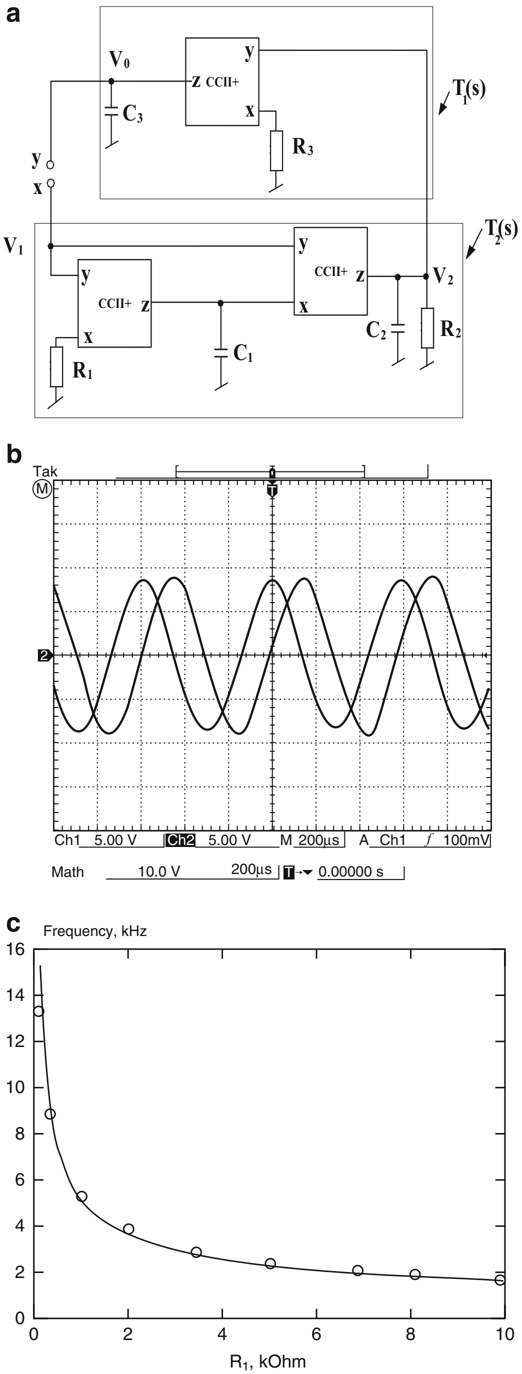


Fig. 4.24 Quadrature sinusoidal oscillator proposed by Horng (adapted from Ref. [45] © 2005 Elsevier) (**a**) The circuit configuration (**b**) Experimental results (**c**) Variation of oscillation frequency with respect to R_1

The CO and FO for this circuit are given by:

$$C_1R_2 = C_3R_3; \quad \omega_o = \sqrt{\frac{1}{C_2C_3R_1R_3}} \tag{4.49}$$

The phase between the two generated signals V_2 and V_1 is given by:

$$\phi = -\tan^{-1}\omega C_1R_1 - \tan^{-1}\omega C_2R_2 \tag{4.50}$$

Using the value of C_1 from Eq. (4.49), the phase different ϕ can be then represented as:

$$\phi = -\tan^{-1}\left(\sqrt{\frac{C_3R_1R_3}{C_2R_2}}\right) - \tan^{-1}\left(\sqrt{\frac{C_2R_2}{C_3R_1R_3}}\right) = -\frac{\pi}{2} \tag{4.51}$$

which shows that V_2 and V_1 are in quadrature.

The workability of this circuit was verified by experimental results with CCII+ realized by AD844 IC biased with power supply ±12 V and component values taken as $C_1 = C_2 = C_3 = 10$ nF, $R_1 = 9.9$ KΩ, $R_2 = 9.88$ KΩ, $R_3 = 9.78$ KΩ. A typical waveform obtained from this circuit is shown in Fig. 4.24b while the variation of the oscillation frequency with the resistance R_1 is shown in Fig. 4.24c.

We now present a voltage-mode (VM) two-phase quadrature oscillator introduced by Minhaj [50] which employs only CCII+s and is shown in Fig. 4.25.

The CO and FO for this circuit are given by:

$$\text{CO}: R_1 \geq R_4 \quad \text{and} \quad \text{FO}: \omega_o = \sqrt{\frac{1}{C_1C_2R_2R_3}} \tag{4.52}$$

Since the second CCII+ is configured as a non-inverting integrator, it is obvious that its input and output, i.e., the two outputs V_2 and V_4, will be in quadrature.

We now present some exemplary circuits capable of generating multiphase (n-phase; $n > 2$) sinusoidal oscillations.

A generalized circuit for realizing a multiphase sinusoidal oscillator which consists of "n" numbers of identical CCII–RC sections was presented by Wu, Liu,

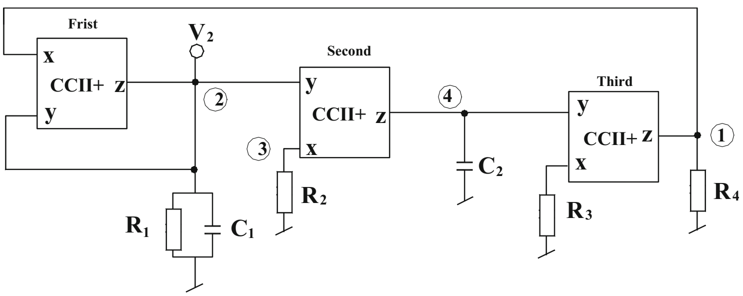


Fig. 4.25 VM two-phase quadrature oscillator proposed by Minhaj [50]

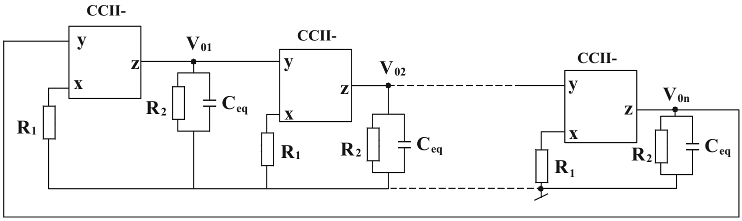


Fig. 4.26 Generalized structure for realizing a multiphase oscillator proposed by Wu, Liu, Hwang, and Wu [27]

Hwang, and Wu [27] and is shown in Fig. 4.26. The condition for which the proposed circuit will provide the sustained oscillations of frequency (ω_0) is given by:

$$(j\omega_0 + \omega_2)^n - (-\omega_1)^n = 0 \tag{4.53}$$

The Eq. (4.53) would have a solution only if the value of n is odd ($n \geq 3$). By equating the real and imaginary parts of Eq. (4.53) to zero, the CO and FO can be then expressed as:

$$\text{CO} : R_2 \geq R_1 \sec\frac{\pi}{n} \quad \text{and} \quad \text{FO} : \omega_0 = \omega_1 \sin\frac{\pi}{n} \tag{4.54}$$

The workability of the circuit was experimentally verified using AD844AN CFOAs for the case of a three phase oscillator. Since FO of each phase is independently controllable by a grounded resistor, the circuit to be easily converted into a voltage controlled multiphase oscillator by replacing the concerned grounded resistors by FET/MOSFET based voltage controlled resistances (VCR). Another noteworthy feature of the circuit is its suitability for IC fabrication because of the use of AGPEs.

Another three phase sinusoidal oscillator using only CCII+s advanced by Skotis and Psychalinos [53] is shown in Fig. 4.27. The internal structure of this configuration can be identified to be a cascade connection of two lossy non-inverting integrators, each made from a single CCII+ and a lossy inverting integrator made from two CCII+s.

An exemplary experimental result of the circuit from Ref. [55] obtained by using AD844 for realizing CCs and using component values as $R_1 = 10$ K, $R_2 = 20$ K, $C = 270$ pF with DC bias supply used for biasing AD844 as ±10 V is shown in Fig. 4.27b. The measured value of the frequency has been found to be 45 KHz and the phase difference was found to be −60° between V_{02} and V_{01} and −114° between V_{03} and V_{01}.

By addition of two inverting amplifies, each implemented from two resistors and a CCII+, the circuit is readily converted into a six-phase quadrature oscillator as shown in Fig. 4.28.

The experimental results of the circuit from Ref. [53], as shown in Fig. 4.28b, demonstrate that by using AD844 for realizing CCs and using component values as

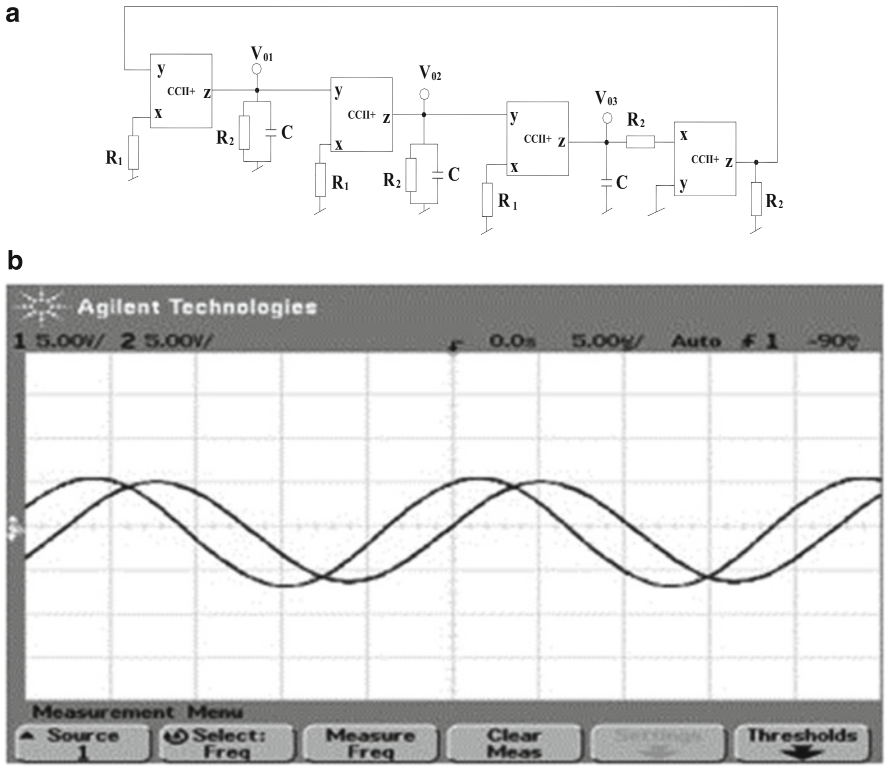


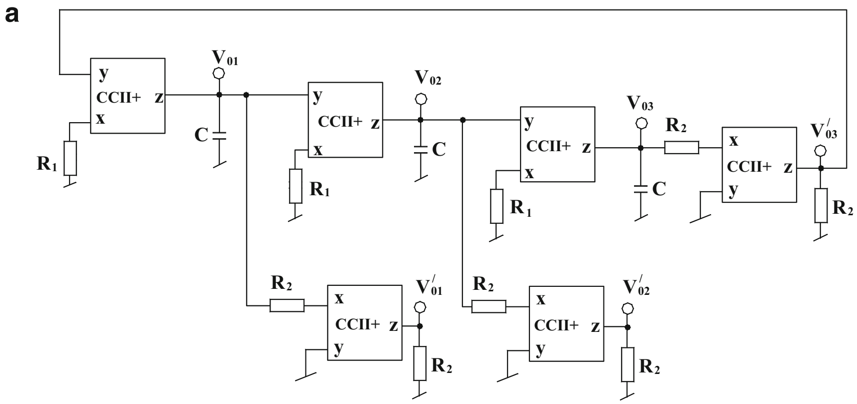
Fig. 4.27 Three-phase oscillator employing CCII+s proposed by Skotis and Psychalinos (adapted from Ref. [53] © 2009 Elsevier GmbH) (**a**) The circuit (**b**) Typical waveforms

$R_1 = 10$ K, $R_2 = 20$ K, $C = 270$ pF with DC bias supply used for biasing AD844 as ± 10 V, the measured value of the frequency has been found to be 46.9 KHz whereas the measured phase difference was found to be 160° between V'_{01} and V_{01}, $-60°$ between V_{02} and V_{01}, 102° between V'_{02} and V_{01}, $-115°$ between V_{03} and V_{01}, and 56° between V'_{03} and V_{01}.

Both the circuits follow essentially from the general block diagram of Fig. 4.29 which shows the basic principle of synthesizing an n-phase oscillator for "n" being odd or even. The second circuit then follows from this schematic by adding two inverting amplifiers.

By a straightforward analysis, the open loop transfer function assuming "n" identical stages is given by:

$$L(s) = -\left(\frac{K}{1 + s\tau}\right)^n \tag{4.55}$$



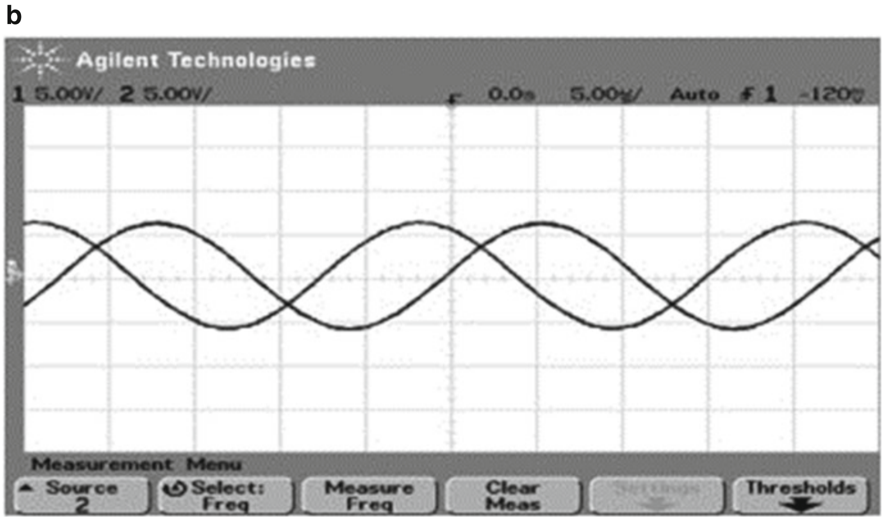


Fig. 4.28 Realization of six-phase oscillator employing CCII + s proposed by Skotis and Psychalinos (adapted from Ref. [53] © 2009 Elsevier GmbH) (**a**) The circuit (**b**) Typical waveforms

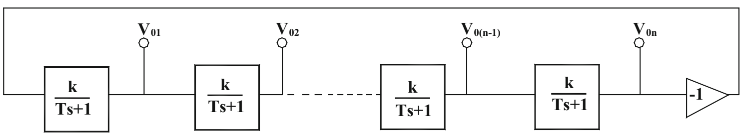


Fig. 4.29 General block diagram of an even and odd phase sinusoidal oscillator proposed by Skotis and Psychalinos (adapted from Ref. [53] © 2009 Elsevier GmbH)

Denoting the maximum gain (R_2/R_1) as K of non-inverting lossy integrators and the time constant $CR_2 = \tau$, the criterion of oscillation at a specific frequency ω_0 requires the following conditions to be fulfilled:

$$\tau = \frac{1}{\omega_0} \tan\left(\frac{\pi}{n}\right) \tag{4.56}$$

$$K = \left(1 + \left[\tan\left(\frac{\pi}{n}\right)\right]^2\right)^{1/2} \tag{4.57}$$

Both the circuits are practically realizable with CCII+ from AD844 ICs with reasonably reduced total components than those required in other design methodologies using CCs [21, 27, 33, 34].

Abuelma'atti and Al-Qahtani [33] introduced a multiphase oscillator using dual output CCs and grounded resistors in which the poles were created by considering the X-port parasitic resistance R_x and Z-port parasitic impedance of each CCII comprised of a resistance $R_z \| 1/sC_z$ with typical values of these components being $R_z = 3$ MΩ, and $C_z = 4.5$pF. It was demonstrated by SPICE simulation that oscillation frequency of the order of 6 MHz with THD 1.4 % was easily achievable for a 3-phase oscillator (Fig. 4.28). The CO and FO are given by respectively (Fig. 4.30):

$$\text{CO} : (R_1 + R_x) = \frac{1}{2}\left(R_z \| R_2\right) \quad \text{and} \quad \text{FO} : \omega_0 = \frac{\sqrt{3}}{C_z(R_z \| R_2)} \tag{4.58}$$

Horng [45] presented six quadrature oscillators such that CO and FO in each oscillator circuit is independently controllable through grounded resistors. All the circuits employ only GCs and grounded resistors as preferred for IC implementation. Two exemplary circuit configurations from Ref. [45] are shown in Fig. 4.31. The CO and FO for these oscillators are given by:

$$\text{CO} : C_1 R_2 = C_3 R_3 \quad \text{and} \quad \text{FO} : \omega_0 = \frac{1}{\sqrt{R_3 C_3 C_2 R_1}} \tag{4.59}$$

Khan and Hasan [49] demonstrated that using multiple output CCIIs (MOCCIIs), it is, possible to devise a novel 4-phase quadrature oscillator in current-mode (CM) while employing all grounded passive components. Their proposition is shown in Fig. 4.32. The CO and FO for this quadrature oscillator are given by:

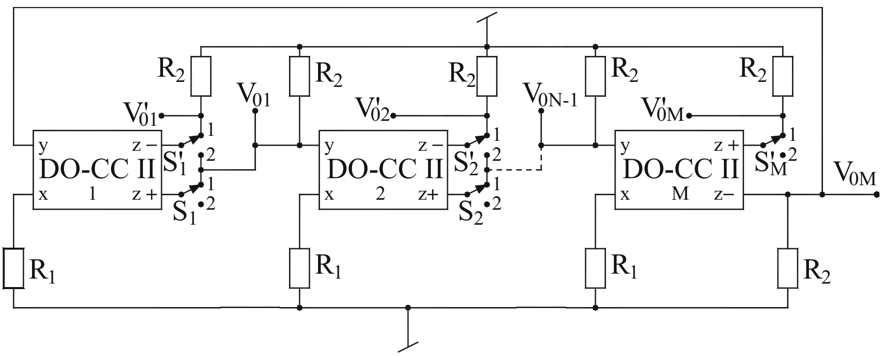


Fig. 4.30 Multiphase sinusoidal oscillator proposed by Abuelma'atti and Al-Qahtani [33]

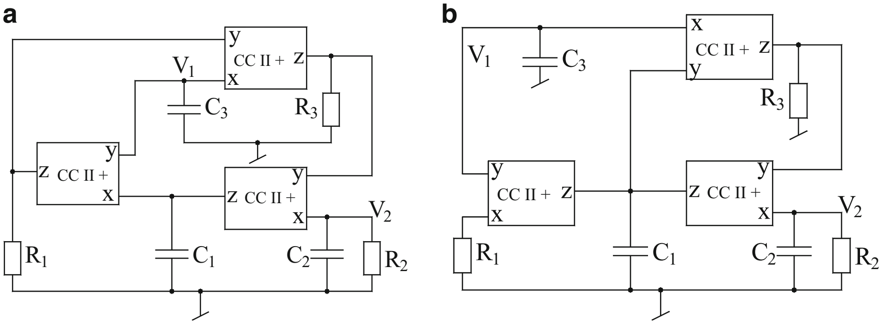
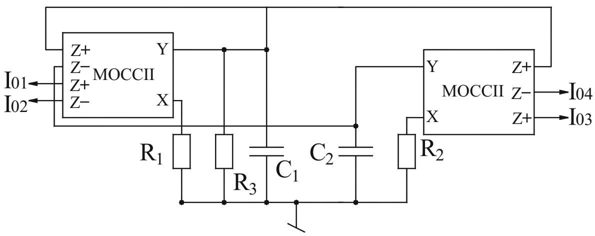


Fig. 4.31 (**a**, **b**) Quadrature oscillators proposed by Horng [45]

Fig. 4.32 Current-mode 4-phase quadrature oscillator proposed by Khan–Hasan [49]



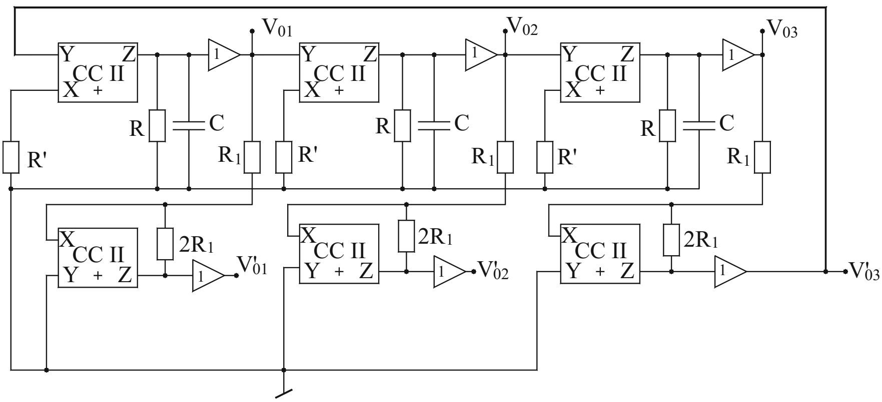


Fig. 4.33 Six-phase sinusoidal oscillator proposed by Hou and Shen [21]

$$\mathrm{CO}: R_1 \leq R_3 \quad \text{and} \quad \mathrm{FO}: \omega_0 = \frac{1}{\sqrt{R_1 C_1 C_2 R_2}} \tag{4.60}$$

Hou and Shen [21] presented a general configuration for synthesizing multiphase oscillator using CCs of which a special case of six phase sinusoidal oscillator is shown in Fig. 4.33. The CO and FO of this 6-phase oscillator are given by:

$$\text{CO}: R = 2R^{/} \quad \text{and} \quad \text{FO}: \omega_0 = \frac{1.732}{RC} \tag{4.61}$$

4.6 SRCOs with Explicit Current Outputs

A large number of current mode signal processing circuits for realizing filters, rectifiers, instrumentation amplifiers etc. have so far been advanced. In view of this, there is a need for test oscillators capable of generating explicit current outputs from high impedance nodes. Therefore, sinusoidal oscillators capable of providing *explicit current output* from high outputs impedance nodes are needed. Several such oscillators have been reported in literature such as those in Refs. [32, 35, 39, 43, 47, 56, 57]. In this section, we outline some important contributions made in this area.

Senani and Gupta [39] proposed two methods for devising such oscillators, resulting thereby in different configurations which are shown here in Fig. 4.34.

The circuit of Fig. 4.34a is based upon a simulated negative resistor with a CCI in parallel with series RLC branch, whose series RL part is created by two unity gain voltage followers B_1 and B_2, resistors R_1 and R_2 and capacitance C_2 thereby leading to a simulated RL impedance with $R_{eq} = R_1$ and equivalent inductance $L_{eq} = C_2 R_1 R_2$. The resulting oscillator is characterized by:

$$\text{CO}: \ R_1 \ = \ R_3; \ \text{FO}: \omega_0 = \sqrt{\frac{1}{C_1 C_2 R_1 R_2}} \tag{4.62}$$

On the other hand, the second circuit (Fig. 4.34b) is derived through the application of the state variable synthesis technique proposed by Senani and Gupta [59–61]. For this circuit, assuming $C_1 = C_2 = C$, CO turns out to be same as in Eq. (4.62) while the FO is given by:

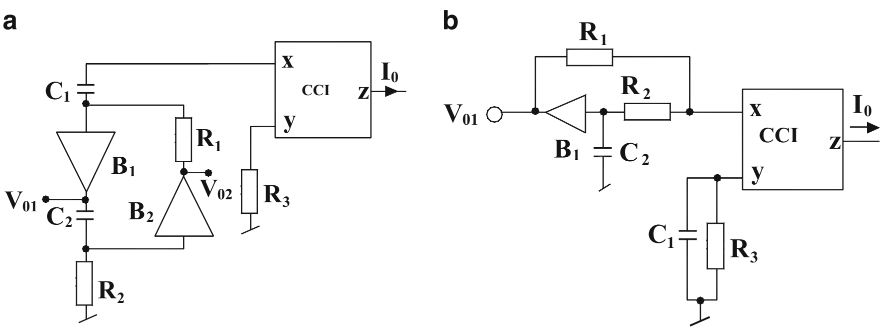


Fig. 4.34 (**a**, **b**) SRCOs proposed by Senani and Gupta [39]

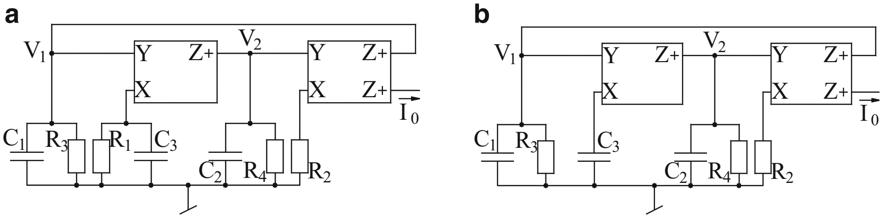


Fig. 4.35 (**a**, **b**) Current-mode CCII-based oscillators proposed by Soliman [32]

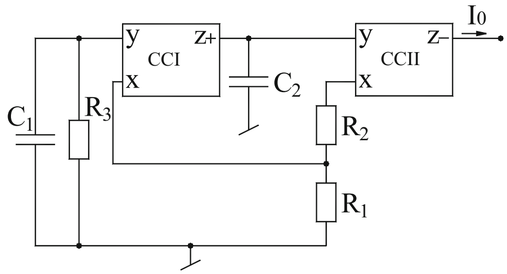


Fig. 4.36 Explicit CM SRCO proposed by Bhaskar–Abdalla–Senani [62]

$$\omega_0 = \frac{1}{C\sqrt{R_2R_3}} \tag{4.63}$$

When a comparison is made with the other alternative AGPE oscillators published earlier, it may be noted that the previously known oscillations suffer from one or more of the following drawbacks: use of more than three resistors and/or more than two capacitors as in Refs. [32, 40]; non-availability of single element controls for CO and FO as in Refs. [35, 56] and requirement of more active elements than those in the circuit of Fig. 4.34 considered here [57].

Soliman [32] presented a number of new realizations of current-mode oscillators with all grounded passive elements. Two exemplary circuits are shown here in Fig. 4.35. The CO and FO for these oscillators are as follows:

For the Fig. 4.35a

$$\text{CO} : \frac{C_1}{R_4} + \frac{C_2}{R_3} = \frac{C_3}{R_2} \quad \text{and} \quad \text{FO} : \omega_0 = \sqrt{\frac{1}{C_1C_2}\left(\frac{1}{R_3R_4} - \frac{1}{R_1R_2}\right)} \tag{4.64}$$

For the Fig. 4.36b

$$\text{CO} : \frac{C_1}{R_4} + \frac{C_2}{R_3} = \frac{C_3}{R_2} \quad \text{and} \quad \text{FO} : \omega_0 = \frac{1}{\sqrt{R_3C_1C_2R_4}} \tag{4.65}$$

The CO and FO for this circuit are given by:

$$\text{CO}: R_1 \leq R_3 \quad \text{and} \quad \text{FO}: \omega_0 = \frac{1}{\sqrt{R_3 C_1 C_2 R_2}} \tag{4.66}$$

A two CCII based SRCO providing explicit current output was proposed by Bhaskar–Abdalla–Senani [62]. This circuit configuration is shown in Fig. 4.36. The circuit offers the advantages of the availability of independent control of CO and FO, use of both the GCs (suitable for IC implementation), low passive and active sensitivities and reasonably good frequency stability.

Subsequently, Abdalla–Bhaskar–Senani [54] proposed a three CCII based CM single input multi output (SIMO) type filter which can be converted into a SRCO, capable of providing both CM and VM outputs by the simple modification of making the input current zero and connecting another z_1-output terminal of $CCII_1$ therein to node "C" in the circuit. This circuit is shown in Fig. 4.37. The CO and FO of this oscillator which provides quadrature outputs both in CM and VM, are given by:

$$\text{CO}: R_1 \leq R_3 \quad \text{and} \quad \text{FO}: \omega_0 = \frac{1}{\sqrt{R_3 C_1 C_2 R_2}} \tag{4.67}$$

So far we have described a number of CC-based oscillator circuits which produce either two quadrature voltage outputs or two quadrature current outputs; Horng [71] came up with a CCII based circuit which provided both of these from a single circuit configuration. His proposition is shown in Fig. 4.38.

The CO and FO of this of this oscillator are given by:

$$\text{CO}: R_3 = \frac{C_1 C_2 R_2}{(C_1 + C_2) C_3} \quad \text{and} \quad \text{FO}: \omega_0 = \frac{1}{\sqrt{R_1 C_1 C_2 R_2}} \tag{4.68}$$

Two additional advantages offered by this circuit are (1) independent controls of CO and FO, the former by R_3 and latter through R_1 (2) use of both GCs as preferred for IC implementation.

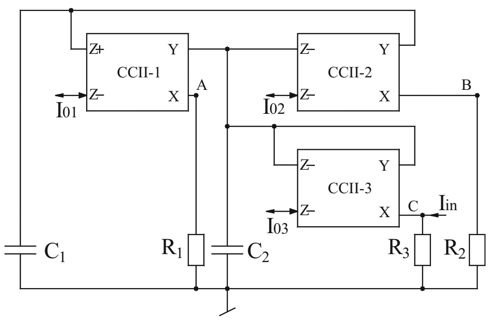


Fig. 4.37 CM and VM quadrature SRCO proposed by Abdalla–Bhaskar–Senani [54]

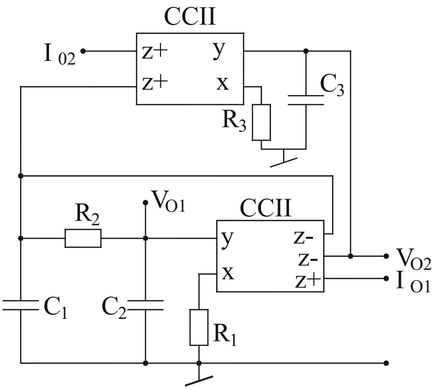


Fig. 4.38 CCIIs based quadrature oscillator proposed by Horng [71]

4.7 SRCOs with Grounded Capacitors and Reduced Effect of Parasitic Impedances of CCIIs

Martinez, Sabadell, Aldea, and Celma [36], derived a number of CC-based oscillators through a formulation based upon two integrator loop biquadratic general structure. Out of the three oscillators derived in Ref. [36] the two circuits which appear to be the better than the remaining ones are shown here in Fig. 4.39.

A notable feature of these circuits is that the parasitic X-terminal input resistance R_x of all the current conveyors can be absorbed in the external resistances connected to terminal-X of the CCs, namely R_A and R_2 in the circuit of Fig. 4.39a and R_A,R_1, and R_2 in case of circuit of Fig. 4.39b. On the other hand, the Z-port parasitic capacitance of the CCs can be merged in the external capacitors C_1 and C_2 as they are connected from the Z-terminals of the CCs only. In view of the above, it is expected that the circuits would exhibit good high frequency performance.

Experimental results of these circuits were demonstrated [36] by using CCIIs implemented from AD844. It was confirmed that these circuits can be satisfactorily employed to generate oscillations up to 10 MHz. It has been possible to obtain signals in the frequency range of 40–400 KHz with average peak-to-peak amplitude of 15 V and THD of less than 1 %.

It may be noted that the circuit of Fig. 4.39a has an additional advantage of providing an explicit current mode output from the Z-terminal of the first CCII+ while the voltage output can be taped from any appropriate node.

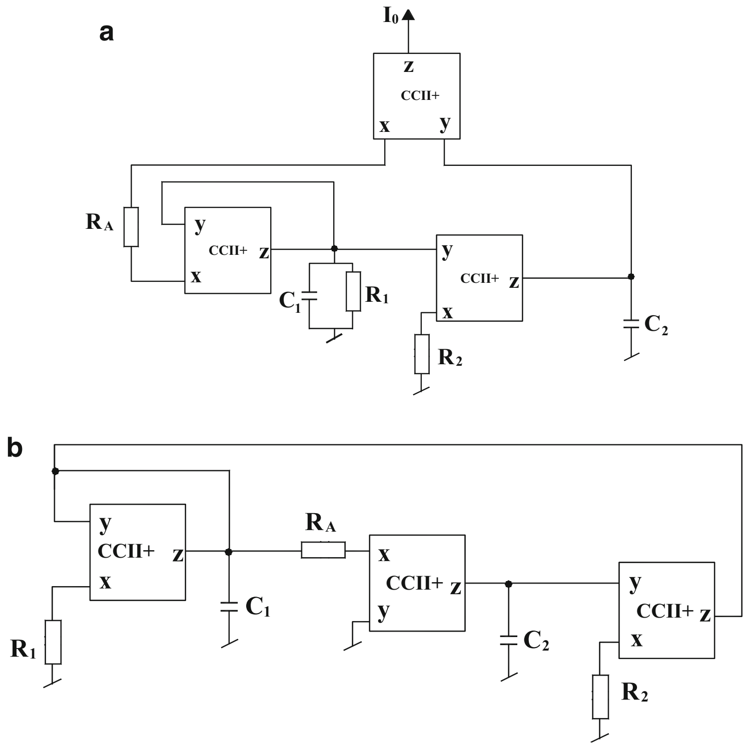


Fig. 4.39 (**a**, **b**) Variable frequency sinusoidal oscillators employing CCII+s proposed by Martinez, Sabadell, Aldea, and Celma [36]

4.8 Sinusoidal Oscillators with Fully uncoupled Tuning Laws

Fully uncoupled sinusoidal oscillators are those in which the components which control the CO do not appear in the expression for FO and on the other hand, the components appearing in FO do not appear in CO. Experience and the observation of the existing CC-based oscillators lets us believe that such oscillators can neither be made from a single CC nor with two CCs and therefore, call for the use of at least three CCs.

One such circuit was proposed by Martinez, Sabadell, Aldea, and Celma [36] and is shown here in Fig. 4.40.

The circuit is characterized by the following equations:

$$\mathrm{CO}: R_1 = R_\mathrm{A};\quad \mathrm{FO}: \omega_0 = \sqrt{\frac{1}{C_1C_2R_2R_3}} \tag{4.69}$$

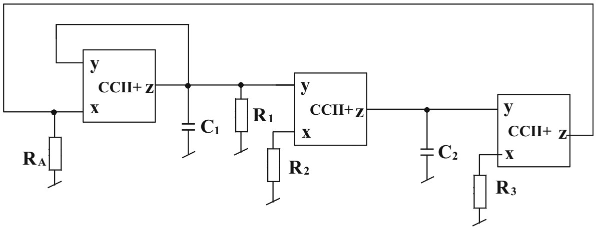


Fig. 4.40 Fully uncoupled oscillator proposed by Martinez, Sabadell, Aldea, and Celma [36]

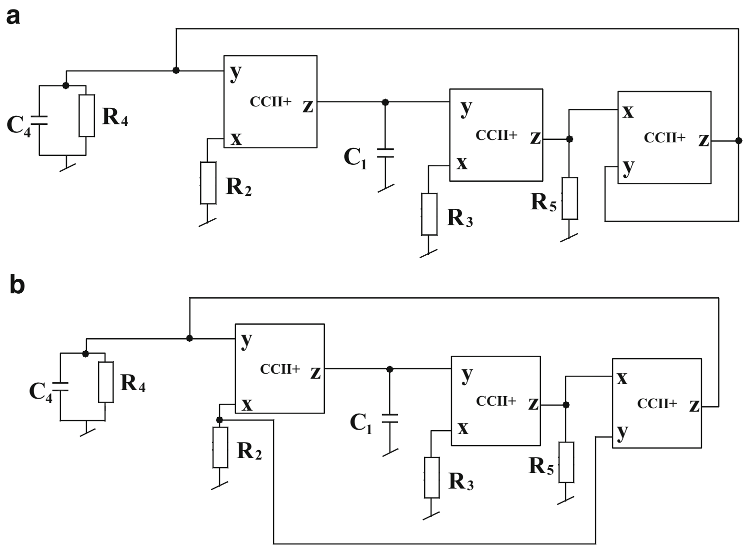


Fig. 4.41 (**a**, **b**) Fully uncoupled oscillators proposed by Abuelma'atti (adapted from Ref. [41] © 2000 Springer)

Abuelma'atti [41] derived a number of three CCII+ based circuits each employing exactly the same number of active and passive components, namely four resistors and two capacitors—all of which are grounded. Two exemplary circuits from Ref. [41] are shown in Fig. 4.41.

Both the circuits are characterized by same CO and FO which are given by:

$$\text{CO} : R_4 = R_5; \;\; \text{FO} : \omega_o^2 = \sqrt{\frac{1}{C_1C_4R_2R_3}} \tag{4.70}$$

In nonideal analysis of these circuits assuming the voltage gains of the three CCII+ as α_i, $i = 1–3$ and current gains as β_i, it is found that the nonideal expressions for the FO of these circuits is exactly same, i.e., $\frac{\alpha_1\alpha_2\beta_1\beta_2\beta_3}{C_1C_4R_2R_3}$ whereas the CO for both the circuits is slightly different as can be seen from the following:

$$\frac{R_5}{\alpha_3\beta_3} = R_4;\ \text{ for the circuit of Fig. 4.41a} \tag{4.71}$$

$$\frac{R_5}{\alpha_1\alpha_3\beta_3} = R_4;\ \text{ for the circuit of Fig. 4.41b} \tag{4.72}$$

From the above expressions, it can be seen that the nonideal, non-unity gains of the CCs may not result in larger errors between the theoretical and experimental frequencies. This has indeed been substantiated by the experimental results of these circuits by using AD844 to implement CCII+s as given in Ref. [41].

4.9 Concluding Remarks

This chapter concerns with the realization of sinusoidal oscillators using basic types of current conveyors, namely CCI± and CCII± only. The literature is abundant with CC-based topologies that realize fixed frequency oscillators as well as variable frequency oscillators. Due to obvious reasons the latter category is more important and therefore, a major emphasis is laid on the so-called single element controlled oscillators which are further divided into two classes the single capacitor controlled oscillators and the single resistance controlled oscillators. The SRCOs which provide frequency control through a grounded resistor are suitable candidates for being converted into VCOs by replacing the frequency controlling resistance by a FET/MOSFET used as voltage controlled resistance. Among the various types of the configurations we deal with single CC SRCO, SRCOs employing grounded capacitances although at the cost of employing one more CC, SRCOs employing all grounded passive elements, quadrature and multiphase oscillators, SRCOs with explicit current outputs, SRCOs with reduced effect of parasitics of the CCs and lastly variable frequency oscillators with fully uncoupled tuning laws. In each of the quoted category, we include only those circuits which in our opinion are better than those omitted from inclusion. The circuits presented in this chapter can be practically implemented using AD 844 CCII+ or CCs constructed from μA741/LF356 type op-amps and CA3096 type mixed transistor arrays or OPA 660/OPA668 or IC CC PA630/PA 630A. It is believed that the present collection of CC based oscillators not only provides an overview of the work done in this area, it could also serve as an useful catalogue for circuit designers, of course, those interested in knowing more may refer to the list of references given at the end of this chapter and those given at the end of this monograph.

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Chapter 5
Realization of Sinusoidal Oscillators Using Current Feedback Op-Amps

Abstract After briefly outlining the significant advantages offered by CFOA-based sinusoidal oscillators, a variety of prominent single-resistance-controlled oscillators (SRCO) have been described which include single CFOA-based SRCOs, two-CFOA-two-grounded-capacitor-based SRCOs, quadrature SRCOs, active-*R* SRCOs, SRCOs with explicit current output, and fully-uncoupled SRCOs. Also included are a variety of voltage-controlled oscillators (VCO) using CFOAs, employing FET-based VCOs as well as analog multiplier (AM)-based VCOs. Various characteristic features and advantages of all the discussed topologies have been highlighted and a number of ideas for further research have been pointed out.

5.1 Introduction

As detailed out in Chap. 3, the area of RC-active oscillators using the conventional voltage-mode op-amps (VOA) had been a very prominent area of analog research before the advent of the current conveyors, CFOA, and other modern active circuit building blocks. A large number of VOA-based sinusoidal oscillators were published in the technical literature during 1976–2001 which have already been dealt in Chap. 3 of this monograph.

With the commercial availability of CFOAs as off-the-shelf integrated circuits (IC), there have been numerous investigations, intuitive as well as systematic, on the generation of a variety of sinusoidal oscillators employing CFOAs [1–99]. Of the various CFOAs available, AD844 has been particularly popular due to the flexibility and versatility offered by it because of the external availability of the compensation pin (*Z*-terminal of the internal CC II+). In this chapter, we present an exposition of some of the prominent CFOA-based sinusoidal oscillators evolved during the past two decades.

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5.2 Realization of Single-Element-Controlled Oscillators Using Modern Circuit Building Blocks

The problem of devising SRCOs has continued to attract the attention and imagination of researchers using a variety of newer active building blocks during the last two decades. The various ABBs considered have been the new variants of second-generation current conveyors (CCII), operational transconductance amplifiers (OTA), four-terminal floating nullors (FTFN), current differencing buffered amplifiers (CDBA), current differencing transconductance amplifiers (CDTA), operational transresistance amplifiers (OTRA), etc. The search for newer topologies of SRCOs is usually aimed at ultimately achieving more and more or all of the following desirable features: employment of grounded capacitors as preferred for IC implementation, use of a minimum possible number of active and/or passive components, suitability for VCO realization, achieving quadrature signal generation, providing explicit voltage mode as well as current-mode outputs, achieving a high-frequency stability, exhibiting higher operational frequency range and minimization of the effects of parasitic impedances or nonideal parameters, etc.

This chapter presents a variety of SRCOs employing CFOAs from the vast amount of literature accumulated during the past two decades in this area (e.g., see [1–16, 19–29, 31–51, 56–59, 69, 73–75] and the references cited therein).

5.3 Wien Bridge Oscillator Using a CFOA

Celma, Martinez, and Carlosena [1] and Martinez, Celma, and Sabadell [2] demonstrated that the use of CFOA, rather than VOA, in the classical Wien bridge oscillator offers improved performance, as compared to its VOA-based counterpart, in terms of frequency accuracy, dynamic range, distortion level, and frequency span.

Consider the Wien bridge oscillator (WBO) using a conventional VOA (see Fig. 5.1a). Straight-forward analysis of this circuit gives the closed-loop characteristic equation (CE) as

$$s^2 + s\frac{(3-k)}{RC} + \left(\frac{1}{RC}\right)^2 = 0 \tag{5.1}$$

from where the condition of oscillation (CO) and frequency of oscillation (FO) are given by

$$\text{CO: } k \geq 3 \text{ and FO: } \omega_0 = \frac{1}{RC} \tag{5.2}$$

Under nonideal conditions, the VOA may be assumed to have a one-pole open-loop gain function which can be approximated by $A_{\mathrm{v}}(s) \cong \frac{\omega_{\mathrm{t}}}{s}$ for $\omega \gg \omega_{\mathrm{p}}$ where ω_{p} is the

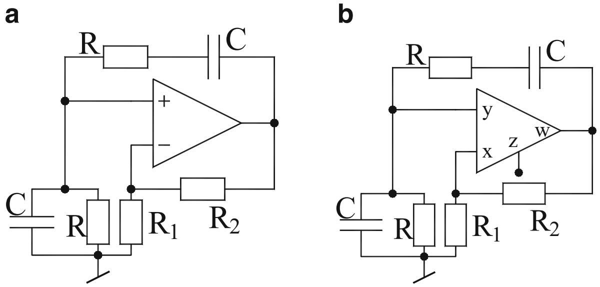


Fig. 5.1 Wien bridge oscillators (**a**) realized with a VOA (**b**) realized with a CFOA

pole frequency and ω_t is the gain-bandwidth product of the op-amp. Through a reanalysis of the circuit, the following nonideal FO $(\hat{\omega}_0)$ and nonideal CO are obtained [1] as

$$\hat{\omega}_0^2 = {\omega_0}^2\left(\frac{1}{1+3\tau k\omega_0}\right) \tag{5.3}$$

$$k \geq 3\left(\frac{1}{1-\tau\omega_0\left(1-\frac{\omega^2}{{\omega_0}^2}\right)}\right) \tag{5.4}$$

$$\text{where } \tau = \frac{1}{\omega_t} \text{ and } k = \left(1+\frac{R_2}{R_1}\right) \tag{5.5}$$

From Eqs. (5.3) and (5.4), it is seen that because the closed loop amplifier gain k appears in the expressions of FO and CO both, therefore, any change in the signal amplitude calibration (distortion) by changing k would disturb the oscillation frequency also and vice versa.

In the case of the CFOA-based) WBO circuit of Fig. 5.1b on the other hand, the nonideal non-inverting amplifier gain is given by

$$k(s) = \frac{k}{1+s\tau} \tag{5.6}$$

where $\tau = R_p C_p$ with $R_p//(1/sCp)$ being the parasitic output impedance looking into the Z-terminal of the CFOA. The nonideal FO and CO are therefore, given by

$$\hat{\omega}^2 = \frac{{\omega_0}^2}{1+3\tau\omega_0} \tag{5.7}$$

$$\text{and } \kappa \geq 3 + \tau\omega_0\left(1-\frac{\hat{\omega}^2}{\omega_0^2}\right) \tag{5.8}$$

From Eqs. (5.7) and (5.8) it can be seen that the behavior of the CFOA version of the WBO is distinctly different than its VOA counterpart in that the CO and FO in the CFOA version are decoupled in the sense that k does not appear in Eq. (5.7) hence, any change in adjusting the CO by changing k does not have any effect on nonideal oscillation frequency given by Eq. (5.7).

A detailed comparative study of the CFOA-based WBO vis-a-vis VOA-based WBO has demonstrated that use of CFO in the WBO provides the following significant advantages: (i) oscillation frequency can be adjusted more accurately, (ii) much wider frequency range of operation is possible, (iii) much higher slew rate of CFOA permits attainment of higher frequencies and/or larger amplitude, (iv) lower sensitivity of the oscillation frequency with respect to variation of bandwidth hence, higher frequency stability, and (v) easy amplitude control by means of a grounded) resistor R_1 without disturbing oscillation frequency.

5.4 Realization of Single-Resistance-Controlled Oscillators Using a Single CFOA

When the research on CFOA-based oscillators started, a number of circuits were initially derived by various researchers either intuitively or by deriving inspiration from the large repertoire of op-amp-based and CC-based oscillators already known in the literature. Systematic formulation of methodologies to enable the generation of all possible circuits belonging to a specific class came into existence much later. Thus, a number of CFOA oscillator circuits were initially reported in a piecemeal manner by various researchers. It was only afterward that systematic approaches started being formulated by a number of research groups for deriving all possible CFOA-based SRCOs belonging to specific classes.

One such systematic approach of deriving all possible single-CFOA canonic SRCOs was formulated in [3, 7] which was based upon the constitution of the most general single-CFOA-based six-node twelve-admittance structure shown in Fig. 5.2a and two converted five-node structures obtained therefrom, shown in Fig. 5.2b, c, whose general characteristic equations were found wherefrom all single-CFOA-based SRCOs using no more than three resistors and two capacitors were enumerated. The class of single-CFOA-based SRCOs resulting from this approach has been shown in Table 5.1.

In the set of eight SRCOs displayed in Table 5.1, oscillators 6 and 8 do not permit independent adjustability of CO although FO can be varied independently; oscillators 1, 2, 3, 5, and 7 provide the control of CO through a single grounded resistor which is attractive from the point of view of ease of incorporating amplitude stabilization/control circuitry. On the other hand, oscillator 4 provides FO control through a grounded resistor thereby making it a suitable choice for easy conversion into a VCO by replacing this frequency-controlling grounded resistor by an FET or linearized VCR [52].

It is worth pointing out that Table 5.1 includes only canonic SRCOs, i.e., those circuits which use no more than three resistors and two capacitors. Indeed, SRCOs using more than three resistors and more than two capacitors have also been

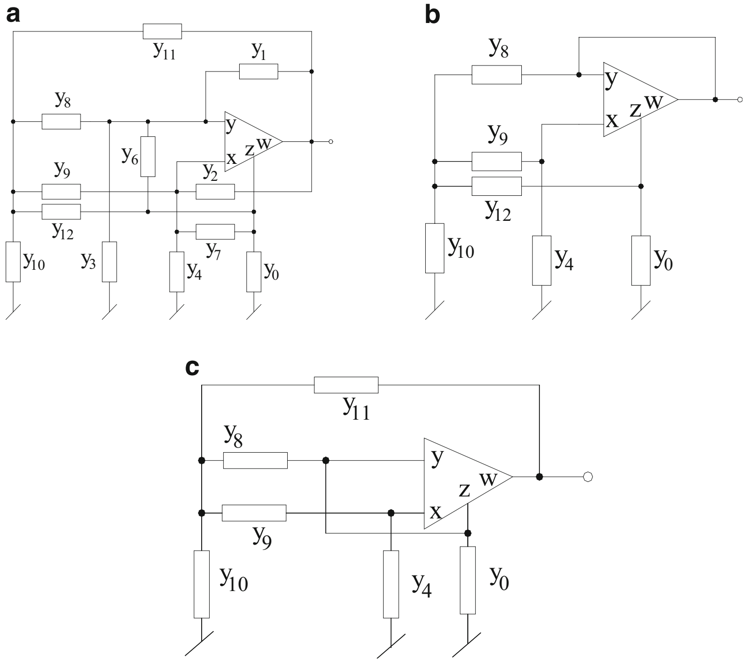


Fig. 5.2 Generalized single-CFOA-based configurations employed for systematic generation of canonic SRCOs (adapted from [77]). (**a**) Generalized six-node structure. (**b**) First converted five-node structure. (**c**) Second converted five-node structure

investigated in [15, 16] and it is found that permitting more than two capacitors makes it possible to realize single-CFOA SRCOs using all grounded capacitors—a feature which is not possible with canonic-single-CFOA oscillators. Two three-GC SRCOs devised by Toker, Cicekoglu, and Kuntman [15] are shown in Fig. 5.3.[1]

It can easily be verified that the CO and FO for the circuits of Fig. 5.3 are given: For the circuit of Fig. 5.3a

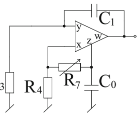
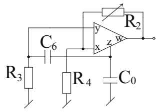
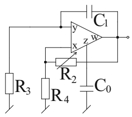
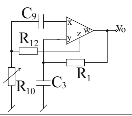
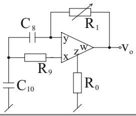
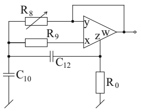
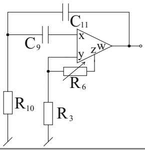
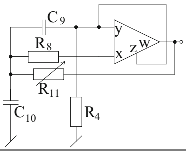
$$\text{CO}: \quad C_2\left(\frac{1}{R_5}+\frac{1}{R_6}\right)-C_1\left(\frac{1}{R_6}\right)+C_3\left(\frac{2}{R_3}\right)\leq 0 \tag{5.9}$$

$$\text{FO}: \quad f_0=\frac{1}{2\pi}\sqrt{\frac{\frac{2}{R_3R_5}-\frac{1}{R_1R_6}}{C_2C_3}} \tag{5.10}$$

and for the circuit of Fig. 5.3b

[1] The circuit of Fig. 5.3a has also been derived independently in [24].

Table 5.1 The class of Single-CFOA-based canonic SRCOs (adapted from [77])

Circuit number and references where appeared	Oscillator circuit	CO and FO
1 [8] (Fig. 3) [3] (Fig. 5) [15] (Fig. 6(c), (f)) [16] (Fig. A1–B4)		$\frac{R_3}{R_4}=\frac{C_0}{C_1}$; adjustable by R_4 $\omega_0=\sqrt{\frac{2}{R_3R_7C_0C_1}}$; controllable by R_7
2 [8] (Fig. 6) [3] (Fig. 6) [15] (Fig. 6(g)) [16] (Fig. A2–B3)		$\frac{R_3}{R_4}=1+\frac{C_0}{C_6}$; adjustable by R_4 $\omega_0=\sqrt{\frac{1}{R_2R_3C_0C_6}}$; controllable by R_2
3 [3] (Fig. 6) [2] (Fig. 3) [11] (Fig. 3) [14] (Fig. 1) [15] (Fig. 6(a), (e)) [16] (Fig. A1–B3)		$\frac{C_0}{C_1}=\frac{R_3}{R_4}$; adjustable by R_4 $\omega_0=\sqrt{\frac{1}{R_3R_2C_0C_1}}$; controllable by R_2
4 [12] (circuit 1 of Table-I) [15] (Fig. 6(k))		$\frac{R_{12}}{R_1}=\frac{C_3}{C_9}$; adjustable by R_{12} $\omega_0=\sqrt{\frac{1}{2R_1R_{10}C_3C_9}}$; controllable by R_{10}
5 [4] (Fig. 8) [15] (Fig. 6(n))		$\frac{R_0}{R_9}=1+\frac{C_8}{C_{10}}$; adjustable by R_0 $\omega_0=\sqrt{\frac{1}{R_1R_9C_8C_{10}}}$; controllable by R_1
6 [14] (Fig. 6(l))		$\frac{R_0}{R_9}=1+\frac{C_{12}}{C_{10}}$; $\omega_0=\sqrt{\frac{1}{R_0C_{10}C_{12}}\left(\frac{1}{R_9}+\frac{1}{R_8}\right)}$ controllable by R_8
7 [7] (Fig. 7)		$\frac{R_3}{R_{10}}=\left(1+\frac{C_{11}}{C_9}\right)$; adjustable by R_3 $\omega_0=\sqrt{\frac{1}{R_6R_{10}C_9C_{11}}}$ controllable by R_6
8 [7] (Fig. 8)		$\frac{R_4}{R_8}=1+\frac{C_9}{C_{10}}$ $\omega_0=\sqrt{\frac{1}{R_4C_9C_{11}}\left(\frac{1}{R_8}+\frac{1}{R_{11}}\right)}$ controllable by R_{11}

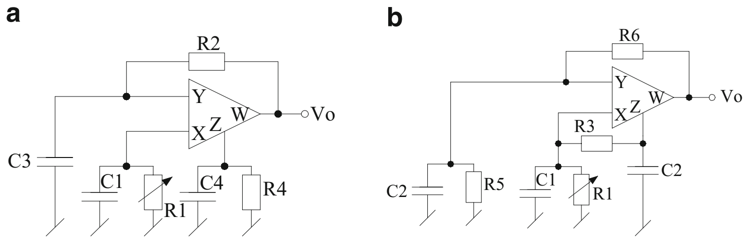


Fig. 5.3 Single-CFOA SRCOs employing all grounded capacitors as proposed by Toker, Cicekoglu, and Kuntman (adapted from [15] © 2002 Elsevier)

$$\text{CO}: \quad \left\{\frac{1}{R_2}(C_4 - C_1) + C_3\left(\frac{1}{R_4}\right)\right\} \leq 0 \tag{5.11}$$

$$\text{FO}: \quad f_0 = \frac{1}{2\pi}\sqrt{\frac{\frac{1}{R_2}\left(\frac{1}{R_4} - \frac{1}{R_1}\right)}{C_3 C_4}} \tag{5.12}$$

Thus, in both the cases, FO can be controlled by a single-variable resistor R_1; however, an independent control of CO is not available.

The oscillator of Fig. 5.3a has the obvious advantage of facilitating absorption of the Z-terminal parasitic impedances $R_p//(1/sC_p)$ in the external components R_4 and C_4. As far as is known, there does not appear to be any single-CFOA canonic SCRO known in the literature which can have only two grounded capacitors while using only a single CFOA. In the next section, we show how two-GC SRCOs can be synthesized systematically, if the use of two CFOAs is allowed.

5.5 A Novel SRCO Employing Grounded Capacitors

A novel SRCO employing two CFOAs and both grounded capacitors (GC) as preferred for integrated circuit implementation was first presented by Senani and Singh in 1996 [13]. This circuit, which was, of course, derived intuitively, is shown in Fig. 5.4.

By a straightforward analysis, the CO and FO for this circuit are given by

$$R_3 = R_2 \tag{5.13}$$

$$\text{and} \quad f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{R_1 R_2 C_1 C_2}} \tag{5.14}$$

From the above it is seen that the CO can be satisfied by adjusting R_3, whereas the FO can be *independently* varied by R_1.

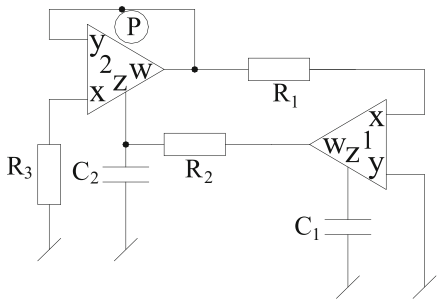


Fig. 5.4 A two-CFOA-GC SRCO proposed by Senani and Singh [32]

The circuit can be converted into a VCO by replacing the resistor R_1 by an FET-based VCR. It may be noted that due to the right-hand-side end of R_1 being at virtual ground potential, the floating nature of resistor R_1 does not pose any difficulty in its replacement by FET-based voltage-controlled resistor (VCR) conventional or otherwise, such as by Senani's floating VCR (FVCR) in [17] to obtain voltage-controlled oscillations.

It is also important to note that the various external RC components are to be selected to have appropriate values so that the parasitic impedances appearing at the x-input and compensation pin-z of the CFOAs have the least effect on the performance of the oscillator. Alternatively, these internal compensating capacitances can be accommodated into the main capacitances, while x-port input resistances r_x of both the CFOAs can be easily accommodated in the external resistors R_1 and R_3.

The frequency stability, which is considered to be an important figure of merit of sinusoidal oscillators, is defined as $S_F = -d\phi(u)/du$ evaluated at $u = 1$ where $u = \omega/\omega_o$. Thus, S_F can be calculated from the open-loop transfer function of the circuit of Fig. 5.4 which, by opening the feedback link at Ⓟ, is given by:

$$T(s) = \frac{s\frac{1}{R_3C_2}}{s^2 + s\frac{1}{R_2C_2} + \frac{1}{R_1R_2C_1C_2}} \tag{5.15}$$

Now choosing $R_2 = R_3 = R$, $C_1 = C_2 = C$, and $R_1 = R/n$, we get $\omega_0 = \sqrt{(n)}/RC = \sqrt{(n)}/\tau$, the open-loop transfer function reduces to

$$T(s) = \frac{s/\tau}{s^2 + s/\tau + \omega_0^2} = \frac{s\frac{\omega_0}{\sqrt{n}}}{s^2 + \frac{s\omega_0}{\sqrt{n}} + \omega_0^2} \tag{5.16}$$

by putting $s = j\omega$, we get

$$T(j\omega) = \frac{j\frac{\omega}{\omega_0}\frac{1}{\sqrt{n}}}{\left[1 - \left(\frac{\omega}{\omega_0}\right)^2\right] + j\frac{\omega}{\omega_0}\frac{1}{\sqrt{n}}} \tag{5.17}$$

From the above, the S_F has been found to be $S_F = 2\sqrt{n}$ which can be kept large by keeping n large, inspite of n being variable.

Lastly, it must also be mentioned that it has been shown in [13] that by breaking the link at "P" the resulting two open-loop transfer functions of the circuit assuming outputs to be taken from W-terminals of the two CFOAs turn out to be low-pass/band-pass filters. Also, by removing external capacitors C_1 and C_2 and incorporating the Z-pin parasitic capacitances into design, the circuit can also be used as an active-R oscillator with ω_0 still controllable through R_1. It has been confirmed that this SRCO works well in generating oscillation frequencies of the order of 500 kHz in normal mode, while in active-R mode it has been possible to extend the generated frequencies till 9.85 MHz.

An alternative circuit exhibiting similar properties was proposed in [4] and has been shown in Fig. 5.5. The CO and FO for this circuit are found to be

$$\frac{R_2}{R_1} = \frac{C_2}{C_1} \tag{5.18}$$

$$\text{and}\ \ f_0 = \frac{1}{2\pi\sqrt{R_0 R_2 C_1 C_2}} \tag{5.19}$$

Thus, in this case also CO and FO are independently adjustable; the former by R_1 and the latter by R_0. This circuit has the advantage of providing FO control through a grounded resistor.

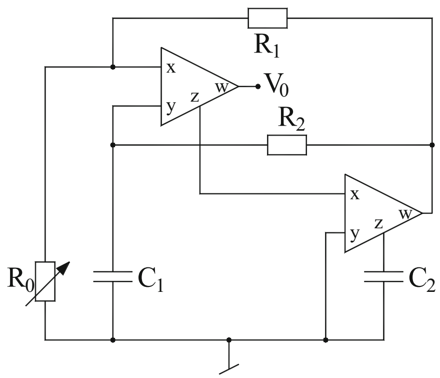


Fig. 5.5 A two-CFOA-GC SRCO proposed by Senani [4]

5.6 A Systematic State-Variable Synthesis of Two-CFOA-Based SRCOs

Subsequent to the publication of the two-CFOA-based SRCOs of [13], it dawned upon the first author of this monograph that if state equations of this circuit are formulated then it should be possible to derive node equations of the circuit which could then be used to synthesize a circuit using CFOAs and RC components. Such an approach appeared to hold the promise that a chosen [A] matrix of the state-variable characterization of an SRCO could, therefore, lead to more than one circuit depending upon how exactly the equations are implemented to formulate a physical circuit. Furthermore, since, for any specified CO and FO, the state-variable characterization is not unique, many different state-space representations leading to different [A] matrices (but all leading to the same characteristic equation (CE)) could be evolved, which appeared to have the potential of generating a large number of SRCO (quite likely all possible) circuits. This, indeed, turned out to be true and the initial results were published in [27–29]. In this section, we present the state-variable methodology from [27–29], which resulted in a family of 14[2] two-CFOA-two-GC SRCOs generated therefrom, which were presented in [27–29].

It is easy to visualize that a canonic second-order (i.e., employing only two capacitors) oscillator can, in general, be characterized by the following autonomous state equation:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \tag{5.20}$$

The characteristic equation (CE) of the circuit is then given by

$$s^2 - (a_{11} + a_{22})s + (a_{11}a_{22} - a_{12}a_{21}) = 0 \tag{5.21}$$

The above CE leads to the following CO and FO:

$$\mathrm{CO}: (a_{11} + a_{22}) = 0 \tag{5.22a}$$

$$\mathrm{FO}: \omega_0 = \sqrt{(a_{11}a_{22} - a_{12}a_{21})} \tag{5.22b}$$

The methodology employed in references of [27–29] involves: (i) a selection of the parameters a_{ij}, $i = 1, 2$; $j = 1, 2$, in accordance with the required features (e.g., noninteracting controls for FO and CO through separate resistors), (ii) conversion of the resulting state equations into node equations (NE), and, finally, (iii) constitution of a physical circuit from these node equations.

[2] Many more variants of the 14 basic oscillators, obtained by using some transformations, terminal interchanges, etc., have been described in [78].

Different circuits are expected to be generated by making different choices of parameters a_{11}, a_{12}, a_{21}, and a_{22}. For noninteractive controls of condition of oscillation and frequency of oscillation, let us assume that the condition of oscillation is to be controlled by R_1 (independent of R_2) and frequency of oscillation is to be controlled by R_2 (independent of R_1, with the third resistor R_3 featuring in both CO and FO). These conditions can be satisfied in a number of ways leading to different [A] matrices. It has been shown in [28, 29] that a set of 14 different matrices can be conceived, which resulted in as many SRCOs, all employing two CFOAs and grounded capacitors.

We now illustrate the procedure by considering the following [A] matrix which satisfies the requirements of independent adjustability of both CO and FO through separate resistors:

$$[A] = \begin{bmatrix} 0 & \dfrac{1}{C_1 R_2} \\ -\dfrac{1}{C_2 R_3} & \dfrac{1}{C_2}\left(\dfrac{1}{R_3} - \dfrac{1}{R_1}\right) \end{bmatrix} \tag{5.23}$$

From the above system matrix, the CO and FO of the oscillator to be synthesized are

$$R_3 = R_1 \tag{5.24}$$

$$\text{and } f_0 = \frac{1}{2\pi\sqrt{C_1 C_2 R_2 R_3}} \tag{5.25}$$

The state equation resulting from substituting Eq. (5.23) in (5.24) can be arranged as the following node equations:

$$C_1 \frac{\mathrm{d}x_1}{\mathrm{d}t} = \frac{x_2}{R_2} \tag{5.26}$$

$$C_2 \frac{\mathrm{d}x_2}{\mathrm{d}t} = \frac{(x_2 - x_1)}{R_3} - \frac{x_2}{R_1} \tag{5.27}$$

The final circuit resulting from the implementation of the node equations (5.26) and (5.27) using CFOAs results in the final oscillator circuit as shown in Fig. 5.6a in which the mechanism of synthesizing the equations is also shown which is self-explanatory.

Following the above explained procedure, a large number of circuits are derived in [78], although a set of 14 basic SRCOs were already described in [28, 29]. Some circuits possessing interesting properties are shown here in Fig. 5.6 (FO is same for all oscillators as given by Eq. (5.25)).

It may be mentioned that single-resistance control (SRC) of the frequency of oscillation through a grounded resistor makes it easier to incorporate FET-based voltage-controlled resistors (VCR) thereby leading to VCO realizations. On the

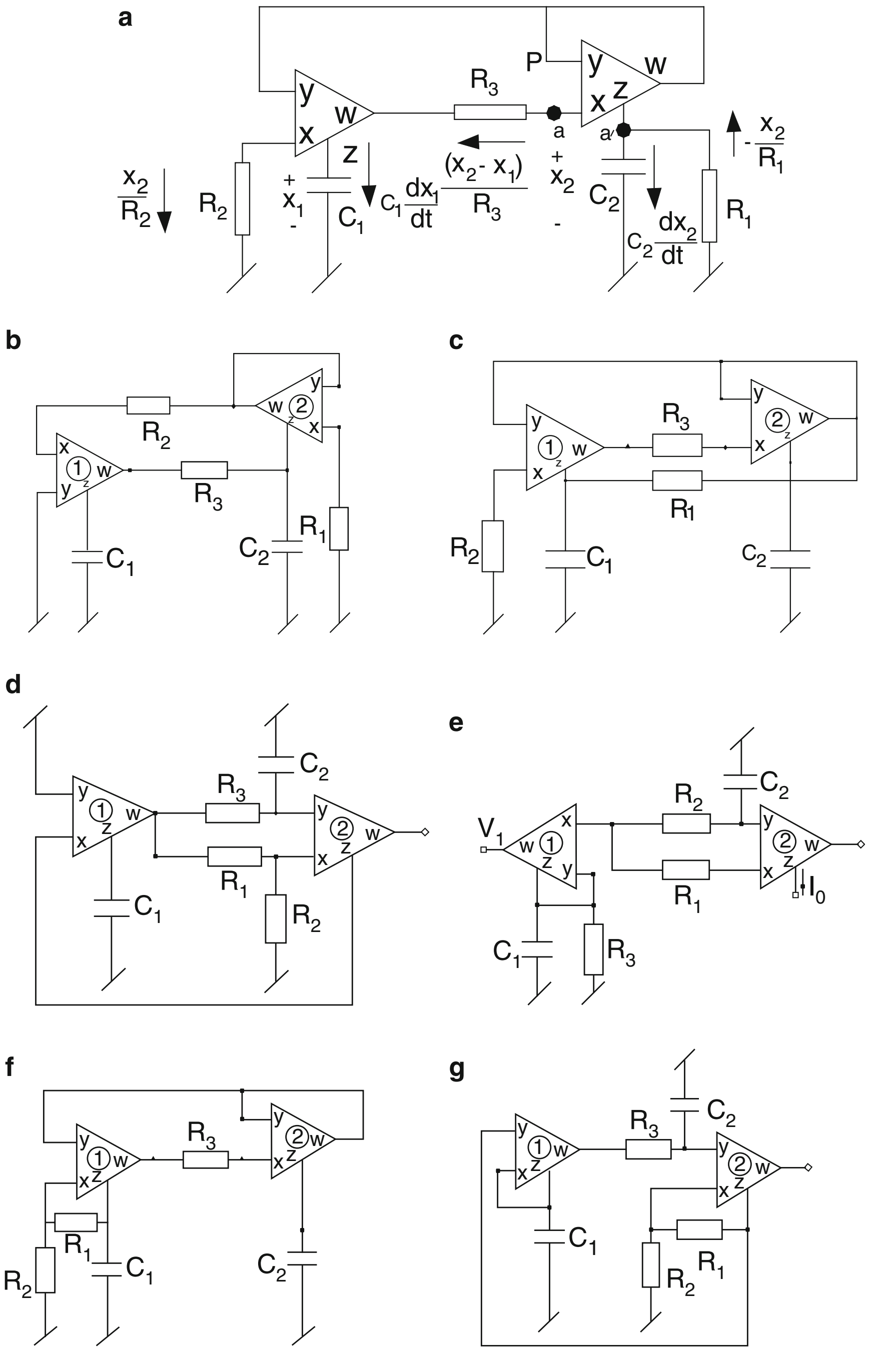


Fig. 5.6 Some exemplary circuits synthesized through the state-variable methodology (adapted from [78]) COs for (**a**) $R_3 = R_1$, (**b**) $R_1 = R_3$, (**c**) $C_1R_1 = C_2R_3$, (**d**) $C_1R_1 = C_2R_3$, (**e**) $R_3 = R_1$ (for $C_1 = C_2$), (**f**) $C_1R_1 = 2C_2R_2$, (**g**) $C_1R_1 = 2C_2R_2$

other hand SRC control of CO through a grounded resistor is desirable from the viewpoint of easy incorporation of amplitude stabilization/control circuitry. In view of the foregoing points, it is seen that the circuit of Fig. 5.6a provides controls of CO and FO both through separate grounded resistors R_1 and R_2, respectively, and is, therefore, superior to the remaining SRCOs of Fig. 5.6 SRCOs from this view point.

In case of the circuits of Fig. 5.6a–c, f the z-pin parasitic capacitances can be easily merged with the main external capacitances and hence, these parasitics do not affect the circuit behavior adversely. In the circuits of Fig. 5.6d, e, h, the capacitor C_1 is connected to the Z-terminal of CFOA1 and no capacitor is connected at the Z-terminal of CFOA2. However, the parasitic capacitance at the Z-terminal of the CFOA2 is made ineffective by the Z-terminal being connected to virtual ground (as in Fig. 5.6d, h) or is ineffective as the current through the Z-terminal of CFOA2 is not coming into picture as in case of Fig. 5.6e. However, in the circuit shown in Fig. 5.6g the z-pin parasitic capacitance at the Z-terminal of the CFOA2 cannot be accounted for.

It is found that with $C_1 = C_2 = C$ and $R_1 = R_3 = R$, $R_2/R = n$ for the circuits of Fig. 5.6a–c, f–h, the frequency stability factor S_F can be made sufficiently large as "n" can always be kept greater than unity, and, therefore, these circuits enjoy excellent frequency stability properties. The circuit of Fig. 5.6e is notable due to the availability of an explicit current output.

The family of 14 two-CFOA-GC SRCOs presented in [28, 29, 78] has been found to work quite well for generating sinusoidal signals up to several hundred kilohertz.

5.7 Some Other Two-CFOA Sinusoidal Oscillator Configurations

In the previous section, a number of SRCOs were shown to be derivable through a systematic synthesis procedure. The circuits were synthesized with the following objectives (a) use of two GCs, (b) use of two CFOAs, and (c) independent control of CO and FO through two separate resistors. All the SRCOs were based upon the tuning laws of the type:

$$\text{CO} : R_1 = R_3 \text{ and FO} : \omega_0 = \sqrt{\frac{1}{C_1 C_2 R_2 R_3}} \tag{5.28}$$

which give in the resulting circuits and the control of CO trough R_1 and that of FO by R_2.

Although major attention has been received in the literature on the above kind of SRCOs, oscillators governed by other types of tuning laws have also been considered by many researchers. To this class belong the circuits which provide CO control through a single-variable capacitor or FO control through a single-variable capacitor or provide an expression for FO containing a difference term. These alternative types of oscillators are also useful due to the following:

1. Oscillators providing single-element control (SEC) of FO through a single-variable capacitor can be used as a transducer oscillator in conjunction with capacitive transducers.
2. Oscillators providing CO control through a capacitor can be used in some capacitance measurement schemes, for instance, see [60–62].
3. Oscillators having a difference term in the expression of FO may be usefully employed as very low frequency oscillators [33].

It has been demonstrated in the literature that given only two CFOAs and only two GCs, along with two to three resistors, a number of *other* sinusoidal oscillator circuits are possible which may have the tuning law different than those considered so far and yet satisfy the single-element-controllability conditions. In Fig. 5.7, we show a number of such two-CFOA-GC SECOs. The CO and FO for these circuits are given in Table 5.2. These circuits too are derived by the state-variable methodology by framing new tuning laws, determining the required [*A*] matrices,

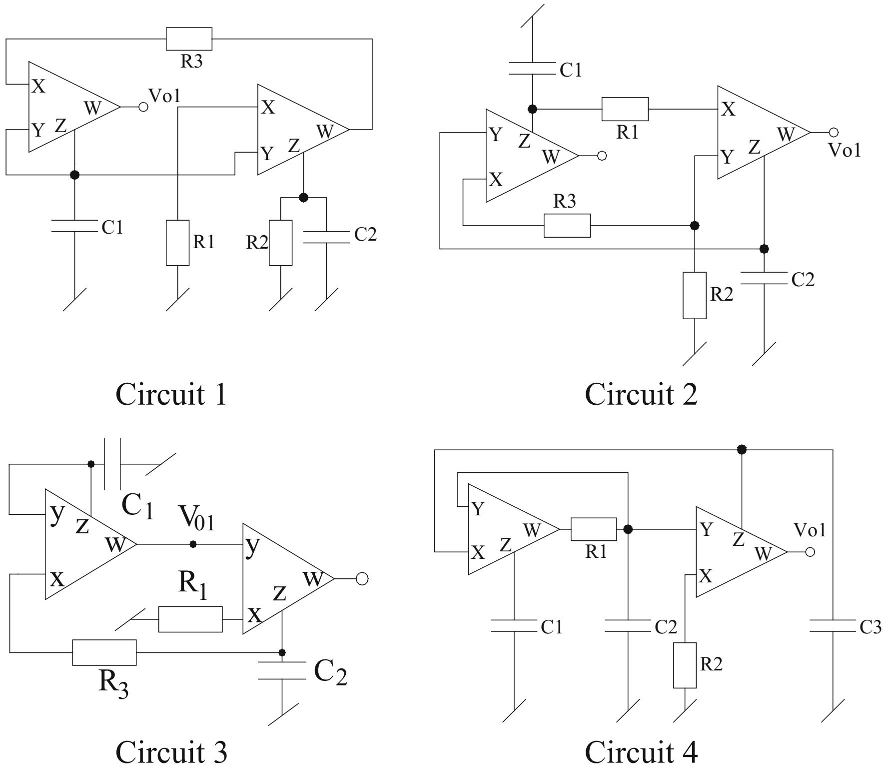


Fig. 5.7 Some two-CFOA oscillators with different tuning laws proposed by Bhaskar and Senani [40]

Table 5.2 CO, FO and design constraints for the oscillator circuits of Fig. 5.7

Circuit no	CO	FO
1	$\frac{R_3}{R_2} = \frac{C_2}{C_1}$	$\omega_0 = \sqrt{\frac{\left(1-\frac{R_1}{R_2}\right)}{C_1 C_2 R_1 R_3}}; \frac{R_1}{R_2} < 1$
2	$\frac{C_1}{C_2} = 1 + \frac{R_3}{R_2}$	$\omega_0 = \frac{1}{\sqrt{C_1 C_2 R_1 (R_2+R_3)}}$
3	$C_1 = C_2$	$\omega_0 = \frac{1}{\sqrt{C_1 C_2 R_1 R_3}}$
4	$C_1 = C_3$	$\omega_0 = \frac{1}{\sqrt{C_1 C_2 R_1 R_2}}$

converting the [A] matrices into node equations, and finally, synthesizing the resulting node equations by physical circuits using CFOAs and RC elements. These oscillator circuits possess the following features:

1. Circuits 1–2 have tuning laws that do not conform to Eq. (5.28) and yet these circuits do possess features (a) and (b).
2. Circuit 3 is the only oscillator circuit realizable with a bare minimum of only four passive components. This circuit, however, can be treated to be the CFOA-version of a similar circuit using CCIIs described earlier in references [63] and [64] but by contrast, this CFOA version has the advantage of providing buffered outlets from the output of either CFOA.
3. Circuit 4 although employs three grounded capacitors but still qualifies for feature (c).

Like most CFOA-based circuits, the influence of the parasitic impedances of the CFOAs can be reduced in the circuits of Fig. 5.6, also by selecting the external resistors to be much larger than the input resistance r_x of the X-terminal and smaller than the parasitic output resistance R_p looking into the compensation Z-terminal of the CFOA and the external capacitances to be larger than the parasitic output capacitance C_p of the CFOAs.

An analysis of the frequency stability properties of the circuits reveals that the frequency stability factors are largest for the circuits shown in Fig. 5.7 similar to other circuits contained in [4, 13, 36].

It may be noted that oscillator 1 contains a difference term in the expression for FO of type $\omega_0 = \frac{\sqrt{1-n}}{\mathrm{RC}}$ where n is the frequency-controlling resistor ratio. In view of the difference term in the numerator of ω_0, this oscillator qualifies to be used as a very low-frequency oscillator (i.e. 1 Hz or lower) by choosing n such that $(1-n)$ can be made as small as possible. On the other hand, oscillators 3 and 4 appear to be suitable for capacitance measurement methods such as those of [60–62]. In such a case, the unknown capacitance can be connected in place of C_1, and then the known variable capacitance C_2 is to be varied until the circuit just starts (or stops) oscillating. For further details of this method of measurement, the reader is referred to references [60–62].

An interesting alternative two-CFOA-two-GC-based SRCO was introduced by Liu and Tsay [10] which is shown in Fig. 5.8. This circuit has the attractive feature of offering an additional scaling factor $\left(\frac{R_3}{R_4}\right)$ in the expression for FO which makes it

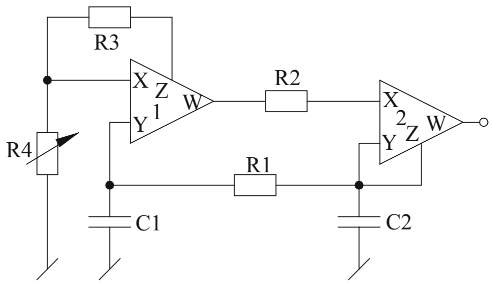


Fig. 5.8 Another SRCO grounded-resistor controlled using GCs proposed by Liu and Tsay [10]

suitable for generation of very low frequency of oscillations, like of [10], without having to require large RC values.[3]

The CO and FO of this circuit are given by

$$\mathrm{CO}:(C_1+C_2)=C_1\left(\frac{R_1}{R_2}\right)\ \text{and FO}:\omega_0=\sqrt{\frac{1}{2C_1C_2R_1R_2}\left(\frac{R_3}{R_4}\right)}\tag{5.29}$$

Thus, the CO can be controlled by R_1, while FO can be varied through R_4 and/or R_3.

Another grounded-resistor controlled sinusoidal oscillator using CFOAs was proposed by Martinez, Sabadell, Aldea in [19] which is shown in Fig. 5.9.

The CO and FO for this circuit are given by

$$R_A=R_1\ \text{and}\ \omega_0=\sqrt{\frac{1}{C_1C_2R_2R_A}}\tag{5.30}$$

In this oscillator also, both CO and FO can be controlled independently through grounded resistors R_1 and R_2, respectively.

Martinez and Sanz [20] presented a method for generation of variable-frequency sinusoidal oscillators based on two-integrator loop and presented a sinusoidal oscillator using two CFOAs. However, this circuit is quite similar to the SRCO circuit of Senani and Singh [13].

Tangsrirat and Surakampontorn [21] proposed a single-resistance-controlled quadrature oscillator using two CFOAs which is shown in Fig. 5.10. The circuit, however, does not have both capacitors grounded as prevalent in circuits described earlier.

[3] It is interesting to point out that although not explicitly mentioned in [10], this circuit can be considered to be derivable from a two-op-amp-GC SRCO published earlier in [18] by realizing the negative-impedance converter (NIC) therein by a CFOA without requiring any resistors and thereby simplifying the circuit as shown in Fig. 5.9.

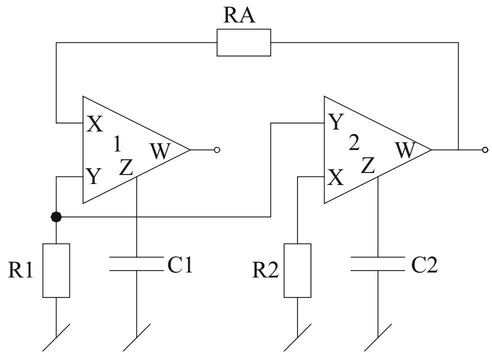


Fig. 5.9 A grounded resistor-controlled sinusoidal oscillator [19]

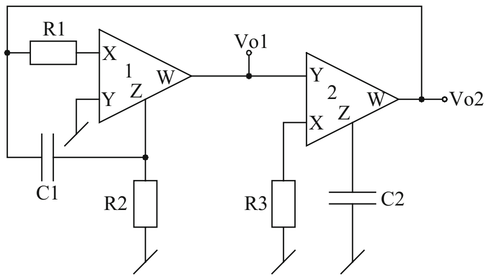


Fig. 5.10 SRCO proposed by Tangsrirat and Surakampontorn [21]

The CO and FO for this circuit are given by

$$R_2C_1 = R_3C_2 \text{ and } \omega_0 = \sqrt{\frac{1}{R_1R_3C_1C_2}} \tag{5.31}$$

In this case, the CO can be adjusted by R_2 without affecting the oscillation frequency, while ω_0 can be adjusted by R_1.

Another quadrature oscillator using two CFOAs has been proposed by Hou and Wang [22], which has been obtained by a circuit transformation proposed by them through which OTA-C circuits can be transformed into CFOA-RC circuits.

Two very interesting two-CFOA-GC SRCOs were shown to result from two novel active gyrators presented in [35] which are reproduced here in Fig. 5.11.

The CO and FO for these circuits are given by

$$R_0 = R_2 \text{ and } f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{C_1C_2R_1R_2}} \tag{5.32}$$

Including the various parasitic impedances of CFOAs into account, the nonideal CO and FO for these circuits are given by

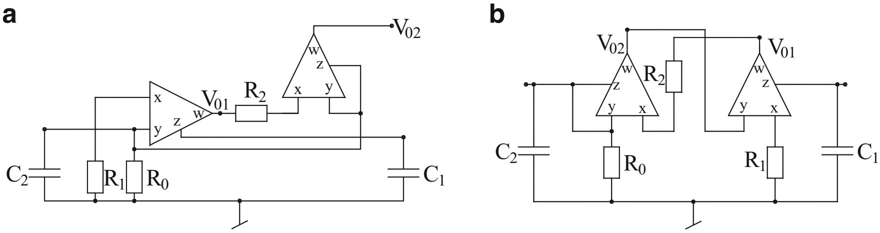


Fig. 5.11 Two new active gyrators and novel active-RC SRCOs [35]

$$\frac{1}{R_0'} = \frac{\left[1 - \frac{C_2' R_2'}{C_1' R_{p1}}\right]}{R_2} \tag{5.33}$$

$$f_0 = \frac{1}{2\pi\sqrt{C_1' C_2' R_1' R_2'}} \left(1 - \frac{R_1' R' C_2'}{R_{p1}^2 C_1'}\right)^{1/2} \tag{5.34}$$

where

$$C_1' = C_1 + C_{p1}, C_2' = C_2 + C_{p2} + C_{y1} + C_{y2},$$
$$R_1' = (R_1 + R_{x1}), R_2' = (R_2 + R_{x2}), \frac{1}{R_0'} = \left(\frac{1}{R_0} + \frac{1}{R_{p2}} + \frac{1}{R_{y1}} + \frac{1}{R_{y2}}\right)$$

for the circuit of Fig. 5.11a and

$C_1' = C_1 + C_{p1}, C_2' = C_2 + C_{p2} + C_{y2}, R_1' = (R_1 + R_{x1}), R_2' = (R_2 + R_{x2}),$ $\frac{1}{R_0'} = \left(\frac{1}{R_0} + \frac{1}{R_{p2}} + \frac{1}{R_{y2}}\right)$ for the circuit of Fig. 5.11b.

It may be noted that CO and FO are still independently controllable through R_0 and R_1, respectively. Thus, the independent single-resistance controllability of CO and FO remains intact even when all the parasitics of CFOAs are accounted! These circuits can, in fact, be regarded to be two interesting new variants of the SRCO of [29] (Fig. 3a therein). However, whereas the circuit of [29] cannot accommodate all parasitics of the two CFOAs, the present circuits are superior in that they can accommodate the influence of all the parasitics!

A systematic method of realization of very low-frequency (VLF) oscillators was proposed by Elwakil [33], which requires a passive resistor and one active (negative) resistor which modifies the expression for oscillation frequency such that it contains a difference term. By keeping this difference term as small as possible, low-frequency oscillations can be achieved.

An exemplary CFOA-based VLF oscillator circuit presented by Elwakil is shown in Fig. 5.12.

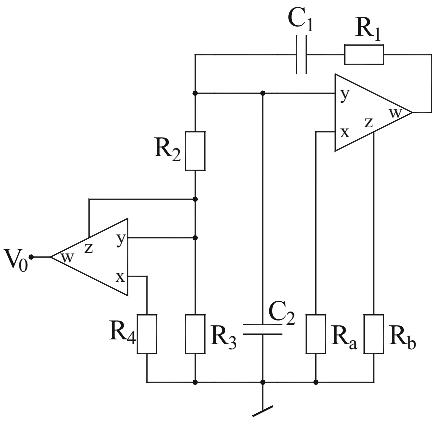


Fig 5.12 A very low-frequency SRCO proposed by Elwakil [33]

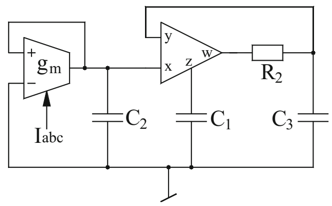


Fig. 5.13 Electronically controllable CFOA-OTA SRCO proposed by Abuelma'atti and Al-Shahrani [37]

It is seen that in the above circuit, which essentially is a Wein bridge oscillator, the usual R_2 of the Wien bridge has been replaced by:

$$R_{eq} = \left(R_2 + \frac{R_3 R_4}{R_3 - R_4}\right) \tag{5.35}$$

which makes the oscillation frequency as

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{(R_3 - R_4)}{C_1 C_2 R_1 [R_2 (R_3 - R_4) + R_3 R_4]}} \tag{5.36}$$

Thus, by keeping the term $(R_3 - R_4)$ as small as possible (making $R_3 > R_4$), very low-frequency oscillations can be achieved.

Lastly, we present an oscillator circuit containing mixed sources; the circuit using a CFOA and an OTA was presented by Abuelma'atti and Al-Shahrani [37] and is shown in Fig. 5.13.

In this circuit, the CFOA along with resistor R_2 and grounded capacitors C_1 and C_3 realizes a parallel combination of negative capacitance and a frequency-dependent positive conductance, so that an oscillator is obtained by terminating the input port into a parallel combination of a positive capacitor and a negative resistor which is simulated by the OTA. As a consequence, the CO and FO of the final oscillator circuit are given by

$$\text{CO}: \ C_1 = C_2 \text{ and } \text{FO}: \omega_0 = \sqrt{\frac{g_m}{C_1 C_3 R_2}} \tag{5.37}$$

Since $g_m = I_B/2V_T$, where I_B is the external dc bias current, the FO is electronically controllable through I_B. Although the circuit is noncanonic due to the employment of three capacitors, this drawback is compensated by the fact that all of them are grounded.

Three types of sinusoidal oscillators each using three CCII+, two to four resistors and two grounded capacitors have been presented by Martinez, Sabadell-Aldea, and Celma [58], whose practical workability has been verified by AD844 type CFOAs.

Apart from the above, Soliman [34] has presented a number of sinusoidal oscillators out of which only one is two-resistor-two capacitor-two CFOA-based circuit which is a quadrature oscillator. The other two sinusoidal oscillators use two CFOA-4R-2C configurations having identical expressions for CO and FO. However, in these circuits, only CO can be controlled by a single resistance.

5.8 Design of SRCOs Using CFOA Poles

Using the traditional voltage-mode op-amp compensation poles as a useful parameter, a class of circuits called "active-R" filters and oscillators were investigated since 1973. Likewise, several researchers have investigated "active-R" circuits using CFOA poles also. Thus, taking into account the parasitic capacitance of the *Z*-terminal of the CFOA into design, a variety of active-R oscillators have been reported in technical literature. Some representative circuits in this category are highlighted in this section.

Some of the first active-R oscillators using CFOAs were investigated by Liu, Chang, and Wu [25]. A two-CFOA oscillator circuit from [25] is shown in Fig. 5.14.

If R_1, $R_3 \gg R_x$ and R_2, $R_4 \ll R_p$, the condition of oscillation of this oscillator is found to be

$$\frac{R_6}{R_3(R_5 + R_6)} = \frac{1}{R_4} + \frac{1}{R_2} \tag{5.38}$$

whereas the frequency of oscillation is found to be

$$\omega_0 = \frac{1}{C_p}\sqrt{\frac{1}{R_2 R_4} + \frac{R_8}{R_1 R_3 (R_7 + R_8)} - \frac{R_6}{R_2 R_3 (R_5 + R_6)}} \tag{5.39}$$

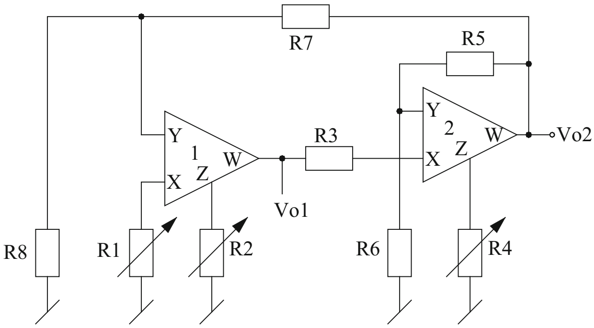


Fig. 5.14 An exemplary active-R sinusoidal oscillator proposed by Liu, Chang, and Wu [25]

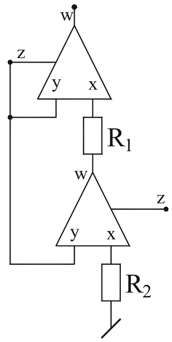


Fig. 5.15 Active-R SRCO proposed by Singh and Senani [35]

It is, thus, found that the FO can be independently controlled by the resistor R_1 which does not appear in the CO.

Using only two resistors, two sinusoidal oscillator circuits, each employing two CFOAs, were proposed by Singh and Senani [35], one of which is shown here in Fig. 5.15.

The CO and FO for this circuit are found to be

$$\text{CO}: R_2' = \frac{R_p}{\left(\frac{2R_p}{R_y}+1\right)+\left(1+\frac{2C_y}{C_p}\right)} \tag{5.40}$$

$$\text{FO}: f_0 = \frac{1}{2\pi C_p R_p}\sqrt{\frac{2R_p}{R_y}+\frac{\frac{R_p}{R_2'}\left(\frac{R_p}{R_1'}-1\right)+1}{\left(1+\frac{2C_y}{C_p}\right)}} \tag{5.41}$$

where $R_1' = (R_1 + R_{x1})$, $R_2' = (R_2 + R_{x2})$, R_p, C_p are the z-port parasitics, R_y, C_y are the y-port parasitics and R_x is the x-port parasitic input resistance. It may be noted that the FO can be varied through R_1 which does not feature in the CO.

A number of two-CFOA active-R SRCO circuits can be easily obtained from those oscillators presented in Sects. 5.6 and 5.7 where each CFOA has a capacitor connected from its Z-pin to ground. Thus, from such circuits active-R VCOs can be

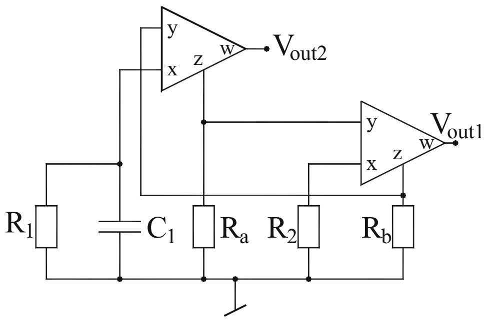


Fig. 5.16 Two CFOA and one GC partially active-R SRCO proposed by Abuelma'atti and Al-Zaher [31]

obtained by deleting the external capacitors connected at the Z-terminals of the CFOAs and employing in their places the Z-pin parasitic capacitances in the design.

We now describe another class of oscillators which incorporate the CFOA pole to reduce the number of external capacitors partially. Such circuits have been called "partially active-R" oscillators in literature. Using CFOA pole, a variety of partially active-R oscillators have been reported using two CFOAs as well as using only a single CFOA.

Abuelma'atti and Al-Zaher [31] derived, from a general two-CFOA configuration, a set of six partially active-R oscillators, one specific case from which is presented in Fig. 5.16.

The resulting circuit employs a single external grounded capacitor and four external grounded resistors. By straightforward analysis, the FO and CO for this circuit are found to be

$$\omega_0 = \sqrt{\frac{\left(\frac{1}{R_a}+\frac{1}{R_{p1}}+\frac{1}{R_{y2}}\right)\left(\frac{1}{R_b}+\frac{1}{R_{p2}}+\frac{1}{R_{y1}}\right)-\frac{1}{R_1R_2}}{C_{p1}C_{p2}}} \tag{5.42}$$

$$\frac{C_1}{R_2} = \left(\frac{1}{R_a}+\frac{1}{R_{p1}}+\frac{1}{R_{y2}}\right)C_{p2} + \left(\frac{1}{R_b}+\frac{1}{R_{p2}}+\frac{1}{R_{y1}}\right)C_{p1} \tag{5.43}$$

Thus, FO is tunable by R_1 without affecting CO.

A low-component oscillator using one external capacitor and the pole of the CFOA was presented by Liu, Chang, and Wu [25] which is shown in Fig. 5.17.

If $R_1 \gg R_x$ and $R_3 \ll R_p$, the CO and FO for this circuit are given by

$$\text{CO}: \frac{C}{R_1} = \frac{C}{R_3} + \frac{C_p}{R_2} \text{ and FO}: \ \omega_0 = \sqrt{\frac{1}{C_pCR_2R_3}} \tag{5.44}$$

Although this oscillator has the advantage of using only four passive elements FO cannot be independently controlled.

On the other hand, four sinusoidal oscillators, each consisting of two capacitors, a single CFOA with its pole accounted in the design, were proposed by

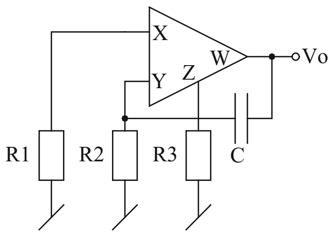


Fig. 5.17 CFOA-pole-based sinusoidal oscillator introduced by Liu, Chang, and Wu [25]

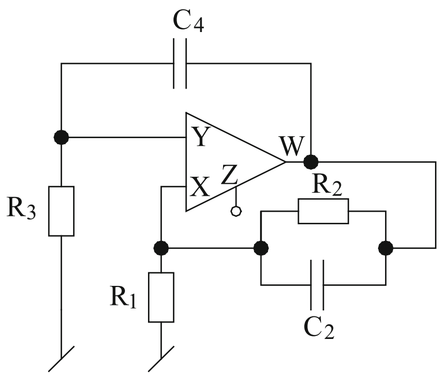


Fig. 5.18 An SRCO using CFOA pole proposed by Abuelma'atti, Farooqi, and Al-Shahrani [9]

Abuelma'atti, Farooqi, and Al-Shahrani [9]. A single-resistance-controlled oscillator out of this set is shown in Fig. 5.18 for which the CO and FO are given by

$$\text{CO} : \frac{1}{C_{\mathrm{p}}}(C_4R_3 - C_2R_1) = R_1\left(1 + \frac{C_4R_3}{C_{\mathrm{p}}R_{\mathrm{p}}}\right) \tag{5.45}$$

$$\text{and FO} : \quad \omega_0 = \sqrt{\frac{1}{C_pC_4R_pR_3}\left(1 + \frac{R_p}{R_2}\right)} \tag{5.46}$$

It is thus seen that in this circuit CO can be controlled by R_1, whereas FO is independently controlled by R_2. It has been demonstrated in [9] that using $R_1 = R_3 = 100\,\Omega$, $C_2 = 4.7$ pF, and $C_4 = 22$ pF and using AD844-type CFOA biased with ± 15 V dc power supplies, variable-frequency oscillations up to 27.5 MHz with peak-to-peak voltage of 2–9 V were successfully obtained using this circuit.

We now present two single CFOA-based partially active-R oscillators. Consider the circuit of Fig. 5.19 which was proposed by Abuelma'atti and Khan [85].

Taking into account the various parasitic impedances of the CFOA, the approximate expressions for the FO and CO for this circuit assuming $C_3 \gg C_{\mathrm{p}}$ and $C_3 \gg C_y$ are given by [85]

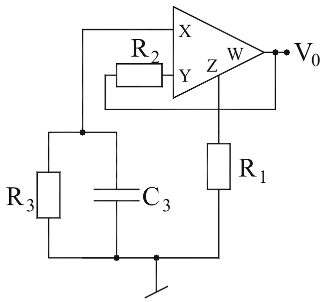


Fig. 5.19 An SRCO using single CFOA and one GC proposed by Abuelma'atti and Khan [85]

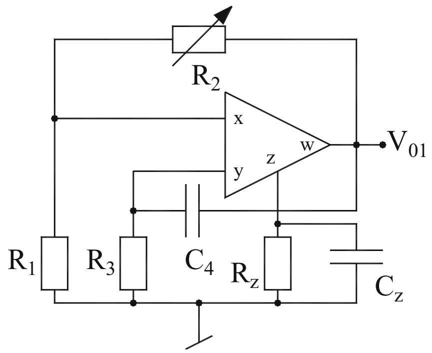


Fig. 5.20 A single-CFOA partially active-R oscillator proposed by Abuelma'atti and Al-Shahrani [11]

$$\mathrm{FO}: \omega \cong \sqrt{\frac{\left(1-\frac{R_1}{R_3}\right)}{C_3 R_x\left(C_y R_2 + C_{\mathrm{p}} R_1\right)}} \tag{5.47}$$

$$\mathrm{CO}: R_1 \cong R_3 \tag{5.48}$$

Thus, FO is adjustable by R_2 and/or the grounded capacitor C_3 without disturbing CO.

A special case of this circuit obtained by shorting R_2 is of interest. In this situation ω_0 reduces to

$$\omega_0 \cong \sqrt{\frac{\left(1-\frac{R_1}{R_3}\right)}{\left(C_3 R_x C_{\mathrm{p}} R_1\right)}} \tag{5.49}$$

This reduced component-count circuit employs only three external passive components where FO can be tuned by the grounded capacitor without affecting CO.

This circuit has been successfully used to generate a frequency of 3.12 MHz by employing AD844-type CFOAs biased with ±15 V supply with $R_1 = 200\,\Omega$, $R_2 = 3\ \mathrm{k}\Omega$, $R_3 = 203\,\Omega$, and $C_3 = 200$ pF (see [85]).

Yet another single-CFOA-based partially active-R oscillator was proposed by Abuelma'atti and Al-Shahrani [11] which is shown here in Fig. 5.20.

A straightforward analysis gives its CO and FO as

$$\mathrm{CO}: \frac{C_4}{R_1C_\mathrm{p}} = \frac{C_4}{R_\mathrm{p}C_\mathrm{p}} + \frac{1}{R_3} \text{ and } \omega_0 = \sqrt{\frac{\left(1+\frac{R_\mathrm{p}}{R_2}\right)}{C_4C_\mathrm{p}R_3R_\mathrm{p}}} \tag{5.50}$$

It is seen that the circuit provides independent controllability of FO through a single-variable resistance R_2. The measured results based upon the realization of the circuit using AD844 have been found to be in good agreement with the theory [11].

For a number of other active-R and partially active-R sinusoidal oscillators using CFOAs, the reader is referred to [2, 11, 14, 41].

5.9 Quadrature and Multiphase Oscillators Using CFOAs

A number of authors have investigated the realization of quadrature and multiphase oscillators. Here, we present some representative circuits.

Toker, Ozcan, Kuntman, and Cicekoglu [80] presented a quadrature oscillator using two all pass sections (Fig. 5.21), out of which one is having transfer function

$$T_1(s) = -\frac{R_2}{2R_1}\left(\frac{1-sCR_1}{1+sCR_2/2}\right) \tag{5.51}$$

with all-pass realizable with $R_1 = R_2/2$, while the second one has its transfer function

$$T_2(s) = \left(\frac{1-sCR_2}{1+2sCR_1}\right), \tag{5.52}$$

needing $2R_1 = R_2$ for all pass realization.

We now present a multiphase active-R oscillator circuit proposed by Wu, Liu, Hwang, and Wu [23] which is shown in Fig. 5.22. This circuit has the following merits: it (i) uses only parasitic poles of CFOAs thereby making it suitable for high-frequency oscillations and monolithic IC fabrication due to complete elimination of

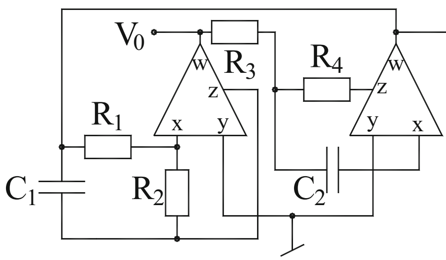


Fig. 5.21 Two all-pass sections cascaded for realization of Quadrature Oscillator proposed by Toker, Ozcan, Kuntman, and Cicekoglu [80]

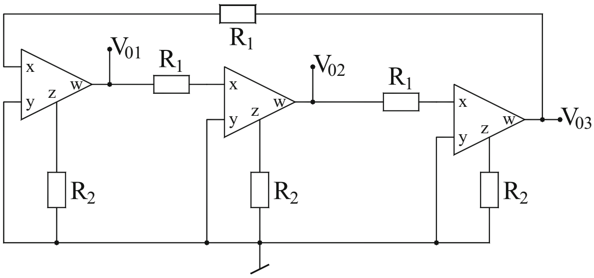


Fig. 5.22 Multiphase oscillator (for $n = 3$) proposed by Wu, Liu, Hwang, and Wu [23]

external capacitors (ii) exhibits large output voltage swing and (iii) has moderately low total harmonic distortion (THD).

Assuming all the CFOAs to be identical, the loop gain for an n-phase oscillator can be expressed as

$$L(s) = \left(\frac{-G_0}{1 + \frac{s}{\omega_b}}\right)^n \text{ where } \omega_b = 1/R_b C_p,\ R_b = R_2//R_p \text{ and}$$
$$G_0 = R_b/R_a,\ R_a = R_1 + R_x \tag{5.53}$$

The frequency and condition of oscillation are given by

$$\omega_0 = \omega_b \tan\frac{\pi}{n} \text{ and } R_b \geq R_a \sec\frac{\pi}{n} \tag{5.54}$$

As a special case, the condition of oscillation and frequency of oscillation of a three-phase sinusoidal oscillator ($n = 3$) can be expressed as

$$\omega_0 = \omega_b\sqrt{3} \text{ and } R_b \geq 2R_a \text{ or } R_1 \leq \frac{R_2 R_p}{2(R_2 + R_p)} - R_x \tag{5.55}$$

Thus, the circuit of Fig. 5.22 produces the maximum and the minimum oscillation frequencies when $R_1 = 0$ and $R_1 = (R_p/2) - R_x$, respectively.

5.10 SRCOs Providing *Explicit* Current Output

The advent of numerous CM filters and other signal processing circuits has obviously created the demand for the design of oscillators providing an explicit current output (ECO) from a high-output impedance node which would be useful as signal generators to test various current-mode circuits. In the literature, there have been a number of investigations [38, 65–67, 86] on realizing oscillators with ECO using other building blocks, such as first-generation current conveyor [38, 65], differential difference current conveyors [66], differential difference complementary current feedback amplifier [67], four-terminal floating nullor [86], unity gain voltage and current followers [68]; however, none of these building blocks are available commercially yet. Since a CFOA of AD844 type does have a current output terminal and is commercially available, ECO oscillators made from CFOAs are obviously important.

In this section, we show the extension of the state-variable approach of synthesis [27] of oscillators to synthesize systematically current-mode sinusoidal oscillators with *explicit current output* using CFOAs as active building blocks. Of course, current-mode oscillators based on CCII+ can also be implemented by AD844; however, oscillators using exclusively CCII+ which have the capability providing explicit current output are known to employ three CCII+, whereas none of the circuits in earlier works [65–67, 86] have been realized with CFOAs.

In the frame work of the state-variable methodology, the various conditions for noninteracting controls of CO and FO as well as to provide ECO have the following requirements:

(a) The expression of $(a_{11} + a_{22})$ should either not have terms containing R_2 or they should be cancelled out. Thus, in $(a_{11} + a_{22})$, there should be two terms left with opposite signs involving R_1 and R_3.
(b) Similarly, to have FO independent of R_1, the expression $(a_{11}a_{22} - a_{12}a_{21})$ should either not have the terms containing R_1 or they should be cancelled out. Thus, FO should be a function of resistors R_2 and R_3 only (along with C_1 and C_2).
(c) The Z-terminal of at least on CFOA must be left unused to enable availability of ECO.

Let us now construct the required $[A]$ matrix by choosing $a_{11} = \frac{1}{C_1R_1}$, $a_{22} = -\frac{1}{C_2R_3}$ which satisfy the requirement (a). Now, choosing $a_{12} = -\frac{1}{C_1}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$, $a_{21} = \frac{1}{C_2R_3}$, we can satisfy the requirement (b). The required $[A]$ matrix, therefore, takes the following form:

$$[A] = \begin{bmatrix} \dfrac{1}{C_1R_1} & -\dfrac{1}{C_1}\left(\dfrac{1}{R_1} + \dfrac{1}{R_2}\right) \\ \dfrac{1}{C_2R_3} & -\dfrac{1}{C_2R_3} \end{bmatrix} \tag{5.56}$$

which results in the following CO and FO:

$$\text{CO} : R_1 = \frac{C_2}{C_1}R_3 \tag{5.57}$$

$$\omega_0 = \frac{1}{\sqrt{C_1C_2R_2R_3}} \tag{5.58}$$

From the Eq. (5.56) the following node equations are obtained:

$$C_1 \frac{dx_1}{dt} = \frac{x_1 - x_2}{R_1} - \frac{x_2}{R_2} \tag{5.59}$$

$$C_2 \frac{dx_2}{dt} = \frac{x_1 - x_2}{R_3} \tag{5.60}$$

For meeting the specific objective of having an *explicit current-mode output*, Eqs. (5.59) and (5.60) are to be implemented using CFOAs keeping in mind that the Z-terminal of at least one of the CFOAs has to be left unutilized. The circuit, thus formulated from Eqs. (5.59) and (5.60), takes the form of circuit (a) of Fig. 5.23. The circuit also clearly shows the various current segments of the Eqs. (5.59) and (5.60) as marked in the circuit 1 of Fig. 5.23a, to enable the understanding of the synthesis procedure.

The CO and FO for the circuit of Fig. 5.23b remain the same as given in Eqs. (5.57) and (5.58), whereas for the circuit of Fig. 5.23c, these are found to be

$$\text{CO} : R_1 = \left(\frac{C_2}{C_1 + C_2}\right) R_3 \text{ and FO} : \omega_0 = \frac{1}{\sqrt{C_1 C_2 R_2 R_3}} \tag{5.61}$$

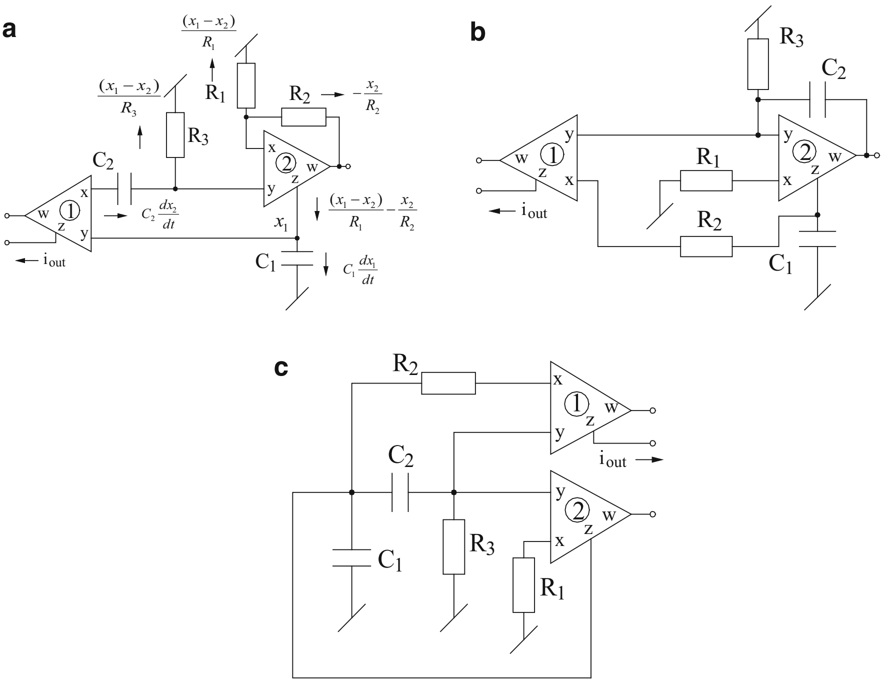


Fig. 5.23 Some exemplary SRCOs providing explicit current output proposed by Gupta, Sharma, Bhaskar, and Senani (adapted from [45] © 2010 John Wiley& Sons, Ltd.)

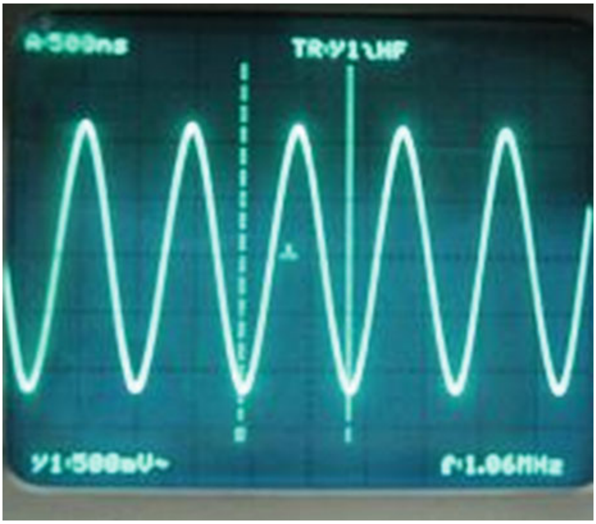


Fig. 5.24 A typical waveform generated from the oscillator of Fig. 5.23c (1.06 MHz, 2.3 Vpp). Component values: $C_1 = C_2 = 100$ pF, $R_1 = 404\ \Omega$, $R_2 = R_3 = 1\ \mathrm{k}\Omega$ (adapted from [45] © 2010 John Wiley& Sons, Ltd.)

In all the three circuits of Fig. 5.23, explicit current output is available from the Z-terminal of the first CFOA.

The parasitics of the CFOA would make the nonideal expression of the oscillation frequencies of the circuits of Fig. 5.23 to be different than their ideal counterparts and would eventually limit the operation of the oscillations at higher frequencies. It has been shown in [45] that with judicious choice of component values, oscillations around 1 MHz range are attainable with these circuits. Figure 5.24 shows a typical waveform (1.06 MHz, 2.3 V (p–p)) obtained from the oscillator of Fig. 5.23c using AD 844 CFOAs biased with ±15 V DC supplies.

Some other interesting explicit current output oscillators (ECO) using CFOA are discussed next.

Two single CFOA-based ECO oscillators were presented by Senani and Sharma [39]. One of the circuits from [39] is shown in Fig. 5.25.

The CO and FO for this circuit are given by

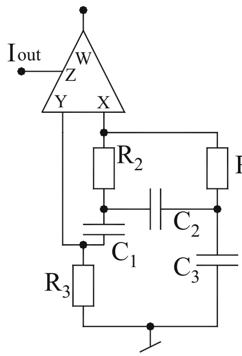
$$\text{CO}:\ R_3 = 6(R_1 + R_2);\ \text{provided}\ C_1 = C_2 = C_3 = C \tag{5.62}$$

$$\text{and FO}: f_0 = \frac{1}{2\pi C\sqrt{3R_1R_2}} \tag{5.63}$$

Although the circuit has the advantage of using a single CFOA, a drawback of this circuit is that it has three capacitors. On the other hand, f_0 can be varied through a potentiometer by changing “n” (which is the ratio R_1/R_2), while their sum $(R_1 + R_2)$ and, hence, the CO remains invariant. However, CO can be adjusted independently through the resistor R_3.

A family of two-CFOA-based SRCOs with explicit current output has been proposed recently by Lahiri, Jaikla, and Siripruchyanun [50], out of which three

Fig 5.25 CM oscillator using a single CFOA proposed by Senani and Sharma [39]



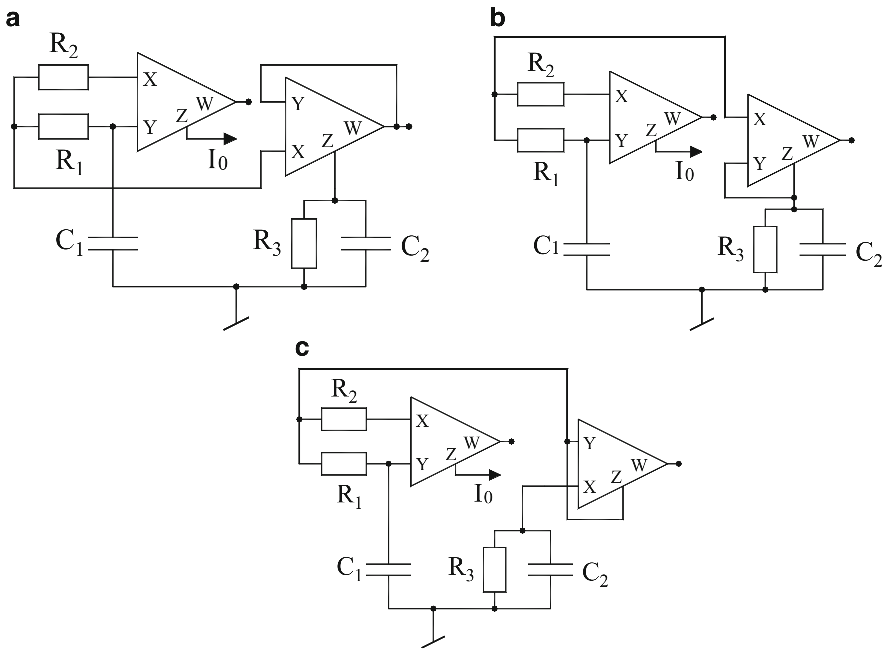


Fig. 5.26 Explicit-current output second order sinusoidal oscillator proposed by Lahiri, Jaikla, and Siripruchyanun (adapted from [50] © 2011 Elsevier)

exemplary SRCOs are shown in Fig. 5.26. These circuits have the advantage of employing both grounded capacitors as desirable for IC implementation.

All the three circuits of Fig. 5.26 are characterized by the following equations:

$$\text{CO} : R_3 \geq R_2 \text{ and FO} : f_0 = \frac{1}{2\pi C\sqrt{R_1 R_3}};\ C_1 = C_2 = C \tag{5.64}$$

It may be noted that CO is controllable by R_2, while FO can be varied through resistor R_1.

In view of the interest on quadrature oscillators, as explained in an earlier section, quadrature oscillators with explicit current outputs are of obvious interest. From any given second-order CFOA oscillator employing one or two CFOAs, such oscillators can be devised by using either or both of the following two simple operations:

(i) Current in a grounded resistor can be sensed by using an additional CFOA as current follower with a virtual ground at *x*-input terminal (with Y-terminal connected to ground) and making an explicit current output from *Z*-terminal of the CFOA.
(ii) A voltage signal can be tapped from an appropriate node by using an additional CFOA connected as a voltage-to-current converter with its Y-terminal connected to the chosen node, with a resistor connected from *X*-terminal to ground and current output taken from its *Z*-terminal.

Two ECO SRCOs were obtained in this manner from two known two-CFOA-based SRCO of Gupta and Senani [29] (see the circuit of Fig. 5.3a therein) by Lahiri, Jaikla, and Siripruchyanun [79] and are shown in Fig. 5.27. It may be noted that both the circuits of Fig. 5.27 utilized both the techniques (i) and (ii) outlined above.

The ongoing search for newer topologies of SRCOs with ECO using CFOAs may lead to the circuits which might be useful as test oscillators for verifying various current-mode signal processing circuits[4] such as current-mode filters, current-mode precision rectifiers, etc., to which the proposed kind of circuits can be interfaced without any additional hardware.

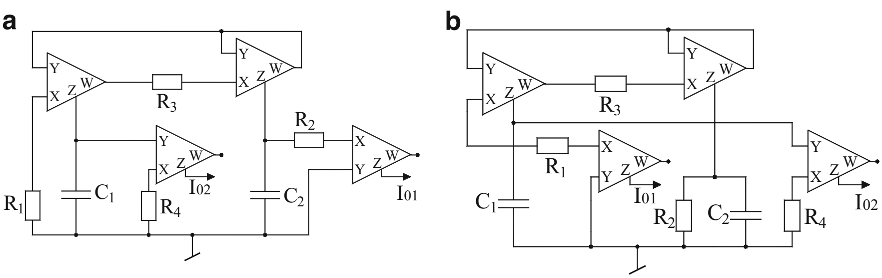


Fig. 5.27 Two ECO SRCOs proposed by Lahiri, Jaikla, and Siripruchyanun [79] derived from the circuit of Gupta and Senani [29]

[4] In spite of the criticism of [30], the class of circuits described in this section is interesting and important in their own right.

5.10.1 CFOA SRCOs Exhibiting Fully Uncoupled Tuning Laws

For CFOA-based canonic realization SRCOs, a minimum of five passive components, namely, three resistors and two capacitors, are needed. Furthermore, for providing tuning laws such that both CO and FO can be controlled/adjusted by two independent resistors and for ensuring both capacitors to be grounded, as preferred for IC implementation, one requires at least two CFOAs. When the various nonidealities/parasitics of the CFOAs are accounted for in the analysis of such circuits, the theoretically derived independence of CO and FO vanishes due to the frequency-controlling resistor also getting involved in the nonideal expression for the CO.

The term fully decoupled has been used for those SRCOs in which CO and FO are decided by two completely different sets of components, such that none of the components involved in CO are also involved in FO and vice versa. Such fully uncoupled SRCOs are characterized by the tuning laws of the following types:

$$\text{CO}: (R_1 - R_2) \leq 0 \tag{5.65}$$

$$\text{and FO } f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{C_1 C_2 R_3 R_4}} \tag{5.66}$$

From the above, it follows that such oscillator circuits would need at least four resistors along with two capacitors. Such "fully-uncoupled" SRCOs, however, are not feasible with only two active elements and call for the employment of at least three active elements as in [34, 36].

The circuits presented by Soliman [34] and the one proposed by Bhaskar [36] appear to be the only known circuits employing CFOAs which provide the type of tuning laws represented in Eqs. (5.65) and (5.66). The Soliman's circuit from [34] is shown in Fig. 5.28.

whereas the circuit presented by Bhaskar [36] is shown in Fig. 5.29.

Both the circuits employ an exactly the same number of active and passive components. The ideal CO and FO for the circuit of Fig. 5.28 are given by

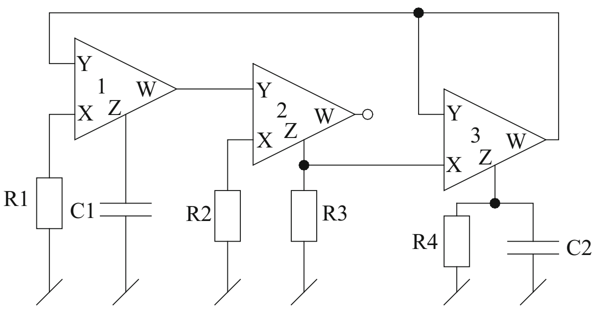


Fig. 5.28 Fully uncoupled oscillator proposed by Soliman (adapted from [34] © 2000 Springer)

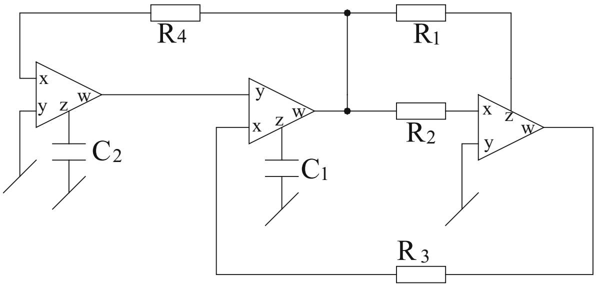


Fig. 5.29 Fully uncoupled oscillator proposed by Bhaskar [36]

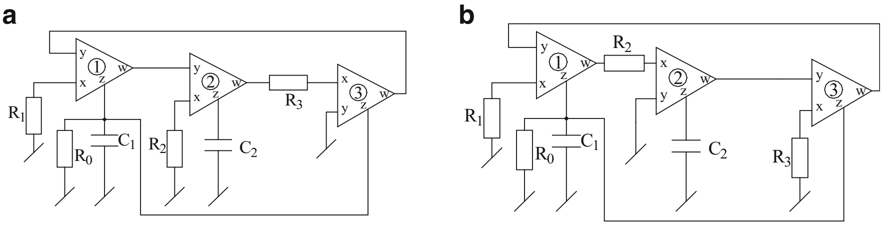


Fig. 5.30 Fully uncoupled SRCOs proposed by Bhaskar, Gupta, Senani and Singh (adapted from [51] © 2012 Springer)

$$\text{CO}: \ R_3 = R_4 \text{ and FO}: f_0 = \frac{1}{2\pi\sqrt{C_1C_2R_1R_2}} \tag{5.67}$$

whereas those for the circuit of Fig. 5.29 are given by

$$\text{CO}: \ R_1 = R_2 \text{ and FO}: f_0 = \frac{1}{2\pi\sqrt{C_1C_2R_3R_4}} \tag{5.68}$$

It has been shown in [51] that the above-described fully uncoupled oscillators from [34, 36] also fail to retain the independent controllability of FO under the influence of nonideal parasitic impedances of CFOAs as all the four resistors employed in the oscillators appear in the nonideal expressions of both CO and FO, thereby completely disturbing the intended property.

In the following, we show two circuits from [51] which retain the independent controllability of FO even under the influence of CFOA parasitic impedances. These circuits are shown in Fig. 5.30.

Assuming the CFOAs to be characterized by $i_y = 0$, $v_x = v_y$, $i_z = i_x$ and $v_w = v_z$, both the circuits are governed by a common characteristic equation (CE) given by

$$s^2 + \frac{s}{C_1}\left(\frac{1}{R_0} - \frac{1}{R_1}\right) + \frac{1}{C_1C_2R_2R_3} = 0 \tag{5.69}$$

From this characteristic equation, the CO and FO are found to be

$$\text{CO} : (R_1 - R_0) \leq 0 \text{ and FO} : \ \omega_0 = \frac{1}{\sqrt{C_1 C_2 R_2 R_3}} \tag{5.70}$$

For an evaluation of the nonideal performance of these circuits, we consider the finite input resistance R_{xi} at the x-port, $i = 1$–3, parasitic components R_{yi} in parallel with $1/sC_{yi}$ at the y-port and parasitic components R_{zi} in parallel with $1/sC_{zi}$ at the z-port of all the CFOAs $i = 1$–3. Analysis reveals that in both cases, the nonideal CE of both the circuits continues to remain second order which can be related to the fact that in both the circuits, the Z-terminals of the two CFOAs are common. The non-ideal CO and FO for both the circuits have been found to be:

For the circuits of Fig. 5.30a, b

CO:

$$\begin{aligned}&(C_2 + C_{z2})\left\{\frac{1}{R_0} - \frac{1}{R_1 + R_{x1}} + \frac{1}{R_{y1}} + \frac{1}{R_{z1}} + \frac{1}{R_{z3}}\right\} \\ &+ \frac{C_1 + C_{z1} + C_{y1} + C_{z3}}{R_{z2}} \leq 0\end{aligned} \tag{5.71}$$

FO:

$$\begin{aligned}f'_0 = {}& \frac{1}{2\pi}\sqrt{\frac{1}{C_1 C_2 R_2 R_3}}\left(\frac{1}{\left(1 + \frac{C_{y1} + C_{z1} + C_{z3}}{C_1}\right)\left(1 + \frac{C_{z2}}{C_2}\right)}\right)^{1/2} \\ &\times \left[\begin{array}{l}\frac{1}{\left(1 + \frac{R_{x2}}{R_2}\right)\left(1 + \frac{R_{x3}}{R_3}\right)} \\ + \frac{R_2 R_3}{R_{z2}}\left\{\frac{1}{R_0} + \frac{1}{R_{y1}} + \frac{1}{R_{z1}} + \frac{1}{R_{z3}} - \frac{1}{R_1 + R_{x1}}\right\}\end{array}\right]^{1/2}\end{aligned} \tag{5.72}$$

From Eqs. (5.71) and (5.72), it is observed that in both the circuits, the frequency-controlling resistors R_2 and R_3 do not come into the nonideal expressions for CO; therefore, the independent controllability of FO remains intact even under the influence of the nonideal parameters/parasitic of the CFOAs employed.

Both the circuits can be converted into voltage-controlled oscillators by replacing the frequency-controlling resistors R_2 and/or R_3 by FET-based or CMOS voltage-controlled-resistors (VCR). In this context, the floating nature of the frequency-controlling resistor does not pose any difficulty since it is well known that grounded/floating VCRs could be realized with exactly the same number of active and passive elements, for instance, see [17, 70–72].

The frequency stability properties of the circuits can be evaluated by using the definition of frequency stability factor (S_F) as where $S_F = \left.\frac{d\phi(u)}{du}\right|_{u=1}$ $u = \frac{\omega}{\omega_o}$ is the normalized frequency and $\phi(u)$ denotes the phase function of the open-loop transfer function. Taking $C_1 = C_2 = C$, $R_0 = R_1 = R_2 = R$ and $R_3 = R/n$, S_F for the proposed oscillators is found to be $S_F = 2\sqrt{n}$. On the other hand, if both the resistors R_2 and R_3 are varied simultaneously, i.e., $R_2 = R_3 = R/n$, then S_F becomes $2n$. This figure appears to be the highest attained so far as compared to all SRCOs known earlier. Thus, both the circuits of Fig. 5.30 offer very high-frequency stability factors for larger values of n.

Lastly, it must be mentioned that the generation of any new three-CFOA-two-GC-four-resistor fully uncoupled oscillators which, apart from retaining independent controllability of FO, can also retain independent controllability of CO even under the influence of the nonideal parameters/parasitic of the CFOAs appears to be an interesting but challenging problem, which is open to investigation.

5.11 Voltage-Controlled Oscillators Using CFOAs and FET-Based VCRs

Voltage-controlled oscillators (VCOs) are important building blocks in many instrumentation, electronic and communication systems. A well-known method of realizing a sinusoidal VCO is to realize a single-resistance-controlled oscillator (SRCO) and then replace the frequency-controlling resistor by an FET-based voltage-controlled-resistor (VCR). This section discusses a number of CFOA-based VCO configurations possessing different advantageous features.

It may be recalled that a number of CFOA-based SRCOs have been described in the earlier sections of this chapter in which FO can be independently controlled through a single-variable resistor without affecting the CO. Thus, in a specific CFOA-based SRCO, this frequency-controlling resistor may be either grounded (having one terminal connected to ground) or floating (none of the resistor terminals connected to ground).

From a careful examination of the family of 14 SRCOs presented earlier in [29], it is found that in all, only seven structures are suitable for being converted into VCOs by replacing frequency-controlling resistors by appropriate grounded and floating VCRs. Out of this set of seven, only five circuits can be realized using no more than two CFOAs, which are shown here in Fig. 5.31. The CO and FO of these VCOs are given in Table 5.3. In all the cases, the CO can be adjusted by R_1 without affecting FO whereas FO is controllable independently by the resistance Rm and hence, by V_C.

Taking into consideration the finite X-terminal input resistance Rx and parasitic impedance at the Z-terminal (consisting of a resistance Rp in parallel with a capacitance Cp) it is found that the influence of CFOA parasitics on the performance of these oscillators can be reduced by choosing external resistances to be much greater than Rx and much smaller than Rp and selecting external capacitors to be much larger than Cp.

From the frequency stability analysis it has been found [42] that all the VCOs of Fig. 5.31 enjoy good frequency stability properties.

Some experimental results for the oscillators of Fig. 5.31a, e from [42] are shown here in Figs. 5.32, 5.33, and 5.34.

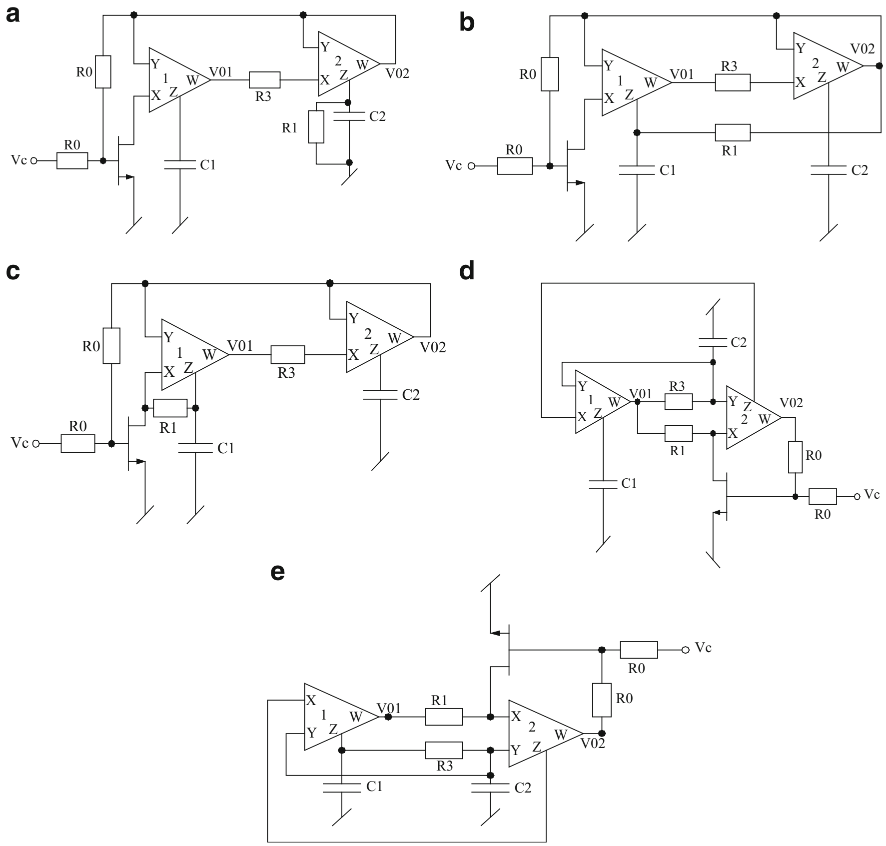


Fig. 5.31 Voltage-controlled oscillators proposed by Gupta, Bhaskar and Senani (adapted from [42] © 2009 Elsevier)

Table 5.3 CO and FO for the oscillators of Fig. 5.31

VCO number	Condition of oscillation	Frequency of oscillation
(a)	$R_1 = R_3$	In all the cases, the frequency of oscillation is given by $f_0 = \frac{1}{2\pi\sqrt{C_1 C_2 R_m R_3}}$
(b), (d)	$R_1 = \frac{C_2}{C_1} R_3$	
(c)	$R_1 = 2\frac{C_2}{C_1} R_3$	
(e)	$R_1 = R_3\left(\frac{C_1+C_2}{C_2}\right)$	
where $R_m = r_{DS} = \frac{2V_p^2}{I_{DSS}\left(V_C - 2V_p\right)}$		

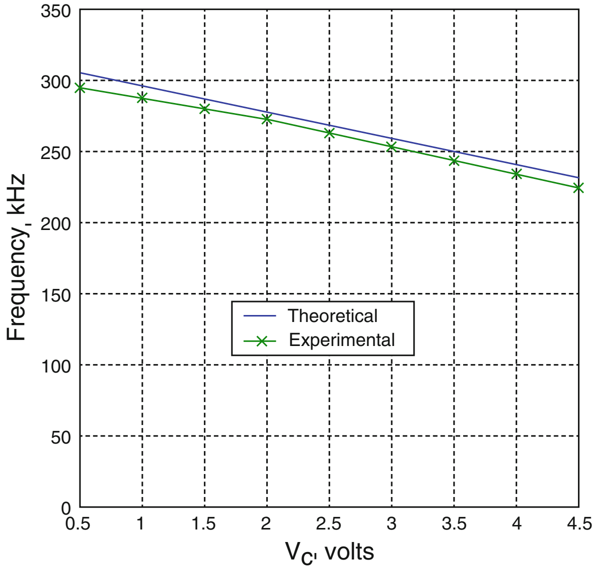


Fig. 5.32 Variation of oscillation frequency with control voltage V_C for the VCO of Fig. 5.31a (adapted from [42] © 2009 Elsevier)

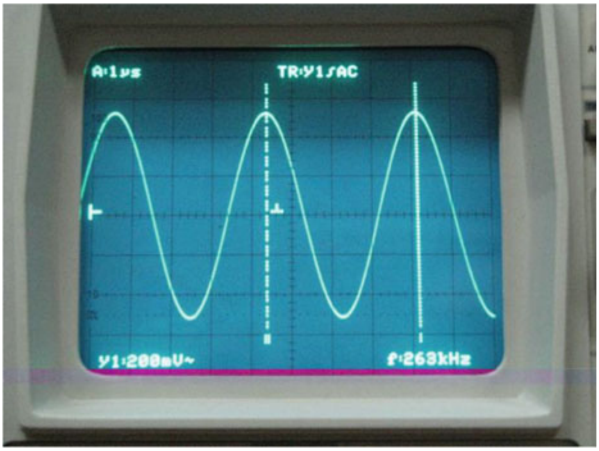


Fig. 5.33 A typical waveform generated from VCO of Fig. 5.31a, $f_0 = 263$ kHz, $V_0 = 1$ V (p–p); $V_{CC} = \pm 6$ VDC (adapted from [42] © 2009 Elsevier)

5.12 Realization of Linear VCOs Using CFOAs

In this section, we present CFOA-based sinusoidal oscillators capable of providing *linear* tuning law of the type $f_0 \propto V_C$ between the oscillation frequency f_0 and an external control voltage V_C.

It may be noted that although the VCOs, presented in Sect. 5.11, are simple, they do not provide a linear tuning law between the control voltage (say, V_C) and the oscillation frequency f_0, since the tuning law for such VCOs is of the form

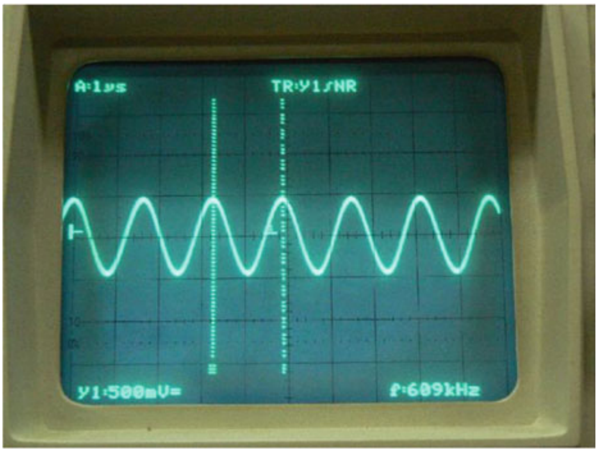


Fig. 5.34 A typical waveform generated from VCO of Fig. 5.31e, $f_0 = 609$ kHz, $V_0 = 1$ V (p–p); $V_{CC} = \pm 5$ V DC (adapted from [42] © 2009 Elsevier)

$$f_0 \propto \sqrt{\frac{1}{r_{ds}}} \tag{5.73}$$

$$\text{where } r_{ds} = \frac{V_p^2}{I_{DSS}(V_c - 2V_p)} \text{(in case of VCR realized by JFET)} \tag{5.74}$$

$$\text{and } r_{ds} = \frac{1}{2k(V_{gs} - V_{th})} \text{(in case of VCR realized by MOSFET)} \tag{5.75}$$

In Eqs. (5.74) and (5.75), V_p is the pinch-off voltage of the JFET and I_{DSS} is the saturated drain current (at $V_{gs} = 0$) of the FET, V_C is the control voltage, and V_{th} is the threshold voltage and $K = \mu_S C_{OX}\left(\frac{W}{L}\right)$ where μ_s is the surface mobility, C_{ox} is the capacitance of the gate electrode per unit area, and $\left(\frac{W}{L}\right)$ is the aspect ratio of the MOSFET.

However, if an oscillator is evolved with two analog multipliers (AM) appropriately embedded into the oscillator configuration, to enable independent control of the oscillation frequency through an external control voltage V_C applied as a common multiplicative input to both the multipliers, this technique may give rise to a linear tuning law of the form

$$f_0 \propto V_C \tag{5.76}$$

Based upon this idea, some VCO configurations have been proposed by various researchers in the past ([53–55] and the reference cited therein) employing traditional voltage-mode op-amps (VOA) and AMs. These circuits, however, require larger number of resistors (5–12) and their usability is limited to low-frequency ranges due to the finite GBP and limited slew rate of VOAs.

Bhaskar, Senani, and Singh [56] have presented a family of eight VCOs each employing two CFOAs which offer the following features: (i) availability of linear

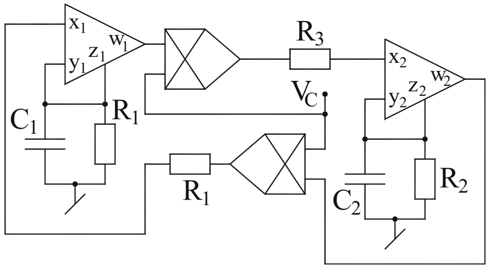


Fig. 5.35 Linear VCO proposed by Bhaskar, Senani, and Singh [56]

tuning law between FO and control voltage, (ii) employment of both GCs which is attractive from the view point of IC implementation, (iii) employment of a small number of resistors (three to four only), (iv) a relatively larger operating frequency range (several hundreds of kHz as compared to a few kHz only in case of VCOs based upon traditional voltage-mode op-amps), (v) the possibility of appropriate scaling of the frequency range independently through a single resistor, and (vi) effective accommodation of the various parasitic impedances of the CFOAs/AMs.

It was found that out of the eight circuits presented therein, circuit H of [56] has least value of percentage error in the frequency resulting due to the effect of the various parasitic impedances of the CFOAs. This circuit from [56] is shown in Fig. 5.35.

The circuit of Fig. 5.35 is characterized by the following CO and FO:

$$\text{CO}: (R_3 - R_2) \leq 0 \text{ and FO}: \omega_0 = \sqrt{\frac{k_1 k_2}{C_1 C_2 R_1 R_3}} \tag{5.77}$$

K_i, $i = 1$–2 either equal to +1 or −1. For more details readers are referred to [56].

In the following we will outline the state-variable technique advanced by Gupta, Bhaskar, and Senani [73], described in detail in Sect. 5.10 of this chapter through which a class of VCOs have been obtained by making different choices of the parameters of the [A] matrix of the state-variable characterization of VCOs.

If we choose the [A] matrix of the oscillator (to be synthesized) in the following form:

$$[A]_1 = \begin{bmatrix} 0 & -\dfrac{1}{C_1 R_2} \\ \dfrac{1}{C_2 R_3} & \dfrac{1}{C_2}\left(\dfrac{1}{R_1} - \dfrac{1}{R_3}\right) \end{bmatrix} \tag{5.78}$$

the CE formulated from the above matrix leads to the following CO and FO:

$$\text{CO}: R_1 = R_3 \tag{5.79}$$

$$\mathrm{FO}: \omega_0 = \frac{1}{\sqrt{C_1 C_2 R_2 R_3}} \tag{5.80}$$

Since the oscillators to be derived have to incorporate at least two analog multipliers (characterized by $V_o = K\left(\frac{V_1 V_2}{V_{\text{ref}}}\right)$, where V_1 and V_2 are two inputs, V_{ref} is the reference voltage set internally, usually at 10 V in case of AD534 and K can be set up +1 or -1 by grounding appropriate input terminals). For enabling linear control of oscillation frequency through an external control voltage V_C (to be applied as a common multiplicative input to both the analog multipliers) the selection of the matrix parameters outlined above needs to be modified to include the term β $\left(\beta = \frac{V_C}{V_{\text{ref}}}\right)$. However, it needs to be done in such a way that the final expression of the CO does not contain the term β and the expression of FO is modified to $f_0 = \frac{\beta}{2\pi\sqrt{C_1 C_2 R_2 R_3}}$ so that we can get $f_0 \propto \beta$ and hence, $f_0 \propto V_{C.}$

The parameters of the matrix [A] given in Eq. (5.77) can now be *modified* in one of the following ways: (i) by including β^2 as a factor of a_{12} or a_{21} (ii) β as a factor a_{12} as well as that of a_{21} (iii) β as a factor in all the parameters of matrix [A]

Using the *modification* (i) we get the following node equations:

$$C_1 \frac{\mathrm{d}x_1}{\mathrm{d}t} = -\frac{\beta^2 x_2}{R_2} \tag{5.81}$$

$$C_2 \frac{\mathrm{d}x_2}{\mathrm{d}t} = \frac{x_1 - x_2}{R_3} + \frac{x_2}{R_1} \tag{5.82}$$

If we employ two AMs and two CFOAs and try to implement above NEs, we can synthesize VCO-1 as shown in Fig. 5.36. On this diagram itself various current components of Eqs. (5.81) and (5.82) make the synthesis clear.

Now if we apply *modification* (iii), we get the following node equations:

$$C_1 \frac{\mathrm{d}x_1}{\mathrm{d}t} = -\frac{\beta x_2}{R_2} \tag{5.83}$$

$$C_2 \frac{\mathrm{d}x_2}{\mathrm{d}t} = \frac{\beta x_1 - x_2}{R_3} + \frac{x_2}{R_1} \tag{5.84}$$

The implementation of Eqs. (5.83) and (5.84) gives us VCO-2 which is shown in Fig. 5.36. It may be noted that if polarity of β is inverted in both the AMs, the synthesized circuit still remains VCO with the same CO and FO. Based on the state-variable methodology explained above, along with any or all the modifications (i)–(iii) suggested above, a number of VCOs have been generated from the suitable matrices in [73]. Two other circuits from the set of 12 VCOs generated in

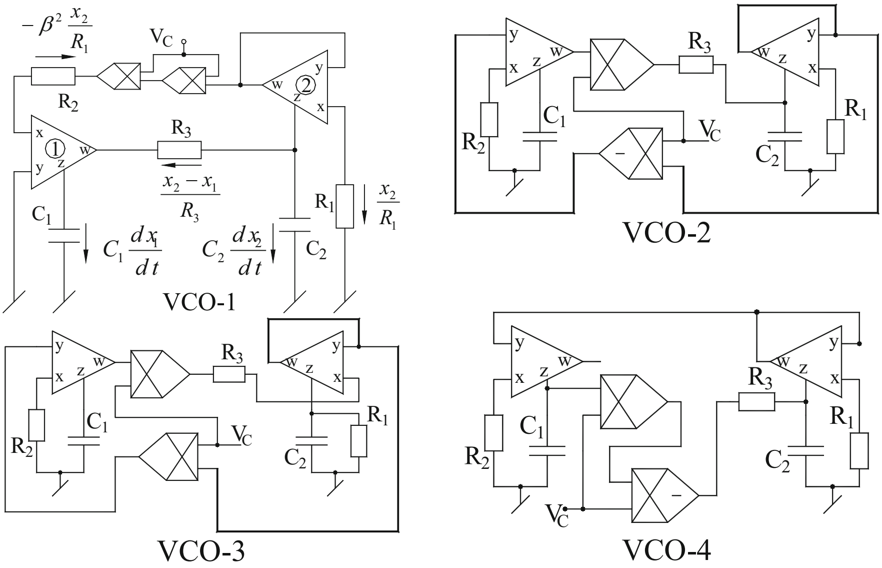


Fig. 5.36 Linear VCOs proposed by Gupta, Bhaskar, Senani, and Singh [73]

[73] are shown here in Fig. 5.36. The expressions for CO and FO of all the circuits are given by

$$\text{CO}: R_1 = R_3 \text{ and FO}: \ f_0 = \frac{\beta}{2\pi\sqrt{C_1C_2R_2R_3}}, \tag{5.85}$$

The expressions for the frequency stability factors for all the VCOs have been evaluated in [73], and it has been found that in all the VCOs, S_F can be made large.

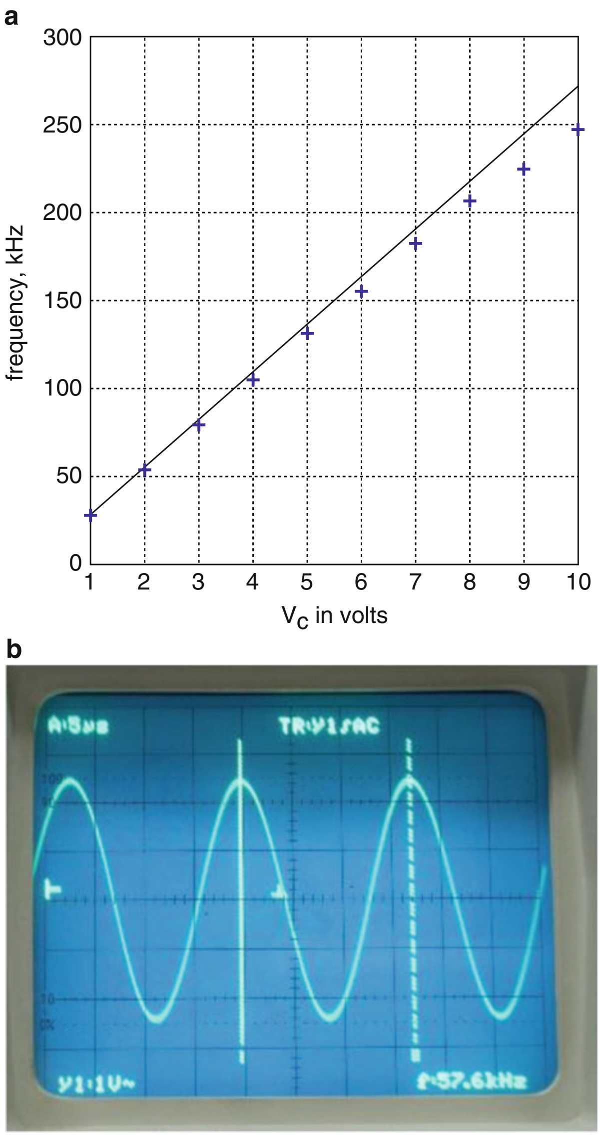
Variation of oscillation frequency with control voltage V_C (for $C_1 = C_2 = 50$ pF, $R_2 = R_3 = 10$ kΩ, with DC biasing $\pm V_{CC} = \pm 6$ V for CFOAs and $\pm V_{CC} = \pm 15$ V for AMs) and a typical waveform generated by VCO-1 of Fig. 5.36 are shown in Fig. 5.37a, b.

Since the above-described synthesis methodology yields circuits all of which requires at least two CFOAs, a question arises whether single-CFOA-based VCOs employing both grounded capacitors are feasible or not. An affirmative answer to this was given by Bhaskar, Senani, Singh, and Gupta [57] who presented two single-CFOA-based linear VCOs which are shown here in Fig. 5.38.

Choosing the same V_{ref} for both the multipliers, both the circuits have the CO and FO given by

$$\text{CO}: (C_1 - C_2) \leq 0 \text{ and FO}: \omega_0 = \beta\sqrt{\frac{1}{C_1C_2R_1R_2}} \tag{5.86}$$

Fig. 5.37 Experimental results for VCO-1 of Fig. 5.36. (**a**) Variation of oscillation frequency with control voltage V_C. (**b**) A typical waveform generated for $V_C = 2$ V



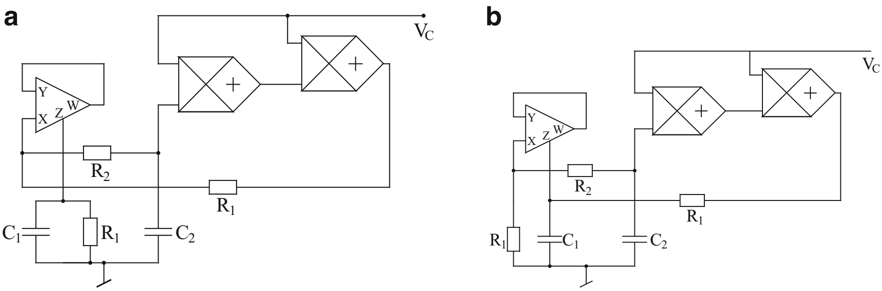


Fig. 5.38 Single-CFOA-based VCOs proposed by Bhaskar, Senani, Singh, and Gupta [57]

In nonideal analysis considering the various parasitic impedances of CFOA, namely, r_x, $R_y \,||(1/sC_y)$, $R_p||(1/sC_p)$, and the finite output resistance of the AMs, it is found that [57] the errors caused by the influence of CFOA and AMs parasitics can be kept small by choosing all resistors to be much larger than r_x but much smaller than R_p and choosing both the external capacitors to be much larger than C_p.

5.13 Synthesis of Single-CFOA-Based VCOs Incorporating the Voltage Summing Property of Analog Multipliers

In this section, we present a family of CFOA-based VCOs which employ a bare minimum number of active and passive components, namely, only one CFOA, only two multipliers (essential for obtaining linear control of FO), two/three resistors, and two capacitors.

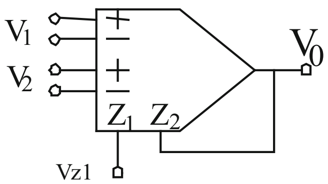
The AD534-type AM is a 4-port building block symbolically shown in Fig. 5.39 with three differential inputs (shown as V_1, V_2, and V_{Z1} in the present case) and one output V_o and is characterized by $V_o = K\left(\frac{V_1 V_2}{V_{ref}}\right) + V_Z$ where V_1 and V_2 are two inputs; V_{ref} is the reference voltage set internally, usually at 10 V in case of AD534, and K can be set up +1 or -1 by grounding appropriate input terminals; and V_Z is the voltage applied at the third input terminal of AM which appears at the output without any multiplying factor.

The VCO circuits presented in this section too are derived using the state-variable methodology.

If we choose the required $[A]$ matrix in the following form [74]:

$$[A] = \begin{bmatrix} \frac{1}{C_1}\left(\frac{1}{R_3} - \frac{1}{R_1}\right) & \frac{1}{C_1 R_2} \\ -\frac{1}{C_2 R_3} & 0 \end{bmatrix} \tag{5.87}$$

the CE formulated from the above matrix gives the following CO and FO:



$$\left\{ V_o = K\left(\frac{V_1 V_2}{V_{ref}}\right) + V_{Z1} \right\}$$

Fig. 5.39 Symbolic notation of an analog multiplier of AD534 type

$$\text{CO}: R_1 = R_3\ \omega_0 = \sqrt{\frac{1}{C_1 C_2 R_2 R_3}} \tag{5.88}$$

Since incorporation of at least two analog multipliers is needed in order to provide linear control of oscillation frequency through an external control voltage V_C which is applied in place of the second input V_2 of the multipliers (to be applied as a common multiplicative input to both the analog multipliers), the selection of the matrix parameters outlined above needs to be modified to include the term β $\left(\beta = \frac{V_C}{V_{\text{ref}}}\right)$. However, this needs to be done in such a way that the final expression of the CO does not contain the term β while the expression of FO is modified to $f_0 = \frac{\beta}{2\pi\sqrt{C_1 C_2 R_2 R_3}}$ so that we can get $f_0 \propto \beta$ and hence, $f_0 \propto V_C$.

Consider now the following [A] matrix

$$[A]_2 = \begin{bmatrix} \frac{\beta}{C_1}\left(\frac{1}{R_2} - \frac{1}{R_1}\right) & \frac{\beta}{C_1 R_1} \\ -\frac{\beta}{C_2 R_2} & 0 \end{bmatrix} \tag{5.89}$$

From matrix [A], we get the following node equations:

$$C_1 \dot{x}_1 = \frac{\beta(x_2 - x_1)}{R_1} + \frac{\beta x_1}{R_2} \tag{5.90}$$

$$C_2 \dot{x}_2 = -\frac{\beta x_1}{R_2} \tag{5.91}$$

Implementation of the above NEs employing two multipliers results in the circuit shown in Fig. 5.40. Various current components of Eqs. (5.90) and (5.91) have been marked in the circuit to make the synthesis clear.

It may be noted that the circuit of Fig. 5.40 requires three CFOAs along with two AMs to implement NEs of (5.90) and (5.91). We now show that by an alternative

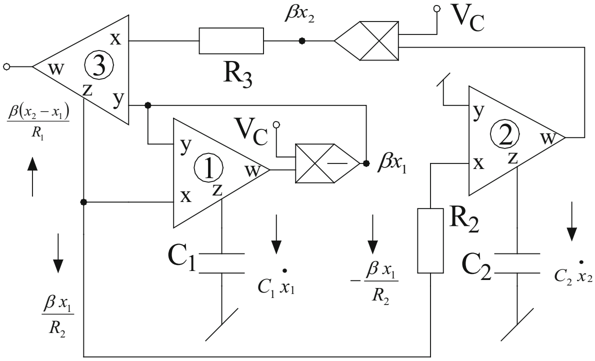


Fig. 5.40 VCO derived from matrix [A] [74] CO : $R_1 = R_2$
FO : $f_0 = \frac{\beta}{2\pi}\sqrt{\frac{1}{C_1 C_2 R_1 R_2}}$

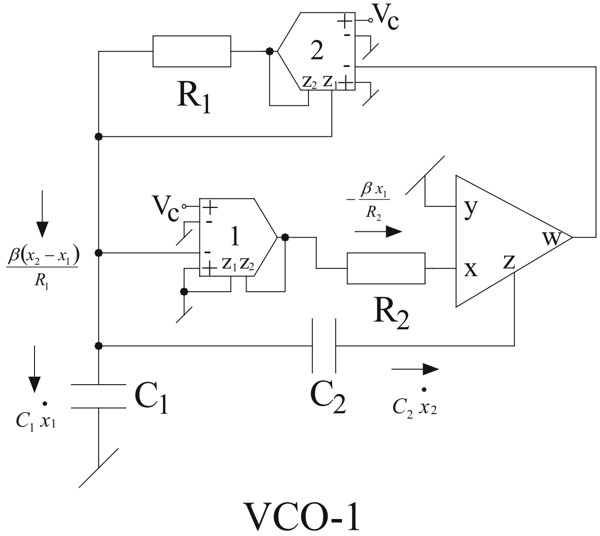


Fig. 5.41 VCO derived from matrix [A] of [74]

representation of Eqs. (5.90) and (5.91) and an appropriate incorporation of the Z-terminal of AM, it becomes possible to implement the modified NEs with only a single CFOA. Let us add (5.90) and (5.91) to create a new equation (Eq. (5.92) in the following), while we keep Eq. (5.91) as it is (shown as Eq. (5.93) in the following):

$$C_1\dot{x}_1 + C_2\dot{x}_2 = \frac{\beta(x_2 - x_1)}{R_1} \tag{5.92}$$

$$C_2\dot{x}_2 = -\frac{\beta x_1}{R_2} \tag{5.93}$$

When the voltage summing property of the AMs is effectively utilized, the implementation of Eqs. (5.92) and (5.93) then leads to a different circuit (shown as VCO-1 in Fig. 5.41) which requires only a single CFOA in contrast to the circuit of Fig. 5.40 needing three CFOAs.

It has been shown in [74] that in addition to Eq. (5.89), the following matrices are also suitable for the synthesis of the intended kind of VCOs:

$$[A]_2 = \begin{bmatrix} \frac{\beta}{C_1}\left(\frac{1}{R_2} - \frac{2}{R_1}\right) & \frac{\beta}{C_1R_1} \\ -\frac{\beta}{C_2R_2} & 0 \end{bmatrix} \tag{5.94}$$

$$[A]_3 = \begin{bmatrix} \dfrac{\beta}{C_1R_1} & -\dfrac{\beta}{C_1R_1} \\ \dfrac{2\beta}{C_2R_2} & -\dfrac{\beta}{C_2R_2} \end{bmatrix} \tag{5.95}$$

$$[A]_4 = \begin{bmatrix} \dfrac{\beta^2}{C_1R_1} & -\dfrac{\beta^2}{C_1R_1} \\ \dfrac{1+\beta^2}{C_2R_2} & -\dfrac{\beta^2}{C_2R_2} \end{bmatrix} \tag{5.96}$$

$$[A]_5 = \begin{bmatrix} 0 & \dfrac{\beta}{C_1R_2} \\ \dfrac{-\beta}{C_2R_3} & \dfrac{1}{C_2}\left(\dfrac{1}{R_3} - \dfrac{1}{R_1}\right) \end{bmatrix} \tag{5.97}$$

The circuits resulting from the synthesis based upon the above matrices are shown in Fig. 5.42. The CO and FO for the five VCOs of Figs. 5.41 and 5.42 are shown in Table 5.4.

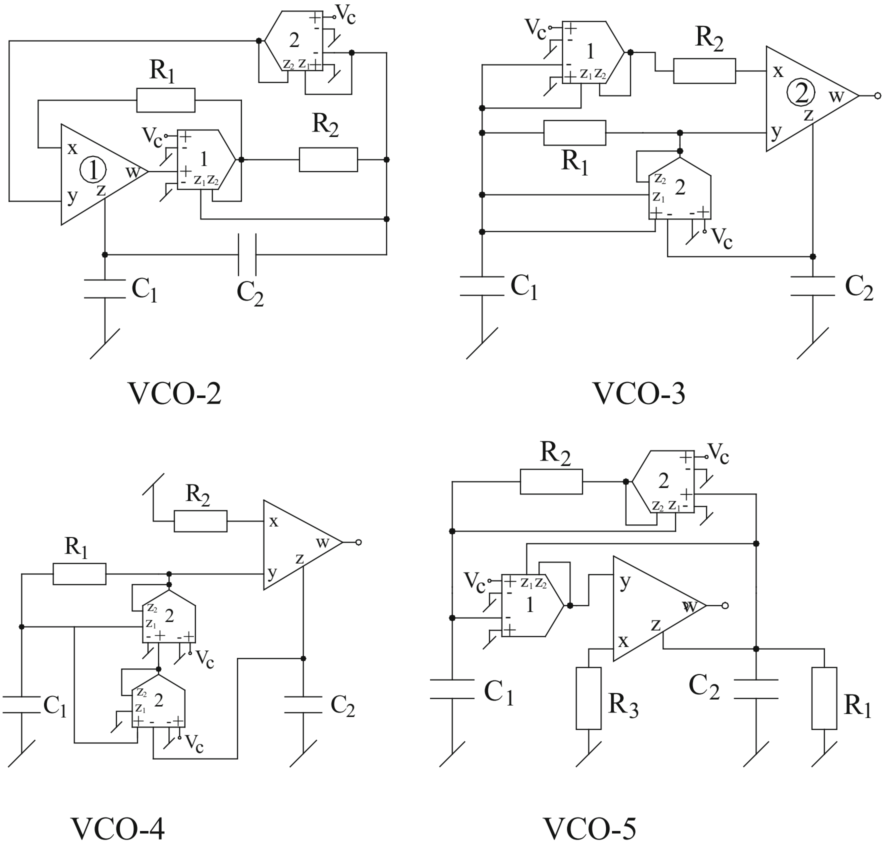


Fig. 5.42 GC-VCOs derived from matrices $[A]_2$ to $[A]_5$ (adapted from [74])

Table 5.4 CO and FO for the VCOs of Fig. 5.41 and Fig. 5.42

VCO	CO	FO
1	$R_1 = R_2$	$f_0 = \frac{\beta}{2\pi}\sqrt{\frac{1}{C_1C_2R_1R_2}}$
2	$R_1 = 2R_2$	$f_0 = \frac{\beta}{2\pi}\sqrt{\frac{1}{C_1C_2R_1R_2}}$
3	$R_1 = \frac{C_2}{C_1}R_2$	$f_0 = \frac{\beta}{2\pi}\sqrt{\frac{1}{C_1C_2R_1R_2}}$
4	$R_1 = \frac{C_2}{C_1}R_2$	$f_0 = \frac{\beta}{2\pi}\sqrt{\frac{1}{C_1C_2R_1R_2}}$
5	$R_1 = R_3$	$f_0 = \frac{\beta}{2\pi}\sqrt{\frac{1}{C_1C_2R_2R_3}}$

It may be noted that VCO-3 and VCO-4 offer the use of both grounded capacitors as desirable for IC implementation and out of these, VCO-4 also has one of the CO controlling resistor R_2 grounded. On the other hand, the VCO-5 possesses simultaneously almost all the desirable features, namely, completely noninteracting control of CO through R_1 (the CO controlling resistor being grounded), the employment of both grounded capacitors, and an additional degree of freedom via R_2 to scale up or down the frequency f_0 which is otherwise linearly controllable by β.

It is well known that the prominent non-idealities of the CFOAs include a finite nonzero input resistance r_x at x-port (typically around 50 Ω), y-port parasitic consisting of a parasitic resistance R_y (typically 2 MΩ) in parallel with a parasitic capacitance C_y (typically 2 pF) and z-port parasitic impedance consisting of a parasitic resistance R_p (typically 3 MΩ) in parallel with a parasitic capacitance C_p (typically, between 4 and 5 pF). In case of an analog multiplier, the finite nonzero output resistance $r_{\mathrm{out},}$ as per datasheet of AD534, is merely 1 Ω and hence, can be ignored in all the cases. On the other hand, the input impedance of the AM, being 10 MΩ, is sufficiently high and hence, its effect can be ignored. The errors caused by the influence of CFOA parasitics can be kept small by choosing all external resistors to be much larger than R_x but much smaller than R_p and choosing both external capacitors to be much larger than C_p.

A nonideal analysis shows that the independent control of CO and FO remains intact for VCO-5 even after consideration of the parasitics. Hence, VCO-5 is the best circuit from this viewpoint.

All the VCOs have been experimentally studied in [74] using AD844-type CFOAs and AD534-type AMs biased with ±12 V DC power supplies. The component values chosen were as under: For VCOs $R_1 = R_2 = 2$ kΩ, those for VCO-2 were chosen as $R_1 = 2$ kΩ and $R_2 = 1$ kΩ and for VCO-5, $R_1 = R_2 = R_3 = 1$ kΩ. Capacitor values for all the VCOs were taken as $C_1 = C_2 = 1$ nF. As per [74] it has been possible to generate oscillation frequencies from tens of kHz to several hundreds of kHz with tolerable errors in the frequency.

In the absence of an automatic amplitude control, it is normally expected that amplitude of oscillation would also vary when the frequency is varied through the external control voltage V_C. This has indeed been the case of VCO-1 and VCO-2. However, in case of VCO-3, VCO-4, and VCO-5, the peak-to-peak output voltage has been found to be constant, 17 $V_{p\text{-}p}$ in case of VCO-3 and VCO-4 and 10 $V_{p\text{-}p}$ in case of VCO-5 when V_C was varied from 1 to 10 V. Thus, VCO-3, VCO-4, and VCO-5 have been found to be superior than the other VCOs in this respect.

Some sample results of the proposed VCOs from [74] are shown in Fig. 5.43a, b which shows the variation of oscillation frequency with control voltage V_C for VCO-1 and VCO-5, respectively.

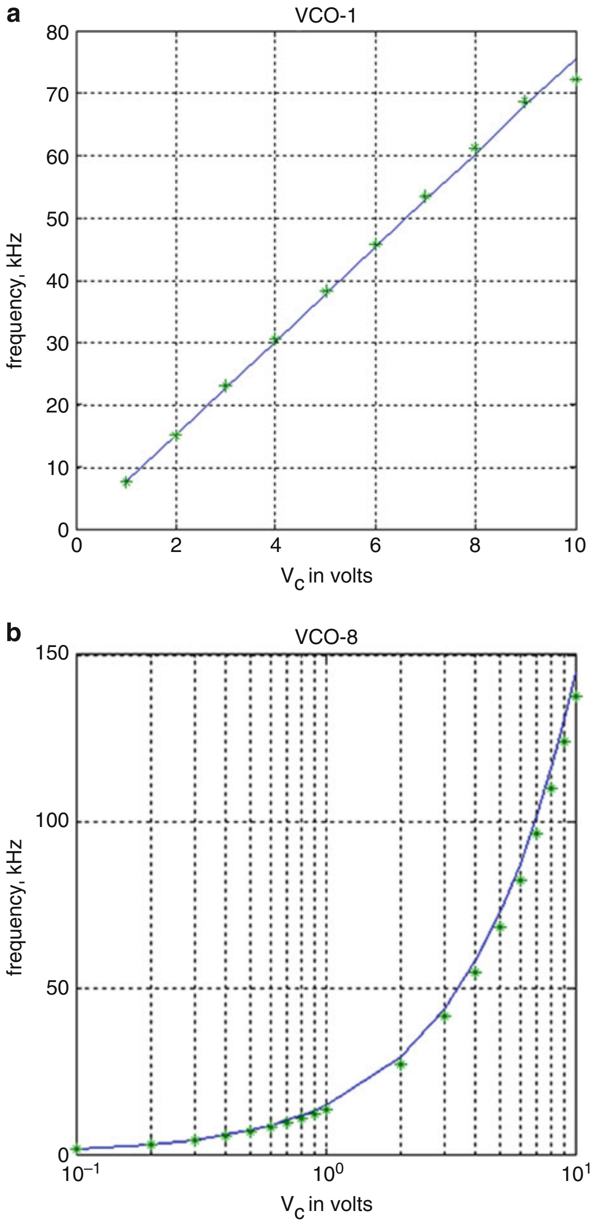


Fig. 5.43 Experimental results of the VCOs. (**a**) Variation of frequency with V_C for VCO-1. (**b**) Variation of frequency with V_C for VCO-5 (adapted from [74])

In view of a number of new CMOS CFOA and CMOS multiplier architectures being evolved in the recent literature, it may be expected that these ideas could possibly be carried over to the design of completely CMOS-based linear VCOs in future.

5.14 Concluding Remarks

A variety of sinusoidal oscillators including canonic SRCOs using a single CFOA, two-CFOA-based oscillators employing grounded capacitors, SRCOs with explicit current output and fully uncoupled SRCOs were discussed in this chapter. This was followed by VCOs employing nonlinearity-cancelled FETs through a general scheme. Subsequently, two different varieties of linear VCOs based upon the use of analog multipliers in conjunction with CFOAs were described.

Two interesting ideas worthy of further investigations and research are now worth mentioning. In Sect. 5.4 a family of eight single-CFOA oscillators was presented. Unfortunately, till date, there has not been any comparative study of all the eight circuits to determine as to which one of these is the best of the entire class and this problem is still open to investigation.

Another problem whose solution has not yet been found is whether or not a single-CFOA SRCO using only two grounded capacitors exists. In view of the fact that such a circuit using a single VOA does exist [76], the existence of a similar or better circuit with a single CFOA may not be ruled out. This constitutes another interesting problem for research.

In the last, it may be mentioned that CFOA oscillators continue to be a popular area of research and a number of new ideas have come into the literature even at the time of finalization of this chapter; these are briefly mentioned in Appendix 1.

Appendix 1: Some Recent Contributions to CFOA-Based Oscillators

Quadrature Oscillators Using Two CFOAs and Four Passive Components

The Circuit 3 of Fig. 5.7 apart from employing two GCs also has the attractive feature of employing minimum number of total passive components namely, only two resistors and capacitors. However, very recently, Chen, Wang, Ku, and Hsieh [84] have presented two new circuits belonging to this class, which are shown in Fig. 5.44.

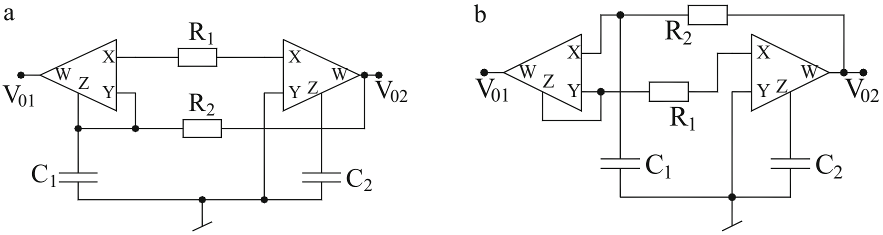


Fig. 5.44 New CFOA-based quadrature oscillators proposed by Chen, Wang, Ku, and Hsieh [84]

Both of these circuits are characterized by the same CE which is given by

$$s^2C_1C_2R_1R_2 + sC_2(R_1 - R_2) + 1 = 0 \tag{5.98}$$

so that the CO and FO are given by

$$\text{CO} : R_1 = R_2 \text{ and FO} : \omega_0 = \sqrt{\frac{1}{C_1C_2R_1R_2}} \tag{5.99}$$

Hence, CO can be adjusted by R_1or R_2, whereas FO can be adjusted by C_1 or C_2. Furthermore, analysis reveals that the two outputs V_{01} and V_{02} are related by

$$\frac{V_{01}(s)}{V_{02}(s)} = -\frac{1}{sC_2R_1} \tag{5.100}$$

Hence V_{01} and V_{02} are 90° apart in phase and hence, both the circuits are quadrature oscillators.

Using a CMOS CFOA biased with ±0.9 V, the circuits have been simulated in SPICE [84] with component values $C_1 = C_2 = 3$ pF, $R_2 = 35$ kΩ and R_1 adjusted 34.5 kΩ. In the CMOS CFOA, TSMC 0.18 CMOS process parameters were utilized. It was found that the oscillation frequency obtained from simulation $f_0 = 1.52$ MHz was quite close to the theoretical frequency 1.527 MHz.

New VLF Oscillators Using a Single CFOA

In the context of CFOA-based oscillators suitable for VLF generation for which circuit 1 of Fig. 5.7 appears to be suitable candidate, it is interesting to point out a circuit recently presented by Srivastava, Singh, and Senani [82] which is shown here in Fig. 5.45. This circuit, in addition to single-resistance-control of FO, also provides single resistance-control of CO but at the cost of one more capacitor but

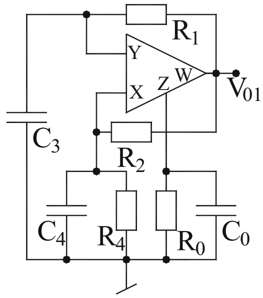


Fig. 5.45 VLF oscillator recently proposed by Srivastava, Singh, and Senani [82]

using only a single CFOA in contrast to the circuit 1 of Fig. 5.7. The CO and FO are found to be

$$\text{CO}:\frac{C_4-C_0}{C_3}=\frac{R_1}{R_0}+\frac{R_1}{R_2}\quad\text{and}\quad\text{FO}:\omega_0=\sqrt{\frac{\left(\frac{R_4}{R_0}-1\right)}{C_0C_3RR_4}}\tag{5.101}$$

From the above equation, it can be seen that FO can be adjusted by varying R_4 while ensuring $(R_4/R_1)>1$, whereas the CO can also be independently adjusted by C_4 and/or R_2. From the expressions of the oscillation frequency (FO), it may be seen that keeping the difference term in the numerator of FO as small as possible, generation of VLF oscillations should be possible. Experimental results using AD844 have demonstrated that it has been possible to generate sinusoidal waveforms of frequency as low as 2 Hz.

Single CFOA-Based Oscillator Capable of Absorbing all Parasitic Impedances

Most of the investigations on CFOA oscillators have focused attention only on canonic realizations thereby it has been largely ignored that some noncanonic circuits may also possess some interesting features which may not be possible from the canonic realization. Recently, Srivastava, Singh, and Senani [83] presented a single CFOA noncanonic oscillator which is shown in Fig. 5.46, which has the interesting property that it can absorb all the parasitic impedances of the CFOA in the various external passive components employed. Analysis shows that CO and FO of this circuit are expressed as

$$\text{CO}:\frac{C_0}{R_3}+\frac{C_1+C_3}{R_0}=\frac{C_1}{R_4}\quad\text{and}\quad\text{FO}:\omega_0=\sqrt{\frac{1}{R_0R_3C(C_1+C_3)}}\tag{5.102}$$

Although this circuit does not provide control of FO, nevertheless the circuit can be adjusted to produce the oscillations by varying the resistor R_4 which does not

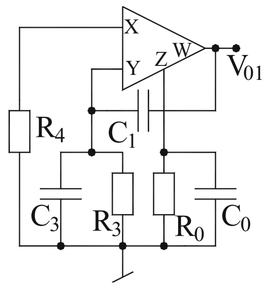


Fig. 5.46 CFOA-based oscillator capable of absorbing all parasitic impedances proposed by Srivastava, Singh, and Senani [83]

feature in FO. Experimental results have shown that the error between the practically observed values of the oscillation frequencies and those calculated from the nonideal formula are indeed extremely small when theoretical values are determined by incorporating all parasitic impedances into external circuit elements [83].

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Chapter 6
Sinusoidal Oscillator Realizations Using Modern Electronic Circuit Building Blocks

Abstract A variety of SRCOs realized with numerous variants of CCs introduced in the literature as well as using a number of other new building blocks have been discussed. Our endeavor here has been on including only some representative circuit configurations (from amongst a large number of oscillator circuits reported in literature) which possess some specific attractive features of practical interest; interested readers can explore other options from the list of references provided at the end of the chapter.

6.1 Introduction

In Chap. 4 of this monograph we presented prominent sinusoidal oscillator circuits using the basic type of current conveyors only (i.e., three-port CC I± characterized by $i_y = i_x$, $v_x = v_y$, $i_z = \pm i_x$ or CC II± characterized by $i_y = 0$, $v_x = v_y$, $i_z = \pm i_x$). Due to the commercial availability of AD844 type CFOA which contains a CCII+ and an on-chip voltage follower, it readily realizes a CCII (omitting buffer which nevertheless used to isolate the output terminal of the oscillator from the load). Also, two CFOAs (hence, two CCII+) can be used to realize a CCII−. Thus, all CCII±-based oscillators could be realized with AD844 ICs. Also, other IC CCs such as PA 630 or OP 660/860 could be used to implement CC-based oscillators. Likewise, in Chap. 5, we elaborated significant work done on realizing various kinds of oscillators using readily available AD844-type CFOAs.

Since a very large number of new active building blocks have been proposed by researchers from time-to-time almost all of them have been employed to realize oscillator structures exhibiting interesting properties, it is extremely essential to deal with such circuits. The purpose of this chapter is, therefore, to deal with oscillators using these newer building blocks [1–73]. A comprehensive treatment of these new active building blocks was given in [74]. A number of other review papers have also dwelt upon the application and implementation of these new build building blocks. Among the new active building blocks introduced so far, the most

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prominent ones, which have been used to realize sinusoidal oscillators are: operational trans-resistance amplifier (OTRA), differential difference amplifiers (DDA), current differencing buffered amplifier (CDBA), current differencing trans-conductance amplifier (CDTA), current follower trans-conductance amplifiers (CFTA), current conveyor trans-conductance amplifier (CCTA), current backward trans-conductance amplifier (CBTA), differential voltage trans-conductance amplifiers (DVTA), voltage differencing inverting buffered amplifier (VDIBA), VD-DIBA, current follower (CF), voltage follower (VF), and their multi-output versions [1–56].

Besides these, a large number of variants of CCs have also been advanced in the literature such as dual/multiple output current conveyors (DOCC/MOCC), differential voltage current conveyors (DVCC), differential difference current conveyors (DDCC), inverting second-generation current conveyors (ICC II), fully differential second-generation current conveyors (FDCC II), third-generation current conveyors (CC III), dual-x current conveyors (DXCC II), controlled current conveyor (CCC II) to name a few [74], which have also found prominent applications in sinusoidal oscillator realization.

It may be mentioned that various researchers have also demonstrated as to how these newer active building blocks can be realized using commercially available ICs such as AD844, LM3080, LM13600, Max435, and OPA660/860. On the other hand, a large number of hardware implementations of these building blocks, suitable for implementation in bipolar and/or CMOS technology, have also been evolved by various researchers from time to time [74].

Thus, on one hand, the oscillators using the quoted building blocks can be employed in discrete designs using commercially available ICs while on the other hand, using evolved CMOS implementations, oscillators using these new building blocks also hold the promise of complete IC implementation of the suggested sinusoidal oscillator topologies in bipolar and/or CMOS technology.

In this chapter, therefore, first we present a brief account of various new building blocks, their characterizations, and hardware implementations including the numerous variants of current conveyors. Subsequently, we would describe the most prominent oscillator topologies using each of the new building blocks, from amongst a large number of oscillator topologies existing in the literature so far. Finally, concluding remarks would be made highlighting the significant contributions made in this area and pointing out the unresolved problems.

6.2 Some Prominent Modern Building Blocks

Before discussing the various oscillator topologies evolved using a variety of modern building blocks it is useful to study the terminal characteristics and circuit realisations of a number of exciting new active building blocks. It is interesting to note that quite a number of them are related to, in some form or the other to the Current Conveyors. Since a popular and convenient means of realizing CCIIs both positive and negative types is through the commercially available current feedback op-amp AD844, it is not surprising therefore that quite a number of new active building blocks can be indeed realized with two (or more) AD844 in conjunction with voltage followers and/or OTAs, both of which are also available commercially as off-the-shelf integrated circuits.

6.2.1 Different Variants of the Current Conveyors

Initially, the applications and hardware implementation of only the basic types of current conveyors introduced by Sedra and Smith namely CCI and CCII were investigated as evidenced by the numerous publications between the periods from 1968 till about 1990 which are well documented in several tutorial/review papers on current conveyors. With the introduction of a number of possible hardware implementations of CCI/CCII, it was soon realized that creating multiple outputs having similar or complementary polarity was quite easy thereby rendering further flexibility to CCI± as well as CCII±. Subsequently, a large number of different varieties of current conveyors with modified terminal characteristics between *X*, *Y*, and *Z* ports and/or increasing the number of *X*, *Y*, and *Z* terminals or some other generalizations came into being over a period of time.

In the following, we outline a variety of modified current conveyors, discuss their models/implementations and also highlight the characteristic features of the new variants which lead to significant advantages over the basic types of current conveyors in a number of applications. However, with a few exceptions, the discussion is limited to only those variants of CCs which have been employed to realize sinusoidal oscillators.

6.2.1.1 Dual/Multiple-Output Current Conveyor (DOCC/MOCC)

Having a multiple-output CCII (MOCC) with multitude of identical as well as complementary outputs is particularly useful in current mode signal processing/signal generation; see [14–18].

An MOCC can be realized from a bipolar or CMOS implementation of a CCII by adding additional transistors and current mirrors. An exemplary bipolar MOCC is shown in Fig. 6.1a, where the additional port creating multiple copies can be easily identified. Its symbolic notation is shown in Fig. 6.1b from where a MOCC is characterized by the terminal equations: $i_y = 0$, $v_x = v_y$, $i_{z1+} = i_{z2+} = i_{z3+} = \ldots = i_{zn+\text{-}} = +i_x$, and $i_{z1-} = i_{z2-} = i_{z3-} = \ldots. = i_{zn-} = -i_x$.

MOCCs are particularly useful for realizing current mode universal biquad filters and also in higher order ladder filter design based upon the simulation of its node equations, for instance, see [15, 16]. A DOCCII can be considered to be a special case of MOCC.

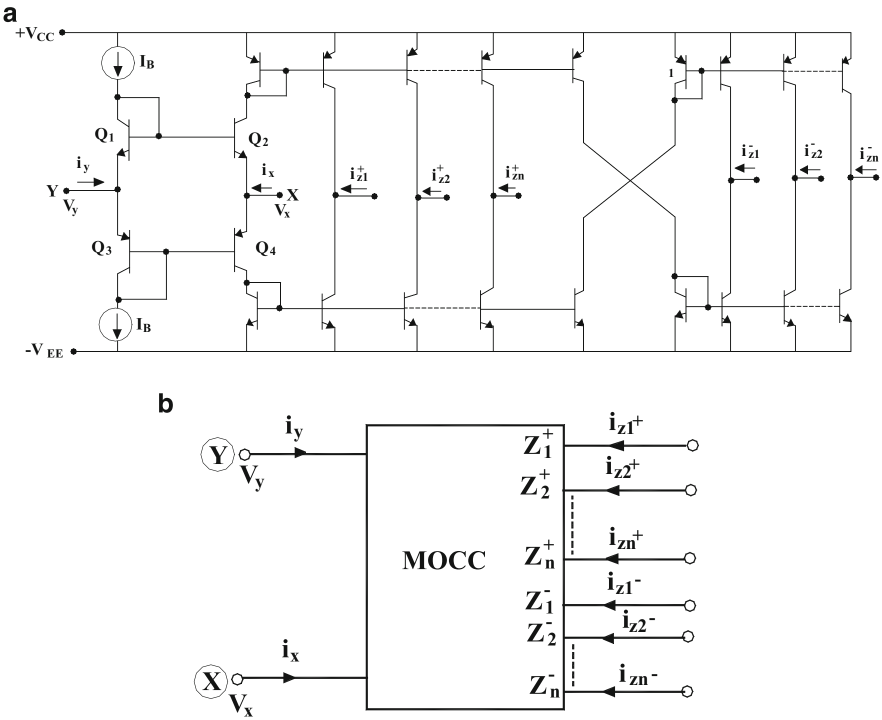


Fig. 6.1 The MOCC: (**a**) An exemplary bipolar implementation. (**b**) Symbolic notation

6.2.1.2 Operational Floating Conveyor

The operational floating conveyor (OFC) was introduced by Toumazou, Payne, and Lidgey [6] in 1991. It is a four-port active building block characterized by the following hybrid matrix:

$$\begin{bmatrix} v_x \\ i_y \\ v_w \\ i_z \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ Z_\mathrm{t} & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} i_x \\ v_y \\ i_w \\ v_z \end{bmatrix} \tag{6.1}$$

The above characterization implies that the output voltage at port-w is obtained by multiplying the current at the x-port by the trans-impedance Z_t, the voltage at port-x follows the voltage at port-y which has infinite input impedance so that the current $i_y = 0$, and finally the current flowing into port-w is conveyed to port-z. An exemplary translinear implementation of the OFC is shown in Fig. 6.2 [1].

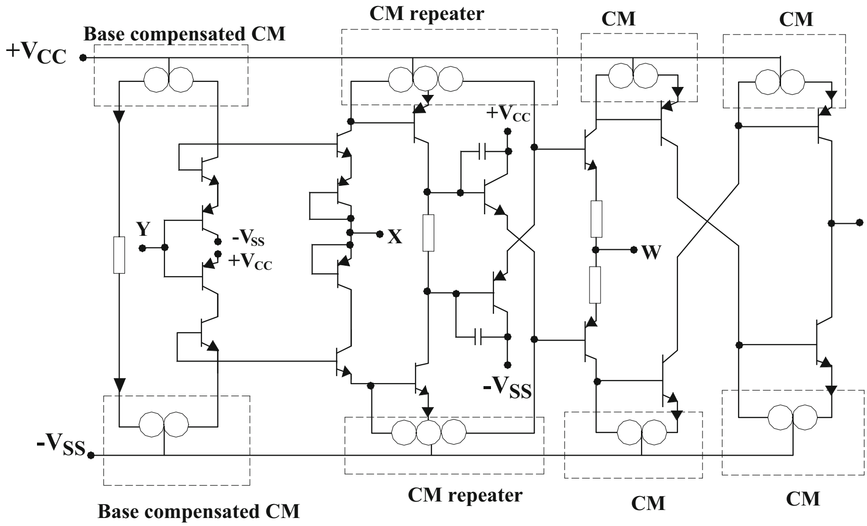


Fig. 6.2 Equivalent circuit of an exemplary CMOS implementation of the OFC proposed by Toumazou, Payne, and Lidgey [6]

6.2.1.3 Differential Difference Current Conveyor

Chiu, Liu, Tsao, and Chen [9] in 1996 introduced differential difference current conveyor (DDCC) which appears to have been inspired by the earlier concept of DDAs [10]. A DDCC has three Y-terminals, one X-terminal, and one Z-terminal leading to the following characterization:

$$\begin{aligned} V_x &= V_{y1} - V_{y2} + V_{y3} \\ i_{y1} &= 0,\ i_{y2} = 0,\ i_{y3} = 0 \\ i_z &= \pm i_x \end{aligned} \tag{6.2}$$

An exemplary CMOS implementation of the DDCC is shown in Fig. 6.3. Other implementations can be devised from existing CMOS CCII structures such as [11]. In this circuit there are the two differential stages which are composed of MOSFETs $M_1 - M_2$ and $M_3 - M_4$ and a current mirror composed of $M_5 - M_6$ which converts the differential current to a single ended output current in the MOSFET M_7. Thus, the circuit between port X and the input terminals Y_1, Y_2, and Y_3 acts as an amplifier.

The relation between the four voltages can be expressed as

$$V_X = \frac{A_0}{A_0 + 1}(V_{Y1} - V_{Y2} + V_{Y3}) \cong V_{Y1} - V_{Y2} + V_{Y3}; \quad \text{for } A_0 \gg 1 \tag{6.3}$$

where A_0 is the open loop gain of the amplifier.

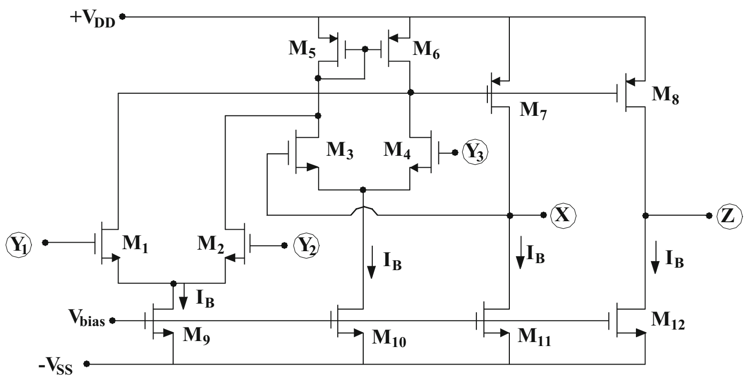


Fig. 6.3 An exemplary CMOS DDCCII+

It can be readily seen that in the specific architecture presented in Fig. 6.3, both i_x and i_z flow simultaneously and equally into or out of the DDCC, thereby leading to $i_z = +i_x$ and the circuit, thus, realizes a DDCC+. It may be mentioned that a DDCC− can be easily realized by appropriate addition of another pair of MOSFET current mirrors.

An attractive feature of the DDCC is its capability of realizing summation or subtraction of the signals without requiring any external resistors as reflected from the equation

$$V_X = (V_{Y1} - V_{Y2} + V_{Y3}) \tag{6.4}$$

For example, a unity gain difference amplifier (UGDA) without using any external resistors is realized if $V_{y3} = 0$; an inverting amplifier sans any resistors is realized by making $V_{y1} = 0 = V_{y3}$ and a unity gain summer is realized by making $V_{y2} = 0$.

That DDCCIIs are particularly attractive for evolving single resistance controlled sinusoidal oscillators employing all grounded passive elements has been demonstrated in [12]. On the other hand, current-controllable versions of DDCC, known as CCDDCC implementable in CMOS as demonstrated in [13], are useful in devising grounded-capacitor-based floating capacitance multiplier and current-mode universal filter [13].

6.2.1.4 Differential Voltage Current Conveyors

Elwan and Soliman introduced a CMOS differential voltage current conveyor (DVCC)[1] in 1997 [20]. The DVCC is a four port building block with there being

[1] It is, however, interesting to point out that it went almost unnoticed that the basic idea of the DVCC was introduced by Pal [19] as early as in 1989 itself who had demonstrated its advantage in realizing lossless floating inductors.

two Y-terminals with flexibility that a differential Y-input can be applied. A DVCC is, thus, characterized by the terminal equations

$$i_{y1} = i_{y2} = 0,\ v_x = \left(v_{y1} - v_{y2}\right)\ \text{and}\ i_z = \pm i_x \tag{6.5}$$

A DVCC can be looked upon as a special case of DDCC with the third Y-input grounded. Hence, a hardware implementation of the DVCC can be easily obtained from the CMOS DDCC realization such as the one in Fig. 6.3 by grounding terminal y_3. For a number of other CMOS realizations of the see [20, 22–25, 27]. Elwan and Soliman in [19] demonstrated that the DVCCs are particularly attractive in realizing grounded to floating positive impedance converter, grounded to floating negative impedance converter, floating generalized negative impedance converter and floating generalized positive impedance inverter/gyrator and MOS transconductor, to name a few. In addition, DVCCs have been found to be equally efficient in realizing instrumentation amplifiers, floating inductors and FDNRs, universal VM/CM filters and relaxation oscillators and waveform generators [19–27].

It is worth mentioning that the adjoint of a DVCC is an element called balance output CCII (BOCCII) [21] characterized by $i_y = 0$, $v_x = v_y$, $i_{z+} = +i_x$ and $i_{z-} = -i_x$.

6.2.1.5 Third-Generation Current Conveyor

The third-generation current conveyor (CCIII) was introduced by Fabre in 1995 [7] as a new helpful active element characterized by the hybrid matrix

$$\begin{bmatrix} i_y \\ v_x \\ i_z \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & \pm 1 & 0 \end{bmatrix} \begin{bmatrix} v_y \\ i_x \\ v_z \end{bmatrix} \tag{6.6}$$

The CCIII was conceived to be an useful element to be used as the input cell of probes and current measuring devices. It can be used to sense the output current flowing to ground or to create a floating current source.

A CCIII can be realized from two CCII+ s as per the schematic of Fig. 6.4.

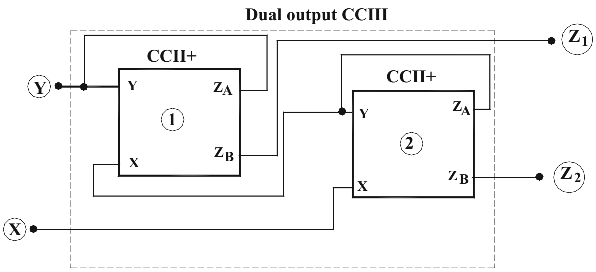


Fig. 6.4 Implementation of a dual-output CCIII [7]

6.2.1.6 Inverting Second-Generation Current Conveyors

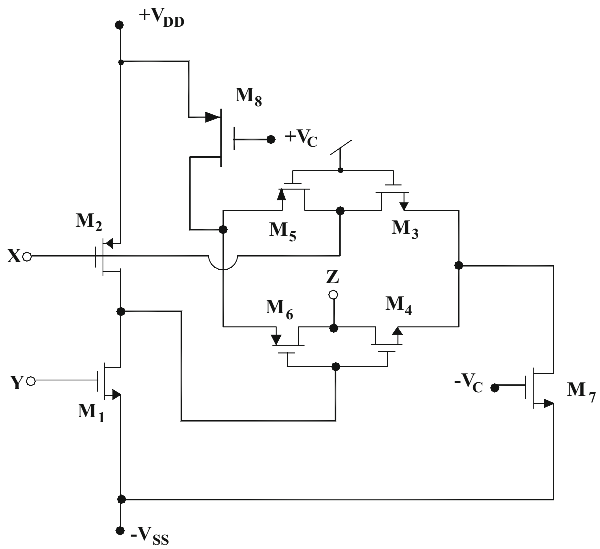
In 1999, Awad and Soliman [28] introduced[2] the *inverting* second-generation current conveyors as the *missing building blocks* characterized by $v_x = -v_y$, $i_y = 0$ and $i_z = \pm i_x$, thereby leading to two varieties of inverting current conveyors, namely ICCII+ and ICCII−.

Nevertheless, Awad and Soliman were the first to demonstrate that the new pathological elements *current-mirror* and *voltage-mirror* introduced by them together with the *nullator* and the *norator* were sufficient to model CCII+, CCII−, ICCII+, and ICCII− without requiring any resistors. They were also the first to present CMOS implementations of the ICCII elements.

The pathological models of ICCII in terms of the pathological elements, nullator, norator, current mirror, and voltage mirrors together with their relationship with CCII has been dealt in [29] while a number of CMOS implementations of the ICCII have been presented in [30, 31]. Novel generations of ICCs have been discussed in [32].

In [28] Awad and Soliman presented two different CMOS realization of the ICCII based upon the employment of two single-input single-output CMOS transconductors and one single-input dual-output CMOS transconductor. A CMOS realization of ICCII from [28] is shown here in Fig. 6.5.

Fig. 6.5 A CMOS realization of the ICCII [28]



[2] It may be pointed out that it went completely unnoticed in [20] as well as in most of the other works on the use of ICCII+ that this variant of the CCII was already conceived much earlier by Chong and Smith in [5] wherein they had called this version of CC as the new type of CC, termed CCIIB which was employed by them in realizing one of the special cases of their single-CC biquads.

6.2.1.7 Inverting Third-Generation Current Conveyors

Extending the concept of CCIII further, inverting CCIII of both polarities, namely ICCIII+ and ICCIII−, were introduced by Sobhy and Soliman in [8]. The ICCIII elements are characterized by $I_y = -I_x$, $V_x = -V_y$, $I_{Z+} = -I_x$ and $I_{Z--} = -I_x$. It has been shown in [7] that ICCIII ± are particularly attractive in realizing V to I converters, integrators, filters, and oscillators-all providing electronic control of various parameters.

6.2.1.8 Differential Current Voltage Conveyor

In 1999, Salama and Soliman [34] introduced a new variant of CC which they chose to call a *differential current voltage conveyor* (DCVC) characterized by the equations $V_{x1} = 0$, $V_{x2} = 0$, $I_z = I_{x1} \_ I_{x2}$, and $V_o = V_z$. They also proposed a CMOS implementation of this block and showed that this is particularly useful for realizing electronically controllable MOS-C quadrature oscillator suitable for designing digitally controlled oscillators.

In the same year 1999, Acar and Ozoguz [73] introduced a new versatile building block suitable for analog signal processing which they called *current differencing buffered amplifier* (CDBA) which is characterized by exactly the same equations as specified above for the DCVC. Thus, DCVC and CDBA are one and the same thing-a fact which does not appear to be so well recognized in the technical literature. The symbolic notation of the DCVC consists of a differential current conveyor (DCC) followed by a voltage buffer as shown in Fig. 6.6a. An exemplary CMOS realization of the DCVC is shown in Fig. 6.6b.

For CMOS implementation and some exemplary application circuits of DCVC, the readers are referred to [33–35] and the references cited therein. Also, yet another variant of DCVC called current-controlled DCVC has been dealt with in [71].

6.2.1.9 Dual-X Current Conveyor

In 2002, Zeki and Toker proposed in [51] a *dual-X current conveyor* (DXCCII) as a new building block suitable for electronically tunable continues time filter. A DXCCII is characterized by the following terminal equation:

$$\begin{bmatrix} I_Y \\ V_{X\mathrm{p}} \\ V_{X\mathrm{n}} \\ I_{Z\mathrm{p}} \\ I_{Z\mathrm{n}} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} V_Y \\ I_{X\mathrm{p}} \\ I_{X\mathrm{n}} \end{bmatrix} \tag{6.7}$$

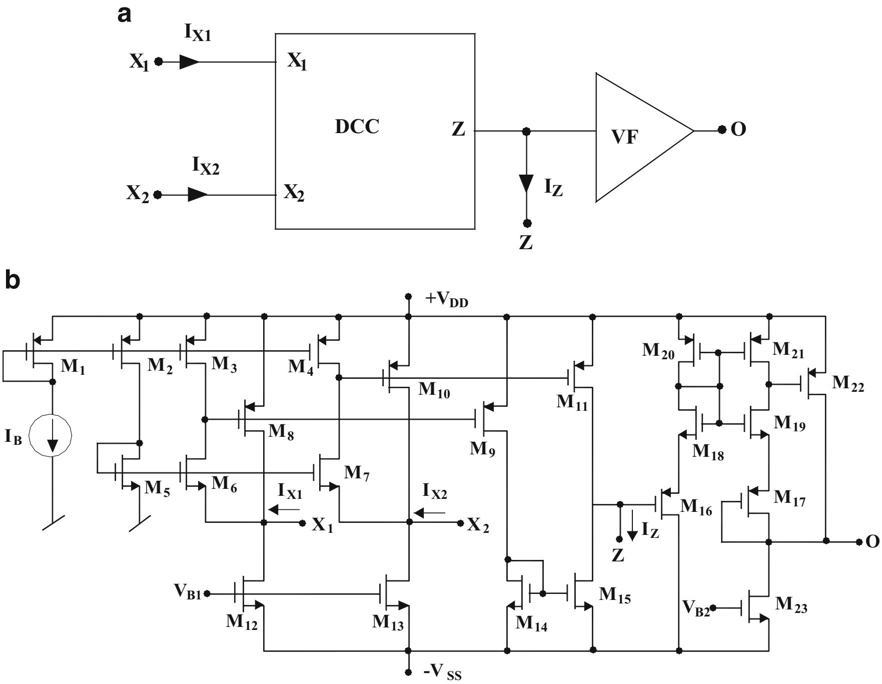


Fig. 6.6 DCVC implementation proposed by Salama, Elwan, and Soliman [33]: (**a**) symbolic notation, (**b**) an exemplary CMOS implementation (adapted from [33] © Springer 2001)

It was shown that since $V_{Xp} = -V_{Xn}$, if a MOSFET is connected between the two *X* terminals, with a control voltage V_c applied at the gate terminal, the square nonlinearities of the MOSFET drain current are cancelled out and the drain current is, therefore, given by

$$I_D = \beta(V_C - V_{Tn})\left(V_{X\,p} - V_{Xn}\right) \text{ where } \beta = \left(\frac{W}{L}\right)\mu_n C_{ox} \tag{6.8a}$$

due to which the MOSFET behaves almost as a linear resistor as long as the MOSFET operates in triode region. The equivalent resistance value is given by

$$R_{eq} = \frac{1}{\beta(V_c - V_{th})} \tag{6.8b}$$

A CMOS realization of the DXCCII was also suggested by Zeki and Toker in [51]. Other interesting variants and applications of the DXCCII can be found in [52–55].

6.2.1.10 Fully Balanced CCII and Fully Differential CCII (FDCCII)

The notion of a fully balanced second-generation current conveyor (FBCCII was introduced by Alzaher, Elwan, and Ismail in 2003 in [56] (Fig. 6.7).

It was demonstrated that such a building block can be constructed from a fully-balanced differential difference amplifier (FBDDA) through power supply current sensing technique. An FBCCII is characterized by the same equations as a CCII+; however, all signals are differential, i.e., $V_x = (V_{xp} - V_{xn})$, $V_y = (V_{yp} - V_{yn})$, $I_x = (I_{xp} - I_{xn})$, and $I_z = (I_{zp}–I_{zn})$.

For many other variants of CCs such as general three-port conveyors, universal conveyors, extended current conveyors, operational conveyors, multiple-input differential CC, multiplication mode CC, balanced output third-generation CC, voltage and current gain CCII, two-*X* and two-*Z* CC, differential CCII, universal voltage conveyors, and floating current conveyors, the reader is referred to [74].

El-Adawy, Soliman, and Elwan in the year 2000 presented [36] another novel modification of the current conveyor termed *fully differential second-generation current conveyor* (FDCCII) which is a eight terminal analog building block characterized by the matrix equation

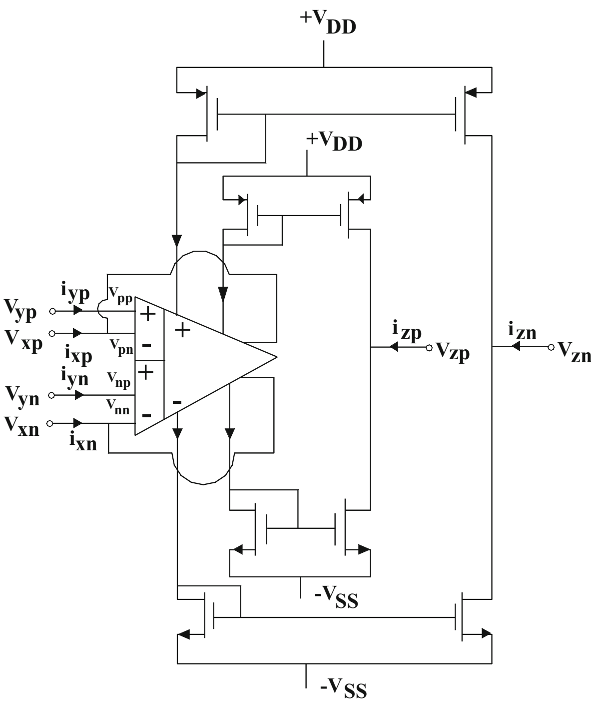


Fig. 6.7 Fully balanced current conveyor an exemplary implementation based upon the one presented by Alzaher, Elwan, and Ismail [56]

$$\begin{bmatrix} V_{X+} \\ V_{X-} \\ I_{Z+} \\ I_{Z-} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} I_{X+} \\ I_{X-} \\ V_{Y1} \\ V_{Y2} \\ V_{Y3} \\ V_{Y4} \end{bmatrix} \tag{6.9}$$

The symbolic notation of the FDCCII has been shown in Fig. 6.8a. The FDCCII has been since then used by a number of researchers, for instance, see [36–42] in various applications and a number of CMOS implementations have also been evolved. An equivalent of the exemplary circuit proposed by Kacar, Metin, Kuntman, and Cicekoglu [38] has been shown here in Fig. 6.8b where the blocks employed have been detailed in Fig. 6.8c, d.

It has been demonstrated by a number of researchers that FDCCII is particularly useful in mixed mode applications where fully differential signal processing is required. As novel applications of the FDCCII, its use in realizing a differential input balanced output transconductor, a four quadrant multiplier, a number of mixed mode universal second-order filters, and a fully differential filter has now been known; see [36–42].

Before we formally introduce a number of new building blocks, it is useful to see how several variants of CCs can be realized using CFOA AD844.

A CCII+ is realizable with only a single CFOA while CCII- requires two of them as shown in Fig. 6.9:

$$\begin{array}{ll} i_y = 0 & i_y = 0 \\ v_x = v_y & v_x = v_y \\ i_z = i_x & i_z = -i_x \end{array} \tag{6.10}$$

A differential voltage CC (DVCC) can be realized with three CFOAs but needs two resistors as well (see Fig. 6.10):

$$\left.\begin{array}{l} i_{y1} = 0 \\ i_{y2} = 0 \\ v_x = \left(v_{y1} - v_{y2}\right) \text{ for } R_1 = R_2 \\ i_z = i_x \end{array}\right\} \tag{6.11}$$

Furthermore, the third-generation current conveyor (CCIII) can also be realized with three CFOAs and two resistors as shown in Fig. 6.11:

$$i_y = -i_x, \ v_x = v_y \ \text{and} \ i_z = i_x \tag{6.12}$$

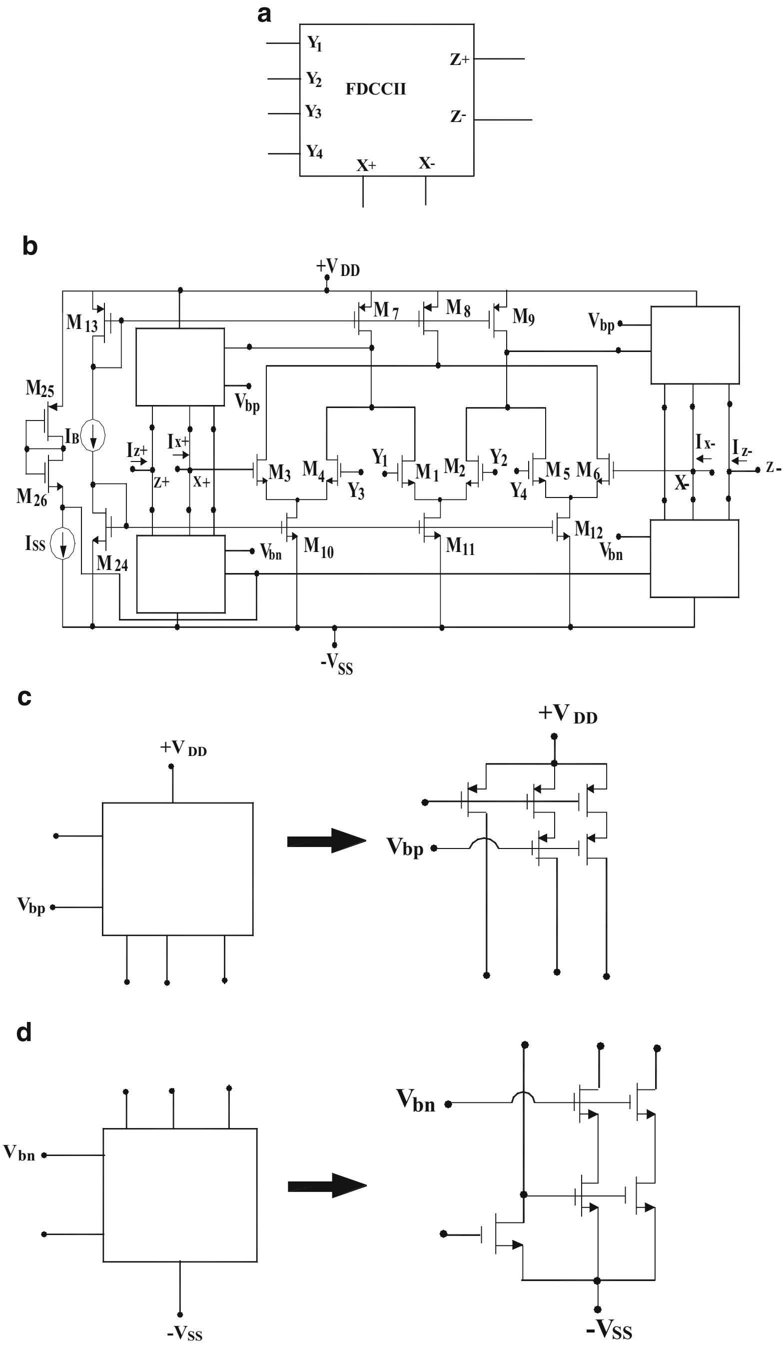


Fig. 6.8 Symbol of FDCCII and an exemplary CMOS implementation. (**a**) The symbolic notation of FDCCII. (**b**) An exemplary CMOS implementation [38]. (**c**) and (**d**) The details of the various blocks used in the circuit of (**b**)

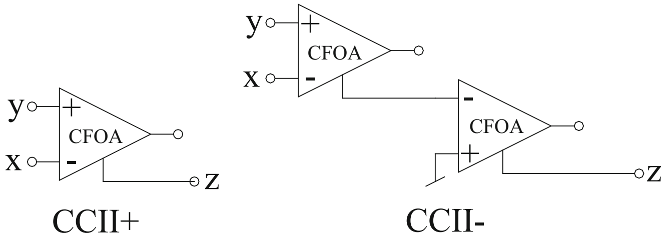


Fig. 6.9 Realization of CCII+ and CCII− using CFOAs

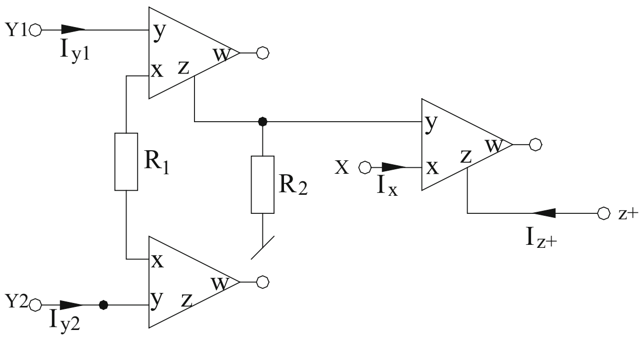
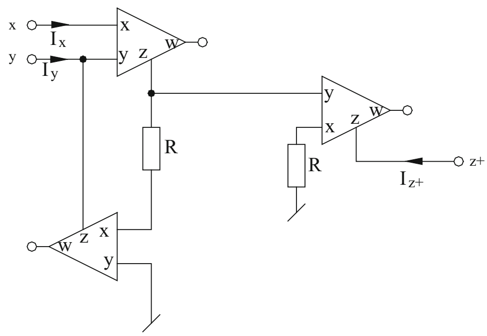


Fig. 6.10 Realization of a DVCC+ using CFOAs (based upon the idea given in [75])

Fig. 6.11 Realization of a CCIII+ using CFOAs (based upon the idea given in [75])



It is worth mentioning that CCII+, CCII−, DVCC, or CCIII-based voltage-mode circuits would invariably require a voltage follower after the *Z*-terminal(s) of those CCs from which a voltage output is being taken since the *Z*-terminal being a current output terminal cannot be connected to the load impedance *directly* as this will modify and change the function realized by the circuit. Realizing the CC-based circuits by CFOAs will easily permit the *Z*-terminal voltage(s) to be available from the *W*-terminal(s) quite easily without requiring any additional external buffer because of the availability of an internal VF between the *Z* and *W* terminals in the CFOA(s) and thereby providing a remedy to this problem.

The CFOA-based CC implementations have been employed by many researchers for the verification of their CC-based circuit proposals; for example see [14, 18, 25, 26, 76–115, 117–129, 132–143, 145–147] and the references citied therein.

We now discuss a variety of other building blocks and would also highlight their hardware implementations in terms of CFOAs, OTAs, and voltage followers—all of which are commercially available as off-the-shelf ICs.

6.2.2 Some Other Modern Active Building Blocks

Other than CCs and their numerous variants, a large number of other new building blocks have also been advanced by various researchers from time to time. In the following, we give a brief account of these exciting new elements and also present their practical realizations using CFOAs, voltage followers, and OTAs—all of which are commercially available as off-the-shelf IC components.

6.2.2.1 Unity Gain Cells

Interest in unity gain voltage follower (VF) and unity gain current follower (CF) is primarily attributed to the relatively larger bandwidth offered by them as well as the theoretical novelty that from several building blocks, VFs and CFs can be realized without requiring any external resistors. For instance, a unity gain VF is known to be realizable by a single VOA with its inverting terminal shorted to the output terminal. Likewise, a non-inverting VF and non-inverting/inverting CFs are also realizable with CFOAs without requiring any external resistances (see Fig. 6.12) In fact, these CFOA-based implementations of non-inverting VF and non-inverting/inverting CF have already been used by a number of researchers to prove the workability of their VFs/CF-based analog signal processing/signal generating circuits.

6.2.2.2 Four-Terminal-Floating-Nullors

It was shown by a number of researchers (such as Nordholt [149], Stevenson [150], Huijsing [151], and Senani [152] who coined the term "four-terminal floating-nullor" (FTFN) to represent a fully floating nullor) that fully floating versions of Op-amps (termed as operational floating amplifier (OFA)) and FTFNs are more versatile and flexible building blocks than the traditional op-amps in several applications.

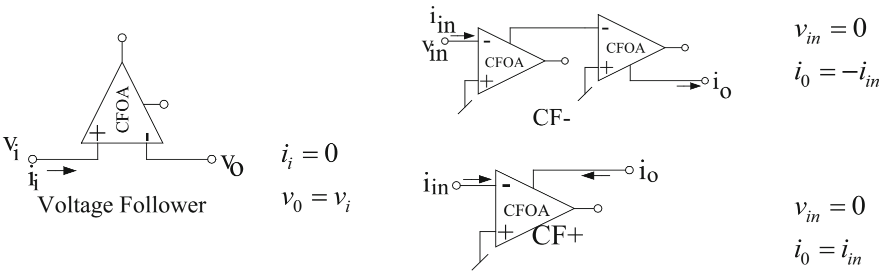


Fig. 6.12 Realization of VF and CF using CFOAs

Fig. 6.13 Nullor equivalence. (**a**) The FTFN, (**b**) FTFN realization using three terminal floating nullors

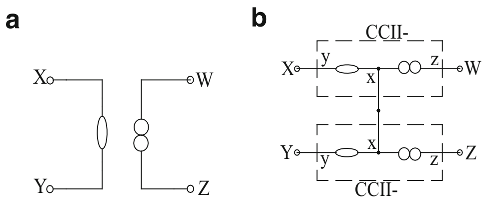
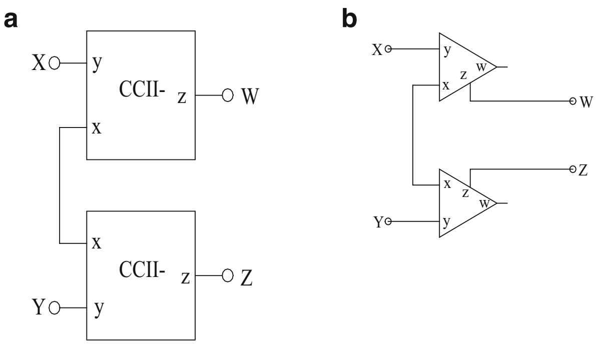


Fig. 6.14 Practical FTFN realizations. (**a**) FTFN using two CCIIϴ, (**b**) FTFN using two CFOAs



It was suggested in [152] that a composite connection of two CCII-s can be used to realize an FTFN (see Fig. 6.13). This follows from the fact that the representation of FTFN of Fig. 6.13a is equivalent to the combination of two 3-terminal nullors as shown in Fig. 6.13b where each 3-terminal nullor is equivalent to a CCII−, thus, finally, leading to the implementation of Fig. 6.14a. In fact, two CCII+ or two current feedback op-amps (CFOA) such as AD844, can also be readily used to realize an FTFN using the same configuration (as in Fig. 6.14b).

A novel application of FTFNs has been in the area of floating impedance simulation; for instance, see [152, 162, 164]. Their use in biquad filters, inverse filters, and oscillator realization has also yielded many interesting results.

The CFOA-based FTFN implementation of Fig. 6.14b has been employed by many researchers for the verification of their FTFN-based propositions, for example, see [149–166] and the references citied therein.

6.2.2.3 Differential Difference Amplifier

DDAs were introduced as versatile active building blocks for fully differential filter design because of the advantage of common-mode noise elimination. DDAs are attractive elements for analog circuit design since when configured in negative feedback DDAs can realize a number of functional circuits like unity gain inverting amplifier, unity gain difference amplifier, unity gain summer, and noninverting amplifier of gain 2—*all without requiring any external resistor!*

6.2.2.4 Modified CFOAs

Several modified forms of CFOAs have been proposed out of which the differential voltage current feedback amplifier (DVCFA), fully differential CFOA (FDCFOA) and the differential difference complementary current feedback amplifier (DDCCFA) are particularly attractive for oscillator realization. A DVCFA is characterized by the terminal equations: $V_x = V_{y1} - V_{y2;} i_{y1} = 0$, $i_{y2} = 0$; $i_z = +i_x$ and $V_w = V_z$. On the other hand DDCCFA is a more generalized modification of the CFOA and is characterized by $V_x = V_{y1} - V_{y2} + V_{y3;} i_{y1} = 0$, $i_{y2} = 0$, $i_{y3} = 0$; $i_z^{+} = +i_{x,} i_z^{-} = -i_x$; $V_{w+} = V_{z+}$ and $V_{z-} = V_{w-}$. DDCCFA also permits to realize a number of basic building blocks as special cases.

6.2.2.5 Current Differencing Buffered Amplifier

In 1999, Acar and Ozoguz [73] introduced a new versatile building block suitable for analog signal processing which they called *current differencing buffered amplifier* (CDBA) which is characterized by exactly the same equations DCVC described earlier. The symbolic notation and an exemplary CMOS implementation of the DCVC/CDBA have already been described earlier.

Although fully integratable circuit implementations of CDBAs have been proposed by a number of researchers, CFOAs have been found to be quite handy in realizing them. Since CDBA is characterized by the terminal equations $V_p = V_n = 0$, $i_z = (i_p - i_n)$, and $V_w = V_z$ it was found that it could be readily implemented by two CFOAs as follows (Fig. 6.15):

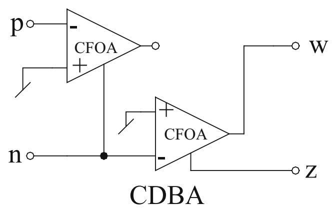


Fig. 6.15 Current differencing buffered amplifier (CDBA) (adapted from [73] © 1999 Elsevier)

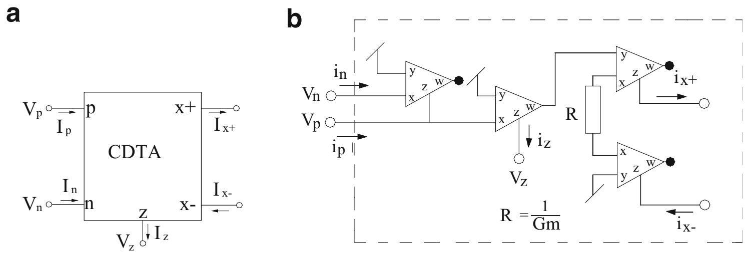


Fig. 6.16 The CDTA and its realization. (**a**) Symbolic notation, (**b**) CFOA implementation

6.2.2.6 Current Differencing Transconductance Amplifier

The current differencing transconductance amplifier was introduced by Biolek in [205] as a new building block suitable for CM analog signal processing. The symbolic notation of the CDTA is shown in Fig. 6.16.

A CDTA is characterized by the following matrix equation:

$$\begin{pmatrix} I_z \\ I_{x+} \\ I_{x-} \\ V_{\mathrm{p}} \\ V_{\mathrm{n}} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 & -1 \\ g_{\mathrm{m}} & 0 & 0 & 0 & 0 \\ -g_{\mathrm{m}} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} V_z \\ V_{x+} \\ V_{x-} \\ I_{\mathrm{p}} \\ I_{\mathrm{n}} \end{pmatrix} \tag{6.13}$$

where g_{m} is the transconductance of the CDTA. An entirely CFOA-based implementation of the CDTA, based upon the idea given in [205], is shown in Fig. 6.16.

CDTAs have received considerable attention in realizing various types of filters, oscillators, impedance simulators and other applications. For those cases, where CDTAs have been implemented with CFOAs, the reader is referred to [206–210] and the references cited therein for further details.

6.2.2.7 Current Follower Transconductance Amplifiers

The current follower transconductance amplifier (CFTA) was introduced by Herencsar, Koton, Vrba, and Misurec in [211]. This current input, current output building block has been shown to be particularly useful in realizing analog signal processing functions requiring explicit current outputs. The symbolic notation of CFTA is shown in Fig. 6.17a and its characterizing matrix equation is given by Eq. (6.14):

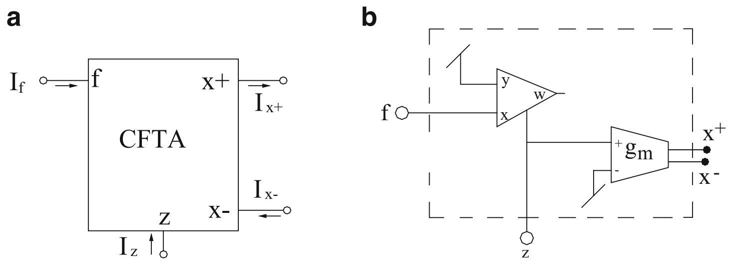


Fig. 6.17 CFTA representation: (**a**) Symbolic notation, (**b**) CFOA implementation of the CFTA

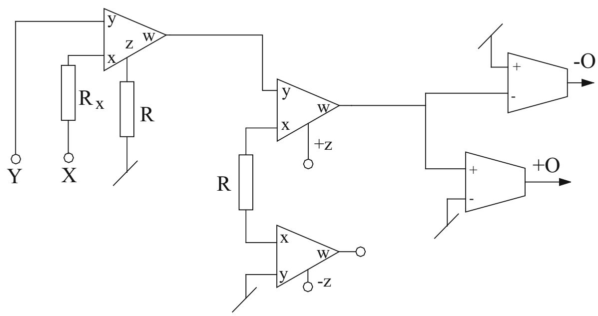


Fig. 6.18 A CFOA-based implementation of CCCC-TA (adapted from [214] © 2011 IET)

$$\begin{pmatrix} I_z \\ I_{x+} \\ I_{x-} \\ V_\mathrm{f} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ g_\mathrm{m} & 0 & 0 & 0 \\ -g_\mathrm{m} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} V_z \\ V_{x+} \\ V_{x-} \\ I_\mathrm{f} \end{pmatrix} \tag{6.14}$$

A discrete version of CFTA can be implemented using one AD844 type CFOA and one balanced output transconductance amplifier such as MAX 435 and is shown in Fig. 6.17b.

This CFOA-based implementation of the CFTA has been employed in the realization of current-mode KHN-equivalent biquad using CFTAs presented in [212].

6.2.2.8 Current Conveyor Transconductance Amplifier

Whereas a CFTA [211, 212, 247, 266, 329] is a composite of a CF (usually realized from a translinear CCII), a current conveyor transconductance amplifier (CCTA) has directly a CCII followed by an OTA. On the other hand, the current-controlled current conveyor transconductance amplifier (CCCC-TA) was introduced as a building block for analog signal processing by Siripruchyanun and Jaikla in [213]. A CFOA-based implementation of this was devised by Maheshwari, Singh, and Chauhan in [214] and is shown here in Fig. 6.18.

6.2.2.9 Voltage Differencing Inverting Buffered Amplifier

A voltage differencing inverting buffered amplifier (VDIBA) is a four-port building block having the terminals V_+, V_-, Z, and W and is characterized by $i_+ = 0$, $i_- = 0$; $i_z = g_m\ (V_+ - V_-)$; $V_w = V_z$. Like other building blocks it has also attracted lot of attention in realizing various analog signal processing/signal generation function more so because of the reason that its internal circuit architecture is extremely simple and is suitable for CMOS implementation.

6.2.2.10 Differential Input Buffered Transconductance Amplifier

The differential input buffered transconductance amplifier (DBTA) was introduced by Herencsar, Vrba, Koton, and Lattenberg in [202]. DBTA has been found to be a useful building block in realizing sinusoidal oscillators, quadrature oscillators, and universal filters. Quite often, all these functions can be carried out effectively using only a single DBTA as in [203, 204]. A DBTA is a six-port building block characterizing by the following equation:

$$v_p = v_y,\ v_n = v_y,\ i_y = 0,\ i_z = \left(i_p - i_n\right),\ v_w = v_z,\ i_x = g_m v_z$$

$$\text{where } g_m = 1/R_E \tag{6.15}$$

A CFOA implementation of this building block is shown in Fig. 6.19 and has been used in [203] to verify their proposed quadrature oscillator.

6.2.2.11 Voltage Differencing Differential Input Buffered Amplifier

The voltage differencing differential input buffered amplifier (VD-DIBA) was introduced by Biolek, Senani, Biolkova, and Kolka in [215]. Although some applications of VD-DIBAs have been reported in the open literature but for the

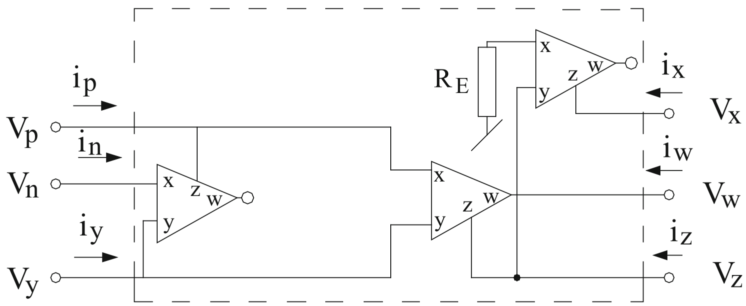


Fig. 6.19 A CFOA implementation of DBTA (adapted from [203] © 2009 IEICE)

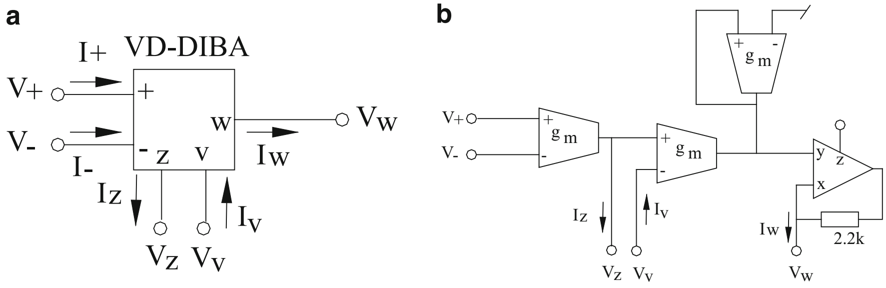


Fig. 6.20 (**a**) Schematic symbol. (**b**) CFOA implementation (adapted from [216] © 2011 Hindawi Publishing Corporation)

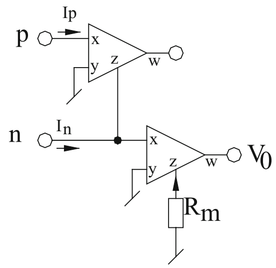


Fig. 6.21 Operational trans-resistance amplifier (OTRA) (adapted from [191] © 2004 Springer)

very first time an application implementing VD-DIBA using OTAs and CFOA was presented in [216].

The schematic symbol of the VD-DIBA is shown in Fig. 6.21a.

A VD-DIBA is characterized by the following matrix equation:

$$\begin{pmatrix} I_+ \\ I_- \\ I_z \\ I_v \\ V_w \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ g_{\mathrm{m}} & -g_{\mathrm{m}} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} V_+ \\ V_- \\ V_z \\ V_v \\ I_w \end{pmatrix} \tag{6.16}$$

The CFOA-implementation of the VD-DIBA as proposed in [216] is shown in Fig. 6.20b.

6.2.2.12 Operational Trans-Resistance Amplifier

An operational trans-resistance amplifier (OTRA) [119, 120] is characterized by the terminal equations:

$$v_{\mathrm{p}} = 0, v_{\mathrm{n}} = 0, v_0 = R_{\mathrm{m}}\left(i_{\mathrm{p}} - i_{\mathrm{n}}\right); R_{\mathrm{m}} \to \infty \tag{6.17}$$

and has been employed as alternative building block to realize a number of functions such as all pass filters, inductance simulators, MOS-C biquads, sinusoidal oscillators, and multivibrators. Although in several publications CMOS OTRA architectures have been employed, in many others, the two-CFOA-based implementation of the OTRA as shown in Fig. 6.21 has been employed; for example, see [33, 189, 191–201] and the references citied therein.

6.3 Sinusoidal Oscillator Realization Using Different Variants of Current Conveyors

In this section, we present some important contributions made in the realization of sinusoidal oscillators using different variants of current conveyors.

6.3.1 A Dual-Mode Sinusoidal Oscillator Using a Single OFCC

An operational floating current conveyor (OFCC), because of being a four terminal building block, provides the flexibility of having a low-output impedance voltage output as well as a high-output impedance current output. This is immediately of interest when a circuit is needed for realizing a fixed frequency oscillator which should be able to provide a voltage mode as well as current mode output. One such circuit was advanced by Celma and Martinez in [217] and is shown here in Fig. 6.22.

The characteristic equation (CE) of the circuit of Fig. 6.22 is given by

$$s^2 + s\left\{\frac{1}{R_1C_1} + \frac{1}{R_2C_2} + (1 - k_0)\frac{1}{R_2C_1}\right\} + \frac{1}{R_1R_2C_1C_2} = 0 \tag{6.18}$$

where $k_0 = \left(1 + \frac{R_B}{R_A}\right)$

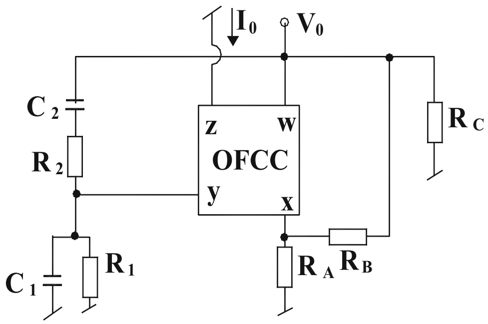


Fig. 6.22 Single OFCC-based oscillator with dual outputs proposed by Celma and Martinez [217]

Thus, the condition of oscillation can be adjusted by varying R_A or R_B. It is seen that both voltage and current outputs are available from this circuit.

The aspect of determining the limit cycle stability in CM oscillators using OFC has been investigated by Celma and Martinez in [218].

The oscillator of Fig. 6.22 was practically tested [217] and its operation verified by realizing the OFC using two CCIIs in the IC CCII01 from LTP Electronics.

6.3.2 DOCCII/MOCCII-Based VM/CM QO

The versatility of CCs is significantly enhanced when multiple copies of the output terminals are made available, thus resulting in either DOCCII or MOCCII. In the filter realizations this is obviously advantageously in the design of single-input multi-output type or multi-input multi-output type universal biquads (UB). In oscillator realization also an obvious advantage of DOCC/MOCC-based oscillators would be to provide ECO and current-mode quadrature/multiphase oscillators.

Several researchers have presented DOCC/MOCC-based oscillators fully exploiting the aforementioned advantageous features. In the following, we outline some prominent results in this direction.

Horng [259] proposed a third-order MOCCIIs based VM/CM QO with two MOCCIIs, three GCs, and three resistors. The presented circuit is shown in Fig. 6.23 which offers controllability of both CO and FO independently.

Assuming ideal MOCCIIs, an analysis of circuit in Fig. 6.23 gives the following CE:

$$s^3C_1C_2C_3R_1R_2R_3 + s^2C_3R_1R_3(C_1 + C_2) + sC_3R_3 + 1 = 0 \tag{6.19}$$

The expressions for CO and FO from the above equation are found to be

$$\text{CO: } R_3 = \frac{C_1C_2R_2}{C_3(C_1 + C_2)} \quad \text{and FO: } \omega_0 = \sqrt{\frac{1}{C_1C_2R_1R_2}} \tag{6.20}$$

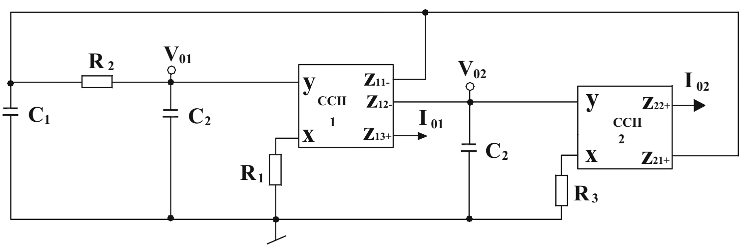


Fig. 6.23 MOCCII-based dual-mode quadrature oscillator proposed by Horng [259]

Thus, both CO and FO can be independently controlled the former by R_3 and the latter by R_1.

The output voltages V_{01} and V_{02} can be expressed as

$$\frac{V_{02}(s)}{V_{01}(s)} = -\frac{1}{sR_1C_3} \tag{6.21}$$

The phase difference (ϕ) between $V_{02}(s)$ and $V_{01}(s)$ is found to be $\phi = \pi/2$; thus, both $V_{02}(s)$ and $V_{01}(s)$ are in quadrature. Similarly, the two output currents $I_{02}(s)$ and $I_{01}(s)$ can be related as

$$\frac{I_{02}(s)}{I_{01}(s)} = -\frac{1}{sR_3C_3} \tag{6.22}$$

The phase difference (ϕ) between $I_{02}(s)$ and $I_{01}(s)$ is seen to be $\phi = \pi/2$, thus ensuring that both $I_{02}(s)$ and $I_{01}(s)$ are also in quadrature.

From the above analysis, it is seen that the oscillator circuit is capable of providing both VM and CM quadrature signals simultaneously.

SPICE simulations were carried out to check the validity of the proposed circuit using MOCC implementations based on those in [11] using 0.18 μm level 3 MOSFET parameters from TSMC. The circuit performance was found to be in good agreement with the theoretical values.

Horng [259] presented a current-mode four-phase quadrature oscillator providing four explicit current out puts while using only two MOCCs. This circuit is shown in Fig. 6.24.

This circuit is characterized by the following equations:

$$\mathrm{CO}: R_1 \le R_3 \text{ and } \mathrm{FO}: \omega_0 = \sqrt{\frac{1}{R_1R_2C_1C_2}} \tag{6.23}$$

Thus, from the above equations it is clear that R_2 controls the FO and R_3 controls the CO. Furthermore, if $R_1 = R_2 = R$ and $C_1 = C_2 = C$ then at the oscillation frequency $I_{03} = -jI_{01}$ since $I_{01} = -I_{02}$ and $I_{03} = I_{04}$, it is obvious that four current outputs will be in quadrature with each other implementing a four-phase oscillator.

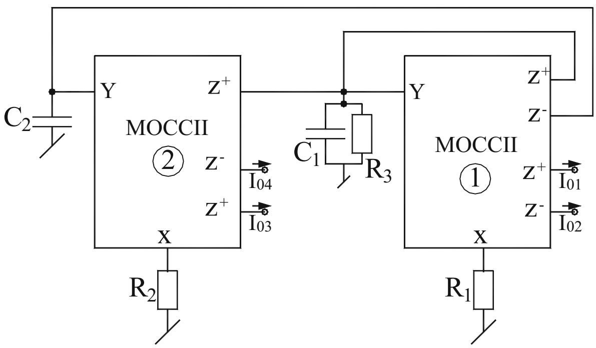


Fig. 6.24 CM four-phase quadrature oscillator proposed by Horng (adapted from [259] 2011)

6.3.3 Oscillators Using DDCCs

A DDCC has already been introduced in an earlier section of this chapter. If we combine DDCC+ and DDCC− into a single unit it can be called a differential difference complimentary current conveyor (DDCCC). It was shown by Gupta and Senani in [12] that a DDCCC is a versatile building block to implement explicit current output SRCOs. Subsequently, a complete catalog of two DDCC-based SRCOs employing all five grounded passive elements was systematically generated through the systematic state variable methodology and as a consequence a family of 24 SRCOs was generated [273]. In the following, we present six best circuits from this sets which were found to have minimum possible errors in frequency of oscillation due to the least possible effects of the parasitic impedances of the DDCCC implantation used. These circuits are shown in Fig. 6.25.

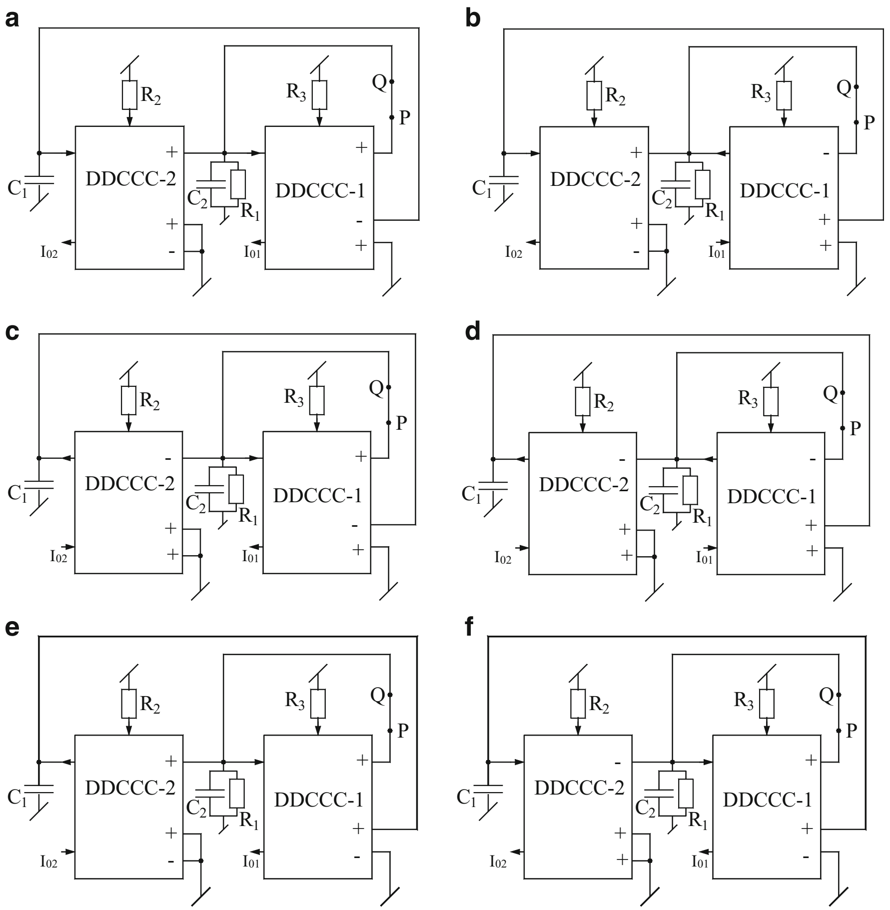


Fig. 6.25 Six DDCC-based SRCO proposed by Gupta and Senani (adapted from [273] 2003 © Frequenz)

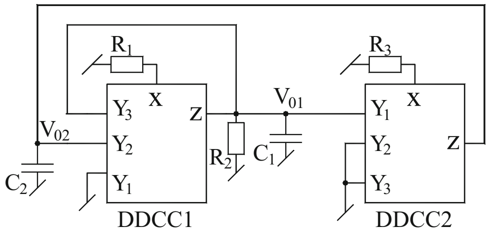


Fig. 6.26 VM quadrature oscillator proposed by Kumngern and Dejhan [243]

Subsequently, several authors have derived sinusoidal oscillator topologies using DDCCs, for instance, see [229, 233, 234, 239]. We present here in Fig. 6.26 a VM QO using two DDCCs, two grounded capacitors, and three grounded resistors which was proposed by Kumngern and Dejhan in [243]. Assuming ideal DDCCs, a routine circuit analysis yields the following CE:

$$s^2C_1C_2R_1R_3 + sC_2R_3\left\{\frac{R_1}{R_2} - 1\right\} + \frac{R_2}{R_1} = 0 \tag{6.24}$$

The CO and FO from the above equation are given by

$$\mathrm{CO}: R_1 \leq R_2 \text{ and } \mathrm{FO}: \omega_0 = \frac{1}{R_1}\sqrt{\frac{R_2}{C_1C_2R_3}} \tag{6.25}$$

It can be observed from the above equations that CO can be controlled by R_1 or/and R_2 and FO can be varied through R_3 without disturbing CO. Hence, CO and FO are orthogonally controllable.

The phase difference ϕ between V_{01} and V_{02} has been found to be

$$\phi = \pi - \tan^{-1}(\omega R_3C_2) \tag{6.26}$$

At $\omega = \omega_0$, from Eq. (6.35), ϕ can be determined as $\phi = \pi/2$; hence, V_{01} and V_{02} are in quadrature.

The circuit operation was verified [243] by a simple DVCC implementation using 0.5 μm CMOS process parameters with the circuit biased from ±2.5 V DC and component values chosen to have oscillation frequency of 649 kHz. The SPICE-generated waveform exhibited a frequency of 640 kHz with % THD as 1.02 % only.

Horng-Hou-Chang-Lin-Shiu-Chiu [233] came up with another circuit for realizing VM quadrature oscillator using only a single DDCC but also using a VF. This circuit uses a bare minimum of only four passive components, namely two capacitors and two resistors with both capacitors being grounded. This circuit is shown in Fig. 6.27 and is based upon the cascade of first-order all-pass filter and an inverting integrator. The CO and FO for this circuit are given by

$$\mathrm{CO}: C_1R_1 = C_2R_2 \text{ and } \mathrm{FO}: \omega_0 = \frac{1}{R_1C_1} \tag{6.27}$$

with voltages V_{01} and V_{02} are in quadrature.

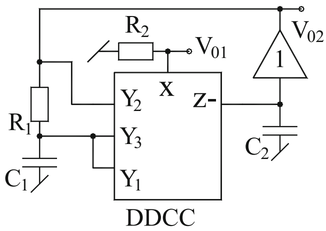


Fig. 6.27 Quadrature oscillator using DDCC proposed by Horng-Hou-Chang-Lin-Shiu-Chiu [233]

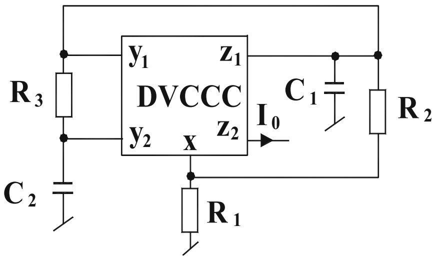


Fig. 6.28 GC-SRCO with explicit-current-output proposed by Gupta and Senani [223]

6.3.4 Oscillators Realized with DVCCs

There has been considerable interest in involving SRCO employing both grounded capacitors which is known to be an attractive feature from the view point of IC Implementation as well as ease of eliminating/accommodating various parasitic capacitances of the active elements. With this motivation in mind, GC-based SRCOs have been attempted using almost every kind of active building block. Among these, the circuits which employ only a single active element are of particular interest because of reason of reduced power consumption, low component count and also the theoretical novelty from various considerations. A GC-SRCO which provides an explicit current output while providing the following other advantageous features would be highly desirable: (i) use of only a single active building block, (ii) employment of only two GCs and no more than three resistors, (iii) provision of non-interacting control of frequency of oscillation (FO) as well as condition of oscillation (CO), and (iv) availability of a simple CO and unconstraint tuning law for the FO.

6.3.4.1 Grounded Capacitors Current Mode SRCO Using a Single DVCCC and Explicit Current Output (ECO

Although there have been several oscillators using DVCCs such as [228, 232], a circuit possessing all the above mentioned desirable properties, while using a single differential voltage complimentary current conveyor (DVCCC), was first proposed by Gupta and Senani in [223] and is shown here in Fig. 6.28.

The CO and FO for this circuit are given by

$$\text{CO}: R_1 \leq R_3 \left[\frac{C_2}{C_1 + C_2} \right] \tag{6.28}$$

$$\text{FO}: f_0 = \frac{1}{2\pi} \sqrt{\frac{2}{C_1 C_2 R_2 R_3}} \tag{6.29}$$

It is, therefore, seen that in this circuit, CO can be adjusted through R_1 without affecting the FO, which also can be independently controlled through R_2 and an explicit current output from high-output impedance terminal Z_2 is available.

For SPICE simulation of the circuit, the DVCCC was realized using the CMOS DVCC of [20] using 1.2 μm level 3 MOSFET parameters obtained from MOSIS. The DVCCC circuit was biased with ±3.3 V DC power supply. The oscillator was found to behave satisfactorily over the entire frequency range of observation (70–470 kHz).

For a fixed-frequency sinusoidal oscillator, with all grounded elements and providing an ECO, see [263] whereas a number of other ECO oscillators using transconductance second-generation current conveyor (TCCII), generalized current conveyor (GCC), and second-generation controlled current conveyor (CCCII) can be found in [45, 227, 246], respectively.

6.3.4.2 CM Quadrature Oscillator (QO) Using DVCCs

In 2011, Maheshwari and Chaturvedi [260] proposed high-output impedance CM QO using three DVCCs and four grounded passive elements. The circuit has the flexibility to generate three more QOs by appropriate connections of input terminals. Assuming ideal DVCCs (DVCC has already been defined in earlier chapter), the CE of the proposed QO, as shown in Fig. 6.29a, is given by

$$s^2 + s\left(\frac{1}{C_1 R_1} - \frac{1}{C_2 R_3}\right) + \frac{1}{C_1 C_2 R_3}\left(\frac{1}{R_2} - \frac{1}{R_1}\right) = 0 \tag{6.30}$$

from which the expressions for CO and FO are obtained as

$$\text{CO}: (C_2 R_3 - C_1 R_1) \leq 0 \tag{6.31}$$

$$\text{FO}: \ \omega_0 = \left[\frac{1}{C_1 C_2 R_3}\left(\frac{1}{R_2} - \frac{1}{R_1}\right)\right]^{1/2} \text{ provided } R_1 > R_2 \tag{6.32}$$

Hence, from the above equations, it is clear that FO can be controlled by R_2 independently. The four quadrature output currents are given by $I_4 = -I_3$,

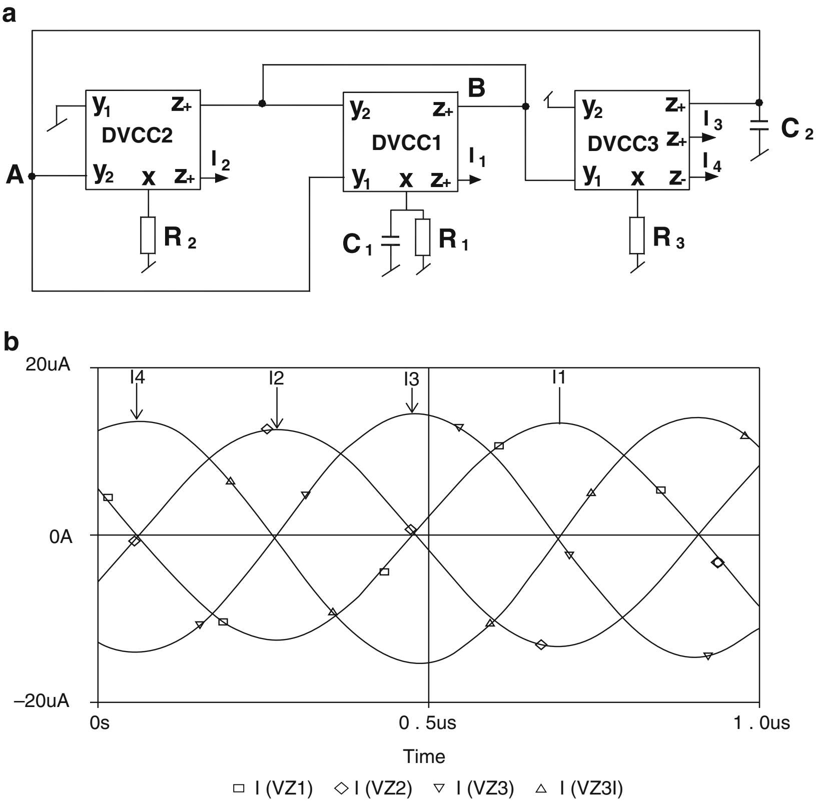


Fig. 6.29 CM QO proposed by Maheshwari and Chaturvedi (adapted from [260] © 2011 John Wiley & Sons, Ltd.). (**a**) The circuit configuration. (**b**) The four quadrature current outputs at 1 MHz

$I_2 = -I_1$, and $I_3 = -j\omega R_2 C_2 I_2$; thus for $\omega R_2 C_2 = 1$, all the quadrature outputs have equal amplitude.

The operation of the circuit was verified [260] by SPICE simulations using CMOS DVCCs realizable in 0.5 μm CMOS technology with level 3 MOSFET parameters and with DC biasing supply taken as ±2.5 V DC, component values chosen to achieve an oscillation frequency of 1.02 MHz. The SPICE-generated frequency was found to be 1.0 MHz. The error in frequency was merely 1.9 %, with % THD of around 2 % and with power consumption of 7.5 mW. Figure 6.37b shows the SPICE-generated waveforms. These results represent good correspondence with theoretical expectations.

Aggarwal-Kilinc-Cam [232] presented two oscillator topologies out of which one employs only five passive components (three resistors and two GCs). This circuit is shown in Fig. 6.30a. This circuit although provides control of FO through a single variable resistor but does not provide independent adjustability of

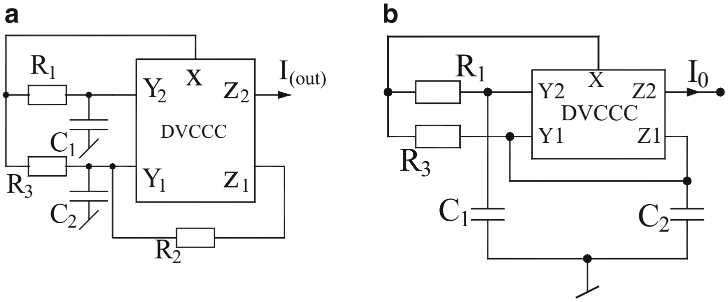
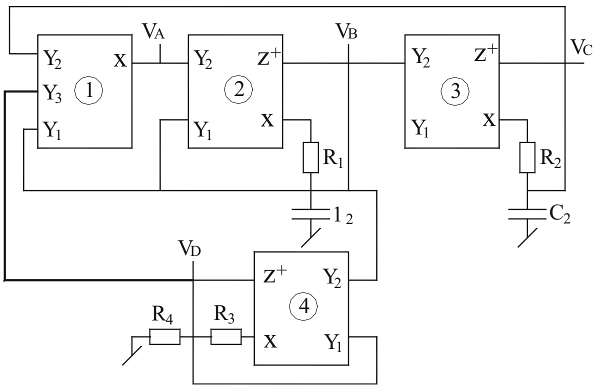


Fig. 6.30 Variable frequency oscillators using DVCCC proposed by Aggarwal-Kilinc-Cam [232]

Fig. 6.31 Four-phase oscillator proposed by Maheshwari [265]



CO. The second circuit, on the other hand, employs only two GCs and two resistors but provide independent control of CO and FO along with availability of ECO which is shown in Fig. 6.30b; the CO and FO of these two circuits are given by

$$\text{Figure 6.30a : CO: } \{C_2(R_2 + 2R_3) - C_1R_3\} = 0; \text{ and FO: } \omega_0 = \sqrt{\frac{2}{R_1R_3C_1C_2}} \tag{6.33}$$

$$\text{Figure 6.30b: CO: } (2C_2 - C_1) = 0; \text{ and FO: } \omega_0 = \frac{2}{C_1\sqrt{R_1R_3}} \tag{6.34}$$

Maheshwari [265] proposed a four-phase quadrature oscillator employing four-DVCCs along with four resistors and two GCs as shown in Fig. 6.31. The circuit provides four VM outputs having 90° out of phase with each other while providing independent control of both CO and FO, that of former by R_3 and/or the later by R_1 and/or R_2:

$$\text{CO: } 2R_4 \geq R_3 \text{ and FO: } \omega_0 = \frac{1}{\sqrt{R_1R_2C_1C_2}} \tag{6.35}$$

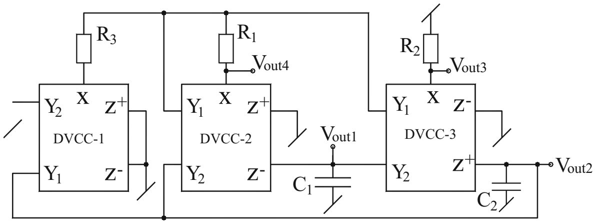


Fig. 6.32 Four-phase oscillator proposed by Maheshwari-Mohan-Chauhan [274]

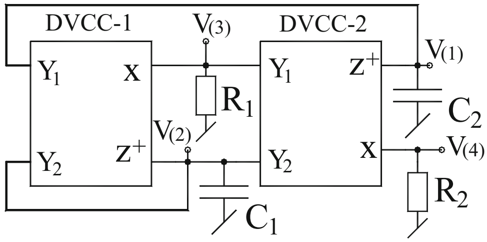


Fig. 6.33 Four-phase oscillator proposed by Maheshwari [384]

Subsequently, Maheshwari-Mohan-Chauhan [274] came up with a circuit performing the same function with a reduced number of only three DVCCs as well as reduced number of passive components namely, only three resistors and two GCs. This circuit is shown in Fig. 6.32.

The CO and FO of this circuit are given by, respectively,

$$\text{CO} : R_3 = R_1 \text{ and } \text{FO} : \omega_0 = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}} \tag{6.36}$$

Again in [384], Maheshwari came up with a further reduced component four-phase quadrature oscillator realized with two DVCCs and only two resistors and two capacitors with all passive components being grounded. This circuit is shown in Fig. 6.33.

CO and FO for this circuit configuration are given by

$$\text{CO} : R_2 C_2 \leq R_1 C_1 \text{ and } \text{FO} : \omega_0 = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}} \tag{6.37}$$

Due to being 2R-2C structure, it does not have the provision for independent adjustability of CO and FO.

For some other QO circuits using DVCCs, the readers may refer to [237, 265].

6.3.5 *Oscillators Using Third-Generation Current Conveyors (CCIII)*

The use of the third-generation current conveyors (CCIII) in oscillator realization was first demonstrated by Un and Kacar in [350] and the circuit devised by them is

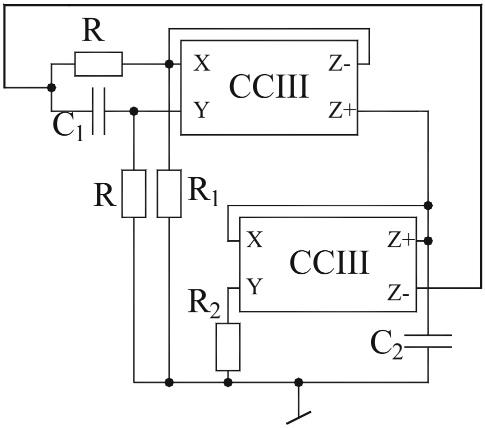


Fig. 6.34 A quadrature sinusoidal oscillator using CCIII proposed by Un and Kacar [350]

shown here in Fig. 6.34. This circuit is based upon the cascade of a first order filter using CCIII and an integrator.

By straight forward analysis, the CO and FO for this circuit are found to be

$$\text{CO}: R_2C_2 = R_1C_1 \ \text{ and } \ \text{FO}: \omega_0 = \sqrt{\frac{1}{C_1C_2R_1R_2}} \tag{6.38}$$

This quadrature oscillator was simulated in SPICE [350] with all resistors of value 50 Ω and all capacitors of value 10 nF resulting in an oscillation frequency of 159 kHz. Except small deviations caused by the parasitic impedances of the MOS CCIII employed, the simulations confirmed the workability of the CCIII-based circuit of Fig. 6.34.

It is surprising that although most of the variants of the CCs have been extensively employed to devise sinusoidal oscillators/waveform generators, the full potential of CCIIIs in this area has not been exploited to the fullest yet. This, therefore, comprises an interesting research area worthy of further investigations.

6.3.6 ICCII-Based Oscillators

ICCII provides an inverted output at terminal-x corresponding to a voltage existing or applied at terminal-Y. This certainly gives some additional flexibility and advantageous characteristic which becomes useful in certain situations/applications. Because of this, the use of ICCIIs has been investigated in number of analog signal processing applications including those of realizing sinusoidal oscillators.

6.3.6.1 ICCII-Based Grounded-Capacitor (GC) SRCO

Like other single building block based oscillators, the generation of oscillator topology using a single ICCII along with five passive components (including as usual three resistors and two capacitors) has been investigated by several authors [241, 242, 248, 249, 257]. An exemplary ICCII-based oscillator has been shown in Fig. 6.35 which employs a single ICCII, two grounded capacitors and three resistors and was proposed by Horng, Wang, and Yang in [257].

The analysis of this oscillator, assuming ideal ICCII, yields the following CE:

$$s^2C_1C_2R_1R_2R_3 + sR_1\{R_2(C_1 + C_2) - C_2R_3\} + 2R_2 = 0 \tag{6.39}$$

From the above CE, CO and FO can be determined as

$$\text{CO}: \left(R_2 - \frac{C_2R_3}{C_1 + C_2}\right) \leq 0 \ \text{ and } \ \text{FO}: \ \omega_0 = \sqrt{\frac{2}{C_1C_2R_1R_3}} \tag{6.40}$$

Thus, from the above equation, it is clear that CO can be adjusted by the grounded resistor R_2 whereas FO is independently controllable through R_1. A completely CMOS version is also possible by replacing all the resistors by CMOS resistors in which case the circuit can be made to function as a voltage controlled oscillator.

It may be noted that an ICCII can be realized from a DVCC by grounding its non-inverting Y-terminal. Thus, the circuit of Fig. 6.35 was simulated in SPICE by using the DVCC of [20] in the manner described above with CMOS DVCC biased with ±1.25 V DC power supply. Using TSMC 1.8 μm CMOS technology, the circuit could produce oscillations of frequency 1.73 MHz with % THD being no more than 2.22 %.

In conjunction with a CMOS ICCII and CMOS floating resistors in place of the three resistors in the circuit, complete CMOS version of this circuit is feasible. The theoretical predictions of this circuit were substantiated by HSPICE simulation using 0.18 μm level 49 MOSFET model parameters for TSMC process with ICC II biased with ±1.25 V DC power supply. Through rigorous simulation, it was verified that circuit would easily produce oscillations from a frequency of 1.73 MHz to about 500 kHz while maintaining THD 2.27 %.

Toker-Kuntman-Cicekoglu-Discigil [225] have carried out a systematic study of generating single ICCII-, two-ICCII-, and three-ICCII-based oscillators by

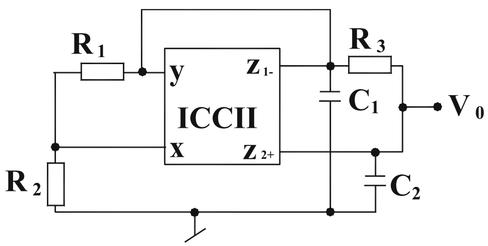


Fig. 6.35 ICCII-based sinusoidal oscillator proposed by Horng, Wang, and Yang [257]

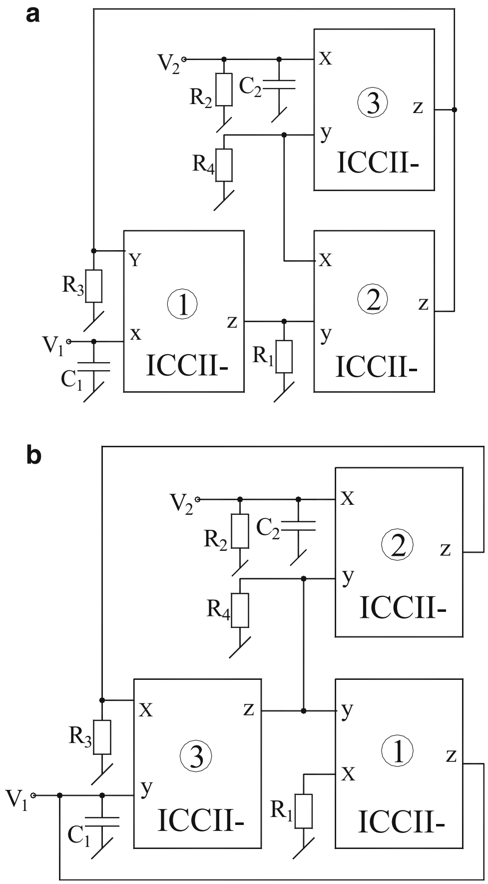


Fig. 6.36 ICCII-based sinusoidal oscillator proposed by Sobhy and Soliman [31]

formulating generalized configuration and enumerating all possible special cases yielding distinctly different circuit configurations in all the three categories mentioned above. It has been found that, what can be accomplished with other varieties of CCs, can also be performed with ICCIIs with a significant advantage being the relatively simplicity of an ICCII internal architecture as compared to other more complex CC variants. Two exemplary three ICCII-based SRCOs providing fully uncoupled tuning laws derived by Sobhy and Soliman [31] are shown in Fig. 6.36.

Both the circuits are characterized by same CO and FO which are given by

$$\mathrm{CO:}\ R_2 = R_4 \ \text{ and } \ \mathrm{FO:}\ \omega_0 = \frac{1}{\sqrt{R_1 R_3 C_1 C_2}} \tag{6.41}$$

With all resistors taken as 1 kΩ and both capacitors taken as 40 pF, the validity of this circuit was checked by SPICE simulation using CMOS ICCII (also proposed by the same authors [31]). The circuit was found to generate 3.98 MHz with CMOS ICCII biased with ±1.5 V DC power supply.

6.3.6.2 ICCII-Based All-Grounded-Passive-Element (AGPE) SRCOs

Of late, there have many attempts for systematic synthesis of oscillators employing all grounded elements, based upon the use of pathological elements and the concept of nodal admittance matrix (NAM) expansions; for instance see [28, 248, 250]. However, in the following, we present a fully uncoupled oscillator supposedly derived by intuitive method by Toker, Kuntman, Cicekoglu, and Discigil in [225]. This fully uncoupled oscillator (i.e., having both CO and FO fully uncoupled) uses three ICCIIs and six grounded passive elements and is shown in Fig. 6.37.

Assuming ideal ICCIIs, the proposed oscillator circuit in Fig. 6.37 has the following CE:

$$y_1 y_4 + y_2 y_5 - y_3 y_5 = 0 \tag{6.42}$$

From the above CE, two different versions of SRCO can be obtained by appropriate choice(s) of elements:

(i) if $y_1 = \frac{1}{R_1}, y_2 = \left(sC_2 + \frac{1}{R_2}\right), y_3 = \frac{1}{R_3}, y_4 = \frac{1}{R_4}$ and $y_5 = sC_5$

$$\text{CO}: \ (R_3 - R_2) \le 0 \ \text{ and } \ \text{FO}: \ \omega_0 = \sqrt{\frac{1}{C_5 C_2 R_1 R_4}} \tag{6.43}$$

(ii) if $y_1 = sC_1, y_2 = \left(sC_2 + \frac{1}{R_2}\right), y_3 = sC_3, y_4 = sC_4$ and $y_5 = \frac{1}{R_5}$

$$\text{CO}: \ (C_2 - C_3) \le 0 \ \text{ and } \ \text{FO}: \ \omega_0 = \sqrt{\frac{1}{C_1 C_4 R_2 R_5}} \tag{6.44}$$

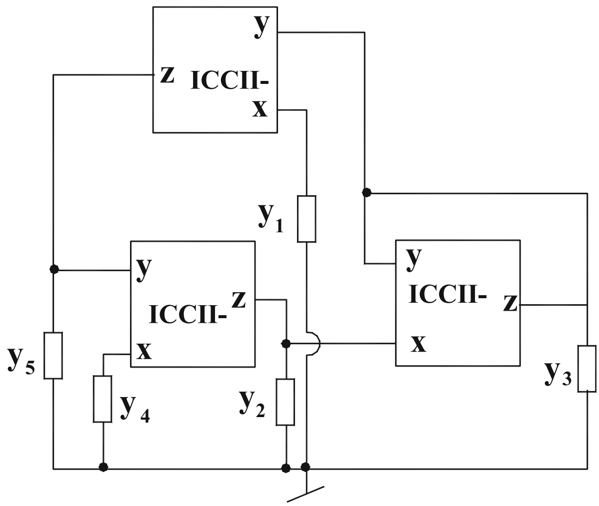


Fig. 6.37 AGPE-based SRCO proposed by Toker, Kuntman, Cicekoglu, and Discigil [225]

Out of the two SRCO versions, version (i) uses a minimum number of capacitors (two) and four resistors while version (ii) employs four capacitors and two resistors. Thus, version (i) of SRCO is superior as compared to version (ii), although both the oscillators qualify for obtaining fully uncoupled CO and FO.

The workability of the circuits described has been verified by using the CMOS ICCII from [28] using 1.2 μm CMOS process model parameters used in SPICE simulations, with component values designed to achieve an oscillation frequency of 39.78 KHz while the CMOS ICCII was biased with ±2.5 V DC power supplies. The frequency generated by SPICE simulations was found to be 38.76 kHz which is seen to be in excellent agreement with theory.

6.3.7 Oscillators Using DXCCII

DXCCII is another variant of the CC family which has two low input impedance *x*-terminals usually referred to as V_{xP} and V_{xN} such that the voltage existing or impressed upon *x*-terminal appears V_{xP} with the same sign and on V_{xN} with inverted sign furthermore two *z*-terminal currents I_{zP} and I_{zN} are, respectively, equal to the current entering at *x*P and *x*N terminals. Like almost all varieties of CCs creating a DXCC II with multiple output terminals thereby leading to V_x-MOCC II is quite feasible. Such DX-MOCC II has been exploited by Beg-Siddiqui-Ansari [256] to devise a four-phase CM quadrature oscillator which is shown in Fig. 6.38.

With $R_1 = R_2 = R$ and $C_1 = C_2 = C$ the CO and FO are given by

$$\text{CO} : \frac{2g_m}{C_3} \geq \frac{1}{RC} \quad \text{and} \quad \text{FO} : \omega_0 = \sqrt{\frac{2g_m}{RCC_3}} \tag{6.45}$$

The circuit has the advantage of employing all grounded passive components but is deficient in providing single element control of FO. Therefore, search for any improved circuit realization overcoming the above quoted limitation of with the function of resistors being accomplished by nonlinearity cancelled MOSFETs appears to be an interesting problem which is open to investigation.

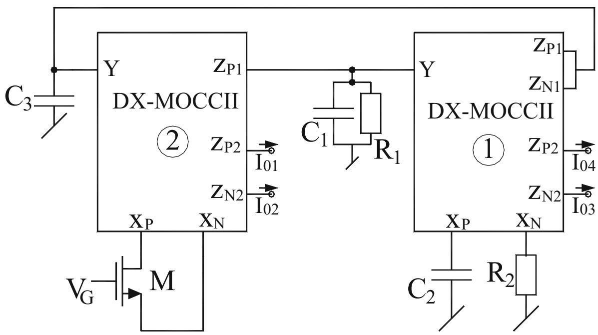


Fig. 6.38 Four-phase oscillator proposed by Beg-Siddiqui-Ansari [256]

6.3.8 FDCCII-Based SRCOs

The fully differential second-generation CCII (FDCCII) is a more generalized variant of the CCII thereby is expected to provide more flexibility and additional degrees of freedom in performing various analog signal processing functions. This was very aptly demonstrated by Chang-Al-Hashimi-Chen-Tu-Wan [226] using a single FDCCII to design CM SRCOs using five grounded passive elements and providing an explicit CM output–feet which could not be accomplished by any of the other varieties of CCs known so far, with all the same advantages.

Although there have been several studies [230, 238, 261], on evolving oscillators using FDCCII, in this section, we present two circuits which attain all the properties of the circuits described in the earlier section but with a difference; the active building block used, instead of a DVCCC, is a FDCCII, as a consequence of which it is possible to have not only both the capacitors grounded but all the resistors grounded too. The two circuits belonging to this class proposed by Chang, Al-Hashmi, Chen, Tu, and Wan [226] are shown in Fig. 6.39.

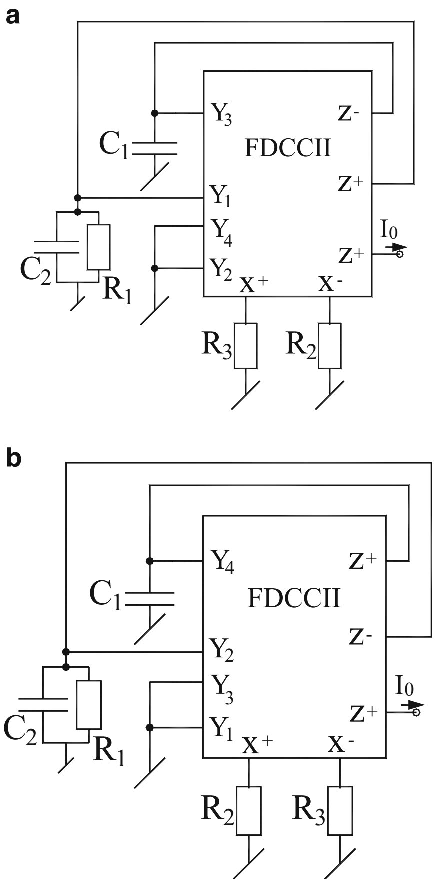


Fig. 6.39 CM SRCOs proposed by Chang, Al-Hashmi, Chen, Tu, and Wan [226]

Both of these circuits have the same characteristic equation from which the resulting CO and FO are found to be

$$\text{CO}: R_1 = R_3 \text{ and FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{C_1C_2R_2R_3}} \tag{6.46}$$

The above equation shows that the FO of these oscillators can be controlled by R_2 while CO is controllable independently by R_1.

The two oscillators described above were simulated in SPICE by realizing the CMOS FDCCII using the implementation from [36] using 0.5 μm level 3 CMOS process parameters. Additional current mirrors were used to produce additional Z-terminals. The biasing used was ±5 V DC. The components were chosen to get the theoretical frequency as 2.25 MHz whereas simulations exhibited the frequency as 2.24 MHz thus, establishing the practical viability of the proposed circuits.

A more flexible and versatile FDCCII-based SRCO was proposed by Horng-Hou-Chang-Chou-Lin-Wen in [235] which is shown in Fig. 6.40 which has all the features of the earlier circuits but in addition, this circuit, with one additional Z-terminal incorporated with the FDCCII architecture, is able to provide two explicit current mode outputs too which are also in quadrature to each other.

The CO and FO for this circuit are given by

$$\text{CO}: \ R_1 = R_2 \text{ and FO}: \ \omega_0 = \sqrt{\frac{1}{C_1C_2R_3R_2}} \tag{6.47}$$

Thus, CO can be controlled by R_1 and FO can be varied independently through R_3.

Analyzing this QO taking into account the various non-idealities such as the voltage tracking errors and current tracking errors, the terminal relationships of voltages and currents are found to be governed by the following equations:

$$V_{xa} = \alpha_{a1}v_{y1} - \alpha_{a2}v_{y2} + \alpha_{a3}v_{y3}, V_{xb} = -\alpha_{b1}v_{y1} + \alpha_{b2}v_{y2} + \alpha_{b4}v_{y4}i_{y1} = i_{y2} = i_{y3}$$
$$= i_{y4} = 0, i_{zai} = \pm\beta_{ai}i_{xa}$$

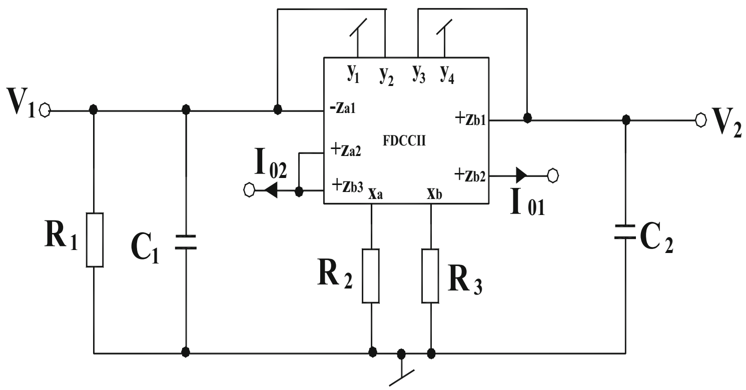


Fig. 6.40 CM QO proposed by Horng, Hou, Chang, Chou, Lin, and Wen [235]

Using the above nonideal parameters, the non-ideal CO and FO become

$$\mathrm{CO}: R_1 = \frac{R_2}{\alpha_{a2}\beta_{a1}} \text{ and FO: } \omega_0 = \sqrt{\frac{\alpha_{b2}\alpha_{a3}\beta_{a1}\beta_{b1}}{C_1 C_2 R_3 R_2}} \tag{6.48}$$

From the above equation, it is seen that both CO and FO differ slightly from their ideal expressions but still they are controllable independently.

The quadrature output voltages can be expressed (under steady state) as

$$V_1 = \omega C_2 R_3 \mathrm{e}^{j90^0} V_2 \tag{6.49}$$

and the quadrature output currents can be given by

$$I_{\mathrm{out1}} = \omega C_2 R_3 \mathrm{e}^{j90^\circ} I_{\mathrm{out2}} \quad \text{for } R_2 = R_3 \tag{6.50}$$

The FDCCII from [36] was used to verify the validity of this circuit. The MOSFET parameters from 0.18 μm CMOS process parameters from TSMC were employed and the FDCCII was biased from ±2.5 V DC power supply. Good agreement between the theoretical and simulation results was observed.

6.4 Sinusoidal Oscillator Realization Using Other Modern Electronic Circuit Building Blocks

In this section we present prominent contributions made in the realization of a variety of sinusoidal oscillators using other modern electronic circuit building blocks.

6.4.1 Unity Gain VF and Unity Gain CF-Based Sinusoidal Oscillators

Current interest in the use of unity gain voltage follower (VF) and unity gain current follower (CF), sometimes also referred as unity gain cells (UGC), in analog signal processing and circuit design is attributed to their relatively wider band width, low power consumption, and relatively simpler internal circuit architecture as compared to other more complex building blocks. In the technical literature, a very important and well-known result indicates that in principle, all the four control sources, namely VCVS, VCCS, CCVS, and CCCS, with single-resistance tunable gains and ideal input and output impedance levels, can be realized using only VFs and CFs along with a bare minimum resistors with the very significant property of all variable gains realizable with the same constant bandwidth.

In the context of oscillator realization, it was demonstrated by Senani in 1985 (see Chap. 2) that it is possible to devise SRCOs using only VFs as active elements. However, all such circuits derived, therein, were noncanonic circuits since they employ three capacitors to realize essentially a second-order dynamics.

Senani and Gupta in [270] demonstrated that *if VFs are combined with CFs* then not only canonic SRCOs are feasible to realize, it is also possible to ground both the capacitors as preferred for integrated circuit implementation.

During the past several years, a number of researchers have worked upon deriving SRCO structures using UGCs. In the following, we present some prominent SRCO configurations realizable with VFs and CFs.

A VF-CF based sinusoidal oscillator and its CFOA implementation is shown here in Fig. 6.41.

The condition of oscillation (CO) and the frequency of oscillation (FO) for both the circuits are given by

$$\mathrm{CO}: \ R_1 = \frac{C_1 R_3}{C_1 + C_2} \ \text{ and } \ \mathrm{FO}: f_0 = \frac{1}{2\pi\sqrt{C_1 C_2 R_2 R_3}} \tag{6.51}$$

The novelty of the CFOA-based SRCO of Fig. 6.26 is that from the same circuit, both VM and CM outputs are available explicitly.

Senani and Gupta in [270] presented two SRCOs shown in Fig. 6.42 both of which employ three resistors and two GCs. In the circuit of Fig. 6.42a, two VFs and

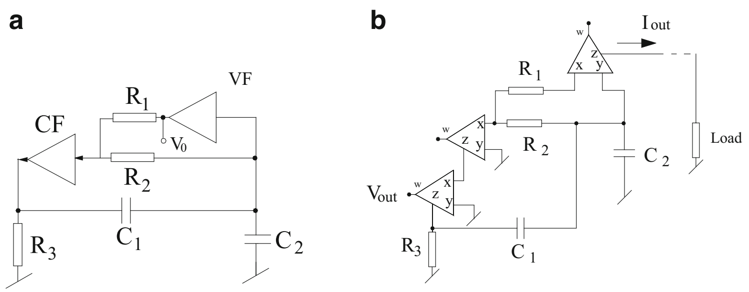
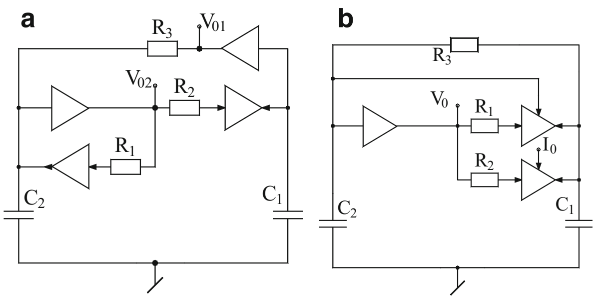


Fig. 6.41 (**a**) A VF-CF-based SRCO. (**b**) CFOA implementation

Fig. 6.42 SRCOs using CFs and VFs proposed by Senani and Gupta [270]



two CFs are employed and the circuit is capable of providing two VM outputs.)In the second circuit of Fig. 6.42b, only three followers are required out of which one is VF and the remaining two are dual output CFs. The second circuit provides one VM output and one CM output.

The CO and FO of these oscillator circuits are given by

$$\text{For the Fig. 6.42a, CO}: R_1 = R_3 \text{ and FO}: \ \omega_0 = \frac{1}{\sqrt{R_2R_3C_1C_2}} \tag{6.52}$$

$$\text{For the Fig. 6.42b, CO}: \ (C_1 + C_2)R_1 = C_1R_3 \ \text{ and } \ \text{FO}: \ \omega_0 = \frac{1}{\sqrt{R_2R_3C_1C_2}} \tag{6.53}$$

Subsequently, Gupta and Senani [271] systematically derived a family of seven circuits, five of which are shown here in Fig. 6.43.

The derivation of these circuits followed the state variable methodology advanced by Senani and Gupta in the context of CFOA-based oscillators which have already been detailed in Chap. 4 of this monograph. The CO and FO for the) circuit of Fig. 6.43a are same as given by Eq. (6.52). The CO and FO for the structures of Fig. 6.43b, c are same and are given by

$$\text{CO}: R_1C_2 = R_3C_1 \ \text{ and } \ \text{FO}: \omega_0 = \frac{1}{\sqrt{R_2R_3C_1C_2}} \tag{6.54}$$

The CO and FO for the circuit of Fig. 6.43d are same as given by Eq. (6.53).

The salient simulation results) of these circuits using CMOS VFs and CFs are shown in Fig. 6.44.

Subsequently, Torres-Papqui-Torres-Munoz and Tlelo-Cuatle [272] presented a modified version of one of the circuits proposed by Senani and Gupta in [270] which is shown herein Fig. 6.45. This circuit however employs three capacitors (along with three resistors); however, all of them are grounded and the circuit employs two VFs and two CFs of normal kind.

More recently, Soliman [253] presented four new UGC-based oscillators all employing two VFs and two CFs, all employing two GCs. These circuits are shown in Fig. 6.46. The CO and FO of these oscillators are given by

$$\text{CO}: R_1 = R_3 \ \text{ and } \ \text{FO}: \omega_0 = \frac{1}{\sqrt{C_1C_2R_1R_2}} \tag{6.55}$$

In [186], Gupta and Senani reported five SRCOs, all of which require no more than two followers each, three resistors and two capacitors which are shown in Fig. 6.47.

By a straight forward)analysis, the CO of the circuits is given by $R_1 = (C_2/C_1)R_3$ for Fig. 6.47a, $R_1 = (C_1/C_2)R_3$ for Fig. 6.47b, $R_1 = \frac{C_2R_3}{C_1+C_2}$ for Fig. 6.47c, and

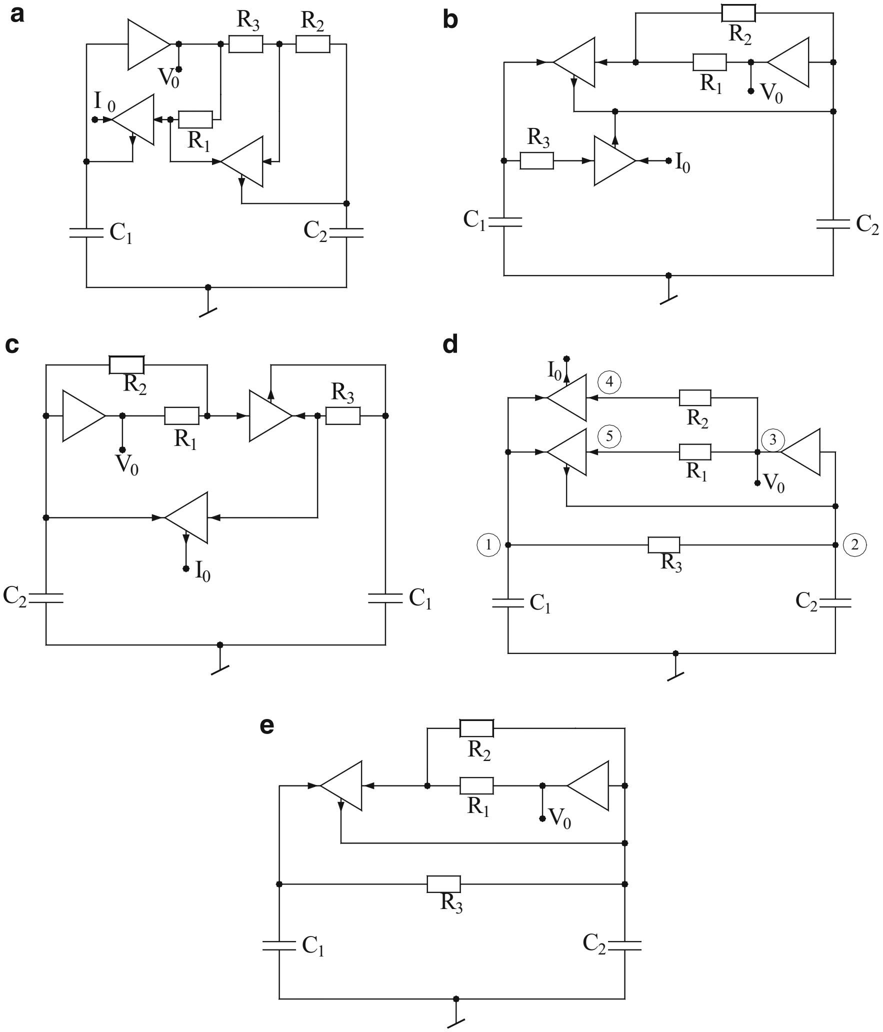


Fig. 6.43 Five SRCOs using CFs and VFs proposed by Gupta and Senani [271]

$$R_1 = \frac{C_2 R_3}{C_1 + C_2} \text{ for Fig. 6.47d, e.} \tag{6.56}$$

whereas FO of all the circuits is given by $\omega_0 = \dfrac{1}{\sqrt{R_2 R_3 C_1 C_2}}$ (6.57)

It is seen that CO and FO are independently controllable through R_1 and R_2, respectively. Furthermore, use of dual-output CF as in Fig. 6.47e permits the use of both grounded capacitors. Lastly, all oscillators are capable of providing explicit VM outputs from low-output impedance node.

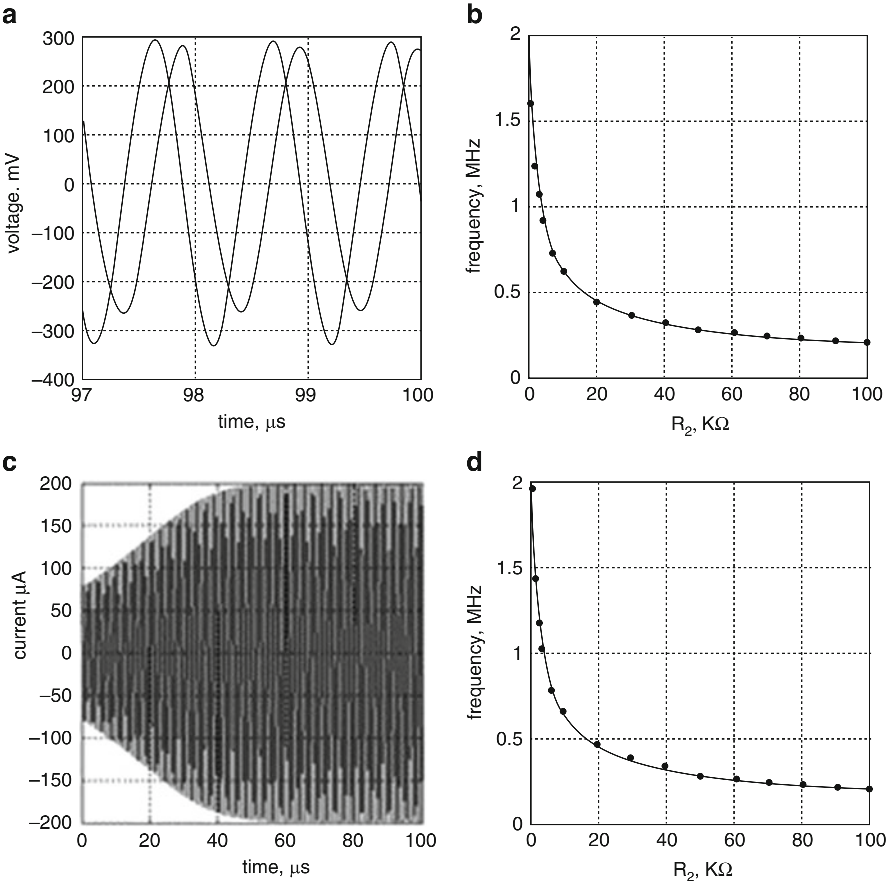


Fig. 6.44 Simulation results. (**a**) VM quadrature response of Fig. 6.42a. (**b**) Variation of VM f_0 with R_2 for Fig. 6.42a (1.62–0.2 MHz). (**c**) CM transient response of Fig. 6.43c ($f_0 = 1.01$ MHz; 400 μA p–p). (**d**) Variation of CM f_0 with R_2 of Fig. 6.43c (1.96–0.2 MHz)

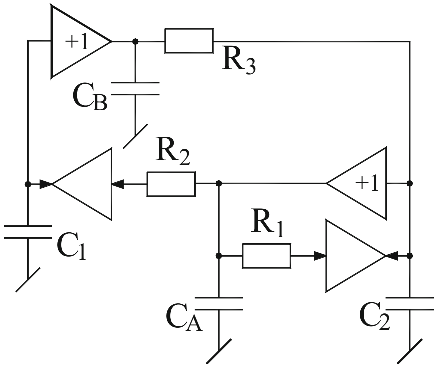


Fig. 6.45 SRCO using VF and CF proposed by Torres-Papqui-Torres-Munoz and Tlelo-Cuatle [272]

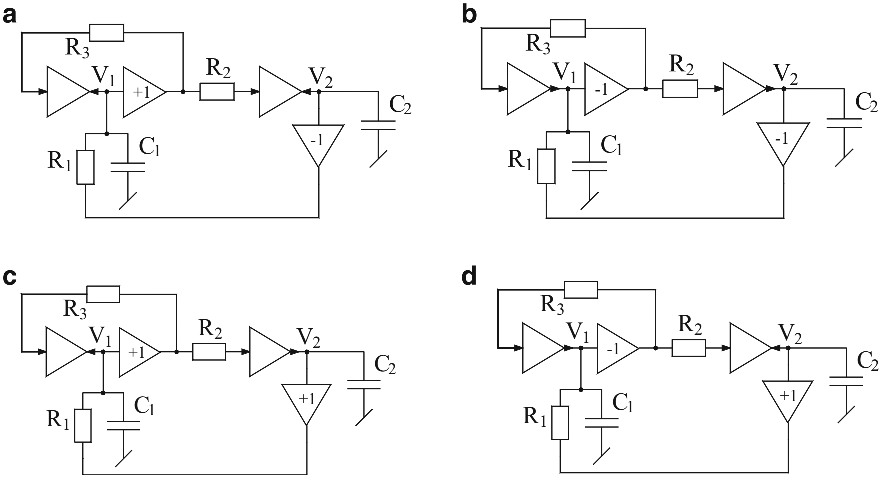


Fig. 6.46 VF-CF based SRCOs proposed by Soliman [253]

An explicit CM output can also be obtained by replacing the CFs by dual-output CFs in the first four cases and by creating one additional terminal in the DOCF in the last case.

6.4.2 *Oscillators Using FTFNs/OMAs*

The FTFN shave been recognized to be the most generalized building blocks since in conjunction with resistors, theoretically they can be used to represent any other building block. In the literature, two varieties of FTFN have been often utilized, namely the NFTFN and PFTFN. However, out of these two only the former is a four-terminal-floating element whereas the latter because of both *Z*-terminal and *w*-terminal currents going into the respective ports cannot be treated to be FTFN. In fact, as argued by a number of authors,[3] a norator with a plus sign cannot be defined as two terminal element at all. Hence, the so-called PFTFN is better identified as an OMA+. In this context, it may also be, therefore, agreed that the other counterpart, i.e., the NFTFN, by the same argument is equivalent to an OMA−. Furthermore, an NFTFN/OMA− and the so-called operational floating amplifier are, in fact, one and the same thing when looked upon from the viewpoint of terminal characteristics of their four terminals, although their internal architectures and their manner of realizations may differ.

[3] For instance, see Senani R (1988) Floating immittance realization: Nullor approach. Electron Lett: 24: 403–405; Soliman AM (2010) On the four terminal floating nullor (FTFN) and the operational Mirror amplifier (OMA). J Active Passive Electron Devices: 5: 209–219.

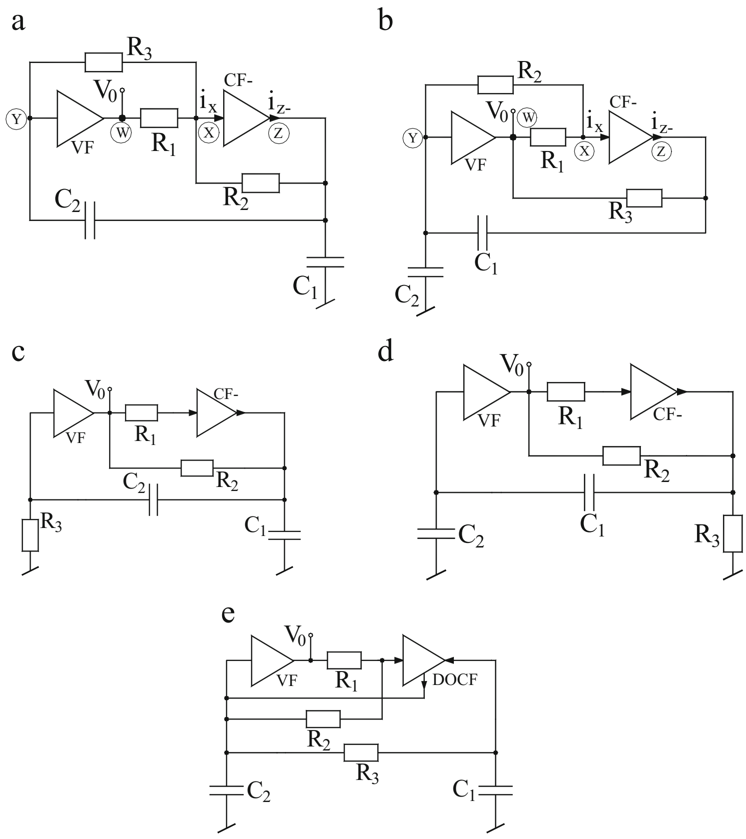
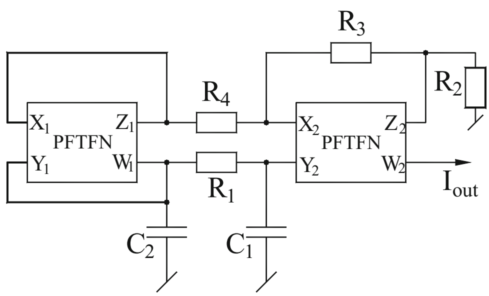


Fig. 6.47 SRCOs using CFs and VFs proposed by Gupta and Senani [186]

Fig. 6.48 Current-mode SRCO using two PFTFNs/OMAs+ proposed by Bhaskar and Senani [275]



NFTFNs and PFTFNs (OMAs) have been used to realize oscillators by a number of researchers and provide the obvious advantage of availability of explicit current output. Although a number of two active building blocks based SRCOs employing both GCs have been reported using a variety of active elements, a circuit using two PFTFNs/OMAs was proposed by Bhaskar and Senani [275] which is shown here in Fig. 6.48.

The CO and FO of this circuit are given by

$$\text{CO}: \left\{\frac{1}{R_1C_1}+\frac{1}{C_2}\left(\frac{1}{R_1}-\frac{1}{R_4}\right)\right\} \leq 0 \text{ and } \text{FO}: \omega_0 = \sqrt{\frac{R_3}{C_1C_2R_2R_4}} \tag{6.58}$$

A systematic derivation of CM sinusoidal oscillators using two FTFNs starting from a previously known op-amp RC circuit by Senani (see reference [36] of Chap. 2) and applying on it a nullor-based transformation method also proposed by Senani (see reference [2] of Chap. 10) was carried out by Abuelma'atti and Al-Zaher [155] which resulted in a complete catalogue of 14 two-FTFN-based circuits, out of which only two employed both GCs. Of these two circuits only one had the provision of providing two explicit current outputs from one of the output terminals of the FTFN. This circuit is shown in Fig. 6.49.

This particular circuit (as well as all other circuits in the quoted catalogue) is characterized by the following CO and FO:

$$\text{CO}: \left(1+\frac{C_2}{C}\right) = \frac{R_1R_5}{R_2R_4} \text{ and } \text{FO}: \omega_0 = \sqrt{\frac{R_1R_3}{C_1C_2R_2R_4R_5R_e}} \tag{6.59}$$

A circuit employing only a single FTFN and five passive elements including two GCs and still capable of providing non-interacting control of CO and FO both was proposed by Bhaskar [157]. This circuit is shown in Fig. 6.50.

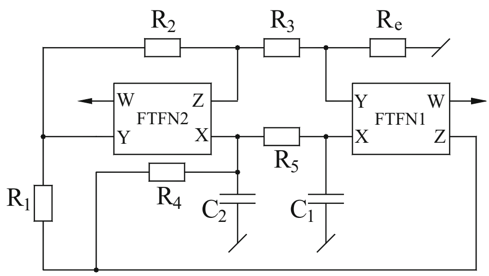


Fig. 6.49 Current-mode SRCO using two PFTFNs/OMAs+ proposed by Abuelma'atti and Al-Zaher [155]

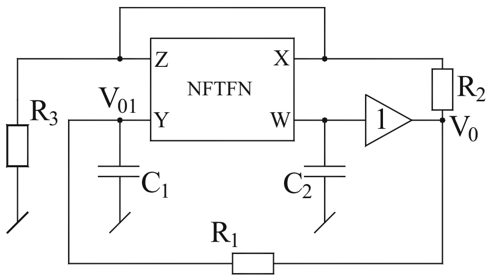


Fig. 6.50 SRCO using single NFTFN proposed by Bhaskar [157]

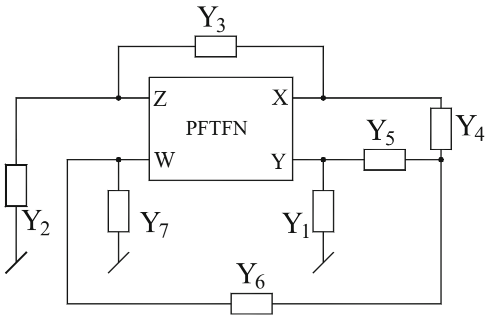


Fig. 6.51 SRCO using single PFTFN (OMA+) proposed by Kumar-Senani [277]

The CO and FO of this circuit are given by the following equations:

$$\mathrm{CO:}\ \left(\frac{C_2}{R_1}-\frac{C_1}{R_2}\right)=0\ \ \text{and}\ \ \mathrm{FO:}\ \omega_0=\sqrt{\frac{1}{C_1C_2R_1R_3}} \tag{6.60}$$

Subsequently, Kumar-Senani in [277] p+)roposed another PFTFN (OMA+)-based circuit which employed five resistors and two capacitors (both grounded) is shown in Fig. 6.51, where Y_1 and Y_2 are selected as capacitors and rest admittances as resistors.

The CO and FO for this circuit are given by

$$\mathrm{CO:}\ \frac{C_2}{C_1}=1+R_5\left(\frac{2}{R_4}+\frac{1}{R_7}\right)+\frac{R_6}{R_7}\left(1+\frac{R_5}{R_4}\right)\ \ \text{and}\ \ \mathrm{FO:}\ \omega_0=\sqrt{\frac{R_4}{C_1C_2R_3R_5R_7}} \tag{6.61}$$

This circuit was experimentally tested by realizing the OMA with two AD844 biased with ±12 V DC power supplies with component values selected as $C_1=1$ nF, $C_2=10$ nF, $R_4=R_5=R_7=10$ kΩ with CO adjusted by 10 kΩ pot connected in series with 20 kΩ resistor. The variation of FO with controlling resistor R_3 is shown in Fig. 6.52a; a typical waveform is shown in Fig. 6.52b and the power spectral density is shown in Fig. 6.52c.

A generalized derivation of all possible single FTFN-based SRCOs based upon a generalized configuration containing as many as eight admittances was carried out by Cam-Toker- Cicekoglu-Kuntman [161] who derived a family of eight SRCOs. These circuits are shown in Fig. 6.53.

The CO and FO of these oscillators have been shown in Table 6.1

6.4.3 Oscillators Using DDAs

The use of differential difference amplifier (DDA) was known in literature in a number of applications for realizing linear and nonlinear functions [286–289,

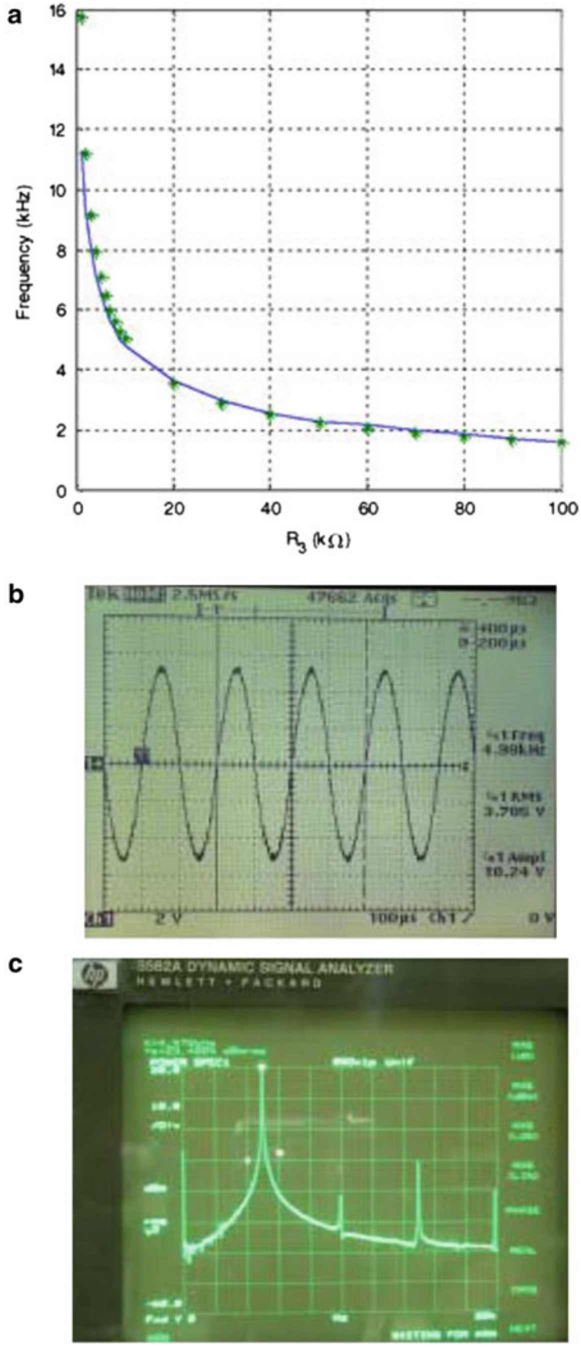


Fig. 6.52 (**a**) Variation of FO with R_3 of Fig. 6.51. (**b**) A typical waveform (4.98 kHz, 3.706 V 9RMS) for $R_3 = 10$ kΩ of Fig. 6.51. (**c**) PSD of the oscillator of Fig. 6.51 observed on HP 3562A Dynamic signal analyzer [277]

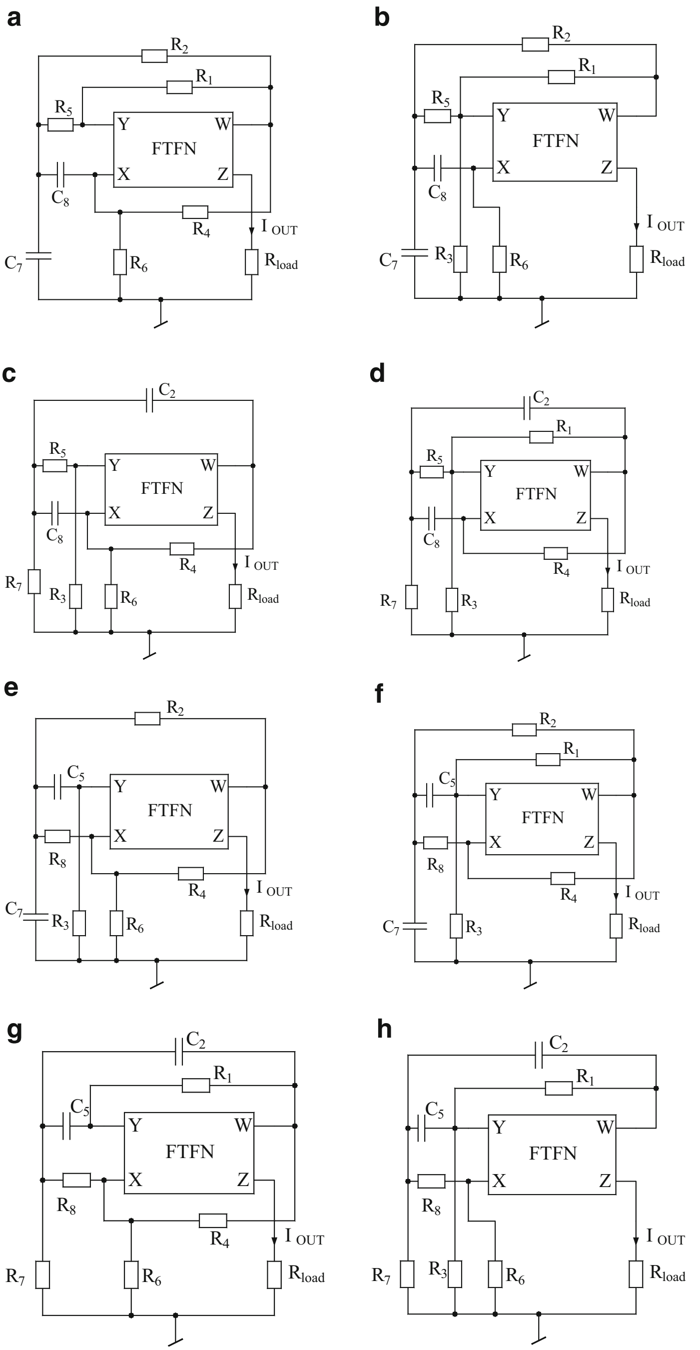


Fig. 6.53 Single FTFN-based SRCOs proposed by Cam-Toker-Cicekoglu-Kuntman [161]

Table 6.1 CO and FO for the FTFN-based SRCOs of Fig. 6.53

Circuit	CO	FO
Figure 6.53a	$1+\frac{C_8}{C_7}=\frac{R_1R_6}{R_4R_5}$	$\omega_0=\sqrt{\frac{R_1}{R_6C_7C_8}\left\{\frac{1}{R_2}\left(\frac{1}{R_1}+\frac{1}{R_5}\right)+\frac{1}{R_1R_5}\right\}}$
Figure 6.53b	$1+\frac{C_7}{C_8}=\frac{R_1R_6}{R_2R_3}$	$\omega_0=\sqrt{\frac{R_1}{R_6C_7C_8}\left\{\frac{1}{R_2}\left(\frac{1}{R_1}+\frac{1}{R_5}\right)+\frac{1}{R_1R_5}\right\}}$
Figure 6.53c	$1+\frac{C_8}{C_2}=\frac{R_3R_4}{R_5R_6}$	$\omega_0=\sqrt{\frac{R_3}{R_4C_2C_8}\left\{\frac{1}{R_7}\left(\frac{1}{R_3}+\frac{1}{R_5}\right)+\frac{1}{R_3R_5}\right\}}$
Figure 6.53d	$1+\frac{C_2}{C_8}=\frac{R_3R_4}{R_1R_7}$	$\omega_0=\sqrt{\frac{R_3}{R_4C_2C_8}\left\{\frac{1}{R_7}\left(\frac{1}{R_3}+\frac{1}{R_5}\right)+\frac{1}{R_3R_5}\right\}}$
Figure 6.53e	$1+\frac{C_7}{C_5}=\frac{R_3R_4}{R_2R_6}$	$\omega_0=\sqrt{\frac{R_4}{R_3C_5C_7}\left\{\frac{1}{R_8}\left(\frac{1}{R_2}+\frac{1}{R_4}\right)+\frac{1}{R_2R_4}\right\}}$
Figure 6.53f	$1+\frac{C_5}{C_7}=\frac{R_3R_4}{R_1R_8}$	$\omega_0=\sqrt{\frac{R_4}{R_3C_5C_7}\left\{\frac{1}{R_8}\left(\frac{1}{R_2}+\frac{1}{R_4}\right)+\frac{1}{R_2R_4}\right\}}$
Figure 6.53g	$1+\frac{C_2}{C_5}=\frac{R_1R_6}{R_4R_7}$	$\omega_0=\sqrt{\frac{R_6}{R_1C_5C_2}\left\{\frac{1}{R_8}\left(\frac{1}{R_6}+\frac{1}{R_7}\right)+\frac{1}{R_6R_7}\right\}}$
Figure 6.53h	$1+\frac{C_5}{C_2}=\frac{R_1R_6}{R_3R_8}$	$\omega_0=\sqrt{\frac{R_6}{R_1C_5C_2}\left\{\frac{1}{R_8}\left(\frac{1}{R_6}+\frac{1}{R_7}\right)+\frac{1}{R_6R_7}\right\}}$

408]. However, before 2013, any circuit for realizing SRCOs using DDAs had not been reported in the literature. This void was filled by Singh-Singh-Senani [283] who presented a class of two-DDA-based SRCOs shown here in Fig. 6.54, each employing two DDAs, three resistors, and two GCs which provided independent control of CO and FO through separate resistors. The COs and FOs of these configurations are given by

$$\text{For the Fig. 6.54a CO: } 2C_1R_1=R_3C_2 \text{ and FO: } \omega_0=\sqrt{\frac{2}{C_1C_2R_2R_3}} \tag{6.62}$$

$$\text{For the Fig. 6.54b CO: } C_1R_1=2R_3C_2 \text{ and FO: } \omega_0=\sqrt{\frac{1}{C_1C_2R_2R_3}} \tag{6.63}$$

$$\text{For the } f \text{ Fig. 6.54c CO: } \frac{R_3}{R_1}=\left(1+\frac{C_1}{C_2}\right) \text{ and FO : } \omega_0=\sqrt{\frac{1}{C_1C_2R_2R_3}} \tag{6.64}$$

$$\text{For the Fig. 6.54d CO: } R_1=R_3 \text{ and FO : } \omega_0=\sqrt{\frac{1}{C_1C_2R_2R_3}} \tag{6.65}$$

The workability of these circuits has been confirmed by both hardware implementations and SPICE simulations. For hardware implementation, video

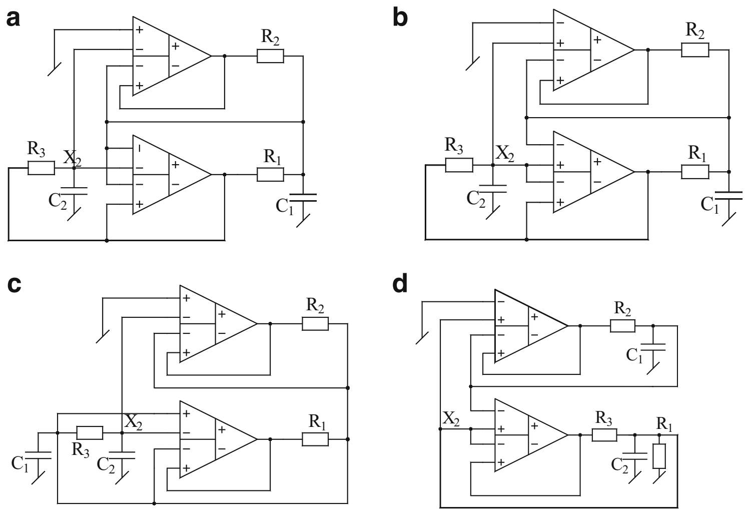
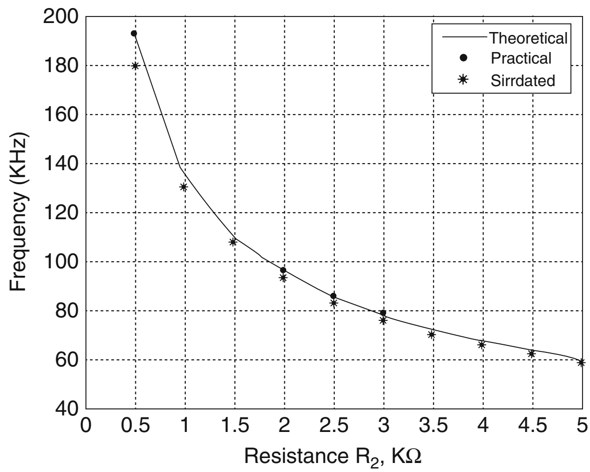


Fig. 6.54 Two DDA-based SRCOs proposed by Singh-Singh-Senani [283]

Fig. 6.55 Variation of FO with R_2



differencing amplifiers AD830 from Analog Devices were used which have gain bandwidth product of 85 MHz and a slew rate of 360 V/μs. All the four oscillators have been tested experimentally using two AD830, biased with ±5 V DC power supplies with passive components taken as $C_1 = C_2 = 1125$ pF, $R_3 = 1.1$ kΩ. In all cases the frequency controlling resistance R_2 was varied from 0.5 to 5 kΩ. Figure 6.55 shows an exemplary result of oscillation frequency (f_0) with R_2 which also shows the variation of f_0 with respect to R_2 obtained by SPICE simulations (using CMOS DDA of Fig. 6.56) of Fig. 6.54d. The CMOS resistors were realized by the 0.18 μm TSMC CMOS models made available through MOSIS. This DDA

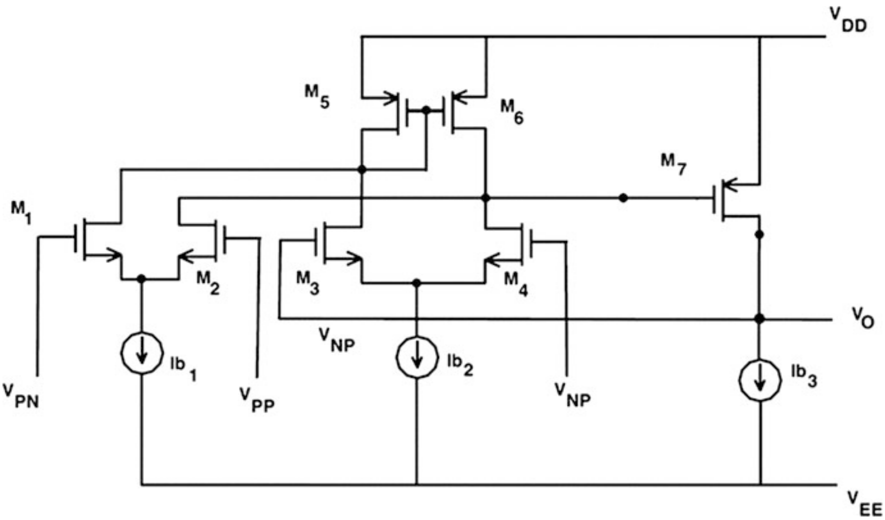
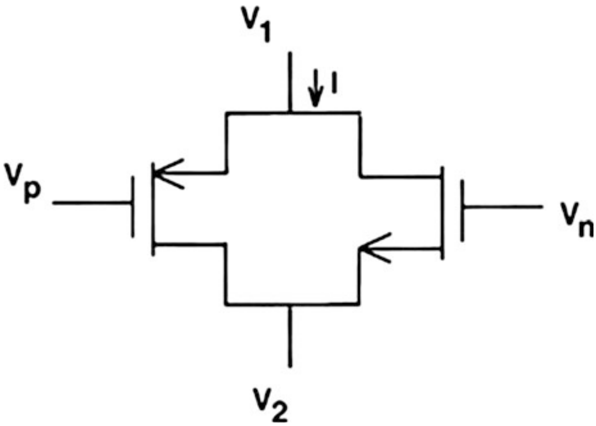


Fig. 6.56 CMOS realization of DDA [284]

Fig. 6.57 Linear voltage-controlled MOS resistor [285]



was biased with $V_{DD} = -V_{EE} = 3$ V DC power supplies and the aspect ratios as per Table 2 of [283] and the biasing currents were taken as $I_{b1} = I_{b2} = 20$ μA and the value of $I_{b3} = 100$ μA. A completely CMOS version of Fig. 6.54a was obtained by replacing resistors R_1, R_2, and R_3 by the two-MOSFET linear voltage-controlled resistor (VCR) of Fig. 6.57. The variation of FO with control voltage V_n (keeping $V_p = -0.6$ V for VCR replacing R_2) with the control voltages 2 V and 1 V for the CMOS VCRs realizing R_1 and R_3, respectively, is shown in Fig. 6.58. These experimental and simulation results thus, confirm the workability of the proposed SRCOs of Fig. 6.54.

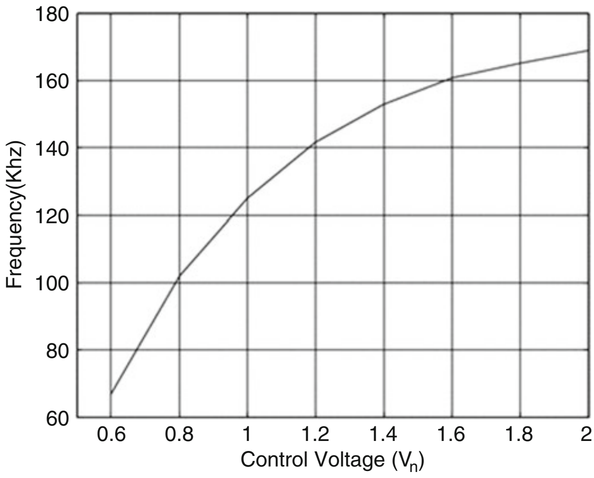


Fig. 6.58 Variation of FO with control voltage V_n

6.4.4 *Oscillators Using Modified CFOAs*

A number of researchers have proposed modified forms of CFOAs such as [278–282]. In the following we describe the use of three specific types of modified CFOAs in oscillator realization whose use has been shown to result oscillator structures possessing interesting properties. In particular we consider the use of differential voltage current feedback amplifier (DVCFA), fully differential CFOA (FDCFOA), and differential difference complementary feedback amplifier (DDCFA) in the synthesis of oscillators.

In Chap. 5 of this monograph we have discussed oscillators realized from normal type of CFOA characterized by: $i_y = 0$, $v_x = v_y$, $\mathrm{i_z} = i_x$ and $v_w = v_z$. It was found that SRCOs requiring GCs need at least two CFOAs. To the best knowledge of the authors no single CFOA circuit has so far been known which can realize an SRCO using both GCs. Gunes and Toker [280] however demonstrated a systematic method by extending the state variable methodology proposed by Senani and Gupta (see reference [35] of Chap. 5 of this monograph) that CFOA-based SRCOs having one GC and one floating capacitor can be converted into both GC SRCOs if DVCFA is employed in place of CFOA. This method is applicable to all those SRCOs which have first order high-pass sections in one of the feedback loops using a DVCFA. This high-pass section conveniently replaced by a low-pass section (thereby containing a GC) by judicious use of the terminal characteristics of the DVCFA which provides $v_x = (v_{y1} - v_{y2})$ which easily creates the complement of first-order low-pass filter as a first-order high pass filter present in the original network. Gunes and Toker [280] have thus derived eight DVCFA-based SRCOs containing both GCs corresponding to eight CFA-based SRCOs none of which had

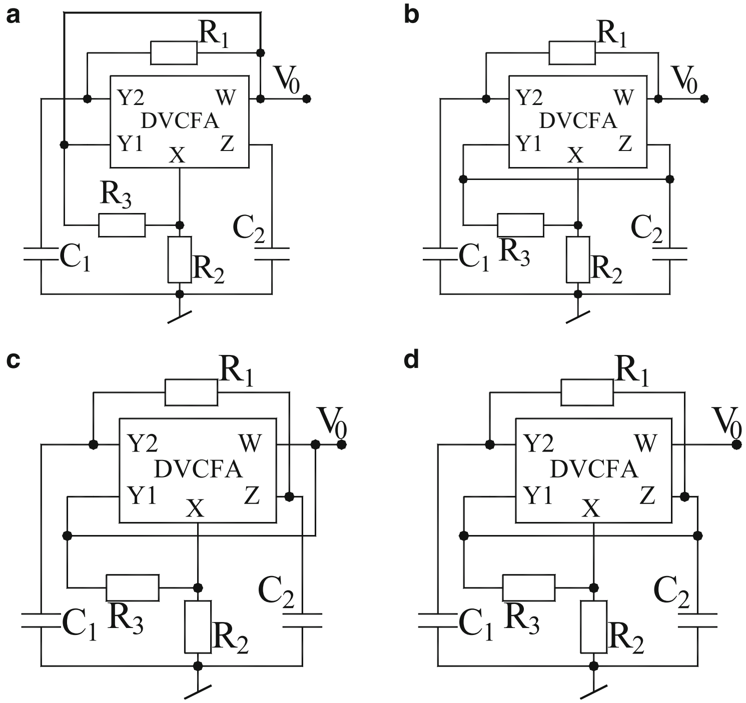


Fig. 6.59 DVCFA-based oscillators proposed by Gunes and Toker [280]

both GCs. Four exemplary SRCOs from the quoted set are shown in Fig. 6.59 whose COs and FOs are as follows:

$$\text{For Fig. 6.59a; CO: } C_1R_1 = R_2C_2 \text{ and FO: } \omega_0 = \sqrt{\frac{1}{C_1C_2R_1R_3}} \tag{6.66}$$

$$\text{For Fig. 6.59b; CO: } C_1R_1 = R_2C_2 \text{ and FO: } \omega_0 = \sqrt{\frac{1}{C_1C_2R_1R_3}} \tag{6.67}$$

$$\text{For Fig. 6.59c; CO: } \frac{1}{C_2R_2} = \frac{1}{C_1R_1} + \frac{1}{C_2R_4} \text{ and FO: } \omega_0 = \sqrt{\frac{(R_3 + R_4)}{C_1C_2R_1R_3R_4}} \tag{6.68}$$

$$\text{For Fig. 6.59d CO: } \frac{1}{C_2R_2} = \frac{1}{C_1R_1} + \frac{1}{C_2R_4} \text{ and FO: } \omega_0 = \sqrt{\frac{(R_3 + 2R_4)}{C_1C_2R_1R_3R_4}} \tag{6.69}$$

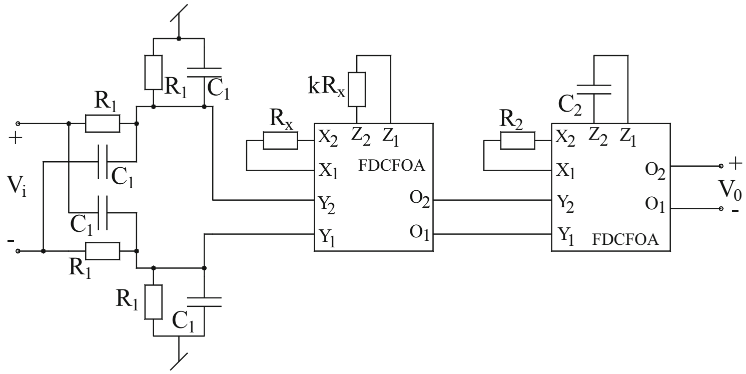


Fig. 6.60 FDCFOA-based QO proposed by Un and Kacar [281]

The FDCFOA has the same characteristics as a normal CFOA, the difference being that the y-input terminals ($y_1 - y_2$) and x-input terminal ($x_1 - x_2$) both are differential and so are the current output terminals ($z_1 - z_2$) and voltage output terminals ($O_1 - O_2$). Thus, FDCFOA has all the advantages of the single ended CFOA and in addition, offers all the advantages of a fully differential signal processing. Un and Kacar [281] presented a VM first-order all-pass filter (APF) and a quadrature oscillator (QO) made by cascading a fully differential integrator in a closed loop along with first-order fully differential APF. This circuit is shown in Fig. 6.60. By a routine analysis, the CO and FO of this QO are found to be

$$\text{CO}: kC_1R_1 = R_2C_2 \ \text{ and } \ \text{FO}: \omega_0 = \sqrt{\frac{1}{C_1C_2R_1R_2}} \tag{6.70}$$

The workability of this proposition was verified by a CMOS realization of the FDCFOA.

Gupta and Senani in [282] demonstrated the usefulness of new variant of CFOA which they called differential difference complementary CFA (DDCCFA). It was shown that using state variable methodology, DDCCFA can be used to realize a number of new SRCOs possessing features all of which were not available simultaneously in any of the earlier known single active building block-based SRCOs. It was also demonstrated that a large variety of CCs and CFOAs could be realized from the DDCCFA as special cases. A family of six SRCOs using a single DDCCFA is presented here in Figs. 6.61 and 6.62. The CO and FO of these oscillators are given by

$$\text{For Fig. 6.61a, b; CO}: C_1R_1 = R_3C_2 \ \text{ and } \ \text{FO}: \omega_0 = \sqrt{\frac{1}{C_1C_2R_2R_3}} \tag{6.71}$$

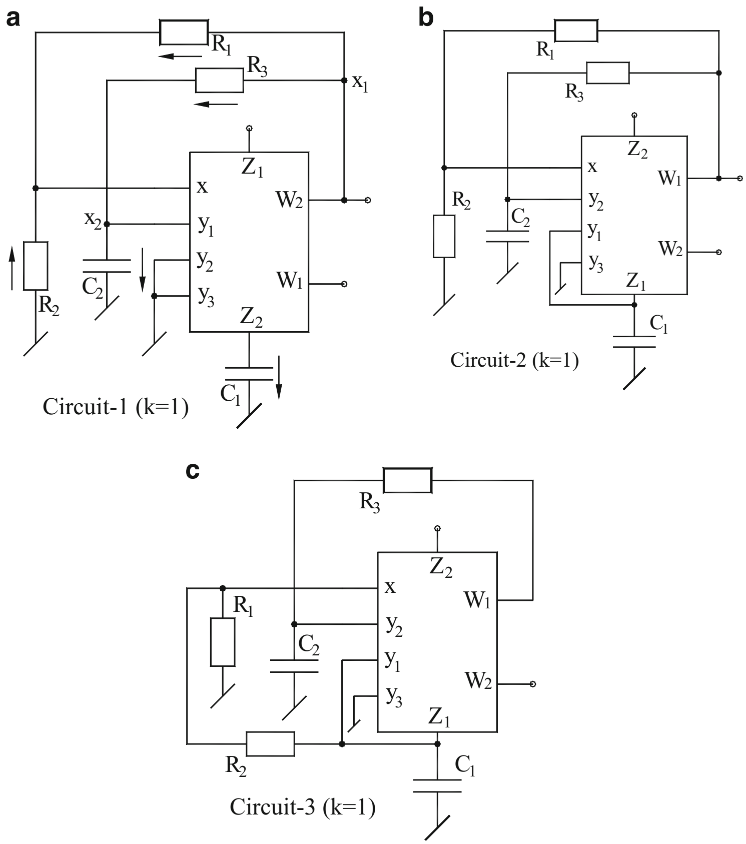


Fig. 6.61 Some SRCOs using the DDCCFA proposed by Gupta-Senani [282]

$$\text{For Fig. 6.61c; CO: } C_1R_1 = R_3C_2 \text{ FO: } \omega_0 = \sqrt{\frac{2}{C_1C_2R_2R_3}} \tag{6.72}$$

$$\text{For Fig. 6.62a, b; CO: } \left(1 + \frac{C_1}{C_2}\right) = \frac{R_3}{R_1} \text{ and FO: } \omega_0 = \sqrt{\frac{2}{C_1C_2R_2R_3}} \tag{6.73}$$

$$\text{For Fig. 6.62c; CO: } \left(1 + \frac{C_1}{C_2}\right) = \frac{R_3}{R_1} \text{ and FO: } \omega_0 = \sqrt{\frac{2}{C_1C_2R_2R_3}} \tag{6.74}$$

These circuits provide the following advantages simultaneously: (i) employment of single active building block (ABB), (ii) employment of both GCs and a minimum number (only three) of resistors, (iii) non-interacting control of CO and FO, (iv) a simple CO and unconstrained tuning law of FO, and (v) simultaneous availability of explicit VM output as well as CM output.

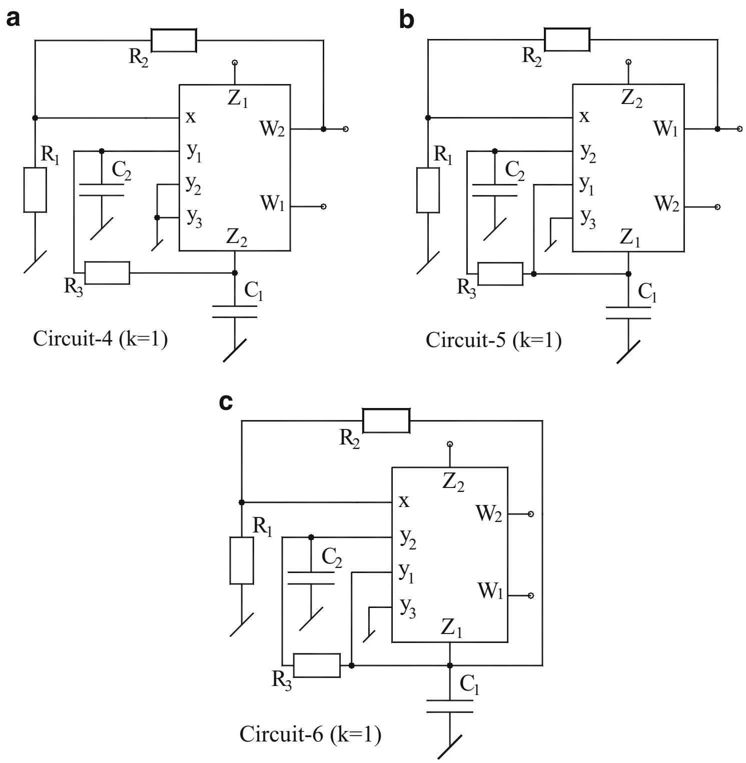


Fig. 6.62 Some more SRCOs using the DDCCFA [282]

When a comparison is made with all previously reported canonic SRCOs employing only a single ABB (which may be an op-amp, a CCII, an OTA, a CFOA, an FTFN, a CDBA, an OTRA, a DVCC, a DVCFA, or an FDCCII) it is found [282] that none of the previously known circuits provide all the above-quoted features *simultaneously*.

6.4.5 Oscillators Using CDBAs

The current differencing buffered amplifier (CDBA) was introduced by Acar and Ozoguz in [73] as a new building block suitable for analog signal processing. Since a CDBA has two current input terminals offering a virtual ground at each input terminal, it was envisaged that the parasitic capacitances from each input terminal to ground will become ineffective in any signal processing circuit made from CDBAs. A number of authors have proposed CDBA-based oscillators. Ozcan-Toker-Acar-Kuntman-Cicekoglu [168] derived a family of six single CDBA canonic oscillators each employing three resistors and two capacitors and providing

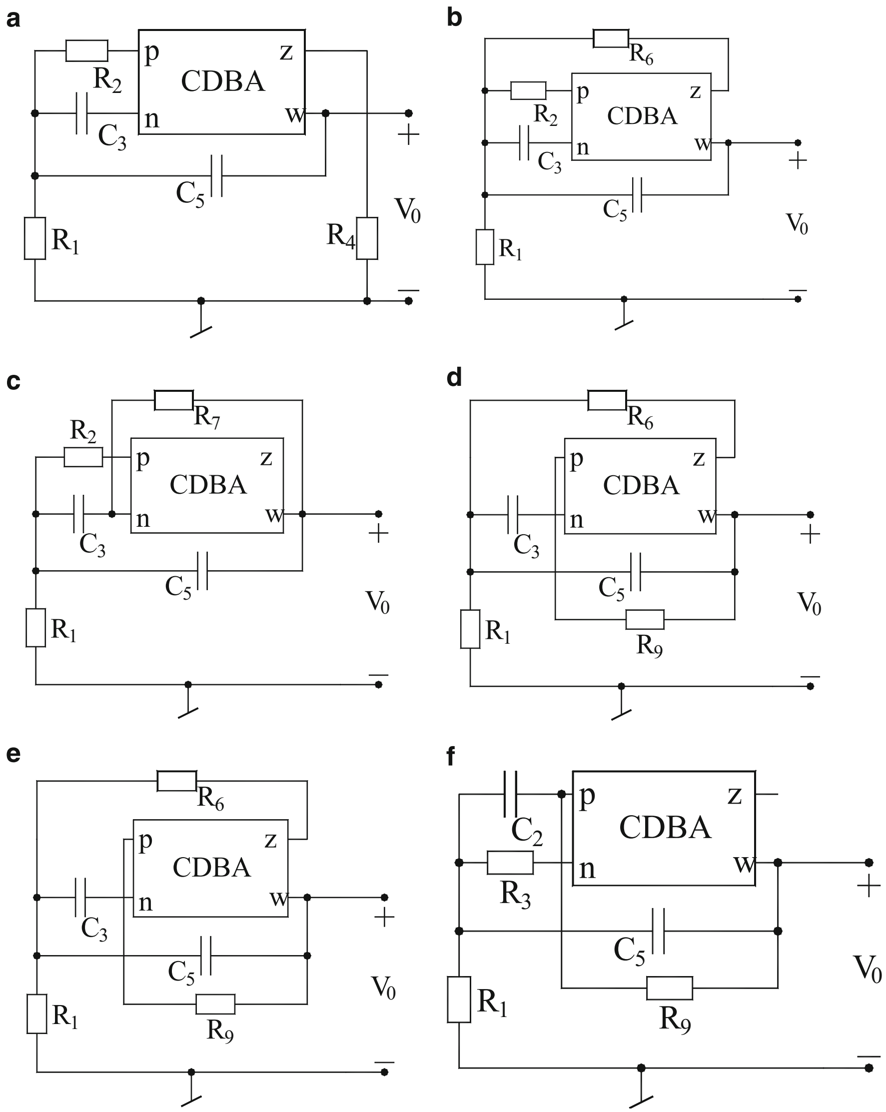


Fig. 6.63 Single CDBA-based SRCOs proposed by Ozcan-Toker-Acar-Kuntman-Cicekoglu [168]

independent tunability of CO and FO. These circuits are shown in Fig. 6.63. The CO and FO of these oscillator circuits are given as follows:

$$\text{Figure 6.63a; CO}: C_5R_4 = R_2(C_3 + C_5) \text{ and FO}: \omega_0 = \sqrt{\frac{(R_1 + R_2)}{C_3C_5R_1R_2R_4}} \quad (6.75)$$

$$\text{Figure 6.63b; CO}: C_5R_6 = 2R_2C_3 \text{ and FO}: \omega_0 = \sqrt{\frac{1}{C_3C_5R_1R_6}} \tag{6.76}$$

$$\text{Figure 6.63c; CO}: C_5R_7 = R_2(C_3 + C_5) \text{ and FO}: \omega_0 = \sqrt{\frac{(R_1 + R_2)}{C_3C_5R_1R_2R_7}} \tag{6.77}$$

$$\text{Figure 6.63d; CO}: C_5R_8 = 2R_2(C_3 + C_5) \text{ and FO}: \omega_0 = \sqrt{\frac{2(R_1 + R_2)}{C_3C_5R_1R_2R_8}} \tag{6.78}$$

$$\text{Figure 6.63e; CO}: 2C_3R_9 = R_6(C_3 + C_5) \text{ and FO}: \omega_0 = \sqrt{\frac{(R_9 + R_6) - R_1}{C_3C_5R_1R_6R_9}};\ R_1 < (R_9 - R_6) \tag{6.79}$$

$$\text{Figure 6.63 f; CO}: C_5R_9 = R_3(C_2 + C_5) \text{ and FO}: \omega_0 = \sqrt{\frac{(R_1 + R_3)}{C_2C_5R_1R_3R_9}} \tag{6.80}$$

6.4.6 Oscillators Using CDTAs

A number of interesting topologies have been evolved using CDTAs as active elements. It has been found very convenient to realize QOs with explicit current output and several authors have derived three-CDTA based QOs out of which two specific circuits both employing two GCs are elaborated here.

Tangsrirat [207] proposed a CM 4-phase oscillator which is shown in Fig. 6.64. The CO and FO of this circuit are given as

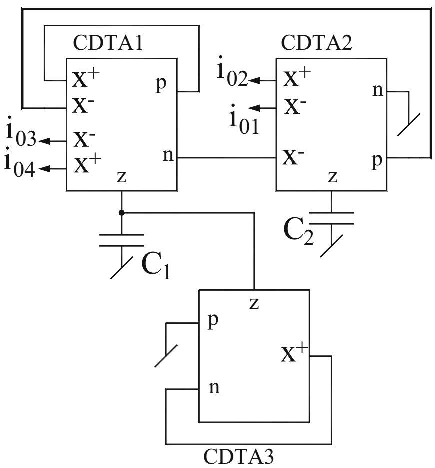
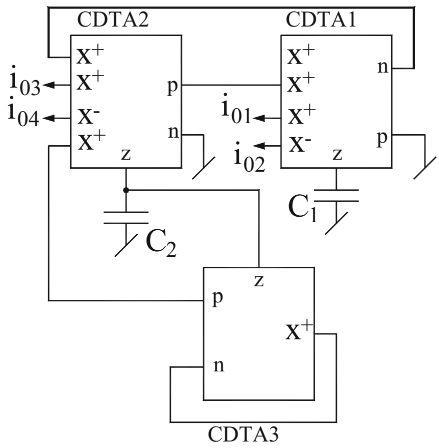


Fig. 6.64 CDTA-based CM four-phase QO proposed by Tangsrirat [207]

Fig. 6.65 CM QO proposed by Jin [297]



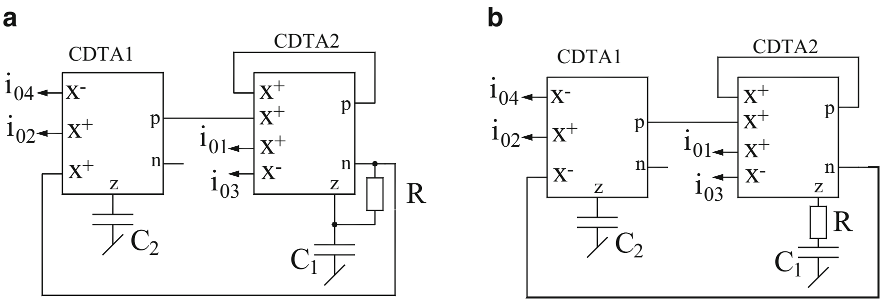


Fig. 6.66 CM QOs proposed by Jin and Wang [298]

$$\mathrm{CO}: g_{\mathrm{m1}} \geq g_{\mathrm{m3}} \;\text{ and }\; \mathrm{FO}: \omega_0 = \sqrt{\frac{g_{\mathrm{m1}} g_{\mathrm{m2}}}{C_1 C_2}} \tag{6.81}$$

Another almost similar circuit realizing 4-phase QO derived from a SIMO-type universal filter is shown in Fig. 6.65 which was proposed by Jin [297]. The CO and FO of this QO are given by

$$\mathrm{CO}: g_{\mathrm{m3}} \leq g_{\mathrm{m2}} \;\text{ and }\; \mathrm{FO}: \omega_0 = \sqrt{\frac{g_{\mathrm{m1}} g_{\mathrm{m2}}}{C_1 C_2}} \tag{6.82}$$

However, subsequently, Jin-Wang [298] and Kumngern-Lamun-Dejhan [299] demonstrated that if one resistor is also incorporated then such quadrature CM oscillators can be realized with only two CDTAs also. Two such circuits proposed by Jin and Wang [298] are shown in Fig. 6.66, for which a straightforward analysis shows that the circuits are characterized by the following equations:

$$\mathrm{CO}: g_{\mathrm{m1}}R \le 2 \;\text{ and }\; \mathrm{FO}:\; \omega_0 = \sqrt{\frac{g_{\mathrm{m1}}g_{\mathrm{m2}}}{C_1C_2}} \;\text{for the circuit of Fig. 6.66a} \tag{6.83}$$

$$\mathrm{CO}: g_{\mathrm{m2}}R \le \frac{C_2}{C_1} \;\text{and FO}: \omega_0 = \sqrt{\frac{g_{\mathrm{m1}}g_{\mathrm{m2}}}{(1-g_{\mathrm{m1}}R)C_1C_2}} \;\text{for the circuit of Fig. 6.66b} \tag{6.84}$$

The current transfer function between I_{01} and I_{02} is given by

$$\frac{I_{01}(j\omega)}{I_{02}(j\omega)} = \frac{g_{\mathrm{m2}}}{\omega C_2}\mathrm{e}^{-j90^\circ} \tag{6.85}$$

From the above equation, it is clear that the two currents are in quadrature. Also $I_{01}(s) = -I_{03}(s)$ and $I_{02}(s) = -I_{04}(s)$ which implies that two QOs can give four quadrature outputs, respectively.

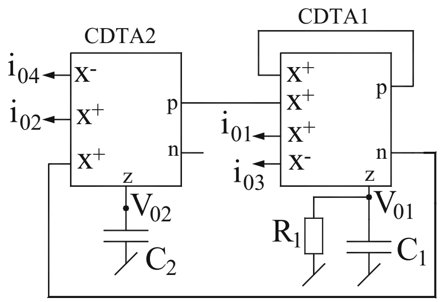
On the other hand, the two CDTA-based 4-phase oscillators proposed by Kumngern-Lamun-Dejhan [299] not only provide 4-CM quadrature signals, but it has the only resistor employed as grounded and also provides two VM quadrature signals. This circuit is shown in Fig. 6.67. The CO and FO of this QO are given by

$$\mathrm{CO}: g_{\mathrm{m1}}R_1 = 1 \;\text{ and }\; \mathrm{FO}: \omega_0 = \sqrt{\frac{g_{\mathrm{m1}}g_{\mathrm{m2}}}{C_1C_2}} \tag{6.86}$$

$$I_{01} = \left(\frac{sC_2}{g_{\mathrm{m2}}}\right)I_{02}, I_{04} = -I_{02} \text{ and } I_{03} = -I_{01} \;\text{ and }\; V_{01} = \left(\frac{sC_2}{g_{\mathrm{m1}}}\right)V_{02} \tag{6.87}$$

A number of researchers have shown that a sinusoidal oscillator exhibiting non-interacting controls of both CO and FO do not however need more than a

Fig. 6.67 CDTA-based CM QO proposed by Kumngern-Lamun-Dejhan [299]



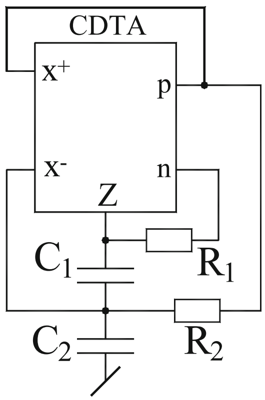


Fig. 6.68 SRCO using a single CDTA proposed by Prasad-Bhaskar-Singh [208]

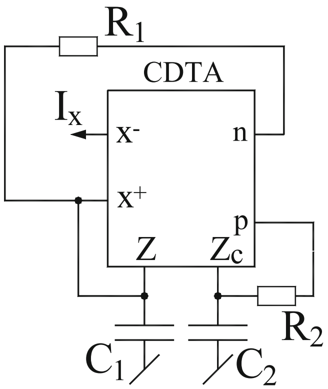


Fig. 6.69 SRCO employing ZC-CDTA proposed by Biolek-Keskin-Biolkova [300]

single CDTA, two resistors, and two capacitors. One such circuit was advanced by Prasad-Bhaskar-Singh [208] which is shown in Fig. 6.68. A routine circuit analysis yields the expressions for CO and FO as below:

$$\mathrm{CO}:\left\{2+\left(1+\frac{C_1}{C_2}\right)-R_1 g_m\right\}\leq 0\ \text{ and }\ \mathrm{FO}:\omega_0=\sqrt{\frac{2}{C_1C_2R_1R_2}} \tag{6.87}$$

Subsequently, Biolek-Keskin-Biolkova [300] proposed an alternative circuit employing a z-copy CDTA which enabled an explicit CM output. This circuit, however, employs both GCs and has the following CO and FO (Fig. 6.69):

$$\mathrm{CO}: g_{m1}R=2\ \text{ and }\ \mathrm{FO}:\omega_0=\sqrt{\frac{1}{C_1C_2R_1R_2}} \tag{6.88}$$

Lastly, Jin and Wang [301] presented another single CDTA-based oscillator which employs a single resistor along with only two capacitors (one grounded and one

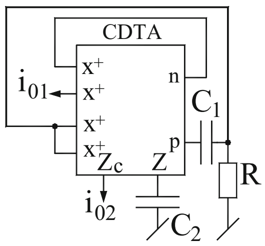


Fig. 6.70 CM quadrature oscillator proposed by Jin and Wang [301]

floating). Other than the non-interacting controls this circuit has the advantage of providing two explicit current outputs in quadrature. This circuit is shown in Fig. 6.70. The CO and FO of this quadrature oscillator are given by

$$\mathrm{CO}: C_2 = g_{\mathrm{m}} C_1 R \ \text{ and FO}: \omega_0 = \sqrt{\frac{g_{\mathrm{m}}}{C_1 C_2 R}} \tag{6.89}$$

The current transfer function of between I_{01} and I_{02} can be obtained as

$$\frac{I_{01}(s)}{I_{02}(s)} = \frac{g_{\mathrm{m}}}{sC_2} = \frac{g_{\mathrm{m}}}{\omega C_2} \mathrm{e}^{-j90^\circ} \tag{6.90}$$

From the above equation, it is clear that the two currents are in quadrature.

6.4.7 Oscillators Using CFTAs

The CFTA, characterized by $V_f = 0$, $I_z = I_f$, and $I_x = \pm g_{\mathrm{m}} V_z$ is, in fact, a translinear current follower (CF) between ports f and z along with an operational transconductance amplifier (TA) between ports z and x and therefore, appears to offer advantages of both CF and TA in analog circuit design as indicated by several publications related to CFTAs. Use of CFTAs in sinusoidal oscillator synthesis has been investigated by a number of researchers; for instance, see [324–328].

Realization of CFTA-based multiphase sinusoidal oscillators (MSO) was first explicitly reported by Uttaphut [324] who described CM-MSOs for odd phase, as well as odd/even phase. These two general topologies of MSOs using CFTAs are shown in Fig. 6.71. The first topology is based upon n-cascaded lossy integrators with CFTAs having dual x-outputs, each providing $I_x = +g_{\mathrm{m}} V_{\mathrm{z}}$. On the other hand, in the second topology, two dual-output currents with negative polarity (i.e., $I_x = -g_{\mathrm{m}} V_z$) have been employed. In each circuit, the basic building block is a first-order lossy integrator characterized by

$$\frac{I_0(s)}{I_{\mathrm{in}}(s)} = \pm \frac{g_{\mathrm{m}} R}{sCR + 1} \tag{6.91}$$

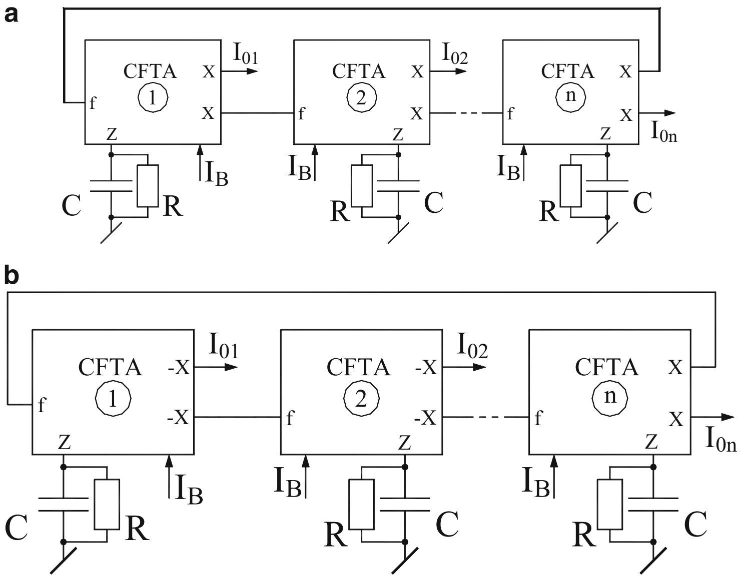


Fig. 6.71 CM MSOs proposed by Uttaphut [324]

where + sign applies if the output current is taken from $+x$ output of CFTA (i.e., $I_x = +g_m V_z$) and the negative sign applies if the output is taken from the $-x$ output of the CFTA (i.e., $I_x = -g_m V_z$).

The CO and FO of the circuit of Fig. 6.71a, b which is an odd phase MSO are given by

$$\text{CO}: g_m R \geq \text{Sec}(\pi/n) \ \text{ and } \ \text{FO}: \omega_0 = \frac{1}{CR} \tan\left(\pi/n\right) \tag{6.92}$$

where n is the number of lossy integrator stages employed in the circuit. Thus, it may be seen that CO is adjustable through g_m (and hence, the external bias current I_B) after which ω_0 can be controlled by varying "R."

Tangsrirat-Mongkolwai-Pukkalanun [325] proposed a mixed-mode (thereby meaning both VM and CM) QO using CFTAs. This circuit shown in Fig. 6.72 employs as many as four CFTAs but has the advantage of employing both GCs and providing two VM outputs V_{01} and V_{02} which are in quadrature and simultaneously also providing two CM outputs I_{01} and I_{02} which are also in quadrature.

The characteristic equation (CE) of the oscillator of Fig. 6.72 is given by

$$s^2 + 2\left(\frac{1-K}{1+K}\right)\left(\frac{g_m}{C}\right)s + \left(\frac{g_m}{C}\right)^2 = 0 \tag{6.93}$$

from which the CO and FO are found to be

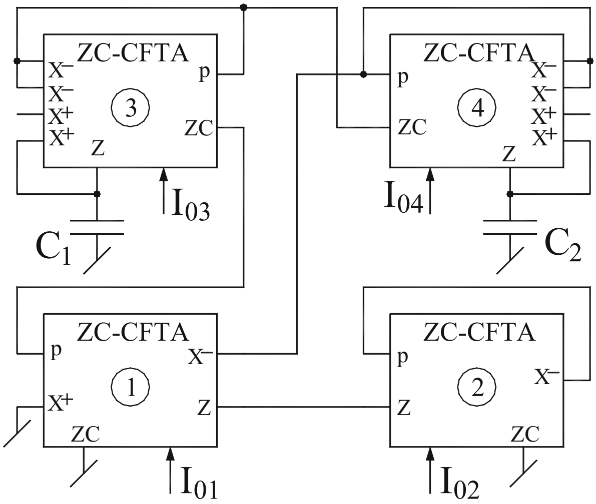


Fig. 6.72 Mixed-mode QO proposed by Tangsrirat-Mongkolwai-Pukkalanun [325]

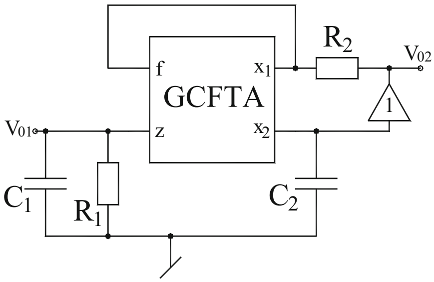


Fig. 6.73 SRCQO proposed by Herencsar, Vrba, Koton, and Lahiri [326]

$$\text{CO}: K = 1 \text{ where } \frac{g_{m1}}{g_{m2}} = \frac{I_{01}}{I_{02}} \text{ and FO}: \omega_0 = \frac{g_m}{C} = \frac{I_0}{2V_T C} \tag{6.94}$$

It is easy to work out that $I_{02} = jI_{01}$ and that $V_{02} = jV_{01}$ which establish the quadrature nature of both current outputs as well as both voltage outputs.

Lastly, we present an SRCO proposed by Herencsar-Vrba-Koton-Lahiri [326] shown in Fig. 6.73 which employs a generalized CFTA (GCFTA) characterized by by $V_f = 0, I_z = \alpha I_f, I_{x1} = b_1 g_m V_z$, and $I_{x2} = b_2 g_m V_z$ where $\alpha, b_1, b_2 \in \{1, -1\}$. From a GCFTA, six different types are possible to be defined [326]. The CE of the oscillator of Fig. 6.73 is found to be

$$s^2(C_1C_2R_1R_2) + sC_2R_2(1 + \alpha b_1 g_m R_1) - \alpha b_2 g_m R_1 = 0 \tag{6.95}$$

Hence, for the circuit to be a valid oscillator circuit, the following conditions apply on the GCFTA:

$$\alpha b_1 = 1, \alpha b_2 = -1, \text{ the CO and FO are then given by}$$

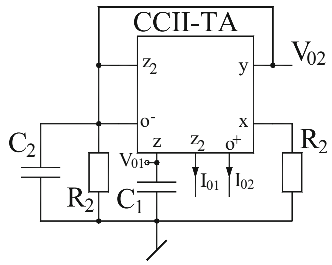


Fig. 6.74 Mixed-mode QO proposed by Lahiri [329]

$$\text{CO}: g_{\text{m}}R_1 \ \text{ and } \ \text{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{g_{\text{m}}}{R_2C_1C_2}} \tag{6.96}$$

Clearly, CO is controllable by R_1 without affecting FO which is also independently controllable preferably by R_2.

In conclusion, the circuit of Fig. 6.73 offers the advantages of suitability of IC implementation (due to the employment of both GCs), non-interactive control of CO and FO through two independent resistors and availability of quadrature signal generation.

6.4.8 Oscillators Using CCTAs

Since a CCTA has directly a CCII followed by an OTA, it appears that a CCTA must be more versatile than a CCII, an OTA or a CFTA- all the three. Indeed, as expected, therefore, it appears relatively easy to devise a sinusoidal oscillator using only a single CCTA which can simultaneously provide the advantages of employment of both GCs, non-interacting tuning laws, availability of both VM and CM outputs explicitly and electronic control of FO. Such a circuit was proposed by Lahiri in [329] and is shown in Fig. 6.74.

6.4.9 Oscillators Using CBTAs

Current-mode and voltage-mode MSOs have been realized using varieties of active building blocks as already dealt with in this chapter. Sagbas-Ayten-Herencsar-Minaei [339] proposed a topology for n-phase VM MSO (Fig. 6.75) and also for CM MSO (Fig. 6.76). In each case only "n" grounded capacitors along with "n" CBTAs and a single resistor are needed.

For the VM MSO of Fig. 6.75, CO and FO are given by

$$\text{CO}: K = \left[1 + \tan^2\left(\frac{\pi}{n}\right)\right]^{\frac{n}{2}} \quad \text{and} \quad \text{FO}: \omega_0 = \frac{\mu_w g_{\text{mi}}}{C_{\text{i}}}\tan\left(\frac{\pi}{n}\right)$$

where CBTAs are characterized by the terminal equations $I_z = g_{\text{m}}(V_p - V_n)$, $V_w = \mu_w V_z$, $I_p = I_w$, and $I_n = -I_w$; and where $K = \mu_w g_{\text{m}} R_{\text{f}}$.

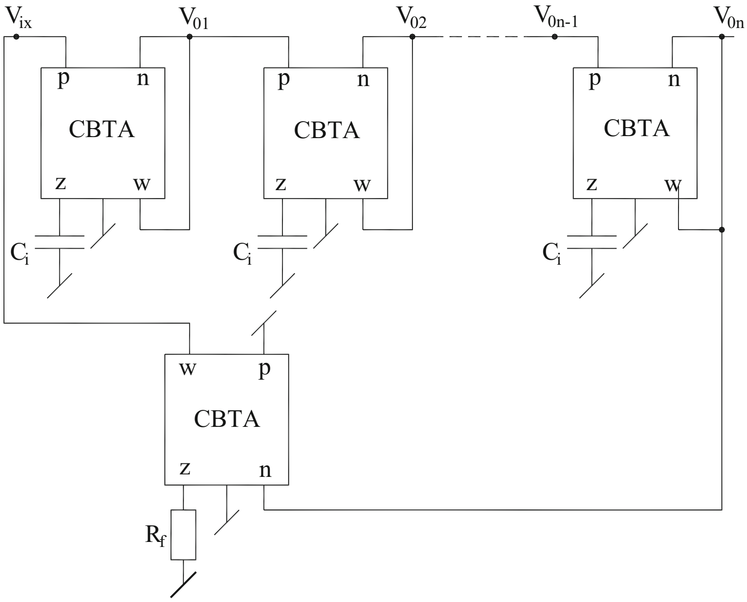
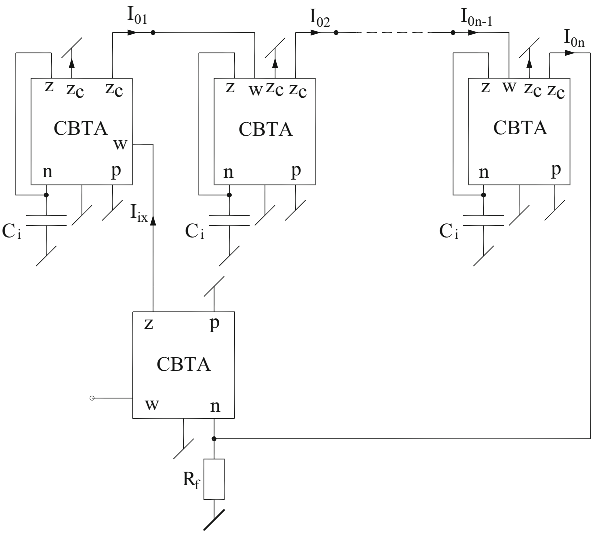


Fig. 6.75 VM CBTA-based MSO proposed by Sagbas-Ayten-Herencsar-Minaei [339]

Fig. 6.76 CM CBTA-based MSO proposed by Sagbas-Ayten-Herencsar-Minaei [339]



On the other hand, the CO and FO for the CM MSO of Fig. 6.76 are given by

$$K = \alpha_n^{-n}\left[1 + \frac{\omega_0^2 C_i^2}{g_{mi}^2}\right]^{\frac{n}{2}} \quad \text{and} \quad \text{FO}: \omega_0 = \frac{g_{mi}}{C_i}\tan\left(\frac{\pi}{n}\right) \tag{6.98}$$

In both cases, oscillation frequency can be controlled through equal-valued g_{mi} parameters which are electronically adjustable by changing the bias currents of the CBTAs.

6.4.10 Oscillators Using DBTAs

A differential input buffered transconductance amplifier (DBTA) [202] has two low-input impedance terminals p and n and high-input terminal-y, a high-output impedance current output terminal-z, a low-output impedance voltage output terminal-w, and a transconductance amplifier between terminal-z and a current output terminal-x. Its characterizing equations are

$$v_p = v_y, v_n = v_y, i_y = 0, i_z = i_p - i_n, v_w = v_z, i_x = g_m v_z \tag{6.99}$$

A number of authors have employed DBTA and VDTA to synthesize oscillators; for instance, see [203, 204, 340–344]; an interesting VM QO was presented by Herencsar-Koton-Vrba-Lahiri [203] which is shown in Fig. 6.77.

6.4.11 Oscillators Using Current-Mode Op-Amps

Although the ubiquitous voltage-mode op-amp (VOA) is universally recognized as the workhorse of analog circuit design since ages and hundreds of variations of VOAs are manufactured as integrated circuits worldwide, its dual the so-called current operational amplifier (COA) has although attracted attention of researchers, to the best knowledge of the authors, no IC COA exists so far.

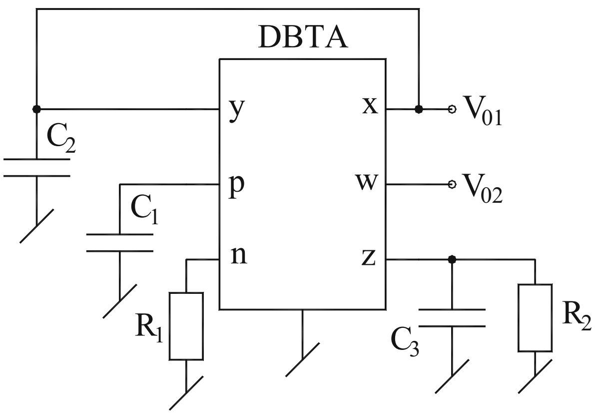


Fig. 6.77 DBTA-based VM QO proposed by Herencsar-Koton-Vrba-Lahiri [203]

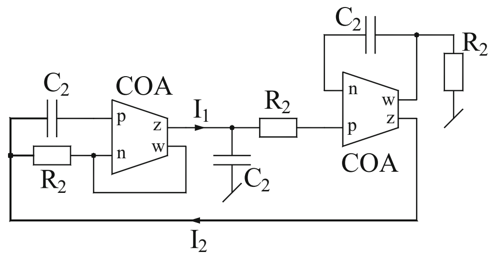


Fig. 6.78 COA-based QO proposed by Kilinc and Cam in [338]

A COA-based QO based upon the use of a first order APF and an inverting amplifier was proposed by Kilinc and Cam in [338]. Their circuit is shown in Fig. 6.78.

The circuit of Fig. 6.78 is characterized by

$$\left(-\frac{s-\frac{1}{C_1R_1}}{s+\frac{1}{C_2R_2}}\right)\left(-\frac{1}{sC_2R_2}\right)=1 \tag{6.100}$$

From the above CO and FO are found to be

$$\mathrm{CO}: C_1R_1=C_2R_2 \quad \text{and} \quad \mathrm{FO}: \omega_0=\sqrt{\frac{1}{C_1C_2R_1R_2}} \tag{6.101}$$

Thus, unfortunately no independent controls of CO and FO are available in this circuit.

Thus, searching of new topologies of SRCOs, SCCOs and other type of oscillators using COAs appears to be an area which is waiting to be explored further!

The circuit is interesting as it employs only a single DBTA, has all the five passive components grounded and yet provides two quadrature outputs. The CO and FO of this circuit are given by

$$\mathrm{CO}: C_1R_2=\frac{C_2}{g_m} \quad \text{and} \quad \mathrm{FO}: \omega_0=\sqrt{\frac{g_m}{C_3C_2R_1}} \tag{6.102}$$

From the above equation it is clear that CO can be controlled by R_2 and /or C_1 without disturbing the FO which is adjustable through $R_1$and /or C_3.

The CO and FO for this circuit are given by

$$\mathrm{CO}: R_2=R_1 \quad \text{and} \quad \mathrm{FO}: f_0=\frac{1}{2\pi}\sqrt{\frac{g_m}{R_1C_1C_2}}=\frac{1}{2\pi}\sqrt{\frac{I_{B2}}{2C_1C_2R_1V_T}} \tag{6.103}$$

Thus, CO can be controlled independently by R_2 while FO can be controlled electronically through the bias current I_{B2}.

6.4.12 Oscillators Using Programmable Current Amplifiers/ Current Differencing Units and Current Mirrors

In one of the previous sections, we highlighted important contributions made in the area of sinusoidal oscillator realization using unity gain voltage followers (VFs) and combinations of unity gain voltage/current followers.

Souliotis and Psychalinos [290] proposed an oscillator circuit based upon a loop consisting of two programmable current amplifiers (PCAs) and two GCs which is shown in Fig. 6.79. A routine analysis of this circuit gives the CO as

$$\mathrm{CO}: K = \frac{C_1}{C_2}\left(\frac{g_{\mathrm{m}1}}{g_{\mathrm{m}2}}\right) + 1 \text{ whereas FO is given by} : \omega_0 = \sqrt{\frac{g_{\mathrm{m}1}g_{\mathrm{m}2}}{C_1C_2}} \tag{6.104}$$

where K is the gain of second PCA.

The circuit provides two CM outputs as shown in Fig. 6.79. The workability of the circuit was tested by the current amplifier topology biased with power supply voltages of ±2.5 V and $I_{\mathrm{bias}} = 50$ μA using BSIM3v3 MOS transistor model parameters for AMS CMOS 0.35 μm C35 process with aspect ratios of NMOS and PMOS transistors as 2 μm/0.85 μm and 8.6 μm/0.85 μm, respectively. With capacitors taken as 5 pF each, an oscillator having frequency 2 MHz was successfully implemented using this approach with noise current spectral density = 138.7pA/$\sqrt{Hz}$. The oscillation frequency was verified to be tunable by changing I_{bias} from 20 μA to about 85 μA.

A QO topology using three PCAs was presented by Herencsar-Lahiri-Vrba-Koton [291] which is shown here in Fig. 6.80. This circuit provides I_{01} and I_{02} in quadrature with CO and FO given by

$$\mathrm{CO}: C_2R_1 \leq n_3R_2C_1 \ \text{ and } \ \mathrm{FO}: \omega_0 = \sqrt{\frac{n_2}{C_1C_2R_1R_2}} \tag{6.105}$$

$$|I_{02}| = jk|I_{01}| \tag{6.106}$$

where n_1, n_2, and n_3 are the gains of PCAs (with $n_1 = 1$). This circuit can provide two quadrature output currents having same amplitude provided $k = \frac{\omega_0 n_3 C_1 R_2}{n_2}$ is taken equal to unity.

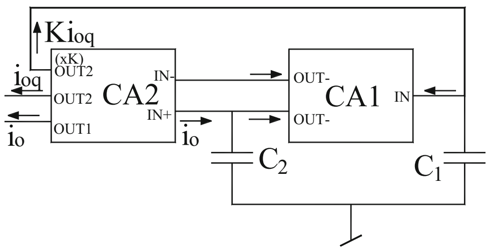


Fig. 6.79 Oscillator-based PCAs proposed by Souliotis and Psychalinos [290]

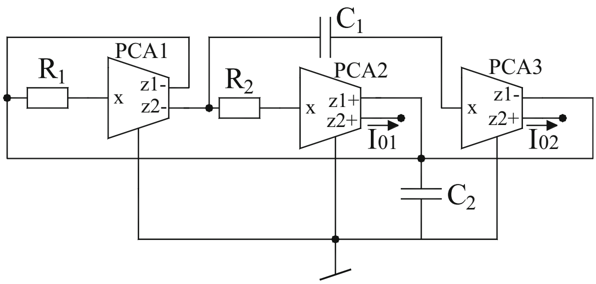


Fig. 6.80 CM electronically tunable QO proposed by Herencsar-Lahiri-Vrba-Koton [291]

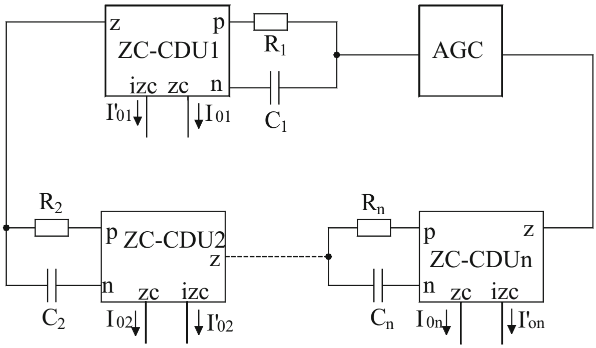


Fig. 6.81 CM ZC-CDU-based multi-phase oscillator proposed by Vavra and Bajer [292]

Vavra and Bajer [292] have employed slightly different current controlled conveyors to realize multi-phase sinusoidal oscillator topology. Their multi-phase sinusoidal oscillator is based upon a first-order APF realizable from a single resistance and single capacitance along with a current differencing unit (CDU). A CDA, in principle, is a building block which provides a current output from a high-output impedance terminal *z*-terminal which is equal to difference of two currents injected into two low-impedance input terminals *p* and *n*. For added flexibility copies of this output current with the same polarity and inverted polarity are also available at two additional output terminals called z_c and i_{zc}.

A general topology for *n*-phase multi-phase oscillator using *n*-number of *z*-copy current differencing units (ZC-CDU) has been shown in Fig. 6.81.

Analysis of such structures can be easily carried out by following the procedure already detailed in this monograph earlier by noting that each first-order AP sections is characterized by a current ratio transfer function:

$$\frac{I_{\text{out}}}{I_{\text{in}}} = \left(\frac{a - s}{a + s}\right); \text{where } a = 1/RC \tag{6.107}$$

In the last, we present an interesting method of producing high frequency low power all current mirror sinusoidal QO. Such a circuit was proposed by Leelasantitham and Srisuchinwong [293]. The topology to be described in the following was devised for wireless communication system having 1.9 GHz

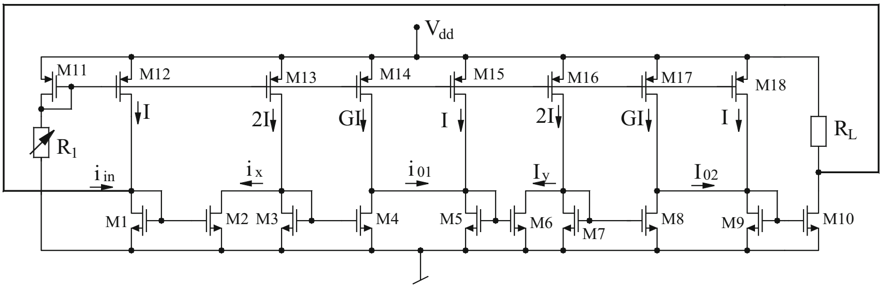


Fig. 6.82 High-frequency low-power CMOS LPF-based all current mirror sinusoidal QO proposed by Leelasantitham and Srisuchinwong [293]

receivers requiring operating frequencies between 1.805 and 1.99 GHz. Thus, to achieve such a high frequency range the inherent time constants of the current mirrors consisting of internal parasitic capacitances of the MOS transistors along with a transconductance of a diode connected NMOS transistor together with a negative resistance using current mirrors have been employed. The circuit diagram of the high-frequency low-power CMOS LPF-based all current mirrors sinusoidal QO is shown in Fig. 6.82.

For a detailed analysis of this circuit, the reader is referred to [293]. It has been reported in [293] that using this approach a 1.9 GHz. 0.45 mW, 2 V CMOS oscillator has been successfully realized with a current tunable frequency range of 370 MHz and THD < 0.3 %.

6.4.13 Oscillators Using VDIBAs

A VDIBA [215, 216] has the advantageous feature of having possibility the simplest type of internal circuit architecture as compared to the most of the active building blocks detailed out in this monograph. An exemplary CMOS implementation of the VDIBA [333] shown in Fig. 6.83 is a testimony to this.

In the absence of a commercially available IC, if however VDIBA is to be implemented using off-the-shelf components, it can be implemented from two OPA860 ICs each of which employs so-called "diamond transistor" (DT) (which is nothing but a current conveyor), two on-chip voltage buffers (VB) along with three resistors [333]. This implementation, however, nearly nullifies the advantage claimed for the CMOS implementation of the VDIBA of Fig. 6.83.

An interesting four-phase oscillator using VDIBA was proposed by Herencsar-Minaei-Koton-Yuce-Vrba [333] which is shown in Fig. 6.84.

This circuit is characterized by the CE:

$$s^2(C_1C_2R) + s(C_1 + C_2g_{m1}R - 2C_1g_{m2}R) + g_{m1} = 0 \tag{6.108}$$

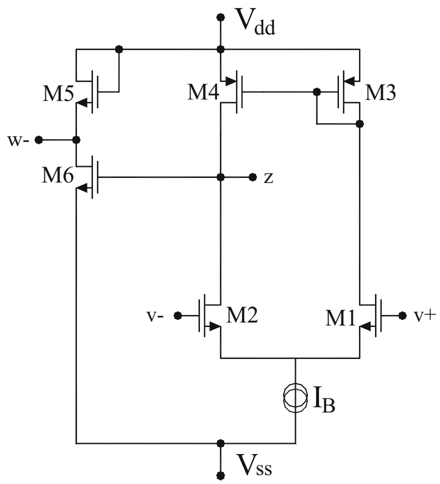


Fig. 6.83 CMOS implementation of VDIBA [333]

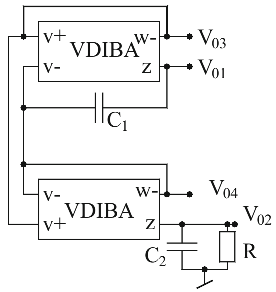


Fig. 6.84 VM four-phase QO proposed by Herencsar-Minaei-Koton-Yuce-Vrba [333]

From the above CE, the following CO and FO are obtained:

$$\mathrm{CO}: g_{\mathrm{m2}} \geq \frac{1}{2}\left(\frac{1}{R} + g_{\mathrm{m1}}\frac{C_2}{C_1}\right) \quad \text{and} \quad \mathrm{FO}: \omega_0 = \sqrt{\frac{g_{\mathrm{m1}}}{C_1 C_2 R}} \tag{6.109}$$

It is suggested in [333] that FO can be controlled by adjusting the value of R and/or by varying the control current I_{B1} of g_{m1}. It is, thus, clear that non-interacting controls of both CO and FO are not available in this circuit.

6.4.14 Oscillator Using VD-DIBA

A fully uncoupled electronically tunable oscillator using two VD-DIBAs was presented by Bhaskar-Prasad-Pushkar [334] which is shown in Fig. 6.85. The CO and FO of this configuration are given by

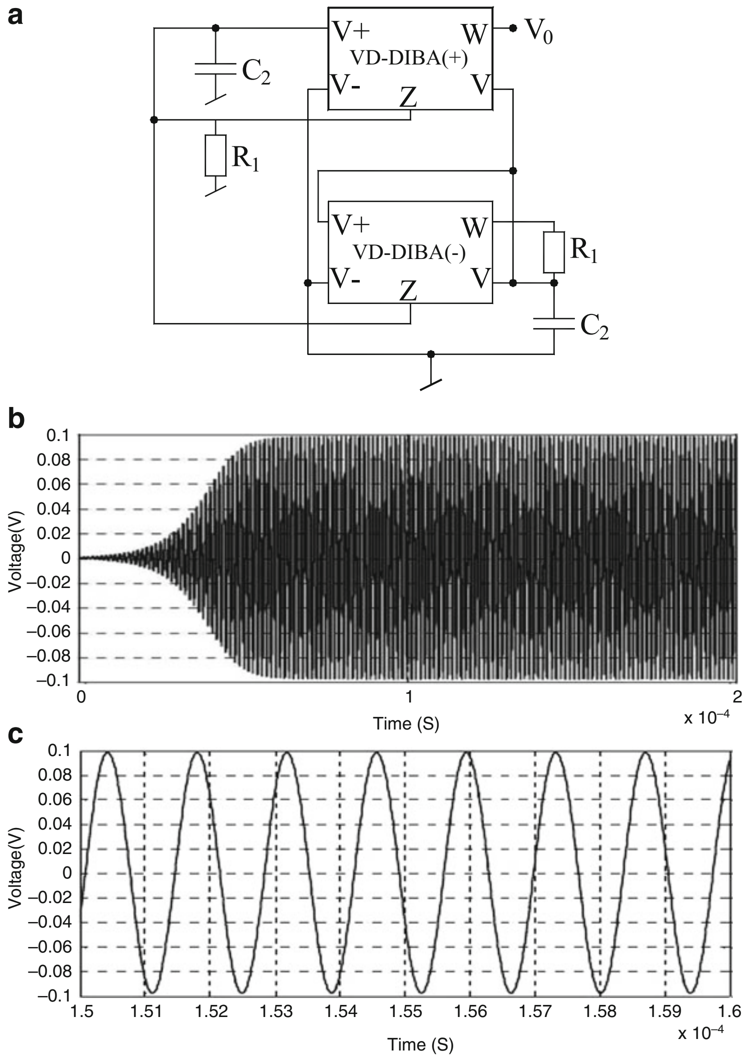


Fig. 6.85 Oscillator configuration using two VD-DIBAs by Bhaskar-Prasad-Pushkar [334]. (**a**) The circuit. (**b**) Transient output waveform of the oscillator of Fig. 6.85a. (**c**) Steady-state response of the output of the oscillator of Fig. 6.85a [334]

$$\text{CO}: \left(\frac{1}{R_1} - g_{\text{m1}}\right) \leq 0 \quad \text{and} \quad \text{FO}: \omega_0 = \sqrt{\frac{g_{\text{m2}}}{C_1 C_2 R_2}} \tag{6.110}$$

Thus, from the above equations, it is clear that both CO and FO are electronically independently controllable i.e., CO through g_{m1} and FO by g_{m2}. This oscillator circuit was simulated using CMOS VD-DIBA [337], CMOS VD-DIBA was biased

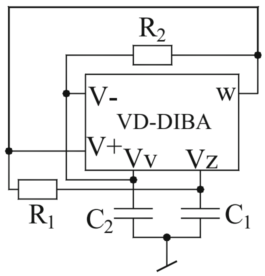


Fig. 6.86 SRCO using a single VD-DIBA proposed by Pushkar-Bhaskar-Prasad [335]

with ±1 V DC power supplies taking the values of $I_{B1} = I_{B2} = I_{B3} = I_{B4} = I_{B5} = I_{B6} = 150$ μA and $I_{B7} = 30$ μA (to control the transconductances of VD-DIBAs). The values of passive components used were $R_1 = 1.67$ kΩ, $R_2 = 10$ kΩ, $C_1 = C_1 = 0.05$ nF. The transient and steady state responses of the circuit of Fig. 6.85a are shown in Fig. 6.85b, c. From Fig. 6.85b, c it is observed that SPICE generated oscillations are quite stable with frequency 731.88 kHz having THD 1.159 %.

Another circuit which provides non-interacting control of both CO and FO using only a single VD-DIBA instead of VDIBA and no more than four passive components with both capacitors being grounded was proposed by Pushkar-Bhaskar-Prasad [335] is shown in Fig. 6.86.

By routine analysis, the CE of this circuit can be obtained as

$$s^2(C_1C_2R_1R_2) + sR_1(2C_1 - C_2g_mR_2) + 1 = 0 \tag{6.111}$$

Thus, the CO and FO from the above CE can be given by

$$\mathrm{CO}: (2C_1 - C_2g_mR_2) \leq 0 \quad \text{and} \quad \mathrm{FO}: \omega_0 = \sqrt{\frac{1}{C_1C_2R_1R_2}} \tag{6.112}$$

Therefore, it is clear from the above equation that FO is independently controllable by the resistor R_1 and CO is electronically adjustable by g_m.

6.4.15 Oscillators Using OTRAs

OTRAs as active elements have been in vogue since about 1992 or so and their hardware implementations suitable for CMOS technology as well as their applications in signal processing have been widely investigated [189, 200, 201, 315–322]. The work on sinusoidal oscillator realization using OTRAs has also attracted some attention in literature; for instance, see [200, 201, 315–320]. Also, work has been done on realizing relaxation oscillators and wave form generators but that is the topic dealt with in a forthcoming chapter of this monograph. A survey of the literature on OTRA-based sinusoidal oscillators reveals that sinusoidal oscillators exhibiting non-interacting controls of both CO and FO can be made using three/two OTRAs whereas SRCOs realizable with a single OTRA are not too many.

In the following, we present some prominent OTRA-based sinusoidal oscillators evolved so far. Two OTRA-based third-order QOs were proposed by Pandey-Pandey-Komanapalli-Anurag [315]; however, only one of the two circuits permits independent adjustment of CO and FO and that is shown in Fig. 6.87. The circuit of Fig. 6.87 is characterized by

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{C_1C_2R_2R_5}} \quad \text{and} \quad \mathrm{CO}: R_4R_5 = R_1R_3 \tag{6.113}$$

Thus, f_0 is adjustable, independent of CO, by R_2 while CO can be adjusted, independent of FO, by R_4 and/or R_1.

OTRA-based sinusoidal oscillators realizable with only two OTRAs were presented by Salama-Soliman [316, 317]. Consider first the circuit of Fig. 6.88a from [316] which is characterized by

$$\mathrm{CO}: R_4 = R_3 \quad \text{and} \quad \mathrm{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{C_1C_2R_2R_1}} \tag{6.114}$$

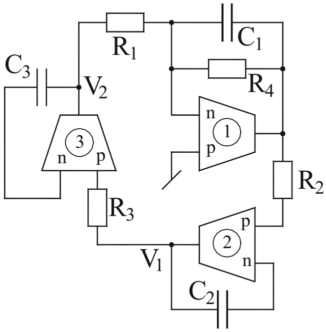


Fig. 6.87 OTRA-based third-order QO proposed by Pandey-Pandey-Komanapalli-Anurag [315]

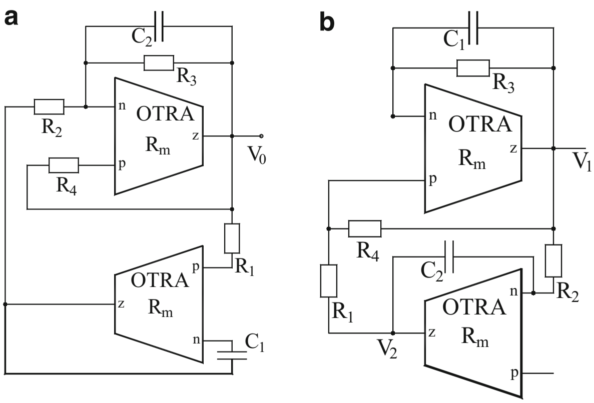


Fig. 6.88 SRCO with independent controls of CO and FO proposed by Salama-Soliman [316, 317]

Independent adjustability of CO and FO is obvious from the above equation.

One of the circuits from [317], on the other hand, is shown in Fig. 6.88b. This circuit is also characterized by exactly the same equation as above.

Gupta-Senani-Bhaskar-Singh [201] presented two oscillators each of which employed one OTRA, one unity gain VF three resistors and two capacitors and provides non-interacting independent controls of CO and FO. The SRCO from [201] is shown here in Fig. 6.89 which is characterized by

$$\mathrm{CO}: R_2 \leq R_1 \quad \text{and} \quad \mathrm{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{C_1C_2R_1R_3}} \tag{6.115}$$

Thus, it is seen that R_2 controls CO while R_3 independently controls FO.

The other circuit from [201] is, in fact, a single-capacitance-controlled-oscillator (SCCO) as well as a SRCO (see Fig. 6.90) and is characterized by

$$\mathrm{CO}: C_2 \leq C_1 \quad \text{and} \quad \mathrm{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{C_1C_3R_1R_2}} \tag{6.116}$$

Thus FO is controllable by C_3 and also by any of R_1or R_2.

Considering nonideal transresistance gain R_M of the OTRA to be approximated as $R_M(s) \cong \frac{1}{sC_p}$, the FO and CO for the circuit of Fig. 6.89 get modified as

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{C_1C_3R_1R_2\left(1+\frac{C_p}{C_1}\right)}};\frac{R_2}{R_1} + \frac{C_pR_2}{C_2R_3} \leq 1 \tag{6.117}$$

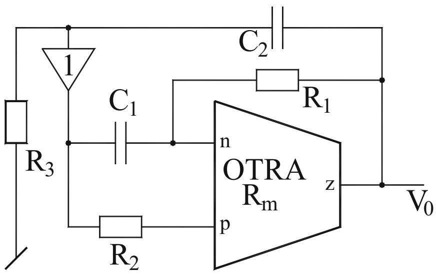


Fig. 6.89 SRCO proposed by Gupta-Senani-Bhaskar-Singh [201]

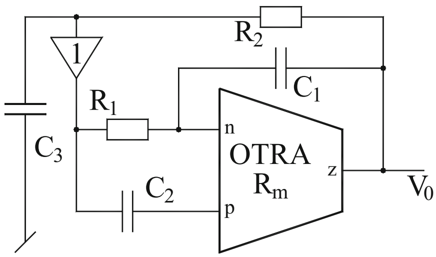


Fig. 6.90 SCCO/SRCO proposed by Gupta-Senani-Bhaskar-Singh [201]

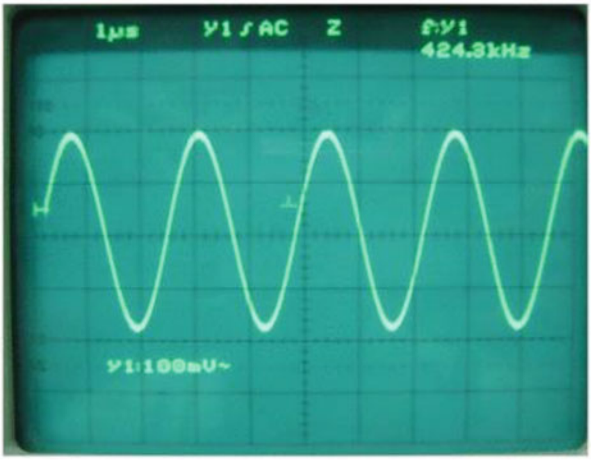


Fig. 6.91 A typical waveform generated by the SRCO of Fig. 6.75 [201]

Similarly, for the circuit of Fig. 6.90, the nonideal FO and CO are given by:

$$\mathrm{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{C_3R_1R_2C_1\left(1+\frac{C_p}{C_1}\right)}} \quad \text{and} \quad \mathrm{CO}: \left(C_1 + C_p\right) \leq C_2 \tag{6.118}$$

From the above nonideal expressions, it is deduced that the parasitic capacitance C_p although disturbs the independent tunability in the case of SRCO of Fig. 6.89, it does not alter these desirable properties in case of the SCCO/SRCO of Fig. 6.89 where C_2 can still be used to adjust the circuit to produce oscillation without affecting FO, which is also independently adjustable by C_3, R_1, and R_2.

It has been shown in [201] that if the effect of finite input and output impedances of the OTRA is also taken into consideration, the errors in the FO of the oscillators of Figs. 6.90 and 6.91 are no more than 3 %.

The circuits of Figs. 6.89 and 6.90 were experimentally tested in [201] by constructing OTRA using two CFOAs (AD844s). The VF has been implemented using a CFOA. The DC bias supply was taken as ±5 V DC and the component values selected were for the circuit of Fig. 6.89, $C_1 = C_2 = 100$ pF, $R_1 = 1$ kΩ, $R_2 = 1$ kΩ (fixed) + 10 kΩ (variable), and $R_3 = 10$ kΩ. A typical waveform observed on the oscilloscope is shown in Fig. 6.91 while the variation of f_0 with respect to R_3 is shown in Fig. 6.92. For the circuit of Fig. 6.90, $C_1 = 600$ pF, $C_2 = 100$–1200 pF (variable), $C_3 = 100$ pF, $R_1 = 1$ kΩ, and $R_2 = 15$ kΩ. A typical waveform observed on the oscilloscope is shown in Fig. 6.93.

In 2002, Cam [318] presented a single OTRA-based oscillator which is shown here in Fig. 6.94. This circuit has CO and FO given by

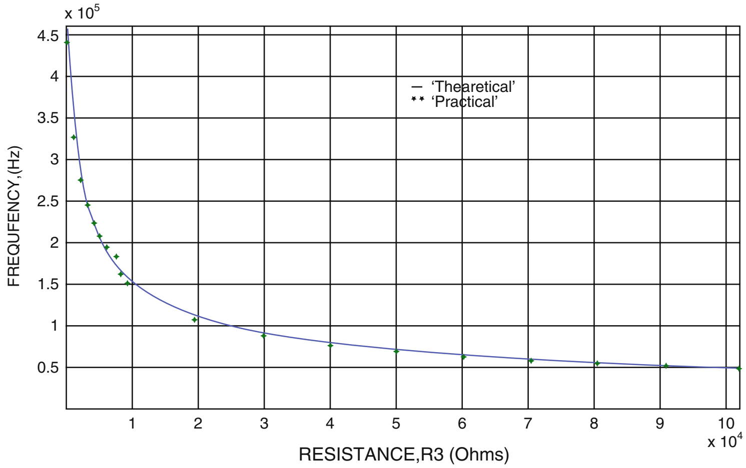


Fig. 6.92 Variation of frequency of oscillation with R_3 of Fig. 6.75 [201]

Fig. 6.93 A typical waveform generated by the SCCO of Fig. 6.76 [201]

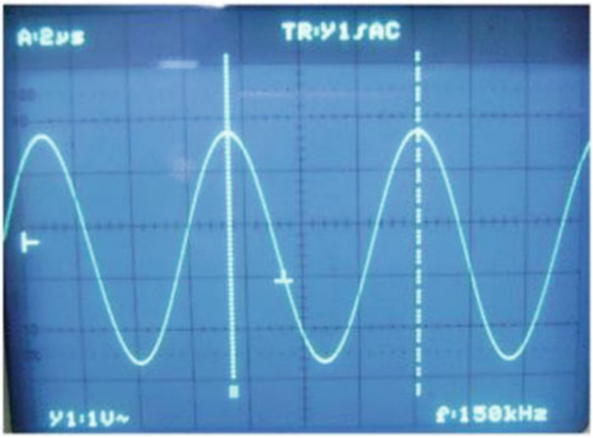
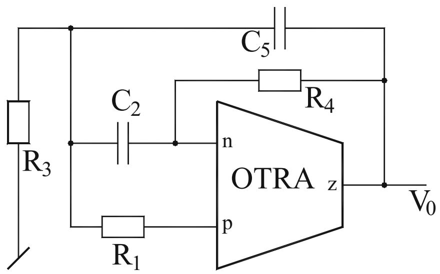


Fig. 6.94 SRCO using single OTRA proposed by Cam [318]



$$\text{CO}: \frac{1}{R_4}(C_2 + C_5) = \frac{C_5}{R_1} \quad \text{and} \quad \text{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{\frac{1}{R_4}\left(\frac{1}{R_1} + \frac{1}{R_3}\right)}{C_2 C_5}} \tag{6.119}$$

It is, thus, seen that after adjusting the CO which depends upon C_2, C_5, and R_1, R_4, f_0 can be independently varied by resistor R_3. Unfortunately, however, CO cannot be adjusted without affecting FO.

More recently, Avireni and Pittala [319] derived a family of five single OTRA sinusoidal oscillators. A close examination of these five circuits, however, reveals that only one of them is really capable of providing variability of FO through a single resistor without affecting CO. This circuit is shown in Fig. 6.95 and is characterized by

$$\text{CO}: \frac{1}{R_2}(C_3 + C_6) = \frac{C_6}{R_4} \quad \text{and} \quad \text{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{\frac{1}{R_2}\left(\frac{1}{R_4} + \frac{1}{R_7}\right)}{C_3 C_6}} \tag{6.120}$$

Thus, FO is adjustable by R_7 without affecting CO which is dependent on the rest of the components.

A similar circuit providing control of FO by a variable capacitor (but no independent adjustability of CO available) was recently proposed by Chien [320] and his circuit is shown in Fig. 6.96 for which analysis yields

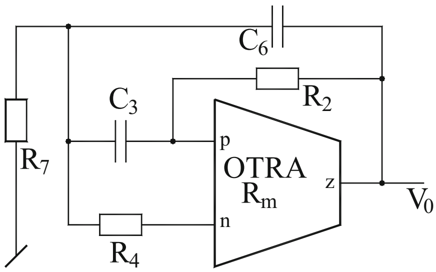


Fig. 6.95 Single OTRA-based SRCO proposed by Avireni and Pittala [319]

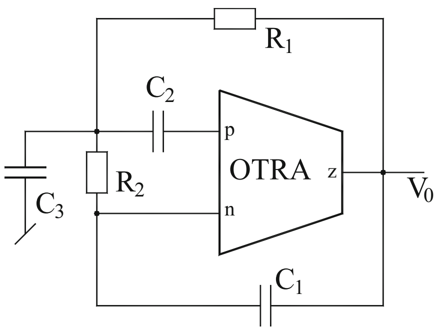


Fig. 6.96 Single OTRA-based SCCO proposed by Chien [320]

$$\text{FO}: f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{R_1R_2C_1(C_3+C_2)}} \quad \text{and} \quad \text{CO}: \left(1+\frac{R_1}{R_2}\right) = \frac{C_2}{C_1} \tag{6.121}$$

and thus the circuit design is an SCCO, with FO controlled by C_3. Here, also CO is not adjustable independently.

In conclusion, it is revealed from the existing literature on OTRA-based sinusoidal oscillators [200, 201, 315–320] that till the time of writing this monograph (June, 2015), not even a single circuit is known which can realize a canonic SRCO (i.e., using no more than three resistors and two capacitors) using a single OTRA with independent controllability of both CO and FO through separate resistors. This, therefore, appears to be interesting open problem, worthy of investigating.

6.5 Concluding Remarks

This chapter has presented a concise compilation of some selected sinusoidal oscillator configurations using a large variety of extended current conveyor types and a number of modern electronic circuit building blocks. For brevity, only a few significant circuits have been included from amongst a large number available in literatures as reflected in the list of references [1–408] at the end of this chapter. For further reading, still more additional references are provided for the interested readers at the end of this monograph.

The inclusion of various circuits in this chapter was guided by considerations such as independent controllability of the condition of oscillation and frequency of oscillation, employment of grounded capacitors, employment of grounded capacitors and /or grounded resistors, availability of explicit voltage and current outputs from low-output impedance and high-output impedance terminals, respectively, availability of quadrature outputs in voltage-mode, current-mode or dual-mode, etc.

It must be emphasized that while some of the circuits discussed can be implemented from the commercially available ICs, several others cannot be actually implemented efficiently in practice, due to the non-availability of the employed building blocks as off-the-shelf ICs; however, since all the building blocks employed are implementable in bipolar and CMOS technologies, such circuits which offer very interesting features should provide good enough motivation for the IC designers to make these exciting varieties of new building blocks as stand-alone ICs. Alternatively, the described topologies appear suitable for being used as an integratable part of a system-on-chip which requires a sinusoidal oscillator to be integrated on the same IC chip using bipolar/CMOS technology.

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Chapter 7
Switched-Capacitor, Switched-Current, and MOSFET-C Sinusoidal Oscillators

Abstract This chapter presents some prominent sinusoidal oscillator configurations from the various switched-capacitor, switched-current, and MOSFET-C sinusoidal oscillators evolved so far. Whereas the first two classes of oscillators are essentially discrete time systems, the third one is, by contrast, a continuous time system. However, all the three types of oscillators have the common features of (a) electronic tunability of the oscillation frequency and (b) the suitability of the circuits for integrated circuit implementation in CMOS technology and have, therefore, attracted considerable attention in technical literature.

7.1 Introduction

With the emergence of fully integratable switched-capacitor filters and other signal processing circuits during 1975–1985, attention of researchers was also diverted to the synthesis and design of switched-capacitor oscillators (SCO) particularly for low- and medium-frequency sinusoidal waveform generation.

Unlike switched-capacitor circuits, switched-current circuits have the advantage of not requiring op-amps or floating capacitors but instead have the advantage of employing MOS current mirrors and MOSFETs as direct elements for synthesizing signal processing circuits. Therefore, together with the developments in the area of switched-current filters, attention was also focused on switched-current oscillators (SIO).

Furthermore, when MOSFET-C filters came into being as continuous-time alternatives to the discrete-time switched-capacitor and switched-current circuits, once again it was logical to think about MOSFET-C oscillators with the expectation that such oscillators would be more natural choice for generating continuous-time sinusoidal signals with the added advantage of providing electronic tunability (by means of external DC voltage applied on the gates of the various MOSFETs) of the oscillating frequency together with the suitability for IC implementation because of employing only MOS-based building blocks and MOS capacitors.

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In this chapter, we take a stock of the work done in this area as contained in references [1–45] and highlight here some representative significant oscillator circuits in each of the three mentioned technologies.

7.2 Switched-Capacitor Oscillators

Although circuit theory literature is flooded with a variety of switched-capacitor filters, comparatively very little attention has been devoted to switched-capacitor oscillators. In the following, we outline some of the switched-capacitor oscillator (SCO) topologies proposed in the literature.

Throughout this monograph, it may be noticed that the classical Wien Bridge oscillator has found place in many chapters wherein it has been realized with different kinds of building blocks in the changing technologies. In view of this, therefore, it is not surprising to observe that among the first few attempts in making a switched-capacitor sinusoidal oscillator was a switched-capacitor Wien Bridge oscillator! This was done by replacing the two resistors of the classical Wien bridge oscillator by periodically switched capacitances implementing the standard bi-linear $s \rightarrow z$ transformation. The resulting circuit is shown in Fig. 7.1.

The circuit of Fig. 7.1 can be analyzed by using the following charge conservation equations:

$$q_1(n+1) = q_1(n) + \Delta_q \tag{7.1}$$

$$q_{R1}(n+1) = -q_{R1}(n) + \Delta_q \tag{7.2}$$

$$q_2(n+1) - q_2(n) + q_{R2}(n+1) + q_{R2}(n) = \Delta_q \tag{7.3}$$

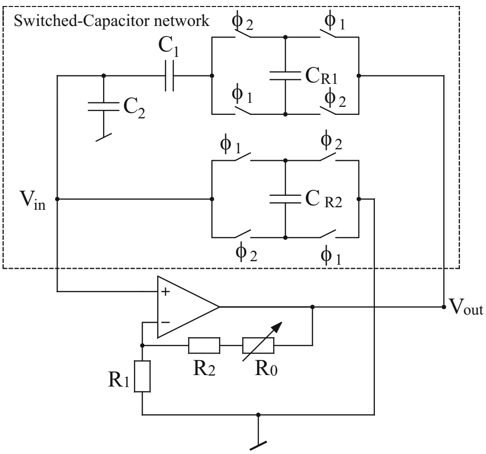

Fig. 7.1 Switched-capacitor Wien Bridge oscillator proposed by Mingguang [1]

By taking Z-transforms and solving, the transfer function of the switched-capacitor circuit is found to be

$$\frac{V_{\text{in}}}{V_0} = \frac{p}{p^2\left(\frac{C_2}{C_{R1}}\right) + p\left(1 + \frac{C_{R2}}{C_{R1}} + \frac{C_2}{C_1}\right) + \frac{C_{R_2}}{C_1}} \tag{7.4}$$

$$\text{where } p = \left(\frac{Z-1}{Z+1}\right) = \tanh\left(\frac{s\tau}{2}\right) \tag{7.5}$$

with τ being the width of the clock being equal to half of the time period of the two phase non-overlapping clock used to periodically switch the capacitors between various positions.

For $C_1 = C_2 = C$, and $C_{R1} = C_{R2} = \alpha C$, it is then found that the gain required to produce and sustain sinusoidal oscillations should be

$$K = \left(1 + \frac{R_2}{R_1}\right) = 3 \tag{7.6}$$

and the oscillation frequency is, then, given by

$$f_0 = \left(\frac{f_c}{\pi}\right) \text{Sin}^{-1}\left\{\frac{C_{R1}C_{R2}}{C_1C_2}\right\} \tag{7.7}$$

Thus, oscillation frequency is directly proportional to the clock frequency and can be varied without having any variable circuit components.

We now present another example to demonstrate that just as the initial work on switched-capacitor filters heavily relied on well-known active-RC/op-amp-RC filters; it is not surprising that the initial work on switched-capacitor oscillators (SCO) too followed several time-tested topologies of active RC oscillators. The SCO presented by Vittoz in [2] was around the well-known RC phase shift (third order) oscillator topology. This circuit is shown in Fig. 7.2.

In this circuit, the phase shift network consists of three capacitors and two resistors composed of periodically switched capacitors. Thus, the capacitors C_0 to C_4 and the p-channel MOSFETs M_1 to M_4 with in-phase switches constitute the phase shifting network. MOSFET M_5 acts as the transconductance gain element of the oscillator. The rest of the circuit constitutes of the amplitude limiter circuit wherein the MOSFETs M_7 and M_{10} are supposed to operate in weak inversion. The diodes D_1 and D_2 were lateral diodes in the polycrystalline gate layer which replace resistors of very high but non-critical values. MOSFET M_{11} behaves as a resistor that limits the gain of the regulating loop.

Taking the three capacitors to be of value C each and the two switched capacitors (simulating the resistors) of value αC, for oscillation frequencies which are much smaller than the clock frequency f_c, each switched capacitor simulates an equivalent

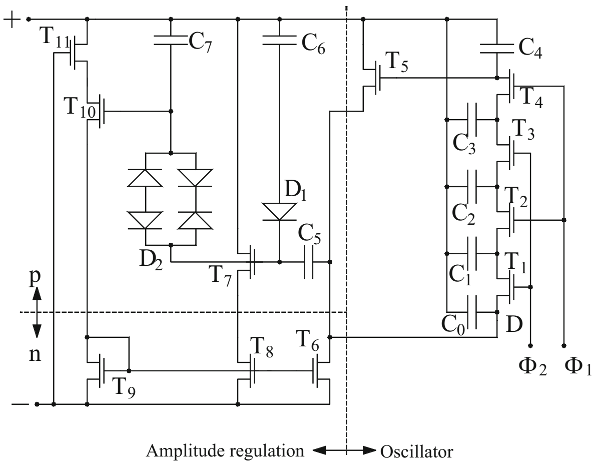

Fig. 7.2 CMOS switched-capacitor third-order phase-shift oscillator proposed by Vittoz [2]

resistance of value $R = 1/(\alpha C f_c)$. For $\alpha \ll 1$, the critical transconductance value required for producing oscillations is found to be

$$g_{mc} = 12 f_c \alpha C \left(1 + \frac{14}{A}\right) \tag{7.8}$$

and the oscillation frequency is, then, given by

$$f_0 = \frac{\sqrt{3}}{2\pi} \alpha f_c \left(1 + \frac{6}{A}\right) \tag{7.9}$$

where A is the DC amplification of the MOSFET acting as the transconductance amplifier.

The value of the current necessary to obtain the critical transconductance g_{mc} may be decreased by increasing the channel width of the transistor until a minimum value given by

$$I_{crit} = \frac{nkT}{q} g_{mc} \tag{7.10}$$

is reached on the verge of weak inversion.

The workability of this circuit was confirmed [2] by actually fabricating the circuits in two different versions. In the first case, the capacitors ($\alpha = 0.1$ and $C = 10$ pF) were realized between n+ polysilicon and aluminum layers with a deposited oxide as a dielectric while in the second version, the capacitor ($\alpha = 1.7$ and $C = 1$ pF) were gate-oxide capacitors to p-well [2]. Both the versions of the fabricated oscillators consumed less than 50 nA at a clock frequency of 100 kHz corresponding to the oscillation frequencies of values 2.4 kHz and 8.5 kHz, respectively.

Sinusoidal oscillators which are capable of providing multi-phase signals, having same amplitude but specific phase relationship between its various outputs, are required in many applications particularly in the communication area.

A very commonly employed idea to design an oscillator of this kind is to employ a two integrator loop. The quadrature sinusoidal oscillator, in fact, is based upon the implementation of the following state equations:

$$\dot{x}_1 = \omega_0 x_2,\ \dot{x}_2 = -\omega_0 x_1 \tag{7.11}$$

Now the continues-time integrator needs to be replaced by a discrete-time integrator for which there are two possibilities, keeping in mind that the discrete integrator must have the transformation property such that the imaginary axis in the s-plane is mapped on to the unit circle in z-plane. This objective can be accomplished either by lossless discrete integrator (LDI) or by bilinear discrete integrator (BDI) as demonstrated in [18–28].

Based on the above, Mikhael and Tu [3] presented a number of switched capacitor oscillator configurations which are shown in Figs. 7.3, 7.4, 7.5, and 7.6.

The SCO of Fig. 7.3 has lossless-discrete-integrators (LDI); hence, the oscillation frequency is given by

$$f_0 = \frac{f_c}{\pi}\mathrm{Sin}^{-1}\left(\frac{C_1}{2C}\right);\ 2C > C_1 \tag{7.12}$$

where f_c is the clock frequency.

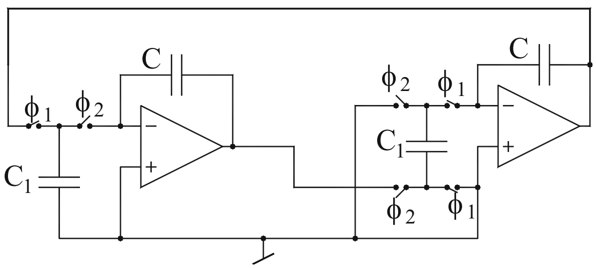

Fig. 7.3 SCO based on LDI proposed by Mikhael-Tu [3]

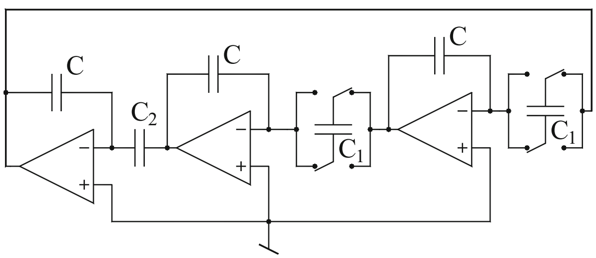

Fig. 7.4 SCO based on BDI proposed by Mikhael-Tu [3]

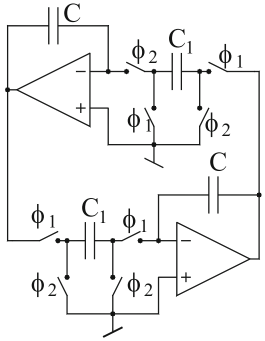

Fig. 7.5 An alternative LDI-based SCO; Mikhael-Tu [4]

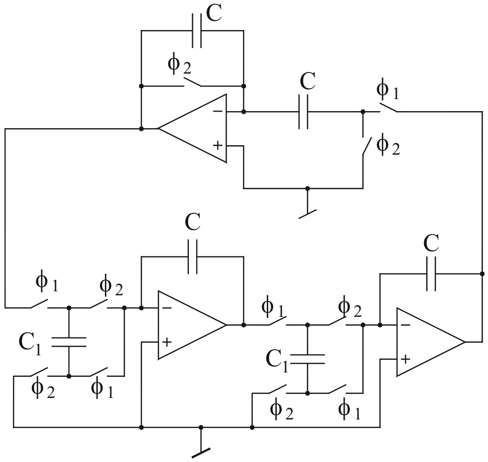

Fig. 7.6 An alternative SCO based upon BDI proposed by Mikhael-Tu [4]

On the other hand, for the circuit of Fig. 7.4 which incorporates bilinear discrete integrators (BDI), the oscillation frequency is given by

$$f_0 = \left(\frac{2f_c}{\pi}\right) \tan^{-1}\left(\frac{C_1}{C}\right) \tag{7.13}$$

Two alternative implementations of the SCOs based upon LDI and BDI transformations are shown in Figs. 7.5 and 7.6.

The expressions for the oscillation frequency for both the abovementioned circuits can be easily derived by formulating the characteristic equation in z-plane. It is then found that for the SCO of Fig. 7.5, based upon LDI, the oscillation frequency is given by

$$f_0 = \frac{f_c}{\pi} \arcsin(C_1/2C) = f_c/K;\ 2C >> C_1 \tag{7.14}$$

where f_c is the sampling frequency which is equal to the clock frequency for the realization in Fig. 7.5 and K is the number of samples per oscillation cycle.

Similarly, for the SCO of Fig. 7.6 based upon BDI, the oscillation frequency is given by

$$f_0 = \frac{f_c}{\pi} \arctan(C_1/C) = f_c/K \tag{7.15}$$

where C_1 is the SC and f_c is the sampling frequency which is equal to the clock frequency for the realization shown in Fig. 7.6

It may be noted that in both the cases, the SCO design offers the interesting property that the oscillation frequency is directly proportional to the clock frequency thus, there is no need to change the components of the oscillators to vary the oscillation frequency over a very wide range.

Mickhael and Tu [4] also presented a topology for multi-phase oscillator based upon the concept of *active sequence discriminator*. The interested readers are referred to [4] for further details.

An alternative switched-capacitor Wien bridge oscillator with automatic gain controller circuit was presented by Horie, Youssef, Miyazaki, and Takeishi [5] which is shown here in Fig 7.7.

The SC Wien bridge oscillator of Fig 7.7 differs from the usual Wien bridge oscillators in two respects: (a) part of the output voltage is fed back as voltage V_d

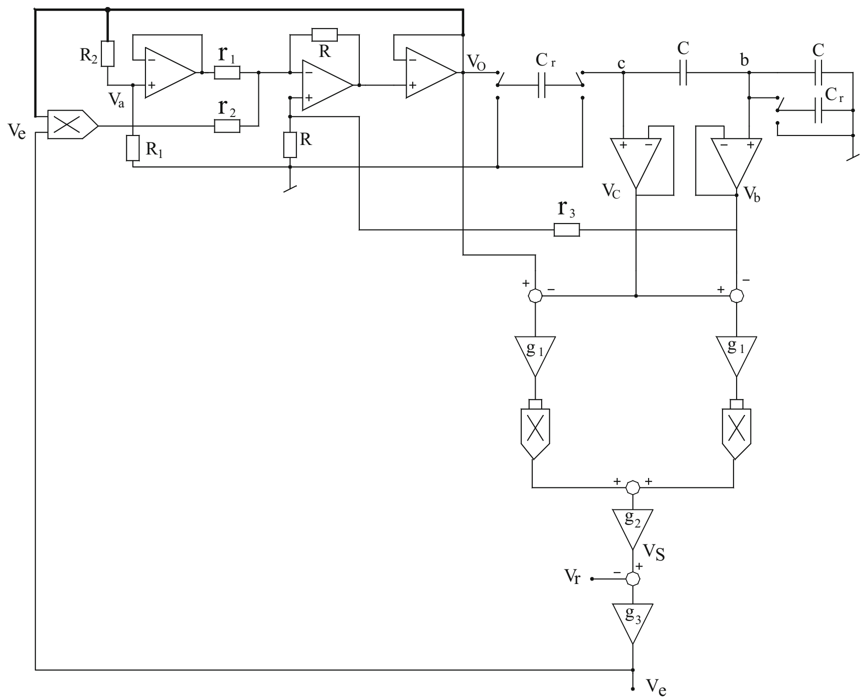

Fig. 7.7 SC Wien-bridge oscillator along with amplitude control circuitry proposed by Horie-Youssef-Miyazaki-Takeishi [5]

which is generated by multiplying the output voltage V_0 with the error signal V_e of the automatic gain control circuit (AGC). The parameter $\gamma = r_1/(r_1 + r_2)$ gives the degree of contribution of V_d to the total voltage applied to the inverting terminal of the amplifier (b) the two resistors of the Wien bridge RC network have been simulated by switched capacitors with the usual assumption of the clock frequency controlling these switches being much larger than the oscillation frequency to be generated by the oscillator. The two simulated resistors have the value $R_{SC} = 1/f_c C_r$. Thus, when $C_R/C \ll 1$, the frequency of oscillations of the Wien bridge oscillator is given by

$$f = \frac{1}{2\pi}\left(\frac{C_r}{C}\right)f_c \tag{7.16}$$

It is seen that the condition $f_c \gg f$ is automatically satisfied, as required.

The working of the AGC can be explained as follows: it may be seen that the potential differences $(V_0 - V_C)$ and $(V_C - V_b)$ have the amplitude proportional to V_0 but will have the phase shift of 90° between them. Using analog multipliers, these potential differences are squared and added, resulting in a DC voltage V_s which is then compared with a fixed reference voltage V_r to create an error voltage V_e which is subsequently applied to another multiplier as shown in Fig. 7.7.

The experimental results of this set up demonstrated that the oscillation frequency may be successfully generated in the range 10 μHz to 100 Hz. The circuit demonstrated good agreement of the proportionality of the oscillation frequency with the clock frequency f_c. The circuit demonstrated the distortion factor of the output voltage to be less than −40 dB over the observed frequency range.

Two different methods of designing switched capacitor sinusoidal oscillator using unity gain amplifiers (UGA) as the active elements are discussed next.

The first method is based upon conversion of UGA-based active RC oscillators into switched-capacitor oscillators. In this context, it may be recalled that single element controlled sinusoidal oscillators employing only unity gain amplifiers as active elements were first proposed in reference [57] of Chap. 2 and subsequently in [127] of Chap. 2. Two exemplary oscillators of this kind are reproduced here in Fig. 7.8.

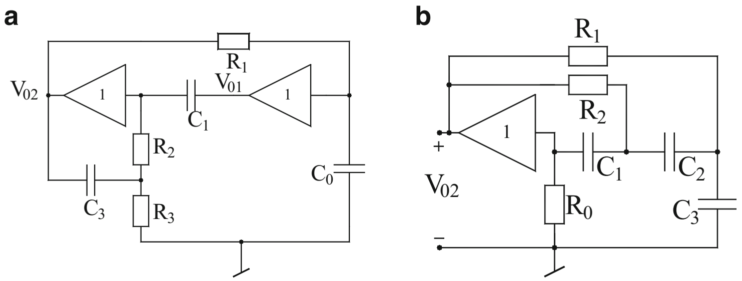

Fig. 7.8 Two exemplary sinusoidal oscillators using only unity gain voltage buffers: (**a**) RC:CR transformed, two unity gain amplifier-based oscillator from reference [57] of Chap. 2. (**b**) RC:CR transformed, single amplifier-based oscillator from reference [127] of Chap. 2

The first circuit is characterized by the condition of oscillation given by

$$\frac{1}{C_0R_1} = \frac{1}{C_3}\left(\frac{1}{R_2} + \frac{1}{R_3}\right) \tag{7.17}$$

and the frequency of oscillation f_o given by

$$f_0 = \frac{1}{2\pi\sqrt{C_1C_3R_2R_3}} \tag{7.18}$$

On the other hand, for the single buffer-based oscillator, the CO and FO are found to be, respectively,

$$R_0 = \left\{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right\}(C_1R_1 + C_2R_2 + C_2R_1 + C_3R_2) \tag{7.19}$$

$$\text{and } f = \frac{1}{2\pi}\sqrt{\frac{1}{R_1R_2(C_1C_2 + C_2C_3 + C_1C_3)}} \tag{7.20}$$

These circuits can be treated as active RC prototypes from which SCOs can be derived using switched-capacitor/resistor equivalence implementing either the bilinear transformation or lossless discrete integrator (LDI) transformation [3]. In Fig. 7.9, we show the SCOs based on the circuits of Fig. 7.8a, b, respectively. As can be seen in the diagram the bottom plates of all the capacitors are either connected to ground or else are voltage driven.

It is, therefore, clear that employing switched-capacitor/resistor equivalence as implemented in these circuits has the novel feature of providing insensitivity to the effect of bottom plate stray capacitances of the MOS capacitors. It may be noted that in these circuits, instead of using op-amps configured as unity gain buffers (UGB) it would be advisable to use IC voltage buffers which would provide the

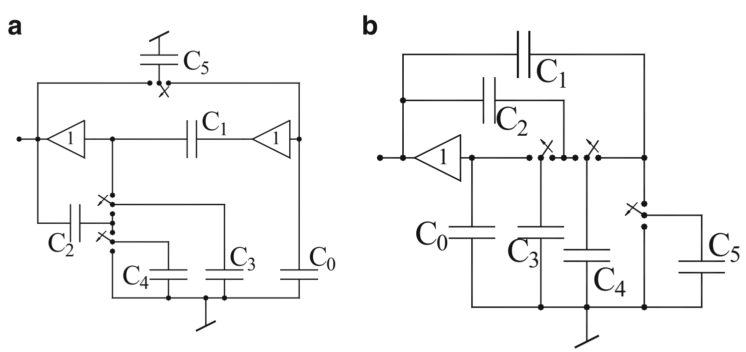

Fig. 7.9 SCOs using only UGBs: (**a**) SCO using two UGBs. (**b**) SCO using only single UGB

advantages of reduced noise, chip area and DC power consumption as well as improved bandwidth and dynamic range. It is also worth mentioning that, as compared to the earlier SCOs of [3, 4] the SCO described here have the advantage of use of UGBs. On the other hand, like the SCOs [3, 4], the present circuits also have the advantage of insensitivity of parasitic capacitances. By routine analysis, the condition of oscillation (CO) and the oscillation frequency (FO) for the SCO of Fig. 7.9a are given by

$$\mathrm{CO:}\ \frac{C_0}{C_5} = \frac{C_2}{C_3 + C_4} \tag{7.21}$$

$$\mathrm{FO:}\ f_0 = \left(\frac{f_c}{\pi}\right) \sin^{-1}\left\{\frac{1}{2}\sqrt{\frac{C_3C_4}{C_1C_2}}\right\} \tag{7.22}$$

where f_c is the clock frequency of the two-phase non-overlapping clock.

On the other hand, for the circuit of Fig. 7.9b, the corresponding equations are given by

$$\mathrm{CO:}\ C_0 = \frac{C_2C_3C_4}{(C_3C_4 + C_4C_5 + C_3C_5)}\left\{\frac{C_3C_4}{C_1} + \frac{C_4C_5}{C_2}\right\} \tag{7.23}$$

$$\mathrm{FO:}\ f_0 = \left(\frac{f_c}{\pi}\right) \sin^{-1}\left\{\frac{1}{2}\sqrt{\frac{C_3C_4 + C_4C_5 + C_3C_5}{C_1C_2}}\right\} \tag{7.24}$$

Following a quite different approach and theory, a simpler switched-capacitor oscillator using two unity gain buffers but only four capacitors was proposed by Huertas, Rodriguez-Vazquez, and Perez-Verdu [6] which is shown here in Fig. 7.10.

The circuit is designed in such a manner that the output V_{01} of the first UGB goes into saturation whereas the other output V_{02} does not reach saturation. This is ensured by choosing capacitor values such that

$$\frac{C_4}{C_1} > \frac{C_3}{C_2} \tag{7.25}$$

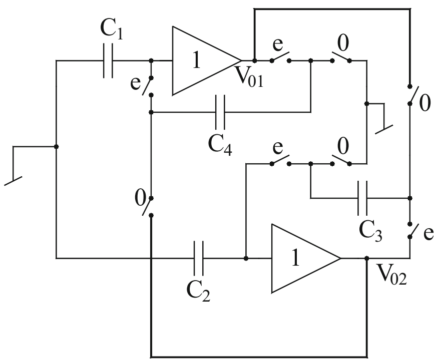

Fig. 7.10 SCO proposed by Huertas, Rodriguez-Vazquez, Perez-Verdu [6]

In view of the above, the circuit can be modeled as a second-order discrete time system having transfer function $H(z)$ along with a saturation type nonlinearity made from the UGB having output V_{01}.

The transfer function of the linear block is found to be [6]

$$H(Z) = \frac{C_4}{C_1 + C_4} \frac{Z^2 - Z - \frac{C_3}{C_2}}{Z^2 + \left(\frac{C_4}{C_1+C_4} - 2\right)Z + \frac{C_1}{C_1+C_4}} \tag{7.26}$$

If a sinusoidal input to the nonlinearity given by

$$V_{\mathrm{i}}(n) = A \sin\left(nw_0 T_{\mathrm{c}}\right) \tag{7.27}$$

is assumed, where T_{c} is the time period of the two phase (having even and odd phases) non-overlapping clock, A is the oscillation amplitude and ω_0 is the angular oscillation frequency. Now representing the static nonlinearity by an equivalent gain $N(A)$ corresponding to the first harmonic of the nonlinearity output, it has been shown in [6] that the oscillation frequency is given by

$$f_0 = \frac{f_{\mathrm{c}}}{2\pi} \cos^{-1}\left[1 - \frac{1}{2\left\{1 + \frac{C_1}{C_4}\left(1 + \frac{C_2}{C_3}\right)\right\}}\right] \tag{7.28}$$

whereas the oscillation amplitude can be evaluated from the following equation:

$$\frac{1}{1 + \frac{C_3}{C_2}} = \frac{2}{\pi}\left[\sin^{-1}\left(\frac{V_{\mathrm{sat}}}{A}\right) + \frac{V_{\mathrm{sat}}}{A}\sqrt{1 - \left(\frac{V_{\mathrm{sat}}}{A}\right)^2}\right] \tag{7.29}$$

From the above expression it is seen that frequency of the sinusoidal signal generated is directly proportional to the clock frequency of the two-phase non-overlapping clock used to control the various switches in the circuit.

The measured results from the experimental set up of the circuit have confirmed the working of this circuit which was found to generate variable frequency oscillations controlled by the clock frequency over a range of three decades [6].

7.3 Switched-Current Sinusoidal Oscillators

The evolution of switched-current filters was a promising development since unlike switched-capacitor filters, the switched-current filters did not use op-amps and did not employ floating capacitors. This resulted in a lot of saving in the chip area and also gave renewed impetus to the mixed signal VLSI implementable in standard digital CMOS technology. While most of the work on switched-current filters

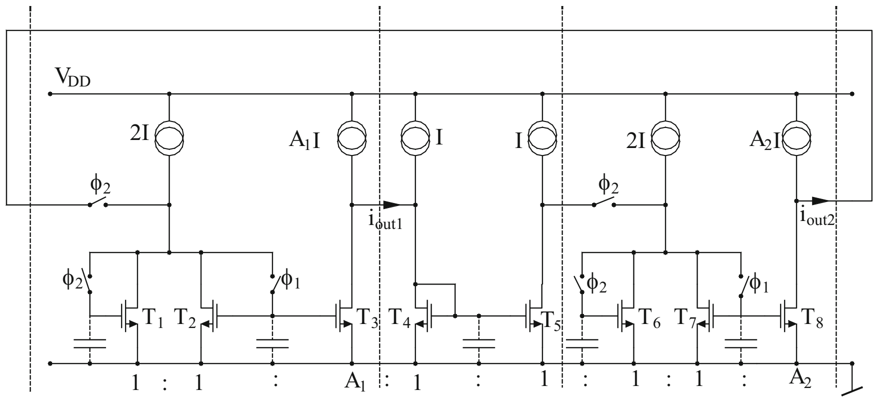

Fig. 7.11 Switched-current oscillator proposed by Jia-Niu-Chen [7]

focused on filters only, only a limited number of works focused on the realization of oscillators using switched-current techniques. In the following, we present some representative developments on switched-current oscillators.

Jia-Niu-Chen [7] presented a novel switched-current sinusoidal oscillator which is shown here in Fig. 7.11. The circuit is composed of three sections out of which the first two sections are non-inverting forward lossless switched-current integrators with scale factors A_1 and A_2 respectively, whereas, the section in the middle is a switched-current inverter made from a simple current mirror. The discrete time transfer function of the integrator is given by

$$H_k(z) = \frac{i_{\mathrm{out}k}}{i_{\mathrm{in}k}} = A_k \frac{z^{-1}}{1 - z^{-1}}, \; k = 1, 2 \tag{7.30}$$

Hence, the loop gain of the circuit in Z-domain is given by

$$-A_1 A_2 \left(\frac{z^{-1}}{1 - z^{-1}} \right)^2 = 1 \tag{7.31}$$

From the above equation, the two roots of the resulting characteristic equation are given by

$$z_{1,2} = 1 \pm j\sqrt{A_1 A_2} = \sqrt{1 + A_1 A_2} \left(\frac{1}{\sqrt{1 + A_1 A_2}} \pm j \frac{\sqrt{A_1 A_2}}{\sqrt{1 + A_1 A_2}} \right) \tag{7.32}$$

which are clearly seen to be located on outside the unit circle in the Z-plane which is indicative of the fact that building up of oscillations is guaranteed. Furthermore, at least one output current would saturate due to which sinusoidal oscillation with stable amplitude can be achieved. By a routine analysis, the FO is a given by

$$f_0 = \frac{f_c}{2\pi} \cos^{-1}\left(\frac{1}{\sqrt{1 + A_1 A_2}}\right) \tag{7.33}$$

which shows that the FO can be controlled by aspect ratio (W/L) of the scaling transistors and is linearly controllable by changing the clock frequency f_c

This switched-current oscillator was simulated [7] using SPICE with level 3 MOS transistor models. The basic aspect ratio chosen was 10 μm/5 μm. A 5 V DC powers supply was used for biasing and the clock signal was 0–5 V pulse with frequency 1 MHz. The bias current I was taken as 50 μA, the MOS switches had aspect ratios equal to 5 μm/1 μm. The oscillation frequency obtained from simulation was found to be 26.954 kHz which is close to the value 27.596 calculated from Eq. (7.33).

The same authors in another publication [16] presented a basic scheme for realizing multi-phase switched-current oscillators both for even-phase oscillators as well as for odd-phase oscillators by proposing switched current inverting bilinear damped integrators. The interested readers may see [16] for further details.

7.4 Sinusoidal Oscillator Using an Alternative Form of Capacitor-Switching

Whereas the normal switched-capacitor oscillators have employed either periodically switched capacitors as resistor-equivalents in some form or the other or employed switched-capacitor networks to create discrete time systems which can be employed to realize sinusoidal oscillators, Karybakas-Sarafis [8] presented a different way of incorporating synchronous periodically operated switches, used in conjunction with capacitors, to create a duty-cycle-dependent sinusoidal oscillator suitable for low-frequency generation. Their proposition is shown here in Fig. 7.12.

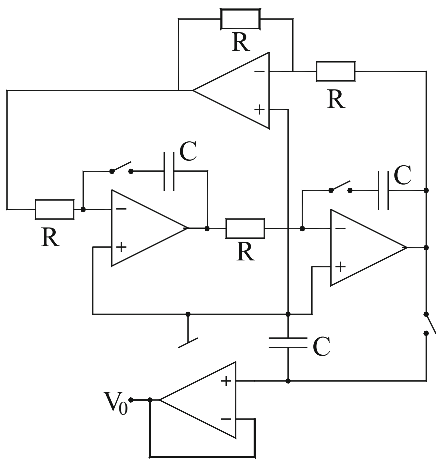

Fig. 7.12 The duty cycle-dependent sinusoidal oscillator proposed by Karybakas-Sarafis [8]

The basic building block of the oscillator shown in Fig. 7.12 is the integrator circuit wherein a periodically switched capacitor is connected in series with the integrating capacitor. If this switch is controlled by a pulse which has time period T and narrow pulse width τ, the switch then closes every T seconds and remains closed during the interval τ. The charge having flowed in the capacitor during the interval τ is given by

$$Q(n) = C[V_c(n) - V_c(n-1)] \tag{7.34}$$

Noting that the voltage across the capacitor is also the output voltage of the circuit, taking Z-transform of the above equation, one finds

$$Q(n) = -C\left(1 - z^{-1}\right)V_0(z) \tag{7.35}$$

If time τ is smaller than the time period of the input signal $V_i = A\sin\omega t$, then V_i can be supposed to be constant during the period τ and hence one can write

$$Q(n) = \int_{t_{n-1}}^{t_n} \frac{V_i}{R}\,dt = \frac{V_i(n)}{R}\tau \tag{7.36}$$

From Eqs. (7.35) and (7.36) the transfer function of the integrator is given by

$$\frac{V_0(z)}{V_i(z)} = -\frac{\tau}{RC(1 - z^{-1})} \tag{7.37}$$

Taking the sampling frequency to be much larger than the signal frequency (i.e., $1/T \gg (\omega/2\pi)$ and further assuming that $\omega T \ll 1$, the transfer function of the circuit under question can be approximated by

$$\frac{V_0}{V_i} \cong -\frac{1}{j\omega R\left(\frac{T}{\tau}\right)C} \tag{7.38}$$

In view of the above, formulation of the circuit of Fig. 7.12 can be easily understood. It can be easily visualized that the FO of the sinusoidal oscillator of Fig. 7.12 would be given by

$$\omega_0 = \left(\frac{\tau}{T}\right)\frac{1}{RC} \tag{7.39}$$

Thus, it is seen that the oscillation frequency can be readily controlled by varying the duty cycle of the pulse waveform applied to control the various switches in the circuit. Also, it should be possible to generate very-low-frequency oscillations without requiring large RC component values because of the additional scale factor featuring in Eq. (7.39).

The workability of this oscillator was demonstrated in [8] by employing the op-amp LF356 and RC components $R = 1$ kΩ, $C = 100$ nF and $\tau = 5$ μs for various values of the switching frequency. The circuit generated good-quality sine waves for frequency 0.5 mHz when a simple first-order filter made from the same technique as described above was used to smoothen the staircase-type sinusoidal oscillator generated by the basic oscillator circuit devised for the purpose.

7.5 MOSFET-C Sinusoidal Oscillators

The technique of designing MOSFET-C filters which were developed as continuous-time alternatives to the discrete-time switched-capacitor filters was also extended to the design of MOSFET-C oscillators which possess (a) the advantage of electronic control of the oscillation frequency through the control voltages applied on the gates of the MOSFETs and (b) suitability for implementation in CMOS technologies because of employing CMOS op-amps, MOS capacitors and MOSFET only as active elements.

A number of researchers employed the basic methodologies to devise circuit configurations for MOSFET-C oscillators using a variety of active elements such as op-amps, current conveyors, current feedback op-amps, DDAs, and others elements of more recent origin. In this section, we highlight the significant contributions made in this area.

7.5.1 MOSFET-C Oscillators Using DDAs

The differential difference amplifier (DDA) introduced by Sackinger and Guggenbuhl [36] has been found to be an useful building block which makes it possible to carry out several interesting analog circuit operations such as realizing inverting amplifier, a unity gain difference amplifier or a unity gain summing amplifier, without requiring any external passive components (resistors). Mahmoud and Soliman in [9] while presenting a new CMOS realization of the DDA demonstrated an interesting application of the DDA in realizing a MOS-C oscillator which is shown here in Fig. 7.13.

In the MOS-C sinusoidal oscillator circuit of Fig. 7.13, one DDA along with two MOSFETs M_3 and M_4 has been employed as a negative impedance converter (NIC). In this part of the circuit, each MOSFET acts as a linear voltage controlled resistor (controlled by the gate voltage V_G) thereby giving unity conversion constant to the NIC. One of the two ports of this NIC is terminated into a series RC branch consisting of capacitor C_2 along with another DDA and the MOSFET M_2 which realize another linear voltage-controlled resistor. On the other port of the NIC lie, a parallel RC branch with capacitor C_1 and another linearized resistor made from the MOSFET M_1 and the third DDA. The circuit can, therefore, be considered

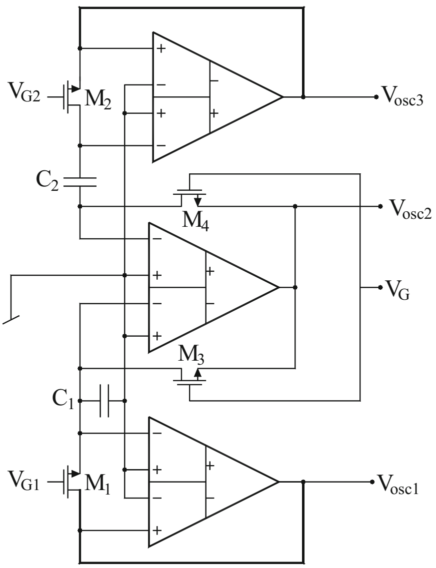

Fig. 7.13 A MOS-C oscillator realized with DDAs proposed by Mahmoud and Soliman [9]

to be the MOS-C version of one of the variants of the classical Wien bridge oscillator.

The realized linear voltage-controlled resistors have the values given by

$$R_1 = \frac{1}{2K_1(V_{G1} - V_{th})}; R_2 = \frac{1}{2K_2(V_{G2} - V_{th})}; R = \frac{1}{2K(V_G - V_{th})} \tag{7.40}$$

where K_1 and K_2 are the transconductance parameters of the MOSFETs M_1 and M_2 and K is the transconductance parameter of the MOSFETs M_3 and M_4 which are assumed to be matched having the same K value.

In terms of the above, it is easy to verify that the CO is given by

$$\frac{R_2}{R_1} + \frac{C_1}{C_2} = 1 \tag{7.41}$$

which can be satisfied by taking $R_2/R_1 = C_1/C_2 = 1/2$. These conditions can be satisfied by taking $K_1 = K$, and $K_2 = K/2$ with $V_{G1} = V_{G2} = V_{G0}$ and $C_1 = C$; $C_2 = 2C$. With these conditions, the FO would then be given by

$$\omega_0 = \frac{2K(V_{G0} - V_{th})}{C} \tag{7.42}$$

Thus, it is seen that the FO can be linearly controlled by varying the control voltage V_{G0} applied to the gates of the MOSFETs M_1 and M_2 both.

The workability of this configuration has been demonstrated in [9] using SPICE simulations wherein a simple amplitude control scheme consisting of two additional MOSFETs and two diodes was also incorporated in the circuit.

7.5.2 MOSFET-C Oscillators Using CFOAs

It has been amply demonstrated in the literature (for instance, see [38]) that the current feedback op-amp, particularly the AD844 variety is quite versatile in that it not only can serve as a replacement of a normal op-amp on a one-to-one basis, it can also be employed to realize second-generation Current Conveyors of both the polarities; CCII+ with one AD844 and CCII- using two AD844s. Furthermore, due to the external availability of the compensation pin (the Z-terminal of the internal CCII+ of the CFOA), it is a four-terminal versatile building block in its own right. Although a number of interesting realisations of MOSFET-C biquad filters have been proposed in literature using CCIIs and CFOAs, in this section we present some prominent MOS-C sinusoidal oscillator topologies using CFOAs.

The MOS-C oscillators proposed by Mahmoud and Soliman [10] are based upon three well-known MOSFET-based circuits of cancelling the nonlinearities of the MOSFETs [37] to realize a linear VCR directly or creating an equivalent resistive effect indirectly. These circuits are shown in Fig. 7.14a–c.

For the first circuit (Fig. 7.14a) it can be easily shown that since $V_{DS}=2V_1$, $V_{GS}=(V_G+V_1)$, assuming the MOSFETs to be operating in triode region, the drain current can be written as

$$I_D=I=\beta\left[(V_G+V_1-V_T)-\frac{2V_1}{2}\right]2V_1 \text{ where } \beta=\mu_s C_{ox}\left(\frac{W}{L}\right) \tag{7.43}$$

It is readily seen that in the above expression the square nonlinearity is effectively cancelled out and the circuit implements a linear voltage controlled resistance whose value is given by

$$R_{eq}=\frac{1}{2\beta(V_G-V_T)} \text{ for } (V_G-V_T)\geq|V_1| \tag{7.44}$$

Similarly, for the circuit of Fig. 7.14b analysis shows that

$$i=I_1-I_2=\beta V_G(V_1-V_2) \text{ for } (V_G-V_T)\geq \max(V_1,V_2) \tag{7.45}$$

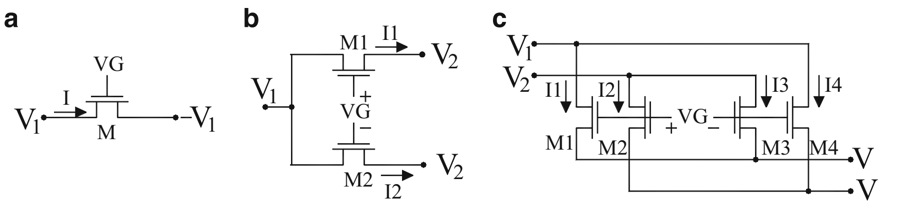

Fig. 7.14 Three techniques of nonlinearity cancellation in MOSFETs [37]

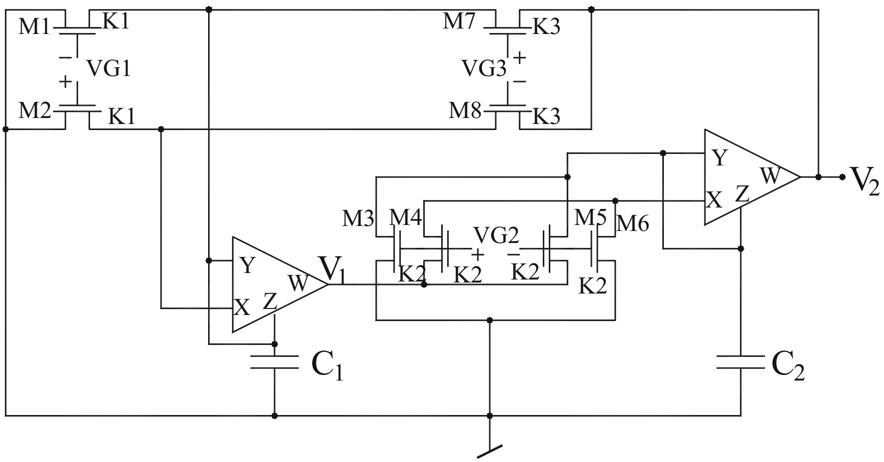

Fig. 7.15 A MOS-C sinusoidal quadrature oscillator proposed by Mahmoud and Soliman [10]

Lastly, for the circuit of Fig. 7.14c it is found that

$$I = (I_1 + I_3) - (I_2 + I_4) = (I_1 - I_4) - (I_2 - I_3) = \beta V_G(V_1 - V_2) \tag{7.46}$$

Using these basic MOS resistive elements, in conjunction with a low-voltage rail-to-rail CMOS CFOA operatable form ±1.5 V power supplies, Mahmoud and Soliman [10] presented three MOSFET-C oscillators, the first one of which is shown in Fig. 7.15. By a routine analysis of this circuit, its state equations, denoting the voltages across the capacitors C_1 and C_2 as X_1 and X_2, respectively, are found to be

$$\dot{x}_1 = \frac{(G_1 - G_3)}{C_1}x_1 + \frac{G_3}{C_1}x_2 \text{ and } \dot{x}_2 = -\frac{G_2}{C_2}x_1 \tag{7.47}$$

where $G_1 = k_i V_{Gi}$; $i = 1,\ 2$ and 3

From the above equations, the CO and FO are given by, respectively,

$$\text{CO}: G_1 = G_3 \text{ and FO}: \omega_0 = \left(\frac{G_2 G_3}{C_1 C_2}\right)^{1/2} \tag{7.48}$$

The same authors [10] presented yet another MOS-C quadrature oscillator which is shown Fig. 7.16. By straightforward analysis, the state equations for this circuit are given by

$$\dot{x}_1 = \frac{G_1}{C_1}x_2 \text{ and } \dot{x}_2 = -\frac{G_2}{C_2}x_1 + \frac{(G_3 - G_4)}{C_2}x_2 \tag{7.49}$$

from where the CO and FO for this circuit are given by

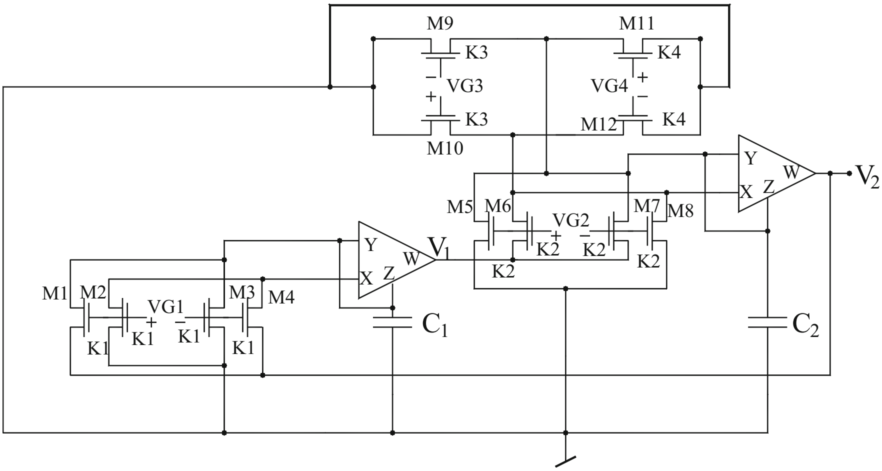

Fig. 7.16 An alternative MOS-C oscillator proposed by Mahmoud and Soliman [10]

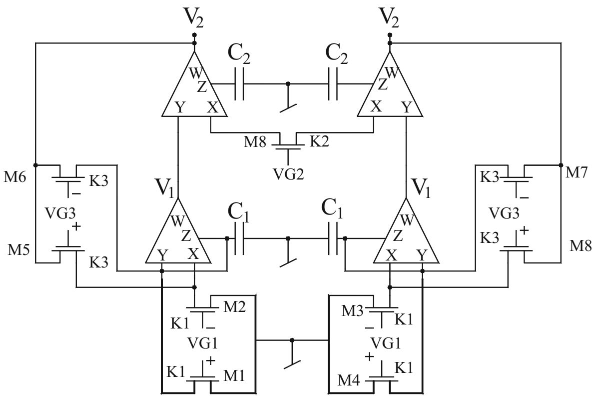

Fig. 7.17 A balanced output sinusoidal oscillator proposed by Mahmoud and Soliman [10]

$$\mathrm{CO}: G_3 = G_4 \text{ and } \mathrm{FO}: \omega_0 = \left(\frac{G_1 G_2}{C_1 C_2}\right)^{1/2} \tag{7.50}$$

A balanced output sinusoidal oscillator was also reported [10] which is shown in Fig. 7.17. The state equations for this circuit are found to be

$$\dot{x}_1 = \frac{(G_3 - G_1)}{C_1} x_1 - \frac{G_3}{C_1} x_2 \text{ and } \dot{x}_2 = \frac{G_2}{C_2} x_1 \tag{7.51}$$

where $G_i = k_i V_{Gi};\ i = 1, 3$ and $G_2 = 2K_2\,(V_{G2} - V_T)$ from where the CO and FO are given by

$$\mathrm{CO}: G_1 = G_3 \ \text{ and } \ \mathrm{FO}: \omega_0 = \left(\frac{G_2 G_3}{C_1 C_2}\right)^{1/2} \tag{7.52}$$

All the three MOS-C oscillators described above have the advantage of providing independent control of the FO as well as CO through separate control voltages. The workability of all the three circuits was confirmed by SPICE simulations using a low voltage rail-to-rail CMOS CFOA operated from ±1.5 V DC power supply with level 3 MOSFET model parameters for 1.2 μm technology obtained through MOSIS. Good-quality sine waves were reported in each case with THD less than 4 % in the first two circuits and less than 0.06 % in the balanced output topology of [10].

7.5.3 MOSFET-C Oscillators Using OTRAs

Another building block which has been found to be quite appropriate for realizing MOSFET-C filters and oscillators is the so-called operational trans-resistance amplifier (OTRA) which has two current input terminals both having virtual ground at the input terminals, thereby easily facilitating cancellation of nonlinearities of two identical MOSFETs whose drain terminals may be held at the same voltage and the source terminals may be connected to, respectively, at the p and n terminals of the OTRA. Once such quadrature oscillator circuit using two OTRAs, only two capacitors, and six MOSFETs was presented by Salama and Soliman in [40] which is shown here in Fig. 7.18. This circuit with the parasitic capacitance of the OTRA C_p accounted (assuming identical OTRAs) is given by

$$s^2 + \frac{G_3}{(C_1 + C_p)} s + \frac{G_1 G_2}{(C_1 + C_p)(C_2 + C_p)} = 0 \tag{7.53}$$

where

$$G_i = K_{Ni}(V_{ai} - V_{bi}) \ \text{ and } \ K_{Ni} = \mu_N K_{Ni} C_{OX} \left(\frac{W}{L}\right)_i ; i = 1 \text{ to } 3 \tag{7.54}$$

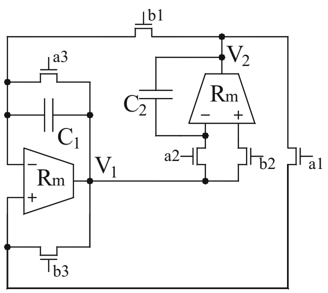

Fig. 7.18 MOSFET-C quadrature oscillator proposed by Salama-Soliman [40]

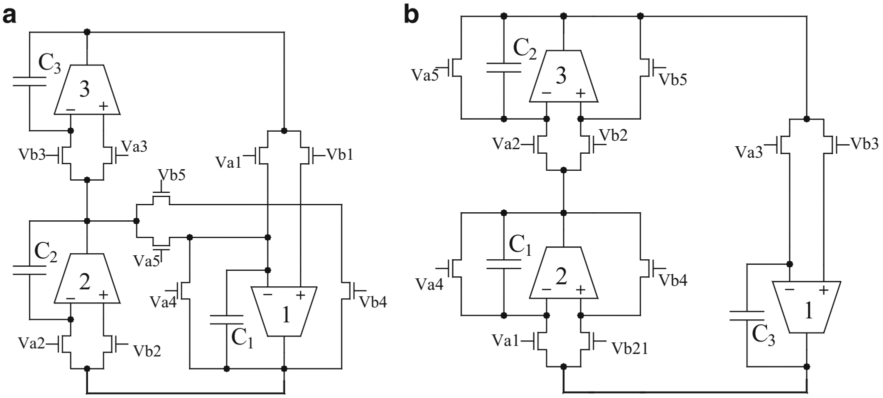

Fig. 7.19 OTRA-based MOSFET-C sinusoidal oscillators proposed by Pandey-Pandey-Komanapalli-Anurag [42]

Thus, the frequency of oscillation can be controlled through G_1 or G_2 through appropriate choice of any of the gate control voltages V_{a1}, V_{a2}, V_{b1}, and V_{b2} without effecting the oscillation condition which is controllable by V_{a3} or V_{b3}.

Using the same principle of nonlinearity cancellation, Pandey-Pandey-Komanapalli-Anurag [42] presented two topologies of third-order quadrature oscillators which are shown here in Fig. 7.19. Taking into consideration the parasitic capacitance of the OTRAs, the oscillation frequency for the first circuit is found to be

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{(C_1 + C_p)(C_2 + C_p)R_2R_5}} \tag{7.55}$$

whereas the corresponding expression for the circuit of Fig. 7.19b is given by

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{(C_1 + C_p)(C_2 + C_p)R_4R_5}} \tag{7.56}$$

From the above, therefore, it is clear that the effect of C_p can be reduced by pre-distorting the values of external capacitors C_1, C_2, and C_3. Thus, achieving self-compensation, the condition of oscillation for the two circuits are given by, respectively,

$$R_4R_5 = R_1R_3 \text{ and} \tag{7.57}$$

$$R_1R_2R_3C_3[R_4C_1 + R_5C_2] = R_4^2R_5^2C_1C_2 \tag{7.58}$$

where

$$R_i = \frac{1}{K_{\mathrm{N}i}(V_{\mathrm{a}i} - V_{\mathrm{b}i})} \tag{7.59}$$

$$K_{\mathrm{N}i} = \mu C_{\mathrm{OX}}\left(\frac{W}{L}\right) \tag{7.60}$$

Through SPICE simulation based upon CMOS OTRA implemented in 0.5 μm CMOS process parameters with supply voltages taken as ±1.5 V the authors [42] have demonstrated that the circuit could work as predicted by theory with total harmonic distortion of the order of 0.57 % for the circuit of Fig. 7.19a and 0.7 % for the circuit of Fig. 7.19b when designed for an oscillation frequency of 159 kHz (the SPICE-generated frequency being 161 kHz).

7.5.4 *MOSFET-C Oscillators Using Inverting Third-Generation Current Conveyors*

Other than the above, there are few more building blocks which facilitate the cancellation of square nonlinearity of the MOSFET by a different mechanism of applying two equal but complementary voltages on the source and drain terminals of a single MOSFET. Two such building blocks are the so-called inverting current conveyors ICCII and ICCIII.

Sobhy and Soliman [41] presented an oscillator using four ICCIIIs, two grounded capacitors, and only four MOSFETs which provided independent adjustability of the CO and FO. This circuit is shown in Fig. 7.20 and is characterized by

$$G_3 = G_4; \quad \omega_0 = \sqrt{\frac{G_1 G_2}{C_1 C_2}} \tag{7.61}$$

It may be noted that in this circuit, all the MOSFETs have complementary voltages on their source and drain terminals since an ICCIII is characterized by $I_y = -I_x$, $V_x = -V_y$, $I_z+ = I_x$, and $I_{z-} = -I_x$. Consequently, the value of the resistance represented by each MOSFET is given by

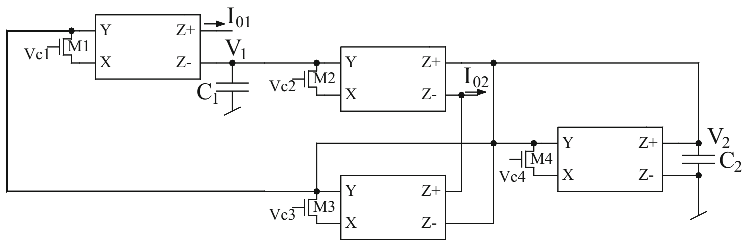

Fig. 7.20 Another MOSFET-C oscillator proposed by Sobhy-Soliman [41]

$$R_i = \frac{1}{2K_i(V_{Ci} - V_T)}; K = \mu C_{OX}\left(\frac{W}{L}\right)_i; i = 1 - 4 \tag{7.62}$$

7.5.5 MOSFET-C Oscillators Using Dual-X CCII

The dual-X current conveyor (DXCCII) is yet another building block which is suitable for creating MOSFET-C oscillators quite easily because of the fact that its two X-terminals also have complimentary voltages existing on them in response to a voltage input applied at the Y-terminal of the DXCCII. This helps in connecting the drain and source terminals of a MOSFET between the two X-terminals, which if operated under triode regime represents a linear voltage-controlled resistor with its even nonlinearities cancelled out. Thus, the MOSFET represents an almost linear resistor, which, in contrast to the other approaches does not require two matched MOSFETs and yet makes an economical linear VCR. With this idea in mind, a number of electronically controllable analog function realization networks were proposed by Zeki and Toker in [39], out of which the DXCCII-based current mode sinusoidal oscillator is shown here in Fig. 7.21. The oscillation frequency and the condition of oscillation for this circuit are given by

$$\omega_0 = \frac{2}{\sqrt{CCR_{M3}(R_{M1}//R_{M2})}} \tag{7.63}$$

$$\text{and } R_{M1} = R_{M2} \tag{7.64}$$

where $R_{Mi} = \frac{1}{k(V_{ci} - V_{th})}$; $i = 1$–3 and symbols have their usual meaning.

It is seen from the above equations that the oscillation frequency can be adjusted via V_{c3} without affecting the condition of oscillation which is also adjustable either by V_{c1} while keeping V_{c2} constant or vice versa. A quadrature oscillator is realizable since the output current shown in the circuit and the current in the capacitor C_2 are in quadrature. The latter current is although not accessible explicitly in the circuit shown but nevertheless can be made available by creating a replica of this by appropriate modification of the DXCCII CMOS architecture appropriately [39].

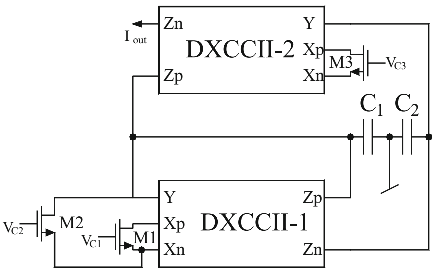

Fig. 7.21 Electronically controllable sinusoidal oscillator using DXCCII proposed by Zeki and Toker [39]

7.6 Switched-Capacitor Voltage-Controlled Relaxation Oscillators

Although a number of switched-capacitor sinusoidal oscillators have been evolved in literature, however, only a few relaxation oscillator circuits have been reported so far. In the following, we describe switched-capacitor relaxation oscillators proposed in [11] and [12].

The relaxation oscillator proposed by Martin [11] is shown in Fig. 7.22. This circuit generates a square wave whose frequency is linearly controllable through clock frequency f_c and can be digitally programmed or voltage-controlled with good linearity. Moreover, the operation of the circuits is insensitive to stray capacitances.

Assuming $K_2 \ll K_1$, the oscillation frequency of the circuit is given by

$$f_0 = \left(\frac{K_2}{K_1}\right) \frac{f_c}{\left(2 + \frac{V_{ss}}{V_{cc}} + \frac{V_{cc}}{V_{ss}}\right)} \tag{7.65}$$

For equal positive and negative supply voltages, the oscillation frequency is only dependent on the ratio K_2/K_1 which can be realized very accurately. Even with unequal supply voltages, the dependence of the oscillation frequency on the variation between them is very small. The oscillation frequency can be digitally controlled by connecting/disconnecting additional capacitors in parallel with K_2C. If 50 % duty cycle is not necessary then the second op-amp can even be replaced by an inverter which however does not have any effect on the oscillation frequency. Lastly, in order to realize a voltage controlled oscillator, an additional switched feed-in is required to be added; see [11] for further details.

In the last, we discuss the SC relaxation oscillator proposed by Jamal, Zafrullah, and Holmes [12], shown here in Fig. 7.23.

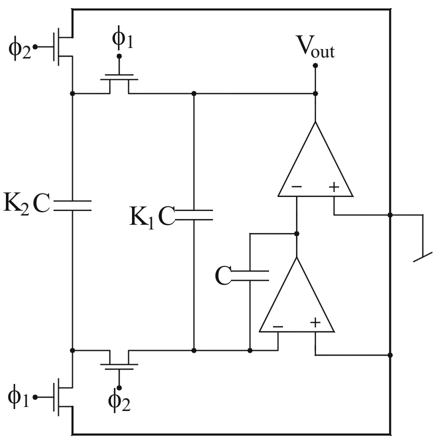

Fig. 7.22 SC stray-insensitive oscillator proposed by Martin [11]

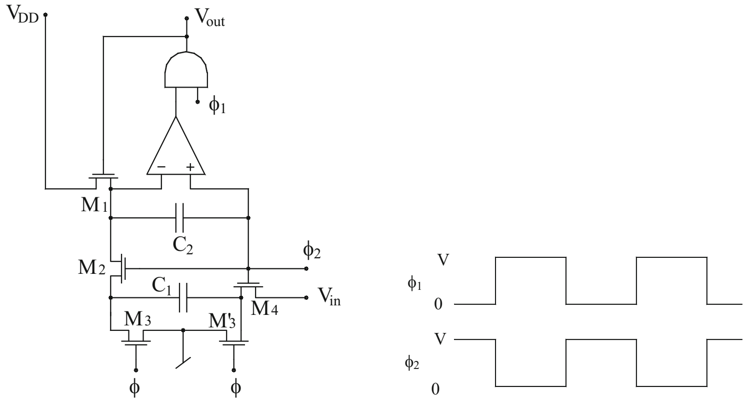

Fig. 7.23 SC relaxation oscillator proposed by Jamal, Zafrullah, and Holmes [12]

In the circuit shown ϕ_1 and ϕ_2 are the pulses generated by the two-phase non-overlapping clock. The circuit operation is based upon charge-balancing principle. Assume that initially there is some charge stored on capacitor C_2. During each clock cycle, some charge is pulled off by the capacitor C_1, the amount being dependent upon the input V_{in}. The comparator keeps on monitoring the voltage across the capacitor C_2. When the voltage across the capacitor C_2 reaches zero, the comparator enables the output gate which provides an output pulse and also recharges the capacitor C_2. Since the comparator samples the voltage across the capacitor only when the clock pulse ϕ_2 makes a transition from HIGH to LOW, the output pulse will always be synchronized with ϕ_2 and the frequency of the output will be related to the clock frequency(f_{c}) by an integer number. The output frequency, in fact, is given by

$$f_0 = \frac{f_{\text{c}}}{N} \tag{7.66}$$

where N is the smallest integer that satisfies the relation

$$\frac{1}{N} \leq \frac{C_1(V - V_{\text{T}} - v_{\text{in}})}{C_2(V - V_{\text{T}})} \tag{7.67}$$

The maximum output frequency occurs for $v_{\text{in}} = 0$ resulting in $N = C_2/C_1$ (assuming that C_1 and C_2 are integer related. The minimum output frequency of zero occurs when V_{in} approaches its maximum value $(V - V_{\text{T}})$. Thus, the frequency range of the VCO is from 0 to $(C_1/C_2)f_{\text{c}}$.

7.7 Concluding Remarks

In this chapter, we considered MOS switched-capacitor, MOS switched-current, and continuous-time MOSFET-C sinusoidal oscillators. From the account given in this chapter, it appears that compared to the other variety of oscillators discussed in this monograph, not much work has been reported in the class of MOS-compatible oscillators. In fact, the considered opinion of the authors is that OTRA-based, ICCII/III-based, DDA-based, and DXCCII-based MOSFET-C oscillators evolved are not necessarily the best realizations in their respective class. Thus, there is ample scope for new ideas leading to better alternative circuits using the building blocks considered in this chapter.

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Chapter 8
Current-Controlled Sinusoidal Oscillators Using Current-Controllable Building Blocks

Abstract In many applications, one requires an oscillator or waveform generator whose frequency can be varied through either a single external voltage or an external current. In this chapter, a variety of such electronically controllable oscillators would be introduced using modern current-controllable building blocks. Electronically controllable quadrature and multiphase oscillators are also included.

8.1 Introduction

Voltage-controlled oscillators (VCO) are known to be important building blocks in several instrumentation, electronic, and communication systems, such as in function generators, in production of electronic music to generate variable tones, in phase-locked loops, and in frequency synthesizers. VCOs can be made in a number of ways, for instance, from single resistance-controlled oscillators by replacing the frequency controlling resistor by a JFET/MOSFET. In another method, two analog multipliers are embedded into a closed loop active RC circuit devised to function as an oscillator such that the imaginary part of the two complex conjugate roots of the close loop characteristic equation becomes a linear function of an external control voltage applied simultaneously to both the analog multipliers in the circuit, thereby resulting in a sinusoidal oscillator whose frequency is a linear function of this external control signal. Such linear VCOs using op-amps have been dealt with in Chap. 2 whereas those realizable with CFOAs have been discussed in Chap. 5 of this monograph.

Besides VCOs, researchers and circuit designers have also been interested in devising and using the so-called current-controlled oscillators (CCO)[1] which are almost as useful as their counterparts, the VCOs. A very popular method of designing CCOs is to employ operational transconductance amplifiers (OTA) and a variety of OTA-C oscillators have been dealt with in Chap. 3. All the OTA-based sinusoidal oscillators described in Chap. 3 are basically CCOs and many of them

[1] In conjunction with appropriate voltage-to-current converters, CCOs can always be converted into VCOs.

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are linear CCOs in the sense that their oscillation frequency is a linear function of some external control current (usually the external DC bias current of the OTAs).

This chapter is concerned with a specific type of CCOs which are made from those active circuit building blocks which have one or more number of its characteristic parameters electronically controllable. The most notable and most basic of such building blocks is the so-called second-generation controlled current conveyor (CCCII), which has a four-transistor mixed translinear cell (MTC) between its ports Y and X such that its characteristic equations are $i_y = 0$, $v_x = (v_y + R_x i_x)$ and $i_z = \pm i_x$ where the parasitic input resistance looking into the X-terminal of the CCCII is given by $R_x \cong V_{\mathrm{T}}/2I_{\mathrm{B}}$ and is therefore adjustable/tunable by the external bias current applied to the CCCII circuit. With CCCII, one can realize a number of electronically controllable function circuits such as current/voltage amplifiers, instrumentation amplifiers, sum/difference circuits, current controlled resistance, inductance and capacitance elements, and current-controlled filters, to name a few.

When CCCIIs are employed to devise sinusoidal oscillator configurations, the resulting circuit would have the oscillation frequency which would be electronically controllable. This applies whether the CCCIIs are realized by bipolar hardware or CMOS hardware; the only difference being in the nature of dependence of the R_x on the external DC bias current I_{B} in the two cases.

In the technical literature a large number of building blocks have been proposed which employ the MTC circuit at the front end in some form or the other for instance the CCI, CCIII, CDBA, OTRA, CFOA, and numerous others. There are also numerous other building blocks which employ an OTA as a sub-circuit such as CDIBA, VDTA, and others. Furthermore, there are yet other building blocks which employ a combination of MTC as well as OTA in some form or the other for instance CFTA, CCTA, and others, which also provide an electronically programmable parameter.

This chapter is concerned with the realization of CCOs using the type of building blocks elaborated in the preceding paragraph. A large number of CCOs have been proposed in the literature using the building blocks elaborated above [1–111]. This chapter presents some prominent CCOs realizable with the modern current mode building blocks enumerated above.

8.2 CCOs Using Second-Generation Controlled Current Conveyors (CCCII)

With the introduction of CCCII by Fabre, Saaid, Wiest, and Boucheron [1] and demonstration of its applicability in realizing electronically controllable BPF in [1, 2] and in realizing current-controlled grounded and floating resistors in [3, 4, 6, 7, 10, 12, 20, 34, 40], these building blocks soon started gaining attention for the realization of various linear and nonlinear analog signal processing and signal generation functions. In this section, we would outline the significant works done on the realization of sinusoidal oscillators using CCCII.

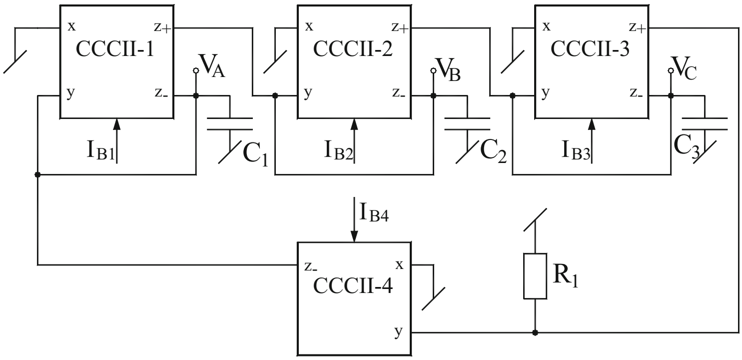


Fig. 8.1 Third-order sinusoidal oscillator proposed by Maheshwari and Verma [74]

A third-order electronically tunable oscillator providing non-interacting electronic tunability of FO, use of grounded passive components, and simultaneous availability of three sinusoidal voltage outputs was presented by Maheshwari and Verma [74] which is shown in Fig. 8.1. By straightforward analysis, the simplified equations for CO and FO are given by

$$\text{CO} : R_1 = 8R_{x4} \ \text{ and } \ \text{FO} : \omega_0 = \frac{\sqrt{3}}{R_x C} \tag{8.1}$$

where it has been assumed that $C_1 = C_2 = C_3 = C$, $R_{x1} = R_{x2} = R_{x3} = R_x$, i.e., $I_{B1} = I_{B2} = I_{B3} = I_B$. It is, therefore, seen from the above that CO is controlled by I_{B4} while FO is independently controllable by I_B provided that the corresponding CCCIIs are driven by a common bias current I_B which can be easily created by an appropriate current mirror arrangement [74].

The utility of this configuration was demonstrated in [74] by SPICE simulations using level 3 model of TSMC MOSIS 0.25 μm CMOS process parameters with a CMOS CCCII. It is shown that R_x is approximately given by $R_x = \frac{1}{\sqrt{8\mu C_{ox}(W/L)I_B}}$, where symbols have their usual meanings. With DC supply voltages as ±1.25 V, $C_1 = C_2 = C_3 = 20$ pF, $R_1 = 16$ kΩ, $I_{B1} = I_{B2} = I_{B3} = 40$ μA, and $I_{B4} = 13$ μA, the circuit produced oscillation frequency around 9.78 MHz against the designed frequency of 10 MHz, thereby giving 0.2 % error.

Yet another third-order translinear-C quadrature oscillator utilizing the same number of CCCIIs capable of providing four-phase current outputs was presented by Maheshwari and Khan [27]. This circuit is shown here in Fig. 8.2. The CO and FO for this circuit are given by

$$\text{CO} : R_{x1}C_1 = C_3R_{x4} \ \text{ and } \ \text{FO} : \omega_0 = \frac{1}{\sqrt{R_{x4}(R_{x2} + R_{x3})C_2C_3}} \tag{8.2}$$

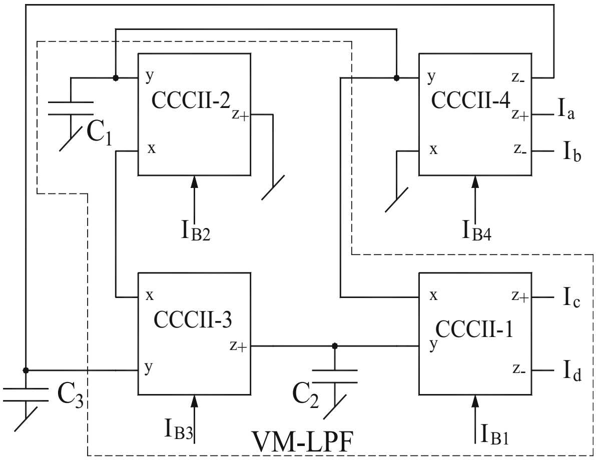


Fig. 8.2 Third-order sinusoidal oscillator proposed by Maheshwari and Khan [27]

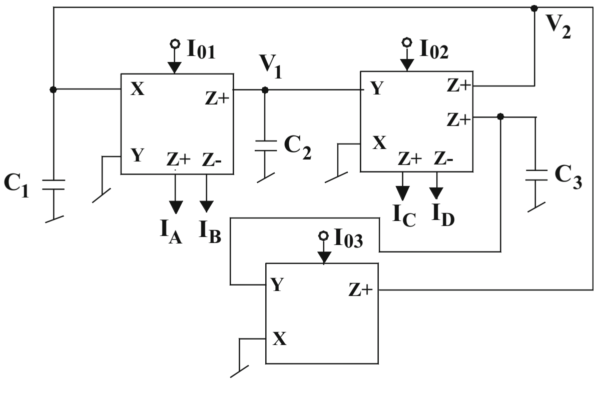


Fig. 8.3 Third-order quadrature oscillator proposed by Maheshwari [58]

It is, therefore, seen that FO can be controlled independent of CO by varying either I_{B2} or I_{B3} whereas CO can be adjusted through I_{B1}. The four current outputs are related by the following equations:

$$I_d = -I_c,\ I_c = jkI_a(\text{where } k = \omega_0 R_{x4}C_1),\ I_b = -I_a \tag{8.3}$$

A third-order QO realizable with only three CCCIIs, each possessing multiple outputs, was also proposed by Maheshwari [58] which is shown in Fig. 8.3. The characterizing equations for this circuit are given by

$$\text{CO}: R_{x1}C_1 = C_3R_{x3} \ \text{ and } \ \text{FO}: \omega_0 = \frac{1}{\sqrt{R_{x2}R_{x3}C_2C_3}} \tag{8.4}$$

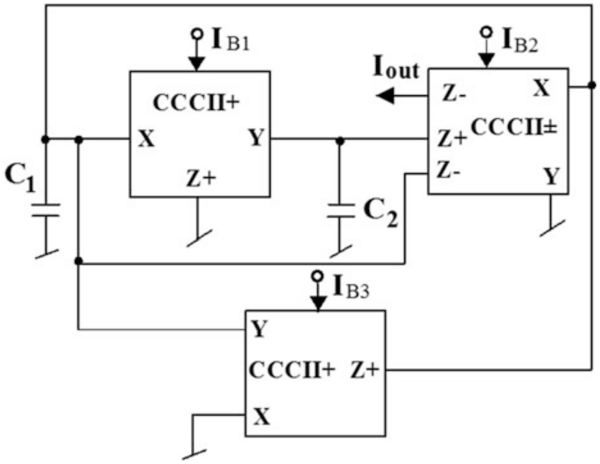


Fig. 8.4 CM sinusoidal oscillator proposed by Abuelma'atti and Tasadduq [14]

For $C_1 = C_2 = C_3 = C$, CO and FO become

$$\text{CO} : R_{x1} = R_{x3} \text{ and FO} : \omega_0 = \left(\frac{1}{C}\right)\frac{1}{\sqrt{R_{x2}R_{x3}}} \tag{8.5}$$

Thus, FO can be controlled by the biasing current I_{O2} and CO then can be established independently by I_{O1}. The quadrature current outputs and voltage outputs of this configuration are related by

$$I_A = jk_1 I_C, I_D = -I_C, I_B = -I_A; k_1 = \omega_0 R_{x2} C_2 \tag{8.6}$$

$$V_2 = -jk_2 V_1;\ k_2 = \omega_0 R_{x1} C_2 \tag{8.7}$$

Further, for $k_1 = 1$ and $k_2 = 1$, equal amplitudes of quadrature currents and quadrature voltages are ensured, respectively.

Another three CCCII-based CM oscillator using two GCs was proposed by Abuelma'atti and Tasadduq [14]. This circuit is shown in Fig. 8.4. This circuit provides independent control of CO and FO as is evident from the following:

$$\text{CO} : R_{x1} = R_{x3} \text{ and FO} : \omega_0 = \left(\frac{2}{V_T}\right)\sqrt{\frac{I_{B1} I_{B2}}{C_1 C_2}} \tag{8.8}$$

Using PNP and NPN transistors PR200N and NR200N and bipolar CCCII, the circuit has successfully generated a CM signal using $C_1 = C_2 = 2$ nF, $I_{B1} = I_{B2} = 100$ µA, $I_{B3} = 105$ µA. With I_{B2} varied from 0.25 µA to 250 µA, the frequency was found to be variable linearly over around three decades [14].

Possibly the simplest second-order electronically controlled oscillator is one proposed by Horng [22] shown in Fig. 8.5 which employs only two CCCIIs and two GCs. The CO and FO are given by

Fig. 8.5 CCCII-based oscillator proposed by Horng [22]

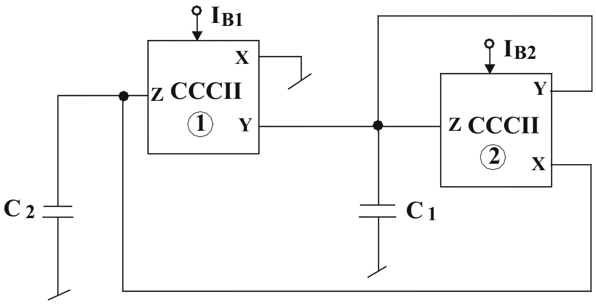
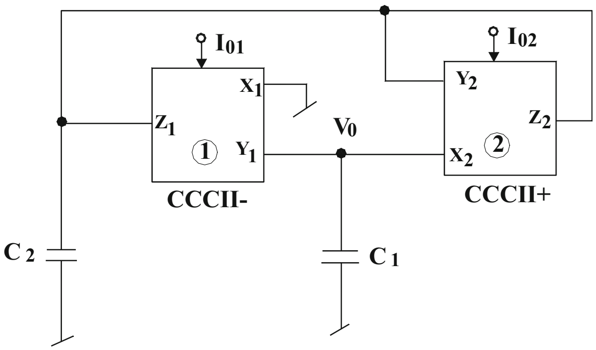


Fig. 8.6 CCCII-based oscillator proposed by Fongsamut, Anuntahirunrat, Kumwachara, and Surakampontorn [28]



$$\text{CO} : C_1 = C_2 \text{ and FO} : \omega_0 = \left(\frac{2}{V_T}\right)\sqrt{\frac{I_{B1}I_{B2}}{C_1C_2}} \tag{8.9}$$

A quite similar circuit using exactly the same number of active and passive components was proposed by Fongsamut, Anuntahirunrat, Kumwachara, and Surakampontorn [28] which is shown here in Fig. 8.6. It may be noted that the two circuits differ only in the manner in which the second CCCII (configured as an NIC) is employed between the two GCs. In view of this, it is, therefore, clear that the influence of parasitic impedances of CCIIs would be different in these circuits. In [28], the performance of both the circuits has been compared using CCCIIs realized from bipolar transistors PR100N and NR100N for PNP and NPN, respectively. The power supplies used were ±5 V DC with $I_{B1} = I_{B2} = I_B$ and $C_1 = 11$ nF and $C_2 = 10$ nF where the current I_B was varied from 0.1 to 100 μA. The variant of Fig. 8.6 [28] appears to have an edge in terms of high frequency performance over the Horng's circuit of Fig. 8.5 of [22].

Sotner-Hrubos-Slezak-Dostal [61] derived an adjustable oscillator based on two CCII− by removing the two external resistors connected to the x-terminal of each CCII− and incorporating R_{x1} and R_{x2} in their places. Since CMOS CCIIs have been employed in which case the parasitic impedances of X-terminals have been incorporated in their places, this results in a resistor-less variant of the circuit which is shown in Fig. 8.7. For further details the reader is referred to [61].

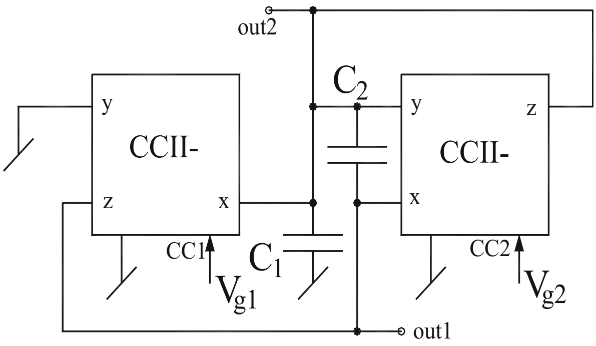


Fig. 8.7 Adjustable oscillator based on two CCII− proposed by Sotner-Hrubos-Slezak-Dostal [61]

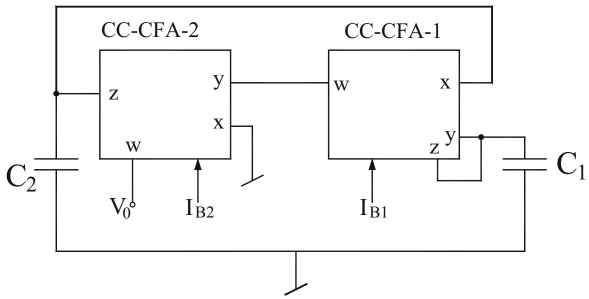


Fig. 8.8 CC-CFA-based oscillator proposed by Siripruchyanun-Chanapromma-Silapan-Jaikla [46]

8.3 CCOs Using CC-CFOAs and Their Variants

In the commercially available CFOAs such as AD 844, the non-ideal input resistance looking into terminal-X is given by $R_x \cong \frac{V_T}{2I_B}$. However, since the DC bias current I_B is fixed by the internal circuitry and cannot be changed, the electronic tunability of R_x cannot be put to any use. However, Siripruchyanun-Chanapromma-Silapan-Jaikla [46] have proposed a BiCMOS current-controlled CFA which makes it possible to vary R_x through an external bias current.

Using such current-controlled feedback amplifier (CC-CFA), they also proposed an oscillator structure employing only two CC-CFAs and two GCs which is shown here in Fig. 8.8. The CO and FO for this circuit are given by

$$\mathrm{CO} : C_1 = C_2 \text{ and FO} : \omega_0 = \frac{1}{\sqrt{R_{x1} R_{x2} C_1 C_2}} \tag{8.10}$$

Thus, FO can be controlled through the external bias currents.

Sotner-Herencsar-Jerabek-Dvorak-Kartci-Dostal-Vrba [87] proposed double current-controlled CFAs (DCC-CFA) which are characterized by the following terminal equations: $i_y = 0$, $V_x = \left(R_x i_x + V_y\right)$, $i_{z+} = Bi_x$, $i_{z-} = -Bi_x$, $V_{0+} = V_{z+}$,

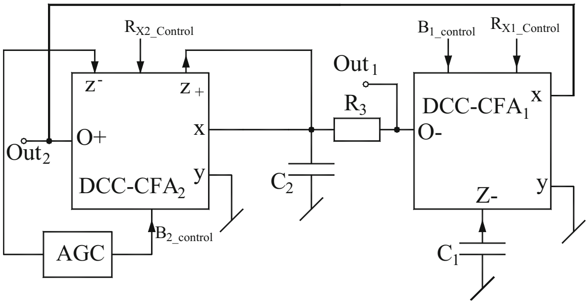


Fig. 8.9 DCC-CFA-based oscillator proposed by Sotner-Herencsar-Jerabek-Dvorak-Kartci-Dostal-Vrba [87]

$V_{0-} = V_{z-}$ with the provision that the intrinsic input resistance R_x is controlled by one external current and the current gain B between ports x and z is controlled by another bias current. Using DCC-CFA, the above-quoted authors have proposed a sinusoidal oscillator circuit which employs two DCC-CFAs, two GCs, and one external resistor while taking into account the intrinsic x-port resistance of both the DCC-CFAs. This oscillator circuit is shown in Fig. 8.9.

The CO and FO of this oscillator structure are given by

$$\text{CO} : B_2 \geq \frac{R_3 + (R_2 + R_{x2})}{R_3} \text{and FO} : \omega_0 = \sqrt{\frac{B_1}{\sqrt{R_3 C_1 C_2 (R_1 + Rx1)}}} \tag{8.11}$$

8.4 CCOs Using CC-CDBAs

Tangsrirat and Surakampontorn [55] exploited the fact that a translinear-based CDBA would have its intrinsic resistances at both P and N terminals controllable through respective external DC bias currents, thereby making it a current-controlled CDBA (CC-CDBA). Such CC-CDBA would, therefore, be characterized by $v_{\text{p}} = r_{\text{p}} i_{\text{p}}, v_{\text{n}} = r_{\text{n}} i_{\text{n}},\ i_z = \left(i_{\text{p}} - i_{\text{n}}\right)$ and $v_w = v_z$ where r_{p} and r_{n} are ideally zero, while, in fact, they are given by $r_{\text{p}} = r_{\text{n}} \cong \frac{V_{\text{T}}}{2I_{\text{B}}}$. Using three such CCCDBAs they presented a circuit to realize electronically tunable QO in which the only other components needed were two GCs placed at z-terminals of two CCCDBAs. This circuit shown in Fig. 8.10 is characterized by

$$\text{CO} : R_{x1} = R_{x3} \quad \text{and} \quad \text{FO} : \omega_0 = \frac{1}{\sqrt{R_{x2} R_{x3} C_1 C_2}} \tag{8.12}$$

For $C_1 = C_2 = C$, CO and FO become

$$\text{CO} :\ I_{\text{B1}} = I_{\text{B3}} \quad \text{and} \quad \text{FO} : \omega_0 = \left(\frac{2}{C V_{\text{T}}}\right) \sqrt{I_{\text{B3}} I_{\text{B2}}} \tag{8.13}$$

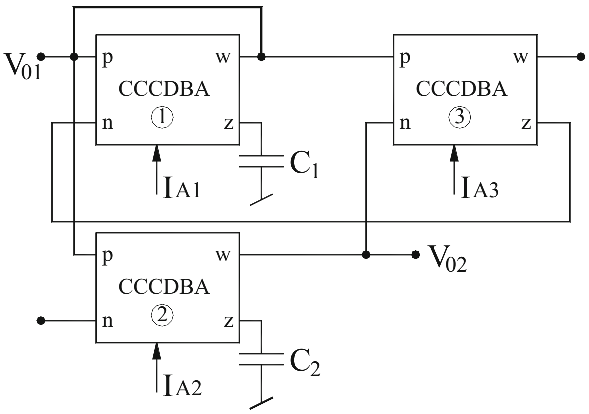


Fig. 8.10 CC-CDBA-based oscillator proposed by Tangsrirat and Surakampontorn [55]

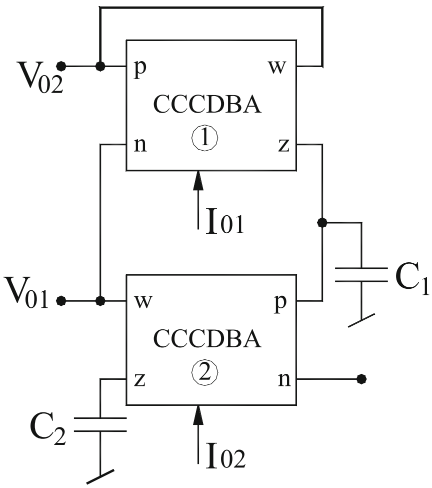


Fig. 8.11 Two CC-CDBA-based oscillator proposed by Tangsrirat [53]

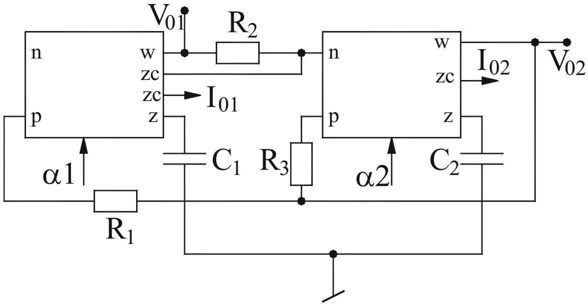
Subsequently, Tangsrirat [53], Khateb-Jaikla-Kubanek-Khatib [96], and Biolek-Lahiri-Jaikla-Siripruchyanun-Bajer [69] demonstrated that oscillators possessing independent controls and GCs can be made from only two CC-CDBAs also. The circuit presented by Tangsrirat [53] is shown in Fig. 8.11 and is characterized by

$$\text{CO}: R_{\text{n1}} = R_{\text{p2}} \quad \text{and} \quad \text{FO}: \omega_0 = \frac{1}{\sqrt{R_{\text{n1}} R_{\text{p2}} C_1 C_2}} \tag{8.14}$$

For $C_1 = C_2 = C$ and letting $I_{\text{O1}} = I_{\text{O2}} = I_{\text{O}}$, FO is given by

$$\text{FO}: \omega_0 = \frac{I_{\text{O}}}{\pi V_{\text{T}} C} \tag{8.15}$$

Fig. 8.12 Two ZC-CG-CDBA-based oscillator proposed by Biolek-Lahiri-Jaikla-Siripruchyanun-Bajer [69]



The two voltage outputs V_{01} and V_{02} are related by $\frac{V_{02}(s)}{V_{01}(s)} = sC_2R_{\mathrm{p}2}$, and thus the phase difference between V_{02} and V_{01} is 90°. Therefore, the two outputs are in quadrature.

Biolek-Lahiri-Jaikla-Siripruchyanun-Bajer [69] adopted a slightly different version of CDBA which was characterized by

$$v_{\mathrm{p}} = v_{\mathrm{n}} = 0, I_{\mathrm{zc}} = \left(I_{\mathrm{p}} - I_{\mathrm{n}}\right), I_z = \alpha I_{\mathrm{zc}} \text{ and } v_w = v_z \tag{8.16}$$

Due to these specified characteristics, they chose to call this a *Z*-copy current gain CDBA (ZC-CG-CDBA). Using this variant of CDBA, they proposed a current/voltage mode QO employing two such devices, two GCs, and also three resistors. This circuit is shown in Fig. 8.12.

The characterizing equations for this configuration are

$$\text{CO}: R_1 \geq R_3 \quad \text{and} \quad \text{FO}:\ \omega_0 = \sqrt{\frac{\alpha_1\alpha_2}{C_1C_2R_1R_2}} \tag{8.17}$$

Thus, CO and FO are independently tunable by R_3 and R_2, respectively. The two quadrature current outputs from high-output impedance ZC-terminals are

$$\frac{I_{02}(s)}{I_{01}(s)} = \frac{sC_2R_1}{\alpha_2} = j\sqrt{\frac{C_2R_1\alpha_1}{C_1R_2\alpha_2}} \tag{8.18}$$

and the two quadrature voltage outputs:

$$\frac{V_{02}(s)}{V_{01}(s)} = \frac{sC_1R_1}{\alpha_1} = j\sqrt{\frac{C_1R_1\alpha_2}{C_2R_2\alpha_1}} \tag{8.19}$$

Now for $C_1 = C_2 = C$, $R_1 = R_2 = R$, and $\alpha_1 = \alpha_2 = \alpha$, the expression for FO becomes

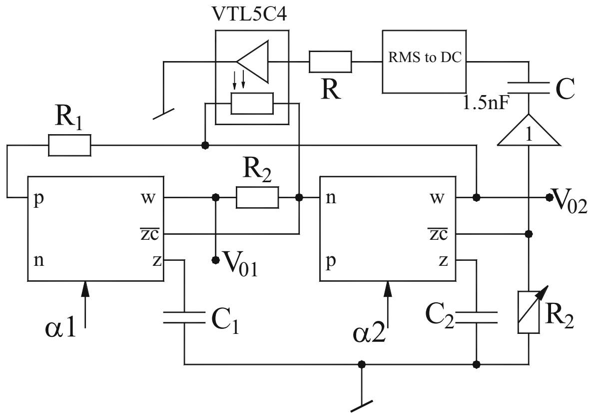


Fig. 8.13 AGC scheme employing opto-coupler VTL5C4 proposed by Biolek-Lahiri-Jaikla-Siripruchyanun-Bajer [69]

$$\omega_0 = \frac{\alpha}{RC} \quad \text{and} \quad \frac{I_{02}(s)}{I_{01}(s)} = \frac{V_{02}(s)}{V_{01}(s)} = j \tag{8.20}$$

Thus, FO is directly proportional to the current gain α and indirectly to C and R values with equal magnitudes of both quadrature currents and voltage signals. An interesting automatic gain control (AGC) scheme has also been incorporated as shown in Fig. 8.13.

In this circuit, the CO is accomplished through the opto-coupler VTL5C4 which contains a photo resistor whose resistance is controlled by a signal proportional to the amplitude of the generated waveform. The opto-coupler is excited from the output of a fast RMC-to-DC converter which gets its input from a potentiometer connected to ZC-terminal. This potentiometer also enables the adjustment of the amplitude of the generated sinewave, apart from acting as sensor for the current output.

Khateb-Jaikla-Kubanek-Khatib [96] presented a two-CC-CDBA-based QO shown in Fig. 8.14a which employs both GCs and also a grounded resistor. The CO and FO of this QO are given by

$$\text{CO} : R_1 = R \quad \text{and} \quad \text{FO} : \omega_0 = \frac{1}{\sqrt{R_1 R_{\text{p2}} C_1 C_2}} \tag{8.21}$$

Thus, CO can be controlled by R and FO by R_{p2}. The resistance R_{p2} is controllable by the bias current I_{B2} and hence FO is electronically controllable. The quadrature VM outputs are given by the following relationship:

$$\frac{V_{02}(s)}{V_{01}(s)} = -j\sqrt{\frac{C_1 R_1}{C_2 R_{\text{p2}}}} \tag{8.22}$$

The amplitudes of the output voltages are equal for $C_1 R_1 = C_2 R_2$.

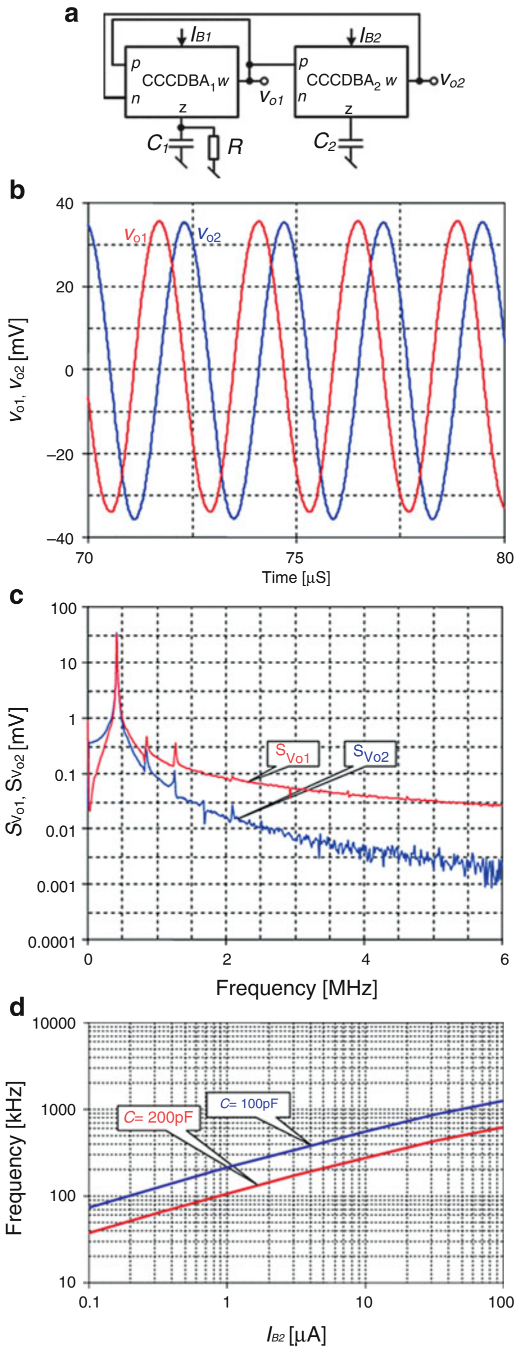


Fig. 8.14 (**a**) CC-CDBA-based QO proposed by Khateb-Jaikla-Kubanek-Khatib [96]. (**b**) Steady-state waveforms of the quadrature output voltages of Fig. 8.13a. (**c**) Spectra of the quadrature output voltages of Fig. 8.13a. (**d**) The oscillation frequency versus I_{B2} for two capacitance values of Fig. 8.13a

The authors demonstrated the workability of this QO using a CMOS structure of CC-CDBA in which case R_p and $R_n$are approximately equal to $R_p = R_n = \frac{1}{\sqrt{8\mu C_{ox}(W/L)I_B}}$. Using 0.18 μm n-well CMOS technology from TSMC, the workability of the circuit was verified [96] by SPICE simulations by taking $C_1 = C_2 = 100$ pF, $I_{B1} = I_{B2} = 5$ μA, and $R = 3.8$ kΩ. The steady-state waveforms of the quadrature outputs are shown in Fig. 8.14b. The spectra of the output oscillation frequency versus I_{B2} are shown in Fig. 8.14c and the variation of the oscillation frequency with respect to I_{B2}, for two capacitance values is shown in Fig. 8.14d. It was shown that this oscillator has minimum THD (varying between 1.2 and 2 %) between 200 and 400 kHz.

Lastly, it is worth mentioning that it has been demonstrated by Koksal-Sagbas-Sedef [44] that a minimum component oscillator can be realized using only a single CC-CDBA and two capacitors. However, such circuit does not have both GCs and also does not have non-interacting controls of CO and FO.

8.5 CCOs Using CC-CDTAs

The principle of current-controlled current differencing transconductance amplifier (CC-CDTA) was published in 2006 by Jaikla-Siripruchyanun [105]. Thus, as expected, a CC-CDTA consists of a translinear CMOS current differencing unit, a CMOS transconductance amplifier, and CMOS current repeater stage to produce multiple current outputs. Using a CMOS CC-CDTA, Jaikla-Prommee [72] presented a circuit configuration for realizing multiphase sinusoidal oscillator (MSO) with current outputs for even phase as well as for odd phase. These circuits are actually based upon a CC-CDTA-based CM all-pass filter (APF), a grounded capacitor, a grounded resistor which is implementable by a two MOSFET-based electronic resistor. The APF is shown in Fig. 8.15a whereas the proposed MSOs are shown in Fig. 8.15b, c.

Since a CMOS CC-CDTA was employed, it is useful to review various parameter values along with the terminal equations. Noting that CC-CDTA is characterized by $V_p = I_p R_p$, $V_n = I_n R_n$, $I_z = (I_p - I_n)$ and $I_x = \pm g_m V_z$, where R_p and R_n are given by

$R_p = R_n = \sqrt{\frac{1}{8\mu C_{ox}\left(\frac{W}{L}\right)I_{B1}}}$, whereas the value of $g_m = \sqrt{\mu C_{ox}\left(\frac{W}{L}\right)I_{B2}}$, it may also be noted that the transresistance $R_k = \frac{1}{2\mu C_{ox}\left(\frac{W}{L}\right)(V_{DD} - V_T)}$. The first-order APF shown in Fig. 8.15a has the transfer function given by

$$\frac{I_0(s)}{I_{in}(s)} = g_m R_k \left[\frac{s - \frac{1}{R_n C}}{s + \frac{1}{R_n C}}\right] \tag{8.23}$$

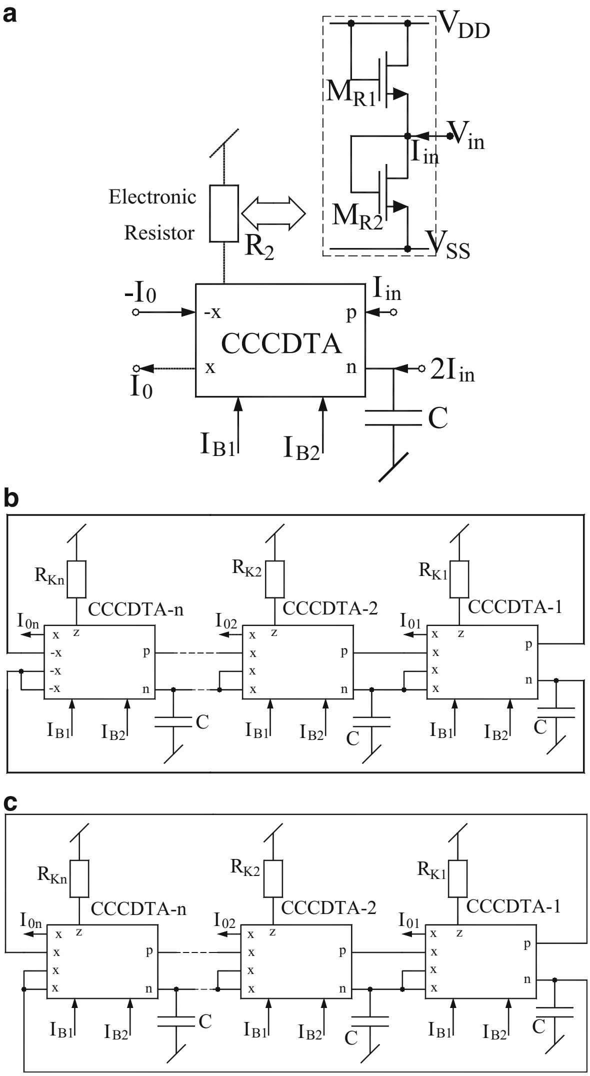


Fig. 8.15 (**a**) CC-CDTA-based APF with grounded capacitor [72]. (**b**) CM MSO for even phase proposed by Jaikla and Prommee [72]. (**c**) CM MSO for odd phase proposed by Jaikla and Prommee [72]

The CO and FO are expressed as

$$\text{CO}: g_{\mathrm{m}} R_{\mathrm{k}i} = 1; i = 1, 2, \ldots, n \text{ and FO}: \omega_0 = \frac{1}{CR_{\mathrm{n}}} \tan\left(\frac{\pi}{2n}\right) \tag{8.24}$$

Sakul-Jaikla-Dejhan [71] and subsequently Jaikla-Lahiri [106] demonstrated that two CC-CDTAs and two GCs are sufficient to realize CM QOs with explicit current outputs if multiple copies of *x*-terminals are made available in the design of CC-CDTAs. The QO circuits proposed by Sakul-Jaikla-Dejhan [71] are shown in

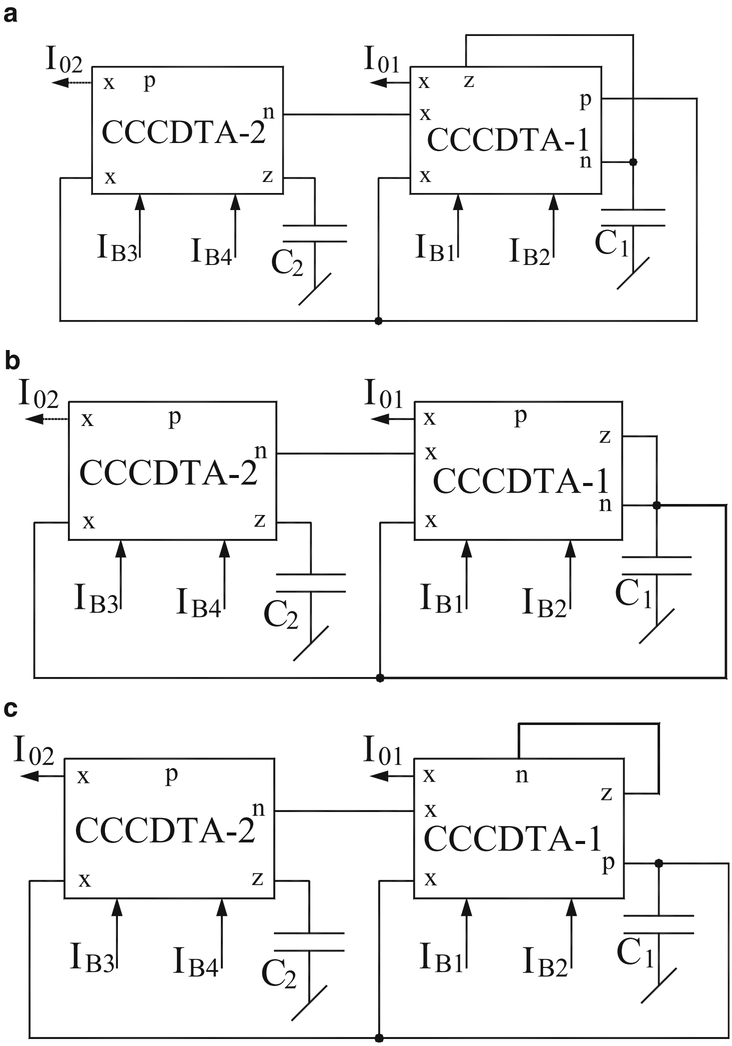


Fig. 8.16 CM QOs proposed by Sakul-Jaikla-Dejhan [71]

Fig. 8.16. Using bipolar CC-CDTAs which have $R_{\mathrm{p}} = R_{\mathrm{n}} \cong \frac{V_{\mathrm{T}}}{2I_{\mathrm{B1}}}$ and $g_{\mathrm{m}} \cong \frac{I_{\mathrm{B2}}}{2V_{\mathrm{T}}}$, all the three circuits of Fig. 8.16 have the following CO: $g_{\mathrm{m1}}R_{\mathrm{n1}} = 2$ while the FO for the circuits of a and b $\omega_0 = \sqrt{\frac{g_{\mathrm{m1}}g_{\mathrm{m2}}}{C_1C_2}}$ and that for the circuit of c is $\omega_0 = \sqrt{\frac{g_{\mathrm{m1}}g_{\mathrm{m2}}}{2C_1C_2}}$. The independence of controllability of CO and FO is obtainable that of the former by R_{n1} and hence by I_{B1}; the latter is then controllable, independently by g_{m2}, and hence by I_{B4}. It has been demonstrated [71] that this independence of tunability remains unaffected even if the input resistances R_{p} and R_{n} are not exactly equal.

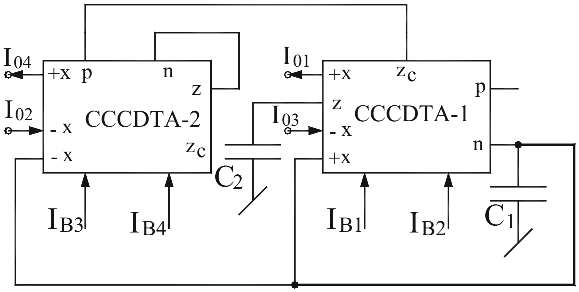


Fig. 8.17 CM QO proposed by Jaikla-Lahiri [106]

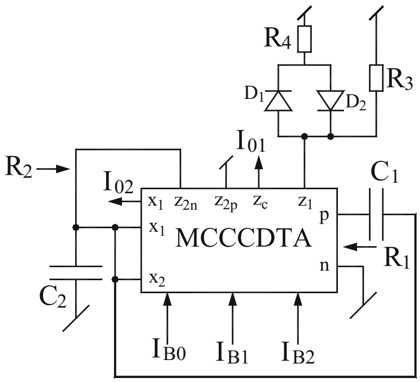


Fig. 8.18 CM QOs proposed by Li [82]

Another two CC-CDTA-based CM QO with both GCs but providing four explicit current outputs was presented by Jaikla-Lahiri [106] and is shown in Fig. 8.17. This circuit is characterized by

$$\mathrm{CO}: g_{\mathrm{m}2}R_2 \geq 2 \text{ and } \mathrm{FO}: \omega_0 = \sqrt{\frac{g_{\mathrm{m}1}}{R_{\mathrm{n}1}C_1C_2}} \tag{8.25}$$

Assuming bipolar CC-CDTA, for which $R_{\mathrm{p}} = R_{\mathrm{n}} \cong \frac{V_{\mathrm{T}}}{2I_{\mathrm{B}1}}$ and $g_{\mathrm{m}} \cong \frac{I_{\mathrm{B}2}}{2V_{\mathrm{T}}}$, these equations reduced to CO: $I_{\mathrm{B}4} = \ \geq 8I_{\mathrm{B}3}$ and FO: $\omega_0 = \frac{1}{V_{\mathrm{T}}}\sqrt{\frac{I_{\mathrm{B}1}I_{\mathrm{B}2}}{\sqrt{C_1C_2}}}$ which shows that CO and FO are fully uncoupled and can be controlled independently.

Subsequently, a number of researchers have demonstrated that oscillators with several interesting properties can be devised with a minimum number of one CC-CDTA and two capacitors. In the following, we present a number of interesting designs.

Li [82] presented a modified current-controlled CDTA (MCC-CDTA) and used it to devise a single MCC-CDTA-based oscillator. It is interesting to note that although the classical Wien-bridge oscillator does not have the independence of controlling the oscillation frequency without affecting the CO, the MCC-CDTA-based version does provide these desirable properties. This circuit is shown in Fig. 8.18, where MCC-CDTA is characterized by

$$V_{\mathrm{p}} = I_{\mathrm{p}}R_{\mathrm{p}},\ V_{\mathrm{n}} = I_{\mathrm{n}}R_{\mathrm{n}},\ I_Z = I_{ZC} = \left(I_{\mathrm{p}} - I_{\mathrm{n}}\right),\ I_{X1} = g_{\mathrm{m1}}V_{Z1},\ \text{and}\ I_{X2} = g_{\mathrm{m2}}\left(V_{Z2\mathrm{p}} - V_{Z2\mathrm{n}}\right) \tag{8.26}$$

where $R_{\mathrm{p}} = R_{\mathrm{n}} = R = \dfrac{V_{\mathrm{T}}}{2I_{\mathrm{B0}}}$ and the transconductance gains of two different OTAs in MCDTA, g_{m1} and g_{m2}, are $g_{\mathrm{m1}} \cong \dfrac{I_{\mathrm{B1}}}{2V_{\mathrm{T}}}$, $g_{\mathrm{m2}} \cong \dfrac{I_{\mathrm{B2}}}{2V_{\mathrm{T}}}$, where I_{B1} and I_{B2} are the DC biasing currents of the MCC-CDTA and V_{T} is the thermal voltage. Now, noting that $R_1 = R_{\mathrm{p}} \cong \dfrac{V_{\mathrm{T}}}{2I_{\mathrm{B0}}}$, $R_2 \cong \dfrac{1}{g_{\mathrm{m2}}} = \dfrac{2V_{\mathrm{T}}}{I_{\mathrm{B2}}}$ and $A_{\mathrm{i}} \cong \dfrac{I_{\mathrm{B1}}}{2V_{\mathrm{T}}}R_{\mathrm{eq}}$, where R_{eq} is the equivalent resistance for the diode-resistor network and A_{i} is the current gain which for a normal Wien-bridge oscillator having a series R_1C_1 and a parallel branch R_2C_2 along with a current amplifier with gain A_{i} could be given by

$$A_{\mathrm{i}} \geq \left(1 + \frac{R_1}{R_2} + \frac{C_2}{C_1}\right) \tag{8.27}$$

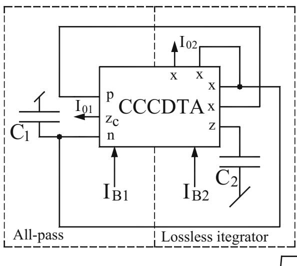
Now, taking $C_1 = C_2 = C$, $R_1 = R_2$ (which can be achieved by setting $I_{\mathrm{B}} = I_{\mathrm{B2}} = 4I_{\mathrm{B0}}$), it is found that the CO, FO, and the current transfer function are given by

$$\mathrm{CO}: I_{\mathrm{B1}} \geq \frac{6V_{\mathrm{T}}}{R_{\mathrm{eq}}},\ \mathrm{FO}: \omega_0 = \frac{1}{V_{\mathrm{T}}}\sqrt{\frac{I_{\mathrm{B0}}I_{\mathrm{B2}}}{C_1C_2}} = \frac{I_{\mathrm{B}}}{2V_{\mathrm{T}}C}\ \text{ and }\ \frac{I_{01}}{I_{02}} = \frac{1}{3} \tag{8.28}$$

Therefore, it is seen that FO is linearly tunable with I_{B} whereas CO is independently controllable by I_{B1}.

The oscillator circuit of Chien-Wang [85], although providing both voltage and current mode outputs, has independent controls of both CO and FO but does not have both GCs. On the other hand, Keawon-Jaikla [107] and Jie [84] proposed single CC-CDTA-based CM QOs as shown in Figs. 8.19 and 8.20, respectively; however, none of these circuits have independence of CO and FO.

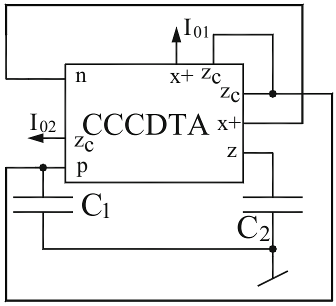
Fig. 8.19 Single CC-CDTA-based CM QOs proposed by Keawon-Jaikla [107]



$$\mathrm{CO}: \frac{C_2}{C_1} = g_m R_n \ \text{ and FO: }\ \omega_0 = \sqrt{\frac{g_m}{R_n C_1 C_2}}$$

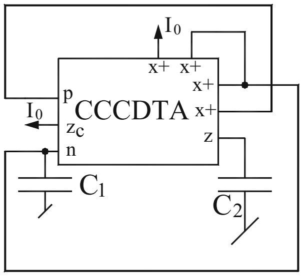
Fig. 8.20 Single CC-CDTA-based CM QOs proposed by Jie [84]

a



b

$$\text{CO: } \frac{g_m}{C_2} = \frac{1}{R_p C_1} \text{ and FO: } \omega_0 = \sqrt{\frac{g_m}{R_p C_1 C_2}}$$



$$\text{CO: } \frac{g_m}{C_2} = \frac{1}{R_n C_1} \text{ and FO: } \omega_0 = \sqrt{\frac{g_m}{R_n C_1 C_2}}$$

It is, therefore, clear that any single CC-CDTA oscillator with both GCs and providing two explicit current outputs in quadrature and also providing non-interacting control of CO and FO is not known so far. This, therefore, constitutes an interesting problem for further investigation.

8.6 CCOs Using CC-CCTAs

A current-controlled CCTA is essentially a composite building block consisting of current-controlled current conveyor (CC-CC) and a transconductance amplifier (TA) and can be characterized by the terminal equations:

$$I_y = 0,\ V_x = \left(V_y + I_x R_x\right), I_z = I_x \quad \text{and} \quad I_0 = \pm g_{\mathrm{m}} V_z \tag{8.29}$$

In some applications, the availability of Z-copy current, i.e., $I_{ZC} = I_Z$, may lead to some additional flexibility and degree of freedom. When implemented with bipolar circuit structure the resistance R_x would be given by

$$R_x \cong \frac{V_T}{2I_{B1}} \text{ whereas } g_m \cong \frac{I_{B2}}{2V_T} \tag{8.30}$$

The applications of CC-CCTA have been investigated in several publications such as [47, 68, 99]. In this section, we highlight some prominent configurations employing CC-CCTAs as active elements. Sa-Ngiamvibool-Jantakun [99] proposed two CM QOs containing both GCs providing non-interacting controls. Figure 8.21a is characterized by the following CO and FO:

$$\text{CO}: g_{m1}R_{x1} = 2 \quad \text{and} \quad \text{FO}: \omega_0 = \sqrt{\frac{g_{m1}}{R_{x2}C_1C_2}} \tag{8.31}$$

On the substitution of parasitic resistance R_x and transconductance g_m as given by Eq. (8.30), the expressions for CO and FO become

$$\text{CO}: 8I_{B1} = I_{B2},\ \text{FO}: \omega_0 = \frac{1}{V_T}\sqrt{\frac{I_{B3}I_{B2}}{C_1C_2}} \tag{8.32}$$

From the above it is clear that CO and FO can be electronically controlled independently; that is, CO can be controlled electronically by I_{B1} without disturbing FO which can be adjusted by I_{B3}. The transfer function between output currents can be expressed as

$$\frac{I_{02}(s)}{I_{01}(s)} = \frac{1}{sC_2R_{x2}} \tag{8.33}$$

Similarly, for the circuit of Fig. 8.21b, the CO and FO are obtained as

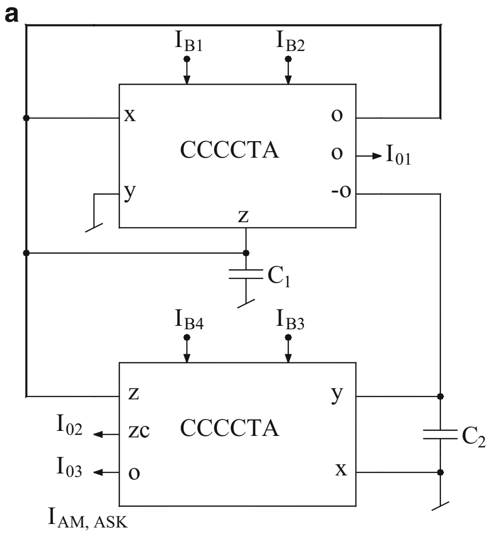
$$\text{CO}: R_{x1} = R_{x2} \quad \text{and} \quad \text{FO}: \omega_0 = \sqrt{\frac{g_{m1}}{R_{x1}C_1C_2}} \tag{8.34}$$

This can be expressed in terms of biasing currents as

$$\text{CO}: I_{B1} = I_{B3} \quad \text{and} \quad \text{FO}: \omega_0 = \frac{1}{V_T}\sqrt{\frac{I_{B2}I_{B3}}{C_1C_2}} \tag{8.35}$$

Thus, CO can be controlled by I_{B1} and FO is controllable through I_{B2} without disturbing CO. The explicit output current ratio is given by

$$\frac{I_{02}(s)}{I_{01}(s)} = \frac{sC_2R_{x1}}{g_{m1}R_{x2}} = \frac{C_2}{g_{m1}}e^{j\frac{\pi}{2}} \tag{8.36}$$



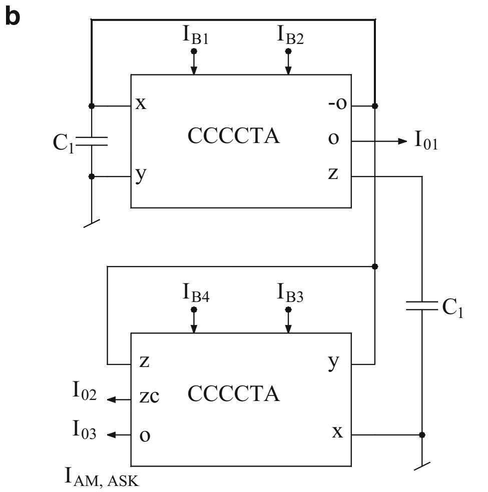


Fig. 8.21 Two CC-CCTA-based CM QOs proposed by Sa-Ngiamvibool-Jantakun [99]

It has also been demonstrated by Siripruchyanun-Jaikla in [47] and Jaikla-Siripruchyanun-Lahiri in [68] that a single CC-CCTA is sufficient to realize a GC oscillator with non-interacting controls. The oscillator circuit proposed by Siripruchyanun-Jaikla in [47] is shown in Fig. 8.22 and is characterized by

$$\text{CO} : C_1 = C_2 \;\; \text{and} \;\; \text{FO} : \omega_0 = \sqrt{\frac{g_\mathrm{m}}{R_x C_1 C_2}} = \frac{2}{V_\mathrm{T}}\sqrt{\frac{I_{\mathrm{B2}} I_{\mathrm{B1}}}{C_1 C_2}} \tag{8.37}$$

From the above, the non-interacting controls are evident.

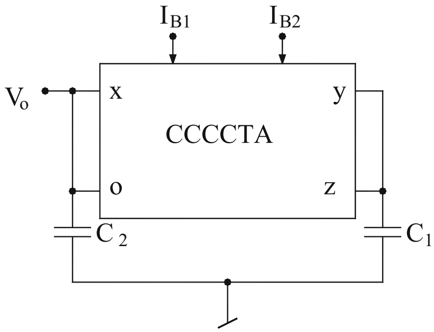


Fig. 8.22 Single CC-CCTA-based oscillator proposed by Siripruchyanun and Jaikla [47]

Jaikla-Siripruchyanun-Lahiri in [68] formulated a differential voltage current-controlled conveyor transconductance amplifier (DV-CCCTA) which is characterized by

$$I_{y1} = I_{y2} = 0, V_x = \left\{\left(V_{y1} - V_{y2}\right) + I_x R_x\right)\}, I_z = I_x, I_{-01} = -g_{\mathrm{m1}} V_z, I_{02} = g_{\mathrm{m2}} V_z \tag{8.38}$$

Since it has two parameters which are electronically controllable, namely R_x of the front end CC and the transconductance gain g_{m}, it follows that any circuit built around this building block can, therefore, take advantage of reducing two external resistors and using in their place *implicitly* R_x and $1/g_{\mathrm{m}}$ as electronically variable resistors.

A QO using a single DV-CCCTA and two GCs has two O-terminals, two –O terminals, and two Z-terminals, proposed by Jaikla-Siripruchyanun-Lahiri [68] shown in Fig. 8.23a.

It may be noted that since a CMOS DV-CCCTA has been employed the parameters R_{x}, g_{m1}, and g_{m2} are given by

$$R_x = \sqrt{\frac{1}{8\mu C_{\mathrm{ox}}\left(\frac{W}{L}\right) I_{B1}}}, g_{\mathrm{m1}} = \sqrt{\mu C_{\mathrm{ox}}\left(\frac{W}{L}\right) I_{\mathrm{B2}}}, g_{\mathrm{m2}} = \sqrt{\mu C_{\mathrm{ox}}\left(\frac{W}{L}\right) I_{\mathrm{B3}}} \tag{8.39}$$

An analysis of this circuit shows that CO and FO are given by

$$\text{CO} : g_{m1} R_x \leq 1 \text{ or } k_2 I_{\mathrm{B2}} < 8 k_1 I_{\mathrm{B1}} \text{ and}$$

$$\text{FO} : f_0 = \frac{1}{2\pi}\sqrt{\frac{g_{\mathrm{m2}}}{R_x C_1 C_2} = \left(\frac{1}{\pi}\right)\frac{\left(8 k_1 k_3 I_{\mathrm{B1}} I_{\mathrm{B3}}\right)^{1/4}}{\left(4 C_1 C_2\right)^{1/2}}} \tag{8.40}$$

Hence, it is seen that CO and FO are independently electronically variable by I_{B2} (g_{m1}) and I_{B3} (g_{m2}), respectively. The various output voltages and currents are related by the following expressions:

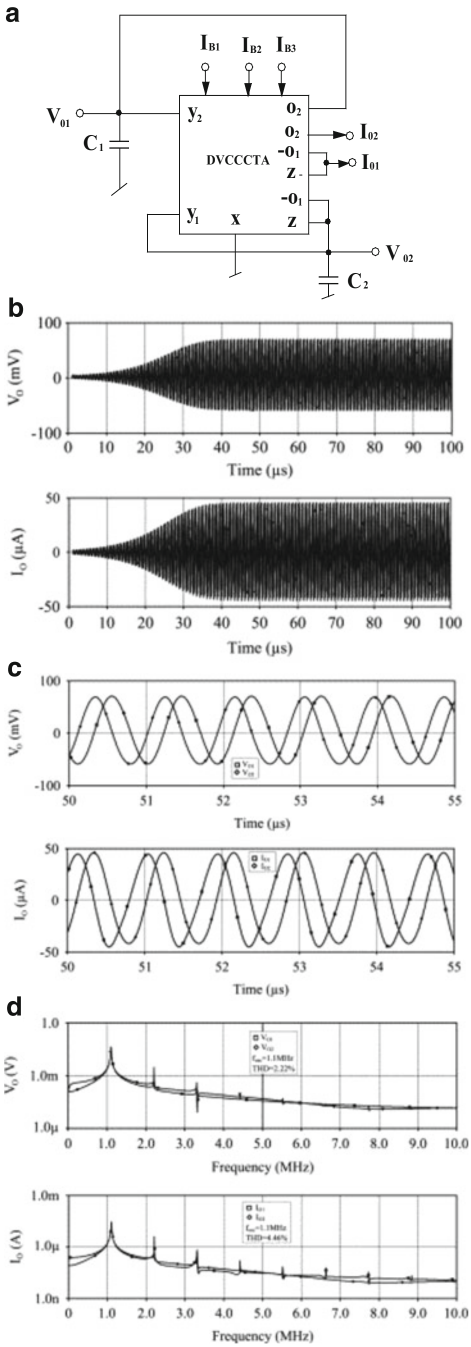


Fig. 8.23 DVCCCTA-based QO proposed by Jaikla, Siripruchyanun, and Lahiri [68] (**a**) Circuit configuration. (**b**) The growing oscillation of quadrature oscillations for VM and CM. (**c**) Steady-state waveforms for the quadrature VM and CM outputs. (**d**) The frequency spectrums for VM and CM outputs

$$V_{02} = j\frac{\omega_0 C_1}{g_{m2}} V_{01} \tag{8.41}$$

$$I_{02} = j\frac{\omega_0 C_2}{g_{m2}} I_{01} \tag{8.42}$$

Thus, from the above equations, it is clear that both voltage output signals and current output signals are in quadrature. It is worth mentioning that the circuit of Fig. 8.23a is inspired from a classical CFOA-based SRCO proposed by Senani and Singh (reference [13] of Chap. 5).

Other than the use of both GCs and non-interacting controls, the circuit provides two explicit CM quadrature outputs as well as two quadrature voltage outputs. However in case of latter two VFs will be needed to make these voltages available from a low-output impedance terminals.

The workability of the circuit of Fig. 8.23a was verified [68] by a CMOS DVCCCTA. The QO was simulated in SPICE using 0.25 μm TSMS CMOS technology. The component values were chosen to give oscillation frequency of 1.1 MHz. Figure 8.23b shows the building up of the oscillations; Fig. 8.23c shows the steady-state waveforms for the CM and VM outputs while Fig. 8.23d shows the frequency spectra of the CM and VM output waveforms. The THD was found to be 3 % and 5 %, respectively, for VM and CM outputs. These results confirm the practical workability of the proposed QO circuit of Fig. 8.23a.

Other than the abovementioned current-controlled building blocks, a number of authors have used either alternative building blocks such as current-controlled current backward TA (CCCBTA) [80], *Z*-copy current gain voltage differencing current conveyor (ZC-CG-VDCC) [108], current-controlled current differencing current copy conveyor (CC-CD-CCC) [102], current gain-controlled CCTA (CG-CCCTA) [70], and *Z*-copy current-controlled current inverting TA (ZC-CCCITA) [77] on one hand and combination of two or more different building blocks such as [89, 97, 104] to realize oscillators of possessing different kind of properties, e.g., employment of GCs and non-interacting controls.

8.7 Concluding Remarks

In this chapter we had described some important contributions made in the area of electronically controllable sinusoidal oscillators using various current-controlled building blocks. A number of configurations were elaborated which the various researchers have devised in order to meet as many as possible of the following desirable properties: (a) use of minimum number of active elements (preferably one), (b) employment of both GCs, (c) completely independent non-interacting electronic control of both CO and FO, (d) quadrature current outputs from high-output impedance node, (e) quadrature voltage output from a low-impedance node, (f) least errors because of non-ideal parasitic impedances/non-ideal parameters of the active building block employed, and (g) highest possible operation frequency response.

A number of two active element-based oscillators as well as single active element-based oscillators have been highlighted in this chapter which appear to achieve several of the desirable properties mentioned above. However, till date no single element-based circuit is known to have been discovered which is capable of meeting all the abovementioned seven objectives, though the circuits of Figs. 8.22 and 8.23a appear to be quite close to the intended objectives. This, therefore, appears to be a worthwhile problem which is open to investigation.

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Chapter 9
Bipolar and CMOS Translinear, Log-Domain, and Square-Root Domain Sinusoidal Oscillators

Abstract Translinear, log-domain, and square-root domain analog signal processing circuits potentially offer high-frequency operation and extended dynamic range at low power supply voltages and have, therefore, been widely investigated in literature due to being also very promising for implementation in IC technology because of employing only transistors (BJTs/MOSFETs) and very low-valued capacitors (having values of the order of a few pF) and providing electronic tunability of the parameters of the realized functional circuits. In this chapter, we discuss some significant contributions made in the realization of fully integratable sinusoidal oscillators evolved during the past two decades using translinear, log-domain, and square-root domain techniques and circuits.

9.1 Introduction

Due to ever decreasing values of DC biasing power supply in modern integrated circuits and systems, analog designers have been faced with a new challenge of designing analog signal processing and signal generating circuits capable of operating from low DC power supply voltages. In BJT technology, a promising approach to meet this challenge was provided by the class of filters known as *translinear* filters made from *dynamic* translinear circuits. Such circuits exploit the exponential characteristics of the BJT between its collector current and base-to-emitter voltage, namely, $I_c = I_{co}$ exp (V_{BE}/V_T) where symbols have their usual meanings, to create circuits which are internally nonlinear but externally linear (ELIN). In CMOS technology, translinear circuits are built using MOSFETs operating in subthreshold or weak-inversion mode wherein the drain current is exponentially related to the gate-to-source voltage. On the other hand, when the MOSFETs are biased in strong inversion region, the drain current of the MOSFET is governed by the so-called square law, i.e., $I_D = k\ (V_{GS} - V_{TH})^2$ where $k = \mu_s C_{ox}$ $(W/2L)$. This forms the basis for the MOS translinear circuits or the dynamic square-root domain circuits.

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While a lot of work has been published on the design of translinear,[1] log-domain, and square-root domain integrators, filters, and other analog signal processing circuits, see [1–26] and the references cited therein, comparatively only a limited number of publications [1–8, 16, 19, 22] have dealt with the synthesis of sinusoidal oscillators using these techniques. The purpose of this chapter is to highlight the significant contributions made in the design of low-voltage, low-power, high-frequency, low-distortion sine-wave generators using translinear, log-domain, and square-root domain circuits.

9.2 Log-Domain Oscillators

Several researchers have developed sinusoidal oscillator topologies using log-domain circuits. In this section, we discuss some prominent topologies of log-domain oscillators from those available in [1–8].

Thanachayanont, Pookaiyaudom, and Toumazou [2] presented a state-space technique of synthesizing a log-domain oscillator circuit from its open-loop transfer function. Considering the following open-loop transfer function

$$T(s) = \frac{\omega_0^2}{s^2 + \omega_0^2} \tag{9.1}$$

where ω_0 is the frequency of oscillation, its state-variable representation can be written as

$$\dot{X}_1 = \omega_0 X_2 - \omega_0 X_1 \tag{9.2a}$$

$$\dot{X}_2 = \omega_0 X_2 - 2\omega_0 X_1 + \omega_0 u \tag{9.2b}$$

$$Y = X_1 \tag{9.2c}$$

According to the methodology developed by Frey [15], the above state equations can be transformed into a set of nodal equations by using exponential mappings on the inputs and the state variables.

Let us now apply the following mappings to obtain the required node equations:

$$X_1 = I_0 \exp(V_1/V_{\mathrm{T}}) \tag{9.3a}$$

[1] Though there have been a number of works published on the so-called *translinear*-C oscillators such as [14], the circuits therein are based upon the electronic tunability of the x-port input resistance of the second-generation current-controlled conveyor (CCCII) whose internal circuit consists of a mixed-*translinear* cell; such circuits are already described earlier in Chap. 8 of this monograph.

$$X_2 = I_s \exp(V_2/V_T) \tag{9.3b}$$

$$u = \left(I_s^2/I_0\right) \exp(V_{01}/V_T) \tag{9.3c}$$

where I_s and V_T are the reverse saturation current and the thermal voltage corresponding to the forward biased $V-I$ characteristics of the BJT.

After some mathematical rearrangements, the following node equations can be written:

$$C_1\dot{V}_1 = I_s\exp[(V_2 - V_1)/V_T] \tag{9.4a}$$

$$C_2\dot{V}_2 = I_0 - 2\frac{I_0^2}{I_s}\exp[(V_1 - V_2)/V_T] + I_0 \exp[(V_{01} - V_2)/V_T] \tag{9.4b}$$

$$Y = X_1 = I_0 \exp(V_1/V_T) \tag{9.4c}$$

where $I_0 = C\omega_0 V_T$.

The required oscillator can be synthesized by observing that the left-hand sides of the Eqs. (9.4a) and (9.4b) represent the currents flowing into two grounded capacitors C_1 and C_2, whereas to create the right-hand sides of these equations, specific BJT circuit implementations are needed. For example, in Eq. (9.4a), the last term I_0 can be realized by a constant current source, whereas the exponential terms in all the three equations with positive coefficient of I_s can be realized by NPN transistors whose base and emitter are connected to the respective first and second voltages in the argument of the exponential functions. The output term of the Eq. (9.4c) can be rewritten as

$$Y = I_0\exp(V_1/V_T) = I_s\exp[(V_1 + V_0)/V_T] \tag{9.5}$$

where $I_0 = I_s \exp(V_0/V_T)$. Therefore, it is readily visualized that the output current Y can be created by extracting the current in a forward biased NPN transistor whose base-emitter voltage has been made equal to $(V_1 + V_0)$. This can be implemented by the sub-circuit in Fig. 9.1a.

On the other hand, the second term of Eq. (9.4b) can be appropriately realized by the sub-circuit in Fig. 9.1b. This circuit can be analyzed as follows:

Applying translinear (TL) principle in Fig. 9.1b, it can be easily determined that

$$I_d = 2\left(\frac{I_0^2}{I_s}\right)\exp[(V_1 - V_2)/V_T] = 2\frac{I_0^2}{I_c} \tag{9.6a}$$

where

$$I_c = I_s\exp[(V_2 - V_1)/V_T] \tag{9.6b}$$

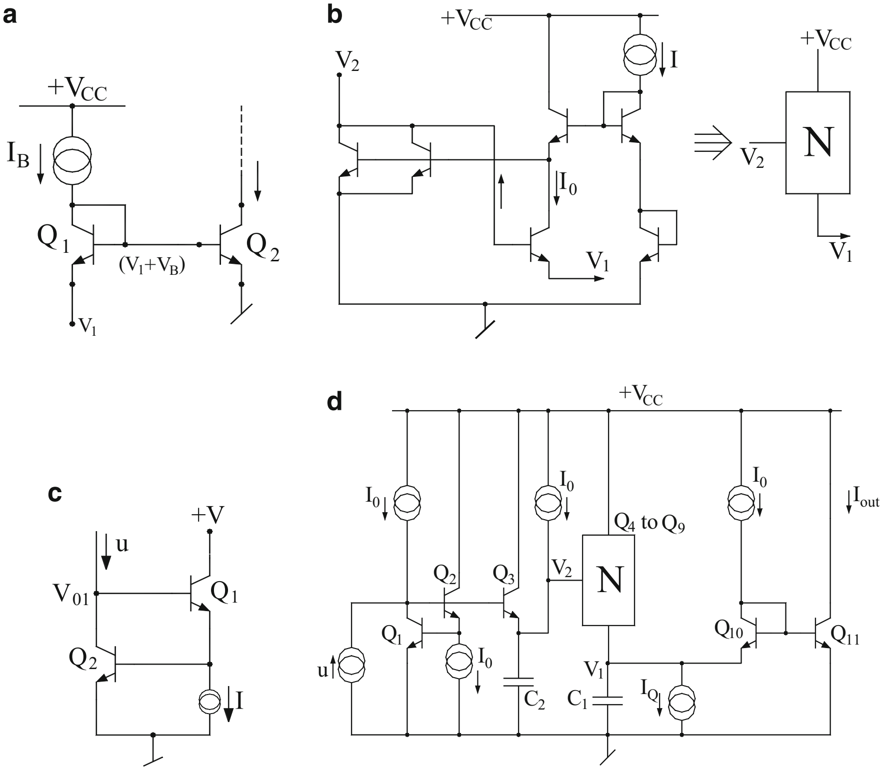

Fig. 9.1 Synthesis of log-domain oscillator as per the methodology proposed by Thanachayanont, Pookaiyaudom, and Toumazou [2]. (**a**) The circuit for realizing Eq. (9.5). (**b**) The circuit for realizing the second term of Eq. (9.4b). (**c**) The circuit for realizing Eq. (9.9). (**d**) The final log-domain oscillator circuit

Therefore,

$$V_{\mathrm{T}}\ln\left(\frac{I_{\mathrm{d}}/2}{I_{\mathrm{s}}}\right) = 2V_{\mathrm{T}}\ln\left(\frac{I_0}{I_{\mathrm{s}}}\right) - V_{\mathrm{T}}\ln\left(\frac{I_{\mathrm{c}}}{I_{\mathrm{s}}}\right) \tag{9.7}$$

The current I_{d} flows out of the node V_2 thereby ensuring negative sign and the input voltage is realized from the input current mapping:

$$u = \frac{I_{\mathrm{s}}^2}{I_0}\exp(V_{01}/V_{\mathrm{T}}) \tag{9.8}$$

Thereby yielding

$$V_{01} = V_{\mathrm{T}}\ln\left(uI_0/I_{\mathrm{s}}^2\right) = V_{\mathrm{T}}\ln(u/I_{\mathrm{s}}) + V_{\mathrm{T}}\ln(I_0/I_{\mathrm{s}}) \tag{9.9}$$

An appropriate circuit to accomplish this is shown in Fig. 9.1c. The final oscillator circuit is obtained by synthesizing the node equations using the various sub-circuits detailed above and is shown in Fig. 9.1d.

The workability of the proposed synthesized oscillator configuration has been demonstrated in [2] by three different implementations: using BJTs and MOS devices (with MOSFETs operating in weak inversion), using HSPICE with 0.8 μm BiCMOS technology, and hardware implementation using ELANTEC high-frequency NPN EN2016C and PNP EP2015C transistor arrays. It was reported in [2] that the BJT oscillator in Fig. 9.1d showed THD of 0.313 % at 10 MHz, while the oscillator implemented in subthreshold MOS technology worked well up to 10 kHz with THD of 0.9 % at 5.34 kHz. The circuit implemented in hardware operated with a supply voltage of ±2.5 V functioned well up to 10 MHz with THD of 0.5 % [2].

9.3 Square-Root Domain Oscillators

Sinusoidal oscillators can be implemented in CMOS using OTA-C structures implemented with CMOS OTAs. However, CMOS OTAs can handle very small signals only because the transconductance of the OTA itself is based upon small signal considerations. By contrast, the square-root domain circuits exploit the large signal characteristics of the MOSFETs operating in saturation and, hence, have been widely investigated. Several authors have also investigated the realizability of oscillators using square-root domain techniques, for instance, see [16–19, 22] and references cited therein. Thanachayanont, Payne, and Pookaiyaudom [16] demonstrated that CMOS sinusoidal oscillators can be systematically synthesized using state-space techniques and implemented as square-root domain oscillators.

The basic methodology of [16] is quite similar to the one employed in the synthesis of log-domain oscillators using BJTs in [2] and is as follows:

The synthesis starts by assuming the open-loop transfer function of the oscillator as given by Eq. (9.1), and from this equation, the state equations can be formulated which can be specified by Eqs. (9.2a, 9.2b, 9.2c). If the input signal u and the state-variable X_1 and X_2 are assumed to be the node voltages U, V_1, and V_2 of the circuit to be synthesized, then the state equation in Eqs. (9.2a, 9.2b, 9.2c) can be expressed as follows:

$$C\dot{V}_1 = C\omega_0 V_2 - C\omega_0 V_1 \tag{9.10a}$$

$$C\dot{V}_2 = C\omega_0 V_2 - 2C\omega_0 V_1 + C\omega_0 U \tag{9.10b}$$

$$Y = V_1 \tag{9.10c}$$

where C is the multiplying factor. Considering the MOSFETs to be operating in the saturation region, one can write the drain current as

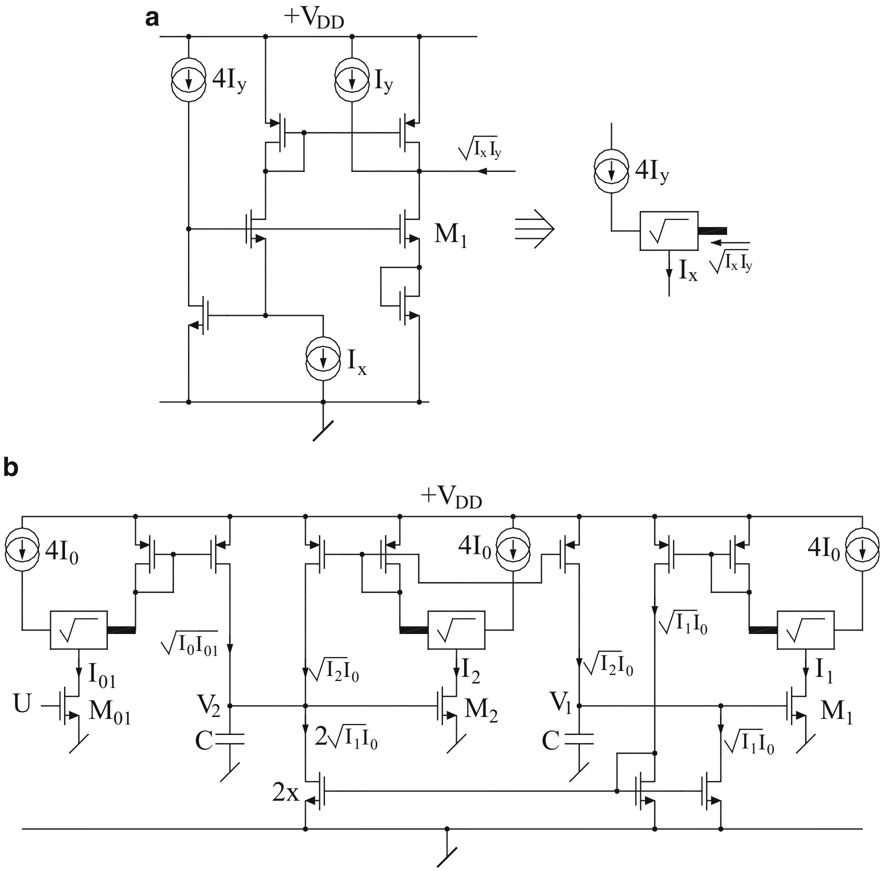

Fig. 9.2 Square-root domain oscillator proposed by Thanachayanont, Payne, and Pookaiyaudom [16]. (**a**) The square-root domain oscillator. (**b**) The geometric-mean circuit

$$I_{DS} = K_{\mathrm{p}}\left(\frac{W}{2L}\right)(V_{GS} - V_{TH})^2 \tag{9.11}$$

where the symbols have their usual meanings. Using Eq. (9.11), three drain currents of the matched CMOS transistors can be defined as follows:

$$I_{01} = K(U - V_{TH})^2 \tag{9.12}$$

$$I_1 = K(V_1 - V_{TH})^2 \tag{9.13}$$

$$I_2 = K(V_2 - V_{TH})^2 \tag{9.14}$$

where $K = K_{\mathrm{p}}\left(\frac{W}{2L}\right)$. Using Eqs. (9.12)–(9.14), the state equations in Eqs. (9.10a, 9.10b, 9.10c) can be now expressed as

$$C\dot{V}_1 = \sqrt{I_0 I_2} - \sqrt{I_0 I_1} \tag{9.15}$$

$$C\dot{V}_2 = \sqrt{I_0 I_2} - 2\sqrt{I_0 I_1} + \sqrt{I_0 I_{01}} \tag{9.16}$$

where

$$I_0 = C^2 \omega_0^2 / K \tag{9.17}$$

It may be seen that a current-mode geometric-mean circuit is needed to implement Eqs. (9.15) and (9.16). Furthermore, it can also be noticed that the oscillation frequency can be tuned by I_0.

The generic circuit following the above node equations is symbolically shown in Fig. 9.2a, whereas the required geometric-mean circuit is shown in Fig. 9.2b [16]. In this circuit, the gate-to-source junctions of the MOSFETs M_1, M_2, M_3, and M_4 are forming *a translinear* loop for which, assuming matched MOSFETs having same aspect ratios, application of *MOS translinear principle* leads to the following equation:

$$\sqrt{I_{d1}} + \sqrt{I_{d2}} = \sqrt{I_{d3}} + \sqrt{I_{d4}} \tag{9.18}$$

Since $I_{d1} = 4I_y$, $I_{d2} = I_x$, and $I_{d4} = I_{d3} = [(I_x/4) + I_y + I_{\text{out}}]$, substituting these values in Eq. (9.18) and simplifying, we finally obtain

$$I_{\text{out}} = \sqrt{I_x I_y} \tag{9.19}$$

Simulations of this circuit were carried out [16] using HSPICE with Nortel 0.8 μm CMOS technology with capacitors as 2.1 pF, $I_0 = 10$ mA and aspect ratios of the MOSFETs M_{01}, M_1, and M_2 taken as 5 μm/5 μm, DC bias voltage U taken as 1.7 V and DC bias power supply taken as 5 V.

It was observed [16] that oscillation frequencies from 750 kHz to 2.3 MHz were realizable with output amplitude varying from 80 mV to 1.1 V peak-to-peak and THD ranging from 0.1 to 3 % with total power consumption being less than 16 mV. These results clearly demonstrated the good potential of CMOS square-root domain oscillators for high-frequency applications.

9.4 Current-Mode Oscillator Employing f_T Integrators

The f_T integrators were introduced by Mahattanakul, Toumazou, and Pookaiyaudom [20] and subsequently by Worapishet and Toumazou [23]. In [20] it was demonstrated that the nonlinear $V - I$ characteristics of the BJT NPN transistor combined with its nonlinear base-emitter capacitance can be directly employed to realize a current-mode integrator exclusively using BJTs only. The essence of the idea is to employ the transistor as the transconductance element along with an integrating capacitor.

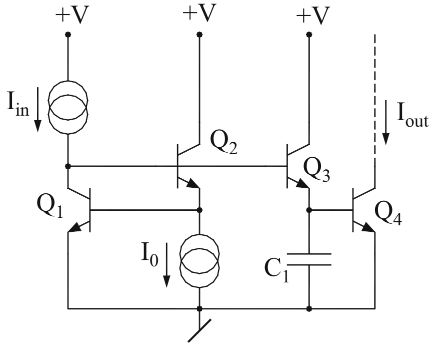

Fig. 9.3 Current-mode integrator proposed by Mahattanakul, Toumazou, and Pookaiyaudom [20]

This circuit is shown in Fig. 9.3. The working principle of this circuit can be explained as below:

If the input current used to charge the capacitor C_1 is taken as

$$i_{C_1} = C_1 V_T \omega_u I_{in}/I_{out} \tag{9.20}$$

such a current can be created from an appropriate translinear circuit, thereby leading to the current-mode integrator circuit. This integrator would be an instantaneous companding or log-domain integrator.

The base-emitter capacitance of BJT, C_π, consists of the base charging capacitance C_b and the emitter-based layer capacitance C_{je} and is given by

$$C_\pi = C_b + C_{je} \tag{9.21}$$

where the value of C_b is given by

$$C_b = \tau_F I_c/V_T \tag{9.22}$$

where I_c is the collector current of the transistor, V_T is the thermal voltage, and parameter τ_F is the base-transit time in the forward direction. Since the collector current is also the output current of the integrator, i.e., $I_{out} = I_c$, it follows that the angular unity gain frequency of the integrator would be given by

$$\omega_u = n/\left(\tau_F + \frac{C_{je} V_T}{I_{out}}\right) \tag{9.23}$$

If the circuit is implemented such that $I_{out} >> (C_{je}V_T/\tau_F)$, then the value of ω_u will be constant and equal to n/τ_F; hence, under such condition, the circuit would function as a linear integrator whose time constant can be tuned by varying the current gain n.

Khumsat, Worapishet, and Payne [13] presented a low-distortion high-frequency oscillator using the concept of f_T integrator technique. Their circuit is shown in Fig. 9.4.

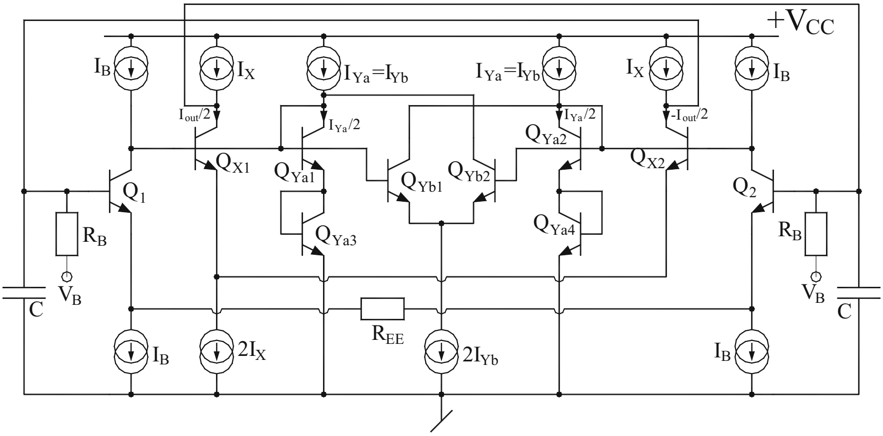

Fig. 9.4 The current-mode oscillator employing f_T integrator proposed by Khumsat, Worapishet, and Payne [13]

This circuit is essentially an implementation of an LC-based resonator employing an active inductor realized from $G_M - F_T$ integrator configuration and is equivalent to parallel combination of two branches, one consisting of a capacitor C and the other consisting of a series combination of a positive inductor L, a positive resistor R_p, and a negative resistor $-R_n$.

Clearly, when $(R_p - R_n) = 0$ the circuit becomes capable of generating sinusoidal oscillations. In this circuit, the transistors Q_1 and Q_2, along with the resistor R_{EE}, function as a differential emitter-degenerated transconductor, whereas the transistors Q_{X1}, Q_{YaI}, and Q_{YbI} constitute the f_T integrator for which the circuit contained in the dotted box serves the purposes of DC gain enhancement and amplitude control.

Simulations were conducted [13] on a prototype oscillator designed with 20 GHz f_T bipolar process, with I_X kept constant at 1 mA and I_{YA}, I_{YB}, and their ratio simultaneously varied for frequency tuning and amplitude control. With DC bias voltage taken as 2.2 V, $V_B = 1.8$ V, $C = 0.5$ pF, $I_B = 1$ mA, $R_B = R_{EE} = 1$ kΩ, the tuning range of this oscillator has been found to be more than one octave from 1 to 2.6 GHz.

9.5 Log-Domain Quadrature/Multiphase Oscillators

Prommee, Prapakorn, and Swamy [7] proposed a log-domain current-mode quadrature sinusoidal oscillator based upon lossless integrators. The required log-domain lossless integrators were realized by using only NPN transistors and grounded capacitors.

The circuit for the non-inverting log-domain lossless integrator is shown in Fig. 9.6a. The exact linear transfer function of this circuit is given by

$$\frac{I_{\mathrm{out1}}(s)}{I_{\mathrm{in1}}(s)} = \frac{\frac{kI_{\mathrm{B}}}{C_1V_{\mathrm{T}}}}{s + \frac{I_{\mathrm{B}}}{C_1V_{\mathrm{T}}}(1-k)} \tag{9.24}$$

On the other hand, an inverting log-domain lossless integrator is shown in Fig. 9.6b which realizes the transfer function

$$\frac{I_{\mathrm{out2}}(s)}{I_{\mathrm{in2}}(s)} = -\frac{I_{\mathrm{B}}}{sC_2V_{\mathrm{T}}} \tag{9.25}$$

The complete quadrature oscillator circuit is then obtained by arranging the preceding two circuits in a cascade connection and then closing the loop. The resulting circuit is shown in Fig. 9.5c.

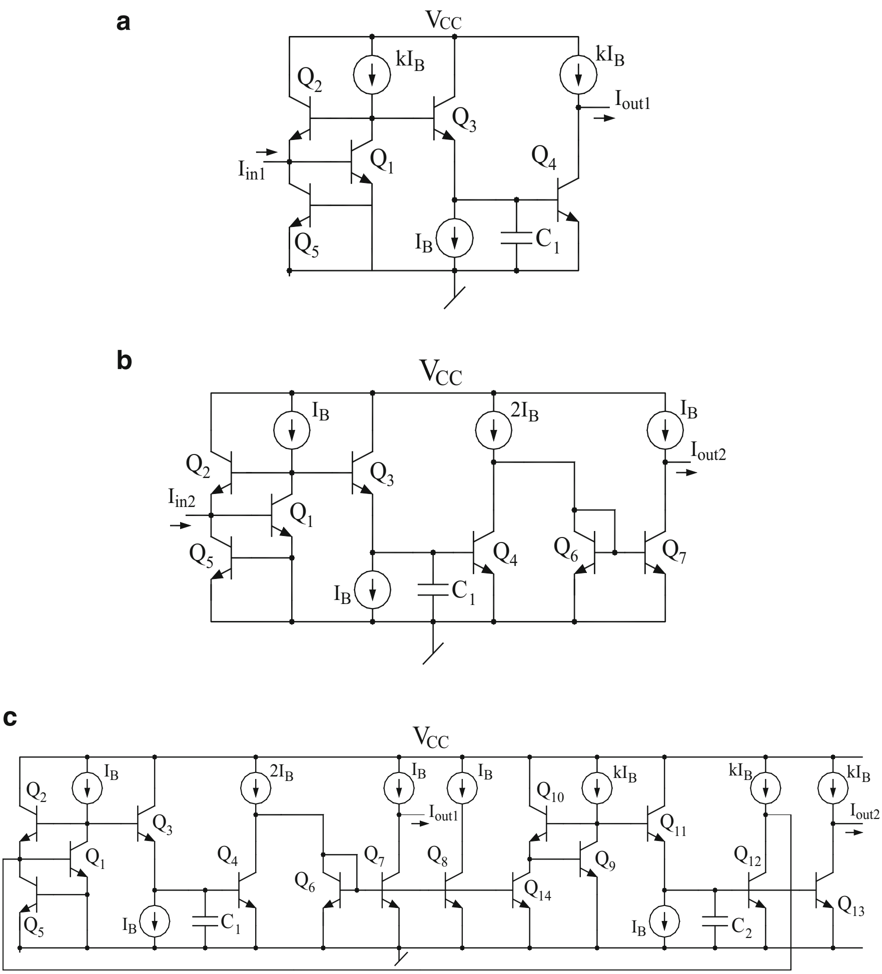

Fig. 9.5 Current-mode log-domain quadrature oscillator proposed by Prommee, Prapakorn, and Swamy [7]. (**a**) Non-inverting lossy integrator. (**b**) Inverting integrator. (**c**) The complete quadrature oscillator circuit

By straightforward analysis, the characteristic equation of this oscillator is given as

$$S^2 + S\frac{I_B}{C_1 V_T}(1-k) + \frac{k}{C_1 C_2}\left(\frac{I_B}{V_T}\right)^2 = 0 \tag{9.26}$$

from which the condition of oscillation is seen to be $k = 1$, whereas the oscillation frequency is given by

$$\omega_0 = \frac{I_B}{V_T\sqrt{C_1 C_2}} \tag{9.27}$$

Therefore, it is clear that the oscillation frequency is a linear function of the external DC bias current I_B.

The SPICE simulation of this circuit with $V_{cc} = 2.5$ V and $k = 1.472$ with capacitors $C_1 = C_2 = 50$ pF demonstrated [7] that wide range oscillation frequencies can be generated from this circuit from 1 kHz to 100 MHz by varying the external DC bias current I_B from 0.01 to 1000 μA with power consumption of the order of 2.46 mW with all NPN transistors modeled as NPN-HSB2 provided by ST Microelectronics and all PNP transistors modeled by PNP-HFA3128 provided by Intersil.

9.6 Log-Domain Multiphase Oscillators Using Exponential Transconductor Cells

A method of designing multiphase oscillators was proposed by Psychalinos and Souliotis [5] in which the method of cascading "*n*" number of lossy integrators (first-order low-pass filters) in a closed loop to generate multiphase signal at the output of these first-order stages was employed. The non-inverting and inverting lossy integrators needed therein, in turn, were realized by employing three exponential conductors and a grounded capacitor in such a way that the angular frequencies and the gain both were electronically controllable through external DC currents.

The technique involved three specific types of exponential-domain transconductors which are shown here in Fig. 9.7.

It is easy to verify that the circuits in Fig. 9.7a, b are characterized by the following equations:

$$i_{OUT} = I_0 e^{\frac{\hat{v}_{IN} - \hat{v}_{OUT}}{V_T}} \tag{9.28}$$

and

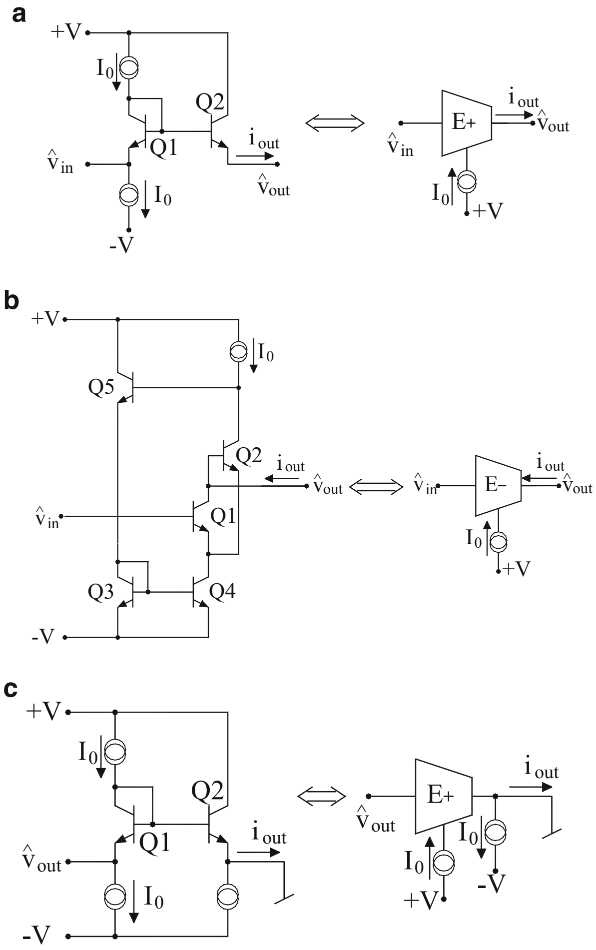

Fig. 9.6 Exponential-domain transconductors [5]. (**a**) Positive exponential transconductor. (**b**) Negative exponential transconductor. (**c**) Implementation of the EXP operator

$$i_{\mathrm{OUT}} = -I_0 e^{\dfrac{\hat{v}_{\mathrm{IN}} - \hat{v}_{\mathrm{OUT}}}{V_{\mathrm{T}}}} \tag{9.29}$$

On the other hand, the circuit in Fig. 9.6c implements the equations

$$v_{\mathrm{OUT}} = V_{\mathrm{T}} \ln\left(\frac{i_{\mathrm{OUT}} + I_0}{I_0}\right) \text{ or } i_{\mathrm{OUT}} = I_0 e^{\frac{v_{\mathrm{OUT}}}{V_{\mathrm{T}}}} - I_0 \tag{9.30}$$

and, hence, can be employed to implement the complementary LOG/EXP operators of the type

$$\widehat{v} = \mathrm{LOG}(i) = V_{\mathrm{T}} \ln\left(\frac{i + I_0}{I_0}\right) \tag{9.31}$$

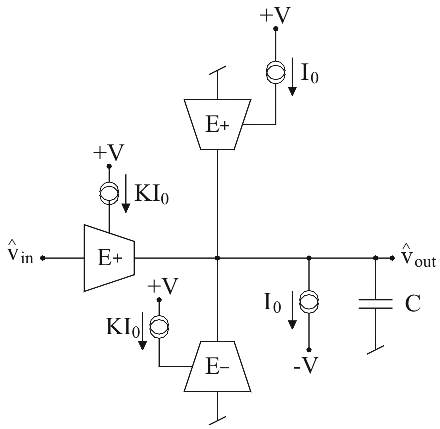

Fig. 9.7 Non-inverting lossy integrator [5]

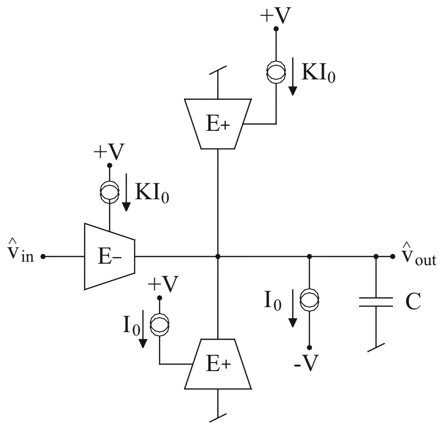

Fig. 9.8 Inverting lossy integrator [5]

$$i = \mathrm{EXP}\left(\widehat{v}\right) = I_0 e^{\frac{\widehat{v}}{V_T}} - I_0 \tag{9.32}$$

Using these transconductors as basic building blocks, a non-inverting log-domain lossy integrator and an inverting lossy integrator can be realized as shown in circuits in Figs. 9.7 and 9.8, respectively. The non-inverting lossy integrator is characterized by the following node equation at the output node of the circuit:

$$C\frac{d\hat{v}_{\mathrm{OUT}}}{dt} + I_0 = KI_0 e^{\frac{\hat{v}_{\mathrm{IN}}-\hat{v}_{\mathrm{OUT}}}{V_T}} + I_0 e^{\frac{-\hat{v}_{\mathrm{OUT}}}{V_T}} - KI_0 e^{\frac{-\hat{v}_{\mathrm{OUT}}}{V_T}} \tag{9.33}$$

In terms of the operators defined, the above equation can be finally put in the following simplified form:

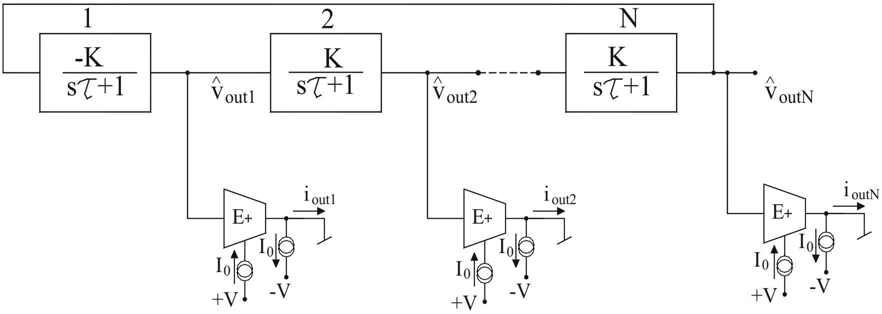

Fig. 9.9 The multiphase oscillator configuration proposed by Psychalinos and Souliotis [5]

$$\left(\frac{CV_{\mathrm{T}}}{I_0}\right)\frac{d}{dt}\left[\mathrm{EXP}\left(\widehat{v}_{\mathrm{OUT}}\right)\right] + \mathrm{EXP}\left(\widehat{v}_{\mathrm{OUT}}\right) = K\left(\mathrm{EXP}\left(\widehat{v}_{\mathrm{IN}}\right)\right) \tag{9.34}$$

Denoting $(CV_{\mathrm{T}}/I_0) = \tau$ as the time constant of the lossy integrator, the corresponding linear domain equation can be finally written as:

$$\frac{i_{\mathrm{out}}}{i_{\mathrm{in}}} = K\left(\frac{1}{1 + s\tau}\right) \tag{9.35}$$

which is the required transfer function of the lossy integrator. Likewise, it is easy to verify that the circuit in Fig. 9.8 realizes an inverting lossy integrator.

Based upon the building blocks explained above, a general n-phase oscillator takes the form as shown in Fig. 9.9.

The open-loop transfer function of this configuration can be easily verified to be

$$T(s) = -K^N\left(\frac{1}{s\left(\frac{CV_{\mathrm{T}}}{I_0}\right) + 1}\right)^N \text{ where } N \geq 2 \tag{9.36}$$

By applying the Barkhausen criterion, the condition of oscillation and frequency of oscillation are found to be

$$K = \sqrt{1 + \tan^2\left(\frac{\pi}{N}\right)} \tag{9.37}$$

and

$$\omega_0 = \frac{I_0}{CV_{\mathrm{T}}} \tan\left(\frac{\pi}{N}\right) \tag{9.38}$$

It is, thus, seen that depending upon "n" the required value of K can be set to generate oscillations whose frequency can then be controlled through the external DC current I_0*linearly*. The circuit was simulated [5] in SPICE by using VBIC BJT model parameters from AMS S35D4 process, with DC bias voltage equal to ±1.5 V and DC bias current I_0 as 50 μA. For a four-phase oscillator design, K was chosen as √2 and the capacitors were taken as 50 pF. For a theoretical value of oscillation frequency as 6.19 MHz, the simulations showed a value of 6.22 MHz, while the phase angles between the four waveforms were found to be quite close to 90° thereby proving the workability of the methodology. Electronic tunability was established by observing the frequency of oscillation to be halved when the bias current was reduced to 25 μA.

9.7 Square-Root Domain Multiphase Oscillators

It is well known that all-pass filters can be used to a construct sinusoidal oscillator. A multiphase oscillator based upon novel approximate square-root domain all-pass filters was proposed by Ozoguz, Abdelrahman, and Elwakil [19]. Two configurations were proposed which were termed as N-Cell and P-Cell and are shown in Fig. 9.10.

In the first circuit, considering the aspect ratio of M_1 to be twice as large as that of M_2, it easy to see that the diode-connected MOSFET M_4 carries a current I_{in} at low frequencies. However, at high frequencies, it carries the current $-I_{\text{in}}$, since M_1 is supposed to be turned off due to the very small voltage across C. Therefore, one can conclude that $I_{\text{out}}/I_{\text{in}}$ ideally retains a unity magnitude while inverting its phase over the frequency range. This is obviously a characteristic of an all-pass filter which can be proved mathematically as follows:

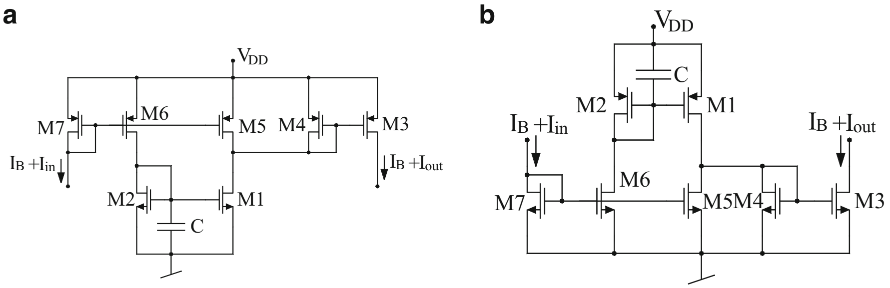

Fig. 9.10 Square-root domain all-pass filters proposed by Ozoguz, Abdelrahman, and Elwakil [19]. (**a**) N-cell. (**b**) P-cell

Note that the basic square law relation for an N-MOSFET is given by

$$I_{\mathrm{DS}} = \beta(V_{\mathrm{GS}} - V_{\mathrm{TH}})^2 \text{ where } \beta = (\mu_n C_{\mathrm{ox}} W/L)/2 \tag{9.39}$$

For the N-Cell, the capacitor current can be written as

$$I_{\mathrm{C}} = C\dot{V}_{\mathrm{GS1}} = C\dot{I}_{\mathrm{DS1}}/\beta(V_{\mathrm{GS1}} - V_{\mathrm{TH}}) \tag{9.40}$$

Because of the companding properties of the MOSFETs, the variation of the capacitor voltage V_{GS1} is very small and the capacitor voltage, therefore, can be assumed to be constant. Therefore, the capacitive current can be written as

$$I_{\mathrm{C}} = \tau\dot{I}_{\mathrm{DS1}} = 0.5\tau\left(\dot{I}_{\mathrm{out}} + \dot{I}_{\mathrm{in}}\right) \tag{9.41}$$

where

$$\tau = C/\beta(V_{\mathrm{GS}} - V_{\mathrm{TH}}) = C/g_m \tag{9.42}$$

In addition, one can also write the node equation

$$I_{\mathrm{C}} = I_{\mathrm{in}} - I_{\mathrm{DS1}} = 0.5(I_{\mathrm{in}} - I_{\mathrm{out}}) \tag{9.43}$$

Finally, from Eqs. (9.41)–(9.43), taking Laplace transform and solving, one obtains the current ratio transfer function to be that of a first-order all-pass function given by

$$\frac{I_{\mathrm{out}}}{I_{\mathrm{in}}} = \frac{1 - \tau s}{1 + \tau s} \tag{9.44}$$

The P-Cell is in fact a compliment of the N-Cell; these two cells together can be used to construct two-phase and three-phase oscillators as shown in Fig. 9.11. In general, an n-phase oscillator can be realized by cascading N-all-pass filter stages in a loop closed by an inverting unity gain amplifier.

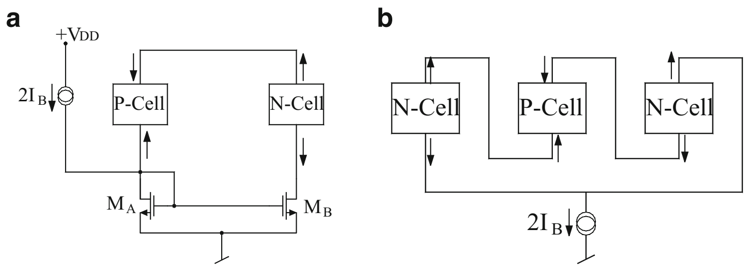

Fig. 9.11 Square-root domain oscillators proposed by Ozoguz, Abdelrahman, and Elwakil [19]. (**a**) Two-phase oscillator. (**b**) Three-phase oscillator

For the case of a two-phase oscillator, the required unity gain amplifier is made from a simple current mirror $M_A - M_B$, whereas for the three-phase oscillator, such an amplifier is not required because of the alternating nature of the cells and the gain can be adjusted by the current mirror $M_3 - M_4$ of the P-Cell. The oscillation frequencies for these two cases are given by

$$\omega_0 = \frac{1}{\sqrt{\tau_1 \tau_2}} \text{ and } \omega_0 = \frac{1}{\sqrt{\tau_1 \tau_2 + \tau_3(\tau_1 + \tau_2)}} \tag{9.45}$$

where τ_i is the time constant of the ith cell.

The measurements carried out in [19] on prototypes built with current mirrors implemented from BC557 PNP bipolar transistors with their emitters connected to V_{DD} via 100 Ω resistors and remaining MOSFETs realized by using CMOS transistor array CD4007 with capacitors of value 10 nF and $I = 800$ μA and, finally, output currents converted into voltages using 1 kΩ resistors confirmed the workability of the proposed method. The two-phase oscillator and three-phase oscillator both were found to work well [19].

9.8 Sinh-Domain Multiphase Sinusoidal Oscillators

Panagopoulou, Psychalinos, Khanday, and Shah [21] presented a method of creating current-mode multiphase oscillators using Sinh-domain and Cosh-domain transconductance and thereby using those to create a lossy integrator or a first-order low-pass filter. It is well known that by cascading "n" lossy integrators in a closed loop, one can create a multiphase oscillator. For the case current-mode multiphase oscillator, one requires current-mode integrators with two similar outputs, one of which becomes the input current to the next stage, whereas the second one becomes the explicitly available output current. Furthermore, if the lossy integrator is realized having three output currents, the third one being the complementary to the first two, then a cascade of "n" such lossy integrators in a closed loop results in a structure which becomes a multiphase sinusoidal oscillator (MSO).

It is easy to visualize that the open-loop transfer function of an "n"-phase oscillator, as per the above description, can be written as

$$L(s) = -K^n \left(\frac{1}{1 + s\,\widehat{\tau}} \right)^n \tag{9.46}$$

From the above, applying Barkhausen criterion, it is easily deduced that the frequency of oscillation and condition of oscillation for such an MSO would be given by

$$\omega_0 = \frac{1}{\widehat{\tau}} \tan\left(\frac{\pi}{n}\right) \tag{9.47}$$

$$K = \sqrt{1 + \tan^2\left(\frac{\pi}{n}\right)} \tag{9.48}$$

The circuit schematic in Fig. 9.12a .implements a lossy integrator/first-order low-pass function using Sinh and Sinh/Cosh transconductance, while the two-quadrant analog divider is shown in Fig. 9.12b.

A Sinh transconductor is characterized by the following input-output expression:

$$i_s = 2I_0 \sinh\left(\frac{\widehat{v}_{IN+} - \widehat{v}\, I_{IN-}}{V_T}\right) \tag{9.49}$$

whereas the output expression for the S/C cell is given by

$$i_C = 2l_0 \cosh\left(\frac{\widehat{v}_{IN+} - \widehat{v}_{IN-}}{V_T}\right) \tag{9.50}$$

where symbols have their usual meanings. On the other hand, the two-quadrant analog divider in Fig. 9.12b is characterized by

$$i_{OUT} = I_{bias}\left(\frac{i_1}{i_2}\right) \tag{9.51}$$

Now from the circuit in Fig. 9.12a, it can be deduced that the current flowing in the integrating capacitor is given by

$$\widehat{C}\frac{d\widehat{v}_{OUT}}{dt} = 2I_0 \frac{i_{in} - 2I_0 \sin\mathrm{h}\left(\widehat{v}_{OUT} - V_{DC}/V_T\right)}{2I_0 \cos\mathrm{h}\left(v_{OUT} - \widehat{V}_{DC}/V_T\right)} \tag{9.52}$$

After appropriate algebraic simplifications and using the constant $\widehat{\tau} = \widehat{C}\, V_T/2I_0$, the transfer function of the circuit in Fig. 9.12a is found to be

$$H^{'}(s) = \frac{i^{'}_{OUT}}{i_{IN}} = \frac{1}{s\,\widehat{\tau} + 1} \tag{9.53}$$

In case a non-unity gain K is required, then the S-cell is required to be biased at current equal to KI_0, and it should produce an output current which should be given by

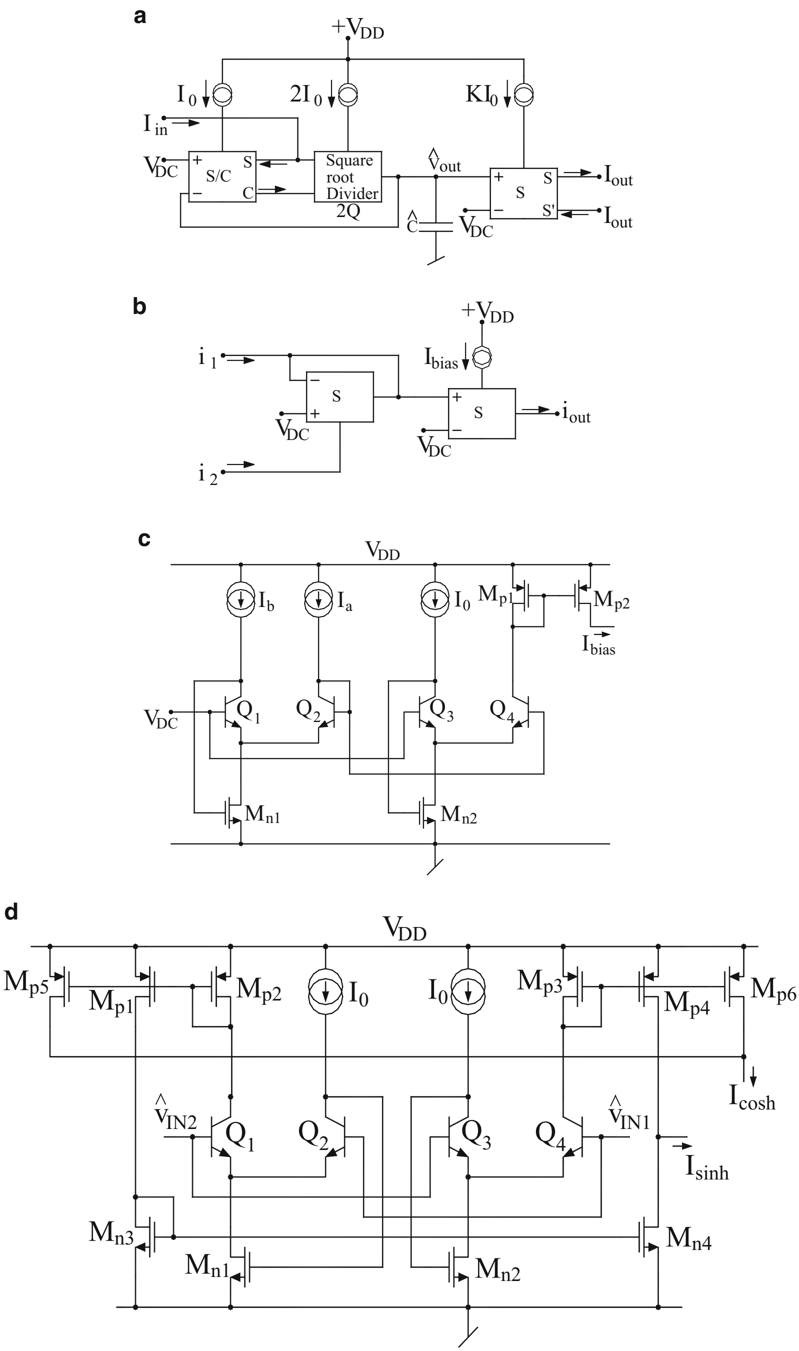

Fig. 9.12 Basic building blocks of Sinh-domain multiphase sinusoidal oscillators in the technique proposed by Panagopoulou, Psychalinos, Khanday, and Shah [21]. (**a**) Electronically controllable Sinh-domain lossy integrator. (**b**) Two-quadrant analog divider using Sinh transconductors. (**c**) The bias current generator with electronic control. (**d**) The Sinh/Cosh cell

$$i_{\mathrm{OUT}} = K2I_0 \sinh\left(\frac{\widehat{v}_{\mathrm{OUT}} - V_{\mathrm{DC}}}{V_{\mathrm{T}}}\right) = K\,\mathrm{Sinh}\left(\widehat{v}_{\mathrm{OUT}}\right) = Ki'_{\mathrm{OUT}} \tag{9.54}$$

In practical implementation, it is desirable that the scale factor K should be realized as a ratio of two DC currents which can be easily accomplished by using the schematic in Fig. 9.12c. By applying static *translinear principle*, it is found that in this circuit, the current I_{bias} would be given by $I_{\mathrm{bias}} = I_0(I_{\mathrm{a}}/I_{\mathrm{b}})$. A circuit which can implement the required Sinh/Cosh cell is shown in Fig. 9.12d.

The workability of a six-phase oscillator, designed by this methodology, was verified [21] by employing AMS 0.35 μm S35 BiCMOS process, with V_{DD} as 1.2 V and V_{DC} as 1 V and bias current chosen as 1 μA and capacitors as 214.5 pF to realize an oscillation frequency of 100 kHz. The observed frequency was 93.77 kHz with errors supposedly caused by transistor imperfections. The tunability of the oscillation frequency was clearly observable. The power dissipation was found to be 240.6 μW. When compared to other log-domain oscillator designs, the six-phase oscillator offers reduced power supply requirement and better power efficiency with the price paid being the limitations on the maximum frequency of operation which gets reduced due to the increased circuit complexity of this technique [21].

9.9 Concluding Remarks

In this chapter, we had highlighted some key developments which have taken place in the domain of realizing fully integratable oscillators in bipolar/CMOS technology based upon the concepts of log-domain and square-root domain circuits. It is seen that although some very potential circuits have been evolved in the various domains, the field is by no means complete. Many meaningful and new ideas appear to be still waiting to be explored and unearthed to yield topologies offering optimum performance in terms of higher operational frequency range, wider electronic tunability range, reduced THD, lowest possible supply voltage, minimum power consumption, minimum phase noise, etc. Thus, there is ample scope in this area for discovering new circuit principle/topologies for log-domain and square-root domain oscillators satisfying *simultaneously* all the desirable properties and characteristics mentioned above. At the time of writing this chapter, a number of new ideas such as sinusoidal oscillators with lower gain requirements at higher frequencies based on explicit tanh(x) nonlinearity [24] and CMOS weak-inversion log-domain glycolytic oscillator [25] have already started emerging as new innovations in this area.

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Chapter 10
Generation of Equivalent Oscillators Using Various Network Transformations

Abstract This chapter discusses a number of transformations which have been proposed by various researchers from time to time to generate several equivalent configurations of a given sinusoidal oscillator.

10.1 Introduction

Generating a number of solutions for a given problem is an important and interesting task for an engineer as it gives him/her a number of alternatives from which the most desirable solution can be chosen in accordance with the requirements of the given application and design constraints. With this motivation in mind, a number of network transformations based on the notion of *adjoints*, *network transposition*, and theories based upon *the nullor representation* of sinusoidal oscillators have been proposed by various researchers from time to time.

It may be mentioned that there are a number of frequency-scaling type transformations which can relate seemingly different oscillator topologies to the same original tank circuit (for instance, see [1]). However, the intent of this chapter is not to discuss such transformations. This chapter is concerned with transformations which involve either the interchange/reallocation of various terminals of an active RC network or replacement of one kind of active element by another different active element or a combination of the two, to result in transformed oscillators which are distinctly different in structure than the original circuit but are governed by the same characterizing equations as the original oscillator.

Thus, in this chapter, we present a number of available methods/transformations of generating equivalents of a given oscillator and bring out the salient features of the various methodologies.

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10.2 Nullor-Based Transformations of Op-Amp-RC Sinusoidal Oscillators

In classical active network theory, it has been known since long that *nullator, norator*, and *nullor* which are considered to be hypothetical and degenerate pathological elements can serve as the means of unifying different circuit realizations into a common framework; see [2] and the references given therein. All the four basic controlled sources, as well as various devices and building blocks like BJT, FET, MOSFET, op-amp, OTA, etc., can be modeled by equivalent networks consisting of either nullators and norators only (as in the case of the ideal BJT, ideal MOSFET, and ideal op-amp) or by combinations of nullators and norators along with resistances (as in the case of OTA).

However, quite often, skeptics have not taken a very serious view of the potential of nullators, norators, and nullors. This skepticism has prolonged since long even after the fact that nullors have been extensively used in the past to generate equivalent gyrator networks realized with not only BJTs but also those realized with ideal op-amps. In the latter case, the pioneering work of Antoniou [3] in generating equivalent networks of op-amp gyrators culminating into the classical two-op-amp generalized impedance converter (GIC) is well recognized and extensively used in practical applications.

Thus, in spite of the powerful application of the nullors in generating equivalent networks, their use for such purposes still did not catch up till it was reinstated by Huijsing [4, 5], Stevenson [6], and Senani [7, 8] that four-terminal floating nullor (FTFN) and the so-called operational floating amplifier (OFA) are, indeed, very versatile active elements for realizing a number of linear active networks much more efficiently than any other methods of doing the same [4–9].

The work quoted above provided further impetus to the use of FTFNs in analog circuit design, and the nullors truly started getting their place in the circuit synthesis and design to the extent that at one point of time, the nullors started getting the recognition as the *universal* active elements. This stature of nullor was further fortified by the emergence of a number of CMOS implementations of the FTFNs. These developments (see [2]) really contributed in finally establishing the true potential of nullors in both analysis and synthesis of active networks.

Further support to the power of nullors in circuit synthesis came from the pioneering work of Haigh [10–12] and his group who developed a novel analytical framework for systematically synthesizing any given function, be it the synthesis of floating impedances or, impedance converters and inverters, based upon the notion of nodal admittance matrix (NAM) stamps of known active circuit building blocks and the operation of pivotal expansion.

The existing knowhow on nullors could be easily extended to model various newly emerging circuit elements such as the current conveyors in which case it was found that while a CCII− can be represented as a 3-terminal floating nullor, unfortunately, its other counterpart CCII+ could not be represented by only nullator and norator; it did require two matched resistances in addition. Likewise, many other modern active circuit building blocks including numerous newer varieties of CCs could also be represented by nullor models, but the use of additional resistances was almost inevitable.

The aforementioned difficulty was ultimately overcome by the introduction of two new pathological elements, by Awad and Soliman [13, 14], namely, the ideal *voltage mirror* (VM) and the ideal *current mirror* (CM), none of which are realizable physically, in isolation. The ideal VM has the voltage inverting property and is a two-port element characterized by $V_1 = -V_2$ and $I_1 = 0 = I_2$, whereas an ideal CM is also a two-port element having current inverting property and is characterized by $I_1 = I_2$ with $V_1 = \text{arb} = V_2$. These elements are, although three-terminal elements, the third terminal being the ground terminal; however, in application and modeling, they are used as two-terminal elements with the ground terminal being considered present *implicitly*.

The real potential of CM and VM was (1) in demonstrating that a CCII+ can be realized in terms of CM and VM without requiring two matched resistors and (2) in demonstrating that eventually, all the building blocks, known so far (excluding, of course, the transconductance-type, and transresistance-type elements), could be realized without any resistance using judicious combination of nullator, norator, CM, and VM only.

This was soon followed by numerous researchers who proposed NAM stamps of various building blocks, generated equivalent models of various active building blocks, and developed systematic synthesis procedures for generating impedance converters and inverters, biquad filters, and families of oscillators in terms of both classical and modern circuit building blocks; see [10–12, 15, 16] and references cited therein.

The work on systematic synthesis of oscillators using all the four pathological elements, namely, nullator, norator, CM, and VM, is still continuing and is far from being complete at this instant of time. In the following, therefore, we present the significant contributions made in the use of well-known pathological elements, i.e., nullator, norator, and nullor in the systematic synthesis and transformation of oscillators which is spread over a number of publications [17–20].

Historically, it appears to be the 1971 paper by Williams [17], where it was demonstrated *for the first time* that by modeling the op-amp by a pair of nullator and norator, the nullor model of the classical Wien bridge oscillator can have four distinctly different equivalent forms having exactly the same condition of oscillation (CO) as well as frequency of oscillation (FO). Williams [17] also demonstrated that the nullor-based equivalents of the same oscillator can also give rise to two-op-amp-based circuits (with each op-amp configured in inverting mode) as well as bipolar junction transistor (BJT)-based oscillators.

More than a decade later, Boutin showed (see Ref. [124] of Chap. 2) that Wien bridge oscillator and other types of sinusoidal oscillators employing a single op-amp have four distinctly different equivalent forms. Subsequently, Senani [18] demonstrated that this is a consequence of a more general property which is applicable to all RC-active sinusoidal oscillators realized with op-amps as well as using any other types of devices or active circuit building blocks which can be represented by nullors. In particular, it was shown in [18, 19] that every oscillator N realized with m-nullors along with an arbitrary number of resistors and capacitors (and/or inductors) has an equivalent companion oscillator N^* which is distinctly different than N but employs exactly the same number of active and passive components and has the same characteristic equation (CE) and, hence, the same CO and FO.

The following theorems were presented in [18, 19]:

Theorem 1 *Suppose there is a sinusoidal oscillator circuit* N *which employs m-nullators, m-norators, and an arbitrary number of RC elements. If* N *is transformed into* N^* *by interchanging all nullators by all norators and vice versa, then* N^* *will have the same CE as that of* N.

Theorem 2 *Corresponding to any RC-nullor oscillator having* n *nodes (excluding the ground node which is taken "external" to the circuit) and consisting of* m-*nullors along with an arbitrary number of passive resistor and capacitors, there are* 2n *grounded nullor-RC equivalent oscillator circuits having the same CE since in an oscillator, because there is no external input, the ground node can be chosen arbitrarily without affecting the CE.*

A logical consequence of Theorem 1 is that corresponding to any single op-amp RC sinusoidal oscillator, at least four equivalent forms, having the same CE and hence the same condition of oscillation and frequency of oscillation, are immediately apparent (see Fig. 10.1). In the figure, the black box represents the passive part in which a single op-amp is embedded. The first two variants shown are obtained by grounding one end of the norator at a time while the remaining two versions are obtained by applying Theorem 1 (i.e., by swapping nullator and norator) and then grounding again, one end of the norator at a time.

Consider now the classical Wien bridge oscillator (WBO) shown in Fig. 10.2, the nullor model of which is shown in Fig. 10.3.

This circuit is characterized by the following condition of oscillation (CO) and frequency of oscillation (FO):

$$\mathrm{CO}: \frac{R_3}{R_4} = \frac{R_1}{R_2} + \frac{C_2}{C_1} \text{ and } \mathrm{FO}: \omega_0 = \sqrt{\frac{1}{C_1 C_2 R_1 R_2}} \tag{10.1}$$

If, apart from applying Theorem 1, we also invoke Theorem 2, then ten[1] distinctly different variants of the WBO are possible which are shown in Fig. 10.4. However, it may be noted that not all of them are realizable with a conventional differential-input grounded output (DIGO) type op-amp; as many as six of them, in cases where none of the terminals of the norator are chosen as the ground node, require a four-terminal floating nullor (FTFN) as the active element. Thus, the circuits in Fig. 10.4a, b, e, h, i, j need FTFN for practical realization. It may be seen that a

[1] This number was further increased to sixteen in [20] by considering the junction of the series RC branch as another node.

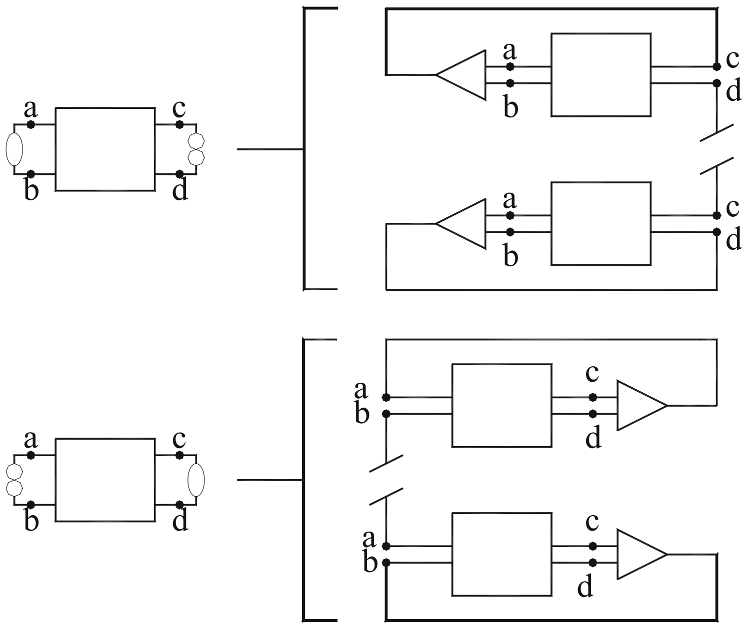


Fig. 10.1 Oscillator N and N^{*} and the four physical oscillators resulting there from, for the special case of N with $m = 1$

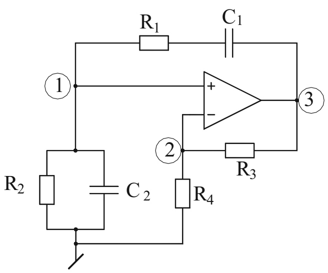


Fig. 10.2 The classical Wien bridge oscillator (WBO)

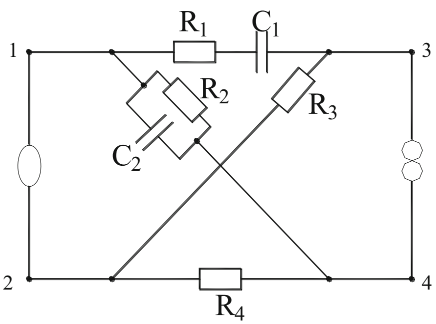
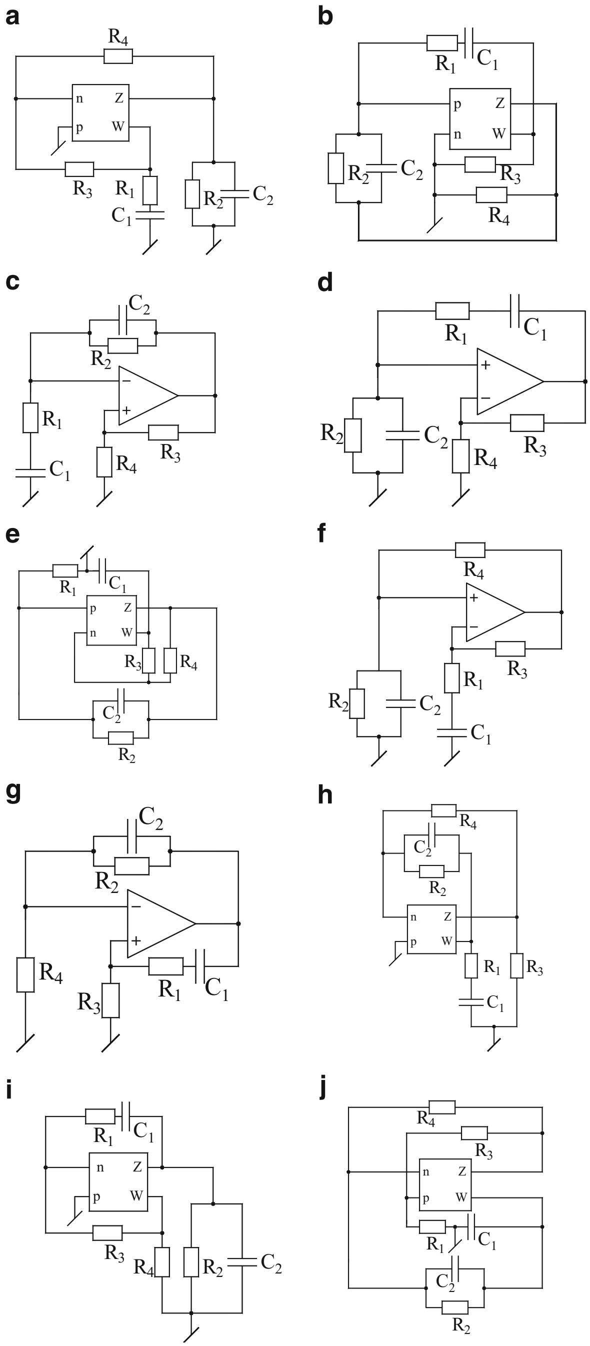


Fig. 10.3 Nullor model derived from the Wien bridge oscillator in Fig. 10.2

Fig. 10.4 Ten equivalents of the Wien bridge oscillator



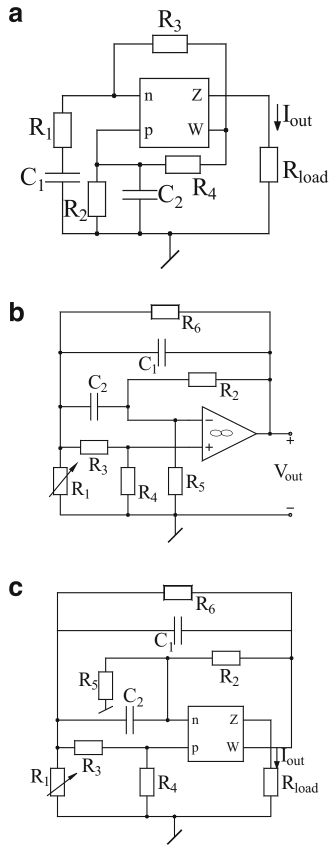


Fig. 10.5 Different equivalents of some known oscillators. (**a**) Explicit current output version of Wien bridge oscillator. (**b**) Senani's SRCO. (**c**) Explicit current output version of Senani's SRCO

number of circuits have the novel feature of employing both *grounded capacitors* as preferred for IC implementation [21, 22], such as those in Fig. 10.4a, f.

It is, thus, seen that on applying Theorems 1 and 2 on existing oscillators, we can derive a number of equivalents having the same characterization such that some of the derived equivalents indeed may have new and interesting properties not available in the original circuit. As another example, in Fig. 10.5a, we show a version of WBO which provides an explicit current output (ECO). Note that this property is not available in the classical WBO. As yet another example, consider Senani's oscillator [23] which is reproduced here in Fig. 10.5b.

For this oscillator, the CO and FO are given by

$$\frac{R_4}{R_6} = 2\frac{R_5}{R_2} + \frac{R_5}{R_6}; C_1 = C_2 = C \tag{10.2}$$

$$f_0 = \frac{1}{2\pi C}\sqrt{\frac{\left(1 + \frac{R_3}{R_6} + \frac{R_3 + R_4}{R_1} - \frac{R_2 R_4}{R_5 R_6}\right)}{R_2 R_3}} \tag{10.3}$$

which reduces to

$$f_0 = \frac{1}{2\pi C}\sqrt{\frac{(R_3 + R_4)}{R_1 R_2 R_3}} \tag{10.4}$$

provided

$$R_5(R_3 + R_6) = R_2 R_4 \tag{10.5}$$

From the above, it is seen that CO is controllable independently by R_5, whereas FO is independently controllable by R_1.

A variant of Senani's oscillator capable of providing ECO is shown in Fig. 10.5c. The circuits in Fig. 10.5a, c are obviously relevant in the context of CM signal processing.

To demonstrate the capability of Theorems 1 and 2 in generating equivalent oscillators possessing properties which are not available in the original circuit, we consider Senani's SRCO shown in Fig. 10.5b once again. A nullor model of this SRCO, with ground node excluded, is shown in Fig. 10.6a, whereas Fig. 10.6b shows an alternative nullor model (having the same CE) obtained by applying Theorem 1 on the model in Fig. 10.6a. Since each model contains five nodes, it follows from Theorem 2 that for each model, ground node can be selected in five different ways, thereby leading to a total of ten equivalent SRCOs having the same CE. It is not difficult to visualize that only four of these would be realizable with a normal op-amp while the remaining six would require an FTFN. Two of the new SRCOs generated by this methodology (resulting from treating node 5 as ground in both cases) are shown in Fig. 10.7a, b and employ grounded capacitors as preferred

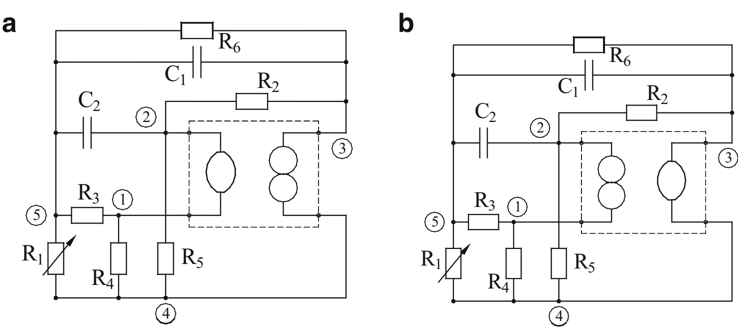


Fig. 10.6 (**a**) A nullor model of the SRCO in Fig. 10.5. (**b**) An alternative nullor model of the same circuit applying Theorem 1

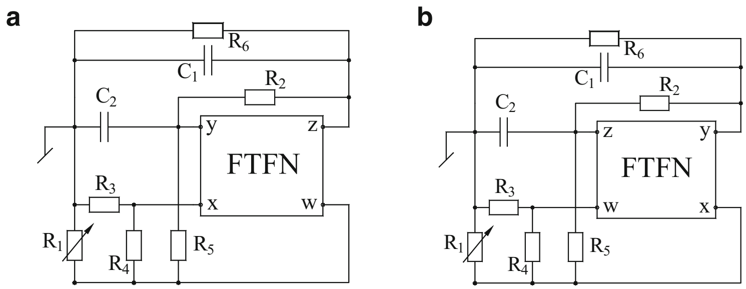


Fig. 10.7 Two FTFN-based grounded-capacitor versions of the SRCO in Fig. 10.5b

for IC implementation [21, 22] and are particularly noteworthy since this feature was not present in the original circuit in Fig. 10.5b. Also, four of the quoted ten circuits can provide explicit current outputs (as demonstrated in the circuits in Fig. 10.5a, c), again a feature which was not there in the original circuit.

10.3 Application of Network Transposition in Deriving Equivalent Forms of OTA-C Oscillators

During the past three decades, there has been a lot of interest in the literature in current-mode circuits and techniques because of their potential advantages. There have also been many methods of transforming the voltage-mode circuits to current-mode circuits, many of which employed the notion of adjoint networks. However, throughout this development, it was somehow overlooked by most of the researchers (the only exception being [18]) that the concept of deriving the current-mode structures from a voltage-mode structure goes back to 1971 when Bhattacharyya and Swamy [24] introduced the concept of network *transposition*. It was shown that through network transposition, a given network N could be easily converted to another network N^T whose admittance matrix is the transpose of that of N. The authors gave the transpose of a number of nonreciprocal elements including the four controlled sources and a number of impedance converters and invertors. It was demonstrated in [24] that the concept of network transposition facilitates the realization of a current transfer function which is identical to the voltage transfer function of a given voltage-mode network. This method was also shown to be useful to derive alternative equivalent structures of one-port networks. In retrospection, it is found that for linear networks, the *transpose* is essentially the same as *adjoint*.

OTA-C oscillators are suitable candidates for realizing fully integratable oscillators in bipolar as well as CMOS technology and have been widely investigated; see [25–32] and numerous other references cited therein and those in Chap. 3 of this monograph. Swamy, Raut, and Tang [33] demonstrated a simple but very useful

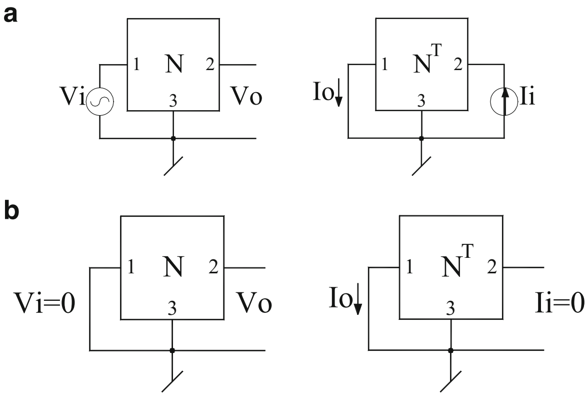


Fig. 10.8 Application of the concept of network transposition

approach based on *network transposition* to derive new CMOS sinusoidal oscillators using OTAs and capacitors.

The methodology is based upon the notion that for a given linear network N, its transposed network N^T can be obtained simply by replacing the nonreciprocal elements by their respective transposes while leaving reciprocal elements as it is as shown in Fig. 10.8a. By doing this, voltage transfer function of N in the forward direction would be the same as the corresponding transfer function of N^T in the reverse direction and vice versa. Thus, a current-mode OTA-C circuit can be obtained in a straightforward manner by simply changing the input and output terminals of each OTA and reversing the input and output ports. Obviously, both the circuits would have the same transfer function. If the given circuit is an oscillator (see Fig. 10.8b), the transposed circuit would also be an oscillator, both having the same CE.

Using this methodology, Swamy, Raut, and Tang [33] derived a number of transposed OTA-C oscillators corresponding to the OTA-C oscillators known earlier in [49] and other works. Some of the derived oscillators are shown in Fig. 10.9.

Obviously, the catalogue of new OTA-C oscillators as given in Fig. 10.9 does not end here. All other available OTA-C oscillators from [25–30] and the references cited therein would lead to as many new transposed OTA-C oscillators.

10.4 Derivation of Equivalent Forms of OTA-RC Oscillators Using the Nullor Approach

Operational transconductor amplifier (OTA)-based oscillators using only capacitors (OTA-C) or using RC elements (OTA-RC) have been widely investigated due to the availability of electronic tunability of the oscillation frequency in such oscillators (for instance, see [25–32, 34, 35] and the references cited therein).

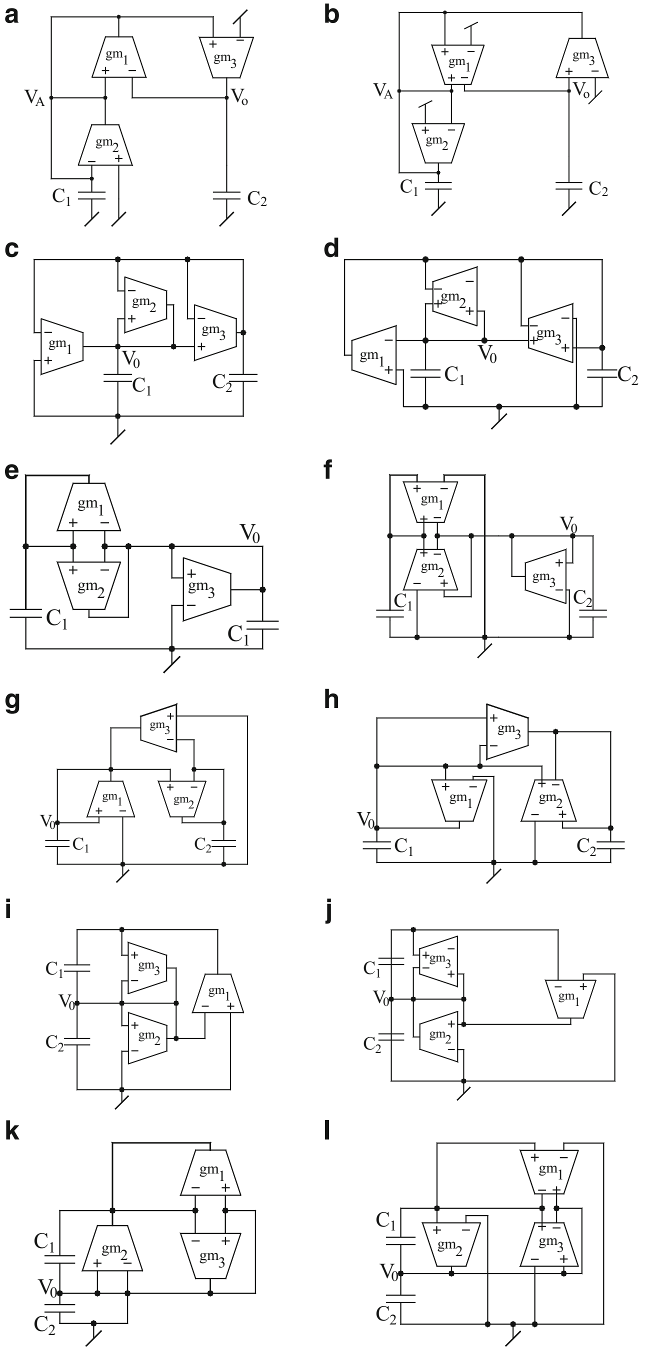


Fig. 10.9 Derivation of the equivalent OTA-C oscillators using network transposition: Circuits **a**, **c**, **e**, **g**, **i** and **k** are known OTA oscillator structures published earlier in [49]. Corresponding new transposed oscillator structures derived in [33] are the circuits **b**, **d**, **f**, **h**, **j** and **l**

In [36], it was shown that corresponding to any given single-OTA-RC sinusoidal oscillator, there are three other structurally distinct equivalent forms (having the same characteristic equation (CE)), one of which employs both grounded capacitors (GC) as preferred for IC implementation [21, 22]. Likewise, in [37], it was demonstrated, through five theorems, that each of the two dual-OTA-RC oscillators considered therein has three other equivalent forms having the same CE, one of which employs both GCs. Swamy, Raut, and Tang [33] showed that from a given OTA-C oscillator, a new OTA-C oscillator structure having exactly the same CE as the original circuit can be derived using the operation of *network transposition* [24] (see [38] for an interesting exposition of the concept of *network transposition* in deriving adjoints of various multiterminal modern active building blocks).

It was demonstrated by Senani, Gupta, Bhaskar, and Singh [39] that the nullor-based theory of op-amp-RC oscillator equivalencies presented earlier in [18, 19] is extendable to OTA-RC oscillators and that by doing so, a much larger number of equivalent OTA-RC oscillators having the same CE can be possibly derived than with the theories presented in [33, 36, 37]. The methodology of [39] is presented here and is shown to lead to several other important consequences, thereby considerably extending the utility of the generated additional OTA-RC oscillator equivalents.

OTAs could be a differential-input single-output (DISO) type or differential-input dual-output (complementary) (DIDO) type. The nullor models for both are shown in Fig. 10.10 which is self-explanatory.

For the present purpose, we assume that, in the most general form, an OTA-RC oscillator can have a DIDO-type OTA. It may be mentioned that an implementation of a DIDO-type OTA can be obtained from the bipolar or CMOS architecture of a DISO-type OTA simply by addition of two pairs of complementary current mirrors to create the second current output terminal providing a current output complementary to the current output of the first output terminal. Furthermore, both types of OTAs are commercially available as off-the-shelf ICs, for instance, LM3080/LM13600/LM13700 being the examples of DISO-type OTAs, while MAX 435 being an example of the DIDO-type OTA.

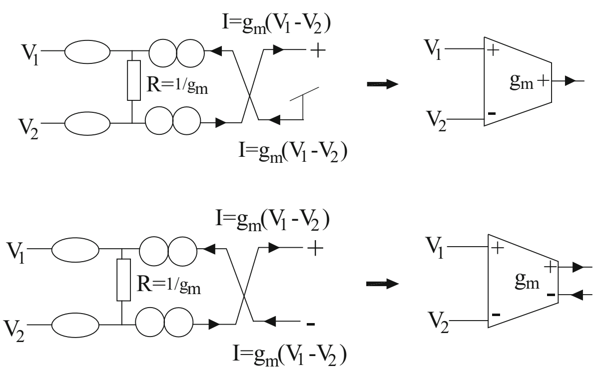


Fig. 10.10 Nullor models of the OTAs. (**a**) DISO-type OTA. (**b**) DIDO-type OTA

In Sect. 10.2 two theorems were presented which were shown to be useful artifices for generating equivalent op-amp RC oscillators, many of which were shown to possess interesting properties not present in the corresponding parent oscillator circuits. Though not demonstrated therein, it was envisaged in [18, 19] that the theory could also be applied to oscillators using any other active building blocks which can be modeled by nullors. In view of this, it follows that since an OTA can be represented by a pair of nullors and a single resistor $R = 1/g_m$, the theory should also be applicable to OTA-RC oscillators of [34, 35] as well. In the following, we demonstrate that it is indeed so and the resulting equivalent OTA-RC oscillator structures exhibit many interesting properties not present in the chosen parent OTA-RC oscillators.

Derivation of equivalent forms of single-OTA-RC oscillators: A single-OTA-RC oscillator, in the general form, can be shown as in Fig. 10.11a, where "*N*" represents a 4-port RC circuit in which a DIDO-type OTA is embedded. Figure 10.11b shows the nullor representation of such an oscillator.

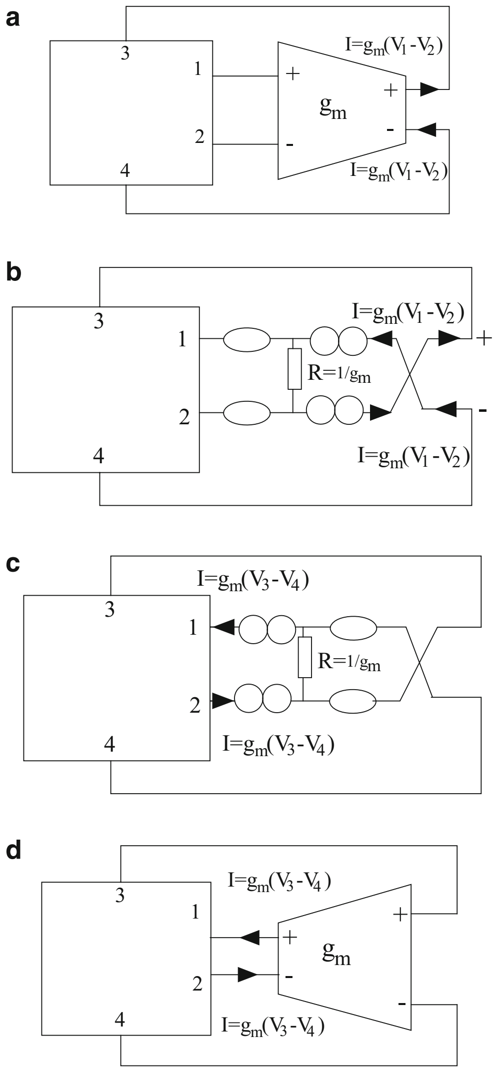
Now applying Theorem 1 on the nullor circuit in Fig. 10.11b, we obtain an alternative oscillator having the same CE as in Fig. 10.11c from which the corresponding OTA-RC configuration having the same CE may be drawn as shown in Fig. 10.11d. It is interesting to note that the configuration in Fig. 10.11d is also derivable from that in Fig. 10.11a by applying the operation of *network transposition* [33].

On each of the circuits in Fig. 10.11a, d, now we can apply Theorem 2. As per Theorem 2, from the circuit in Fig. 10.11a as well as that in Fig. 10.11d, four additional circuits can be generated by *relocating* the ground terminal in each case, thereby resulting in a family of ten single-OTA-RC oscillators having exactly the same CE. Eight of these circuits are symbolically shown in Figs. 10.12a–d and 10.13a–d, while the remaining two are the circuits in Figs. 10.11a–d. It is now interesting to note the following.

In the circuits in Figs. 10.12d, c and 10.13b, a, one of the output terminals of the DIDO-type OTA becomes redundant due to being grounded. Clearly, such circuits, therefore, can be realized practically by a normal DISO-type OTA as well, such as LM3080/LM13600/LM13700. However, it may be further noted that the oscillator circuit d in Fig. 10.12 and circuit b in Fig. 10.13 will in such a case be exactly the same as those in Fig. 1a and c of [36]; while on the other hand, the circuit c in Fig. 10.12 and circuit a in Fig. 10.13 are analogous to those in Fig. 1 b and d of [36] but will become exactly the same if in both cases (i.e., in the circuits in Figs. 10.12c and 10.13a) the polarities of the input terminals are reversed which will automatically result in the current going out of the output terminal which is the usual case in a DISO-type OTA.

Consider now the two single-OTA-RC circuits depicted in Figs. 3a and 4a of [36] which are reproduced here, for convenience, in Fig. 10.14a, b which also shows the DIDO-OTA-based version of these oscillators, with ground node assumed external to the circuits. Now, starting from the DIDO-OTA RC oscillator circuits in Fig. 10.14a as well as Fig. 10.14b, by successive application of Theorems

Fig. 10.11 Nullor model of a single-OTA-RC oscillator and its transform according to Theorem 1



1 and 2, ten single DIDO-OTA-RC oscillators can be derived in each case, the details of which are omitted to conserve space. Out of the ten derivable circuits corresponding to the circuits in Fig. 10.14, the four circuits derived would be exactly the same as in [36], subject to the compliance of the condition specified earlier; however, there would be six new oscillator circuits in this set which, however, have not been known earlier. In a similar manner, using the present theory, a total of ten equivalent OTA-RC oscillators are derivable from the oscillator circuit in Fig. 4a of [36], out of which there would be six new circuits (not shown here).

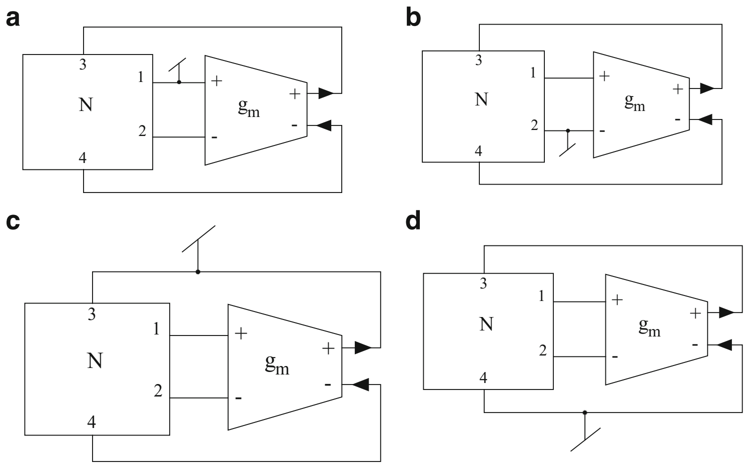


Fig. 10.12 General representation of the four additional equivalent forms of a given single-OTA-RC oscillator based on the circuit in Fig. 10.11a

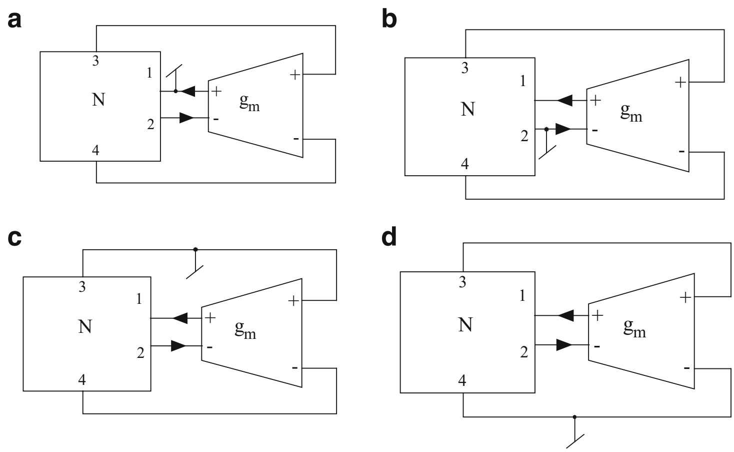


Fig. 10.13 General representation of the remaining four additional equivalent forms of the same oscillator *but* based upon the model in Fig. 10.11d

Derivation of equivalent forms of dual-OTA-RC oscillators: Using five theorems, three equivalent OTA-C oscillators having the same CE corresponding to each of the two oscillator circuit examples taken were derived in [37]. We now show that if we derive equivalent forms of dual-OTA-RC oscillators from those exemplified in [37], by using the method presented here, we will get more equivalent oscillator circuits than those derived in [37].

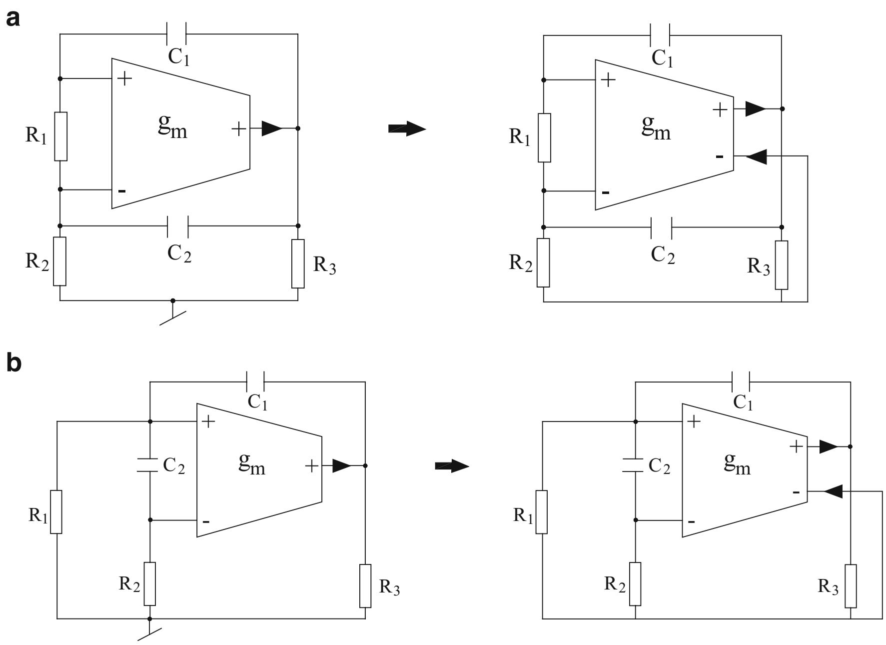


Fig. 10.14 (**a**) Single-OTA-RC oscillator in Fig. 3a of [36] and its representation as a single DIDO-OTA-RC oscillator. (**b**) Single-OTA-RC oscillator in Fig. 4a of [36] and its representation as a single DIDO-OTA-RC oscillator

By using Theorems 1 and 2, we first show that there are a total of eight equivalent oscillator circuits having the same CE corresponding to the oscillator circuit of example 1 of [37] (reproduced here in Fig. 10.15).

The CE of this oscillator is given by

$$C_1C_2R_1s^2 + (C_1 + C_2 - g_{m1}C_2R_1)s + g_{m2} = 0 \tag{10.6}$$

Note that in this oscillator, the condition of oscillation (CO) is controllable through g_{m1}, while frequency of oscillation (FO) can be independently controllable through $g_{m2.}$

The starting point of the required derivation is the configuration shown in Fig. 10.16a, where the circuit in Fig. 10.15 has been redrawn with dual-output OTAs (instead of single-output OTAs) and the ground node has been considered external to the entire circuit. By applying Theorem 1 on this circuit, we obtain the corresponding circuit in Fig. 10.17a (which can also be obtained by applying the operation of *network transposition* [33] on the circuit in Fig. 10.16a). Now the generation of three additional circuits, i.e., circuits in Fig. 10.16b, c, d from that in Fig. 10.16a and circuits in Fig. 10.17b, c, d from that in Fig. 10.17a, is derivable by invoking Theorem 2 and choosing the ground node as node 3, 2, and 1 respectively, in each case. It may now be noted that the equivalent circuits in

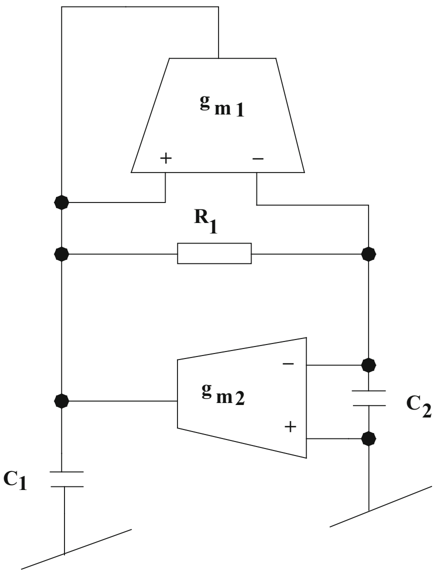


Fig. 10.15 Dual-OTA-RC oscillator of example 1 of [32]

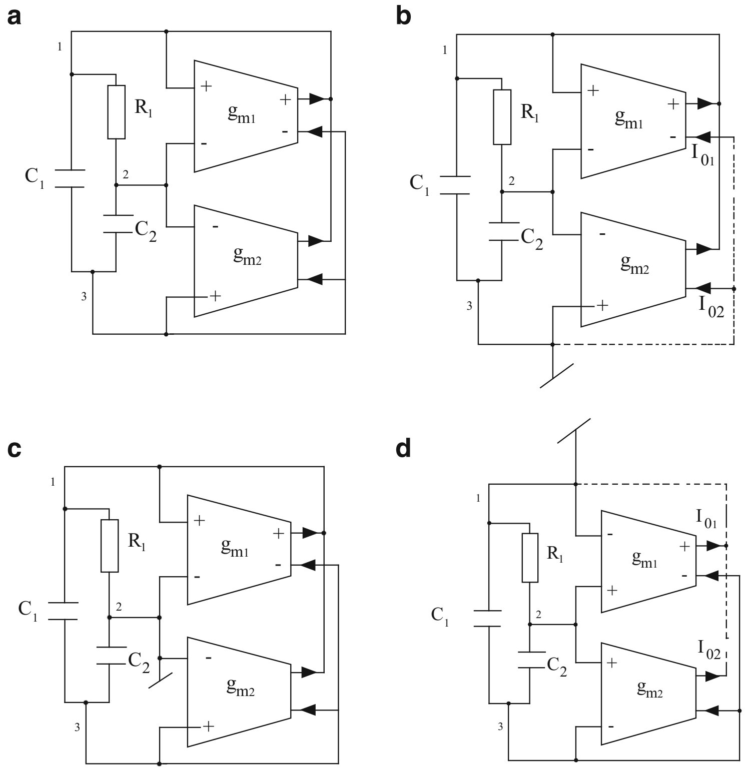


Fig. 10.16 Four equivalent forms of the dual-OTA-RC oscillator in Fig. 3a of [37]

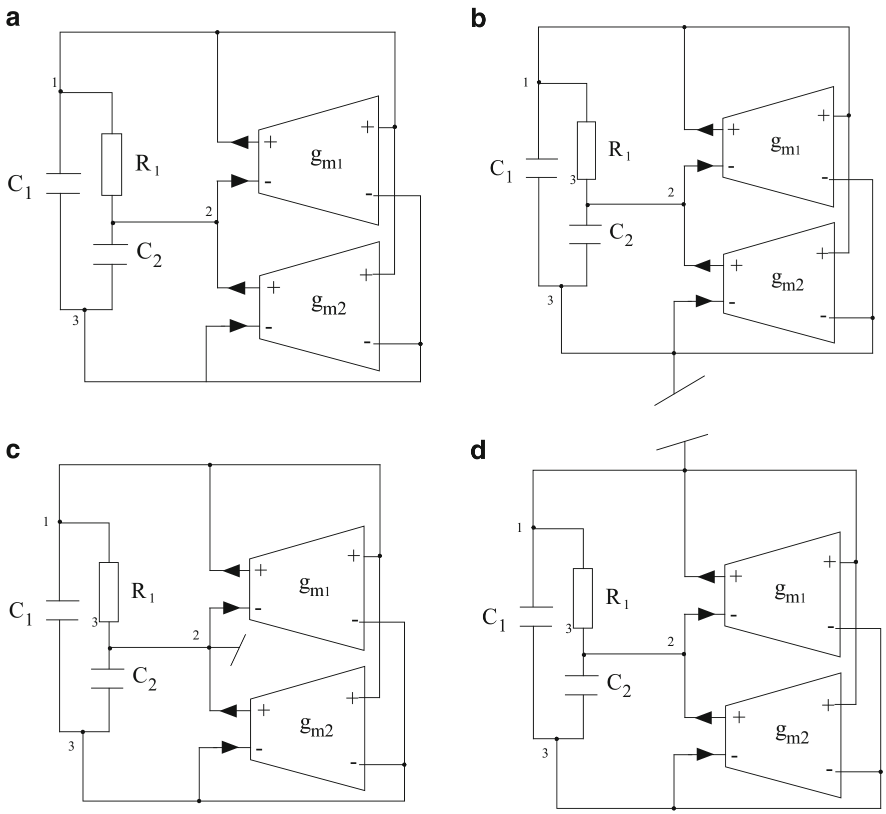


Fig. 10.17 Remaining four equivalent forms of the dual-OTA-RC oscillator in Fig. 3a of [37]

Figs. 10.16b and 10.17c are the same as derived in example 1 of [37] and Fig. 10.16d will become exactly the same if polarities of the input terminals of both the OTAs are reversed, while the remaining five circuits, namely, those in Figs. 10.16a, c and 10.17a, b, d, *are entirely new*.

The following are now worth considering.

In the context of grounded-capacitor-based oscillators, as preferred for IC implementation [40], it is interesting to note that, contrary to the disclosures of [36], there would result two *both-GC* single-OTA-RC oscillators in the set of equivalent oscillators resulting from the application of the procedure explained in this paper on all single-OTA-RC oscillators considered in [36]. Similarly, there are two *both-GC* oscillator equivalents as shown in Figs. 10.16b and 10.17b, here in contrast to only a single *both-GC* oscillator derivable from the theorems of [37] (see Fig. 3a therein).

In case of single-OTA-RC oscillators in Figs. 10.12d, c and 10.13a, b, one output terminal of the OTA is directly connected to ground. From an inspection of the circuits, it can be readily realized that this grounded output terminal can be readily

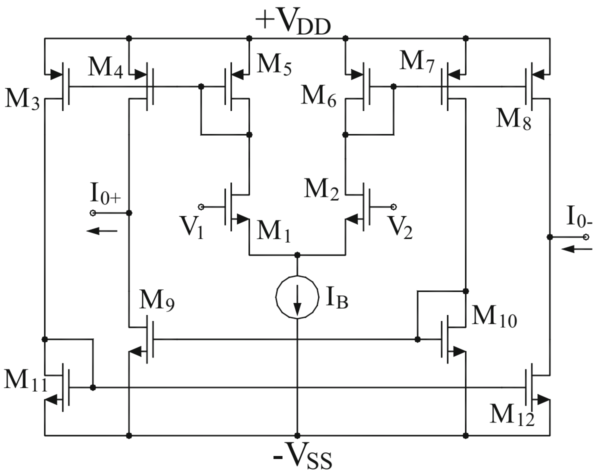


Fig. 10.18 Multi-output Operational Transconductance Amplifier (MO-OTA)

ungrounded without altering the CE, and hence, these single-OTA-RC oscillators are capable of providing an *explicit current output from high output impedance nodes*. This is important in view of the fact that OTA oscillators providing explicit current outputs are the subject of considerable contemporary interest in literature (for instance, see [31] and [32]) due to their possible applications as signal generators for testing current-mode signal processing circuits.

In the class of two-OTA-RC oscillators exemplified in this paper, those in Figs. 10.16 b, d and 10.17 b, c, d have one or two outputs directly connected to ground. Due to this, as explained above, the oscillators in Fig. 10.17b and d have one explicit current output available in each OTA (after ungrounding), while in case of the oscillators in Figs. 10.16b, d and 10.17c, two explicit current outputs are available (one each from each OTA).

It is further interesting to note that all those oscillators which have two explicit current outputs, i.e., oscillators in Fig. 10.16b and d, have the possibility of having the two available output currents in quadrature, thereby making these circuits suitable for quadrature output generation. A reanalysis of these circuits reveals that, indeed, this is true. It has been found that the two-OTA-RC oscillators in Figs. 10.16b, d are both characterized by

$$\frac{I_{o2}(s)}{I_{o1}(s)} = \frac{g_{m2}}{R_1 g_{m1} s C_2} \tag{10.7}$$

without any condition, and thus, these circuits are current-mode *quadrature oscillators*.

The workability of the OTA-RC oscillators has been confirmed in [39] by simulating all the circuits in SPICE using DIDO-type CMOS OTA shown here in Fig. 10.18. The process parameters of 0.18 μm CMOS technology were employed in the simulations. The DC bias supply used was ±1.0 , and the DC bias current I_B

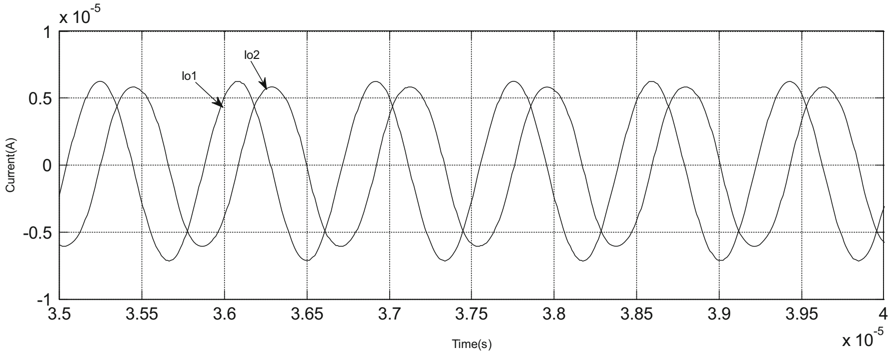


Fig. 10.19 SPICE-generated waveforms for the oscillator in Fig.10.16b

was chosen appropriately in different cases. All the single-OTA-RC oscillators as well as all dual-OTA-RC oscillators described in this paper were simulated and were found to work as predicted by theory. However, to conserve space, we present the simulation results of current-mode quadrature oscillator in Fig. 10.16b only. The circuit was tested with $C_1 = C_2 = 10$pf, $R_1 = 30\text{K}\Omega$, $I_{B1} = 8.4\ \mu\text{A}$, $I_{B2} = 35\ \mu\text{A}$ which resulted in oscillation frequency of 1.2 MHz exhibiting the THD of 0.8 %.

Figure 10.19 shows the waveforms obtained from the circuit in Fig. 10.16b. The measured value of the phase shift between the two waveforms has been found to be 88.916°. Thus, the waveforms are *almost* in quadrature, with the error in the phase difference being merely 1.2 %.

Thus, the nullor-based theory of deriving op-amp RC equivalent oscillator structures [18, 19] as extended to OTA-RC oscillators as well is more general than that presented in [36, 37] (through a number of theorems presented therein). This has been confirmed by demonstrating the following:

(a) The presented nullor-based theory shows the existence of ten equivalent forms for each of the two single-OTA-RC oscillators considered in [36] for which only four equivalents had been derived therein.
(b) When applied to the dual-OTA-RC oscillators of [37], eight to ten equivalent forms of the dual-OTA-RC oscillators are derivable as against only three equivalent OTA-C oscillators corresponding to each of the two circuits derived through five theorems in [37].
(c) The theory presented in [39] shows that, in contrast to what has been demonstrated in [36, 37], corresponding to a given OTA-RC oscillator, there are *two equivalents* (having the same CE) that employ *both grounded capacitors rather than only one, as shown in* [36, 37].
(d) In case of single-OTA-RC oscillators in Figs. 10.12 and 10.13, some of the single-OTA-RC oscillators are capable of providing an explicit current output at high output impedance nodes. This is important in view of the fact that OTA oscillators providing explicit current outputs are the subject of considerable contemporary interest in literature (for instance, see [31] and [36]) due to their

possible applications as signal generators in current-mode signal processing circuits.

(e) In the class of two-OTA-RC oscillators discussed in this paper, some of these have one or two outputs connected directly to ground. Due to this, a number of generated equivalent circuits are capable of providing explicit current outputs.

(f) It is further interesting to note that among all oscillators enumerated above which have two explicit current outputs, oscillators in Fig. 10.16b, d have the two available output currents in quadrature.

It is worth mentioning that the disclosure of these interesting properties has been possible because the presented unified theory of deriving equivalent structures starting from OTA-RC oscillator models is based on differential-input- and complementary dual-output-type OTAs rather than single-output OTAs. This aspect of OTA-RC oscillators does not appear to have been dealt with in the earlier literature.

In conclusion, the nullor-based theory [18, 19] for deriving equivalent OTA-RC sinusoidal oscillators, as presented above (which also embodies the *network transposition* or adjoint *transformation* of [33]), is more general than the methods of deriving equivalent OTA-RC oscillators presented in [36, 37] since it not only produces more number of equivalents but also generates a number of new topologies having practically useful properties which have not been available in the chosen parent OTA-RC oscillator(s).

10.5 Derivation of Oscillators Through Network Transformations Based on Terminal Interchanges

Rathore and Bhattacharyya [40] demonstrated that by appropriate transformations on two subnetworks of a sinusoidal oscillator feeding each other and consisting of linear passive elements and ideal op-amps, five additional equivalent oscillators (having the same characterization in terms of condition of oscillation and frequency of oscillation) can be achieved.

The transformations quoted above were presented by Rathore and Singhi [41] and are summarized in the following. Assuming that an op-amp RC network having a voltage ratio transfer function $T(s)$ is available, from this op-amp RC networks, implementing the other related transfer function (to follow) can be obtained by applying the following network transformations on the given op-amp-RC oscillator network N:

$$\tau_{IE}(T) = (1 - T) \tag{10.8a}$$

$$\tau_{OE}(T) = T/(1 - T) \tag{10.8b}$$

$$\tau_{OEI}(T) = (1 - T)/T \tag{10.8c}$$

$$\tau_{OIE}(T) = 1/(1 - T) \tag{10.8d}$$

$$\tau_{OI}(T) = 1/T \tag{10.8e}$$

In the above equations, T represents the transfer function of the original op-amp-RC oscillator network. On the other hand, the various operators on the left-hand side of the equations can be explained as follows. The operator symbolically denoted as T_{abc} means that the voltages at the terminals *a, b, c* of the transformed network should be brought to the voltages of terminals *b, c, a* of the original op-amp RC oscillator network *N*. Thus, as an example, T_{OIE} implies that the output, input, and earth (ground) terminals of the transformed network must be brought to the potentials of the terminals *I*, *O,* and *E* of *N*, respectively.

When these transformations are applied on a given oscillator, such as the one symbolically shown in Fig. 10.20a, five new oscillator structures having exactly the same CE as the original oscillator are obtained, as given in Fig. 10.20b–f.

To the best knowledge of the authors, this elegant theory of generating new equivalences has still not been applied by anybody on any specific two-op-amp oscillators, and this aspect deserves further exposition.

10.6 Transformation of Biquadratic Band-Pass Filters into Sinusoidal Oscillators

It has already been elaborated in the previous section how some network transformations [41] based upon interchange of input, ground, and output terminals of a given oscillator network can lead to a number of alternative equivalent circuit structures characterized by the same CE [40]. This methodology proposed by Rathore and Bhattacharya [40] was mainly tailored toward op-amp RC oscillators.

Recently, Wang, Tran, Nguyen, Yien, and Lie [42] rediscovered[2] nearly the same methodology in the context of current conveyors taking due cognizance of the fact that a CCII− is equivalent to a three terminal floating nullor. In the quoted methodology [42], there are two sequences of operations which are symbolically denoted as follows:

1. First transformation sequence

$$T \rightarrow (1 - T) \rightarrow \frac{1}{(1 - T)} \rightarrow \frac{T}{(T - 1)} \tag{10.9}$$

[2] Ironically, the author of [42] has not taken any cognizance of the very closely related earlier works of [40] and [41] while doing so!

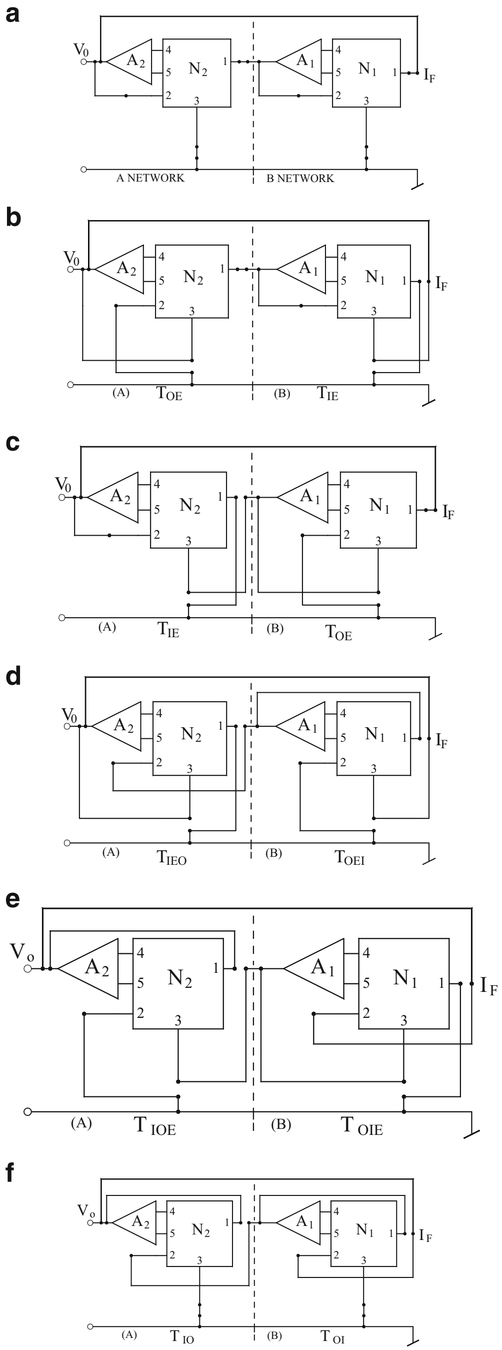


Fig. 10.20 Equivalent oscillator network derived in [40] resulting from the application of *network transformations* introduced by Rathore and Singhi [41] on a given op-amp RC oscillator

2. Second transformations sequence

$$T \rightarrow \frac{1}{T} \rightarrow \frac{(T-1)}{T} \rightarrow \frac{T}{(T-1)} \tag{10.10}$$

From the above representations, it is clear that in both cases, there is a successive use of *complementary* and *inverse* transformations. It is well known that complementary transformation applied on a voltage-mode network is implemented by interchanging the input and output of a network which converts T into $(1-T)$. On the other hand, inverse transformation involves interchanging the output norator and the input voltage sources of a circuit in which case a circuit having transfer function T gets transformed into a circuit having transfer function $(1/T)$. Furthermore, inverse network can also be obtained by interchanging the output pathological current mirror and the input source of a circuit.

The applicability of this approach was demonstrated by two examples in each of which starting from an appropriate band-pass filter using CCII; the authors derived a CCII-based oscillator. One of these examples is shown in Fig. 10.21a in which the starting circuit is a two-port CCII-RC network characterized by the transfer function

$$\frac{V_{out}}{V_{in}} = T(s) = \frac{s\left(\frac{C_2}{R_2}\right)}{s^2C_1C_2 + s\left(\frac{C_2}{R_1}\right) + \frac{1}{R_2R_3}} \tag{10.11}$$

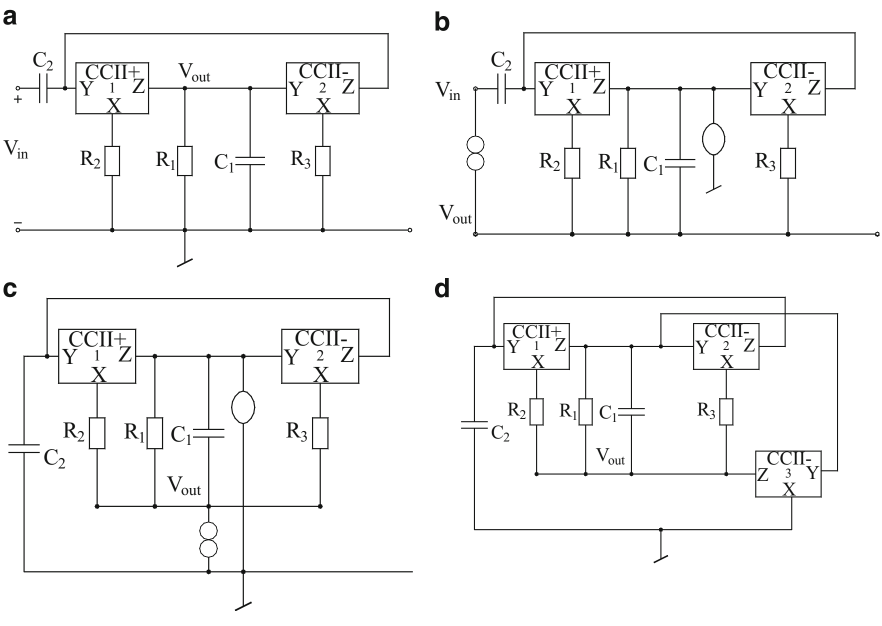


Fig. 10.21 The various steps involved in the transformation of a CCII-based band-pass filter into an oscillator as per the method proposed by Wang, Tran, Nguyen, Yien, and Lie [20]

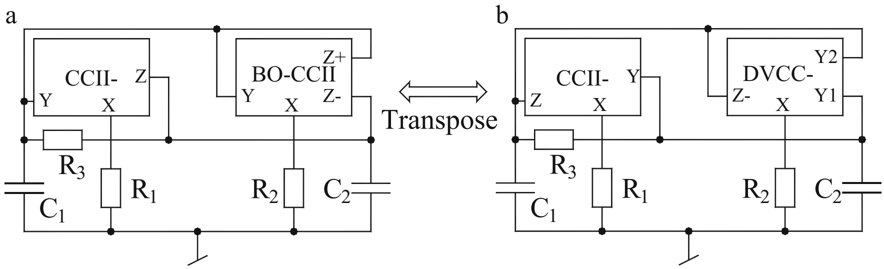


Fig. 10.22 Transformation of a CCII- and BO-CCII-based oscillator into an equivalent oscillator having the same CE by Swamy's method of network transposition [38]

Now by applying the first transformation sequence, the resulting CCII-based circuit having transfer function ($T/(T-1)$) is shown in Fig. 10.21b. However, in doing so, a series nullator-norator pair has been added during performing the inverse transformation for which the actual transfer function is given by

$$T'(s) = -\frac{s\left(\frac{C_2}{R_2}\right)}{s^2C_1C_2 + \left(\frac{1}{R_1} - \frac{1}{R_2}\right)C_2s + \frac{1}{R_2R_3}} \tag{10.12}$$

From the above equation, it is clear that an oscillator can be realized by setting the input voltage to zero which results in the circuit in Fig. 10.21c. It may be noted that the nullator-norator pair can be easily realized by a CCII−. This finally results in the transformed oscillator circuit in Fig. 10.21d.

The condition of oscillation and the frequency of oscillation for the final circuit obtained are given by

$$R_1 = R_2 \,\text{and}\, \omega_0 = \sqrt{\frac{1}{C_1C_2R_2R_3}}. \tag{10.13}$$

It appears that interesting oscillator topologies might result if this method is applied to convert other CC-based band-pass circuits to generate oscillator topologies.

10.7 Transformation of Oscillators Involving Device Interchanges

Recently, Swamy [38] presented a simple direct method of finding the *transpose* or *adjoint* of a multiterminal network element without having to represent it in terms of nullators and norators. The new theory was related to a very earlier work on *network transposition* published in early eighties [24]. The new procedure involves expressing the voltage and the current variable relation as a hybrid matrix and then deriving a simple relation between the hybrid matrices of the original and the *transposed* element.

Through the proposed methodology, Swamy [38] demonstrated twelve different transposed pairs of elements. Some prominent transposed pairs of elements are as follows: voltage-mode op-amp (VOA) and current-mode op-amp (COA), differential-input (DI) VOA and balanced-output (BO) COA, DI-COA and BO-VOA, DIBO-VOA and DIBO-COA, DI-OTA and BO-OTA, CCII+ and ICCII−, DVCC+ and BOICCII, and DVCC− and BO-CCII; furthermore, the DIBO-OTA and its transposed are the same; CCII− and its transpose are the same; and finally, balanced-output DVCC and its transposed are also the same.

Swamy [38] demonstrated the application of the proposed theory in a number of applications which included derivation of alternate structures of simulated impedances and CM filter realization and deriving alternate structures for oscillators. In the last category, it was demonstrate that from a CCII−- and VO-CCII-based oscillator proposed by Horng, Lim, and Yang [43] shown here in Fig. 10.9a the corresponding transposed oscillator can be easily drawn to be the one shown in Fig. 10.9b by noting that the transpose of CCII− is a CCII− itself while the transpose of BO-CCII is a DVCC−.

It is obvious that both circuits in Fig. 10.10 will have the same characteristic equation, which is given by

$$s^2C_1C_2R_1R_2R_3 + sR_1(C_1R_2 + C_2R_2 - C_2R_3) + R_2 = 0 \tag{10.14}$$

from which the oscillation condition and the frequency of oscillation are given by

$$R_2 = (C_2R_3)/(C_1 + C_2) \tag{10.15}$$

$$\omega_0 = 1/\sqrt{C_1}C_2R_1R_3 \tag{10.16}$$

In both circuits, ω_0 can be independently adjusted using the grounded resistor R_1 and oscillation condition by the grounded resistor R_2. Finally, it can also be verified that both circuits have the same sensitivities with respect to corresponding elements.

10.8 Concluding Remarks

In this chapter, we discussed a number of network transformations applicable to sinusoidal oscillators realized with various active building blocks. It was observed that the application of such transformations not only generates a number of alternative configurations, but some of the generated equivalent realizations, indeed, possess some characteristic features or properties not available in the original starting circuit. This confirms that the exercise of generating a number of equivalent circuits of a given oscillator is not merely an academic exercise, but it does have practical significance.

A number of different techniques were highlighted through representative examples from the relevant works. However, after going through the various techniques applied, it would appear that this aspect of the research on sinusoidal oscillators is also not completely exhausted yet and a number of other suggestions such as those in [44–48] have enough scope for possible further work. It is, therefore, hoped that many new meaningful ideas and further applications have both enough scope of being discovered in the future.

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Chapter 11
Various Performance Measures, Figures of Merit, and Amplitude Stabilization/Control of Oscillators

Abstract In this chapter, the various performance criteria and figures of merit related to oscillators are highlighted and some prominent circuits and techniques of amplitude stabilization/control of oscillators are elaborated. Amplitude stabilization/control of current-mode oscillators is also considered.

11.1 Introduction

The performance evaluation of oscillators involves the consideration of many nonideal/nonlinear effects. These include the various types of noises and there effect, the most notable of these being the phase noise, the jitter noise and $1/f$ noise. The other important issues are the amplitude stability and frequency stability. The spectral purity of the signal generated is decided by the total harmonic distortion (THD). Once an oscillator has been designed a very critical issue is to ensure that when powered with a source of energy, the oscillations actually take place. Lastly, when the circuits have started generating oscillations, stabilization of the amplitude at the intended constant amplitude as well as the flexibility to continuously vary the amplitude of oscillation through a variable circuit element or an external voltage/current signal is another desirable property which practical oscillators are required to possess. This chapter addresses all the abovementioned aspects and the related issues [1–52].

11.2 Start-Up of Oscillations

The conventional theory of sinusoidal oscillators is by an large based upon the Barkhausen criterion according to which if an oscillator is made up of a linear amplifier of gain A and frequency selected feedback network of gain β, then the necessary (but not sufficient) criterion for the circuit to produce sustained oscillation is given by;

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$$|A\beta \geq 1| \text{ and } \angle A\beta = 0 \quad \text{or an integral multiple of } 2\pi \tag{11.1}$$

It may, however, be noted that since the condition given by Eq. 11.1 is only necessary and not sufficient there are bound to be counter examples, i.e., there could be circuit in which a $A\beta > 1$ but the roots of the characteristic equation (CE) may not be in the right half of the s-plane which is necessary to ensure that when power is switched on the noise present in the circuit is able to trigger action which results in an exponentially building sinusoidal signal. Thus, in view of the recent controversy[1] on the over reliance on the Barkhausen Criterion, it is clear that the more reliable method of ensuring the startup would be to examine the roots of the close loop characteristic equation of the oscillator and determine the condition under which the two complex conjugate roots can be placed slightly into the right half of the s-plane.

11.3 The Various Figures of Merit and Characterizing Parameters of Oscillators and Waveform Generators

In this section, we shall discuss about several important practical aspects related to oscillators which include harmonic distortion, frequency stability, phase noise, jitter, and $1/f$ noise.

11.3.1 Harmonic Distortion

Total harmonic distortion (THD) is the summation of all harmonic components of the voltage or current waveform compared against the fundamental component of the voltage or current wave:

$$\text{THD} = \frac{\sqrt{V_2^2 + V_3^2 + \ldots\ldots + V_n^2}}{V_1^2} \times 100 \tag{11.2}$$

The formula above shows the calculation for THD on a voltage signal. The end result is a percentage comparing the harmonic components to the fundamental component of a signal. The higher the percentage, the more distortion that is present on the main signal.

[1] For a discussion on this, see Sect. 15.2.5 of Chap. 15 of this monograph.

11.3.2 Frequency Stability

In classical oscillator theory, the frequency stability of a sinusoidal oscillator is related to the slope of the phase function of the open loop transfer function $T(s)$ of the oscillator, evaluated at the oscillation frequency.

Mathematically, the frequency stability factor is defined as:

$$S^{\mathrm{F}} = \left.\frac{d\phi}{du}\right|_{u=1} \tag{11.3}$$

where $\phi = \angle T(j\omega)$ and u represents the normalized frequency, i.e., $u = \frac{\omega}{\omega_0}$ where ω_0 is the oscillation frequency. For most of the classical oscillators like Wien bridge, RC-phase shift, Twin-T, and bridge-T oscillators, this figure is less than unity. However, for several novel classes of oscillators discussed in several chapters of this monograph, particularly those providing non-interacting single-element-controls for both the frequency of oscillation as well as condition of oscillation, it has been shown that this figure can assume a value proportional to $\sqrt{n}$ where n is the ratio of two resistors one of which is the frequency controlling resistor. In all such oscillators, therefore, S^{F} can be made quite high even if the parameter n is variable.

*11.3.3 Phase Noise, Jitter Noise and 1/*f *Noise in Oscillators*

Noise is one of the major concerns in oscillators, because of presence of even small noise in the oscillator may lead to dramatic changes in its frequency spectrum and timing properties.

In any physical oscillator, noise contributed by the passive circuit components and active devices used would perturb both the magnitude as well as the phase of the output signal generated by the oscillator. However, since every practical oscillator would have some kind of automatic gain control circuitry hence any amplitude fluctuations caused by the noise would be greatly attenuated by the amplitude stabilization/control circuitry but the phase noise present would still dominate.

Phase noise is a topic of great theoretical and practical interest in the context of oscillators. Phase noise is usually characterized in the frequency domain. It may be noted that the frequency spectrum of an ideal oscillator would be a single impulse at oscillation frequency ω_0. In case of a practical oscillator however, the spectrum exhibits *skirts* around the center (carrier) frequency. To quantify the phase noise we consider unit bandwidth at an offset $\Delta\omega_0$ with respect to ω_0, calculate the noise power in this bandwidth and divide the result by the carrier power.

From the numerous studies on the phase noise carried out [29–52], it is known that the oscillators based upon LC tank circuits and the classical Colpitts oscillator have better phase noise performance as compared to the RC oscillators although studies have also been made to determine minimum achievable phase noise in RC oscillators [49]. On the other hand, a number of authors have investigated methods of phase noise reduction in FET-based oscillators [29] and microwave oscillators [42].

In the time domain representation of the output of relaxation oscillators (such as a clock or square wave generator) the spacing between the output transitions is ideally expected to be constant. In practice, however, the transition spacing is variable due to *noise fluctuations*. This uncertainty is known as *timing jitter*.

Although *phase noise* is defined in frequency domain and the *jitter* is defined in time domain, they characterize the same phenomenon and methods are known to convert one to another. For more detailed information, the readers are referred to Refs. [187–235] of the list of *Additional references for further reading*, given at the end of this monograph.

11.4 Amplitude Stabilization and Control

The linear theory of oscillators based upon Barkhausen criterion or on state space analysis (assuming the amplifier or any other active elements employed as linear) although spells out the required conditions of oscillation and the frequency of oscillation however, it does not throw any light on the amplitude of the oscillations. If the roots of the closed loop characteristic equation of the oscillator are located in slightly right half of *s*-plane, the oscillations would be building up exponentially and in the absence of any deliberate arrangement for containing them, they would ultimately be limited by some inherent non linearity of the active device (such as the saturation type nonlinear characteristics of the op-amp in case of op-amp oscillators). In other oscillators, an external hard limiter is often used to limit the output of the sinusoidal waveform generated. This may also distort the output waveform and be manifested as harmonic distortion. Thus, an elaborate feedback mechanism to keep the magnitude at some specified level through either an automatic gain control (AGC) method or any other appropriate method would be necessary so that the generated waveform does not reach the saturation levels of the active device employed and thus harmonic distortion can be reduced/avoided.

More elaborate arrangements employ an additional feedback circuit which creates a dc signal by rectifying and filtering the oscillator output, compares it with the reference DC voltage and the error signal, thus generated, is amplified and is put to a voltage controlled resistance (VCR) realized with a FET used as a voltage variable resistor directly or through an integrator such that this VCR changes its resistance, which in turn modifies the value of the resistor controlling the condition of oscillation (thereby the location of the roots of the CE). If the amplitude of oscillator output is increasing due to any reason (such as when the frequency is being varied through a variable element in the circuit), this feedback arrangement operates in such a way that the change in the value of the condition setting resistor moves the roots of the CE to slightly into left half of the *s*-plane. On the other hand, when the oscillation amplitude decreases below the desired value, the feedback circuit ensures that the roots of the CE are moved to slightly in the right half of s-plane. When the loop action settles down, the amplitude of oscillations is stabilized at a value equal to the set reference level.

Although a very large number of sinusoidal oscillators have been evolved during the past four decades, on very few occasions the authors have addressed to the general problem of maintaining a constant amplitude (stabilization) of oscillations when frequency is varied by employing some variable passive element(s) (as in SRCOs) or through an external control voltage (as in VCOs) or making it possible to change the amplitude of the generated sine waves by varying external control/reference voltage (amplitude control). Even then, a number of methods for amplitude stabilization and control for both voltage mode and current mode oscillators have indeed been proposed by a number of researchers some of which are indeed general in nature in that they can be applied to any given oscillator by appropriate design. In the following, we highlight some methods of stabilizing as well as controlling the amplitude of oscillations in voltage-mode as well as current-mode oscillators.

11.4.1 Amplitude Stabilization/Control Using Analog Multipliers

It is well known that ensuring fast amplitude transients and reducing the distortion level to as small as possible are contradictory requirements which call for new concepts for designing efficient amplitude stabilization and control methods. A number of techniques have been advanced to meet this object out of which we discuss in the following a method based upon analog multipliers. The main basis of this method by Filanovsky–Fortier–Taylor [14] who applied their proposed techniques on the classical twin-T oscillators is identification that in this particular circuits there are two nodes a and b such that for sinusoidal signal generated by the circuit to be:

$$v_0 = V_m \sin\left(\omega_0 t + \frac{\pi}{4}\right) \tag{11.4}$$

these inner voltages are found to be given by

$$v_a = \frac{V_{m\sqrt{2}}}{4} \sin \omega_0 t \tag{11.5}$$

$$v_b = \frac{V_{m\sqrt{2}}}{4} \cos \omega_0 t \tag{11.6}$$

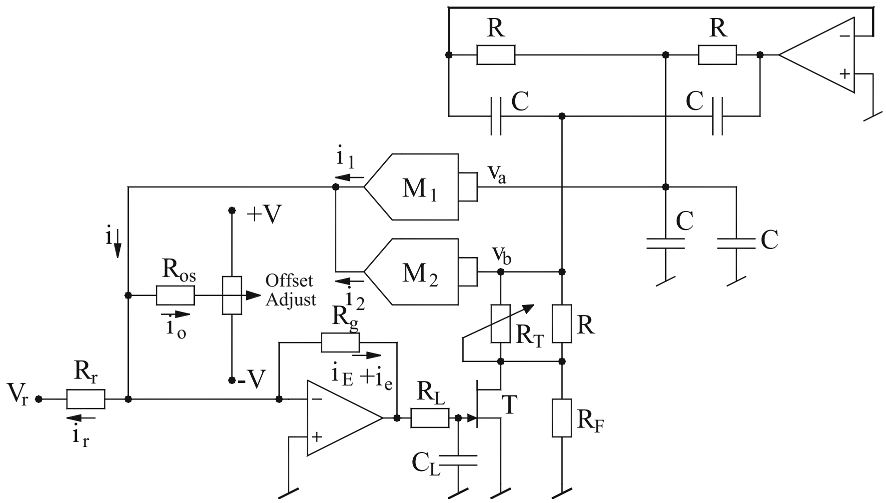


Fig. 11.1 The method of amplitude of stabilization and control using analog multipliers, as proposed by Filanovsky–Fortier–Taylor [14]

Therefore, V_a and V_b have equal amplitudes but are shifted in phase by exactly 90°. This property is exploited by Filanovsky–Fortier–Taylor [14] to produce a DC current signal proportional to the peak amplitude of the generated sine wave by employing to analog multiplier as shown in Fig. 11.1 from which it is also observed that through and op-amp a reference current i_r is generated such that error current i_e is converted into a voltage which is then employed to control the drain to source resistance of a FET used as a voltage control resistor (VCR). Obviously this VCR is part of a resistance in the circuit which actually controls the condition of oscillation such that under balanced condition, the output signal generated by the oscillator would be equal to some function of V_r. Alternatively, if the oscillation amplitude goes up or down from the predecided value, the error current forces the FET to offer a resistance value which will act to control the condition of oscillation in such manner that the amplitude will tend to decrease or increase accordingly.

By an inspection of the circuit, the following equations can be derived:

$$v_g \cong -i_e R_g \text{ where } i_e = i_0 + i - i_r \text{ and } i_r = \frac{V_r}{R_r} \tag{11.7}$$

$$i = K_1 v_a^2 + K_2 v_b^2 \tag{11.8}$$

where K_1 and K_2 are the transconductance coefficients of the multiplier. If $K_1 = K_2 = K$, then current i can be written as:

$$i = \frac{K V_m^2}{8} \tag{11.9}$$

To attain low THD, the drain voltage of the FET is made small by appropriate addition of a resistance of value 1.1 kΩ across it. On the other hand, to optimize the output amplitude transient response to step changes in V_r, the 50 kΩ is adjusted such that the FET in parallel with 1.1 kΩ resistor appears to be about 550 Ω. With this arrangement, the FET is driving hard ON or OFF and the total resistance effectively changes symmetrically about ±550 Ω. Lastly, if the offset adjustment current i_0 is adjusted such that $i_r = i$, then from the equation given above it can be easily calculated that the output voltage amplitude would be given by:

$$V_m = \sqrt{\frac{8V_r}{KR_r}} \tag{11.10}$$

The workability of this scheme was tested experimentally [14] by realizing the circuit using μA741 type op-amps, 1494L type analog multipliers and MPF 102 FET. The circuit was tested at the oscillation frequency of 1.2 kHz. From the experimental studies it was confirmed that the proposed circuit, indeed, has low distortion; THD = 0.06 % for $V_m = 6$ V and fast amplitude transient response. Also, it was confirmed that the THD can be further reduced by the low pass filter consisting of R_f and C_f resulting in a longer amplitude transient.

Lastly, although not so claimed by the authors [14], but it looks feasible to apply this method on any other oscillators which have two internal nodes where the amplitude of the node voltages are equal and phase shift is exactly 90°. This requirement is easily fulfilled by a number of quadrature oscillators, in general and those based upon a cascade of an integrator and a first order all pass filter in particular. However, as far as is known, this extension has not been attempted by anybody in the open literature so far and is, therefore, open to investigation.

11.4.2 Amplitude Control Through Control of Initial Conditions

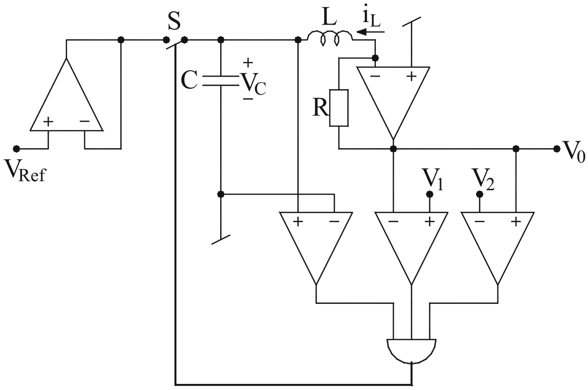
A method of controlling the amplitude of sinusoidal oscillators based upon the restoration of initial conditions at the energy storage elements was presented by Filanovsky [10, 11]. In Fig. 11.2, the basic idea of the method has been explained by applying it on a conceptual LC oscillator where both inductor and capacitor are supposed to be lossless.

By a routine analysis, it can be shown that if both inductor and capacitor are lossless then taking into account the initial conditions $V_C(0)$ and $I_L(0)$, the following can be written:

$$v_C = V_{Cm} \sin(\omega_0 t + \phi) \tag{11.11}$$

$$V_{Cm} = V_{Cm}\sqrt{\left\{V_C^2(0) + [I_L(0)/\omega_0 C]^2\right\}} \tag{11.12}$$

Fig. 11.2 LC oscillator with control of initial conditions and additional control circuit for restoration of initial condition proposed by Filanovsky [11]



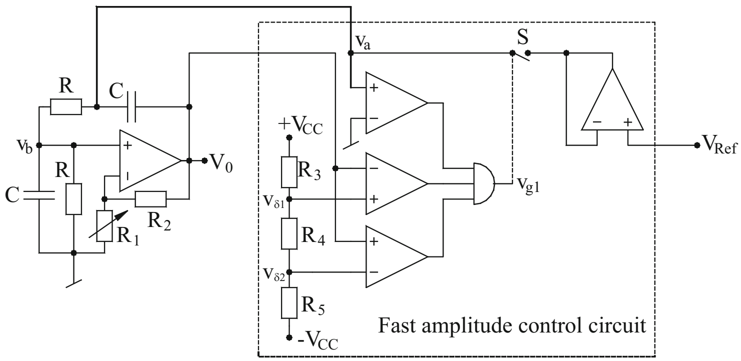


Fig. 11.3 Wein bridge oscillator with fast amplitude control proposed by Filanovsky [10]

$$\tan \phi = \frac{V_C(0)}{I_L(0)} \sqrt{\frac{C}{L}} \tag{11.13}$$

From the above equations, it turns out that if the original oscillator is modified by incorporating additional control circuit which restores the initial conditions $V_c(0) = V_{\mathrm{Ref}}$ and $I_L(0) = I_{\mathrm{ref}}$, then even if the oscillation magnitude falls because of losses in the components, in each oscillation period the oscillator will produce sustained oscillations of magnitude given by Eq. (11.12).

The required additional circuit consist of three comparators, a unity gain amplifier made from an op-amp to produce a replica of reference voltage and an additional op-amp used as a current to voltage convertor which creates a voltage proportional to the inductive current I_L and a gate which controls the switch S. Out of the three comparators, two comparators are connected as a window comparator whereas the third one acts as a polarity comparator.

In Fig. 11.3 the application of this technique to the classical Wein Bridge oscillator is shown. In this case the window comparator is directly connected to the output of the Wien bridge oscillator and it is set at $V_{ref} = 0$. The resistive voltage divider comprised of R_3-R_4-R_5 provides two small voltages $V_{\delta 1} > 0$ and $V_{\delta 2} < 0$. The control pulse appears when the output and input voltage of the amplifier is zero. At this instant the capacitor C is discharged, i.e., $V_{c1}(0) = 0$ and when the switch S is momentary closed then, $V_{c2}(0) = V_{Ref}$ is obtained. Through a routine analysis [10], the steady state value of the voltages v_o and v_a are found to be:

$$v_0 = 3V_{ref} \sin \omega_0 t \tag{11.14}$$

$$v_a = 5V_{Ref} \sin(\omega_0 t + \phi) \text{ where } \omega_0 = 1/RC \text{ and } \phi = \tan^{-1}(1/2) \tag{11.15}$$

The detailed analysis and the results of hardware implementation as given in Ref. [10] for the Wein bridge oscillator mentioned above as well as for Twin-T oscillator have demonstrated the workability of this method of amplitude control. It has been demonstrated that THD smaller than 0.2 % for Wein bridge oscillator and less than 0.1 % for Twin-T oscillator were achievable when the circuits were designed to generate oscillation frequency of 1.5 kHz in the amplitude range varying from 2 to 8 V with op-amps biased with ±10 power supply.

11.4.3 Amplitude Control Through Biasing-Voltage Control

It is a common knowledge that in any sinusoidal oscillator circuit, the amplitude of oscillations grows exponentially until it reaches a level beyond which some kind of nonlinearity comes into picture. In op-amp oscillators, the amplitude gets limited by the saturation type nonlinearity of the transfer characteristic of the op-amp which shows a small linear region around the origin (for differential input voltage of the order of a few micro volts outside which the output voltage of the op-amp saturates to $\pm V_{sat}$ where these saturations levels themselves depend upon the dc biasing power supply $\pm V_{cc}$. Thus, saturation voltage is directly proportional to the power supply voltage. It follows, therefore, that in op-amp oscillators, the amplitude of the output wave form would change if the power supply voltage changes. This idea was exploited by Sundaramurthy–Bhattacharyya–Swamy [7] to devise new scheme of amplitude stabilization and control which is shown in Fig. 11.4.

In the proposed scheme a DC signal proportional to the amplitude of the sine wave generated from a given oscillator is obtained through a rectifier and filter. This DC voltage, which is proportional to the peak amplitude of the output wave form, is compared with a fixed reference V_{R1} in differential amplifier. The error between these two is amplified through a non-inverting amplifier and applied to another comparator made from differential amplifier where this is compared with another reference V_{R2}. The output of this difference amplifier and negative of this created by an additional amplifier are used to create two DC supply voltages to the oscillator for which this amplitude control mechanism is devised. Thus, while changing the frequency of the output signal, if the amplitude goes down, this feedback

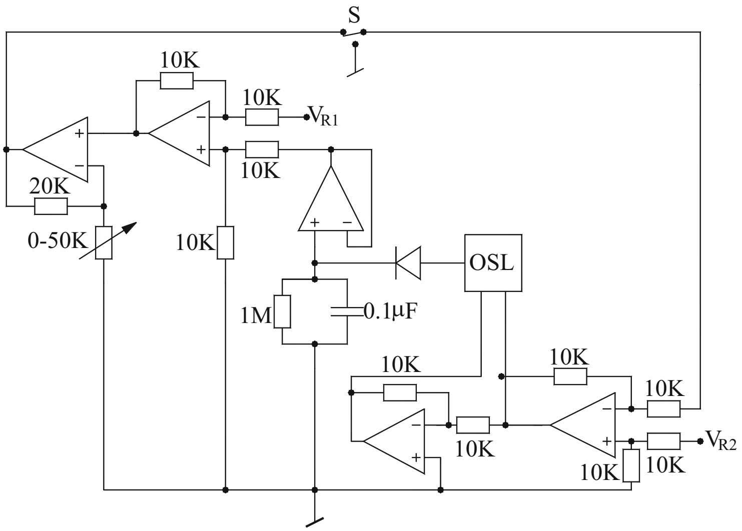


Fig. 11.4 The feedback scheme of amplitude stabilization proposed by Sundaramurthy–Bhattacharyya–Swamy [7]

arrangement acts in a direction which increases the DC biasing voltage to the oscillator thereby bringing back the reduced amplitude to its original level. Conversely, if the amplitude of the oscillations goes up, the feedback mechanism generates a reduced power supply voltage to the oscillator thereby again bringing back the amplitude to its desired value.

11.4.4 Fast Control of Amplitude of Oscillations

Hou–Lin [20] proposed a technique of automatic gain control (AGC) capable of providing wide oscillation variations which is an important issue for the design of sinusoidal oscillators. The scheme presented by Hou–Lin [20] and as implemented on a CFOA-based Wein bridge oscillator, is shown in Fig.11.5. In this circuit, it can be easily observed that CFOA AD844 along with R_3, R_4 constitutes a non-inverting amplifier of gain R_4/R_3 whereas the components R_1, C_1, R_2, C_2 constitute the usual frequency selective network. The AGC circuit consists of three high speed comparators made from CCs implemented with AD844 CFOAs and AD7541 multiplying D/A converter. From the w-terminal of CFOA employed to make the sinusoidal oscillator core, a signal V_{ref} is obtained which is applied to the D/A converter as a reference, the 12bit up-down counter produces a digital value "dig"; where $0 \leq \text{dig} \leq 2^{12}$.

The three comparators generate the trigger pulses rapidly on the transitions when v_{out} crosses the reference voltages V_L, V_M, V_H. When $\max[v_{\text{out}}(t)] \leq V_L$, the circuit

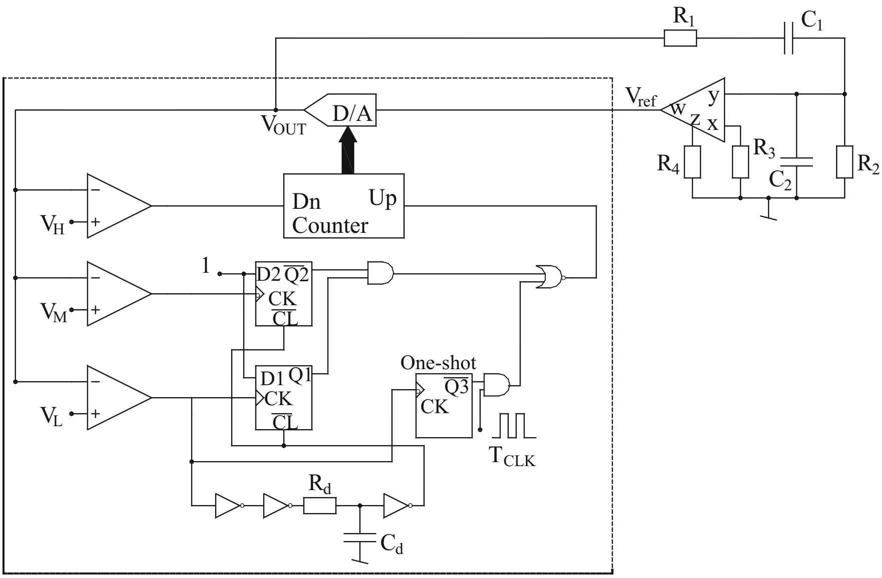


Fig. 11.5 The method of fast control of the amplitude of an oscillator proposed by Hou–Lin [20]

acts in such a way that the complex roots of characteristic equation (CE) get shifted to right hand side of the s-plane when the oscillations grow exponentially. Similar corrective action is taken by the AGC circuit automatically in case the oscillator amplitude goes down and the same needs to be automatically restored to its intended value. For further details of the mechanism and the relevant waveforms at the different locations in the AGC circuit the reader is referred to [20].

It may however, be mentioned that with CFOA parasitic at Z-node ($R_p \| 1/sC_p$) accounted for the oscillator bandwidth is governed by the third-order CE given by:

$$s^3 + \left(\frac{1}{\tau} + c\omega_0\right)s^2 + \omega_0\left\{\frac{c}{\tau} + \omega_0 - \frac{c\,h\,N(\mathrm{dig})}{\tau}\right\}s + \frac{\omega_0^2}{\tau} = 0 \qquad (11.16)$$

with

$$N(\mathrm{dig}) = \frac{R_4 \| R_p}{R_3}\left(\frac{\mathrm{dig}}{2^{12}}\right);\ \tau = \left(R_4 \| R_p\right)C_p;\ \omega_0 = \frac{1}{\sqrt{C_1R_1C_2R_2}} \qquad (11.17)$$

$$c = \frac{C_1R_1 + C_2R_2 + R_2C_1}{\sqrt{C_1C_2R_1R_2}};\ h = \frac{C_1R_2}{C_1R_1 + C_2R_2 + C_1R_2} \qquad (11.18)$$

From the above, it can be seen that the location of the roots of the CE is decided by the digital value of dig. The CO and FO derived from the above CE are found to be:

$$\mathrm{CO}: N(\mathrm{dig}) \geq k_m \quad \text{and} \quad \mathrm{FO}: f_0 = \frac{\omega_0}{2\pi\sqrt{1 + c\tau\omega_0}} \tag{11.19}$$

where

$$k_m = \frac{1}{h}\left[1 + \left(\frac{\tau^2\omega_0^2}{1 + c\tau\omega_0}\right)\right] \tag{11.20}$$

11.4.5 Amplitude Control in Current-Mode Oscillators

With the emergence of a number of CM oscillators which provide one or more explicit current outputs, it has become imperative to develop appropriate amplitude stabilization/control schemes which can be applied to aforementioned CM oscillators. An interesting circuit principle was incorporated in a current differencing oscillator by Pookaiyaudom–Thanachayanont–Sitdhikorn [16]. The method proposed by them incorporates an elegant technique in which the current gain decreased in well-defined manner as the oscillation amplitude increases.

The basic circuit topology is shown in the doted box of Fig. 11.6. This circuit is essentially modified form of a well-known Gilbert gain cell using additional diode connected transistors Q_3 and Q_5. In this circuit, a differential input current is taken as input which is logarithmically compressed by the transistors Q_3, Q_4, Q_5 and Q_6 and applied to the bases of the transistors Q_1, Q_2. Using a routine analysis using the exponential relation between the collector current and base to emitter voltage, it is

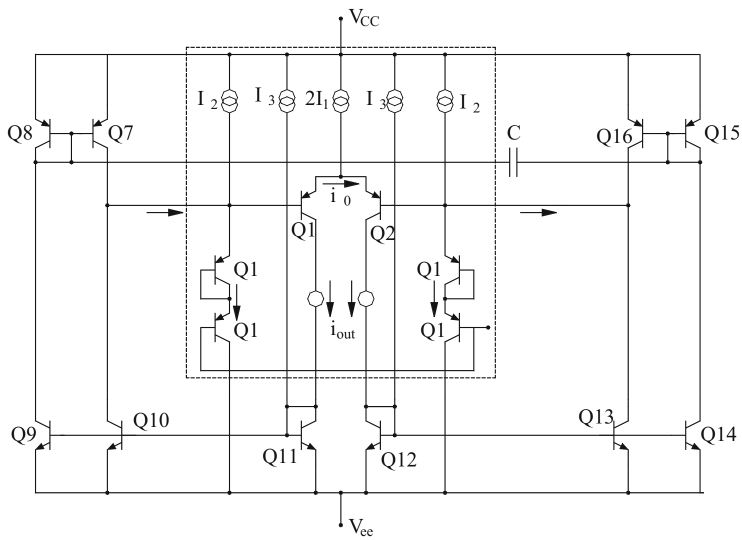


Fig. 11.6 Current-differencing oscillator with amplitude control circuit proposed by Pookaiyaudom–Thanachayanont–Sitdhikorn [16]

found that the small signal current gain between the differential current output and the differential current input is given by $\frac{i_0}{i_1} \cong \frac{2I_1}{I_2}$, thus, the small signal current gain is set by the ratio of bias currents $\frac{I_1}{I_2}$ and can be easily set at a value greater than unity. In the circuit of Fig. 11.6, the part of the circuit other than that contained in the dotted box, is actually the oscillator circuit in which the frequency setting network of the oscillator is composed of the transistor circuitry comprising of BJTs Q_7 to Q_{16} in conjunction with an external capacitor C_1 and the base-emitter capacitance of the npn transistor symbolically represented as C_π. This circuit is essentially a band pass filter with the circuit in the doted box acting as a current amplifier. The FO of the oscillator circuit is given by:

$$f_0 = \frac{(I_1 + I_3)}{2\pi V_t \sqrt{3C_1 C_\pi}} \tag{11.21}$$

Thus, in this circuit, the currents I_1 and I_2 control the loop gain and oscillation amplitude respectively whereas the current I_3 control the FO.

SPICE simulations using 0.8 μm Nortel BiCMOS technology showed that this circuit in conjunction with the proposed scheme of amplitude control, could generate oscillation frequency up to 30 MHz with an output current THD less than 1 % with oscillation current amplitude at 50 % of the bias current I_1. The workability of the proposed scheme was confirmed from the transient analysis of the oscillator. Reference [16] also described a single ended current amplitude control circuit; the readers are referred to [16] for further details.

11.5 Concluding Remarks

In this chapter we discuss a number of issues related to practical design of oscillators. A chosen oscillator is required to perform satisfactorily exhibiting lowest possible % THD, least phase noise, good frequency stability and must be amenable to easy incorporation of an appropriate amplitude stabilization/control circuitry. The various issues related to the aforementioned aspects are briefly considered in this chapter. We also highlight a number of practical circuits and techniques for amplitude stabilization and control for voltage-mode as well current-mode sinusoidal oscillators.

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Part III
Non-Sinusoidal Waveform Generators and Relaxation Oscillators

Chapter 12
Non-sinusoidal Waveform Generators and Multivibrators Using OTAs

Abstract This chapter presents various waveform generators using OTAs. This includes a variety of Schmitt triggers, astable multivibrators, monostable multivibrators, and pulse-width-modulators. Most of the circuits can be practically implemented using the off-the-shelf IC OTAs. The advantages and limitations of the various circuits are highlighted and a number of promising ideas for further work are also pointed out.

12.1 Introduction

Although the OTAs have been extensively employed in analog filters, sinusoidal oscillators and impedance synthesis, their use in the design of non-sinusoidal wave form generators/relaxation oscillators has rather been limited. In this chapter, we bring out the prominent works done on the design of non-sinusoidal waveform generators using OTAs based upon the published works [1–26] in this area.

12.2 Current-Controlled Oscillators Using Op-Amps and OTAs

Like all linear active elements, OTA also functions as a linear voltage controlled current source only for a limited range of input voltage beyond which the OTA moves into saturation mode. In devising circuits which can generate square/triangular waveforms, either combinations of op-amps and OTAs or only OTAs are forced to operate in saturation. In the following, we discuss the operation of OTAs in saturation mode.

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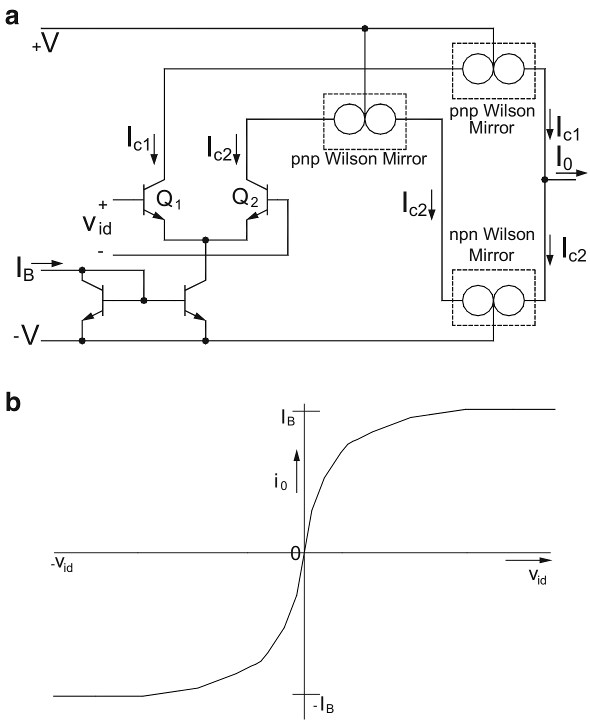


Fig. 12.1 (**a**) A simplified schematic of the OTA (**b**) transfer characteristics of the OTA

12.2.1 Operation of the OTA in Saturation

Consider the internal circuit schematic of the LM3080/CA3080 OTA shown in Fig. 12.1a. By a straightforward analysis, the output current i_0 of the OTA can be expressed as a function of the differential input voltage (v_{id}) by the following equation:

$$i_o = (I_{\mathrm{c1}} - I_{\mathrm{c2}}) = I_{\mathrm{B}}\tanh\left(\frac{v_{\mathrm{id}}}{2V_{\mathrm{T}}}\right) = I_{\mathrm{B}}\left(\frac{\mathrm{e}^{\frac{v_{\mathrm{id}}}{2V_{\mathrm{T}}}} - \mathrm{e}^{-\frac{v_{\mathrm{id}}}{2V_{\mathrm{T}}}}}{\mathrm{e}^{\frac{v_{\mathrm{id}}}{2V_{\mathrm{T}}}} + \mathrm{e}^{-\frac{v_{\mathrm{id}}}{2V_{\mathrm{T}}}}}\right) \tag{12.1}$$

It, therefore, follows from the above equation that for large values of v_{id} ($v_{\mathrm{id}} \gg 2V_{\mathrm{T}}$), the output current i_{o} approaches a value equal to $+I_{\mathrm{B}}$ while for large negative values of v_{id} the output current will approach the value $-I_{\mathrm{B}}$.

The same conclusion can be reached by looking into the large signal behavior of the various transistors in the OTA circuit of Fig. 12.1a. Note that for large positive differential inputs the transistor Q_1 will be saturated and thereby carrying the entire current I_{B} while Q_2 would be in cutoff thereby carrying almost zero current. As a consequence, the output current which is $i_o = (I_{\mathrm{c1}} - I_{\mathrm{c2}})$ would be equal to $+I_{\mathrm{B}}$. On the other hand, for large negative v_{id}, the transistor Q_2 will be in saturation and Q_1

will be in cutoff, and therefore $I_{c1} \approx 0$ while $I_{c2} \approx I_B$ thereby yielding $i_o = -I_B$. A plot of Eq. (12.1) has been shown in Fig. 12.1b which exhibits this behavior.

12.2.2 Linear Current-Controlled Square/Triangular Wave Generator

In the classical astable multivibrator, if the resistance R is simulated by OTAs one can control the frequency of the output square wave through the external DC bias current I_B of the two OTAs. However, such an arrangement will suffer from the drawback that the frequency of the resulting square wave would be temperature sensitive because of the presence of the term V_T. Secondly, it would be difficult to maintain the OTAs simulating the resistor, operating in linear region because of requiring the differential input voltage of the OTAs to be much smaller than $2V_T$.

The better alternative was proposed by Haslett [1] who presented the circuit shown in Fig. 12.2a wherein the OTA is connected in such a way that the differential input voltage across it is definitely much larger than $2V_T$ thereby ensuring that it operates only in saturation in which case the output current $i_0(t)$ delivered by it is either $+I_B$ or $-I_B$ depending upon whether the comparator output is HIGH, $V_0(t) = +V_{sat}$ or LOW $V_0(t) = -V_{sat}$.

The bias current of the OTA can be made a function of external voltage V_{in} by connecting the resistor R_{B1} to the controlled voltage V_{in} directly through the switch in which case the DC bias current injected into OTA is given by

$$I_B = \frac{V_{in} - V_{BE} + V_{cc}}{R_{B1}} \tag{12.2}$$

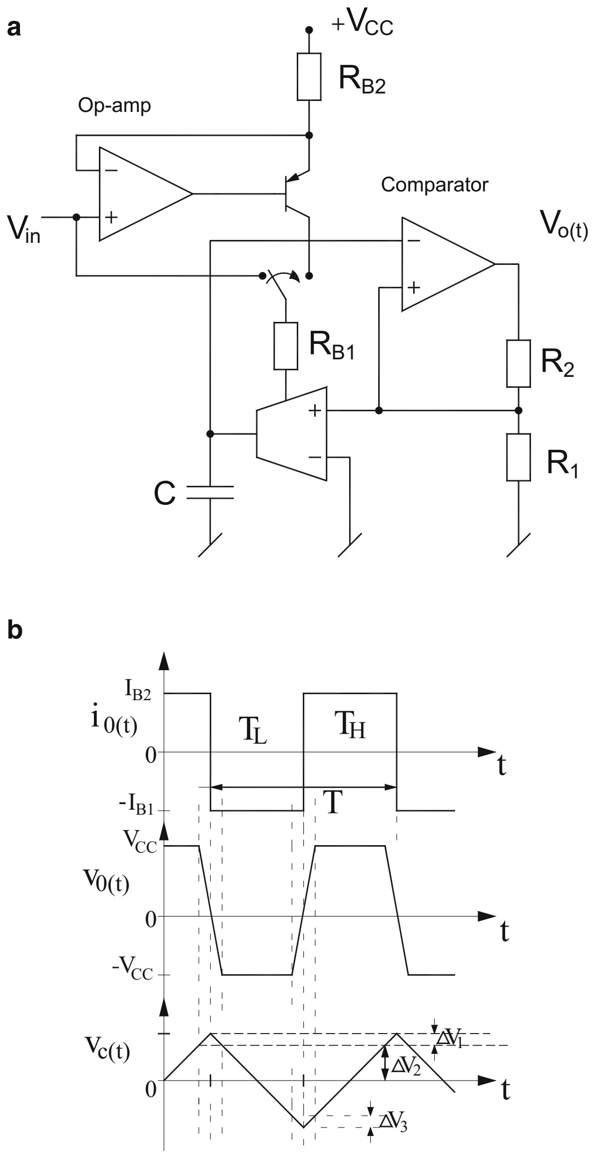
Alternatively, the DC bias current of the OTA can be derived from the collector current of the transistor by alternative positioning of the switch in which case the corresponding value of the DC bias current of the OTA is given by

$$I_B = \alpha_F \left(\frac{V_{cc} - V_{in}}{R_{B2}} \right) \tag{12.3}$$

where α_F is the transistor common base forward current gain.

The operation of the circuit can be explained as follows. When the comparator's output is HIGH, the differential input to the OTA is positive and large which saturates the OTA which forces a current $+I_B$ into the capacitor which charges the capacitor linearly and hence, the voltage across the capacitor is a positive going ramp. However, the capacitor voltage is continuously being monitored (compared with the threshold voltage $+\beta V_{sat}$). As soon as the capacitor voltage is slightly more than this upper threshold voltage, the comparator output switches to the LOW state $-V_{sat}$ due to which the reference voltage existing on the non-inverting terminal of

Fig. 12.2 Linear current-controlled oscillator (CCO) (**a**) the CCO proposed by Haslett [1] (**b**) the relevant waveforms



the comparator also changes and now becomes $-\beta V_{\text{sat}}$. The OTA now receives a large negative differential input and therefore, saturates again but delivers an output current $= -I_{\text{B}}$. The capacitor voltage, which was already charged up to $+\beta V_{\text{sat}}$, now discharges linearly from $+\beta V_{\text{sat}}$ to zero and then would try to reach up to a level $-\beta V_{\text{sat}}$*linearly*. It may be noted that the slope of the positive ramp is I_{B}/C whereas that of the negative ramp is $-I_{\text{B}}/C$. Now, as soon as the capacitor voltage becomes slightly more negative than $-\beta V_{\mathrm{sat}}$ the comparator output again switches to the HIGH state. This process continues resulting in a square wave at the output of the comparator and a triangular wave at the output of the OTA. The relevant waveforms are shown in Fig. 12.2b.

The following may now be noted:

$$f = \frac{1}{T} = \frac{1}{T_{\mathrm{H}} + T_{\mathrm{L}}} = \frac{1}{2T_{\mathrm{H}}} \quad \text{since} \quad T_{\mathrm{H}} = T_{\mathrm{L}} \tag{12.4}$$

The value of

$$T_{\mathrm{H}} = \frac{\beta V_{\mathrm{sat}} - (-\beta V_{\mathrm{sat}})}{I_{\mathrm{B}}/C} = \frac{2\beta V_{\mathrm{sat}} C}{I_{\mathrm{B}}} \tag{12.5}$$

Hence,

$$f = \frac{I_{\mathrm{B}}}{4\beta C V_{\mathrm{sat}}} \tag{12.6}$$

Thus, the frequency of the generated wave forms is a linear function of the external DC bias current I_{B} and the circuit functions as a linear Current-Controlled Oscillator.

It is worth pointing out that the main source of error in the operation of the circuit is the finite slew rate of the op-amp comparator forming the Schmitt trigger due to which the output square wave will be, in practice, a trapezoidal signal rather than a perfect square wave. Another source of error could be the current imbalances between the positive and negative saturation currents of the OTA which would result in the asymmetry in the output waveform.

With finite slew rate (SR) of the comparator (op-amp), accounted for and taking $\beta = \frac{1}{2}$ with $R_1 = R_2$, the expression for the frequency of the output square wave is modified to [1]:

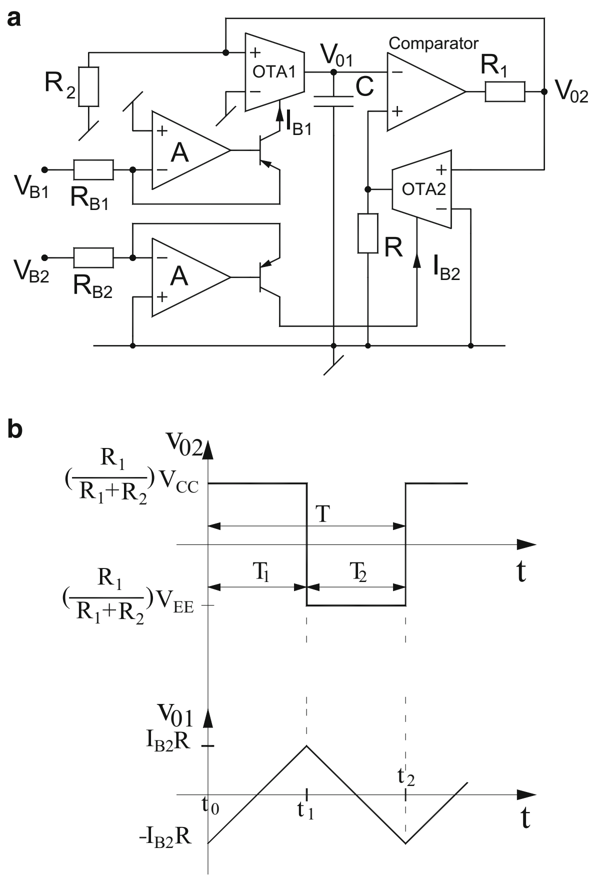
$$f = \frac{I_{\mathrm{B}}}{2CV_{\mathrm{cc}}}\left(1 + \frac{2I_{\mathrm{B}}}{C\,\mathrm{SR}}\right) \tag{12.7}$$

With nominal values of the components, the DC bias current and SR, the error caused in frequency due to finite SR has been found [1] to be of the order of about 2 %.

12.2.3 Improved Temperature-Insensitive VCO

An improved circuit which realizes a *temperature stable* VCO using two OTAs, one comparator and two op-amps was advanced by Chung–Cha–Kim [2] and is shown

Fig. 12.3 Temperature stable VCO (**a**) circuit of proposed VCO by Chung–Cha–Kim [2] (**b**) relevant wave forms



in Fig. 12.3a. In this circuit, OTA_1 along with the capacitor makes an integrator whose time constant is proportional to DC bias current I_{B1}. On the other hand, another comparator along with OTA_2 and the resistance R constitute a Schmitt trigger. The threshold voltage of the Schmitt trigger is proportional to DC bias current I_{B2}. The voltage divider circuit consisting of the resistors R_1 and R_2 in the Schmitt trigger part has been incorporated to prevent zenering the inputs of the OTAs. Besides this two op-amps with a transistor each in the negative feedback path have been so arranged that the two DC bias currents of the OTAs, namely I_{B1} and I_{B2}, are respectively equal to V_{B1}/R_{B1} and V_{B2}/R_{B2}.

The relevant waveforms for the two outputs of the circuit, namely V_{01} and V_{02}, are shown in Fig. 12.3b. By a routine analysis, it is easy to find that the frequency of the output waveforms generated by this circuit is given by

$$f = \frac{1}{4RC}\left(\frac{I_{B1}}{I_{B2}}\right) \tag{12.8}$$

The temperature insensitivity of this design has been demonstrated [2] by a nonideal analysis, incorporating the large signal transistor model and then calculating half periods of oscillation, T_1 and T_2 which have been found to be

$$T_1 = RC\left(\frac{I_{B2}}{I_{B1}}\right)\left[1 - \frac{A_2B_1}{A_1B_2}\right]C_1 \tag{12.9}$$

$$T_2 = RC\left(\frac{I_{B2}}{I_{B1}}\right)\left[1 - \frac{A_1B_2}{A_2B_1}\right]C_1 \tag{12.10}$$

where

$$A_1 = 1 + e^{\left(\frac{R_1}{R_1+R_2}\right)\frac{V_{EE}}{V_T}} \tag{12.11}$$

$$A_2 = 1 - e^{\left(\frac{R_1}{R_1+R_2}\right)\frac{V_{EE}}{V_T}} \tag{12.12}$$

$$B_1 = 1 + e^{\left(\frac{R_1}{R_1+R_2}\right)\frac{V_{CC}}{V_T}} \tag{12.13}$$

$$B_2 = 1 - e^{\left(\frac{R_1}{R_1+R_2}\right)\frac{V_{CC}}{V_T}} \tag{12.14}$$

and

$$C_1 = 1 + \frac{1}{\left(1 - \frac{2}{\beta_T}\right)} \tag{12.15}$$

If the magnitude of the two DC biasing supplies is identical then the expressions in Eqs. (12.7) and (12.8) are simplified to the following

$$T_1 = 2CR(I_{B2}/I_{B1})\left[1 + \left(1 - \frac{2}{\beta_T}\right)\right] \tag{12.16a}$$

$$T_2 = 2CR(I_{B2}/I_{B1})\left[1 + \frac{1}{\left(1 - \frac{2}{\beta_T}\right)}\right] \tag{12.16b}$$

where V_T is the thermal voltage and $\beta_T = \left(\beta_0^2 + 2\beta_0 + 2\right)$; β_0 is the transistor common emitter current gain.

It can be seen that since β_T is >>1, the above expressions will lead to nearly same frequency of oscillation as that given by Eq. (12.6), as shown in [2].

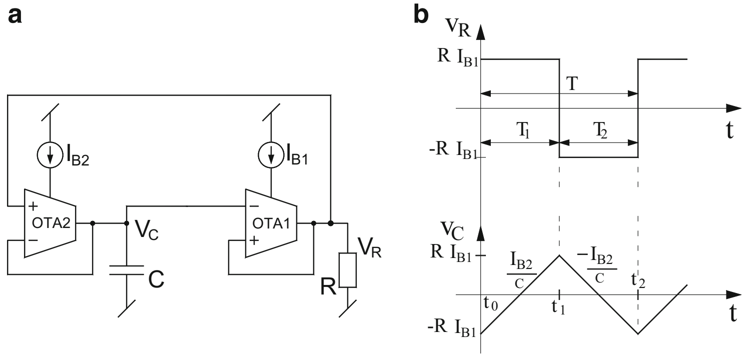


Fig. 12.4 VCO using only OTAs proposed by Jeong–Won–Chung [3] (**a**) the circuit schematic (**b**) output waveforms

12.2.4 A Triangular/Square Wave VCO Using Two OTAs

Jeong–Won–Chung [3] proposed an alternative VCO circuit which employs only two OTAs along with a grounded resistor and a grounded capacitor. In this circuit, which is shown in Fig. 12.4a, OTA_1 along with resistor R forms the Schmitt trigger whose threshold voltage is proportional to the DC bias current I_{B1}. On the other hand, OTA_2 along with grounded capacitor forms an integrator whose time constant is proportional to the DC bias current I_{B2}. The operation of the circuit can easily be understood from the waveforms shown in Fig. 12.4b.

First of all it may be noted that two output levels of the Schmitt trigger would be $+RI_{B1}$ and $-RI_{B1}$ depending upon whether its differential input voltage is positive and large or negative and large. Let us assume that the output of the Schmitt trigger is in its positive saturation level $+RI_{B1}$. This makes OTA_1 saturated which delivers an output current $+I_{B2}$ which charges the capacitor C *linearly* with a slope equal to $+I_{B2}/C$. However, this positive going ramp will continue only till this voltage reaches a value RI_{B1}, after which the Schmitt trigger will switch its state to $-RI_{B1}$. As a consequence, the output current of OTA_2 reverses its direction and a constant current $-I_{B2}$ therefore discharges the capacitor linearly but now the slope of the negative ramp is equal to $-I_{B2}/C$. This negative ramp will continue only till the integrator's output reaches a value $-RI_{B1}$ at which time the Schmitt trigger again switches back to the positive saturation state RI_{B1}. This process continues thereby resulting in a square wave at the output of the Schmitt trigger formed by OTA_1 and triangular wave at the output of OTA_2.

The time period T_1 (Fig.12.4b) during which the output of OTA_1 is high can be determined as follows

$$
\begin{aligned}
T_1 &= \text{Time taken by the capacitor voltage in changing from } -RI_{B1} \text{ to } +RI_{B1} \\
&= \text{Total change in voltage/rate of change of the voltage} = \frac{RI_{B1} - (-RI_{B1})}{(I_{B2}/C)} \\
&= 2RC\left(\frac{I_{B1}}{I_{B2}}\right)
\end{aligned}
$$

Similarly, it can be shown that $T_2 = 2RC\left(\frac{I_{B1}}{I_{B2}}\right)$

Thus, the frequency of oscillation would be given by

$$f = \frac{I_{B2}}{4CRI_{B1}} \tag{12.17}$$

Thus, the frequency of the output waveforms can be controlled by changing I_{B2}, while the magnitudes of the generated waveforms are adjustable by I_{B1}.

The workability of this circuit has been verified [3] by building the circuit using LM13600 OTAs along with a capacitor of 1 nF and a resistor of 100 KΩ. In the experiment, I_{B1} was set to 10 μA whereas the bias current I_{B2} was varied over a range of 1–50 μA. The experimental results established wide range tunability of the oscillation frequency in the ratio 1:50 with a reduced linearity error as compared to the conventional VCO, thereby proving the superiority of this structure.

12.2.5 Current-Controlled Oscillator Using Only a Single OTA

Filanovsky [4] proposed an astable bridge multivibrator using an OTA along with four resistors and a grounded capacitor (see Fig. 12.5). It may be noted that the maximum output current of the OTA, in general, is equal to $\pm KI$, where the current I is setup by the resistor R and the positive supply voltage $+V_{cc}$.

The operation of the circuit can be explained as follows: Assume that steady state exists and the circuit is at its initial instant when the voltage across the capacitor C is at its minimum value $-V_{c0}$ and the timing capacitor starts to recharge by the current $+KI$ flowing out of the OTA. It may be seen that the voltage across the capacitor v_c would try to increase towards $KI\ (R_1 + R_2)$. On the other hand, the voltage v_r across R_2 just have jumped to the value given by the following equation;

$$V_{ro} = KI\frac{R_2R_c}{(R_1 + R_2 + R_c)} - V_{co}\frac{R_2}{(R_1 + R_2 + R_c)}, \tag{12.18}$$

and it starts to increase towards $+KIR_2$ (refer to Fig. 12.5b) and thus, the curves of v_r and v_c should intersect. But before this actually happens, at the instant of time $t = T/2$ when the differential input voltage of the OTA v_i is very close to ΔV, the

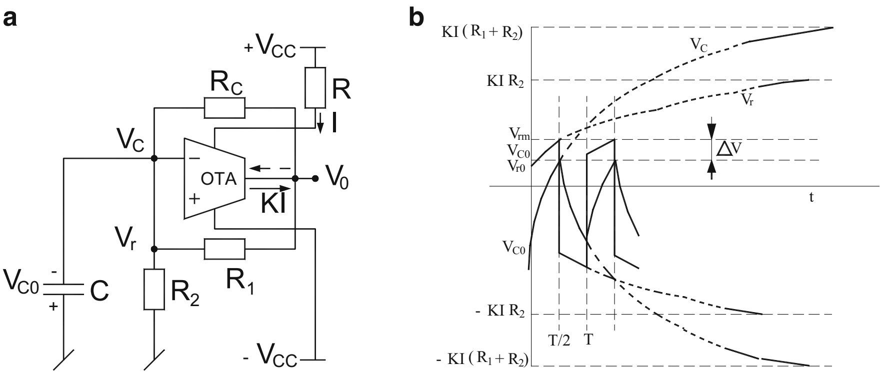


Fig. 12.5 Current-controlled bridge multivibrator (**a**) circuit diagram (**b**) relevant wave forms

output current of the OTA would start changing its polarity (direction). Since there is a positive feedback in the circuit via the potential divider consisting of R_1 and R_2, this process will be accelerated. When v_{i} becomes exactly equal to ΔV and the voltage v_{c} has achieved its maximum value V_{co}, the output current of the OTA will change its direction. As a consequence, the voltage v_{r} jumps to the value $-v_{\mathrm{r0}}$ and the capacitor C now starts recharging towards $-V_{\mathrm{c0}}$. Now the voltage v_{c} would be decreasing towards a value $-KIR_2$. When the voltage v_{c} is close to its minimum value $-V_{\mathrm{c0}}$ and the differential input voltage of the OTA is near to $-\Delta V$, the output current of the OTA would again start changing its direction. The positive feedback in the circuit will make it possible to complete this switching at $t = T$, the output current of the OTA will quickly achieve its maximum value KI flowing out of the OTA output terminal. As a consequence, the voltage v_{r} would jump to the value V_{r0} again and the voltage v_{c} would start increasing. This process continues repeatedly thereby generating a periodical waveform.

12.2.6 *An Entirely OTA-Based Schmitt Trigger and Square/Triangular Wave Generator*

Although the Schmitt trigger is realizable from a single OTA and a single resistor, a current controllable Schmitt trigger capable of providing independently controllable threshold and output voltage levels was proposed by Kim–Cha–Chung [5]. This circuit is shown in Fig. 12.6a and its transfer characteristic is shown in Fig. 12.6b. With the assumption that R_1 and R_2 have been appropriately selected to ensure that both the OTAs operate in saturation, it is easy to derive the transfer characteristic of this Schmitt trigger which clearly shows that the two threshold levels are given by, respectively, $-R_2I_{\mathrm{B2}}$ and $+R_2I_{\mathrm{B2}}$. Whereas the two saturation levels are given by $-RI_{\mathrm{B1}}$ and $+RI_{\mathrm{B1}}$, respectively, from where it is clearly seen that the upper

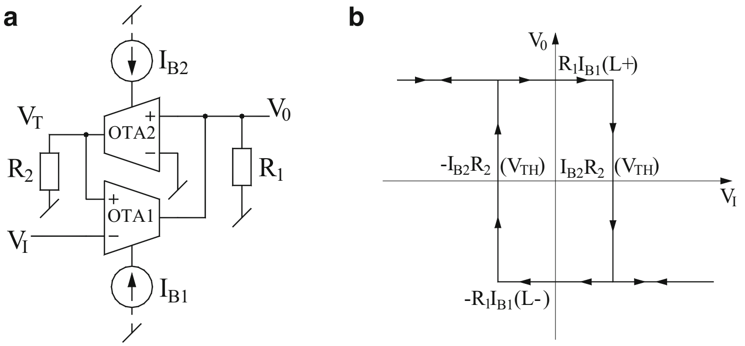


Fig. 12.6 OTA-R Schmitt trigger independently controllable threshold and output voltage levels proposed by Kim–Cha–Chung [5] (**a**) the circuit schematic (**b**) transfer characteristic

threshold levels as well as positive and negative saturation levels both are electronically controllable through I_{B2} and I_{B1}, respectively.

This Schmitt trigger can be readily employed to realize a triangular/square wave generator as was demonstrated by Chung–Kim–Cha–Kim [6] where an additional OTA and grounded capacitor were connected as shown here Fig. 12.7a.

The operation of this circuit can be readily visualized by taking cognizance of the explanation given in the context of earlier circuit. It is a routine matter to confirm that the frequency of the waveforms generated by this circuit would be given by [6]:

$$f = \frac{1}{4R_2C}\left(\frac{I_{B3}}{I_{B2}}\right). \tag{12.19}$$

From the above, as well as from the waveforms shown in Fig. 12.7c, it may be observed that the interesting features of this circuit are that the amplitude of the square wave is controllable through external bias current I_{B1}, whereas the amplitude of triangular wave can be set by I_{B2} and finally, the frequency of oscillation is linearly controllable through I_{B3}.

12.2.7 Square Wave Generator Using a DO-OTA

Lo–Chien–Chiu [7] proposed two novel electronically tunable waveform generators using a single dual output operational transconductance amplifier (DO-OTA) (see Fig. 12.8). The first of these circuits is capable of producing symmetrical square and triangular waveforms simultaneously whereas the second circuit can generate pulse waveform on the application of a negative trigger signal. The oscillation frequency of the first circuit and the pulse width in the second circuit are adjustable by external passive components, whereas the output levels can be

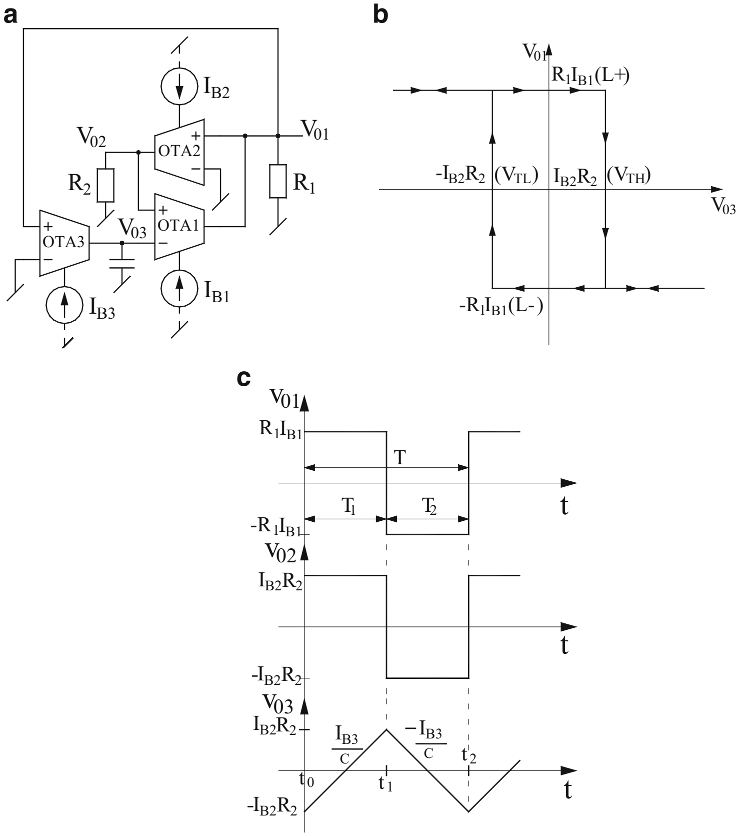


Fig. 12.7 A triangular/square wave generator proposed by Chung–Kim–Cha–Kim [6] (**a**) the circuit configuration (**b**) transfer characteristic (**c**) the relevant waveforms

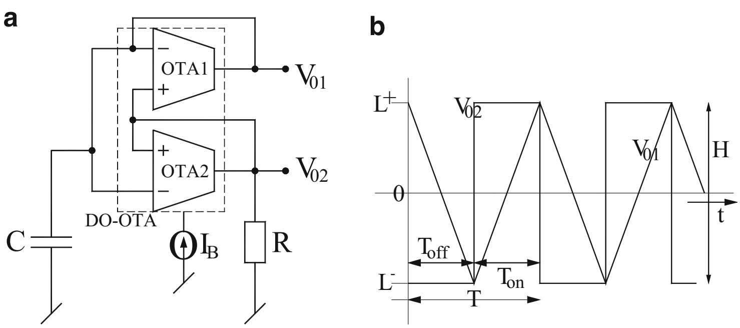


Fig. 12.8 Square/triangular waveform generator using DO-OTA proposed by Lo–Chien–Chiu [7] (**a**) the circuit configuration (**b**) the relevant waveforms

controlled by changing the external DC bias current of the DO-OTA. It may be noticed that through positive feedback around the DO-OTA and the resistor a part of the circuit acts like a Schmitt trigger, whereas the capacitor in conjunction with the other output of DO-OTA constitutes the integrator.

The operation of this circuit can be explained as follow:

When V_{02} is high, the capacitor is charged by the current $+I_B$ and simultaneously an equal current I_B flows through the resistor R to keep V_{02} in the high state. The capacitor obviously charges linearly and this positive going ramp continues until V_{01} is more positive than V_{02} thus, the output voltage in the charging state $(0 < t < T_{on})$ can be expressed as

$$V_{01}(t) = -RI_B + \left(\frac{I_B}{C}\right)t \tag{12.20}$$

Similarly, when the output is in low state, the following equation applies

$$V_{01}(t) = +RI_B - \left(\frac{I_B}{C}\right)(t - T_{on}) \tag{12.21}$$

At the ends of the charging state and discharging state, $V_{01} = V_{02}$.

From which it follows that $T_{on} = 2RC = T_{off}$ and therefore the oscillation frequency is given by $f = \frac{1}{4RC}$.

whereas the peak-to-peak output level of the square wave as well as triangular wave is given by $H = 2I_BR$. Thus, it is seen that the frequency is controllable by R while the amplitudes of the two waveforms are controllable by I_B.

This circuit was constructed [7] from two identical CA 3080 type OTAs biased with ±10 V supply. It has been concluded in Ref. [7] that the highest frequency of waveforms which can be attained by this circuit is limited to several hundred KHz only due to the finite slew rate effects of the OTAs. The second proposition of pulse waveform generator has been dealt in Sect. 12.4.

12.3 Current-Controlled Saw-Tooth Generators

Saw-tooth and pulse generators find a number of applications in many different areas such as in instrumentation and measurement and control. A common methodology to generate saw-tooth waveform is to arrange a grounded capacitor to be charged through a constant current source for specified time and then rapidly discharge the capacitor through a transistor employed as a switch. It is desirable that both the frequency and the amplitude of the saw-tooth waveform be electronically controllable independently. One such circuit, employing IC 555 timer, OTAs and the analog switch was devised by Abuelma'atti–Al-Absi [8] and is shown here in Fig. 12.9.

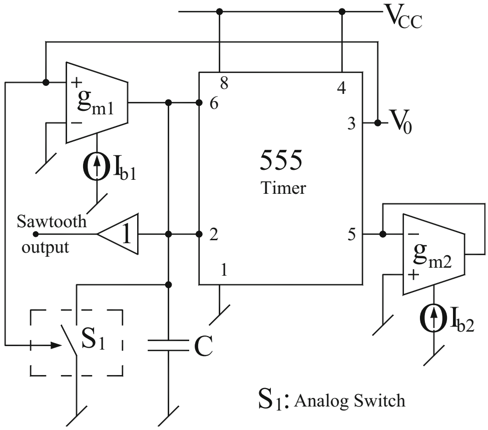


Fig. 12.9 The current controlled saw-tooth generator proposed by Abuelma'atti and Al-Absi [8]

The operation of this circuit can be explained as follows:

Assume that output of the timer is HIGH ($\approx +V_{CC}$). This, being much larger than the linear input voltage range of the OTA, forces OTA_1 into saturation thereby the output current of the OTA_1 equals to $+I_{b1}$ which charges the capacitor linearly with the rate of change of capacitor voltage being given by I_{b1}/C. This charging of the capacitor will stop when the voltage across the capacitor reaches voltage V_5. At this instant of time, the output of the timer goes LOW and the switch S_1 will close thereby forcing the capacitor voltage to discharge almost immediately through the very low ON resistance of the switch. When voltage across the capacitor drops below $V_5/2$ timer output again goes HIGH, switch S_1 will open and another cycle of charging starts. During the charging cycle, therefore, the amplitude of the saw-tooth will increase from $V_5/2$ to V_5. If the transconductance of OTA_2 (g_{m2}) even though it is variable is confined to have value such that ($1/g_{m2}$) is <<5 KΩ, it has been worked out that the approximate value of the voltage at pin5 of the timer will be $V_5 = \frac{V_{CC}}{5Kg_{m2}}$, where

$$g_{m2} = \frac{I_{b2}}{2V_T} \quad \text{and} \quad K = 1000 \tag{12.22}$$

Through a routine analysis [8] it has been found that the amplitude (A) of the saw-tooth wave form can be expressed as $A \cong \frac{V_{CC}V_T}{5KI_{b2}}$, whereas its frequency is given by

$$f \cong \frac{5KI_{b1}I_{b2}}{CV_{CC}V_T} \tag{12.23}$$

From the above, it is seen that the amplitude of the saw-tooth waveform generated by this circuit can be controlled by the external bias current I_{b2}, whereas the frequency can linearly be controlled through the other DC bias current I_{b1}.

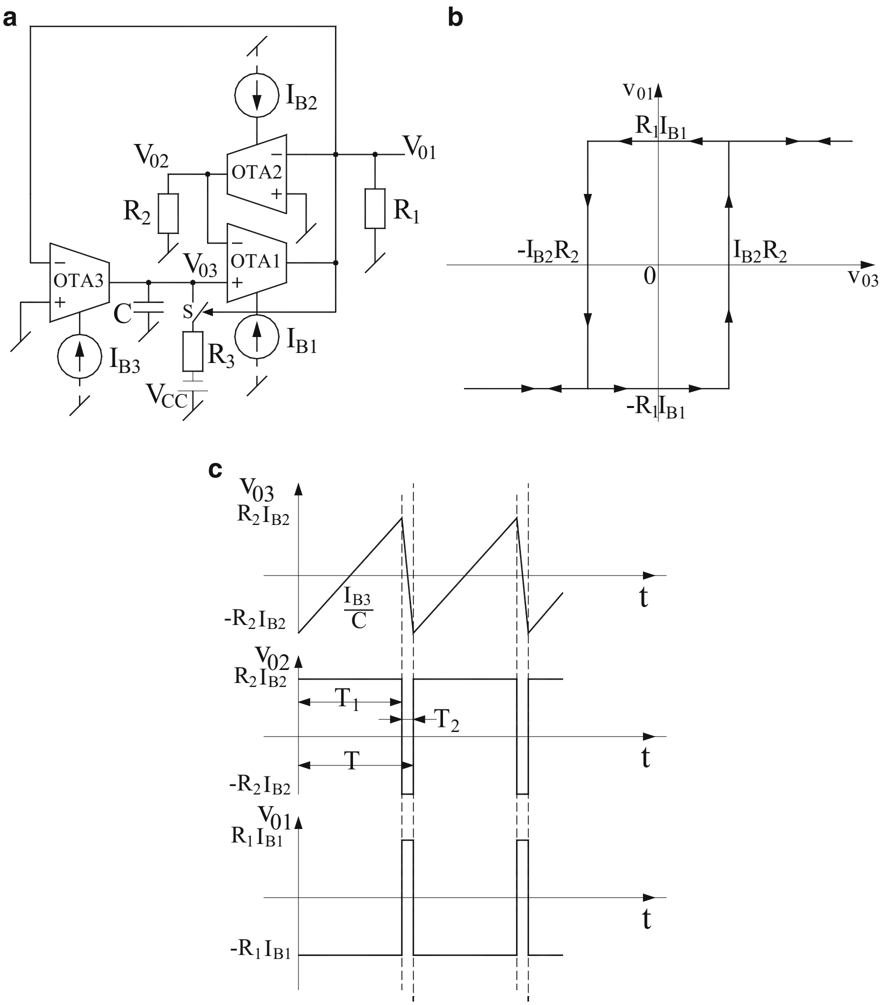


Fig. 12.10 Current controllable saw-tooth waveform generator proposed by Chung–Kim–Cha–Kim [6] (**a**) circuit configuration (**b**) transfer characteristic (**c**) relevant waveforms

Chung–Kim–Cha–Kim [6] demonstrated that their proposed three OTAs-based square/triangular wave form generator can be readily modified to a current controllable saw-tooth waveform generator as shown in Fig. 12.10a. It may be noticed that in this circuit there are two modifications over the earlier circuit presented in Fig. 12.7a. (1) The polarity of OTA_1 and OTA_2 has been reversed (2) a series connection of analog switch and a resistor R_3 and a DC supply voltage V_{CC} has been added. As a consequence of first modification the transfer characteristics of the Schmitt trigger exhibits an anticlockwise hysteresis as shown here in Fig. 12.10b.

The relevant waveforms for this circuit are shown in Fig. 12.10c. The time period T_1 during which the capacitor charges linearly with the rate of the change of the voltage given by I_{B3}/C is found to be

$$\begin{aligned} T_1 &= \frac{\text{Total change in capacitor voltage}}{(\mathrm{d}V_c/\mathrm{d}t)} = \frac{R_2 I_{B2} - (-R_2 I_{B2})}{(I_{B3}/C)} \\ &= \frac{2CR_2 I_{B2}}{I_{B3}} \end{aligned} \tag{12.24}$$

During the time interval T_2 the switch is closed and the capacitor is forced to discharge via R_3 and a negative supply $-V_{CC}$. Assuming that the time taken in the discharge process is much smaller than T_1, the frequency of saw-tooth wave form then can be approximated as [6],

$$f \cong \frac{1}{2CR_2}\left(\frac{I_{B3}}{I_{B2}}\right) \tag{12.25}$$

When implemented with LM13600 type OTAs and analog switch MC14066, it has been found [6] that the nonlinearity in the measured characteristic between the bias current I_{B3} and oscillation frequency was less than 5.2 % within the frequency range 500 Hz to 15 KHz.

12.4 Pulse Wave Form Generator

Lo–Chien–Chiu [7] also proposed a pulse wave-form generator as an extension to their square/triangular wave form generator. This circuit is shown in Fig. 12.11a.

The operation of this circuit can be explained as follows: This circuit has a stable state and a quasi-stable state. In the stable state, the capacitor voltage is held at zero volts through the N-MOS transistor kept in ON state. This leads to the output level of the OTA $L^+ = V_0$ in the HIGH state, i.e., $V_0 = +RI_B$. On the application of a trigger pulse, when V_{in} falls from $+V_{DD}$ to $-V_{DD}$, the output current of the OTAs have reverse polarities thereby forcing V_0 to jump abruptly from $+RI_B$ to $-RI_B$. The

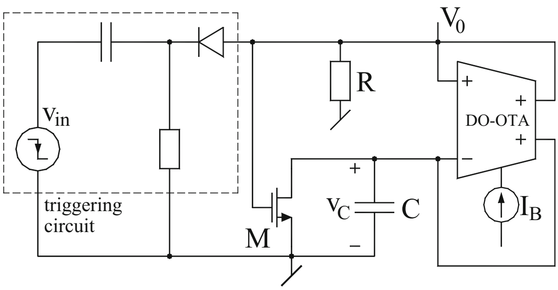


Fig. 12.11 Current-controlled pulse generator proposed by Lo–Chien–Chiu [7]

capacitor now charges linearly yielding a negative going ramp having a slope equal to $-I_B/C$. This is obviously quasi-stable state of the circuit which continues until the capacitor voltage drops from ground to a level $-RI_B$ at which instance of time the OTA output V_0 returns back to its permanent stable state $+RI_B$.

It is, therefore, concluded [7] that the amplitude of the pulse generated by the circuit would have an amplitude equal to $2RI_B$ and a time period $T = RC$.

12.5 Monostable Multivibrators Using OTAs

There are many applications where pulse of known height and width is required to be generated in response to a trigger signal. Such pulses are generated by monostable multivibrators or one-shot timers. There are numerous circuits available to implement monostable multivibrators using op-amps, IC 555 timers and even IC multivibrators such as 74121. In this section, we describe a number of interesting monostable multivibrator circuits based upon the use of OTAs. The obvious motivation to realize monostable multivibrators using OTAs is to exploit the current controllability of the OTA's transconductance for obtaining electronic control over the height and/or the width of the output pulse generated by the circuit.

12.5.1 Current-Controlled Monostable Multivibrator

Chung–Cha–Kim [9] presented a monostable multivibrator using OTAs which is capable of generating a pulse of temperature stable height and width with small power dissipation. In addition, the pulse width and the amplitude are independently and linearly controllable by means of external DC bias currents of the OTAs. Their preposition is shown in Fig. 12.12a.

In this circuit, OTA_1 and OTA_2 along with the resistor R_1 and R_2 constitute the Schmitt trigger whereas OTA_3 along with C_1 constitutes the integrator and the trigger signal is applied through the R_C network composed of capacitor C_2 and resistor R_3. It may be noted that two saturation levels of Schmitt trigger are controlled by the DC bias current I_{B1} and would be $+I_{B1}R_1$ and $-I_{B1}R_1$ whereas the two threshold voltages of the comparator will be proportional to I_{B2} and would be $-I_{B2}R_2$ and $+I_{B2}R_2$ respectively. On the other hand, the capacitor C_1 would be charged by a constant current I_{B3}. The operation of this circuit can be explained as follows:

It has been derived [9] that in the absence of any trigger signal the permanent stable state of the circuit will be $V_{01} = +I_{B1}R_1$ and would be maintained because $I_{B2}R_2 > 0$. Since the HIGH state of V_{01} closes the switch S thereby causing the voltage on the capacitor C_1 as well as the voltage on the inverting input of OTA_1 both to be zero, when a positive going step is applied at the trigger input terminal, the resulting trigger signal V_{trg} would be a positive going spike as shown in

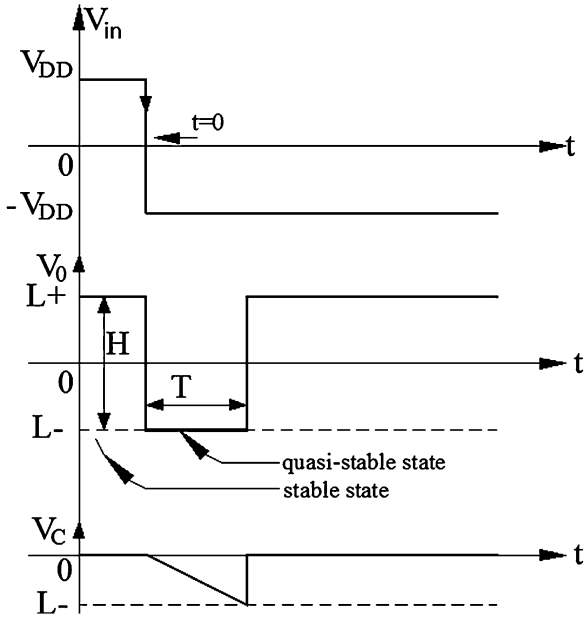


Fig. 12.12 The relevant waveforms of the circuit of Fig. 12.11

Fig. 12.13c. If the trigger signal is greater than $I_{B1}R_1$ the differential input of OTA_2 becomes negative and thereby making its output equal to $-I_{B2}R_2$. As a consequence of this, OTA_1 switches to the LOW state (i.e., $V_{01} = -I_{B1}R_1$). This causes the switch **S** to be opened and OTA_3 to deliver an output current $= -I_{B3}$ which charges the capacitor C_1 leading to a negative going ramp with slope $-(I_{B3}/C_1)$. However, as soon as V_{03} goes lower than $-I_{B2}R_2$, OTA_1 switches back to the HIGH state $+I_{B1}R_1$.

It may be observed that the time period T of the pulse generated is actually the time taken by the capacitor in changing from 0 V to a value $-I_{B2}R_2$. It, therefore, follows that T can be calculated as follows:

$$T = \frac{\text{Net change in capacitor voltage}}{\text{Rate of change of capacitor voltage}} = \frac{I_{B2}R_2}{I_{B3}/C_1} \tag{12.26}$$

Thus, from above it is clear that the pulse width can be linearly varied through I_{B2} and height by I_{B1}.

12.5.2 Monostable Multivibrators with Current Tuning Properties

A positive-edge triggering circuit structure employing two OTAs, one grounded resistor, one grounded capacitor and an NMOS transistor is shown in Fig.12.14a while its relevant waveforms are shown in Fig. 12.14b. This circuit was proposed

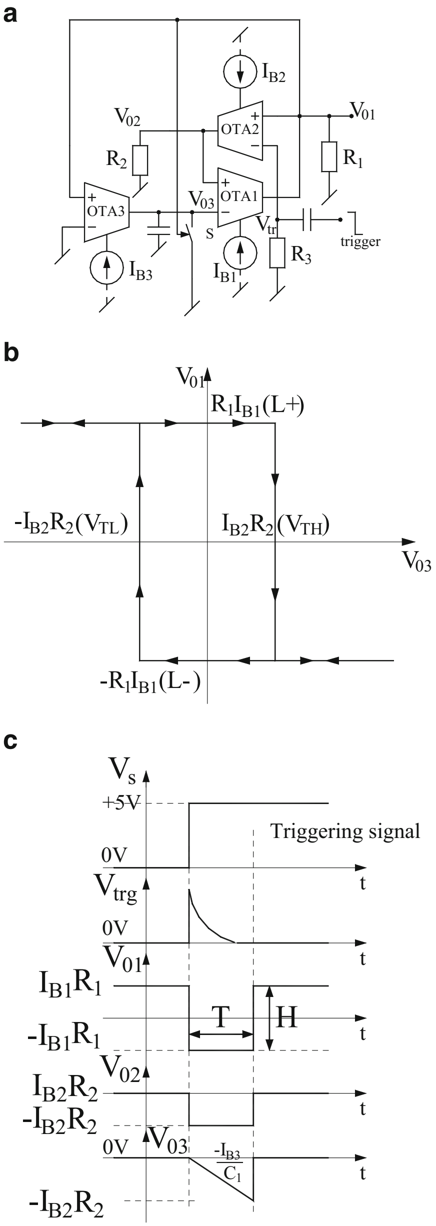


Fig. 12.13 Current controllable monostable multivibrator proposed by Chung–Cha–Kim [9] (**a**) circuit schematic (**b**) transfer characteristics of Schmitt trigger (**c**) output wave forms

by Chien–Lo [10]. It is assumed that initially, before the application of any trigger signal, it is so arranged that the capacitor voltage is zero and the analog switch is turned ON so that capacitor C is clamped at zero voltage which is possible if OTA_2 outputs a current I_{B2} into the resistor R thereby leading to $V_{02} = I_{B2}R$.

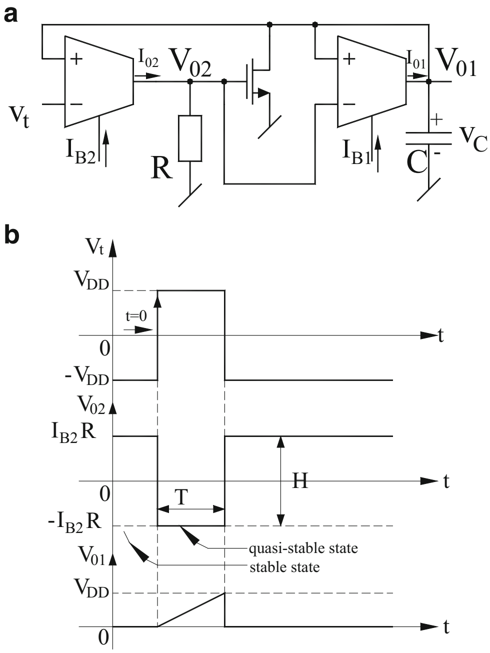


Fig. 12.14 Monostable multivibrator with positive edge triggering proposed by Chien–Lo [10] (**a**) circuit diagram (**b**) waveforms

When a trigger pulse V_t rises from $-V_{DD}$ to $+V_{DD}$, a negative output current $I_{02} = -I_{B2}$ is created thus, V_{02} abruptly jumps from $+I_{B2}R$ to $-I_{B2}R$ and the circuit is in quasi-stable state in which switch is turned OFF and I_{01} charges the capacitor C linearly. This capacitor voltage can rise only up to a value $+V_{DD}$ as shown in Fig. 12.14b. It is, therefore, concluded [10] that the pulse width T and peak-to-peak pulse amplitude H are given by

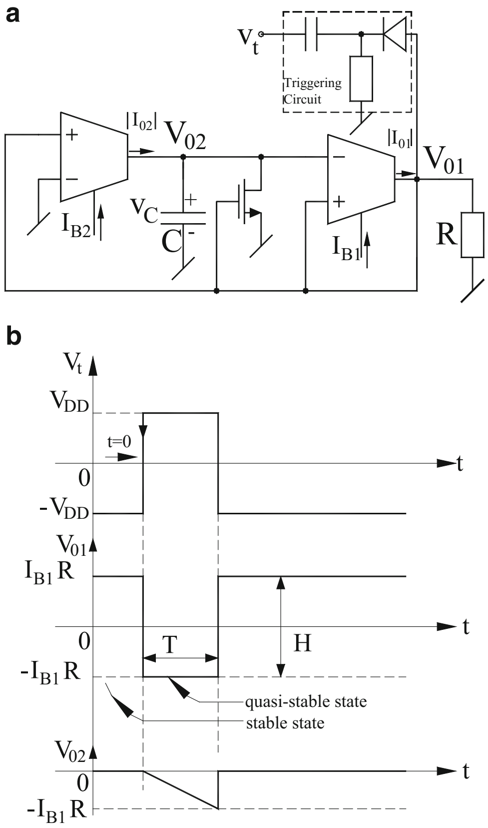
$$T = \frac{V_{DD}}{I_{B1}/C} = \frac{V_{DD}C}{I_{B1}} \quad \text{and} \quad H = 2I_{B2}R \tag{12.27}$$

Chien–Lo [10] have proposed an alternative circuit which operates in negative-edge triggering mode. This alternative circuit is shown in Fig. 12.15. The pulse width and the peak-to-peak pulse amplitude level for this circuit are given by

$$T = \left(\frac{I_{B1}}{I_{B2}}\right)RC \quad \text{and} \quad H = 2I_{B1}R \tag{12.28}$$

Both the circuits have been found to work [10] as predicted by theory when implemented with commercially available off-the-shelf OTAs LM13600 and analog switches CD4066. These circuits may find interesting applications in phase detector circuits, time delay circuits, and pulse width modulation controlled circuits.

Fig. 12.15 Monostable multivibrator with negative edge triggering proposed by Chien–Lo [10] (**a**) circuit diagram (**b**) relevant waveforms



12.5.3 *Current-Controlled Monostable Multivibrator with Retriggerable Function*

Lo and Chien [11] presented a current-controlled monostable multivibrator with retriggerable characteristic using three OTAs, one grounded capacitor, two grounded resistors, and an NMOS (as an analog switch) which is shown in Fig. 12.16a. The corresponding waveforms are given in Fig. 12.16b. The circuit operation can be described as follows:

In the absence of triggering input (v_t) the circuit remains in stable state, the analog switch is turned OFF and OTA_2 outputs a current I_{B2} into the resistor R_2, thereby leading to $V_{02} = I_{B2}R_2$. Thus, the voltage across the capacitor becomes $I_{B2}R_2$. This forces the output current of OTA1 to be zero and hence the output voltage v_{01} sets at zero volts. When a positive rising-edged triggering input signal is applied at $t = 0$, the capacitor discharges to zero volts since analog switch is turned ON. At this instant of time, the differential input voltage of OTA3 has become positive thereby the capacitor is charged linearly through the constant bias current I_{B3}. Similarly, with the

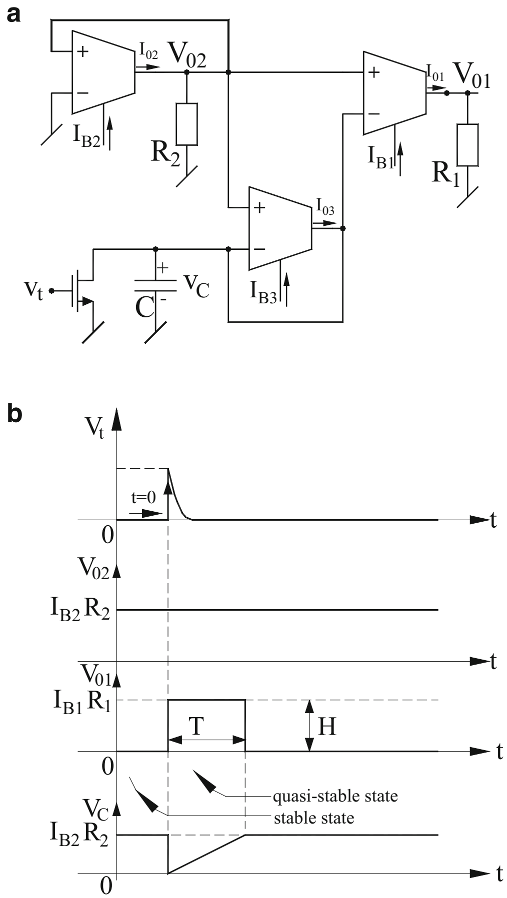


Fig. 12.16 Current-controlled monostable multivibrator with retriggerable characteristic [11] (**a**) circuit diagram (**b**) relevant waveforms

same reasoning, the output current of OTA1 becomes I_{B1} and hence the output $v_{01} = I_{B1}R_1$ and the capacitor voltage $v_c(t) = (I_{03}/C)\ t = (I_{B3}/C)$. As soon as the capacitor voltage reaches the value $I_{B2}R_2$ and if capacitor tries to charge beyond this voltage by a fraction of volts, the differential input voltage of OTA1 becomes negative and hence v_{01} drops to zero volts and remains in this state. In this way, a positive tunable mono pulse of width T and height H is generated as shown in Fig. 12.16b. The values of T and H can be expressed as

$$T = CR_2\left(\frac{I_{02}}{I_{03}}\right) = CR_2\left(\frac{I_{B2}}{I_{B3}}\right) \quad \text{and} \quad H = R_1 I_{01} = R_1 I_{B1} \tag{12.29}$$

Thus, during the quasi-stable state, whenever a positive going-edged input signal is applied a retriggerable pulse will be generated.

12.5.4 Current-Tunable Monostable Multivibrator Using Only a Single OTA

Chien [12] presented a novel current tunable monostable multivibrator using a single differential input single output (DISO) OTA which was a modification over his earlier preposition of a square/triangular wave generator and is shown in Fig. 12.17a.

As usual, there are two possible modes of operation of this circuit. In mode one, which is the permanent stable state of the circuit, it is assumed that the capacitor is clamped at the forward conduction voltage drop of the diode (V_D). This condition is maintained until a negative edged triggering signal V_{tn} is applied. Consequently, the circuit enters into a quasi-stable state in which the output voltage and the capacitor voltage can be expressed as

$$V_0(t) = \frac{R_1 R_2}{R_1 + R_2} I_B + \frac{R_2}{R_1 + R_2} V_D \quad \text{and} \quad v_C(t) = V_D \tag{12.30}$$

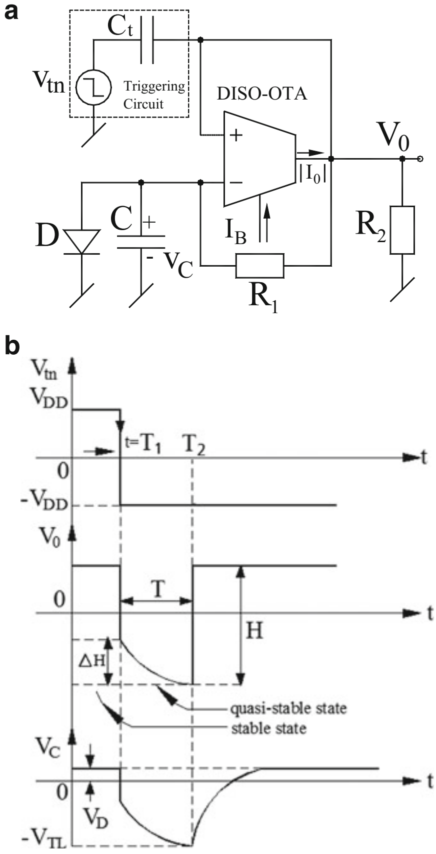


Fig. 12.17 Single OTA based monostable multivibrator proposed by Chien [12] (**a**) the circuit (**b**) the concerned waveforms

When a triggering signal is applied which falls from $+V_{DD}$ to $-V_{DD}$ at $t = T_1$ the OTA delivers reversed current and V_0 jumps from positive level to negative level. As a consequence, the diode becomes reverse biased and open circuited and thus, the capacitor C charges exponentially with a time constant R_1C. This negative going charging can go on only until capacitor voltage becomes slightly more negative than the negative threshold level V_{TL}. As a consequence, the OTA now delivers a current $I_0 = +I_B$ and its output voltage V_0 returns to its positive level the output voltage and the voltage across capacitor in this state can be expressed as [12],

$$v_0(t) = -\frac{R_1R_2}{R_1+R_2}I_B + \frac{R_2}{R_1+R_2}v_C(t) \tag{12.31}$$

$$v_C(t) = (V_D + I_BR_2)\,e^{\frac{-(t-T_1)}{(R_1+R_2)C}} - I_BR_2 \tag{12.32}$$

From the Eq. (12.28), it is, therefore, clear that the output pulse would be asymmetric. When v_c goes more negative from V_{TL} at $t = T_2$, the output v_0 becomes

$v_0(t = T_2) = (V_{TL} - \Delta v)$ where Δv is the input saturated voltage of DISO-OTA. The value of threshold voltage is given by [12],

$$V_{TL} = \left(1 + \frac{R_2}{R_1}\right)\Delta v - I_BR_2 \tag{12.33}$$

The expressions for pulse height (H), the time period (T) and the pulse height error (ΔH) are derived to be [12],

$$H = \left(\frac{R_1R_2}{R_1+R_2} + R_2\right)I_B + \left(\frac{R_2}{R_1+R_2}\right)V_D - \left(\frac{R_2}{R_1}\right)\Delta v \tag{12.34}$$

$$T = T_2 - T_1 = (R_1+R_2)C\ln\left(\frac{V_D + I_BR_2}{\left(1+\frac{R_2}{R_1}\right)\Delta v}\right) \tag{12.35}$$

$$\Delta H = I_BR_2 - \left(\frac{R_1R_2}{R_1+R_2}\right)I_B + \left(\frac{R_2}{R_1+R_2}\right)V_D - \left(\frac{R_2}{R_1}\right)\Delta v \tag{12.36}$$

12.6 Pulse Width Modulation Circuits Using OTAs

Kim–Kim–Chung [14] presented a number of pulse width modulation circuits employing CMOS OTAs. The first circuit operating on single power supply is shown in Fig. 12.18a which is a saw-tooth carrier-based PWM modulator. This circuit employs a positive feedback Schmitt trigger, an inverter, an integrator (consisting current source I_T and a capacitor C_1), a MOS switch and a comparator. The associated relevant waveforms are shown in Fig. 12.18b. The operation of the circuit can be explained as follows:

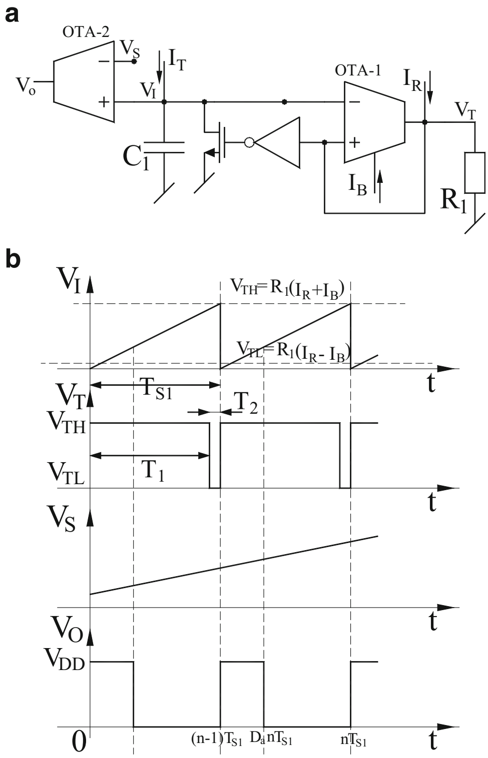


Fig. 12.18 Saw tooth carrier PWM proposed by Kim–Kim–Chuang [14] (**a**) circuit diagram (**b**) relevant waveforms

Assuming initially that the Schmitt trigger (OTA1) saturates positively at $V_{TH} = (I_B + I_R)R_1$, the MOS switch is OFF thereby charging capacitor C_1 by current I_T with a slope I_T/C_1 and the input voltage v_I of the Schmitt trigger will continue to increase linearly until v_I reaches its upper threshold voltage V_{TH}. Thus, the time interval T_1 (as shown in Fig. 12.18b) can be obtained by the following equation

$$\begin{aligned} &\frac{(I_R + I_B)R_1 - (I_R - I_B)R_1}{T_1} = \frac{I_T}{C_1} \quad \text{which yields} \\ &T_1 = 2R_1C_1\left(\frac{I_B}{I_T}\right) \end{aligned} \tag{12.37}$$

Now, when v_I tries to charge beyond V_{TH} by a fraction of volts the output of Schmitt trigger v_T will switch to its lower threshold $V_{TL} = (I_R - I_B)R_1$ and the MOS switch will be closed, forcing the voltage across the capacitor to be zero because capacitor will discharge exponentially through the very small ON-resistance of the MOS switch. Therefore, the period (T_{s1}) of the saw tooth wave, thus generated is almost same as T_1 which is directly proportional to the current ratio, I_B/I_T, thus, canceling the temperature dependent terms. The pulse width modulation (PWM) signal is generated periodically by comparing the input signal v_S at the inverting input terminal of the OTA2 with the signal v_I. The output signal v_0 of PWM as shown in Fig. 12.18b has the duty cycle (D_a)

$$D_a = \frac{v_S(D_a n T_{s1})}{2I_B R_1} + \frac{1}{2}\left(\frac{I_B - I_R}{I_B}\right) \tag{12.38}$$

Fig. 12.19a shows a circuit schematic of triangular carrier-based PWM modulator. The circuit employs a comparator (OTA4), a Schmitt trigger (OTA1) and an integrator (OTA3). The working principle of this circuit is as follows:

Let us assume that the Schmitt trigger saturates positively at the voltage V_{DD} thereby saturating OTA3 too and hence charging capacitor C_2 linearly with a slope of I_{B3}/C_2. The voltage across the capacitor which is same as the input (v_I) to the Schmitt trigger continues to increase up to the upper threshold voltage, $V_{TH} = (I_{B2} + I_R)R_2$. As soon as v_I tries to increase beyond V_{TH} by an small amount, the output voltage of OTA2 (v_T) will switch to its lower threshold voltage $V_{TL} = (I_R - I_{B2})R_2$, which is much lower than the reference voltage V_R and hence I_{B3} will get its sign changed. Therefore, v_I will decrease with a negative slope of

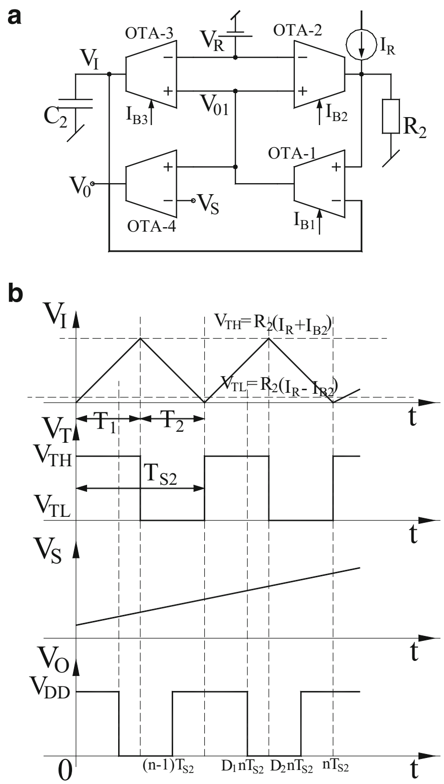


Fig. 12.19 Triangular carrier-based PWM modulator proposed by Kim–Kim–Chung [14] (**a**) Circuit schematic (**b**) relevant waveforms

$-I_{B3}/C_2$. The charging and discharging time intervals of v_I from Fig. 12.19b can be determined as:

$$T_1 = 2R_2C_2\left(\frac{I_{B2}}{I_{B3}}\right) = T_2, \quad \text{therefore the period of triangular waveform}$$
$$\mathrm{T}_{s2} = (\mathrm{T}_1 + \mathrm{T}_2) = 4R_2C_2\left(\frac{I_{B2}}{I_{B3}}\right) \tag{12.39}$$

From the above equation, it is obvious that the period T_{s2} is temperature insensitive because temperature dependent terms of I_{B2} and I_{B3} have been canceled out.

Thus, the continuous comparison of the signal voltage v_S with v_I, yields a periodically generated PWM signal v_0. From Fig. 12.19b, the expression for the duty cycle D_b has been obtained as in Ref. [14]:

$$D_b = \frac{v_S(D_b nT_{s2})}{4I_{B2}R_2} + \frac{1}{2}\left(\frac{I_{B2} - I_R}{I_{B2}}\right) \tag{12.40}$$

Yet another circuit of current-sensing PWM modulator was also presented by Kim–Kim–Chung [14]; interested readers are encouraged to refer Ref. [14] for further detailed analysis.

12.7 Concluding Remarks

In this chapter, various non-sinusoidal waveform generators and multivibrators employing OTAs are dealt with. The chapter starts with the discussion of a class of astable multivibrators made from op-amp-OTA combination in which both the devices operate in saturation. These circuits are capable of generating current-controlled square and triangular waveforms; this is followed by triangular/square waveform generators using exclusively OTAs. Novel configurations using a single dual-output OTA are also discussed. This is followed by OTA-based circuits capable of generating saw tooth waveforms and pulse waveforms.

In the category of monostable multivibrators, a number of current-controlled multivibrators with current tuning properties using only OTAs are discussed. Finally, two pulse width modulators are described.

Whereas some of the authors have established the workability of their propositions using commercially available devices, in most of the cases, the proposed configurations using OTAs have been verified in simulation only wherein CMOS architectures of the OTAs have been employed. It, therefore, appears that feasibility of realizing a large number of Schmitt triggers, astable multivibrators, and monostable multivibrators actual hardware implementation using modern OTAs which having very high slew rate such as OPA 860 (having slew rate 900 V/μs) appears to be a worthwhile area which has not been explored yet.

Secondly, with the continued proposals coming from various researchers in realizing astable and monostable multivibrators with interesting properties, it does appear that this area of research still holds promise for many new ideas and innovations.

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Chapter 13
Waveform Generators Using Current Conveyors and CFOAs

Abstract Whereas a large amount of work has been done on the synthesis of Sinusoidal oscillators using Current Conveyors and CFOAs, comparatively very little has been done on the realization of relaxation oscillators and non-sinusoidal wave form generators using the mentioned building blocks. The intention of this chapter is, therefore, to present the prominent work done in this direction till date.

13.1 Introduction

After the introduction of current Conveyors as the new circuit building blocks, in the beginning by and large, the major emphasis of the circuit designers and researchers used to be on devising new application circuits for analog filters, sinusoidal oscillators, precision rectifiers, high performance instrumentation amplifiers, etc. However, unfortunately, very little focus was on realizing nonlinear waveform generators using CCs or CFOAs.

This chapter presents the developments which have taken place since the inception of the CCs and CFAOs into the domain of nonlinear relaxation oscillators [1–20].

13.2 Schmitt Trigger and Waveform Generators Using CCs

Because a translinear CCII+ has its input stage comprised of a four transistor mixed translinear cell and the output current at the Z-terminal is a replica of the current flowing into or out of X-terminal, which is obtained by using a pair of current mirrors, it turns out that the output current is a Sinh function of the differential input $V_y - V_x$. As a consequence, the maximum charging current for the compensating capacitance connected at Z-terminal (which, in the absence of an external capacitor,

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is simply the parasitic output capacitance looking into terminal-Z which is typically in the range of 4–5 pF), is unlimited and theoretically infinite. This results in a very high slew rate in case of CCII as compared to the conventional op-amps. As an example, for CCII+ implemented from AD844, the slew rate is as high as 2000 V/μs as against a very modest figure of 0.5 V/μs for the popular μA 741 op-amp.

In view of the above, it is obvious that a square wave generator made of CCII+ is expected to be much superior to that made from an op-amp; alternatively, the non-sinusoidal waveform generators made from CCII+ can be expected to perform adequately for much higher frequency ranges than possible with op-amps. Motivated by this superiority of CCII+ over the conventional op-amp, several researchers have paid attention on devising novel realizations of Schmitt triggers and square/triangular wave form generators using CCIIs. In this section, we discuss some prominent ideas and circuits in this direction.

13.2.1 Schmitt Trigger by Di Cataldo, Palumbo, and Pennisi

The first ever Schmitt trigger using a single CCII+ along with two resistors was introduced by Cataldo, Palumbo, and Pennisi [1] which is shown in Fig. 13.1a.

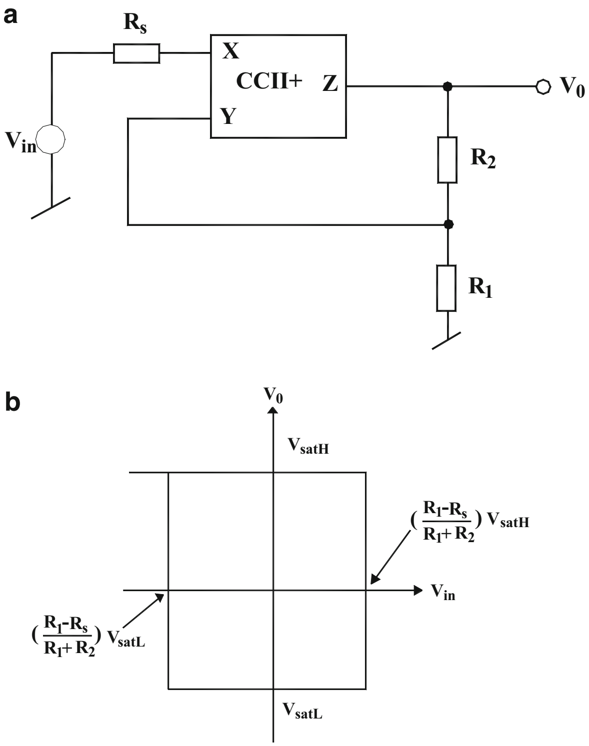


Fig. 13.1 (**a**) Schmitt trigger using CCII+ introduced by Cataldo, Palumbo and Pennisi (**b**) its transfer characteristic (adapted from Ref. [1] © 1995 John Wiley & Sons, Ltd.)

In this circuit, the regenerative feedback required to make a Schmitt trigger is created by the potential divider made from the two resistors. The operation of this circuit can be explained as follows. A decrease in the output voltage leads to a decrease in v_y; since $v_x = v_y$ therefore, v_x also decreases. Since $i_x = (V_{\text{in}} - V_x)/R_s$, this implies that, as a consequence i_x will be increased. In turn i_z also increases by the same amount as i_x which results in the increase in the voltage v_z. However, the maximum voltage V_0 is limited by the two power supply voltages hence, the circuit output will exhibit only two stable states, V_{satH} and V_{satL}, corresponding to which the two output saturation currents are found to be [9]:

$$I_{\text{satH}} = -\frac{V_{\text{satH}}}{R_1 + R_2} \tag{13.1}$$

$$I_{\text{satL}} = \frac{V_{\text{satL}}}{R_1 + R_2} \tag{13.2}$$

The two threshold voltages can be found by assuming first that v_z is set to the stable state V_{satH}. To change this stable state, the current i_x must satisfy the condition $i_x > i_z$, which implies:

$$\frac{v_{\text{in}} - v_y}{R_s} > -\frac{V_{\text{satH}}}{R_1 + R_2} \tag{13.3}$$

Hence, higher threshold voltage (V_{TH}) is given by:

$$V_{\text{TH}} = \frac{R_1 - R_s}{R_1 + R_2} V_{\text{satH}} \tag{13.4}$$

Using a similar procedure, it can be determined that the lower threshold voltage (V_{TL}) is given by:

$$V_{\text{TL}} = \frac{R_1 - R_s}{R_1 + R_2} V_{\text{satL}} \tag{13.5}$$

The transfer characteristics of this Schmitt trigger using the above derivation can be easily deduced to be as shown in Fig. 13.1b.

13.2.2 Square Wave Generator Proposed by Abuelma'atti and Al-Absi

This Schmitt trigger leads to a square wave generator when a capacitor is connected from terminal-X to ground and the resistor R_s is eliminated and in its place the internal parasitic resistance of the CCII+ is accounted for. This square wave generator was proposed by Abuelma'atti and Al-Absi [2] and is shown in Fig. 13.2.

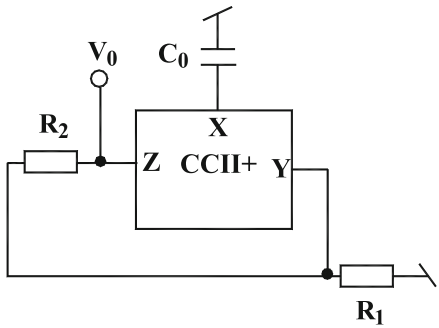


Fig. 13.2 CCII+ based square wave generator proposed by Abuelma'atti and Al-Absi [2]

It is straightforward to determine that the frequency of square wave generated from this circuit would be given by:

$$f = \frac{1}{2C_0 R_x l_n \left(2\frac{R_1}{R_x} - 1\right)} \tag{13.6}$$

13.2.3 Srinivasulu's Schmitt Trigger/Pulse Squaring Circuit

A modified two CCII+ based Schmitt trigger/pulse squaring circuit was proposed by Srinivasulu [3] which is shown in Fig. 13.3.

The operation of this circuit can be explained as follows. Assume that the input V_{in} is positive rising triangular wave changing from $-V_{in}$ to $+V_{in}$. Also assume during this transaction the current in the terminal-X_1 changes from $-i_{x1}$ to $+i_{x1}$ whereas the Z-port current rises from $-i_{z1}$ to $+i_{z1}$. It is further assumed that at node "b" the current changes from $+V_T/R_3$ to $-V_T/R_3$ and current at port-X_2 of CCII+ (2) is rising from $+i_{x2}$ to $-i_{x2}$, i.e., $+V_{x2}/R_2$ to $-V_{x2}/R_2$ and furthermore the Z-port current of CCII+ (2) rising from $+i_{z2}$ to $-i_{z2}$. At this instant the output voltage at node "a" is $+V_0$ (during the previous cycle) when the input current $i_{x1} = (V_{in} - V_{x1})/R_s \geq (R_2/R_1R_3)V_0$ at node "a", the output current direction at port Z_1 changes to $+i_{z1}$ this is the upper trigger point. Now the current at node "b" $-V_T/R_3$ and also $-i_{x2} = -V_{x2}/R_2$ and output current at port Z_2 is $-i_{z2}$. The output voltage at node "a" is $-V_0$ which remains negative until the input triangular waveform reaches a value $-i_{x1} \geq (-V_{in} - V_{x1})/R_s$.

When input current $-i_{x1} \geq (-R_2/R_1R_3)V_0$ at node "a", the direction of the port Z_1 current of CCII+ (1) changes to $-i_{z1}$ this is the lower trigger point. Now the node "b" current is $+V_T/R_3$ and current at Port X_2 of CCII+ (2) is $+i_{x2} = +V_{x2}/R_2$ and the output current at port Z_2 for CCII+ (2) is $+i_{z2}$. The net output voltage at node "a" is $+V_0$ which continues to remain positive until the input triangular waveform reaches a value of $+i_{x1}$ $(\geq (V_{in} - V_{x1})/R_s)$ at port X_1 of CCII+ (1). This cycle goes on

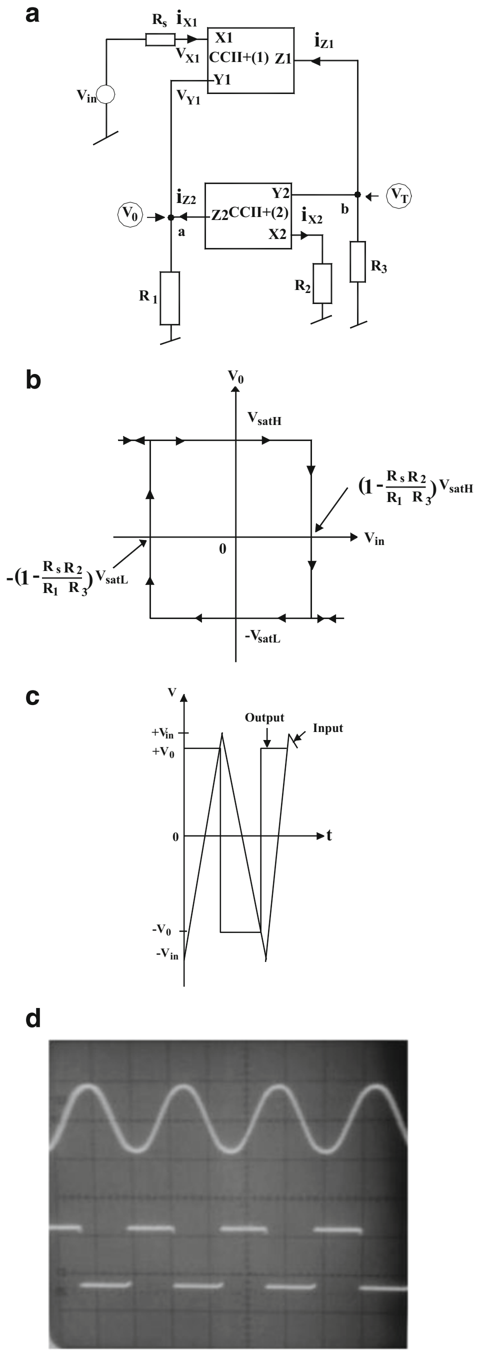


Fig. 13.3 Schmitt trigger/pulse squaring circuit proposed by Srinivasulu (adapted from Ref. [3] © 2010 John Wiley & Sons, Ltd.) (**a**) the circuit (**b**) its transfer characteristic (**c**) input/output waveforms of the trigger circuit (**d**) sine wave form input and square wave output: *x*-axis 5 ms/div and *y*-axis 5 V/div

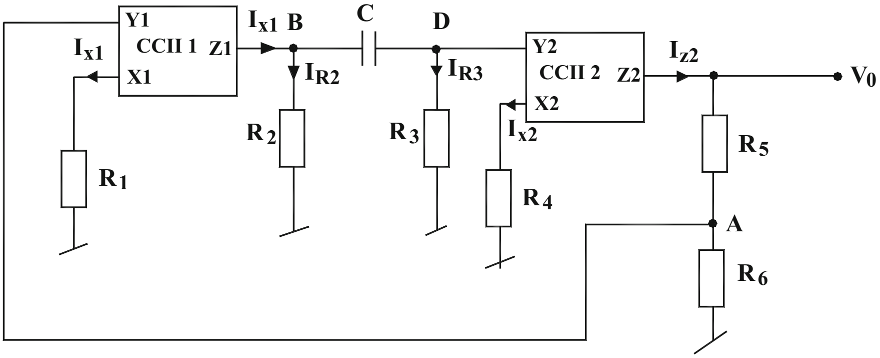
repeating itself thereby generating a square waveform at the output as shown in Fig. 13.3c.

Experimentally observed result for the circuit is shown in Fig. 13.3d for which AD844 were used to realize CCII+ biased with ±6 V supply with component values taken as $R_1 = R_2 = 10$ KΩ, $R_s = 8$ KΩ, $R_2 = 7$ KΩ.

13.2.4 Square Wave Generator Proposed by Marcellis, Carlo, Ferri, and Stornelli

The circuit of Fig. 13.4a was proposed by Marcellis–Carlo–Ferri–Stornelli [4] and consists of a voltage to current converter using $CCII_1$ along with resistor R_1 and a

a



b

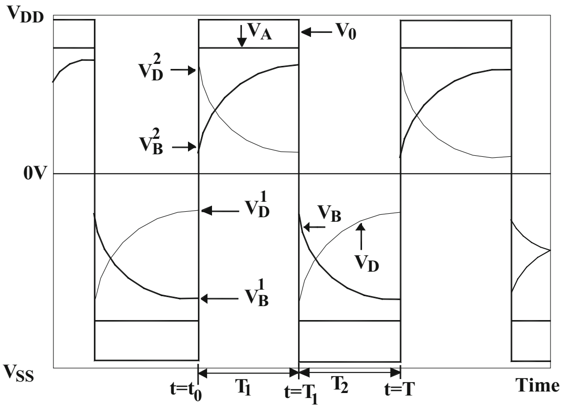


Fig. 13.4 CCII-based oscillator proposed by Marcellis–Carlo–Ferri–Stornelli (**a**) the circuit (**b**) voltage signals behavior at circuit nodes (adapted from Ref. [4] © 2011 John Wiley & Sons, Ltd.)

hysteresis current comparator consisting of $CCII_2$ along with resistor R_5 and R_6. The circuit operation can be explained as follows.

The $CCII_2$ generates a square wave signal V_0 converting the saturated current I_{z2} through R_5 and R_6 into saturation voltage $V_0 = \pm V_{sat}$. A reduced version of this output, through the potential divider consisting of R_5 and R_6, goes as input to Y_1 of $CCII_1$. This provides a current I_{z1} part of which charges the capacitor C. In fact, the circuit composed of C and R_3 acts as a differentiator which differentiates the square wave current I_{z1} and thereby generates an exponential voltage at node D (V_D). This signal is converted into a similar shaped current I_{x2} which compared with the saturation current I_{z2} thereby produces the square wave output V_{out}. Using a straightforward but lengthy analysis [22], it is found that the oscillation frequency of this circuit is given by:

$$f_0 = \frac{1}{2C(R_2 + R_3)\ln\left[\frac{2R_2R_3R_6 - R_1R_4(R_2+R_3)}{R_1R_4(R_2+R_3)}\right]} \tag{13.7}$$

The workability of this circuit has been verified in Ref. [4] using a CMOS CCII+ topology implemented in 0.35 μm technology and has also been verified by realizing CCII+ using AD844 biased with ±15 V DC power supplies. From the experimental results it has been found that a frequency range from about 390 kHz to 780 MHz is feasible exhibiting a good linearity over a frequency range of about five decades when the capacitor C was changed from 1 pF to 1 μF.

13.2.5 Square/Rectangular Wave Generator Proposed by Almashary and Alhokail

A simpler circuit capable of generating a square wave as well as rectangular wave is obtainable using square wave generator of Fig. 13.4a followed by a non-inverting integrator made from a CCII+ along with CCII+ based integrator. This circuit was proposed in Ref. [5] by Almashary and Alhokail and is shown in Fig. 13.5.

The expression of the oscillation frequency of this circuit is given by the same expression as in equation (13.6) with C_0 therein replaced by C_1.

Some sample experimental results from Ref. [5] based upon the implementation of the circuit using AD844 for CCII+, with passive component values as: $R_1 = 280\ \Omega$ and $C_1 = 1\ \mu F$ are shown in Fig. 13.5d, e. The frequency of the output waveform was found to be changeable through C_1 and/or R_1 while the amplitudes can be varied through C_2 and/or R_3.

Interested readers are referred to Refs. [9–12] for some other contributions on this topic such as, design of low-power relaxation oscillators [9], the design of Schmitt triggers with controllable hysteresis [10], tunable CC-based relaxation oscillator [11], and square wave generator with voltage-controlled frequency [12].

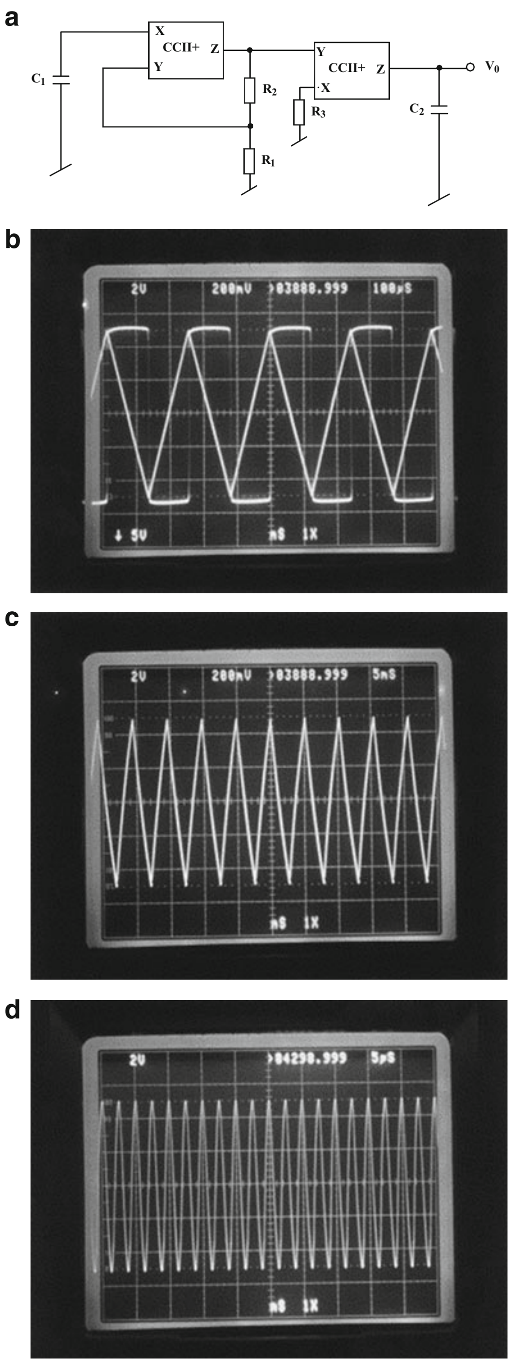


Fig. 13.5 CM triangular wave generator proposed by Almashary and Alhokail (adapted from Ref. [5] © 2000 Elsevier Science Ltd.) (**a**) the circuit (**b**) square wave and triangular wave outputs; 4.1 KHz (**c**) a typical low frequency triangular waveform; 260 Hz (**d**) a typical high frequency triangular wave form; 410 KHz

13.3 Schmitt Trigger and Non-Sinusoidal Waveform Generators Using CFOAs

At the root of any non-sinusoidal signal generator lies either a comparator or a comparator with hysteresis (using positive feedback) often referred as Schmitt trigger. As mentioned earlier, the first Schmitt trigger using a CCII+ was presented by Di Cataldo, Palumbo, and Pennisi [1] which, of course can also be implemented with a CFOA.

13.3.1 CFOA Version of the CCII+ Based Schmitt Trigger of Di Cataldo, Palumbo, and Pennisi

Consider now the Schmitt trigger of Fig. 13.6 which is, in fact, a CFOA version of the CCII based Schmitt trigger of Cataldo, Palumbo, and Pennisi [1]. In a CFOA, the output voltage is ultimately limited to $V_{\text{sat}+}$ and $V_{\text{sat}-}$ with the current flowing into the Z-terminal being

$$I_{\text{sat}+} = -\frac{V_{\text{sat}+}}{R_1 + R_2} \tag{13.8}$$

$$I_{\text{sat-}} = \frac{V_{\text{sat-}}}{R_1 + R_2} \tag{13.9}$$

If the two threshold voltages are V_{TL} and V_{TH}, they can be determined as follows:

If we assume that V_0 is in the state $V_{\text{sat}+}$ then to change this stable state, the current i_x must satisfy the condition $i_x \geq i_z$ which means

$$\frac{V_{\text{in}} - V_y}{R_s} \geq -\frac{V_{\text{sat}+}}{R_1 + R_2} \tag{13.10}$$

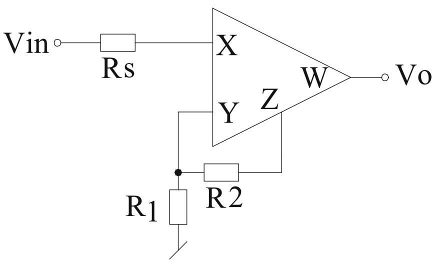


Fig. 13.6 Schmitt trigger circuit using a CFOA (adapted from Ref. [1] © 1995 John Wiley & Sons Ltd.)

The higher threshold level V_{TH} is, therefore, given by

$$V_{\mathrm{TH}} = \frac{R_1 - R_s}{R_1 + R_2} V_{\mathrm{sat}+} \tag{13.11}$$

Similarly, it can be found that the lower threshold level V_{TL} is given by

$$V_{\mathrm{TL}} = \frac{R_1 - R_s}{R_1 + R_2} V_{\mathrm{sat}-} \tag{13.12}$$

From the above analysis, the transfer characteristic of this Schmitt trigger can be drawn as shown in Fig. 13.7.

The circuit can be easily converted into a relaxation oscillator by connecting a capacitor from the input terminal-X to ground. With this addition, the circuit would generate a square wave output at V_o. Figure 13.8 shows the resulting relaxation oscillator incorporating the nonideal model of the CFOA AD844 where the nonideal parameter values are typically given by $r_x = 50\,\Omega$, $R_y = 10$ MΩ, $R_z = 3$ MΩ, $C_x = C_y = 2$ pF, and $C_z = 4.5$ pF. In Ref. [2], it has

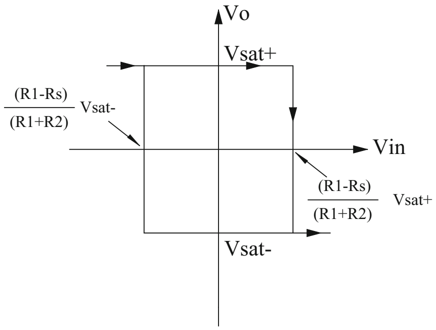


Fig. 13.7 Transfer characteristics of the Schmitt trigger of Fig. 13.6 (adapted from Ref. [1] © 1995 John Wiley & Sons Ltd.)

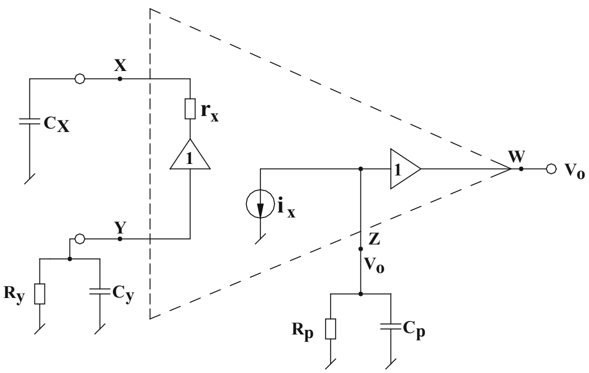


Fig. 13.8 A relaxation oscillator incorporating nonideal model of the CFOA showing various parasitic impedances (adapted from Ref. [2] © 2005 Taylor & Francis)

been shown that the oscillation period of the waveform generated by this circuit is given by

$$T = 2C_{\mathrm{T}}r_x \; l_n\left(2\frac{R_1}{r_x} - 1\right); \quad \text{where} \quad C_{\mathrm{T}} = C + C_x \tag{13.13}$$

Thus, the time period T is a function of the external capacitor C and the resistors r_x and R_1.

13.3.2 Srinivasulu's Schmitt Trigger

An improved CFOA-version of the CCII-based Schmitt trigger proposed by Srinivasulu [3] is shown here in Fig. 13.9.

In this circuit, the two threshold voltage levels are given by

$$V_{\mathrm{TH}} = \left(1 - \frac{R_2R_{\mathrm{s}}}{R_1R_3}\right)V_{\mathrm{sat}+} \tag{13.14}$$

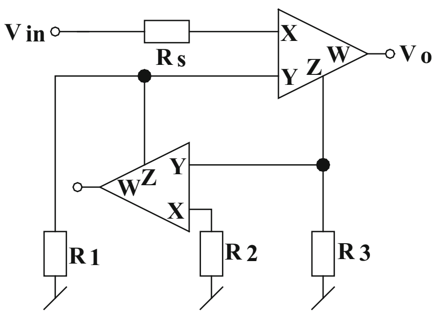
$$V_{\mathrm{TL}} = -\left(1 - \frac{R_2R_{\mathrm{s}}}{R_1R_3}\right)V_{\mathrm{sat}-} \tag{13.15}$$

Based upon the above, the transfer characteristics of the circuit can be drawn as follows (Fig. 13.10).

A square wave/triangular wave generator using the Schmitt trigger of Fig. 13.9 is shown in Fig. 13.11.

In this circuit, the resistors R and R_4 together with the capacitor C constitute an integrator. A straightforward analysis of this circuit shows that the time period (T) of the waveforms generated (a square wave at V_{01} and triangular wave at V_{02}) is given by

Fig. 13.9 An improved CFOA-version of the CCII-based circuit of Schmitt Trigger using two CFOAs



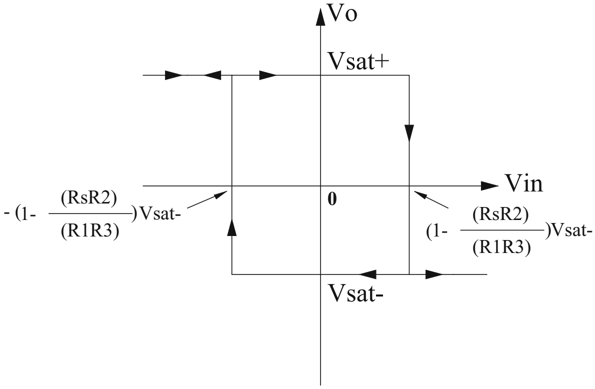


Fig. 13.10 Transfer characteristic of the Schmitt trigger of Fig. 13.9 (adapted from Ref. [3] © 2011 John Wiley & Sons Ltd.)

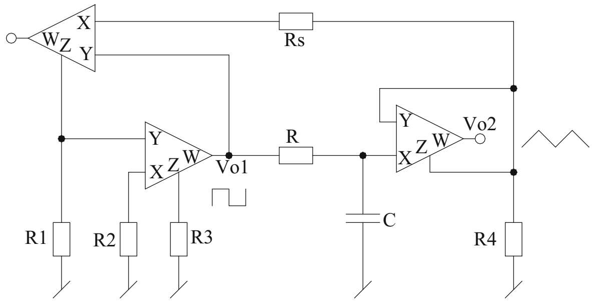


Fig. 13.11 A square/triangular wave generator using Schmitt Trigger of Fig. 13.9 proposed by Srinivasulu (adapted from Ref. [3] © 2011 John Wiley & Sons Ltd.)

$$T = 2\pi RC\left[1 - \frac{R_2 R_s}{R_1 R_3}\right] \tag{13.16}$$

Another two-CFOA-based triangular/square wave generator was advanced by Haque, Hossain, Davis, Russell, and Carter [6]. The circuit, however, requires two CFOAs, four resistors, one capacitor and $2n$ number of diodes to stabilize the Schmitt trigger output levels at $\pm nV_{\mathrm{D(on)}}$.

The frequency of oscillation for triangular/square wave generator by the circuit of Fig. 13.12 is given by

$$f_0 = \frac{\frac{V_R}{R_1} - \frac{V_N}{Z_T}\left(1 + \frac{R_2}{R_3}\right) + \frac{V_R R_2}{R_3 Z_T}}{4C_1\left[V_N\left(1 + \frac{R_2}{R_3}\right) - \frac{V_R R_2}{R_3}\right]} \tag{13.17}$$

where Z_T is the open loop transimpedance of the CFOA, V_N is the peak voltage at X-input terminal of $CFOA_2$, and V_R is the peak voltage of the square wave form.

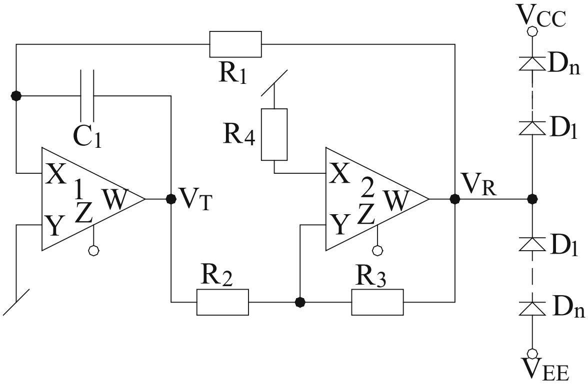


Fig. 13.12 Another two CFOA based triangular/square wave generator (adapted from Ref. [6] © 2008 IEEE)

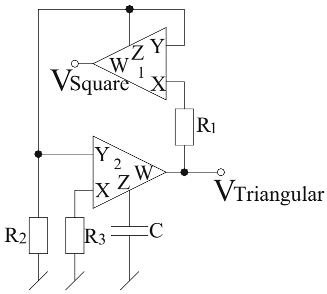


Fig. 13.13 A low-component-count CFOA-based square/triangular wave generator proposed by Minaei and Yuce (adapted from Ref. [7] © 2012 Springer)

13.3.3 Minaei–Yuce Square/Triangular Wave Generator

A novel two CFOA and one grounded capacitor based square/triangular wave generator[1] was proposed by Minaei and Yuce [7]. This circuit is shown in Fig. 13.13.

The operation of this circuit can be explained as follows. Both the CFOAs in this circuit operate as voltage saturated elements. If we assume $V_{\text{square}} = V_{\text{sat}+}$, the capacitor charges by a constant current $V_{\text{sat}+}/R_3$ so that a positive ramp appears at the output of CFOA$_2$ consequently, current flowing through R_1 decreases. When i_x becomes $\leq i_z$ then output voltage of CFOA$_1$ switches to other stable state $V_{\text{sat}-}$. Accordingly, we can write

$$\frac{V_{\text{sat}+}}{R_2} = \frac{V_{\text{sat}+} - V_{\text{tri(Peak+)}}}{R_1} \tag{13.18}$$

[1] For an alternative CFOA-based square wave generator using the same number of CFOAs and passive components which also provides electronic control through an external current/voltage signal, see Ref. [18]

From the above equation, the positive peak voltage of the triangular wave (higher threshold voltage) and the negative peak voltage (lower threshold voltage) are respectively given by

$$V_{\text{tri(Peak+)}} = \left(1 - \frac{R_1}{R_2}\right) V_{\text{Sat+}} \tag{13.19}$$

$$V_{\text{tri(Peak-)}} = \left(1 - \frac{R_1}{R_2}\right) V_{\text{Sat-}} \tag{13.20}$$

Assuming the two saturation voltages to be equal in magnitude, the time period of the waveforms generated by this circuit is given by:

$$T = 4CR_3\left(1 - \frac{R_1}{R_2}\right) \tag{13.21}$$

13.3.4 Abuelma'atti and Al-Shahrani Circuit

Finally, we show a triangular/square wave generator made from a single CFOA as shown in Fig. 13.14a. This circuit was proposed by Abuelma'atti and Al-Shahrani

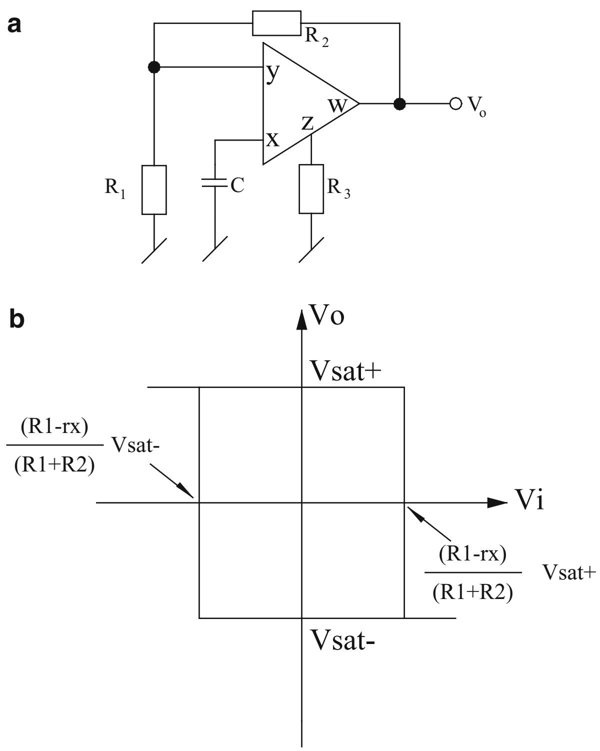


Fig. 13.14 Relaxation oscillator proposed by Abuelma'atti and Al-Shahrani [8] (**a**) triangular/square wave generator (**b**) transfer characteristic of the Schmitt trigger composed of CFOA along with R_2 and R_1

[8]. In this circuit, the CFOA behaves as a Schmitt trigger with the input–output characteristic shown in Fig. 13.14b where the two threshold voltages are given by:

$$V_{TH} = \frac{R_1 - r_x}{R_1 + R_2} V_{sat+} \text{ and } V_{TL} = \frac{R_1 - r_x}{R_1 + R_2} V_{sat-} \tag{13.22}$$

where $V_{\mathrm{sat}+}$ and $V_{\mathrm{sat}-}$ are two stable states decided by the DC biasing power supply voltages of the CFOA and r_x is the input resistance of the CFOA looking into terminal-*X* of the CFOA.

The circuit can be analyzed by starting from any one of the two stable states of the output voltage V_0 (for details, the reader is referred to Ref. [7]). The circuit generates a square wave signal at V_0 and a triangular wave signal at V_x. The frequency of the generated waveforms is given by:

$$f = \frac{1}{\left(2CR_3\left(\frac{R_1 - r_x}{R_1 + R_2}\right)\right)} \cong \frac{1}{2CR_3}\left(1 + \frac{R_2}{R_1}\right); \text{ for } R_1 >> R_x. \tag{13.23}$$

Out of the various circuits presented, the one in Fig. 13.14 is appealing due to its lowest-component-count whereas those of Figs. 13.11, 13.12, 13.13 have the advantage of providing low-output impedance outputs for both square and triangular wave outputs.

13.4 Concluding Remarks

This chapter presents the important contributions made on the realization of Schmitt trigger circuits, square and triangular wave form generators employing both Current Conveyors and CFOAs as active elements. A variety of circuits are discussed all of which are realizable with the commercially available AD844 type of CFOAs (with CCII+ realized from one CFOA while CCII− realizable with the composite connections of two CFOAs). Thus, all the circuits considered in this chapter are practically implementable using off-the-shelf ICs. In view of the high slew rate and wide-bandwidth of AD844 CFOAs, it is obvious that the circuits described herein provide useful alternatives to those realizable by commercially available IC op-amps, particularly for relatively higher frequency applications wherein op-amp-based circuits would be obviously plagued with the limitations caused by finite gain bandwidth effects as well as slew-induced distortion owing to the very modest slew rate (merely 0.5 V/μs as against that of CFOA AD844 which is around 2000 V/μs).

In retrospection, it may be seen that a lot of work still needs to be carried out on evolving CC-CFOA-based circuits for realizing saw tooth waveform generators, with variable amplitude and variable slope and square/rectangular waveform generators with variable amplitude, variable duty-cycle, variable time-period, etc. These are, however, open to investigation.

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Chapter 14
Nonsinusoidal Waveform Generators/Relaxation Oscillators Using Other Building Blocks

Abstract In this chapter, we present various multivibrator circuits and nonsinusoidal waveform generators realized with active building blocks of relatively more recent origin. In particular, we present circuit configurations for the functions mentioned above using OTRAs, current differencing buffered amplifiers, current conveyor transconductance amplifiers, and current differencing transconductance amplifiers.

14.1 Introduction

In the earlier two chapters, we have outlined important work done in the design of relaxation oscillators using OTAs, CCs, and CFOAs all of which are either commercially available as ICs (such as OTA and CFOAs) or can be implemented from off-the-shelf ICs (as in the case of CCs). In recent past, a large number of new active building blocks have been introduced in literature out of which the differential voltage current conveyors (DVCC), operational transresistance amplifiers (OTRA), current differencing buffered amplifiers (CDBA), current conveyor transconductance amplifiers (CCTA), and current differencing transconductance amplifiers (CDTA) have received considerably more attention in realizing various linear and nonlinear signal processing and signal generation applications.

Although, unfortunately, none of these building blocks are yet available as commercial integrated circuits, nevertheless, all of them can indeed be realized in terms of most of these new building blocks available as ICs, such as the CFOA AD844 and the OTA LM13600. Since a number of fully integrable bipolar and CMOS implementations of the mentioned building blocks have also been advanced by a number of researchers, the circuits involving these building blocks could also be possibly made available as integrated circuits in near future.

Thus, the aim of this chapter is to present a number of prominent waveform generators/relaxation oscillators using alternative building blocks, from amongst those presented in [1–22] with particular emphasis upon circuits employing

R. Senani et al., *Sinusoidal Oscillators and Waveform Generators using Modern Electronic Circuit Building Blocks*, DOI 10.1007/978-3-319-23712-1_14

OTRAs, DVCCs, CDBAs, CCTAs, and CDTAs, with the hope that these circuits are not only important in the context of discrete circuit designs but also hold the potential for being integrated in both bipolar and CMOS technologies.

14.2 Relaxation Oscillators Using OTRAs

The OTRA is a three-port active building block which can accept two current input signals at its two inputs which offer ideally zero input resistance. Its output is a voltage and its gain has dimension of resistance and is, therefore, called a *transresistance* represented by R_M. Thus, it is conveniently characterized by the following equation:

$$\begin{bmatrix} V_+ \\ V_- \\ V_0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ R_m & -R_m & 0 \end{bmatrix} \begin{bmatrix} I_+ \\ I_- \\ I_0 \end{bmatrix} \tag{14.1}$$

Although a number of CMOS implementations of the OTRA have been advanced by the various researchers, a typical CMOS OTRA architecture is shown here in Fig. 14.1a. For discrete implementations, however, it is quite convenient to realize an OTRA using two commercially available IC CFOAs of AD844 type as shown in Fig. 14.1b.

In this section, we describe a number of OTRA-based relaxation oscillators of various kinds proposed by a number of researchers during the last decade.

14.2.1 Schmitt Trigger Using OTRA

Lo–Chien–Chiu in [1] proposed a simple current input OTRA Schmitt trigger which offers dual-hysteresis mode operations from the same topology. In view of its importance and frequent appearance it looks appropriate to have a look at the working of this basic circuit shown in Fig. 14.2a.

Because of the positive feedback around the OTRA, the output voltage of the OTRA would be driven to either the positive saturation level V_0^+, when the current I_p is more positive than I_n. Alternatively, OTRA will be in negative saturation mode V_0^- when I_p is more negative than I_n.

The two types of hysteresis curves, which this circuit is capable of implementing, can be derived by the following arguments.

When the analog switch is connected to terminal 1, the non-inverting current and the inverting current are, respectively, equal to $I_p = V_0/R$ and $I_n = I_{in}/R$. From the characteristic of the OTRA described earlier, it follows that the transition of output voltage V_0 occurs when $I_p = I_n$. Thus, if V_0 is V_0^+ initially, it will change to V_0^- only when I_n becomes more positive than I_p. In view of this, the upper threshold current

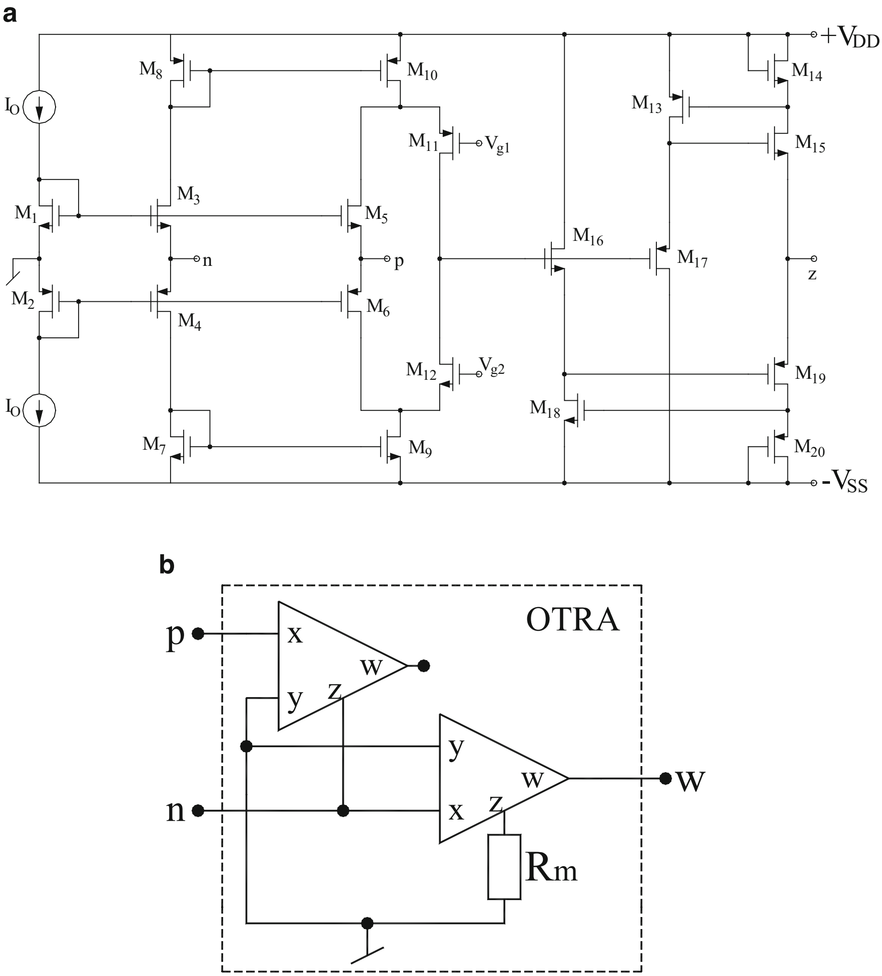


Fig. 14.1 (**a**) CMOS implementation of OTRA based upon the circuit of [23] (**b**) realization of OTRA using two CFOAs

I_{TH} can be defined as $I_{TH} = V_0^+/R$. Similarly, it can be shown that the lower threshold current would be given by $I_{TL} = V_0^-/R$. From the above considerations, the transfer characteristic exhibits a clockwise (CW) orientation as shown in Fig. 14.2b.

For the counterclockwise (CCW) mode of operation, the circuit configuration is changed by connecting the analog switch terminal 2. In this case, the non-inverting and inverting currents are given by $I_p = I_{in} + (V_0/R)$ and $I_n = 0$, respectively. In this case, if the output of the Schmitt trigger is V_0^+ initially, it will change to V_0^- only when I_p is more negative than I_n. In view of this, it follows that the lower threshold current I_{TL} can be expressed as $I_{TL} = -V_0^+/R$. Similarly, if the output of the

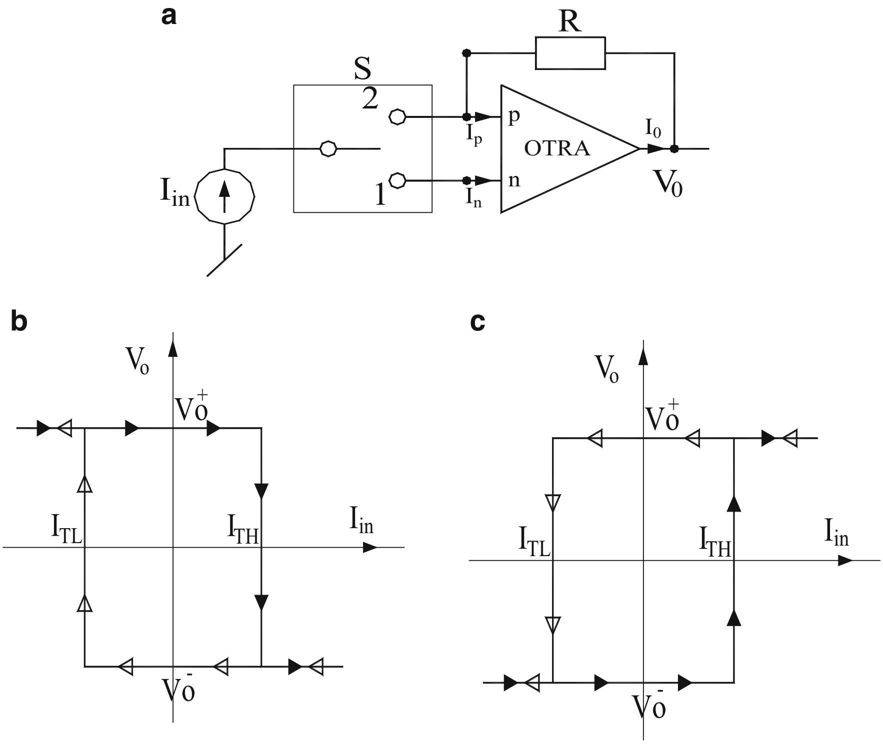


Fig. 14.2 Current input OTRA-based Schmitt trigger proposed by Lo–Chien–Chiu (adapted from [1] © 2009 John Wiley & Sons, Ltd.): (**a**) circuit arrangement, (**b**) hysteresis curve in clockwise mode operation, (**c**) hysteresis curve in counterclockwise mode operation

Schmitt trigger is V_0^- initially, it will switch over to V_0^+ only when I_p is more positive than I_n. From the above considerations, it follows that the upper threshold current I_{TH} is given by $I_{TH} = -V_0^-/R$. This explains the CCW hysteresis curve of Fig. 14.2c.

Thus, it has been demonstrated that the same circuit can function as Schmitt trigger with CW hysteresis curve or CCW hysteresis curve depending upon whether the analog switch is connected to terminal 1 or terminal 2.

14.2.2 Square Wave Generator Using a Single OTRA

In [2] Hou–Chien–Lo presented two simple-looking square wave generators, each employing only a single OTRA which are shown here in Fig. 14.3a, b, whereas the relevant waveforms generated by these circuits are exhibited in Fig. 14.3c.

The operation of these circuits can be explained as follows: The output voltage V_0 of these circuits can attain one of the two possible saturation levels, L^+ and L^-

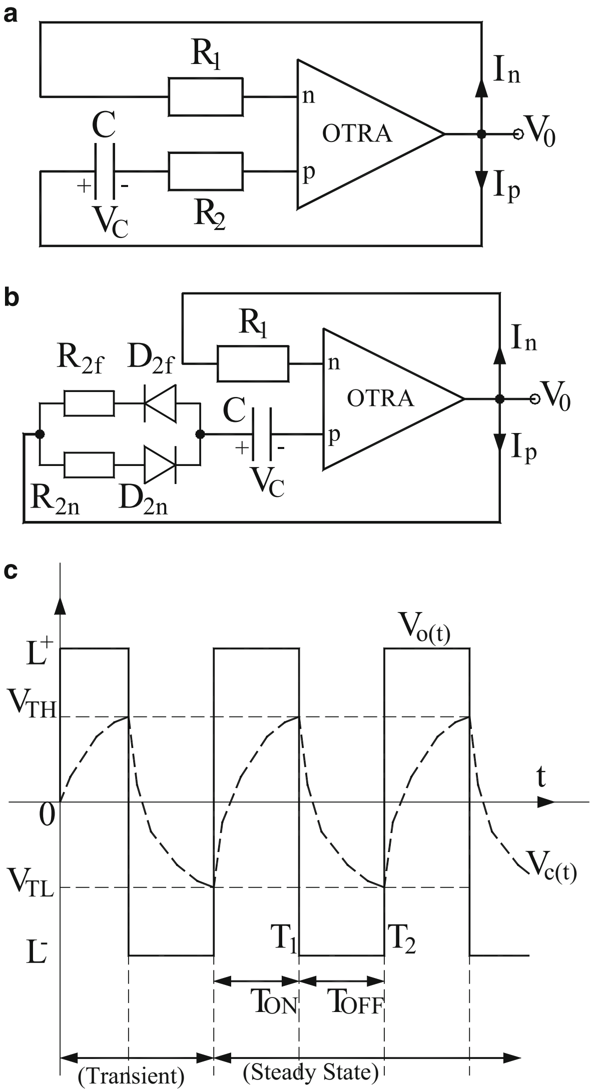


Fig. 14.3 Square wave generator proposed by Hou–Chien–Lo [2]: (**a**), (**b**) two possible implementations, (**c**) output waveforms applicable to both the circuits

(where $|L^+| = |L^-|$). The steady-state operation of the circuit of Fig. 14.2a has two modes of operation. In the first mode (from 0 to T_1), let us assume that V_0 is switched from L^- to L^+ at $t = 0$. Therefore, the capacitor C starts charging from its lower threshold value V_{TL} towards the final value L^+. The expression for capacitor voltage can be written as [2]

$$v_C(t) = (V_{TL} - L^+)\,e^{\frac{-t}{R_2 C}} + L^+ \tag{14.2}$$

At the end of this mode, the capacitor voltage will eventually reach to only up to the upper threshold voltage V_{TH}. When $t = T_1$, the current flowing into the non-inverting terminal (I_p) becomes just slightly less than I_n and hence the output of the OTRA V_0 switches to the lower saturation level L^-. The two input current of the OTRA are given by

$$I_n = \frac{V_0}{R_1} \tag{14.3}$$

$$I_p = \frac{V_0 - v_C}{R_2} \tag{14.4}$$

The two threshold levels are determined by equating I_p and I_n and are found to be

$$V_{TH} = \left(1 - \frac{R_2}{R_1}\right)L^+ = -V_{TL} \tag{14.5}$$

The time period T_1 can be determined from Eq. (14.2) by setting $v_c(T_1) = V_{TH}$ and is found to be

$$T_1 = R_2 C_{in}\left(\frac{V_{TL} - L^+}{V_{TH} - L^+}\right) = R_2 C_{in}\left(\frac{2R_1}{R_2} - 1\right) = T_{on} \tag{14.6}$$

In the second mode of operation, which is applicable for the time duration T_1 to T_2, the output of the OTRA V_0 remains at the level L^- and the capacitor is discharged towards the level L^- until the input current I_p becomes slightly larger than I_n which happens at $t = T_2$ at which time the capacitor voltage equals V_{TL}. The expression for the capacitor voltage for this mode of operation is expressed as

$$v_C(t) = (V_{TH} - L^-)e^{\frac{-(t-T_1)}{R_2 C}} + L^- \tag{14.7}$$

The corresponding time duration T_2 to T_1 is found to be

$$T_2 - T_1 = R_2 C_{in}\left(\frac{V_{TH} - L^-}{V_{TL} - L^-}\right) = R_2 C_{in}\left(\frac{2R_1}{R_2} - 1\right) = T_{off} \tag{14.8}$$

From Eqs. (14.6) and (14.8), it follows that for the proper operation of the circuit of Fig. 14.2a, it is necessary to choose the two resistors such that $R_1 > R_2$.

From the above description, it is clear that the output of the OTRA will be a symmetrical square wave alternating between the two amplitude levels L^+ and L^- and that T_{on} is equal to T_{off} and therefore the frequency of the resulting square wave would be given by

$$f_0 = \frac{1}{2R_2 C_{in}\left(\frac{2R_1}{R_2} - 1\right)} \tag{14.9}$$

A variant of the circuit of Fig. 14.2a is shown in Fig. 14.2b where the resistor R_2 has been split into two parts R_{2f} and R_{2n}. The basic operation of this circuit is similar to that of its predecessor but the frequency of the resulting rectangular wave in this case is given by

$$f = \frac{1}{C\left[R_{2n}\ln\left(\frac{2R_1 - R_{2f}}{R_{2n}}\right) + R_{2f}\ln\left(\frac{2R_1 - R_{2n}}{R_{2f}}\right)\right]} \tag{14.10}$$

From the Eq. (14.10), it is seen that in this circuit, the resistors have to be chosen such that $R_1 > R_{2n}$, R_{2f} and that the on-period and off-period can be independently set by adjusting the resistors R_{2n} and R_{2f}, respectively.

14.2.3 Current-Mode Monostable Multivibrators Using OTRAs

Lo–Chen [3] proposed a number of single OTRA-based monostable multivibrators, two of which are shown in Fig. 14.4a, b. The first circuit is triggered by a rising edged signal to produce an output pulse with a predetermined width whereas the second one is designed to operate with negative triggering signal. Fig. 14.4c displays the corresponding wave forms of the two circuits. The two saturation levels of the OTRA output are L^+ and L^-, respectively. The initial voltage across capacitor C_2 (v_{C2}) is clamped at V_{D2}. In the waveforms shown, T is the pulse width and T_r is the recovery time. By a routine analysis of the circuit, the lower threshold voltage V_{TL} is found to be $V_{TL} = v_{C2}(T) = (1 - R_2/R_1)L^-$. It has been shown in [3] that the permanent stable state of these monostable multivibrators is $V_0 = L^+$ and in this condition, I_p is more positive than I_n. When a trigger pulse is applied at $t = 0$, the output voltage switches to L^- state. Correspondingly, the capacitor voltage discharges from a level $+V_{D2}$ towards L^- but in the process, soon as it crosses the lower threshold voltage V_{TL}, a change of state occurs and output switches back to L^+ state. By a standard analysis, it has been found that the time period of rectangular wave generated by this circuit is given by

$$T = C_2R_2\ln\left[(1+K)\frac{R_1}{R_2}\right] \tag{14.11}$$

whereas the recovery time T_r is given by

$$T = C_2R_2\ln\left[\frac{2 - \frac{R_2}{R_1}}{1 - K}\right] \tag{14.12}$$

where K is given by $K = \frac{V_{D2}}{|L^-|} = \frac{V_{D2}}{-L^-} = \frac{V_{D2}}{L^+}$.

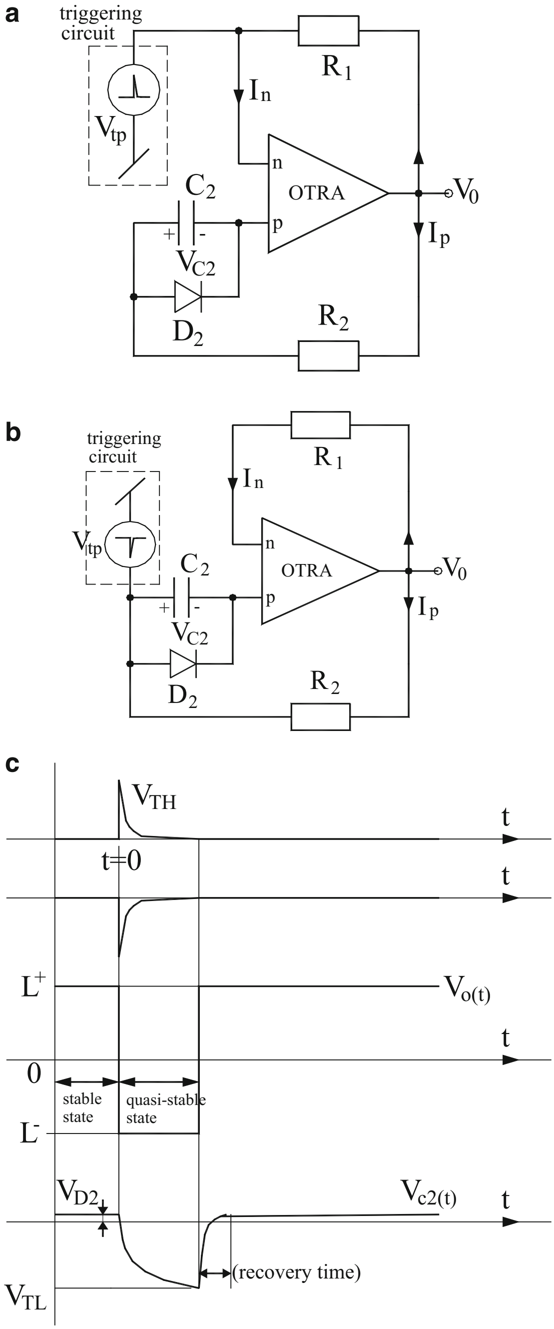


Fig. 14.4 Current-mode monostable multivibrators using OTRAs [3]: (**a**) positive triggering mode, (**b**) negative triggering mode, (**c**) relevant wave forms

The same authors in [4] proposed yet another square/triangular wave generator which however requires two OTRAs and three switches. This circuit can be operated under both inverting and non-inverting hysteresis operation by controlling three SPDT switches therein with the feature that the DC level of the triangular wave form can be adjusted through an external current source. For further details, the readers are referred to [4].

14.3 Multivibrators and Square/Triangular Wave Generators Using DVCCs

Apart from the relaxation oscillators based upon normal type of CCs as described in Chap. 13, there have been a number of investigations on implementing square/triangular wave generators using one specific variety of current conveyors, namely the differential voltage current conveyor (DVCC). In this section, we give a brief account of a variety of relaxation oscillators using DVCC.

14.3.1 Square/Triangular Wave and Saw-Tooth Wave Generator Using DVCC

Chien [5] proposed a DVCC-based voltage-controlled dual-slope square/triangular wave generator which is shown in Fig. 14.5. The basic principle of this circuit can be explained as follows:

The connection Y_1-Z provides a positive feedback path in the circuit of Fig. 14.5a which leads to the saturation of DVCC-2 due to which its output voltage would be either in the positive saturation level V_0^+ or in the negative saturation level V_0^-. The relevant wave forms associated with this circuit are shown in Fig. 14.5b where V_{TH} and V_{TL} represent the upper and lower threshold levels. Assuming that $V_{01} = V_0^+$ and $R_2 > R_1$ it follows that DVCC-I makes it possible to force a constant current flowing through the capacitor C so that the output voltage V_{02} is positive going ramp whose slope is given by

$$\frac{\mathrm{d}V_{02}(t)}{\mathrm{d}t} = \frac{V_{TH} - V_{TL}}{T_1} = \frac{V_{01} - V_B}{R_3 C} = \frac{V_0^+ - V_B}{R_3 C} \tag{14.13}$$

The positive ramp continues until V_{02} reaches the upper threshold level V_{TH} at which point the output voltage V_{01} of DVCC-II switches to negative saturation level V_0^-.

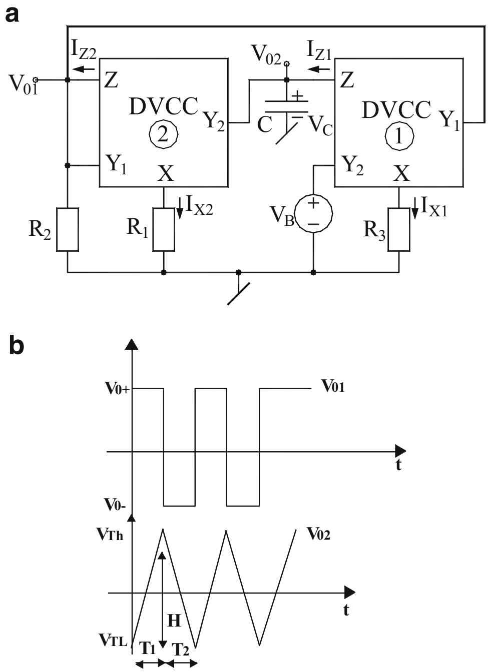


Fig. 14.5 DVCC-based voltage-controlled dual-slope square/triangular wave generator proposed by Chien (adapted from [5] © 2012 Elsevier Ltd.)

The circuit then enters off-duty cycle operation and the expressions for I_{x2} and I_{z2} are given by

$$I_{x2} = \frac{V_{01} - V_{02}}{R_1} = \frac{V_0^+ - V_{02}}{R_1} \tag{14.14}$$

$$I_{z2} = \frac{V_{01}}{R_2} = \frac{V_0^+}{R_2} \tag{14.15}$$

Because of V_{01} being at V_0^-, this reverses the output current of DVCC-I and hence V_{02} is a negative going ramp. As soon as V_{02} reaches the lower threshold level V_{TL}, V_{01} switches back to positive saturation level V_0^+ and this cycle goes on repeating itself. V_{TH} and V_{TL} can be calculated by using the following equations:

$$\frac{dV_{02}(t)}{dt} = \frac{V_{TL} - V_{TH}}{T_2} = \frac{V_{01} - V_B}{R_3 C} = \frac{-V_0^+ - V_B}{R_3 C} \tag{14.16}$$

$$I_{x2} = \frac{V_{01} - V_{02}}{R_1} = \frac{-V_0^+ - V_{02}}{R_1} \tag{14.17}$$

$$I_{z2} = \frac{V_{01}}{R_2} = \frac{-V_0^+}{R_2} \tag{14.18}$$

and setting $I_{x2} = I_{z2}$, which leads to the following values of V_{TL} and V_{TH}:

$$V_{TH} = -V_{TL} = \left(1 - \frac{R_1}{R_2}\right) V_0^+ \tag{14.19}$$

The amplitude (peak-to-peak) of the triangular wave form is given by:

$$V_{02} = (V_{TH} - V_{TL}) = 2\left(1 - \frac{R_1}{R_2}\right) V_0^+ \tag{14.20}$$

The periods of the square and triangular waves are given by

$$T_1 = \frac{2R_3C(1 - (R_1/R_2))}{V_c} \tag{14.21}$$

$$T_2 = \frac{2R_3C(1 - (R_1/R_2))}{2 - V_c} \tag{14.22}$$

Finally, the oscillation frequency (f_0) and duty cycle (D_a) are given by

$$f_0 = \frac{1}{T_1 + T_2} = \frac{V_C(2 - V_C)}{4R_3C(1 - (R_1/R_2))} \tag{14.23}$$

$$D_a = \frac{T_1}{T_1 + T_2} = \left(1 - \frac{V_C}{2}\right) \tag{14.24}$$

where

$$V_C = \left(1 - \frac{V_B}{V_0^+}\right) \tag{14.25}$$

It is interesting to note that f_0 can be independently controlled by R_3, thereby resulting in single resistance-controlled oscillator. Also, the amplitude of the triangular wave is proportional to the ratio (R_1/R_2) whereas the duty cycle is controllable by the control voltage V_c. Furthermore, if one takes $V_c << 1$ the interval T_2 is negligibly small and can be ignored and the circuit can then be used as saw-tooth waveform generator for which the frequency of oscillation is given by

$$f_0 = \frac{1}{T_1} = \frac{V_C}{2R_3C(1 - (R_1/R_2))} \tag{14.26}$$

14.3.2 Switch-Controllable Bistable Multivibrator

Chien in [6] demonstrated that a simple switch-controllable bistable multivibrator with dual-hysteresis modes can be made from only a single DVCC and two external resistors along with a double-pole double-throw (DPDT) switch. This circuit is

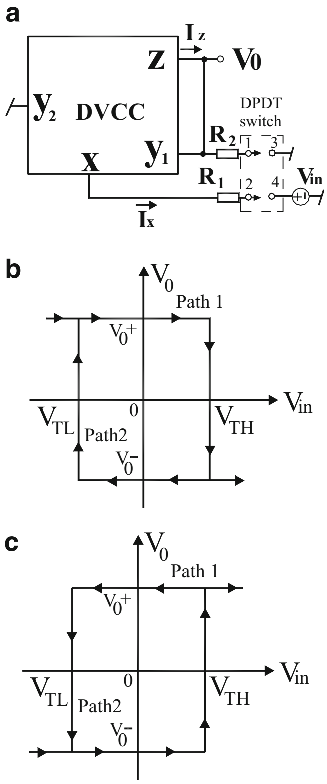


Fig. 14.6 (**a**) Switch-controllable DVCC-based bistable multivibrator, (**b**) transfer characteristics in CW mode, (**c**) transfer characteristics in CCW mode (adapted from [6] © 2011 Elsevier Ltd.)

shown in Fig. 14.6a. The operation of this circuit can be explained as follows. It is assumed that this circuit will be operated with an external voltage signal V_{in} which can be an existing square/triangular wave generator. The shorting of the terminals Y_1 and Z along with the resistor R_2 provides a positive feedback in the circuit. As a consequence, the DVCC is saturated and the output V_0 would be either at positive saturation level V_0^+ or at negative saturation level V_0^-. The transfer characteristics of the circuit are shown in Fig. 14.6b. By proper connection of the DPDT switch, this circuit exhibits hysteresis characteristics of two different kinds which have been termed as clockwise (CW) mode or counter-CW (CCW) mode which are shown, respectively, in Fig. 14.6b, c.

For CW mode, the DPDT switch is made to connect node 1 to node 3 and node 2 to node 4. It is easy to see that the terminal currents I_X and I_Z are, respectively, given by

$$I_X = \frac{V_0 - V_{in}}{R_1}; \quad \text{and} \quad I_Z = \frac{V_0}{R_2} \tag{14.27}$$

The hysteresis characteristics for CW mode can be explained as follows. Assume that V_0 is at positive saturation level V_0^+ and also let initial input $V_{in} = 0$. Further, let

$R_2 > R_1$; then from Eq. (14.25) it follows that I_x would be larger than I_z. Thus, it is ensured that V_0 will be in the level V_0^+ initially. If a positive V_{in} is applied now, the current I_x will be decreasing gradually. Now because I_z is larger than I_x, V_0 switches from the state V_0^+ to V_0^-. This explains the path 1 in Fig. 14.6b. Similarly, in path 2 the output is initially at the level V_0^- and hence current I_z is greater than current I_x. Now if input signal V_{in} decreases from zero and is negative, the current I_x increases until it becomes larger than I_z, due to which V_0 switches from V_0^- to V_0^+; this explains path 2 of the transfer characteristics of Fig. 14.6b. The upper and lower threshold voltages V_{TH} and V_{TL} can be determined from the instant of time when I_x and I_z are equal. The mathematical expressions for V_{TH} and V_{TL} in CW mode are given by

$$V_{TH} = \left(1 - \frac{R_1}{R_2}\right)V_0^+ ; \quad \text{and} \quad V_{TL} = \left(1 - \frac{R_1}{R_2}\right)V_0^- \tag{14.28}$$

For CCW mode, the DPDT switch enables the connections of terminals 1 to 4 and 2 to 3. Obviously, in such a case, the expressions for the currents I_x and I_z change as follows:

$$I_x = \frac{V_0}{R_1}; \quad \text{and} \quad I_z = \frac{V_0 - V_{in}}{R_2} \tag{14.29}$$

The transfer characteristics for the CCW mode can be explained similar to the case of CW mode [6] which is omitted to conserve the space. The expressions of two threshold voltages in this case are given by

$$V_{TH} = \left(\frac{R_2}{R_1} - 1\right)V_0^+; \quad \text{and} \quad V_{TL} = \left(\frac{R_2}{R_1} - 1\right)V_0^- \tag{14.30}$$

A voltage-controlled bistable multivibrator is obtained by connecting an external voltage source V_B at the Y_2 terminal of the DVCC as shown in Fig. 14.7. With the introduction of an additional controlled voltage, it is obvious that the threshold voltages in both CW and CCW modes will become a function of this control voltage also. By routine analysis, the modified expressions for the threshold voltages are determined by the following expressions [6]:

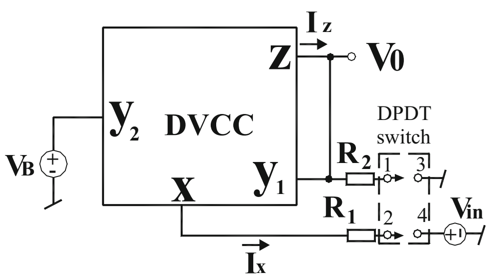


Fig. 14.7 DC voltage-controlled DVCC bistable multivibrator (adapted from [6] © 2011 Elsevier Ltd.)

In CW mode:

$$V_{\mathrm{TH}} = \left(1 - \frac{R_1}{R_2}\right)V_0^+ - V_{\mathrm{B}}; \quad \text{and} \quad V_{\mathrm{TL}} = \left(1 - \frac{R_1}{R_2}\right)V_0^- - V_{\mathrm{B}} \tag{14.31}$$

In CCW mode:

$$V_{\mathrm{TH}} = \left(\frac{R_2}{R_1} - 1\right)V_0^+ + \left(\frac{R_2}{R_1}\right)V_{\mathrm{B}}; \quad \text{and}$$
$$V_{\mathrm{TL}} = \left(\frac{R_2}{R_1} - 1\right)V_0^- + \left(\frac{R_2}{R_1}\right)V_{\mathrm{B}} \tag{14.32}$$

The effectiveness of the above bistable multivibrators has been confirmed by implementing the DVCC by AD844AN ICs. From the breadboard prototypes, the operating frequency achievable from the proposed bistable circuits was found to be around 500 KHz.

14.3.3 Single DVCC-Based Monostable Multivibrators

Chien–Lo in [8] presented two novel single DVCC-based monostable multivibrators which require fewer components than, for example, a traditional op-amp-based multivibrator. These circuits are shown in Fig. 14.8.

The operation of these circuits can be explained as follows:

Firstly, it may be noted that the connection from Z to Y_1 along with R_2 constitutes the required positive feedback path which facilitates the saturation of the DVCC due to which its output would be only in one of the two saturation levels V_0^+ and V_0^-. The trigger input V_{trg} is a rising edge signal which can be provided by an external function generator. The permanent stable state of the circuit before the application of the trigger pulse can be easily shown [7] to be V_0^+ (Fig. 14.8c) by noting that in the stable state ($0 < t < T_1$), the capacitor C is open circuited and V_{c} is clamped by the diode D or the analog switch M in the circuit of 14.8b. If R_2 is much greater than R_1 and since $I_{\mathrm{x}} = \frac{V_0}{R_1} = \frac{V_0^+}{R_1}$, $I_{\mathrm{z}} = \frac{V_0}{R_2} = \frac{V_0^+}{R_2}$, it follows that I_{x} is more positive than I_{z} which ensures that output voltage V_0 will be in positive saturation level V_0^+.

When a positive edge triggering signal is applied at $t = T_1$ the circuit enters into a quasi-stable state ($T_1 < t < T_2$). When this happens, I_{z} becomes more positive than I_{x} and hence the output voltage V_0 changes abruptly from V_0^+ to V_0^-. This results in the discharge of the capacitor through R_1 from $t = T_1$. In the quasi-stable state, the expressions for I_{x}, I_{z}, and V_{c} are given by

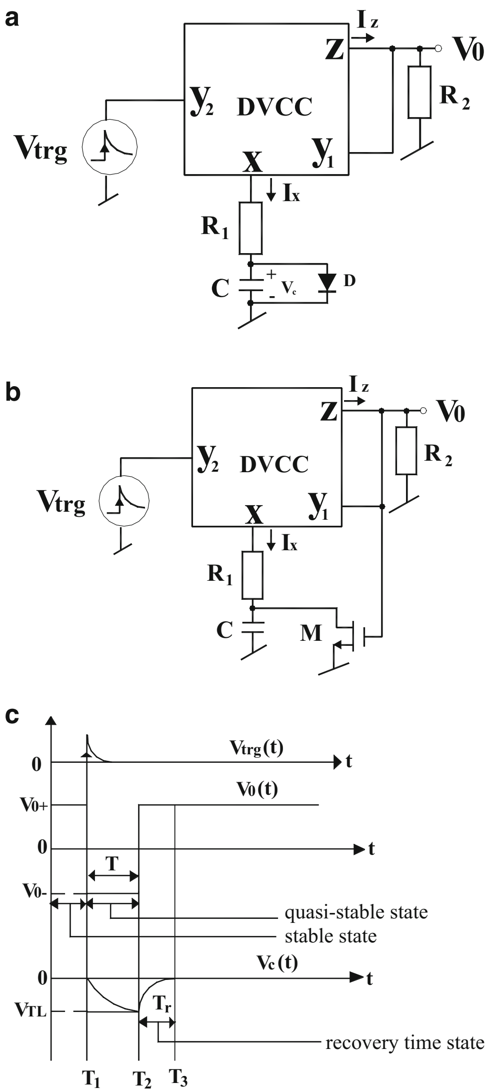


Fig. 14.8 DVCC-based monostable multivibrators: (**a**) the basic circuit, (**b**) the modified circuit shortened recovery time, (**c**) corresponding waveforms (adapted from [7] © 2011 Elsevier Ltd.)

$$V_c(t) = V_0^-\left(1 - e^{-(t-T_1/R_1C)}\right) \tag{14.33}$$

$$I_X = \frac{V_0 - V_C}{R_1} = \frac{V_0^- - V_C}{R_1} \tag{14.34}$$

$$I_z = \frac{V_0}{R_2} = \frac{V_0^-}{R_2} \tag{14.35}$$

When at a time $t = T_2$ the capacitor voltage drops to the lower threshold voltage V_{TL} the expression for which is determined when $I_x = I_z$ and is given by

$$V_{TL} = V_c(T_2) = \left(1 - \frac{R_1}{R_2}\right) V_0^- \tag{14.36}$$

By substituting the value of $V_c(T_2)$ in the general expression for the capacitor voltage time period T is found to be

$$T = T_2 - T_1 = R_1 C \ln\left(1 - \frac{V_{TL}}{V_0^-}\right) = R_1 C \ln\left(\frac{R_2}{R_1}\right) \tag{14.37}$$

By a routine analysis [7], the recovery times T_r for both the circuits are defined as $T_r = (T_3 - T_2)$ and are found to be

$$T_r = T_3 - T_2 \cong R_1 C \ln\left(2 - \frac{R_1}{R_2}\right) \text{ if } \; R_{DR} >> R_1 \tag{14.38}$$

where R_{DR} is the turn-off resistance of the diode D:

$$T_r = T_3 - T_2 \cong R_{DS} C \ln\left[1 + \frac{R_1(R_2 - R_1)}{R_2 R_{DS}}\right] \text{ if } \; R_{DS} << R_1 \tag{14.39}$$

From the above, it is easy to visualize that, since R_{DS} (turn-on resistance of the analog switch) is usually much larger than R_1, the recovery time of the second monostable circuit will be much shorter than the first one.

The effectiveness of both the circuits has been confirmed [7] through SPICE simulations as well as hardware implementations by realizing DVCC with AD844AN ICs and using CD4066 as the analog switch where from the effectiveness of the second circuit in shortening the recovering time has been confirmed.

14.3.4 Relaxation Oscillators Using DVCCs

Two circuits for realizing relaxation oscillators using DVCC were presented by Chien in [8]. The first circuit can generate a symmetrical square wave with 50 % duty cycle while the second circuit can further control the duty cycle by adjusting the tuning factor of the potentiometer. These circuits are shown in Fig. 14.9.

The operation of the circuit of Fig. 14.9 a can be explained as follows: Let us assume that V_0 is in the high state, i.e., $V_0 = V_0^+$ if we take $R_1 > R_2$ and then I_X is

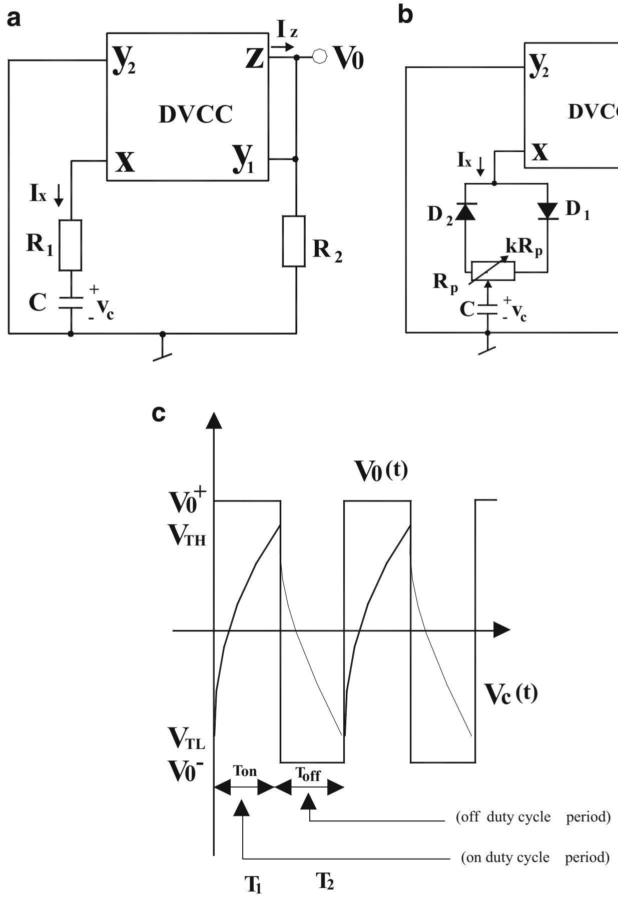


Fig. 14.9 The relaxation oscillators using a single DVCC and their relevant waveforms [8]: (**a**) circuit with 50 % duty cycle, (**b**) circuit with variable duty cycle, (**c**) corresponding waveforms

more positive than I_Z and consequently V_0 is guaranteed to be in the state $V_0{}^+$. The capacitor charges exponentially from low threshold (V_{LT}) to high threshold (V_{HT}). When capacitor voltage becomes slightly greater than V_{TH}, this also implies that now I_X would have become slightly less than I_Z, as a consequence of which V_0 switches from $V_0{}^+$ state to $V_0{}^-$ state. From a routine analysis [8], the two threshold voltages are found to be

$$V_{TH} = \left(1 - \frac{R_1}{R_2}\right) V_0{}^+ \quad \text{and} \quad V_{TL} = -\left(1 - \frac{R_1}{R_2}\right) V_0{}^+ \tag{14.40}$$

If we consider the total time period in which the output waveform remains high as T_1, from a straightforward analysis [8], its value is given by

$$T_1 = R_1 C \ln\left(\frac{2R_2}{R_1} - 1\right) \tag{14.41}$$

Similarly, the time period in which the output wave remains in low state, i.e., V_0^-, is found to be

$$T_2 = R_1 C \ln\left(\frac{2R_2}{R_1} - 1\right) \tag{14.42}$$

Thus, the frequency of the output square wave is found to be

$$f_0 = \frac{1}{T_1 + T_2} = \frac{1}{2R_1 C \ln\left(\frac{2R_2}{R_1} - 1\right)} \tag{14.43}$$

A slightly modified form of this circuit to facilitate non-50 % duty cycle (Fig. 14.9b) is obtained by splitting the resistance R_1 into two parts through a potentiometer (kR$_P$) and ensuring that during the charging and discharging of the capacitor C only one of these two parts is coming into picture. From the diagram, it can be seen that the diode D_1 is ON and diode D_2 is OFF during the charging of the capacitor (for the duration T_1) whereas on the other hand, during the discharging of the capacitor, the roles of the diodes are reversed, i.e., D_2 is ON and D_1 is OFF, due to which the lower part of the potentiometer comes into picture. With a routine but detailed analysis, it has been shown that duty cycle (δ) turns out to be

$$\delta = \frac{1}{1 + \left(\frac{1-k}{k}\right)\left(\frac{\ln\left(\frac{2R_2}{(1-k)R_P} - \frac{k}{k-1}\right)}{\ln\left(\frac{2R_2}{R_P} + k - 1\right)}\right)} \tag{14.44}$$

where k is the tuning factor of the potentiometer. Thus, the value of δ other than 50 % can be achieved by proper selection of the parameter k.

14.3.5 DO-DVCC-Based Square/Triangular Wave Generator

A novel square/triangular waveform generator using single DO-DVCC and three grounded passive components was advanced by Chien in [9] and is shown here in Fig. 14.10

The operation of this circuit can be explained as follows. The Y_1-Z^+ connection along with the resistor R_2 forms the positive feedback path due to which the DVCC saturates and hence its output could be either in state V_0^+ or V_0^-. In the on-duty cycle period (T_1), V_{01} is at positive saturation level in the beginning. In order for this to happen, the current I_x must have a strong positive charge than I_z. To accomplish this

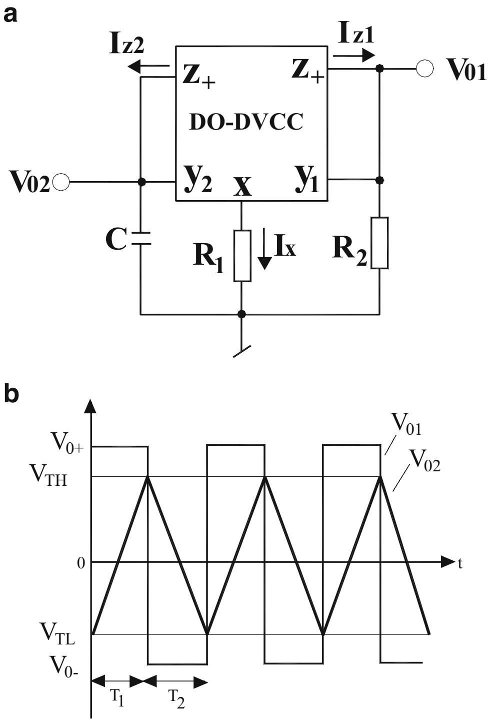


Fig. 14.10 DO-DVCC-based square/triangular waveform generator employing grounded passive components [9]: (**a**) proposed circuit, (**b**) its relevant waveforms

R_2 needs to be larger than R_1. Thus, capacitor C is charged, causing V_{02} to be a positive going ramp. The rate at which the capacitor charges linearly along with the values of I_x and I_z are given by the following equations:

$$\frac{dV_{02}(t)}{dt} = \frac{V_{01}}{R_2C} = \frac{V_0^+}{R_2C} = \frac{V_{TH} - V_{TL}}{T_1} \tag{14.45}$$

$$I_X = \frac{V_{01} - V_{02}}{R_1} = \frac{V_0^+ - V_{02}}{R_1} \tag{14.46}$$

$$I_z = I_{z1} = I_{z2} = \frac{V_{01}}{R_2} = \frac{V_0^+}{R_2} \tag{14.47}$$

The positive ramp continues until V_{02} reaches the upper threshold level (V_{TH}) at which time the output V_{01} switches to low state V_0^-. Now the capacitor voltage V_{02} starts discharging linearly resulting in a negative going ramp at V_{02}. During this period, when the capacitor voltage crosses the lower threshold level (V_{TL}), the circuit resumes the high state and this operation goes on repeating. The equations for the capacitor voltage following a negative ramp and the currents I_x and I_z in this state are given below:

$$\frac{dV_{02}(t)}{dt} = \frac{V_{01}}{R_2C} = \frac{-V_0^+}{R_2C} = \frac{V_{TL} - V_{TH}}{T_2} \tag{14.48}$$

$$I_X = \frac{V_{01} - V_{02}}{R_1} = \frac{-V_0^+ - V_{02}}{R_1} \tag{14.49}$$

$$I_z = I_{z1} = I_{z2} = \frac{V_{01}}{R_2} = \frac{-V_0^+}{R_2} \tag{14.50}$$

Using the above equations and setting $I_x = I_z$, the expressions for the threshold voltages are found to be

$$V_{TH} = -V_{TL} = \left(1 - \frac{R_1}{R_2}\right)V_0^+ \tag{14.51}$$

The oscillation frequency of the waveforms generated by this circuit is found to be

$$f_0 = \frac{1}{T_1 + T_2} = \frac{1}{4R_2C(1 - (R_1/R_2))} \tag{14.52}$$

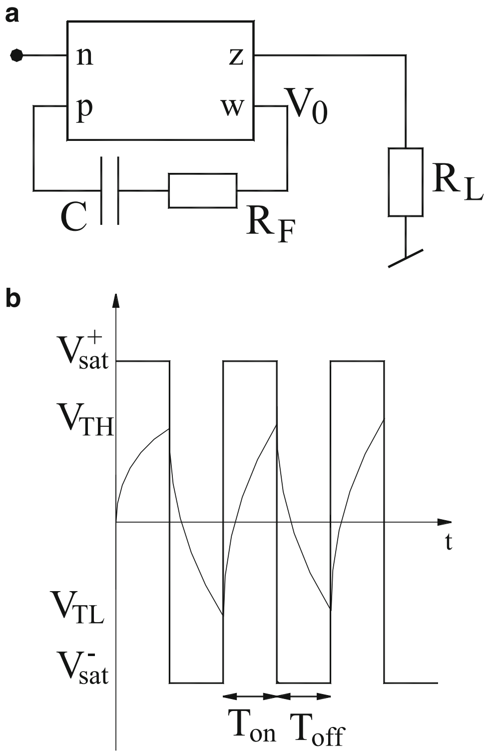
From the experiments based upon realizing DVCC using commercially available AD844AN ICs, it has been found that in the discrete breadboarded versions of the circuit, the maximum attainable oscillation frequency of the circuit has been found to be several hundred KHz (500–800 KHz) which is still larger than that attainable from square/triangular waveform generators using op-amps or OTAs (which were found to be well below 100 KHz in case of OTAs; will be much smaller in case of op-amp-based circuits). In view of the simplicity of the structure coupled with its good performance, it appears that the circuit may find applications in instrumentation, communication, and signal processing systems.

14.4 Multivibrators Using CDBA

A CDBA-based astable multivibrator was proposed by Pandey–Pandey–Paul–Anand–Gautam [10]. Consider the basic stable multivibrator exhibiting 50 % duty cycle shown in Fig. 14.11. In this circuit the required positive feedback to create an astable multivibrator is provided by the series-RC branch whereas by choosing a large enough value of R_L it is ensured that the CDBA is driven into saturation so that the output V_0 would be one of the two possible saturation levels V_{sat}^+ and V_{sat}^-.

If we assume the output state to be V_{sat}^+, the capacitor charges exponentially. When the capacitor voltage reaches a value at which the current through R_L is not large enough to maintain the output voltage at V_{sat}^+, the output switches to V_{sat}^-. Now the capacitor starts charging in the opposite direction as shown in Fig. 14.11b.

Fig. 14.11 Astable multivibrator with fixed 50 % duty cycle proposed by Pandey–Pandey–Paul–Anand–Gautam [10]: (**a**) circuit diagram, (**b**) relevant wave forms



This operation is repeated as a consequence of which the output waveform will alternate between V_{sat}^{+} and V_{sat}^{-}. The two threshold voltages shown in the diagram are found to be [10]

$$V_{\text{TL}} = \frac{(R_{\text{L}} - R_{\text{F}})}{R_{\text{L}}} V_{\text{sat}}^{-} \tag{14.53}$$

$$V_{\text{TH}} = \frac{(R_{\text{L}} - R_{\text{F}})}{R_{\text{L}}} V_{\text{sat}}^{+} \tag{14.54}$$

By a straightforward analysis [10], the time period during which the output remains high is given by

$$T_{\text{on}} = R_{\text{F}} C \ln\left[\frac{2R_{\text{L}}}{R_{\text{F}}} - 1\right] \tag{14.55}$$

Since $T_{\text{H}} = T_{\text{L}}$, it follows that the total time period would be $2T_{\text{H}}$ and hence the frequency of the generated square wave would be given by [10]

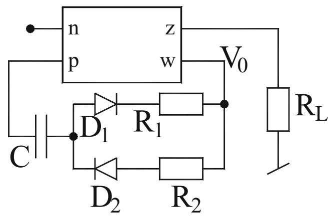


Fig. 14.12 Astable multivibrator having resistor-controlled duty cycle proposed by Pandey–Pandey–Paul–Anand–Gautam [10]

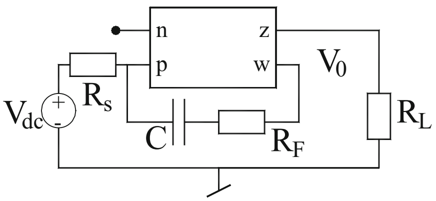


Fig. 14.13 Astable multivibrator having electronically controlled duty cycle proposed by Pandey–Pandey–Paul–Anand–Gautam [10]

$$f_0 = \frac{1}{T_{on} + T_{off}} = \frac{1}{2R_F C \ln[2R_L/R_F - 1]} \tag{14.56}$$

From the above, it is clear that for proper operation of the circuit one needs $R_L > R_F$.

The modification of the circuit to obtain adjustable duty cycle, other than 50 %, is straightforward; it simply requires replacement of the resistor $R_F$by a parallel combination of two series branches, each containing a diode and a resistor as shown in Fig. 14.11b. Following the same analysis as earlier, the time periods T_H and T_L are now modified to (Fig. 14.12)

$$T_{on} = R_2 C \ln\left(\frac{2R_L - R_1}{R_2}\right) \tag{14.57}$$

$$T_{off} = R_1 C \ln\left(\frac{2R_L - R_2}{R_1}\right) \tag{14.58}$$

It is, thus, seen that duty cycle other than 50 % can be obtained by judicious choice of the resistors R_1 and R_2.

By adding an external control signal V_{dc} along with a series resistor R_s as shown in Fig. 14.13, the circuit is easily converted into an astable multivibrator having electronically controllable duty cycle. Since the introduction of the voltage source would modify the two threshold voltages, the time periods T_H and T_L would also become functions of this external control voltage. Therefore, the duty cycle would also become function of this external control voltage.

The various claims made about the three circuits based on the CDBA described in this section have been verified by SPICE simulations and hardware implementation by realizing the CDBA by AD844-based implementation [10].

14.5 Electronically Controllable Schmitt-Trigger and Waveform Generators Using MO-CCCCTA

Of all the Schmitt triggers described in this monograph in general and in this chapter in particular, the most interesting one appears to be the one proposed by Siripruchyanun [11] employing a multiple output current-controlled current conveyor transconductance amplifier (MO-CCCCTA) which is shown in Fig. 14.14a. The most striking feature of this circuit is the complete absence of any external passive components which means that the entire circuit is physically implementable exclusively using transistors only, thereby making it suitable for IC implementation in both bipolar and CMOS technology.

It may be recalled that the MO-CCCCTA is characterized by the following hybrid matrix:

$$\begin{bmatrix} I_y \\ V_x \\ I_z \\ I_{0+} \\ I_{0-} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & R_x & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & +g_{m1} \\ 0 & 0 & 0 & 0 & -g_{m2} \end{bmatrix} \begin{bmatrix} V_y \\ I_x \\ V_o \\ V_{-o} \\ V_z \end{bmatrix} \tag{14.59}$$

where

$$R_x = \frac{V_T}{2I_{B1}}; \quad g_{m1} = \frac{I_{B2}}{2V_T}; \quad g_{m2} = \frac{I_{B3}}{2V_T} \tag{14.60}$$

If X-terminal is connected as a positive feedback to the Z-terminal, the voltage V_z can be found to be: $V_z \cong V_{cc}$ if $I_x \geq I_{0^-}$, and $V_z \cong V_{EE}$ if $I_x \leq I_{0^-}$. Since in an MO-CCCCTA, the output current is tanh function of the voltage V_z, it follows that

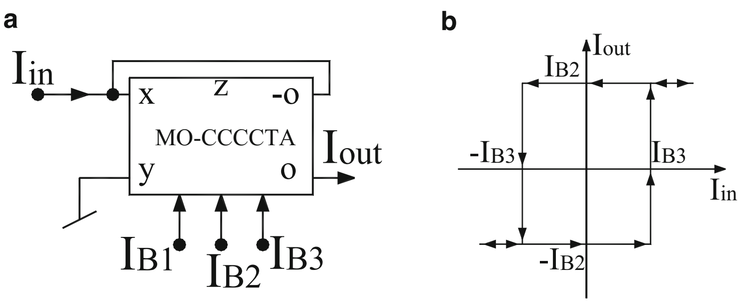


Fig. 14.14 Current-mode Schmitt trigger proposed by Siripruchyanun [11]: (**a**) circuit diagram, (**b**) transfer characteristics

if one assumes V_z to be >> than $2V_T$, the MO-CCCCTA would operate in the saturation mode so that the output currents would be given by

$$I_{out} \cong \begin{cases} I_{B2} & \text{for} \quad I_x \geq I_{0^-} \\ -I_{B2} & \text{for} \quad I_x \leq I_{0^-}, \end{cases} \tag{14.61}$$

Furthermore, the upper and lower threshold currents are given by the following equation and they are obviously linearly adjustable by the external current I_{B3}:

$$I_{TH} = -I_{B3}; \quad I_{TL} = I_{B3} \tag{14.62}$$

Under positive feedback, therefore, the MO-CCCCTA can be seen to be operating as a Schmitt trigger (see Fig. 14.14a) for which the transfer characteristic can be shown as in Fig. 14.14b.

This Schmitt trigger can be easily used to realize a square/triangular wave generator as shown in Fig. 14.15.

It may be noted that in this circuit, the MO-CCCCTA employed has one additional current output terminal 0^- where $I_{0^-} = -g_{m2}V_z = -\frac{I_{B4}}{2V_T}V_z$; therefore, looking into Eq. (14.60), it is clear that the employed MO-CCCCTA has four external control currents, namely I_{B1}, I_{B2}, I_{B3}, and I_{B4}.

From a routine analysis of this circuit [11], it is found that the peak-to-peak magnitude of the output current and the negative time interval T_1 are given by

$$I_{out(p-p)} = 2I_{B2}; \quad T_1 = \frac{2RCI_{B4}}{I_{B3}} \tag{14.63}$$

Since $T = 2T_1$, it follows that the frequency of the output waveforms would be given by

$$f = \frac{1}{T} = \frac{I_{B3}}{4RCI_{B4}} \tag{14.64}$$

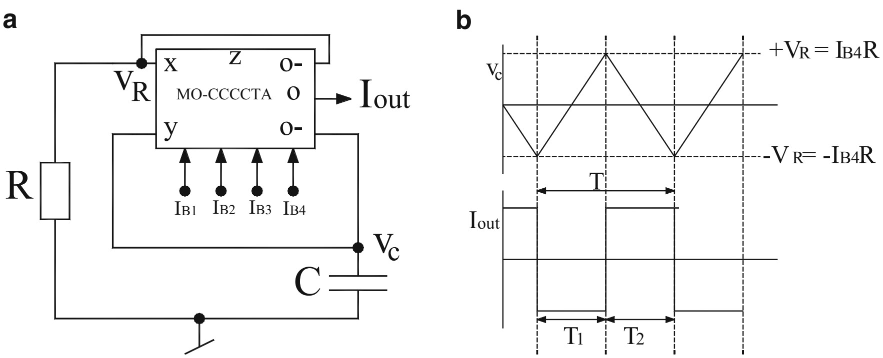


Fig. 14.15 Schmitt trigger used as a square/triangular wave generator proposed by Siripruchyanun [11]: (**a**) circuit diagram, (**b**) relevant wave forms

Thus, the circuit generates a square wave current output I_{out} with a triangular wave voltage output V_c across the capacitor. Furthermore, it is seen that frequency of the output waveforms is a linear function of the external current I_{B3} whereas the magnitude of the square wave is adjustable by another external current I_{B2}.

14.6 Electronically Controllable Current-Mode Schmitt Trigger and Relaxation Oscillators Using MO-CCCDTA

Silapan–Siripruchyanun in [12] carried forward the ideas of the previous section to develop equally versatile configuration of the Schmitt trigger, a square/triangular wave generator, and a monostable multivibrator, using a more versatile building block known as multiple output current-controlled current differencing transconductance amplifier (MO-CCCDTA). In this section, we highlight the main features of the various circuits and relaxation oscillators using MO-CCCDTA presented in [12].

The terminal relationship of the MO-CCCDTA can be defined by the following hybrid matrix:

$$\begin{bmatrix} V_p \\ V_n \\ I_z \\ I_x \\ I_{x-} \end{bmatrix} = \begin{bmatrix} R_p & 0 & 0 & 0 & 0 \\ 0 & R_n & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & g_{m1} \\ 0 & 0 & 0 & 0 & -g_{m2} \end{bmatrix} \begin{bmatrix} I_p \\ I_n \\ V_x \\ V_{x-} \\ V_z \end{bmatrix} \tag{14.65}$$

where

$$R_p = R_n = \frac{V_T}{2I_{B1}}; \quad g_{m1} = \frac{I_{B2}}{2V_T}, \quad g_{m2} = \frac{I_{B3}}{2V_T} \tag{14.66}$$

It is obvious that to make a Schmitt trigger, the active device has to operate in saturation; therefore one requires considering the value of the two output currents I_x and I_{x-} subject to a large voltage signal existing at port Z (voltage V_z). Since any circuit implementation of the MO-CCCDTA will have two transconductance amplifiers between ports-Z and X and Ports-Z and X^- and recalling that a differential transconductance is characterized by tan-hyperbolic characteristic between its output current and input voltage, it follows that under saturation, subject to a large input voltage, the output current delivered will be equal to the DC bias current supplied to the differential transconductance amplifier. From the above description, it follows that when MO-CCCDTA operates in saturation, the output currents I_x and I_{x-} would be governed by the following expressions:

$$I_{\mathrm{x}} = \begin{cases} I_{\mathrm{B2}} & \text{for} \quad I_{\mathrm{p}} \geq I_{\mathrm{n}}, \quad \text{when} \quad V_{\mathrm{z}} >> 2V_{\mathrm{T}} \\ -I_{\mathrm{B2}} & \text{for} \quad I_{\mathrm{p}} \leq I_{\mathrm{E}}, \quad \text{when} \quad V_{\mathrm{z}} >> 2V_{\mathrm{T}} \end{cases} \tag{14.67}$$

and

$$I_{\mathrm{x}-} = \begin{cases} I_{\mathrm{B3}} & \text{for} \quad I_{\mathrm{p}} \geq I_{\mathrm{n}}, \quad \text{when} \quad V_{\mathrm{z}} >> 2V_{\mathrm{T}} \\ I_{\mathrm{B3}} & \text{for} \quad I_{\mathrm{p}} \leq I_{\mathrm{n}}, \quad \text{when} \quad V_{\mathrm{z}} >> 2V_{\mathrm{T}} \end{cases} \tag{14.68}$$

Consider now the topology of the Schmitt trigger shown in Fig. 14.16a.

This circuit can be employed to create a transfer characteristic in current mode which could be either counterclockwise (CCW) hysteresis curve or a clockwise (CW) hysteresis curve depending upon whether the input signal I_{in} is applied to node "a" or node "b" of the circuit. If the input circuit is applied to node "b," it would function as CCW Schmitt trigger which can be explained as follows. The output current I_{out} has two possible saturation levels, either $+I_{\mathrm{B2}}$ or $-I_{\mathrm{B2}}$. If we assume that I_{out} is at its negative saturation level $-I_{\mathrm{B2}}$, the current $I_{\mathrm{x}-}$ would be equal to I_{B3}. At this time, input current I_{in} increases from zero but I_{out} would remain unchanged until I_{in} extends to be equal to $I_{\mathrm{x}-}$. When I_{in} becomes more than $I_{\mathrm{x}-}$, I_{out}

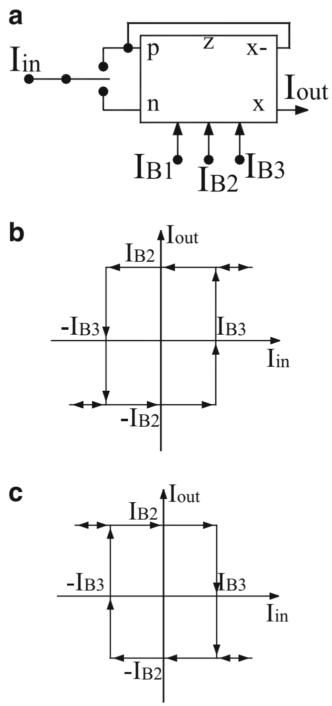


Fig. 14.16 Current-mode Schmitt trigger proposed by Silapan–Siripruchyanun (adapted from [12] © 2011 Springer): (**a**) circuit diagram, (**b**) transfer characteristic in CCW, (**c**) transfer characteristic in CW

will switch from $-I_{B2}$ to $+I_{B2}$. Meanwhile, I_{x-} level is now changed to $-I_{B3}$. The output current will not alter to $-I_{B2}$ until I_{in} is more negative than $-I_{B3.}$ This leads to the CCW transfer characteristic as shown in Fig. 14.16b. Likewise, it can be shown that if the input I_{in} is applied to node "a" the same circuit arrangement would realize a CW Schmitt trigger whose transfer characteristic is shown in Fig. 14.16c (for details the reader is referred to [12]).

This Schmitt trigger can be converted into a current mode relaxation oscillator by connecting a capacitor and resistor as shown in Fig. 14.17a. The wave forms for the voltage across the capacitor and the output current are shown in Fig. 14.17b.

Consider now the current-mode relaxation oscillator shown in Fig. 14.17a. For this circuit, the peak-to-peak magnitude of the output current is found to be $I_{out(p-p)} = 2I_{B2}$ and the time period T_1 is given by

$$T_1 = \frac{CV_T}{2I_{B1}} \ln\left(\frac{V_T + 4I_{B1}R}{V_T}\right) \tag{14.69}$$

Since $T_1 = T_2$, the frequency of the generated square wave is given by

$$f = \frac{1}{T} = \frac{I_{B1}}{CV_T \ln\left(\frac{V_T + 4I_{B1}R}{V_T}\right)} \tag{14.70}$$

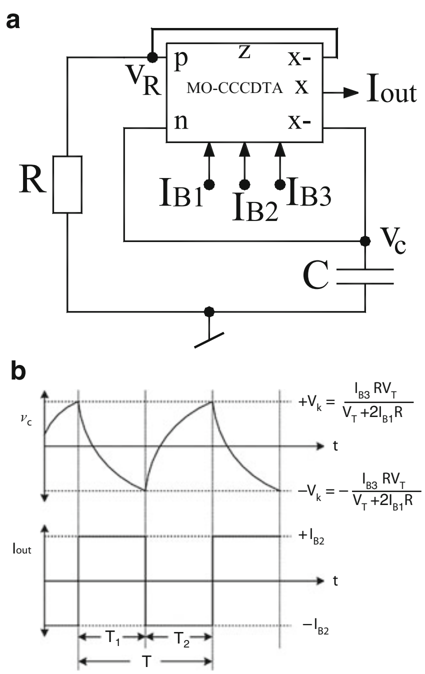


Fig. 14.17 Current-mode relaxation oscillator proposed by Silapan–Siripruchyanun (adapted from [12] © 2011 Springer): (**a**) circuit diagram, (**b**) relevant wave forms

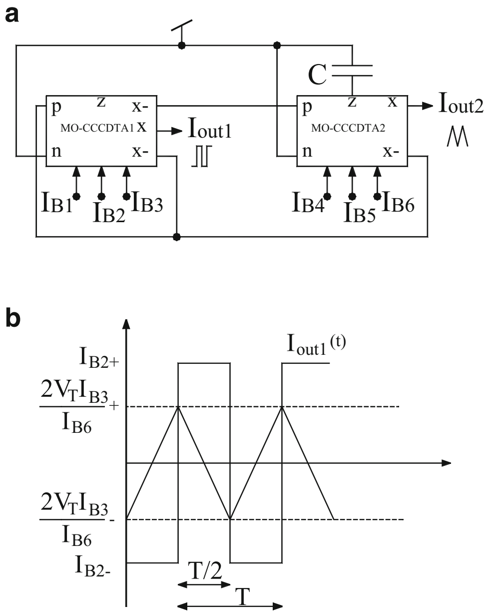


Fig. 14.18 Current-mode triangular/square wave generator proposed by Silapan–Siripruchyanun (adapted from [12] © 2011 Springer): (**a**) circuit diagram, (**b**) relevant waveforms

It is, therefore, seen that the magnitude of the square wave is electronically controllable by the external current I_{B2} whereas the frequency of the output waveform can be electronically controlled by another external current I_{B1}.

A modification of this circuit, by appropriate incorporation of an integrator realized from another MO-CCCDTA, is shown in Fig. 14.18a. In this circuit, I_{out1} is square wave whereas I_{out2} is a triangular wave. The relevant waveforms of the circuit are shown in Fig. 14.18b.

Analysis shows [12] that the peak amplitude of the two output currents is given by

$I_{out1} = I_{B2}$ and $I_{out2} = \frac{I_{B3} I_{B5}}{I_{B6}}$ whereas the frequency of the output waveforms is given by

$$f = \frac{1}{T} = \frac{I_{B6}}{8 C V_T} \tag{14.71}$$

The astable multivibrator described earlier is readily converted [12] into a monostable multivibrator circuit as shown in Fig. 14.19a.

This circuit requires an electronical switch "S" and also needs a current input trigger pulse as shown in Fig. 14.19b. By a routine analysis [12] and looking into the relevant waveforms shown in Fig. 14.19b, the peak-to-peak amplitude of the circuit and the time period of the pulse generated, T, are given by

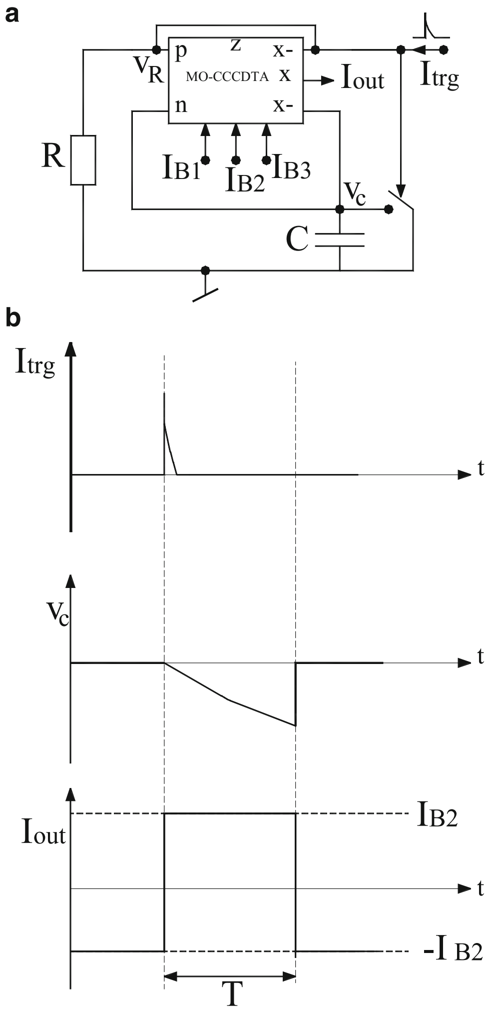


Fig. 14.19 Current-mode monostable multivibrator proposed by Silapan–Siripruchyanun (adapted from [12] © 2011 Springer): (**a**) circuit diagram, (**b**) relevant waveforms

$$I_{\text{out(p-p)}} = 2I_{\text{B2}} \quad \text{and} \quad T = \frac{V_{\text{T}}C}{2I_{\text{B1}}} \ln\left(\frac{4I_{\text{B1}}R + V_{\text{T}}}{2I_{\text{B1}}R + V_{\text{T}}}\right) \tag{14.72}$$

14.7 Miscellaneous Other Waveform Generators Using Other Building Blocks

Kongnun–Aurasopon [13] have represented an electronically controlled current-mode level shifted multicarrier pulse with modulator (PWM) based on multiple output current follower transconductance amplifier (MO-CFTA) whose operating principle is quite similar to that described above in the context of MO-CCCCTA- and MO-CCCDTA-based relaxation oscillator.

Chavoshisani–Hashemipour [14] have proposed a new differential second-generation current conveyor (DCCII)-based current comparator. A new DCCII architecture implementable in CMOS has been presented which is claimed to have smaller propagation delay (approximately 0.6 nS) and high accuracy. However, its application in realizing any relaxation oscillator/waveform generator does not appear to have been explored in the literature yet.

Sotner–Jerabek–Herencsar [15] have demonstrated the application of the active building block voltage differencing buffered amplifier (VDBA) and voltage differencing inverted buffered amplifier (VDIBA) in realizing triangular and square wave generators whose workability has been demonstrated by computer simulations with low-voltage TSMC 0.18 μm CMOS technology model.

In another publication, Sotner–Jerabek–Herencsar–Dostal–Vrba [16] introduced a new active building block called Z-copy controlled-gain voltage differencing current conveyor (ZC-CG-VDCC) which has the availability of three mutually independent and electronically adjustable parameters. This building block has been used to provide voltage-mode and current-mode square wave outputs with the feature of electronic controllability of various parameters.

Kubanek–Khateb–Vrba [17] have employed two universal current conveyors (UCC) to realize a novel square wave form generator with electronic tunability feature. The preposition is supplemented with SPICE simulation results using a CMOS UCC architecture implemented in 0.18 μm CMOS technology.

14.8 Concluding Remarks

In this chapter, we have dealt with Schmitt triggers, multivibrators, and waveform generators/relaxations oscillators using electronic circuit building blocks of recent origin which included OTRA, DVCC, CDBA, MO-CCCCTA, and MO-CCCDTA. Of course there are other building blocks also such as multiple output CFTA, differential current conveyor, differential buffered/inverted amplifier, voltage differencing current conveyor, and universal current conveyor which have been employed in the very recent literature to construct the circuits of the kind described in this chapter; the interested readers are referred to references [18–22] for further details.

From the brief account of the work presented in this chapter and looking into the fact that several of the references have appeared while this chapter was being written (July 2015), it is apparent that the work is still being carried out on the realization of various functional circuits dealt with in this chapter, as outlined in Sect. 14.7. In view of this, it is expected that a number of interesting and more efficient circuits might be waiting to be discovered in near future.

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Part IV
Current directions, Concluding remarks and additional references for further reading

Chapter 15
Current Directions of Research and Concluding Remarks

Abstract This chapter highlights current directions of research by presenting some key contributions of more recent origin which are of interest from the view point of oscillators/waveform generators employing modern electronic circuit building blocks. In the last, concluding remarks are made on the material presented in this monograph; while doing so, we also point out some open problems which some readers may find interesting for pursuing further.

15.1 Introduction

Since the object of this monograph has been to discuss the sinusoidal oscillators and waveform generators using modern electronic circuit building blocks, with the exception of log-domain, translinear, and square root domain oscillators (which also have some kind of building blocks employed in their realizations), we have not dealt with the oscillator circuits using discrete BJTs or MOSFETs directly as circuit elements. Also not included are papers dealing with the oscillator analysis methods, and those dealing with LC-VCOs, ring oscillators, crystal oscillators, and microwave oscillators. Besides this, a lot of work which has been done during the past two decades on investigation and analysis of the phase noise in oscillators has also not been dealt with *explicitly* in this book. Lastly, the important class of chaotic oscillators was also considered to be outside the scope of this monograph and chaotic oscillators too, and therefore have not been dealt with. Nevertheless, a number of references related to the topics which have been omitted have been provided in the *Additional list of references for further reading* given at the end (after this chapter), for the curious and interested readers.

R. Senani et al., *Sinusoidal Oscillators and Waveform Generators using Modern Electronic Circuit Building Blocks*, DOI 10.1007/978-3-319-23712-1_15

15.2 Current Directions of Research on Oscillators and Waveform Generators

In this section, we outline some developments which have taken place during the last few years only and on which only some opening has been made by researchers but a lot of work still appears to be done which might be explored in years to come. Since it is physically impossible to include everything in any monograph of this kind, these topics have not been dealt with in this monograph. In the following, we briefly present some of these more recent innovations related to sinusoidal oscillators and relaxation oscillators/waveform generators.

15.2.1 Oscillator Synthesis Using Pathological Elements

Nullators, norators, and nullors are well-established pathological elements whose unifying role in relating circuits using different devices, their applications in generating equivalent circuits of a given circuit, and their CMOS-compatible hardware realizations in the form of *four terminal floating nullors* (*FTFN*) are well known and well documented in literature; for instance, see [1].

Sometime back, Haigh and his co-workers did pioneering work in developing systematic procedures for synthesizing active networks using nullators and norators; see [2–5] and the references cited therein. Note that whereas a nullator is a two terminal element characterized by $V = 0$, $I = 0$, norator is also a two terminal element but is characterized by $V =$ arbitrary, $I =$ arbitrary. None of the two elements are physically realizable in isolation. However, a pair of nullator and norator as a three terminal nullor is physically realizable by an ideal BJT and an ideal FET and MOSFET and also by a CCII− whereas a three terminal nullor with one of the norators grounded is equivalent to an ideal op-amp. The more general FTFN is realizable by an interconnection of two CCII− or two CCII+ and therefore also by an interconnection of two CFOAs.

In 2002, Awad and Soliman [6] added two more network elements, namely the *voltage mirror* and the *current mirror* to the category of pathological elements. In contrast to nullator and norator, which are two terminal elements, a *voltage mirror* is a lossless two-port network element which embodies a voltage inversion action and is characterized by equations $V_1 = -V_2$ and $I_1 = I_2 = 0$ while on the other hand a *current mirror* has a current reversing action and is a two-port element characterized by $I_1 = I_2$ and $V_1 = V_2 =$ arbitrary. Like nullator and norator, none of these two new pathological elements are physically realizable. Furthermore, although these are two-port elements they are generally shown and used as two terminal elements with the third terminal implicitly assumed to be connected to ground. The importance of these mirror elements lies in the fact that they can be used to model most of the active elements (with the exception of an operational transconductance amplifier, an operational transresistance amplifier, and any other transconductance or transresistive elements) without using any resistors which cannot be modelled by nullators and norator without using matched resistors. The most common examples are those of a CCII+ and an inverting unity gain amplifier, both of which require two matched resistors if modelled by using nullators and norators but can be modelled by voltage and current mirrors without requiring any resistors.

Thus, with the introduction of these two new pathological elements, namely the current mirror and voltage mirror, to the set of two earlier known elements, namely the nullator and norator, it has been possible to construct pathological models of most of the active circuit building blocks in terms of these four elements only. As a consequence, there has been lot of activity in recent literature in evolving systematic synthesis procedures for realizing a variety of analog circuits such as impedance converters and inverters, mutators, and also oscillators using *nodal admittance matrix* (NAM) formulations based upon models of these elements in terms of these pathological elements and then synthesizing physical circuits from these NAMs employing pivotal expansion techniques [7].

The method of synthesizing circuits using NAM templates and the operation of pivotal expansion had long been evolved by Haigh and his group [2–5]. A number of researchers have employed and extended this methodology by adding current mirrors and voltage mirrors to the existing set of pathological elements. Thus, many modern electronic circuit building blocks such as DVCC, OFC, ICC II, BOCC II, DOICC II, FDVCC, DCVC (also known as CDBA), and MDCC, inverting/non-inverting current followers and voltage followers, balanced output op-amps (BOOA), differential difference amplifiers (DDA), DDOFA, and DDOMA are all representable by judicious combination of aforementioned four pathological elements. Soliman (see [8–13]) has made significant contribution on developing systematic synthesis procedures for generating complete families of oscillators using some of the abovementioned active building blocks. The interested reader is referred to [6–23] to know more about this particular methodology and its use in oscillator circuit synthesis. This area of oscillator synthesis, therefore, holds lot of scope for future work.

15.2.2 Fractional-Order Sinusoidal Oscillators

Recently, considerable attention is being given to fractional order circuits and systems, as a consequence of which a lot of work is being carried out on the realization of fractional order integrators, differentiators, inductors, and capacitors based upon the ideas from fractional calculus. A number of researchers have come up with the ideas leading to the design of fractional order oscillators too. In this area, the first reported contribution is probably the fractional order Wien bridge oscillator proposed by Ahmad–El-Khazali–Elwakil in 2001 [24]. Subsequently, there have been other investigations, both on mathematical aspects and on the circuital aspects of the fractional order oscillators. Radwan–Elwakil–Soliman [25] and Radwan–Soliman–Elwakil [26] have presented a number of variants of the classical Wien bridge oscillator and twin-T oscillators to derive the corresponding fractional order oscillators. Maundy–Elwakil–Gift [27], on the other hand, have considered a number of topologies of quadrature/multiphase oscillators using first-order all-pass sections along with inverting/non-inverting amplifiers in conjunction with a specific passive RC realization of the *fractional capacitors*. An advantage of this method appears to be that fractional order oscillators made from fractional order filters require lower time constant for the same frequency of oscillation than an integer order realization of the same function. More recently, in 2015, Said–Radwan–Madian–Soliman [28] have derived a family of fractional order oscillators based upon OTRA.

In view of the above, it is seen that the work on the fractional order synthesis using modern circuit building blocks has just begun and this constitutes an interesting area for further exploration.

15.2.3 Memristor-Based Oscillators

The *Memristor* was hypothesized by Chua in 1971, as the missing fourth basic element of circuit theory. Till 2008, no physically realizable device exhibiting memristor characteristics was known. Hence, the entire circuit theory of circuits composed of resistor, capacitor, inductor, and *memristor* so elegantly formulated by Chua in his seminal 1971 paper remained dormant in the literature for quite a long time. The topic, however, suddenly became alive and active again when HP labs reported the first practical implementation of the memristor in 2008.

In recent years, the memristors have attracted renewed attention in many pioneering applications such as in the design of DRAMs, signal processing, neural networks, programmable logic, and control systems. A number of researchers have also investigated the possible use of *memristors* in realizing both sinusoidal oscillators and relaxation oscillators.

Talukdar–Radwan–Salama [29] demonstrated the application of memristors in a family of four classical Wien bridge oscillators by replacing one or two of the resistors therein by memristors. This paper appears to be the first one where any memristor-based oscillators were presented. The same authors in [30] discussed nonlinear dynamics of memristor-based third-order oscillator system which was essentially an RC-phase shift oscillator with all the three resistors of the RC ladder therein replaced by memristive elements. Jun and Cheng [31] presented another memristor-based oscillator based on twin-T network employing a flux-controlled memristor. In between, Corinto–Ascoli–Gilli [32] made a general nonlinear dynamical study of the memristive oscillators. In [33], Mosad–Fauda–Khatib–Salama–Radwan presented a memristor-based relaxation oscillator and demonstrated that such oscillators provide higher frequency and wider tuning range than the existing reactance-less oscillators. More recently, Yu–Iu–Fitch–Liange [34] have presented a flux-controlled memristor emulator with floating terminals by making use of four CCs. This was used to design a typical relaxation oscillator which is nothing but the classical single op-amp-based astable multivibrator with a resistor replaced by memristor. Thus, three memristive relaxation oscillators were realized by replacing one of the resistors at a time by the proposed memristive emulator. The most attractive advance made by these formulations is the finding that the memristive relaxation oscillator can easily provide the adjustability of the duty cycle of the generated waveform which the classical astable multivibrator is not capable of.

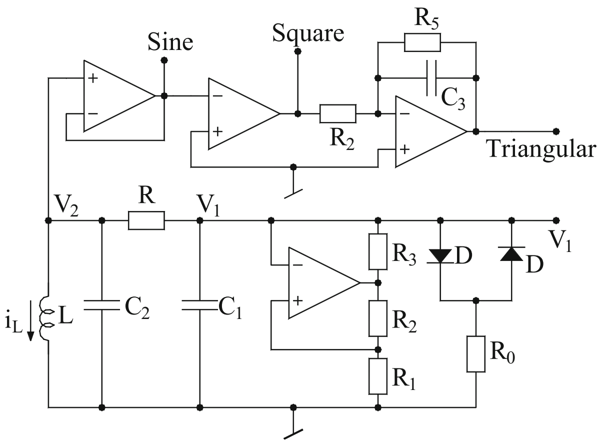
From the above brief discussion, it follows that there is enough scope of new ideas and new kind of circuit implementations of sinusoidal oscillators and relaxation oscillators using memristive elements.

15.2.4 Sine Wave, Square Wave, and Triangular Wave Generation from Chua's Chaotic Oscillator

A very unconventional way of creating a signal generator has recently been advanced by Campos-Canton, Campos-Canton, and Castellanos-Velasco in [35]. They have shown that the classical Chua's chaotic oscillator circuit, apart from generating the usual chaotic signal, can also be *tamed* in such a manner that it can also generate sinusoidal, triangular, and square wave signals. This, therefore, makes an interesting contribution in that a single function generator can be designed which can serve the purpose of producing sine, square, triangular, as well as chaotic waveforms as shown in Fig. 15.1.

The circuits of Fig. 15.1 contain, in fact, the Chua's circuit consisting of a third-order RLC circuit (C_1-C_2-L-R) and nonlinear negative resistor known as Chua's diode comprised of an op-*amp* along with resistors R_1, R_2, R_3, back-to-back parallel-connected diodes along with a resistor R_D. As is well known, the state equations of this circuit can be written as follows:

Fig 15.1 A general-purpose function generator proposed by Campos-Canton, Campos-Canton, and Castellanos-Velasco [35], capable of generating chaotic waveforms, as well as sinusoidal, square, and triangular waveforms



$$\frac{\mathrm{d}V_1}{\mathrm{d}t} = \frac{1}{RC_1}V_2 - \frac{1}{RC_1}V_1 - \frac{1}{C_1}I_\mathrm{x} \tag{15.1}$$

$$\frac{\mathrm{d}V_2}{\mathrm{d}t} = \frac{1}{RC_2}V_1 - \frac{1}{RC_2}V_2 - \frac{1}{C_2}I_\mathrm{L} \tag{15.2}$$

$$\frac{\mathrm{d}I_L}{\mathrm{d}t} = \frac{1}{L}V_2 \tag{15.3}$$

On the other hand, the Chua's diode which generates a three-segment piece-wise linear negative resistance is characterized by the following equations:

$$I_\mathrm{x} = m_1V_1 - \frac{1}{2}(m_0 - m_1)\left(\left|V_1 + V_\mathrm{D}\right| - \left|V_1 - V_\mathrm{D}\right|\right) \tag{15.4}$$

with

$$m_0 = -\frac{R_2}{R_1R_3} \quad \text{and} \quad m_1 = -\frac{R_2}{R_1R_3} + \frac{1}{R_0} \tag{15.5}$$

It has been shown in [35] that if the circuit parameters are chosen to be $L = 18$ mH, $C_1 = 10$ nF, $C_2 = 10$ nF, $R = 1.6$ KΩ, $R_0 = 1.2$ KΩ, $R_1 = 750$ Ω, $R_2 = 220$ Ω, and $R_3 = 200$ Ω, the circuit generates chaotic signals V_1 and V_2. The phase plane trajectory, which exhibits the well-known *double-scroll attractor*, is shown in Fig. 15.2a while the chaotic waveforms of the signals V_1 and V_2 are shown in Fig. 15.2b, c. On the other hand, if the parameters are to be chosen such that the circuit functions as a normal third-order periodic oscillator generating a sine wave, these have been found to be $L = 18$ mH, $C_1 = 10$ nF, $C_2 = 100$ nF, $R = 1.54$ KΩ, $R_0 = 1.2$ KΩ, $R_1 = 750$ Ω, $R_2 = 220$ Ω, and $R_3 = 220$ Ω. In this case, the signal V_2 has been found to be a stable sinusoidal waveform as shown in Fig. 15.3a which, when applied to a comparator (used as zero crossing detector), generates a square wave as shown in Fig. 15.3b. Finally, the square wave, when applied to an integrator, generates a triangular waveform as shown in Fig. 15.3c. In the entire setup of Fig. 15.1, the op-amps used were TL081. It has been shown that the oscillating frequency can be varied from 100 to 5 kHz.

From this unusual example it is clear that in view of a wide variety of implementations of the Chua's oscillator prevalent in the literature, many of them with inductor replaced by simulated inductors using op-amps, CFOAs, and a number of other building blocks, together with the availability of a large number of other op-amp-based chaotic oscillators, the possibility of discovering a more economical, and more efficient circuit capable of performing the same tasks, cannot be ruled out.

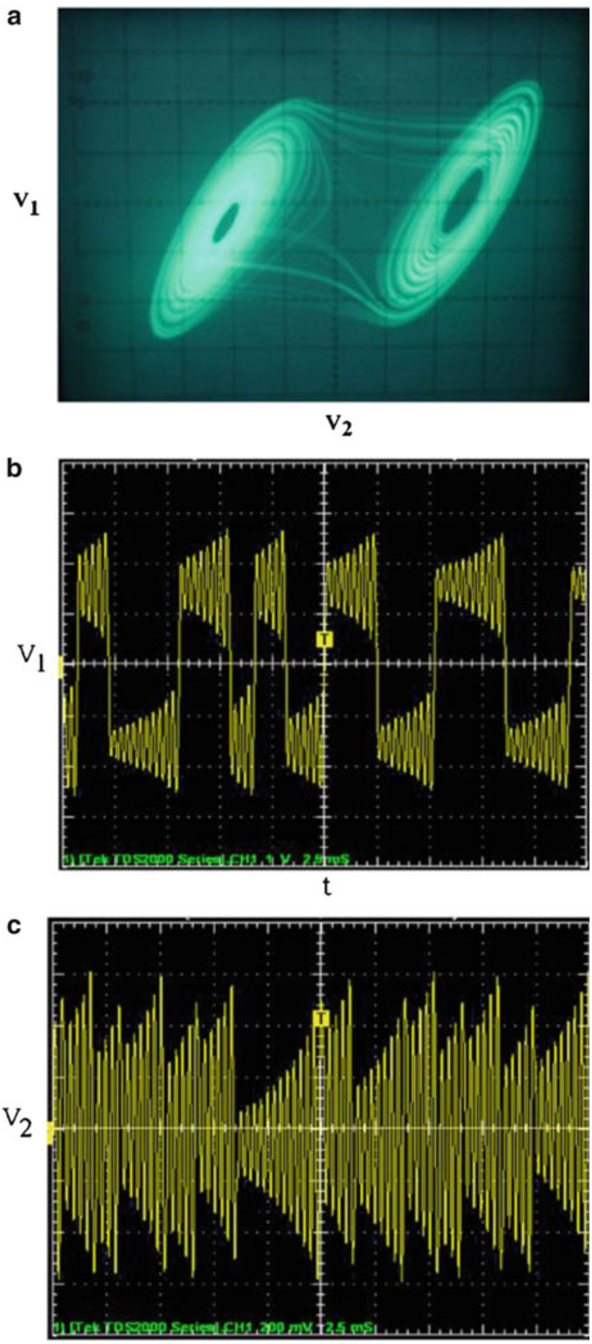


Fig. 15.2 (**a**) Double-scroll attractor, (**b**) chaotic waveforms of the signal V_1, (**c**) chaotic waveform of the signal V_2

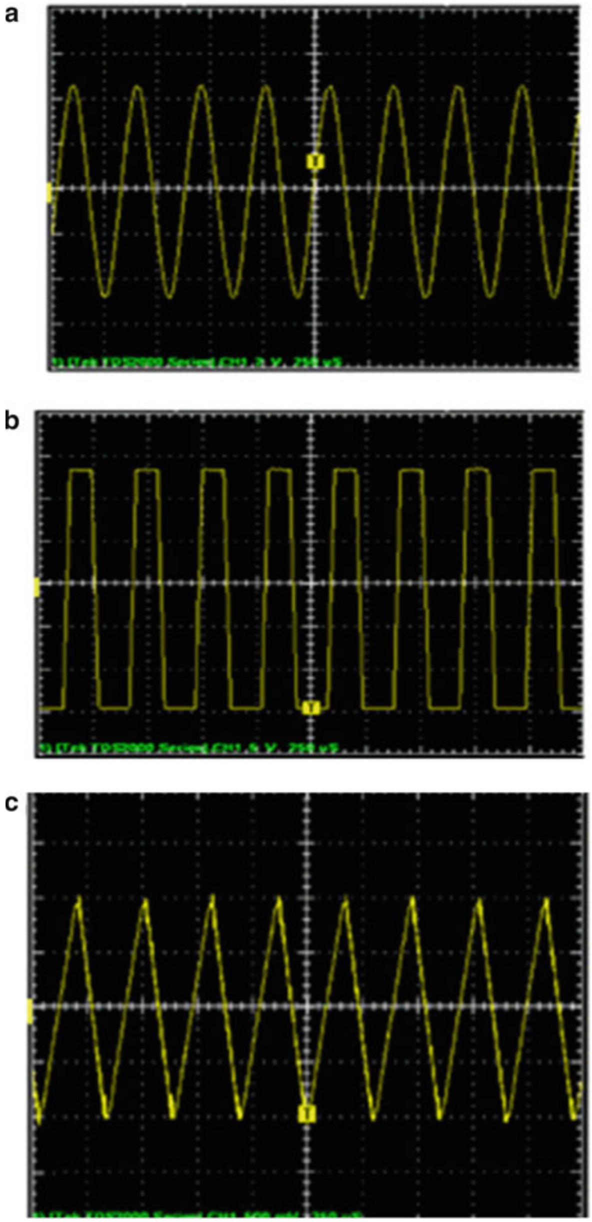


Fig. 15.3 (**a**) Stable sine wave generated at node V_2, (**b**) square wave generated at the output of A_2, (**c**) triangular wave generated at the output of A_3

15.2.5 Counter Examples to Barkhausen Criterion and Oscillator Start-Up Issues

Recently, there have been a number of comments on Barkhausen criterion including a number of counter examples to the applicability of the Barkhausen criterion; see [36–41]. Also, a number of authors have attempted to relate Barkhausen criterion and Nyquist stability criterion [42–44]. Questions have also been raised recently in [40, 41] (particularly in the light of the doubts about the validity of Barkhausen criterion) whether single-element-controlled oscillators and fully uncoupled oscillators (hundreds of varieties of which have been described in the literature so far by numerous researchers around the world (including the first author[1] of [40, 41]!) for over four decades now), are indeed what they are claimed to be?

While the comments in [36–41] make an interesting reading, the authors of this monograph believe that the issues raised in these communications are not settled *clearly*, *un-ambiguously*, and *conclusively* yet either in favor or against the arguments given by the various commentators of [36–41].

It appears that a lot of analysis or experimentation/simulation or a combination of both is still needed to derive correct inference from the publications [36–41] before any satisfactory resolution, widely acceptable to the research community working on oscillators, is reached. On the other hand, the discussions in [36–41] undoubtedly have given some food for thought and need to be addressed to, *appropriately*, *judiciously*, and *seriously*.

15.3 Concluding Remarks

This monograph has covered a wide variety of sinusoidal oscillators and relaxation oscillators/waveform generators using a large number of active building blocks (ABB).

All the available ABBs can be broadly classified into two main categories: (1) those which are commercially available as off-the-shelf integrated circuits such as op-amps (hundreds of varieties of which are produced by the IC manufacturers around the world), operational transconductance amplifiers (OTA), current conveyors (CC), and current feedback operational amplifiers (CFOA) and (2) circuit building blocks including numerous varieties of CCs (such as CCIII, DVCC, DDCC, ICCII, DXCC, FDCC, and others), DDA, FTFN, OTRA, CDBA, CCTA, CDBA, CFTA, DBTA, VDIBA, VD-DIBA, etc., a large variety of bipolar/CMOS/BiCMOS implementations of whom have been advanced by various researchers

[1] Curiously, the first author of [40, 41] has authored/coauthored hundreds of research papers on oscillators, including single-element-controlled oscillators and fully uncoupled oscillators. Ironically, *his contention in his hundreds of earlier publications* and his recent arguments in [40, 41] are in *complete contradiction!*

and presented in the literature but which are not yet available as off-the-shelf ICs. A brief outline of the material given in this monograph is as follows:

- The first chapter contains a detailed coverage of well-known classical as well as some lesser known op-amp-based sinusoidal oscillators. In the category of nonsinusoidal oscillators, a variety of circuits have been dealt with. Thus, we have covered a number of classical op-amp-based sinusoidal oscillators as well as a variety of nonsinusoidal wave form generators using op-amps, IC 555 timers, and op-amp-timer combinations. Since the presentation is almost in a tutorial review format, it is believed that the material of this chapter can serve as appropriate tutorial on sinusoidal oscillators for a wider domain of audience.
- An exclusive coverage of a large variety of op-amp-based single-element-controlled oscillators, active-R and partially active-R oscillators, and linear VCOs has been dealt with in Chap. 2.
- An important class of electronically tunable oscillators is obtained by devising oscillators based upon OTAs as active elements. Since commercially available IC OTAs such as LM3080/LM13600 provide electronic controllability of its transconductance over a wide range of external DC bias current (usually variable over four decades from 0.1 uA till 1 mA), a wide range of variable frequency is attainable with OTA-C oscillators. Thus, OTA-C oscillators realizable from the off-the-shelf OTAs make it possible to realize linear current-controlled oscillators. On the other hand, since bipolar/MOS OTAs do not have any resistor inside the chip and since OTA-C oscillators can always be devised with no external resistors, such oscillators are also suitable for IC implementation. A wide variety of such electronically controllable sinusoidal oscillators, employing OTAs, were discussed in Chap. 3.
- Chapter 4 dealt with a large variety of sinusoidal oscillators employing the basic current conveyors as active elements.
- Chapter 5 elaborated a large number of sinusoidal oscillators using the current feedback operational amplifiers with particular emphasis on a very popular and flexible CFOA, namely the AD844 from Analog Devices Inc. An easy way of obtaining electronically tunable oscillators is based upon SRCOs by replacing controlling resistance with MOSFET-based linear VCRs. On the other hand, oscillators providing linear control of frequency of oscillations can also be designed by appropriately embedding two analog multipliers. Thus, VCOs realizable with CFOAs and FET-based linear VCRs with nonlinearities cancelled and linear VCOs realizable with IC CFOAs and AMs both have been dealt with, in this chapter.
- In Chap. 6, we dealt with sinusoidal oscillators made from a large variety of modern electronic circuit building blocks which included ICCII, DVCC, DDCC, FDCCII, CDBA, OTRA, and several others.
- The switched-capacitor (SC) oscillators, which provide electronic tunability of oscillating frequency by controlling the frequency of the clock used to control

the MOS switches, were considered in Chap. 7 which also dealt with switched-current (SI) oscillators and the continuous-time MOSFET-C oscillators, which are as suitable for full integration in MOS technology as MOS switched-capacitor oscillators. However, in contrast to the SC and SI oscillators, which are discrete-time systems and therefore suffer from several characteristic limitations like effects of aliasing and clock-feed-through, the MOSFET-C oscillators are continuous-time circuits and therefore do not have these drawbacks. The MOSFET-C oscillators employ CMOS op-amps, MOS capacitors, and MOFETs (as equivalent of linear resistors in one form or the other). Thus, in MOSFET-C oscillators also, the frequency of oscillation is controllable by external control voltages applied to the gate of the MOSFETs. MOSFET-C oscillators using a variety of active elements such as CFOAs, DDAs, and OTRAs were presented.

- A fourth category of electronically controllable oscillators emerges from the use of those building blocks which have some parameters electronically controllable. Among such building blocks come the elements such as CCCII, CC-CFOA, CC-CDBA, CC-CDTA, CC-CCTA, and numerous others. A common factor in all these building blocks is that their front end invariably consists of one or more mixed-translinear-cell (MTC) which has its input resistances electronically controllable through an external DC bias current. There is another class of electronically controllable ABBs in which some other parameter of the building block, quite often the current gain between two port currents, is made electronically tunable. The building blocks like ZC-CG-CDTA belong to this latter class of ABBs. In Chap. 8 of this monograph, attention has been devoted to such electronically controllable sinusoidal oscillators synthesized using the quoted type of active building blocks.
- Log domain, translinear, and square-root domain oscillators are another class of oscillators which provide fully integratable circuits capable of providing electronically controllable sinusoidal oscillators. Prominent developments in this important class of oscillators were dealt in Chap. 9.
- Generating a number of solutions for a given problem is an important and interesting task for an engineer as it gives him/her a number of alternatives from which the most desirable solution can be chosen in accordance with the requirements of the given situation and design constraints. With this motivation in mind, a number of techniques using some network transformations based upon the notion of adjoints, network transposition, and theories based upon the nullor representation of sinusoidal oscillators have been proposed by various researchers from time to time. In Chap. 10, we have presented several methods of generating a number of equivalents of a given oscillator and have brought out the salient features of the various proposed methodologies.
- The issue of the amplitude stabilization and control of oscillators and some prominent general methods of stabilizing the amplitude of voltage-mode as well as current-mode oscillators were highlighted in Chap. 11.
- Other than the IC op-amp, the IC OTA is the other building block which has been prominently employed to realize square wave, triangle wave, and other kinds of waveform with the attendant advantage of providing electronic control

(by means of external current signals) of the time period, duty cycle, and sometimes even the magnitude of the generated waveform. Various waveform generators and relaxation oscillators/multivibrators based on OTAs were discussed in Chap. 12. This includes a variety of Schmitt triggers, astable multivibrators, monostable multivibrators, and pulse-width-modulators. Most of the circuits can be practically implemented using off-the-shelf IC OTAs. The advantages and limitations of the various circuits have been highlighted and a number of promising ideas for further work have also been pointed out.

- Contributions to the domain of relaxation oscillators/nonsinusoidal oscillators using IC CC and CFOAs, both of which are conveniently realizable with the commercially available AD 844 type IC, have been elaborated in Chap. 13. This includes several circuits which may appear to be either analogous or derived from the corresponding op-amp-based circuits known earlier but also includes a number of configurations which can be realized only with CCs and/or CFOAs and which do not have any op-amp-based counterparts.
- In the Chap. 14, we have described a number of configurations of relaxation oscillators/waveform generators using modern building blocks such as DVCCs, OTRAs, CDBAs CDTAs, CCTAs, and others. Although these building blocks are not commercially available yet, nevertheless, some of these can be realized using other commercially available IC components such as IC CFOA, IC VF, IC OTA, and others and thus several circuits out of the present lot can be put to use. However, the full potential of these circuits would be attainable only when these building blocks are commercially made available as off-the-shelf ICs.

15.4 Epilogue

In view of the vast repertoire of the circuits presented in this monograph, it is believed that this monograph should serve as a good reference for both academicians and research scholars. Besides this it should also serve as comprehensive reference for practicing engineers as a useful catalog of sinusoidal oscillators and waveform generators. Lastly, for the research-minded readers, we have quite often outlined a number of ideas and problems in most of the chapters of this monograph which will provide a good food for thought.

It is obvious that an area which has witnessed the publication of over 1500 research papers cannot be contained in single monograph while keeping the material limited to a judicious number of pages. Moreover, since the areas dealt in this monograph are very dynamic in nature in which research work is still continuing, it is impossible to cover all areas and all aspects completely. Although an extensive list of references is provided at the end of each chapter, after the end of this chapter, we have provided over 400 *additional references for further reading* related to the types of oscillators covered in this monograph as well as those types of oscillators not covered herein. We do hope that this should be useful to the readers wanting for more!

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